Gasoline Profiles and Impact of Ethanol Blending in Europe

September 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

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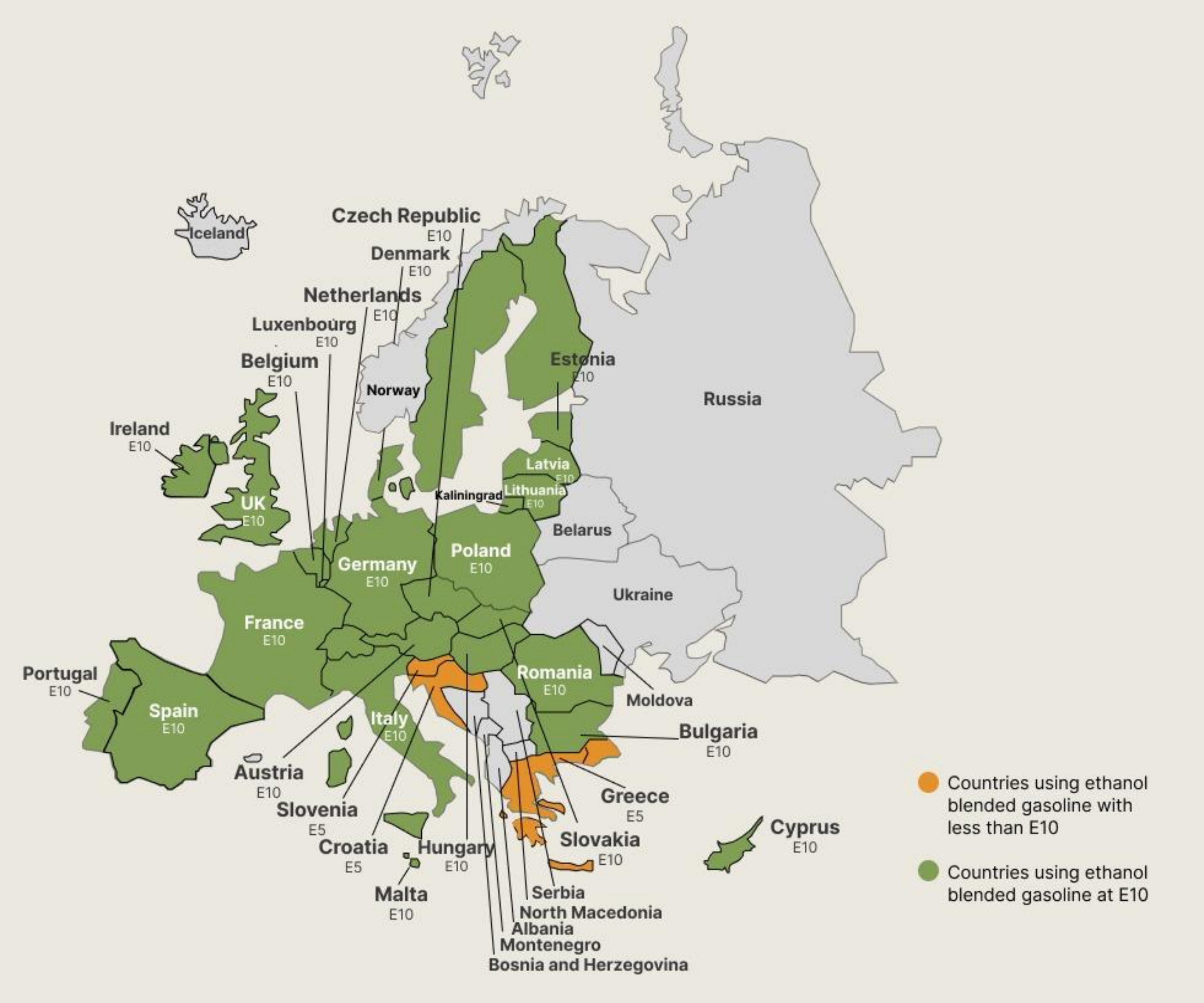
Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

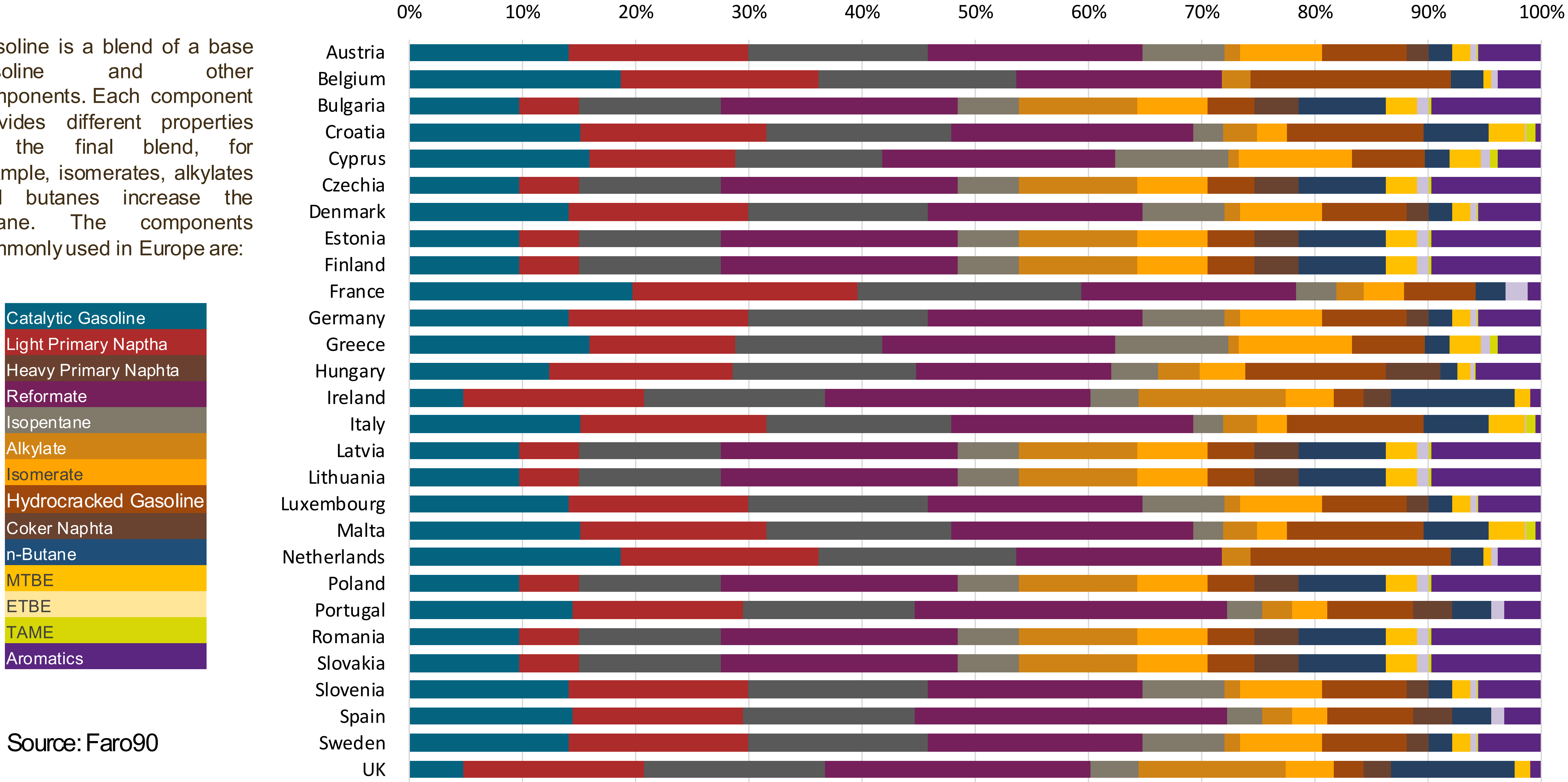
- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

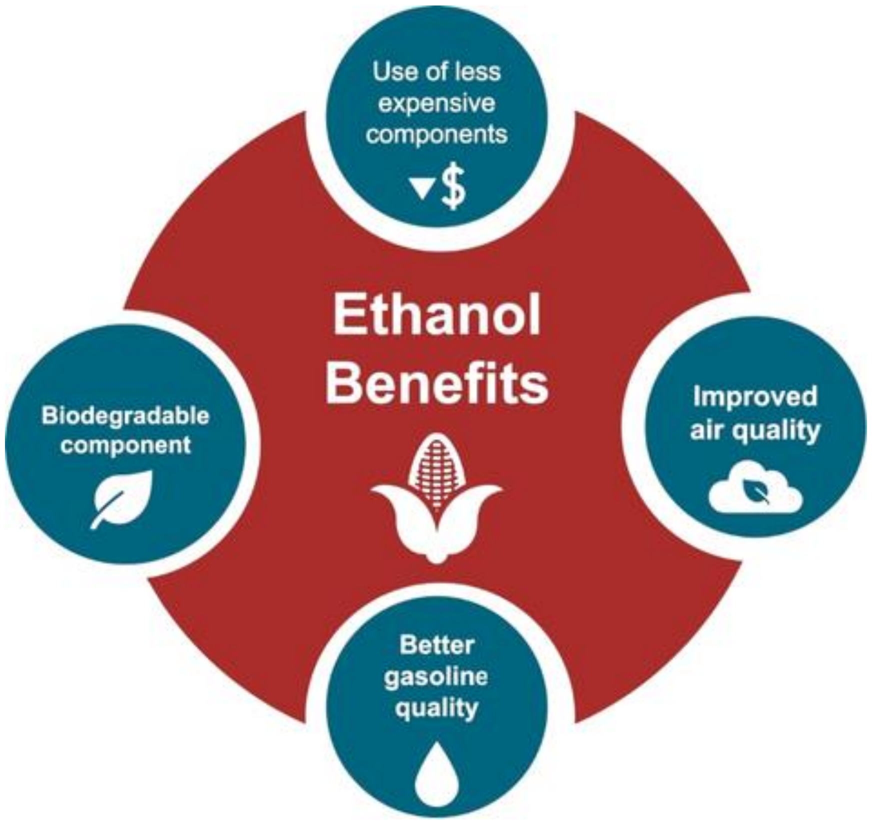
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Vehicle Emission Impact for Ethanol Gasoline Blending

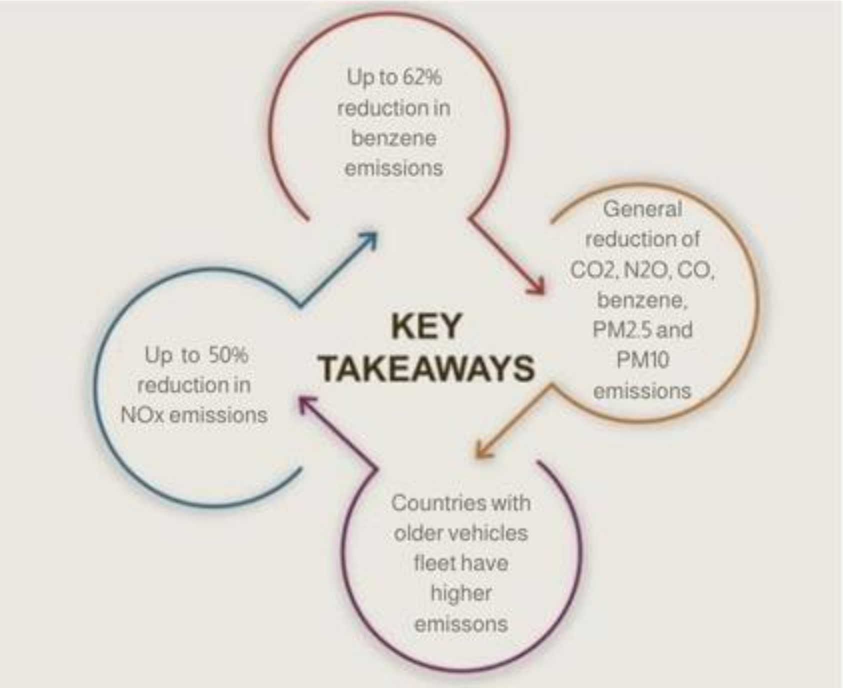
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

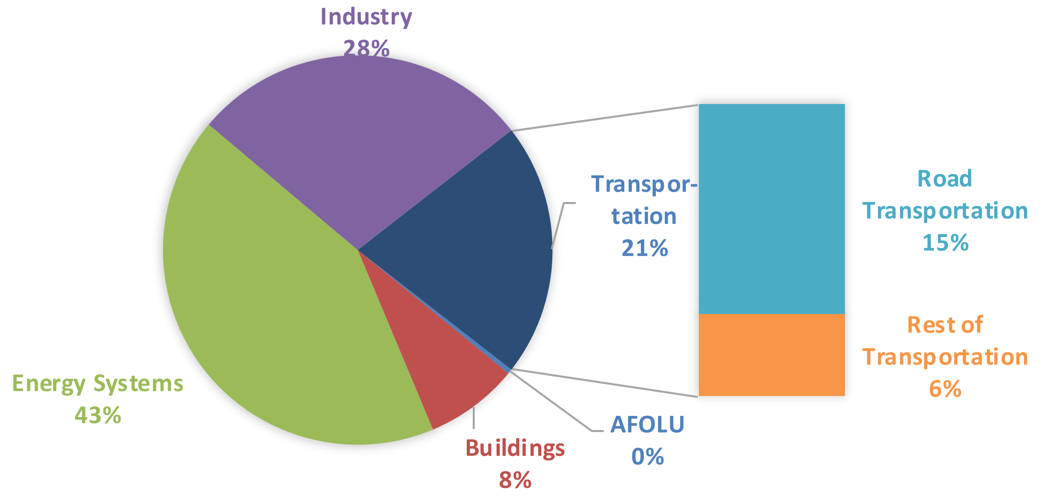
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

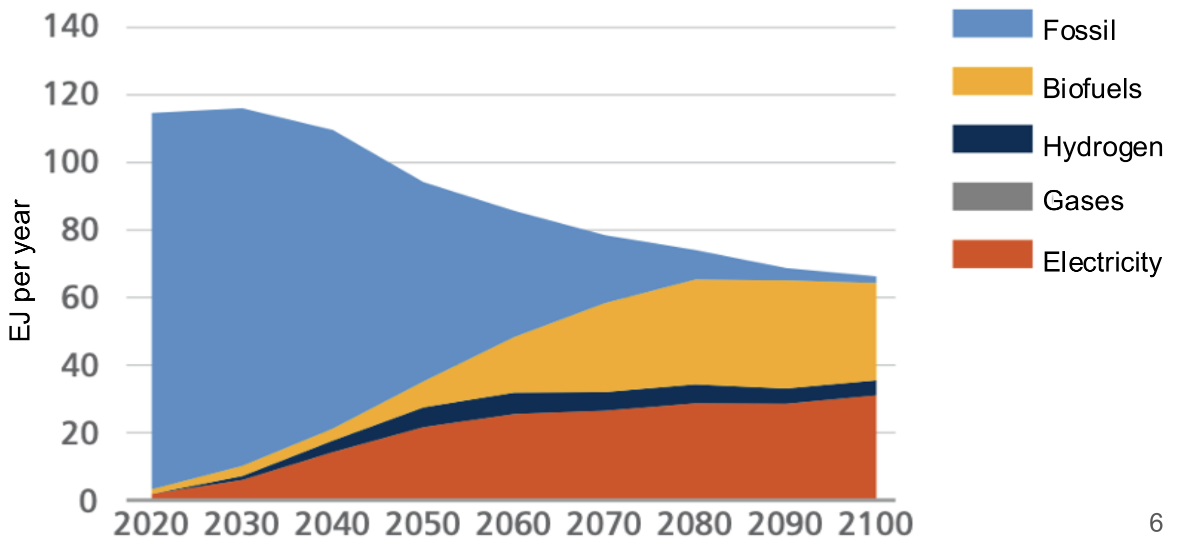
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Case Studies

1. Austria
2. Belgium
3. Bulgaria
4. Croatia
5. Cyprus
6. Czechia
7. Denmark
8. Estonia
9. Finland
10. France
11. Germany
12. Greece
13. Hungary
14. Ireland
15. Italy
16. Latvia
17. Lithuania
18. Luxembourg
19. Malta
20. Netherlands
21. Poland
22. Portugal
23. Romania
24. Slovakia
25. Slovenia
26. Spain
27. Sweden
28. United Kingdom

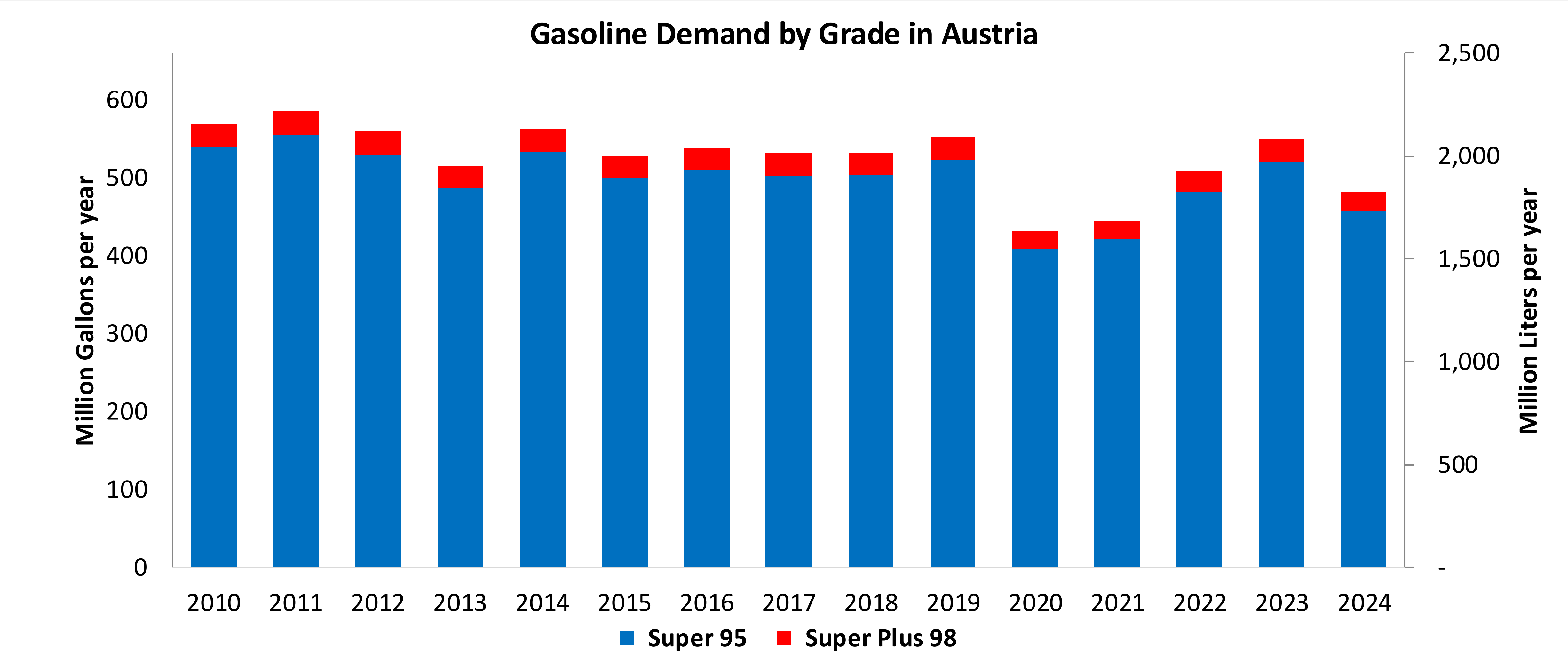
Ethanol Gasoline Blending in Austria

In 2024, gasoline consumption in Austria was 1,829 million liters (483 million gallons) and growth rate was -2.7%. Gasoline production and net imports were 2,317 and -392 million liters (612 and -103 million gallons) correspondingly. Austria complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Austria was 185 million liters (49 million gallons) representing 10.1% of gasoline demand. Ethanol production and Net Imports were 297 and -112 million liters (49 and -30 million gallons) correspondingly. Ethanol sources are Corn 75%, Cereals 15%, and Sugar Beet 10%.

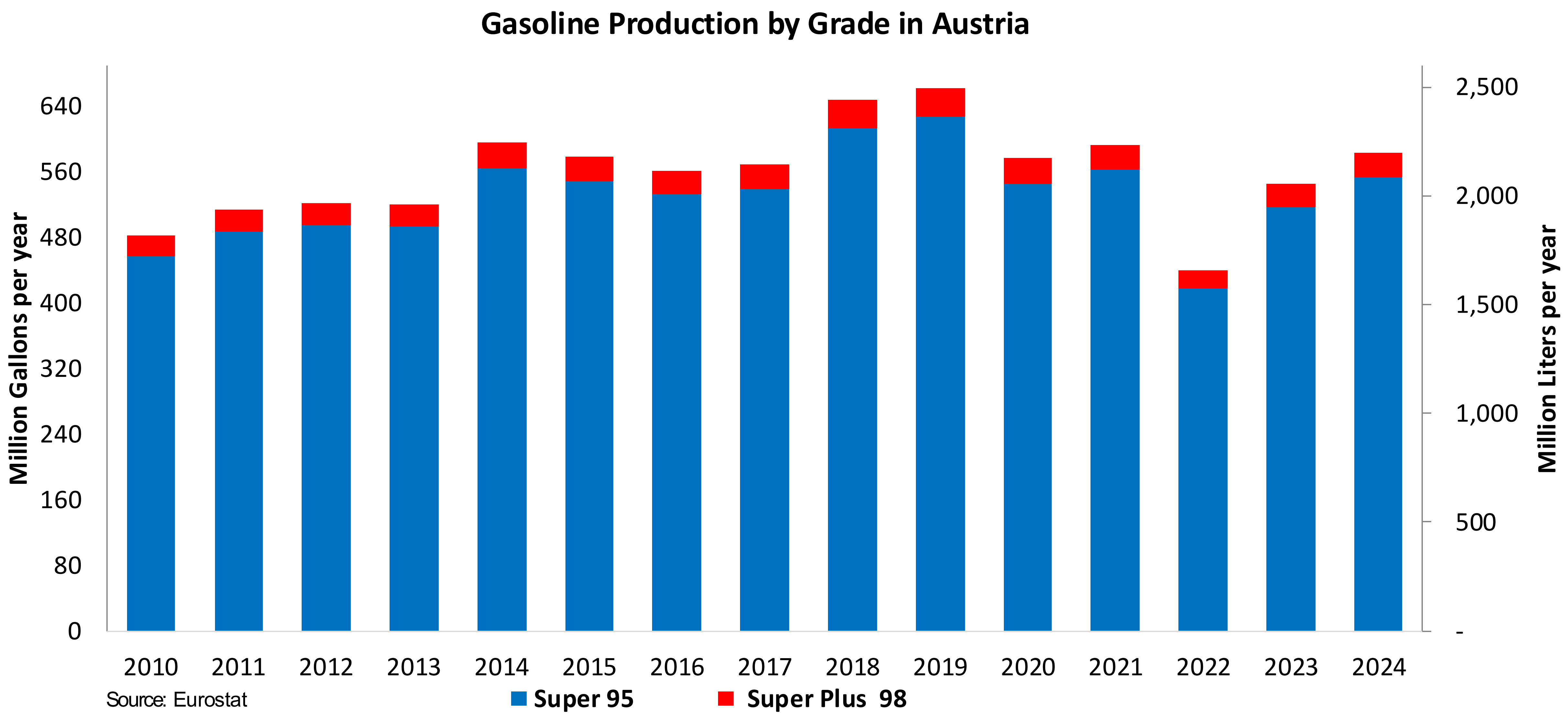
Source: eurostat

Gasoline Demand in Austria



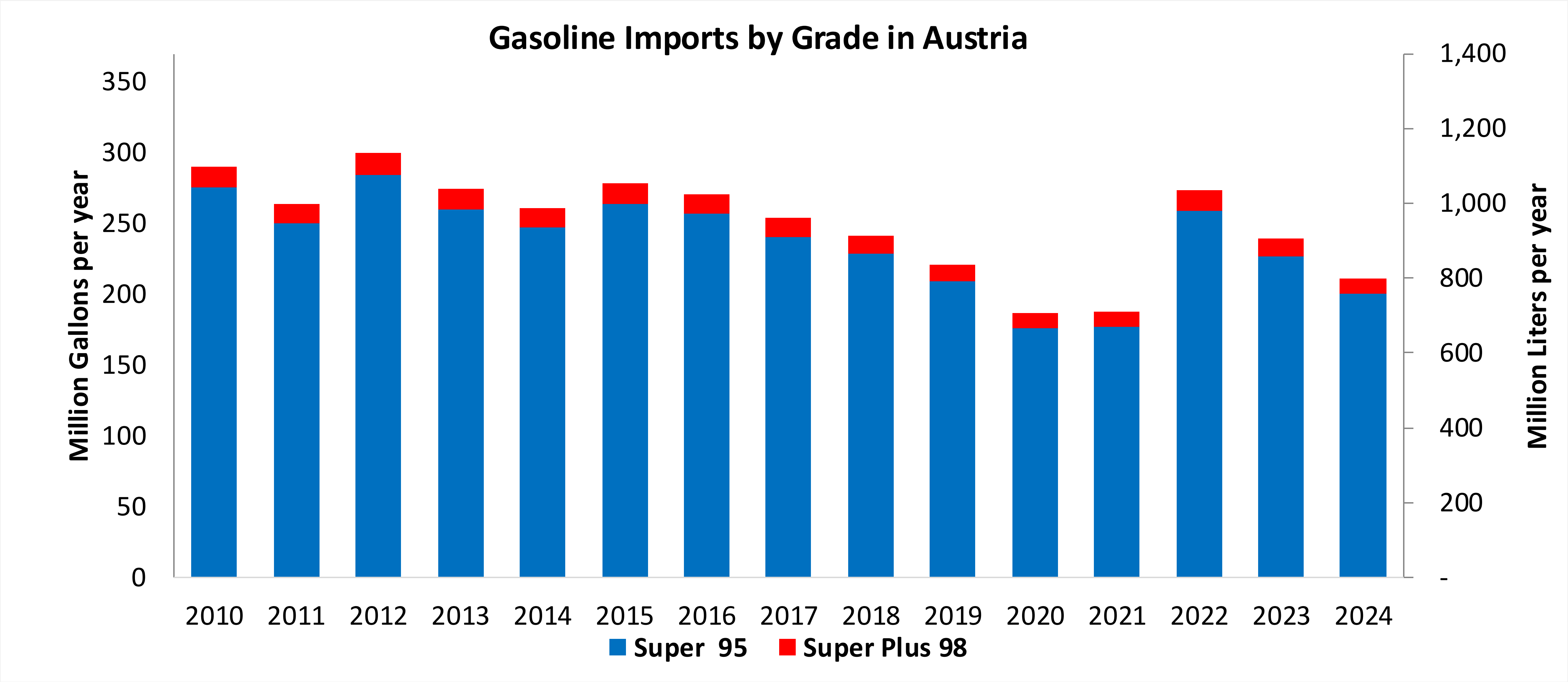
Source: Eurostat

Gasoline Production in Austria



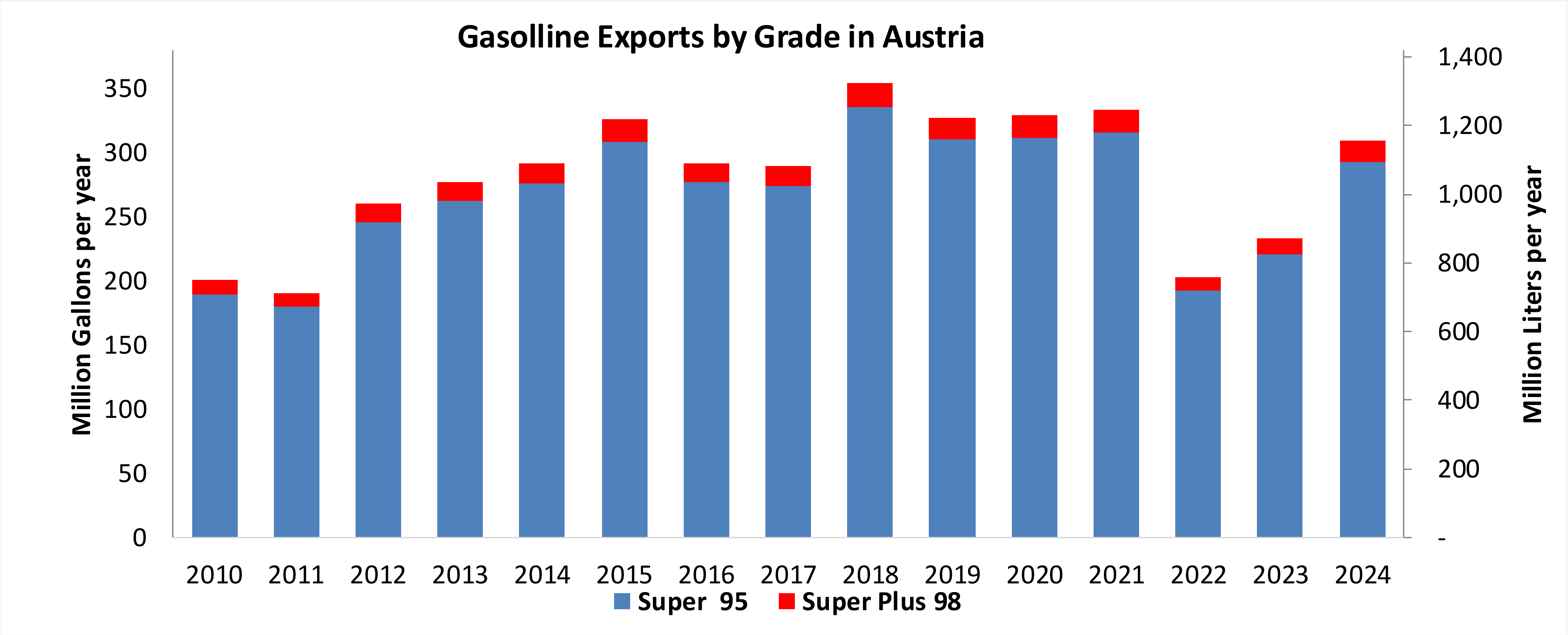
Source: Eurostat

Gasoline Imports in Austria



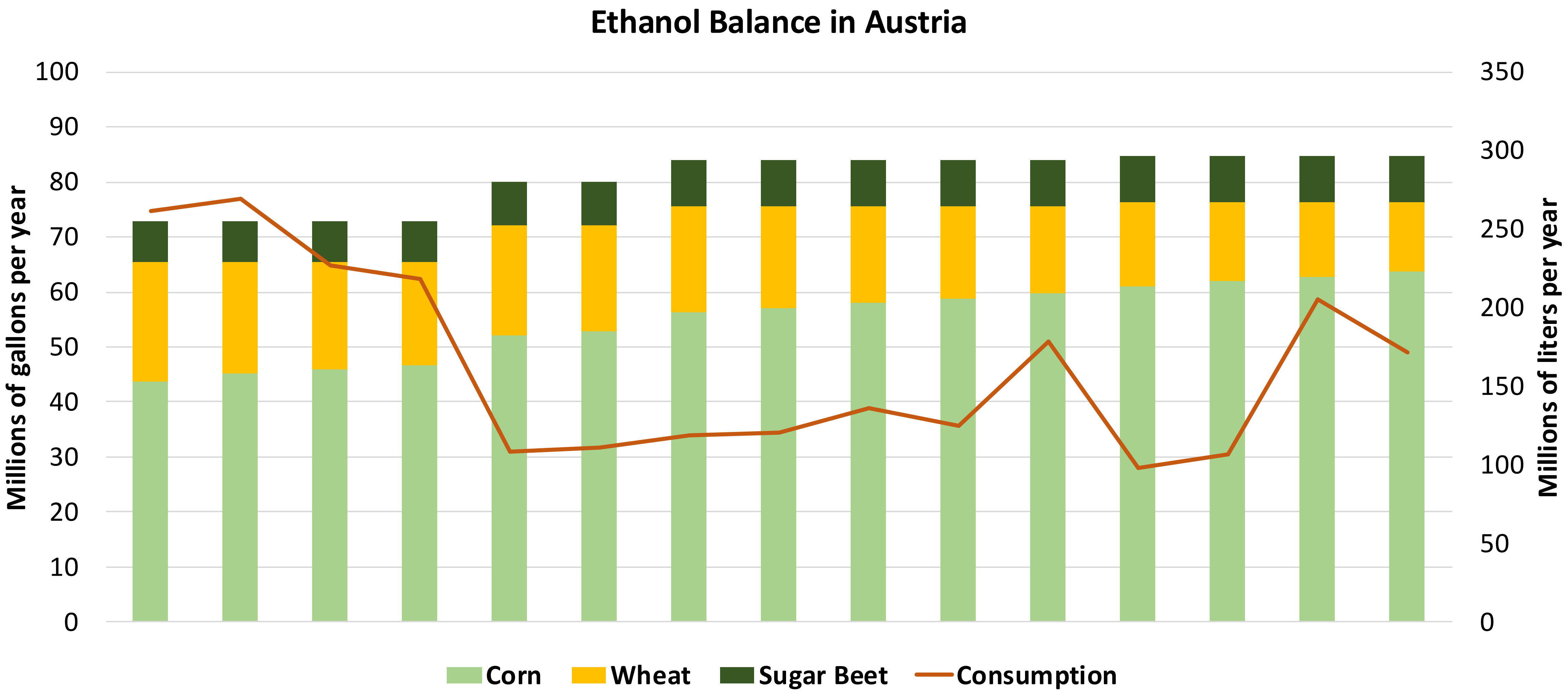
Source: Eurostat

Gasoline Exports in Austria



Source: Eurostat

Ethanol Balance in Austria



Source: Eurostat

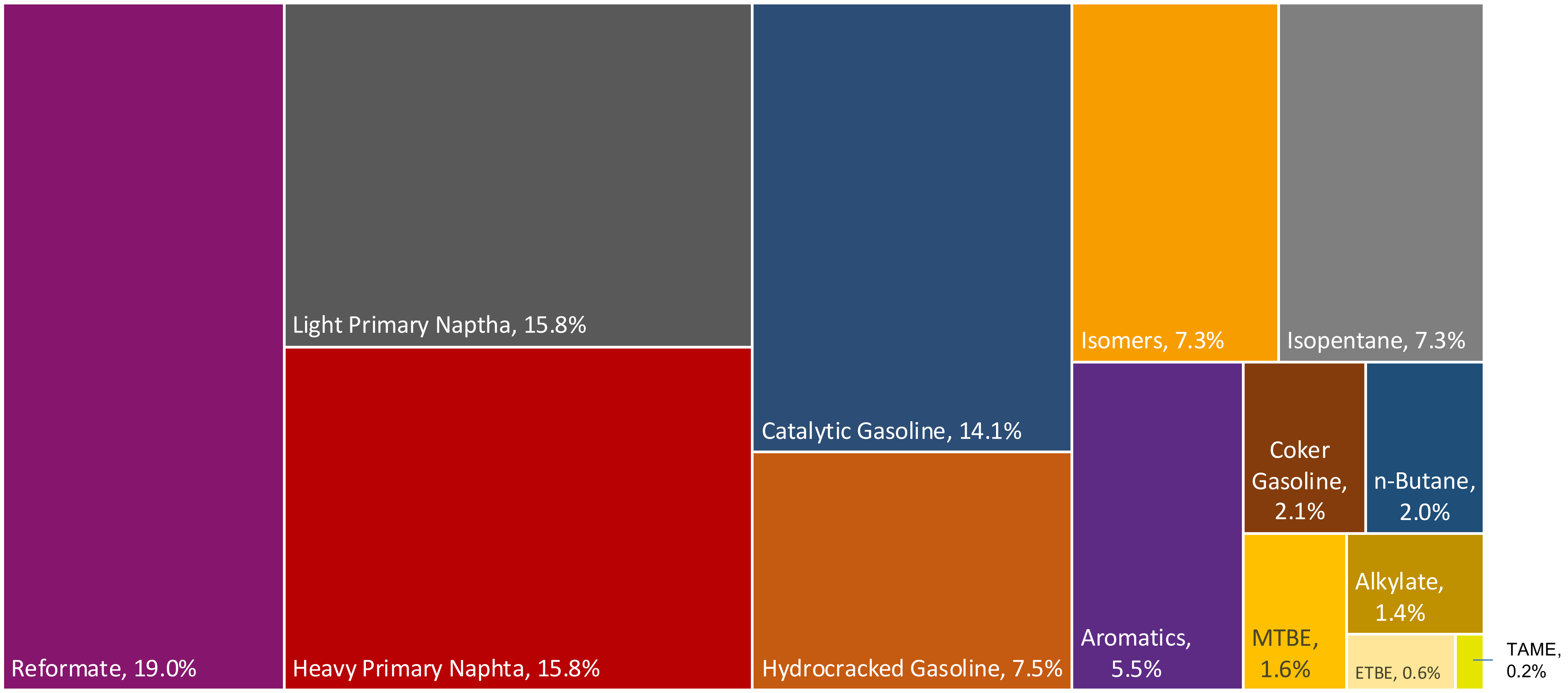
Gasoline Quality in Austria

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	3.4							
GHG Emission Reduction (%)	7.5							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

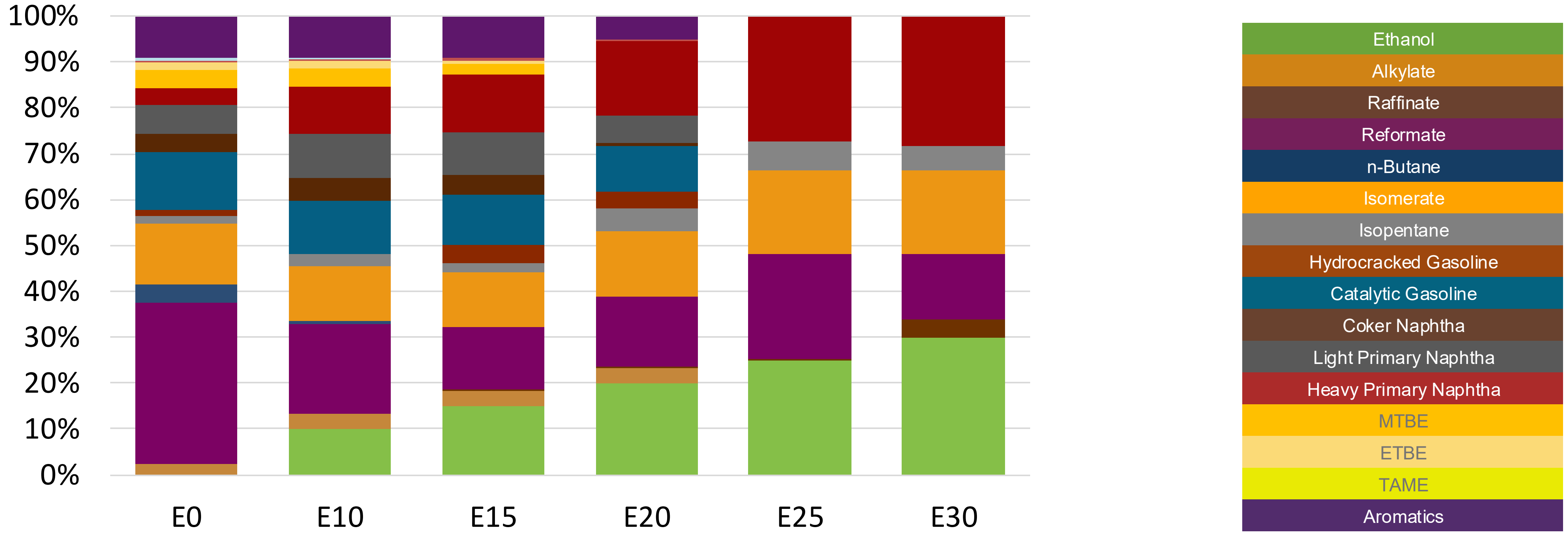
Source: European Commision

Austria Available Blending Components



Source: Faro90

Austria – Gasoline 95 – Constant Octane

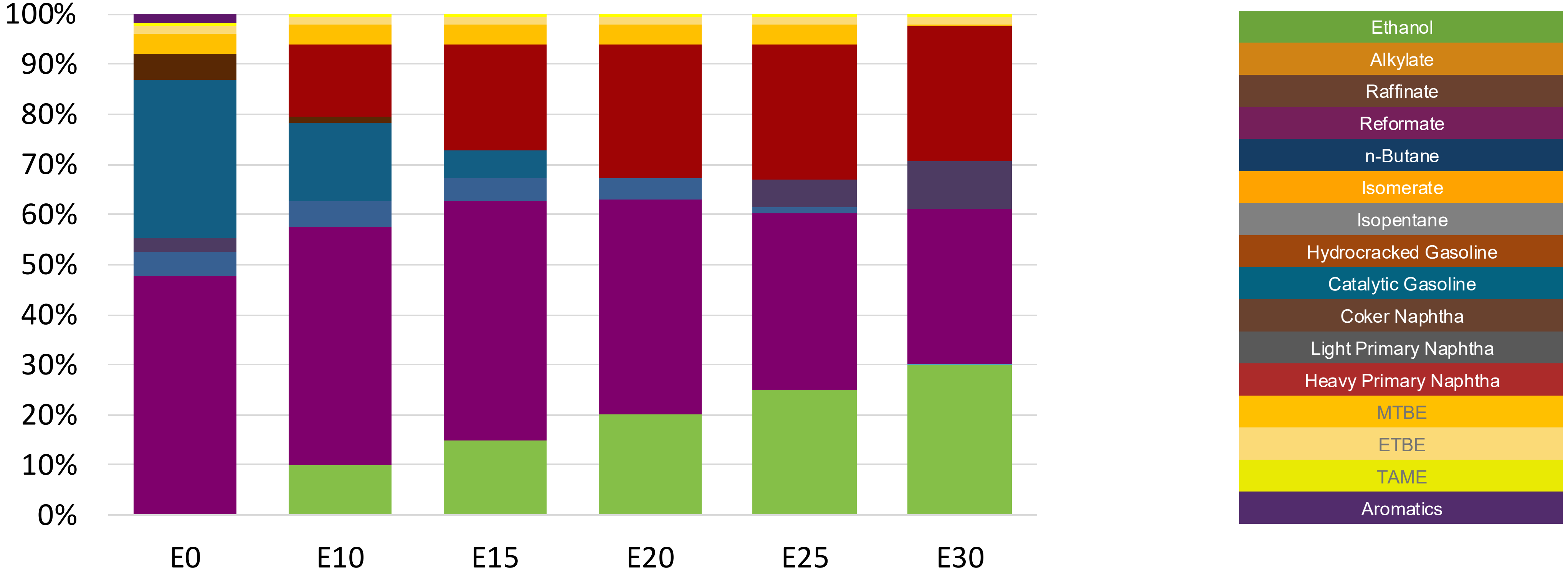


Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US$/gal)	$ 2.272	$ 2.144	$ 2.083	$ 2.002	$ 1.880	$ 1.814

Average CIF 2024 Prices

Source: Faro90

Austria - Gasoline 98 - Constant Octane

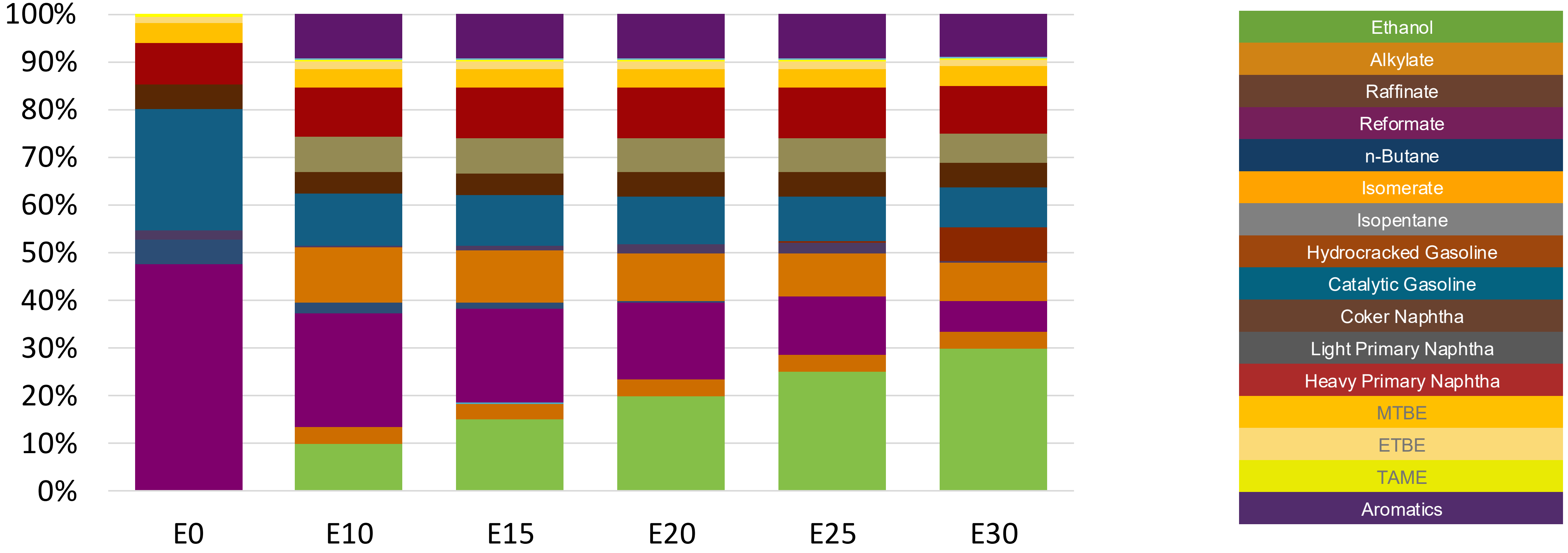


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.347	$ 2.218	$ 2.146	$ 2.069	$ 1.992	$ 1.925

Average CIF 2024 Prices

Source: Faro90

Austria – Gasoline 95 – Increased Octane

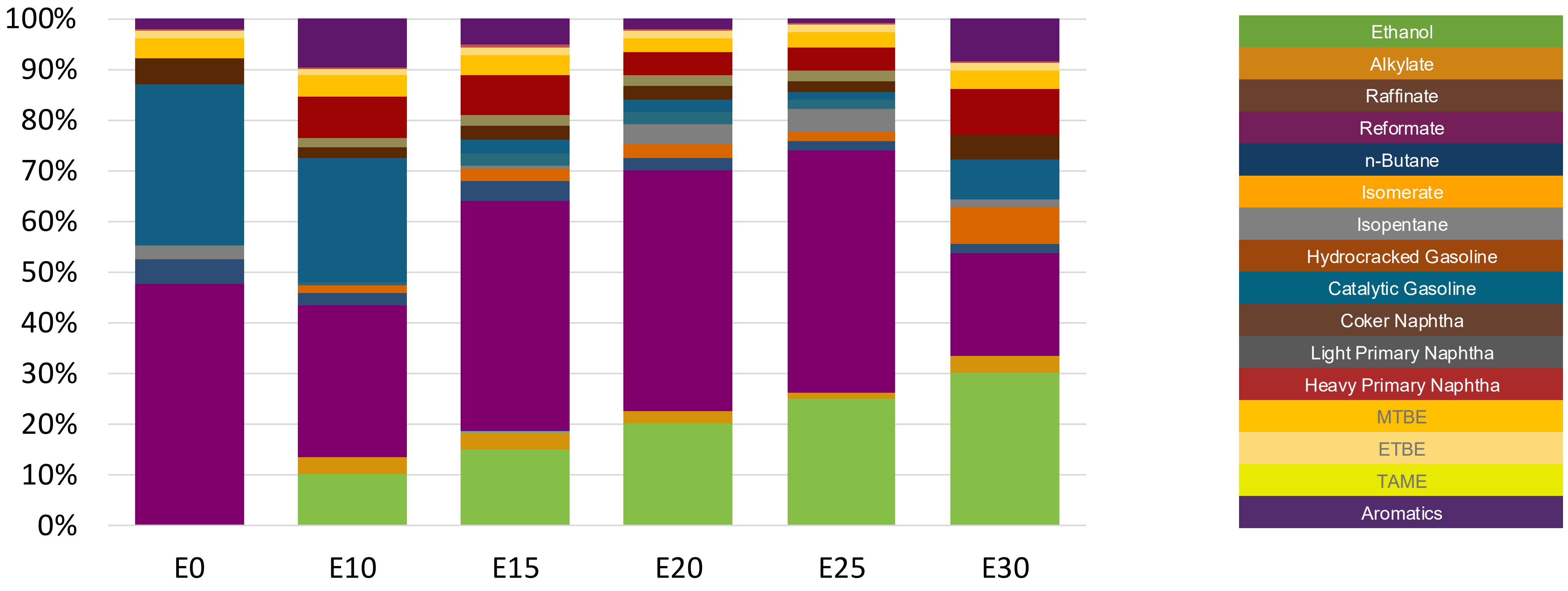


Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US$/gal)	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209

Average CIF 2024 Prices

Source: Faro90

Austria – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US$/gal)	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347

Average CIF 2024 Prices

Source: Faro90

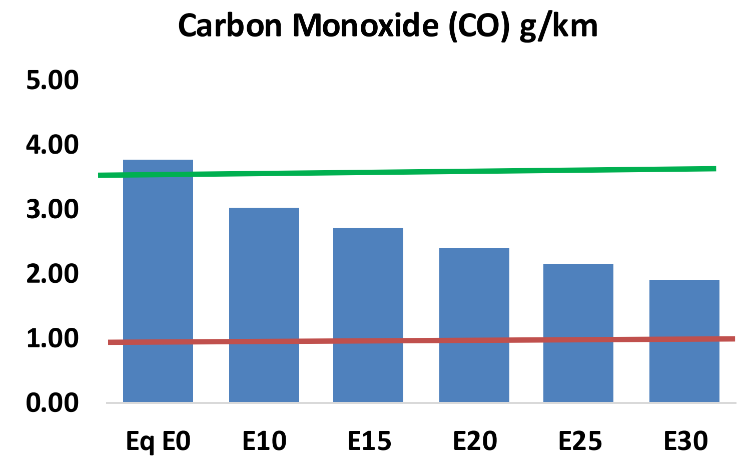
Gasoline Vehicle Fleet - Austria

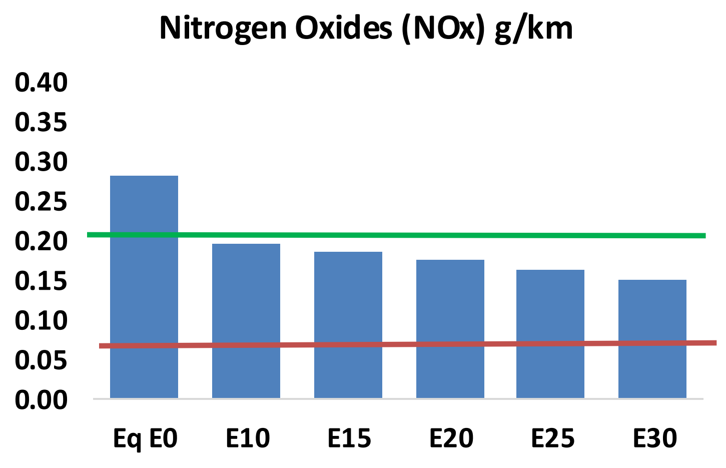
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	2,199
		Single FI	3-Ways		>8 years	45,951
		Multi FI	3-Ways		>8 years	525,647
			Euro III		>8 years	657,739
			Euro IV		>8 years	687,307
			Euro V		>8 years	300,486
			Euro 6		>8 years	638,328
					4-8 years	673,812
					<4 years	412,187
			Hybrid		All	60,203
		Electric			All	149,908
	Medium	Carburator	3-Ways	PCV Tank	>8 years	218
		Single FI	3-Ways		>8 years	1,382
		Multi FI	3-Ways		>8 years	71,247
			Euro III		>8 years	120,321
			Euro IV		>8 years	242,775
			Euro V		>8 years	92,349
			Euro 6		>8 years	212,776
					4-8 years	305,111
					<4 years	327,205
			Hybrid		All	42,528
		Electric			All	105,366
Sml Engines	Light	4-Cycle	None	PCV	>8 years	24,824
			Euro III		>8 years	80,432
			Euro IV		>8 years	603,608
			Euro V		>8 years	174,836
			Euro 6		>8 years	450,562
					4-8 years	623,207
					<4 years	754,983
			Hybrid		All	52,408
		Electric			All	130,226
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	-
		Fuel-Injection	3-Ways		>8 years	749,356
			Euro III		>8 years	951,622
			Euro IV		>8 years	1,363,585
			Euro V		>8 years	240,214
			Euro 6		>8 years	602,646
					4-8 years	792,932
					<4 years	632,043
			Hybrid		All	1,972
		Electric			All	141,946

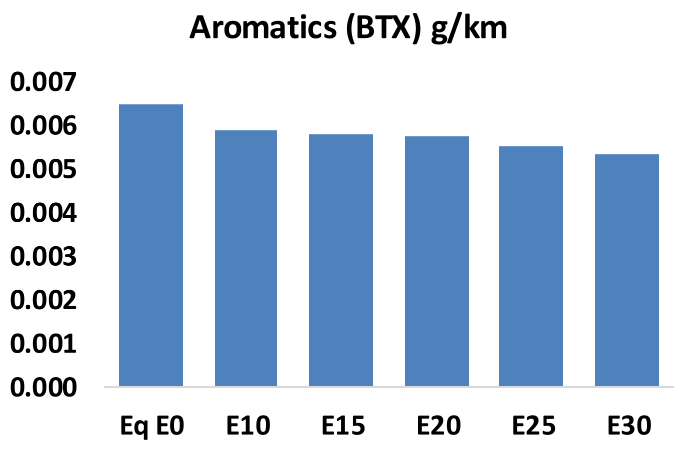
Vehicle Fleet: 14,046,44
Gasoline Vehicle Fleet: 3,251,450

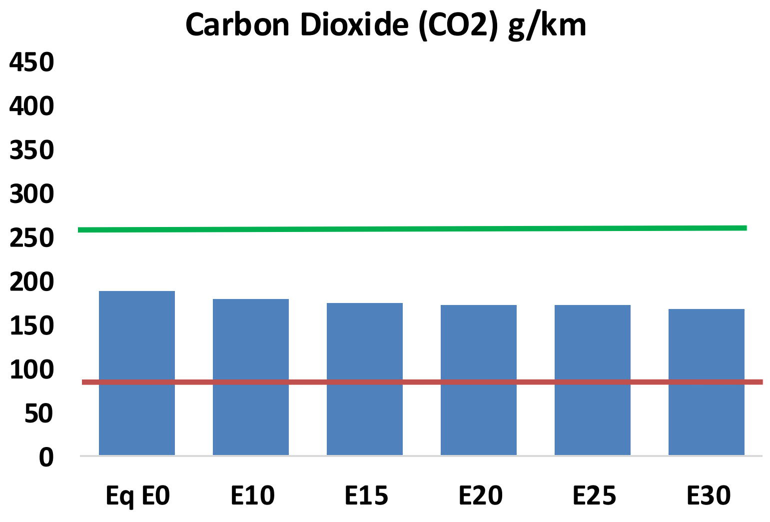
Source: Eurostat, ACEA

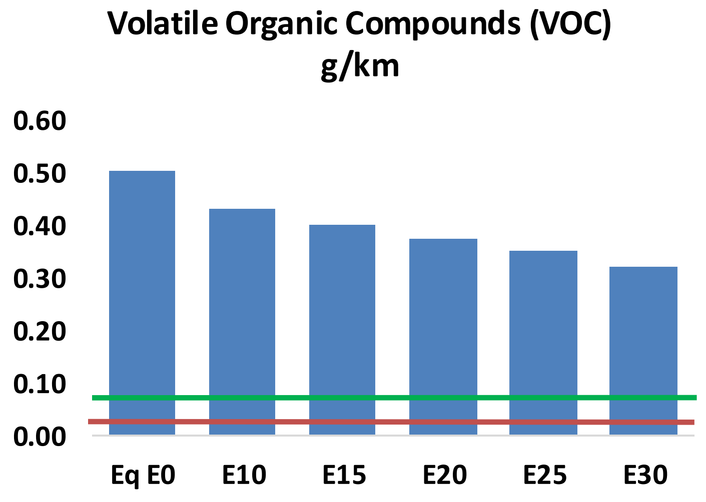
Austria – Vehicular Emissions

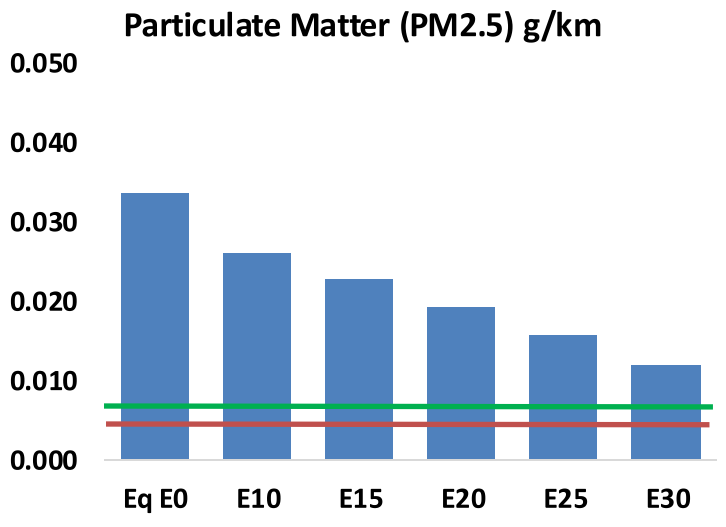












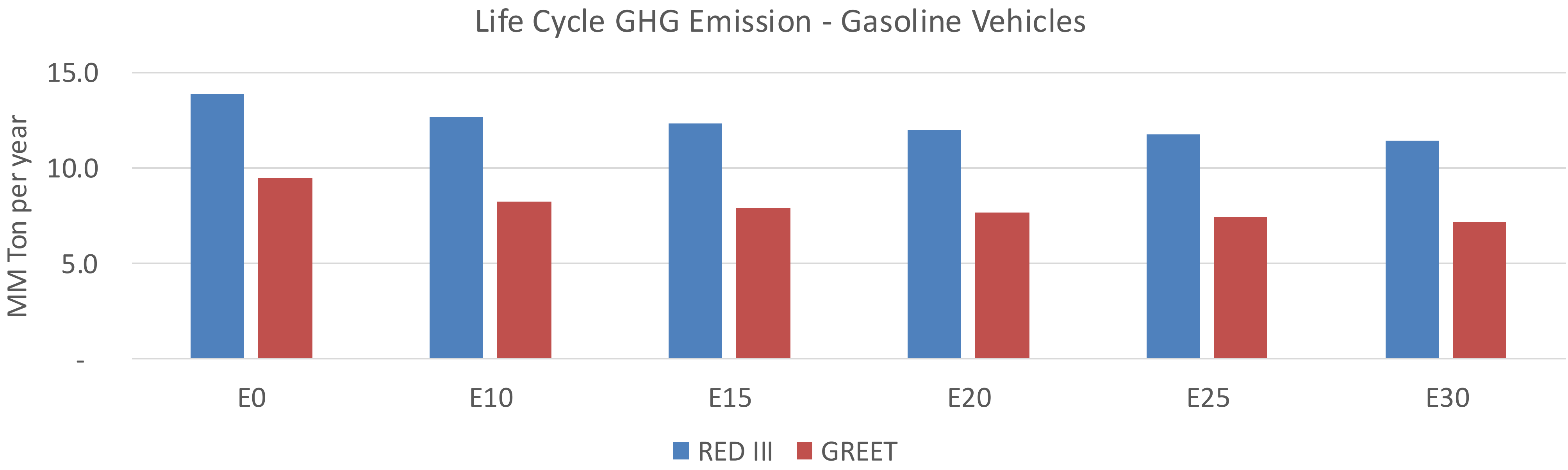
Source: Faro90

Austria – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	315,891	285,353	254,815	228,570	201,597	180,687	158,890	-19%	-36%
CO2	15,777,119	15,371,421	14,988,263	14,686,920	14,537,804	14,452,599	14,186,196	-5%	-8%
VOC	42,267	39,233	36,199	33,755	31,314	29,444	26,927	-14%	-26%
NOx	23,733	17,800	16,613	15,658	14,767	13,784	12,745	-30%	-38%
SOx	178	165	151	140	129	117	105	-15%	-28%
NH3	5,682	5,626	5,569	5,596	5,659	5,703	5,740	-2%	0%
N2O	251	249	248	250	255	258	261	-1%	2%
CH4	9,822	9,822	9,822	9,969	10,186	10,372	10,549	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	313	297	281	270	259	251	239	-10%	-17%
Acetaldehyde	819	940	1,379	2,100	2,861	3,270	3,863	68%	249%
Formaldehyde	2,974	2,892	3,371	3,958	4,146	4,587	4,994	13%	39%
Benzene	545	523	496	488	484	463	448	-9%	-11%
PM2.5	1,014	900	787	683	580	471	357	-22%	-43%
PM10	1,813	1,610	1,407	1,222	1,037	842	638	-22%	-43%

Source: Faro90

Austria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	67.5
Target @2035 (MMT/year)	38.0
Delta	29.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.9%	16.1%	18.6%	21.0%	24.2%
RED II/III	0.0%	9.3%	11.7%	13.6%	15.4%	17.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.1%	5.1%	6.0%	6.7%	7.7%
RED II/III	0.0%	4.4%	5.5%	6.4%	7.3%	8.4%

Source: Greet, RED II/III, climateactiontracker.org, F90

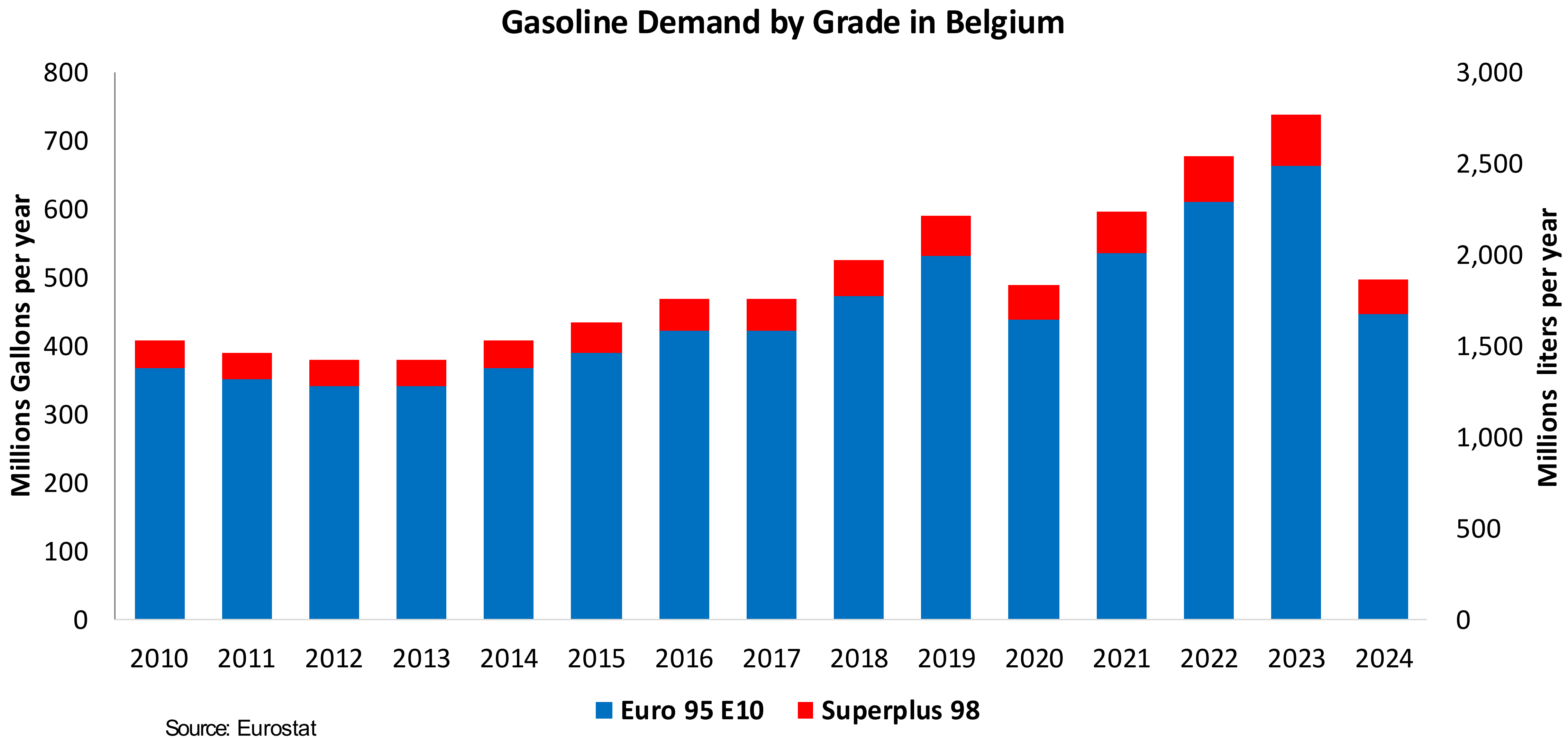
Ethanol Gasoline Blending in Belgium

In 2024, gasoline consumption in Belgium was 1,865 million liters (493 million gallons) and growth rate was -3.4%. Gasoline production and net imports were 5,257 and -4,825 million liters (1,389 and -1,275 million gallons) correspondingly. Belgium complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Belgium was 239 million liters (63 million gallons) representing 12.8% of gasoline demand. Ethanol production and Net Imports were 554 and -315 million liters (63 and -83 million gallons) correspondingly. Ethanol sources are Sugar Cane 20% and Cereals 80%.

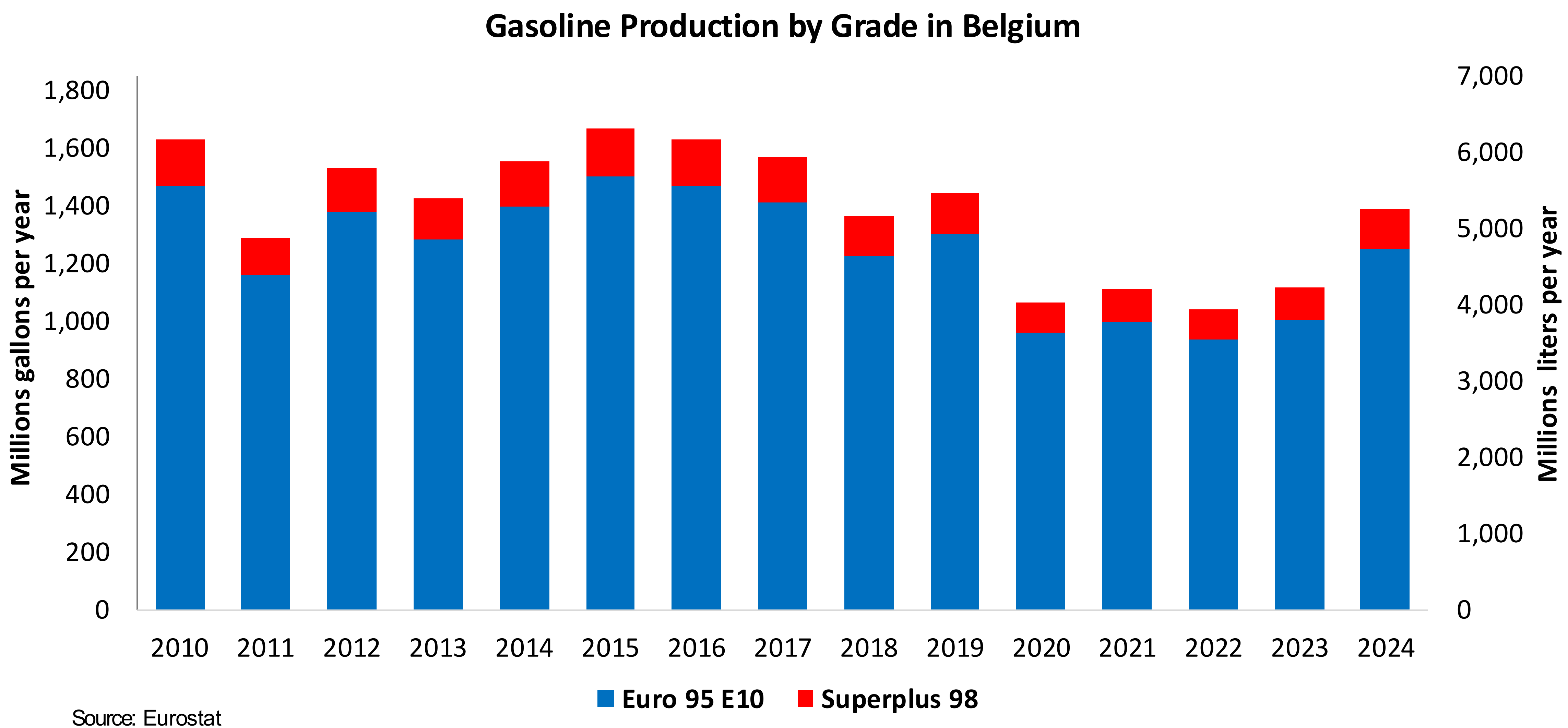
Source: Eurostat, European Commision

Gasoline Demand in Belgium



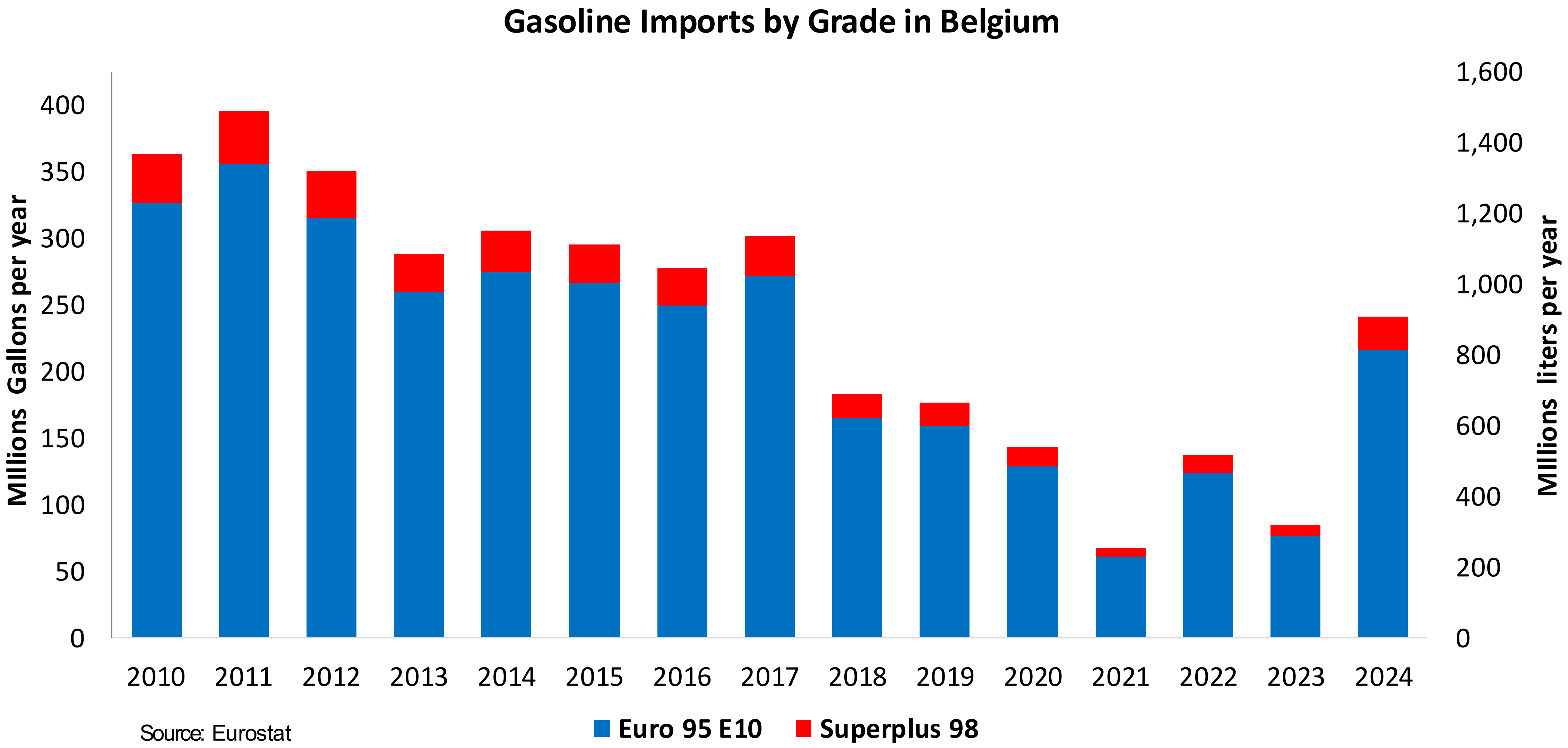
Source: Eurostat

Gasoline Production in Belgium



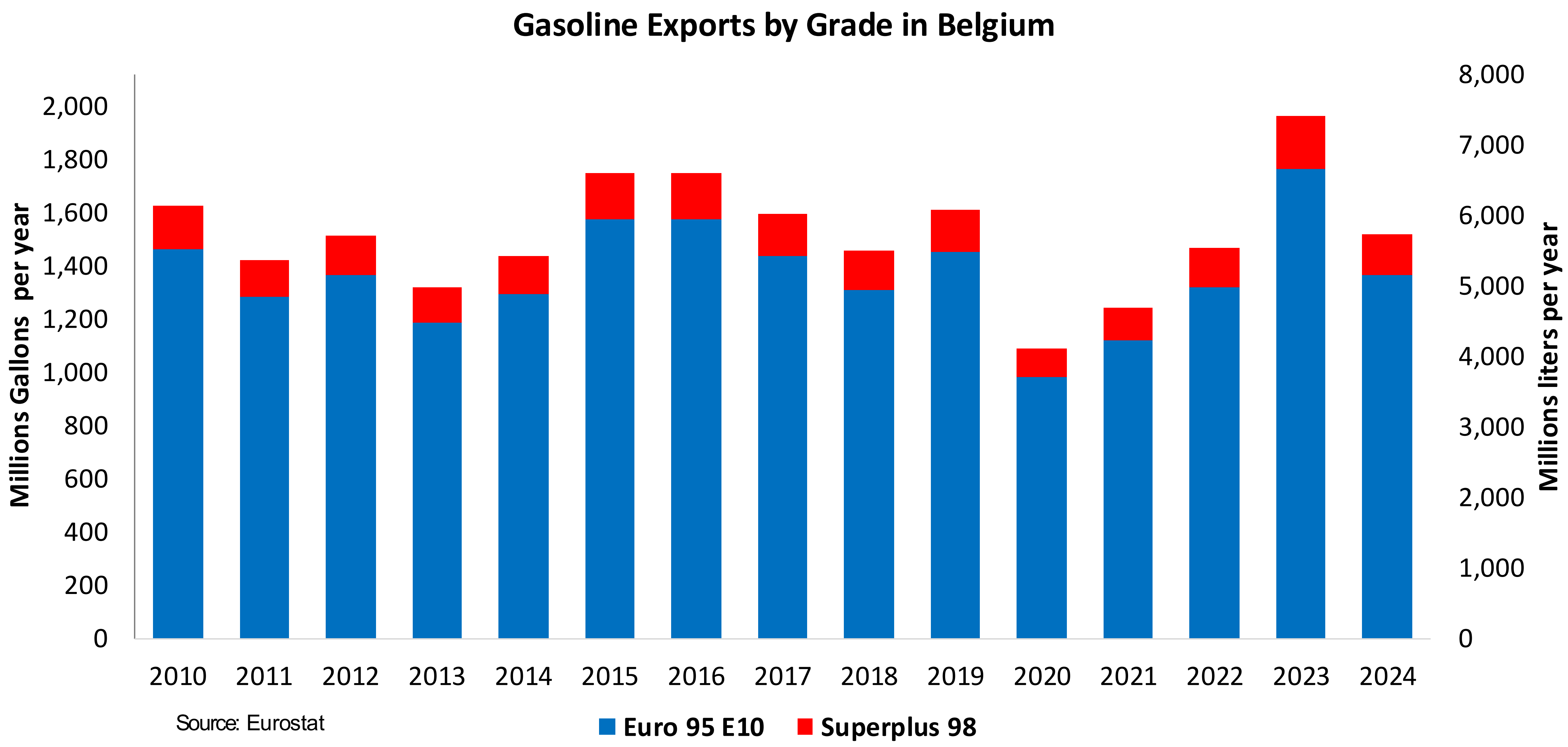
Source: Eurostat

Gasoline Imports in Belgium



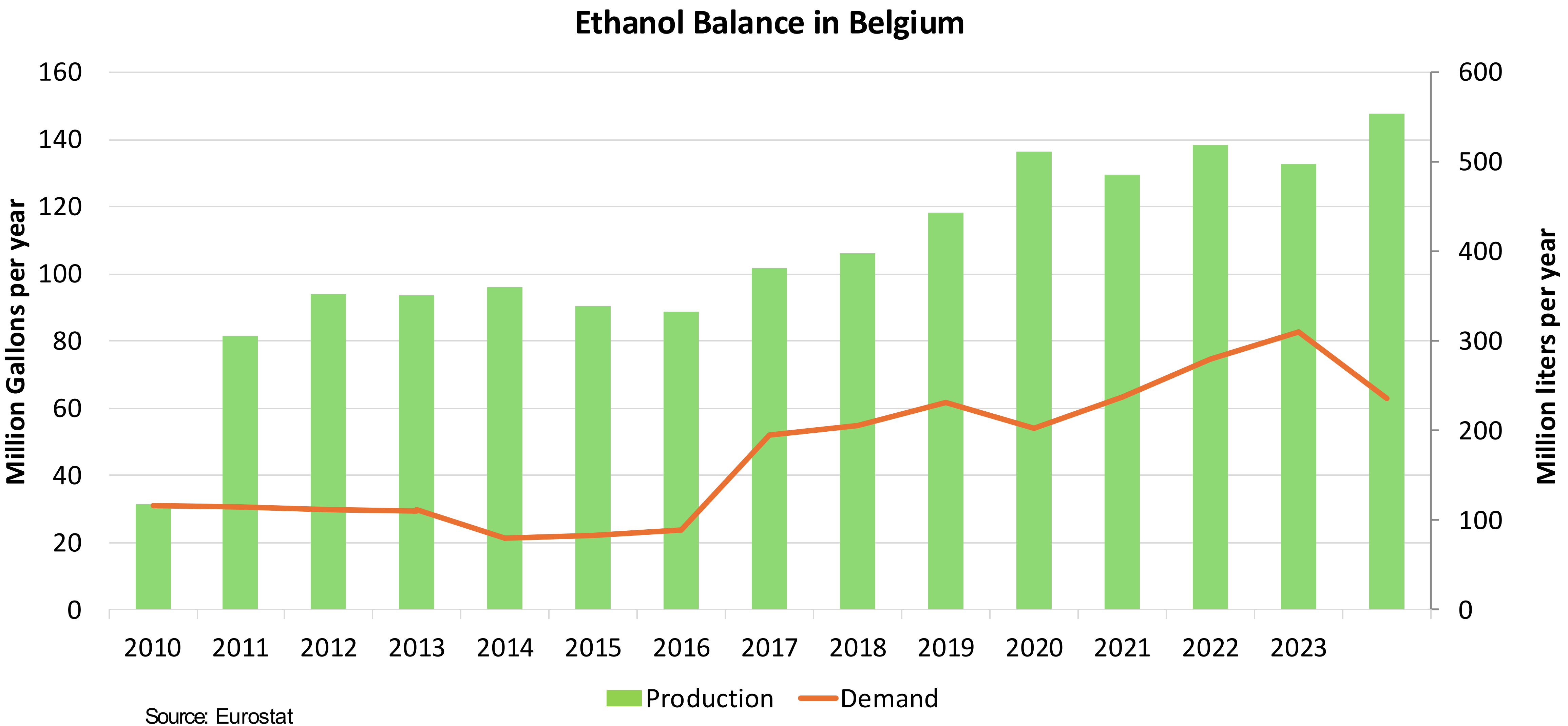
Source: Eurostat

Gasoline Exports in Belgium



Source: Eurostat

Ethanol Balance in Belgium



Source: Eurostat

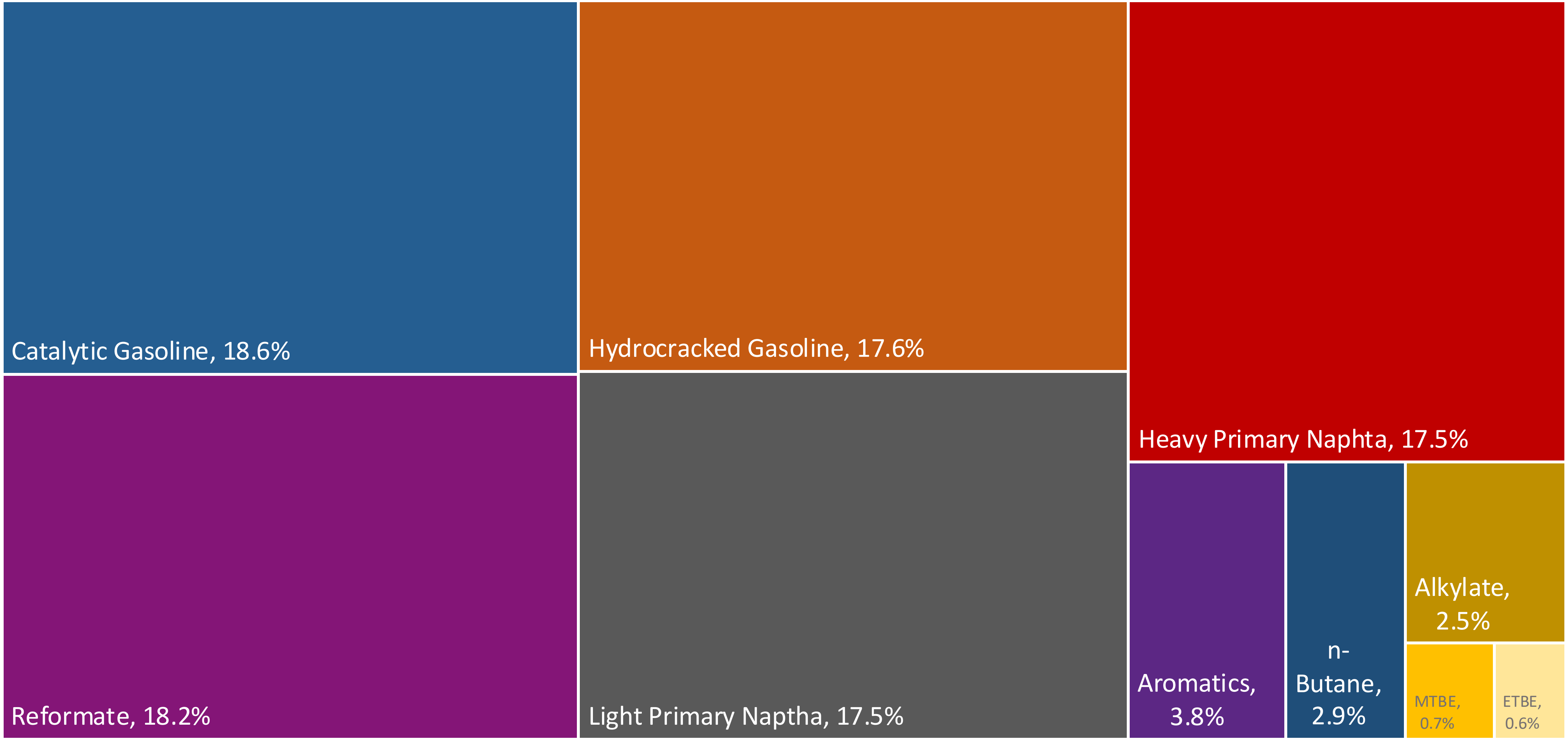
Gasoline Quality in Belgium

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.5							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	5.7							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

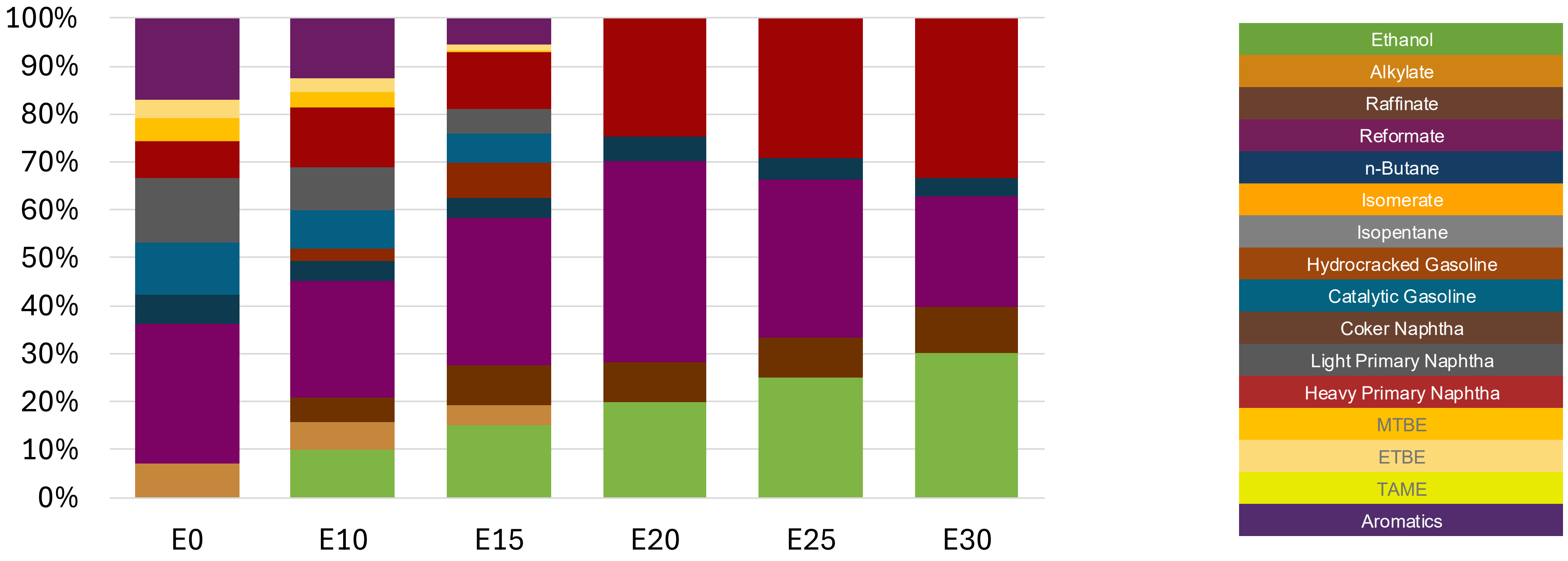
Source: European Commision

Belgium Available Blending Components



Source: Faro90

Belgium – Gasoline 95 – Constant Octane

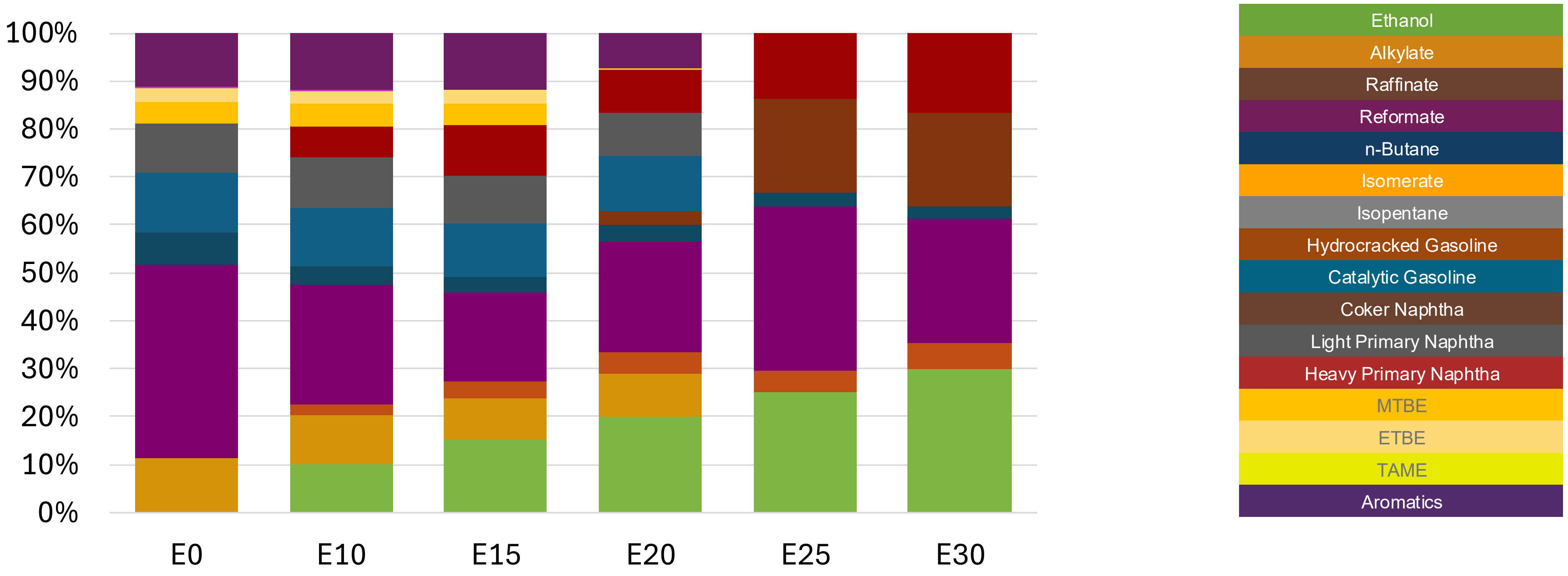


Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US$/gal)	$ 2.281	$ 2.147	$ 2.072	$ 1.951	$ 1.875	$ 1.780

Average CIF 2024 Prices

Source: Faro90

Belgium - Gasoline 98 - Constant Octane

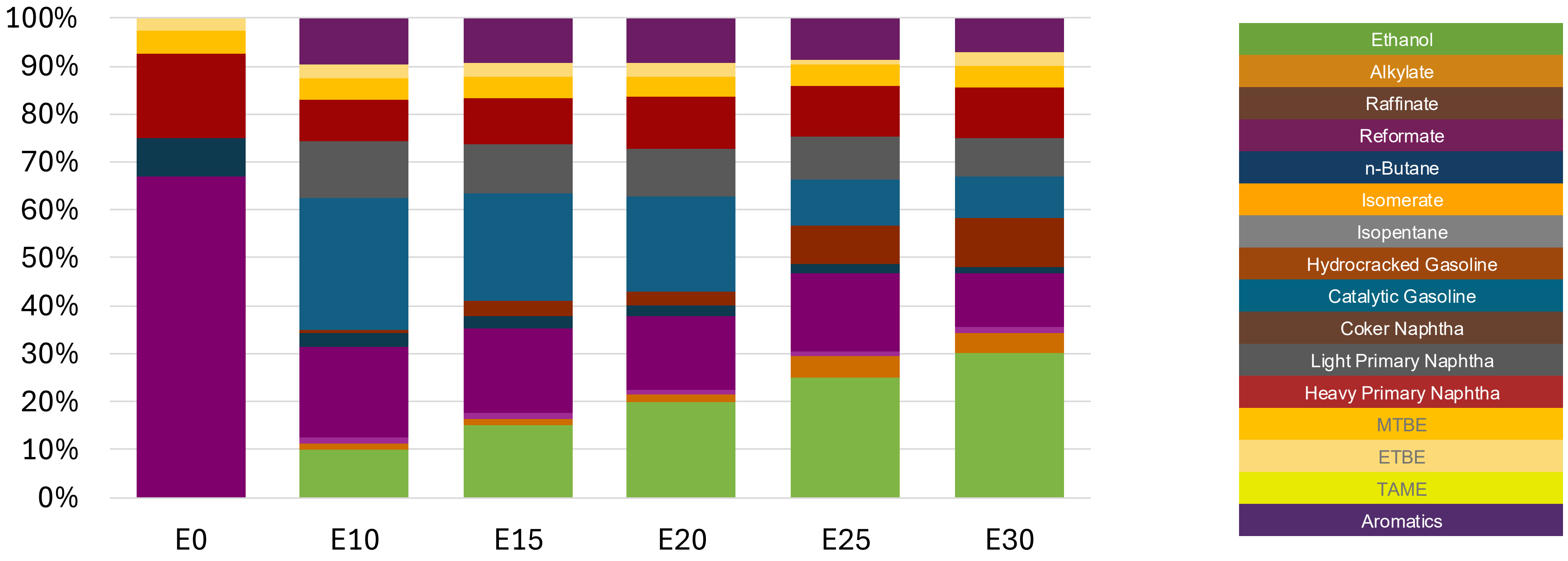


Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US$/gal)	$ 2.409	$ 2.290	$ 2.216	$ 2.164	$ 2.057	$ 1.994

Average CIF 2024 Prices

Source: Faro90

Belgium – Gasoline 95 – Increased Octane

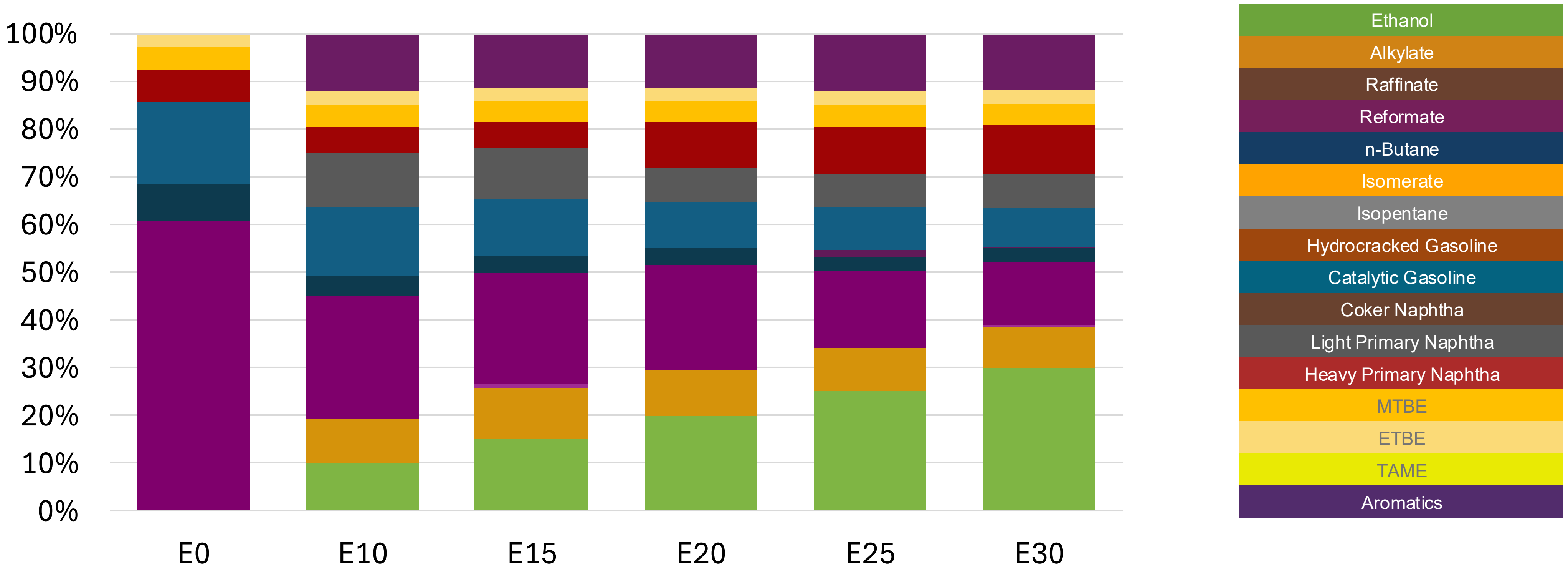


Octane (RON)	95.0	96.5	97.9	99.4	100.5	102.1
Price (US$/gal)	$ 2.193	$ 2.193	$ 2.193	$ 2.193	$ 2.193	$ 2.193

Average CIF 2024 Prices

Source: Faro90

Belgium – Gasoline 98 – Increased Octane



Octane (RON)	98.0	98.8	100.1	101.6	102.9	104.2
Price (US$/gal)	$ 2.329	$ 2.329	$ 2.329	$ 2.329	$ 2.329	$ 2.329

Average CIF 2024 Prices

Source: Faro90

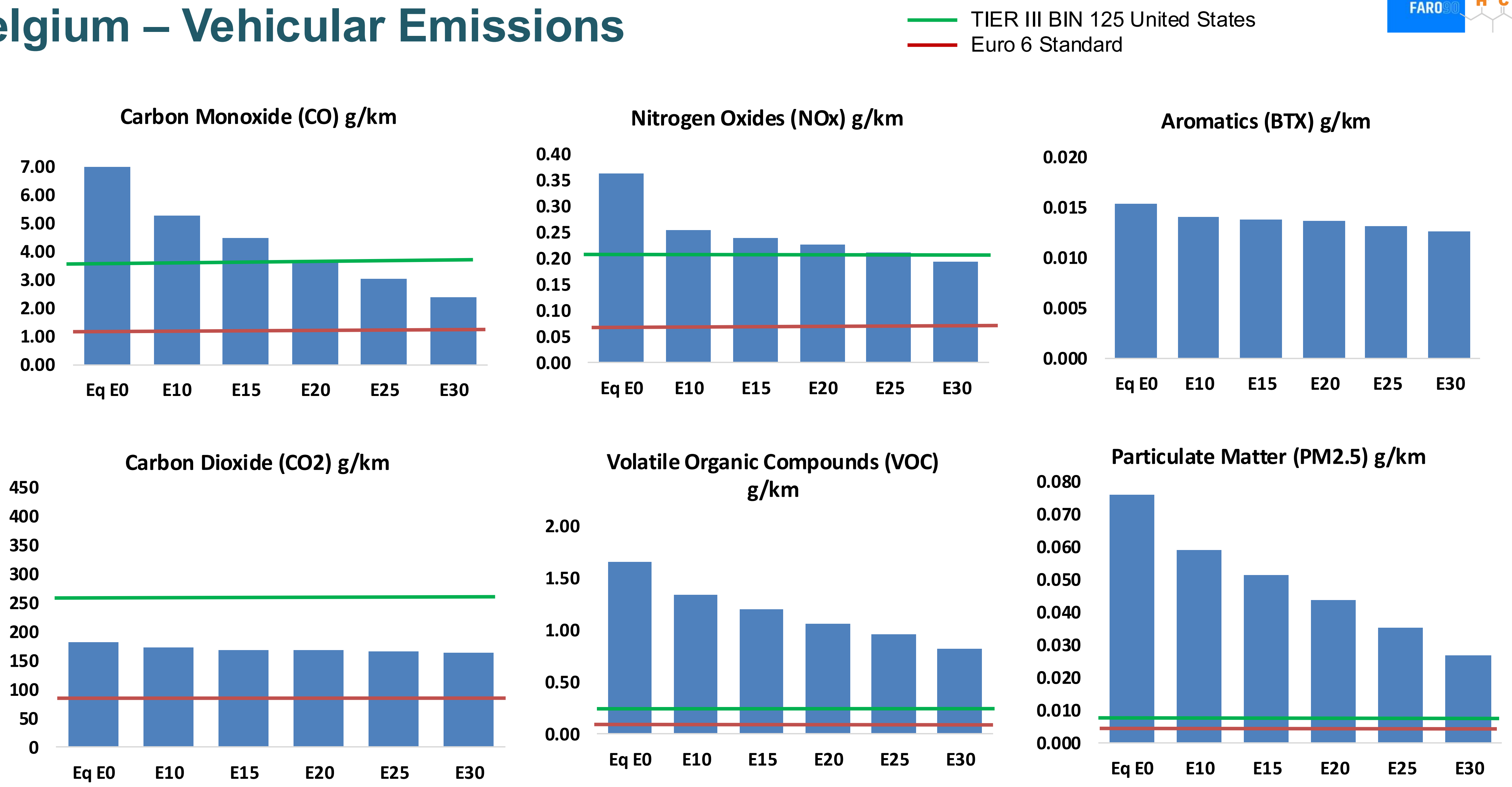
Gasoline Vehicle Fleet - Belgium

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	1,325
		Single FI	3-Ways		>8 years	7,652
		Multi FI	3-Ways		>8 years	157,086
			Euro III		>8 years	65,974
			Euro IV		>8 years	172,196
			Euro V		>8 years	3,547
			Euro 6		>8 years	126,866
					4-8 years	71,607
					<4 years	33,930
			Hybrid		All	142,544
		Electric			All	55,595
	Medium	Carburator	3-Ways	PCV Tank	>8 years	131
		Single FI	3-Ways		>8 years	172
		Multi FI	3-Ways		>8 years	14,786
			Euro III		>8 years	14,481
			Euro IV		>8 years	58,779
			Euro V		>8 years	383
			Euro 6		>8 years	42,289
					4-8 years	32,581
					<4 years	25,981
			Hybrid		All	85,156
		Electric			All	37,058
Sml Engines	Light	4-Cycle	None	PCV	>8 years	7,871
			Euro III		>8 years	13,387
			Euro IV		>8 years	51,663
			Euro V		>8 years	26,359
			Euro 6		>8 years	143,683
					4-8 years	36,132
					<4 years	598,985
			Hybrid		All	653,450
		Electric			All	265,894
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	16,574
		Fuel-Injection	3-Ways		>8 years	187,313
			Euro III		>8 years	105,052
			Euro IV		>8 years	121,096
			Euro V		>8 years	33,899
			Euro 6		>8 years	46,800
					4-8 years	22,064
					<4 years	2,867
			Hybrid		All	230
		Electric			All	271

Vehicle Fleet: 3,483,706
Gasoline Vehicle Fleet: 2,367,805

Source: Eurostat, ACEA

Belgium – Vehicular Emissions



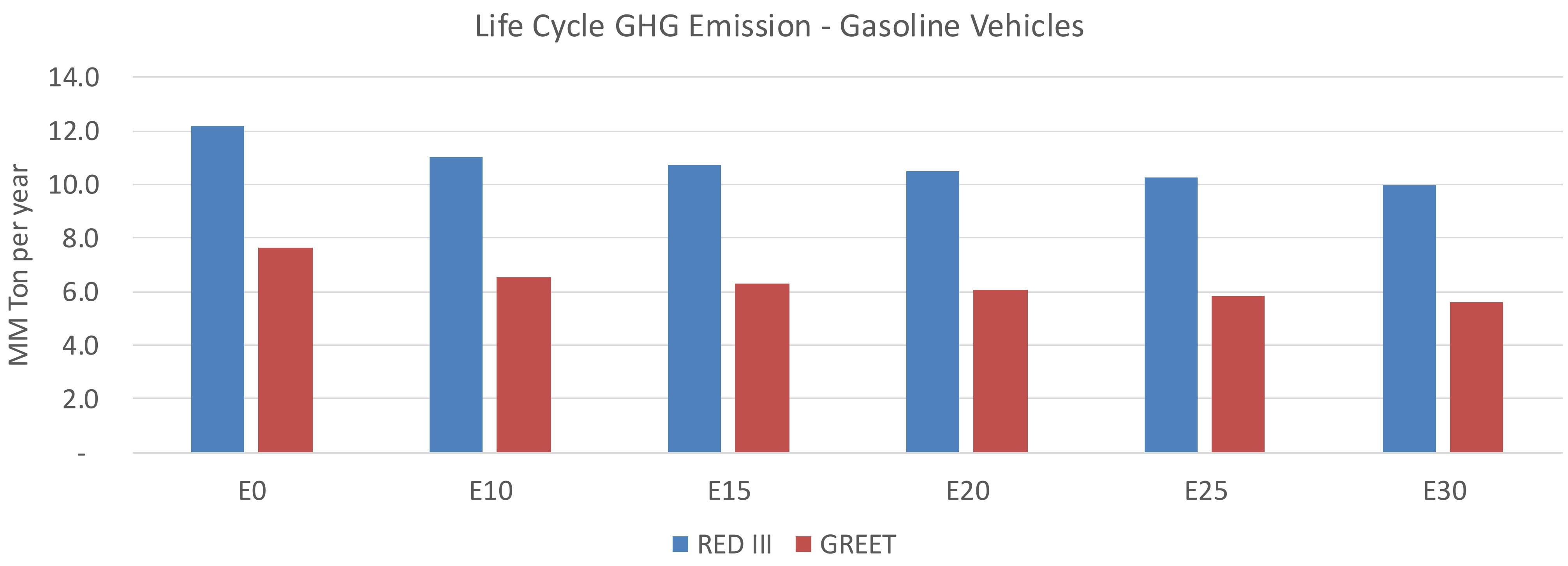
Source: Faro90

Belgium – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	424,560	371,491	318,422	270,948	221,338	182,637	143,912	-25%	-48%
CO2	10,967,384	10,685,366	10,419,015	10,209,538	10,105,881	10,047,600	9,862,394	-5%	-8%
VOC	99,964	90,398	80,831	72,624	64,196	57,665	49,129	-19%	-36%
NOx	21,975	16,481	15,382	14,498	13,673	12,763	11,801	-30%	-38%
SOx	124	115	105	97	90	81	73	-15%	-28%
NH3	4,858	4,810	4,762	4,785	4,839	4,877	4,908	-2%	0%
N2O	207	205	205	206	210	212	215	-1%	2%
CH4	19,986	19,986	19,986	20,285	20,725	21,105	21,465	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	403	375	347	324	302	285	261	-14%	-25%
Acetaldehyde	1,575	1,808	2,653	4,040	5,504	6,290	7,432	68%	249%
Formaldehyde	6,299	6,124	7,138	8,382	8,781	9,715	10,575	13%	39%
Benzene	934	897	850	837	830	793	768	-9%	-11%
PM2.5	706	627	547	475	404	328	248	-22%	-43%
PM10	3,897	3,461	3,024	2,626	2,230	1,809	1,371	-22%	-43%

Source: Faro90

Belgium – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	105.6
Target @2035 (MMT/year)	59.4
Delta	46.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.3%	17.7%	20.5%	23.3%	26.8%
RED II/III	0.0%	9.5%	12.0%	14.0%	15.9%	18.3%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.4%	2.9%	3.4%	3.8%	4.4%
RED II/III	0.0%	2.5%	3.2%	3.7%	4.2%	4.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

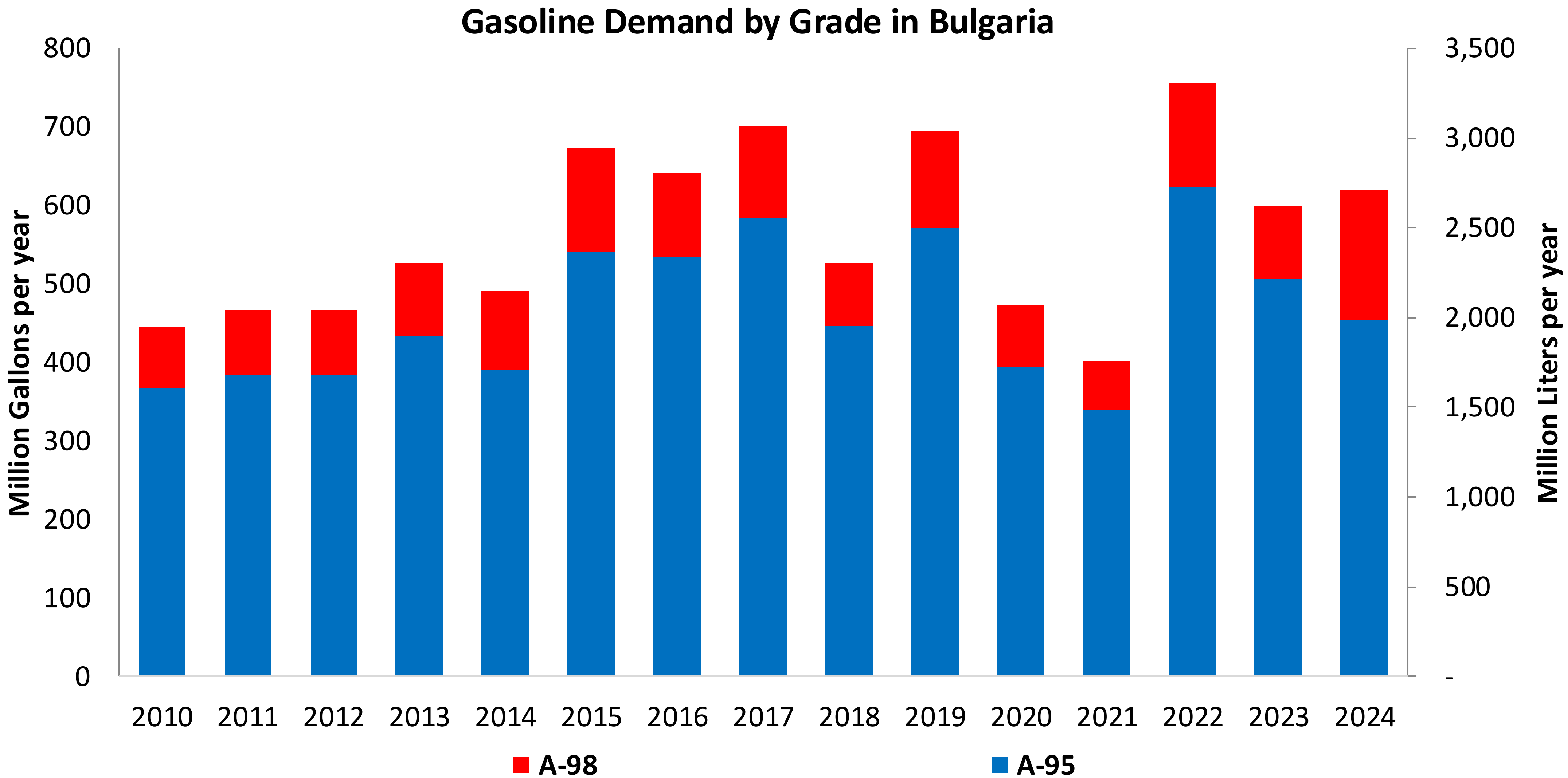
Ethanol Gasoline Blending in Bulgaria

In 2024, gasoline consumption in Bulgaria was 2,342 million liters (619 million gallons) and growth rate was -2.3%. Gasoline production and net imports were 2,199 and 143 million liters (581 and 38 million gallons) correspondingly. Bulgaria complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

In 2024, ethanol consumption in Bulgaria was 54 million liters (14 million gallons) representing 2.3% of gasoline demand. Ethanol production and Net Imports were 40 and 14 million liters (14 and 4 million gallons) correspondingly. Ethanol sources are Corn 45% and Cereals 55.

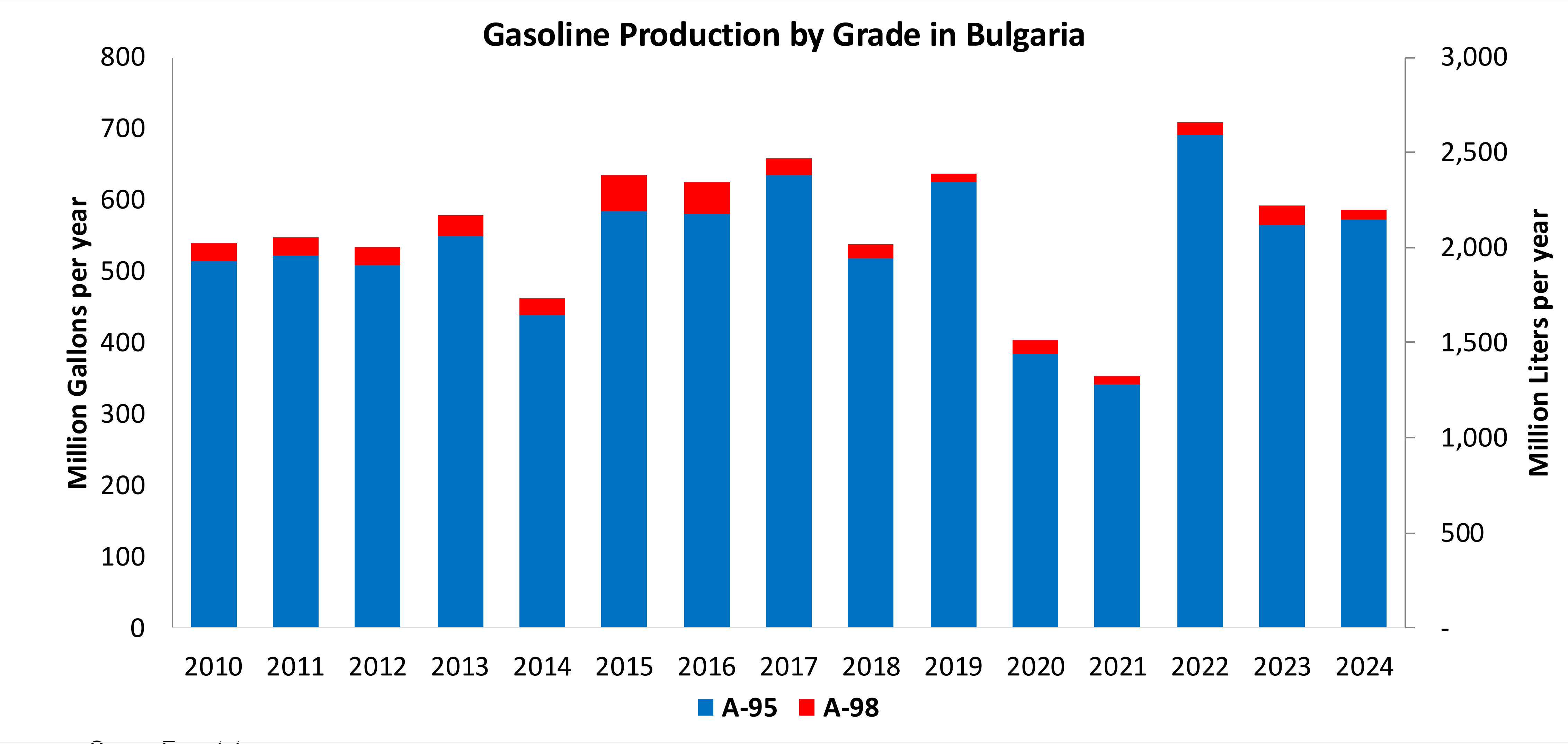
Source: Eurostat, European Commision

Gasoline Demand in Bulgaria



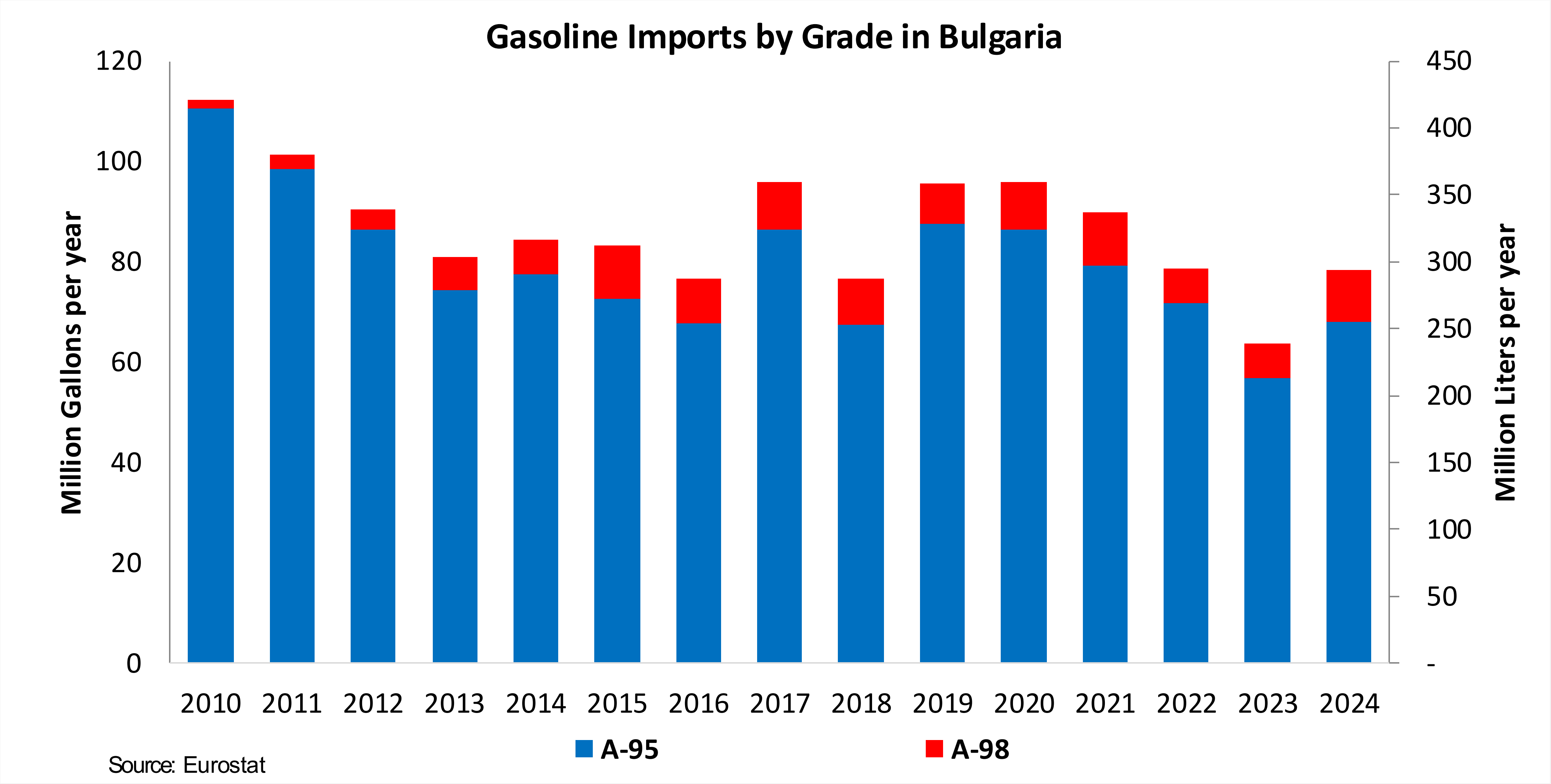
Source: Eurostat

Gasoline Production in Bulgaria

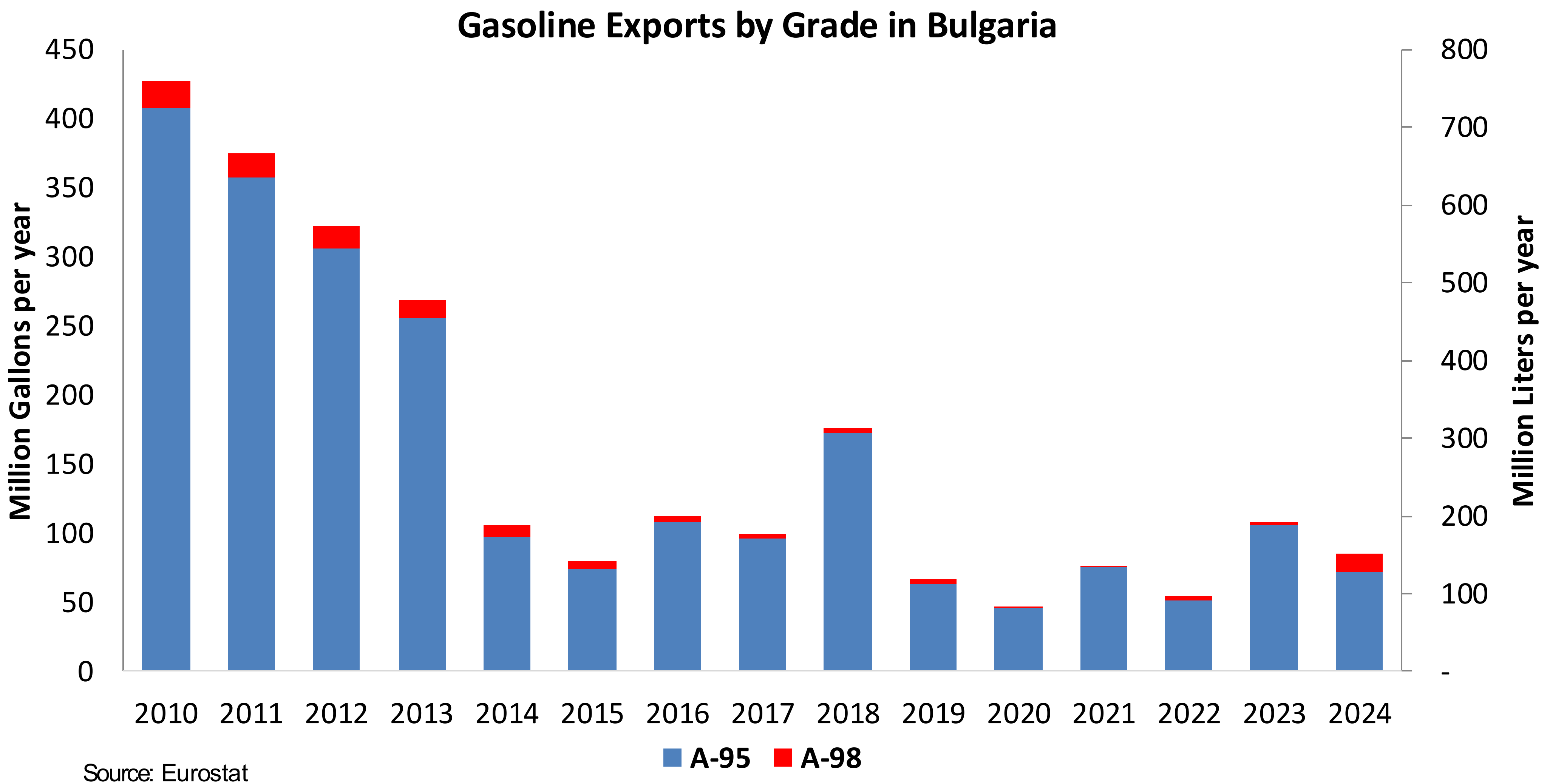


Source: Eurostat

Gasoline Imports in Bulgaria

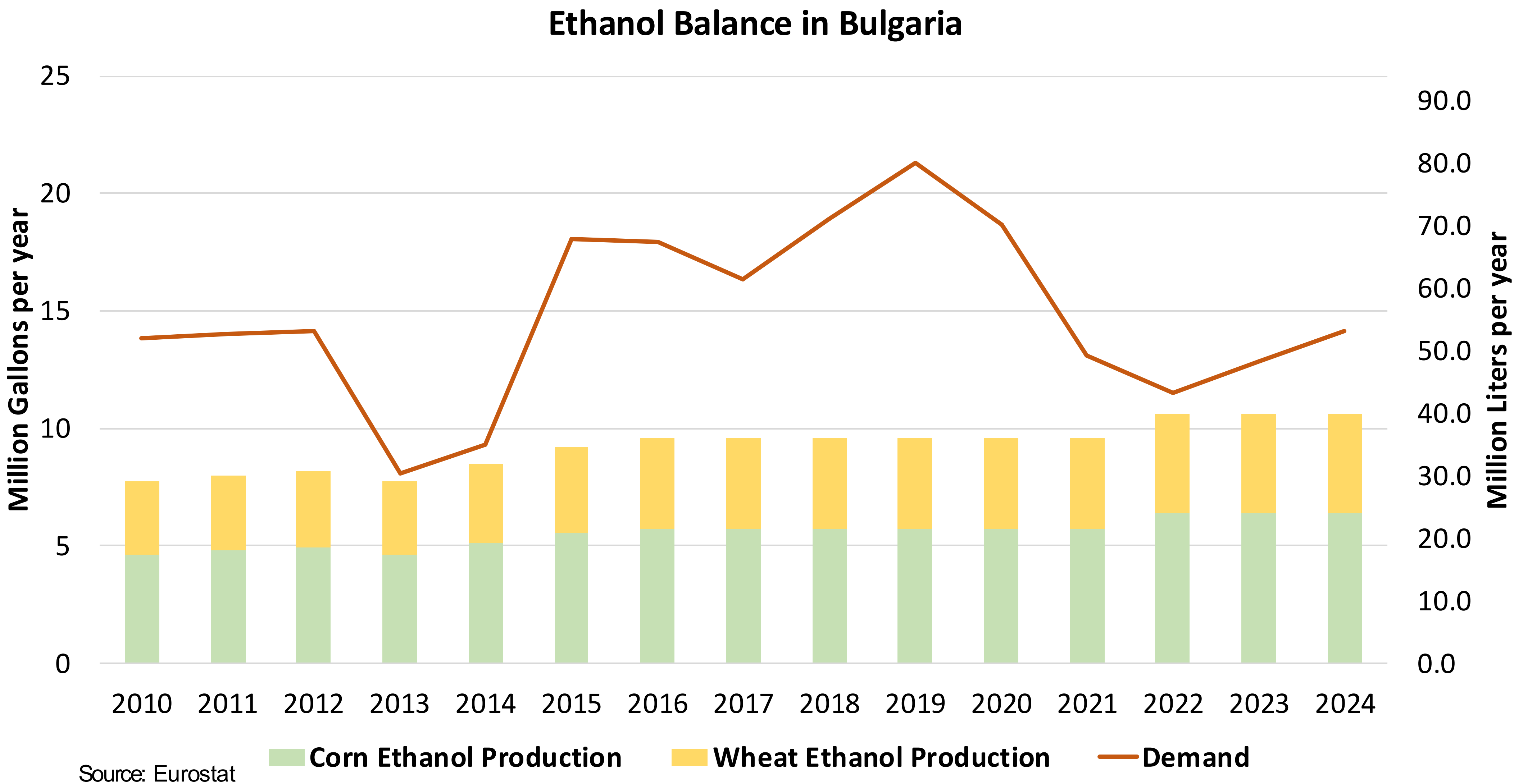


Gasoline Exports in Bulgaria



Source: Eurostat

Ethanol Balance in Bulgaria



Source: Eurostat

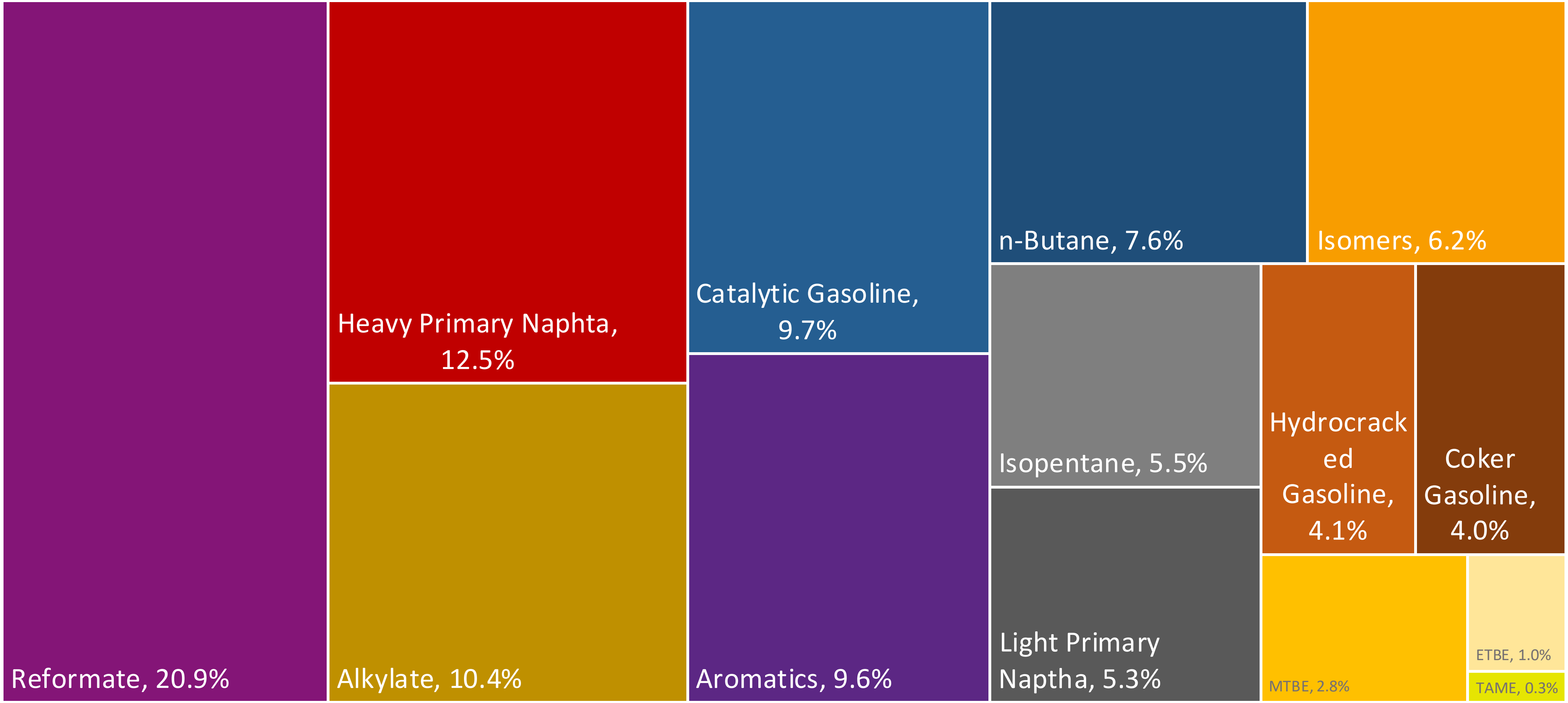
Gasoline Quality in Bulgaria

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

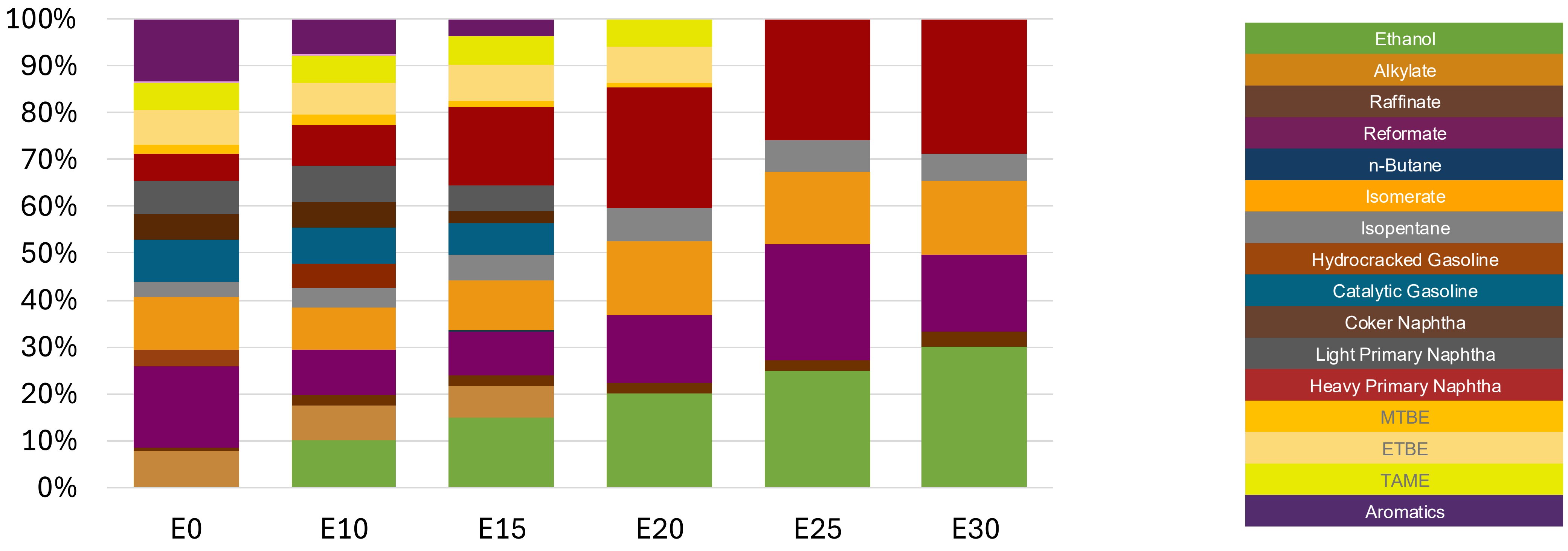
Source: European Commision

Bulgaria Available Blending Components



Source: Faro90

Bulgaria – Gasoline 95 – Constant Octane

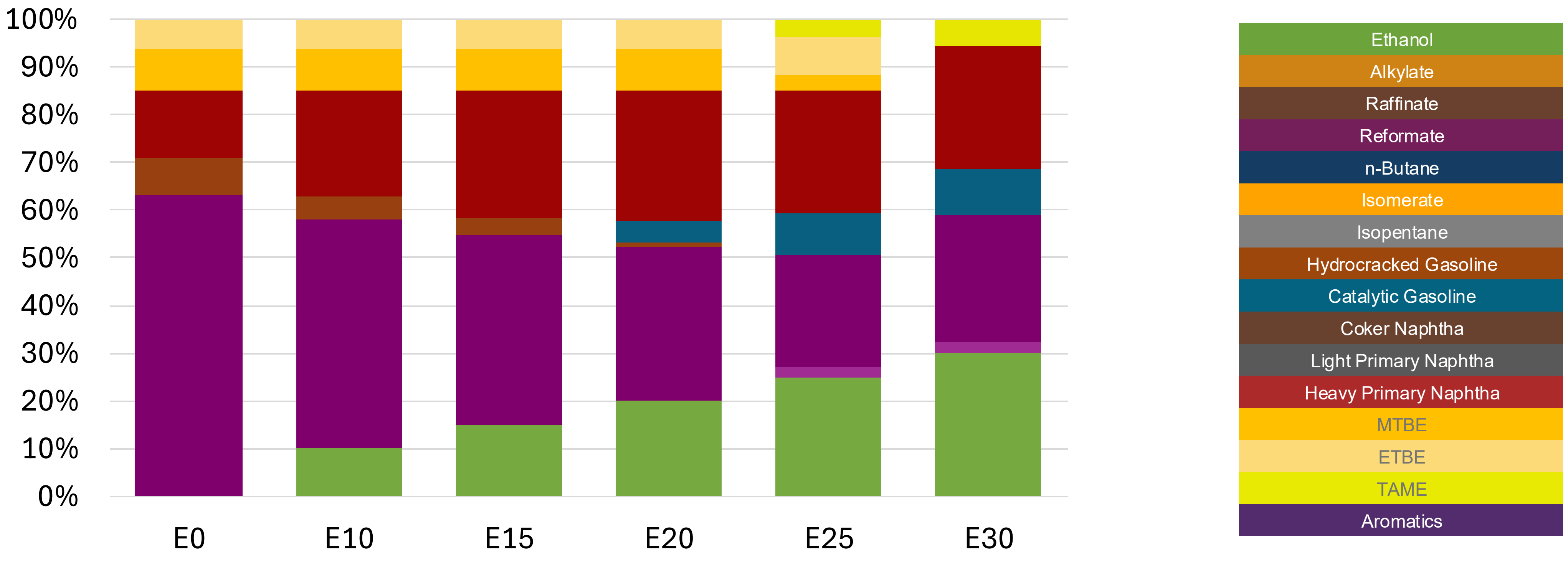


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.294	$ 2.156	$ 2.056	$ 1.941	$ 1.895	$ 1.842

Average CIF 2024 Prices

Source: Faro90

Bulgaria - Gasoline 98 - Constant Octane

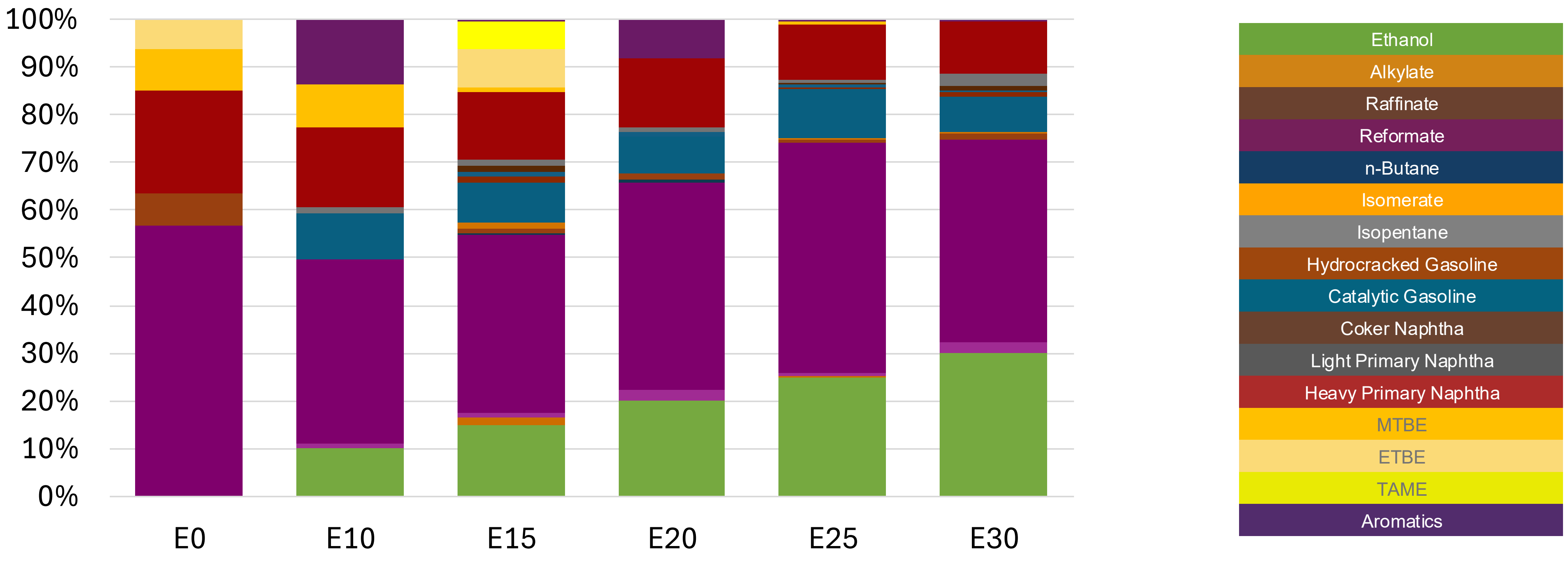


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.314	$ 2.197	$ 2.128	$ 2.056	$ 1.991	$ 1.941

Average CIF 2024 Prices

Source: Faro90

Bulgaria – Gasoline 95 – Increased Octane

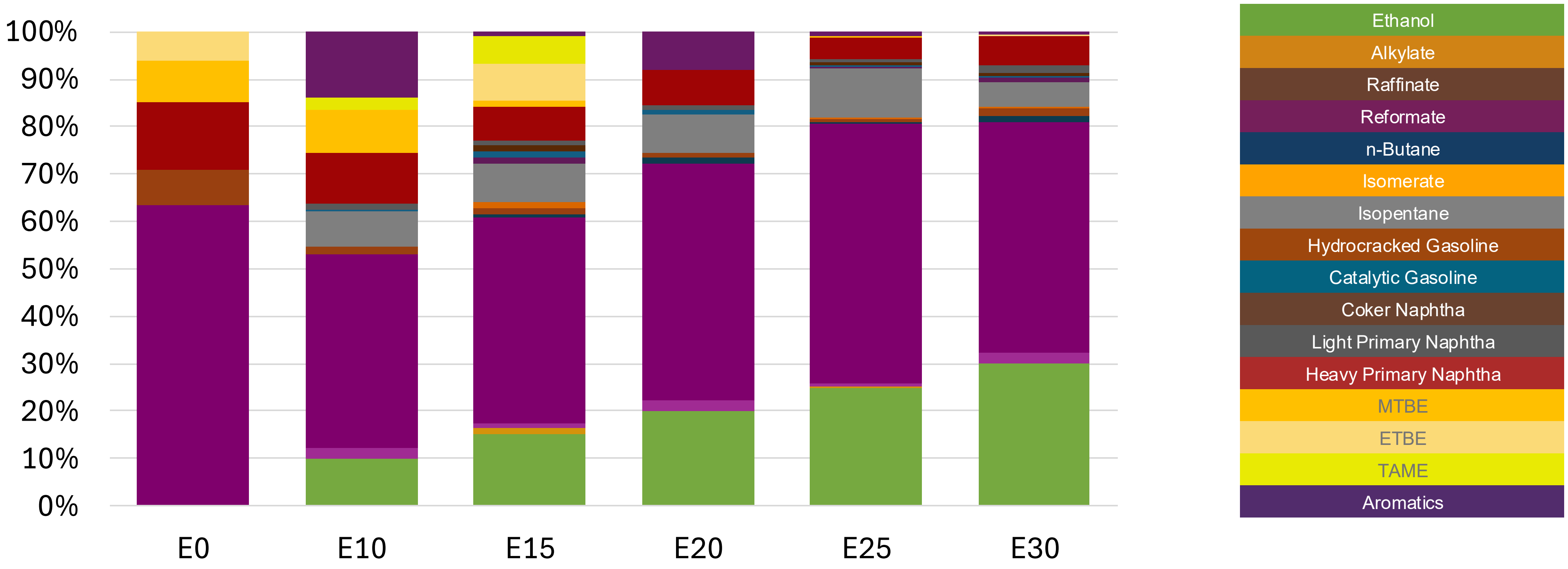


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.182	$ 2.182	$ 2.182	$ 2.182	$ 2.182	$ 2.182

Average CIF 2024 Prices

Source: Faro90

Bulgaria – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.314	$ 2.314	$ 2.314	$ 2.314	$ 2.314	$ 2.314

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Bulgaria

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	1,016
		Single FI	3-Ways		>8 years	22,579
		Multi FI	3-Ways		>8 years	316,355
			Euro III		>8 years	485,118
			Euro IV		>8 years	419,978
			Euro V		>8 years	281,049
			Euro 6		>8 years	551,943
					4-8 years	121,575
					<4 years	316,881
			Hybrid		All	86,307
		Electric			All	22,574
	Medium	Carburator	3-Ways	PCV Tank	>8 years	100
		Single FI	3-Ways		>8 years	652
		Multi FI	3-Ways		>8 years	37,915
			Euro III		>8 years	92,509
			Euro IV		>8 years	151,461
			Euro V		>8 years	86,495
			Euro 6		>8 years	183,981
					[illegible]	[illegible]
					<4 years	263,705
			Hybrid		All	67,747
		Electric			All	18,329
Sml Engines	Light	4-Cycle	None	PCV	>8 years	25,337
			Euro III		>8 years	36,683
			Euro IV		>8 years	156,292
			Euro V		>8 years	100,979
			Euro 6		>8 years	155,824
					4-8 years	178,140
					<4 years	391,093
			Hybrid		All	47,289
		Electric			All	12,556
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	16,574
		Fuel-Injection	3-Ways		>8 years	187,313
			Euro III		>8 years	105,052
			Euro IV		>8 years	121,096
			Euro V		>8 years	33,899
			Euro 6		>8 years	46,800
					4-8 years	22,064
					<4 years	2,867
			Hybrid		All	732
		Electric			All	2,005

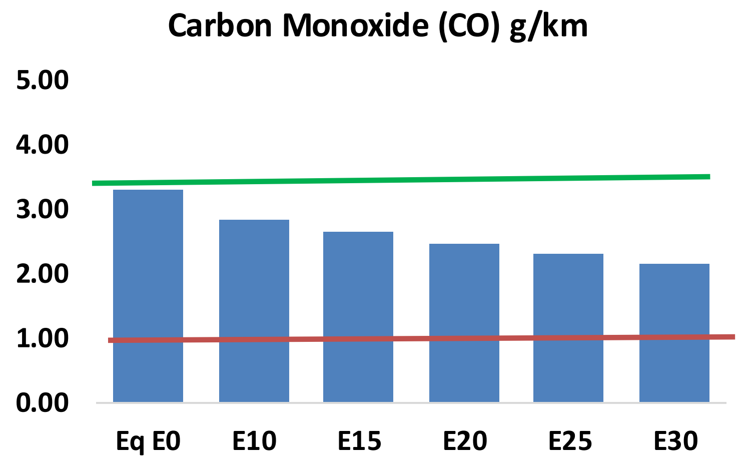
3,368,231

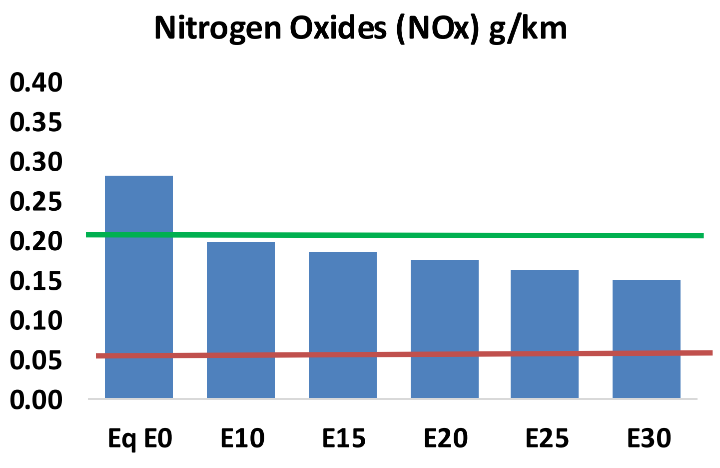
Vehicle Fleet: 5,239,744
Gasoline Vehicle Fleet: 3,368,231

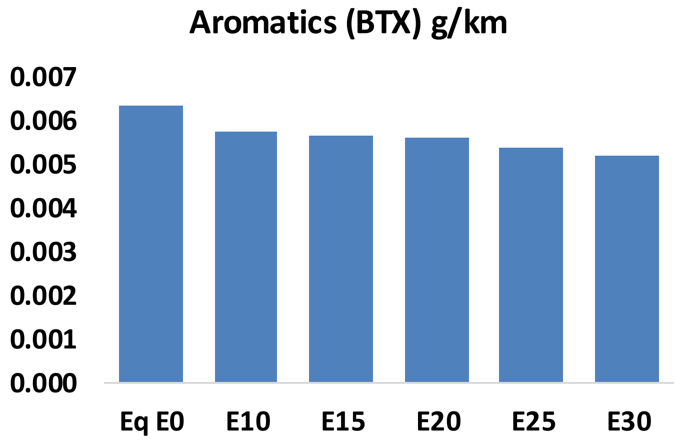
Source: Eurostat, ACEA

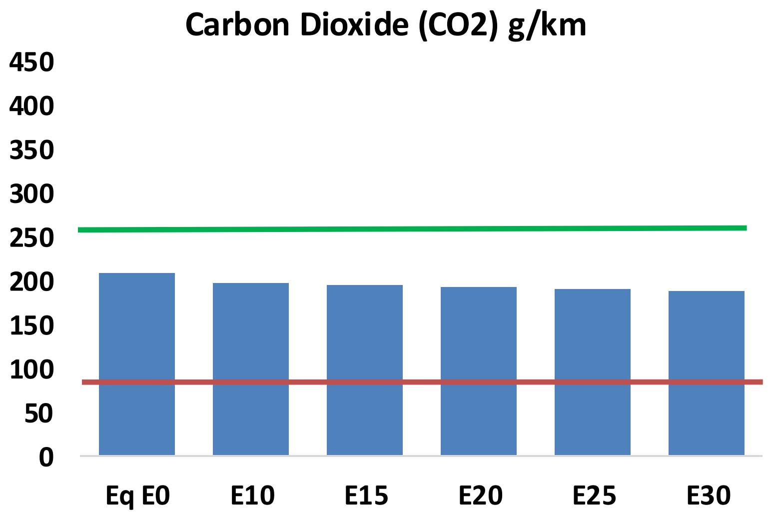
Bulgaria – Vehicular Emissions

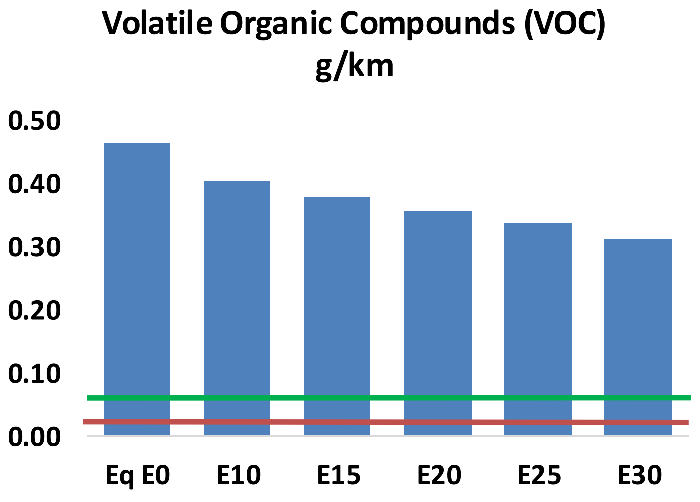
TIER III BIN 125 United States
Euro 6 Standard

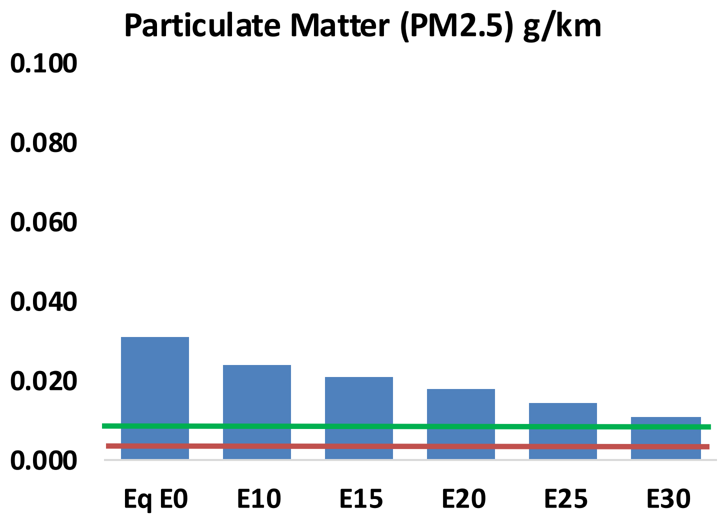












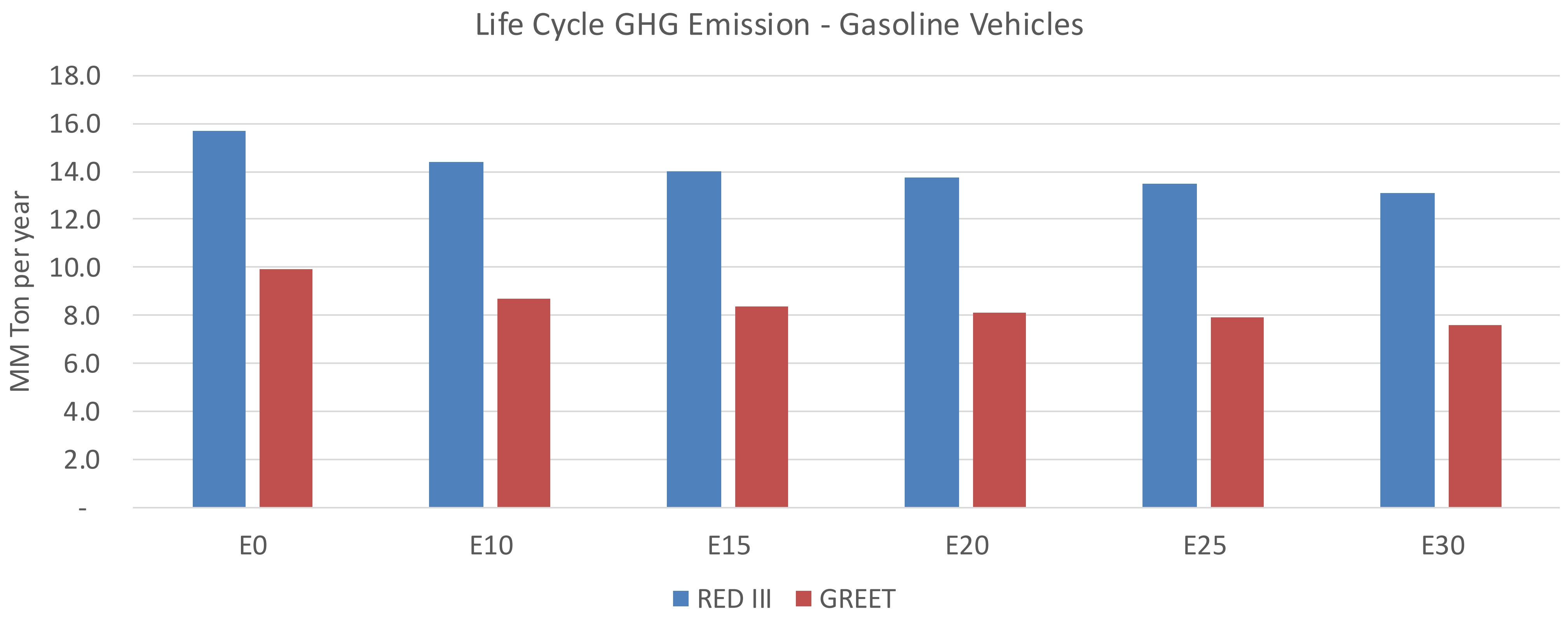
Source: Faro90

Bulgaria – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	263,559	244,735	225,910	210,765	195,653	184,072	171,172	-14%	-26%
CO2	16,709,105	16,279,442	15,873,649	15,554,505	15,396,581	15,309,889	15,027,684	-5%	-8%
VOC	37,225	34,766	32,308	30,372	28,460	27,000	25,014	-13%	-24%
NOx	22,636	16,977	15,845	14,934	14,084	13,147	12,156	-30%	-38%
SOx	199	184	169	156	144	131	117	-15%	-28%
NH3	5,918	5,859	5,800	5,828	5,894	5,940	5,978	-2%	0%
N2O	304	302	301	302	309	312	316	-1%	2%
CH4	8,123	8,123	8,123	8,245	8,423	8,578	8,724	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	271	261	252	247	243	240	235	-7%	-10%
Acetaldehyde	701	804	1,180	1,797	2,449	2,798	3,306	68%	249%
Formaldehyde	2,445	2,377	2,771	3,254	3,409	3,772	4,106	13%	39%
Benzene	507	487	461	454	450	431	417	-9%	-11%
PM2.5	1,033	917	801	696	591	480	363	-22%	-43%
PM10	1,437	1,276	1,115	968	822	667	506	-22%	-43%

Source: Faro90

Bulgaria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	15.3
Target @2035 (MMT/year)	3.1
Delta	12.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.5%	15.7%	18.2%	20.7%	23.9%
RED II/III	0.0%	8.4%	10.6%	12.4%	14.2%	16.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.2%	12.8%	14.9%	16.9%	19.6%
RED II/III	0.0%	10.8%	13.7%	16.1%	18.3%	21.1%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

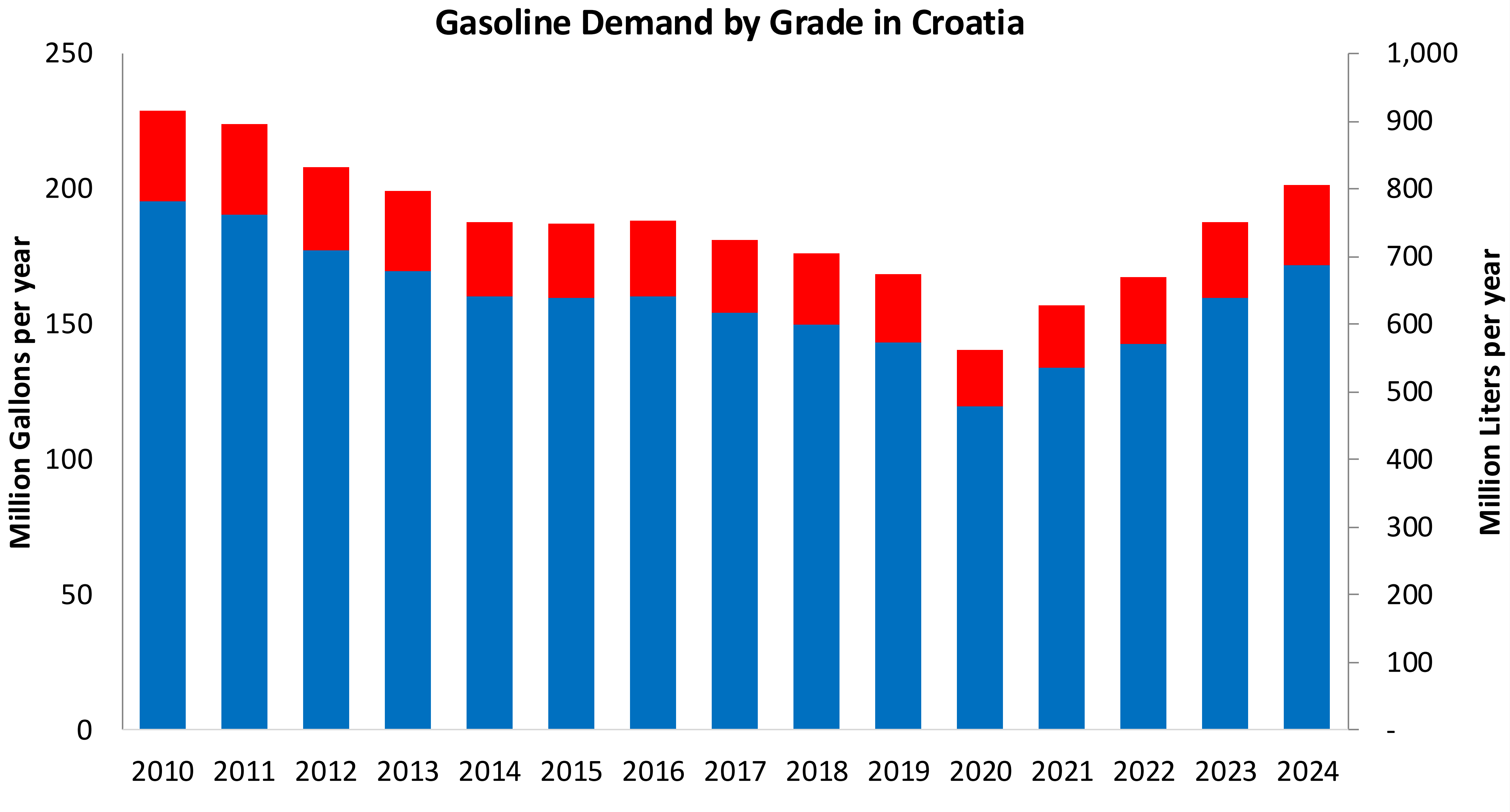
Ethanol Gasoline Blending in Croatia

In 2024, gasoline consumption in Croatia was 763 million liters (202 million gallons) and growth rate was 3.7%. Gasoline production and net imports were 709 and 82 million liters (187 and 22 million gallons) correspondingly. Croatia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

Croatia does not produce or consumes bioethanol in a commercial level. There is a small production of generic ethanol that is exported in minimal quantities to Italy, Slovenia, Austria and Bosnia.

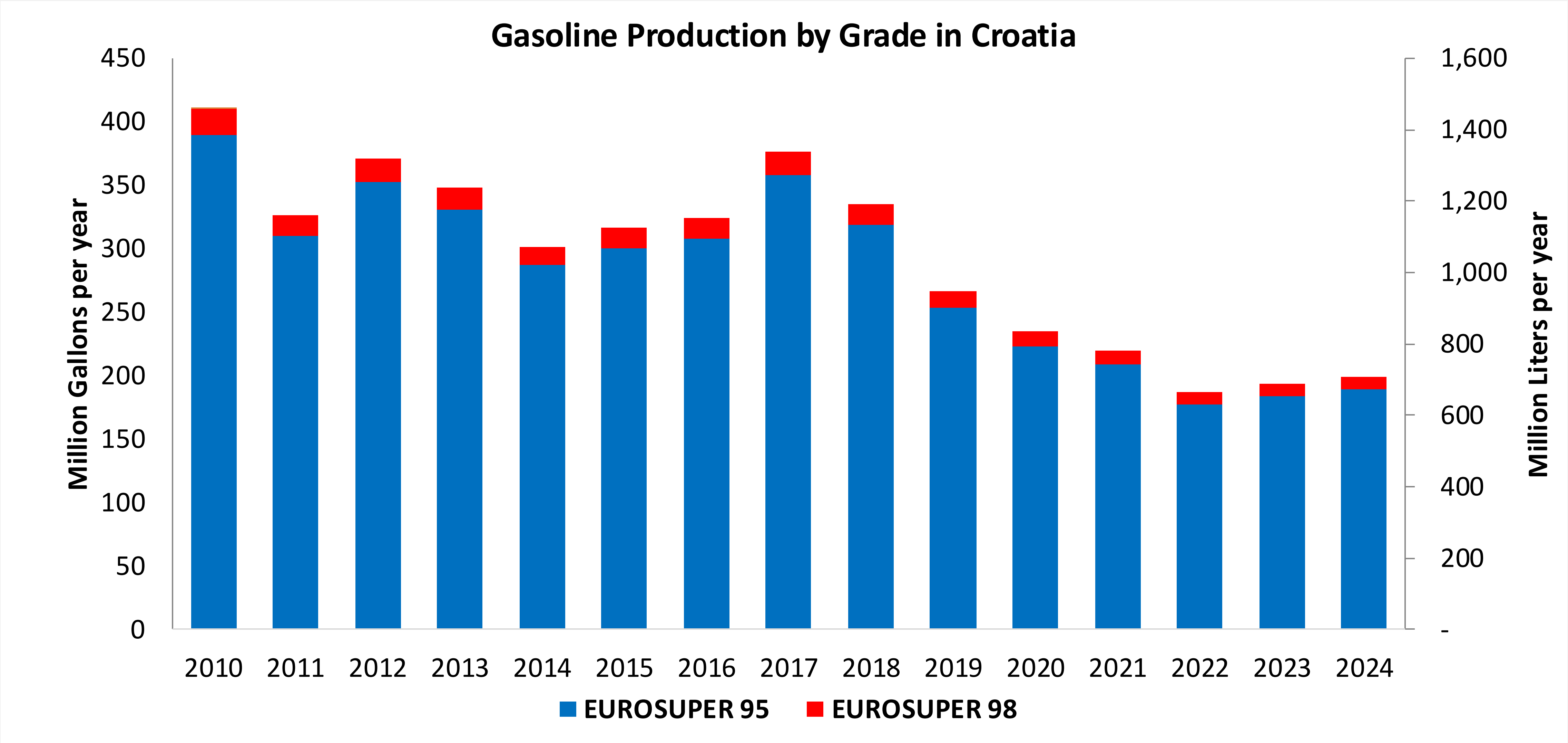
Source: Eurostat, European Commision

Gasoline Demand in Croatia



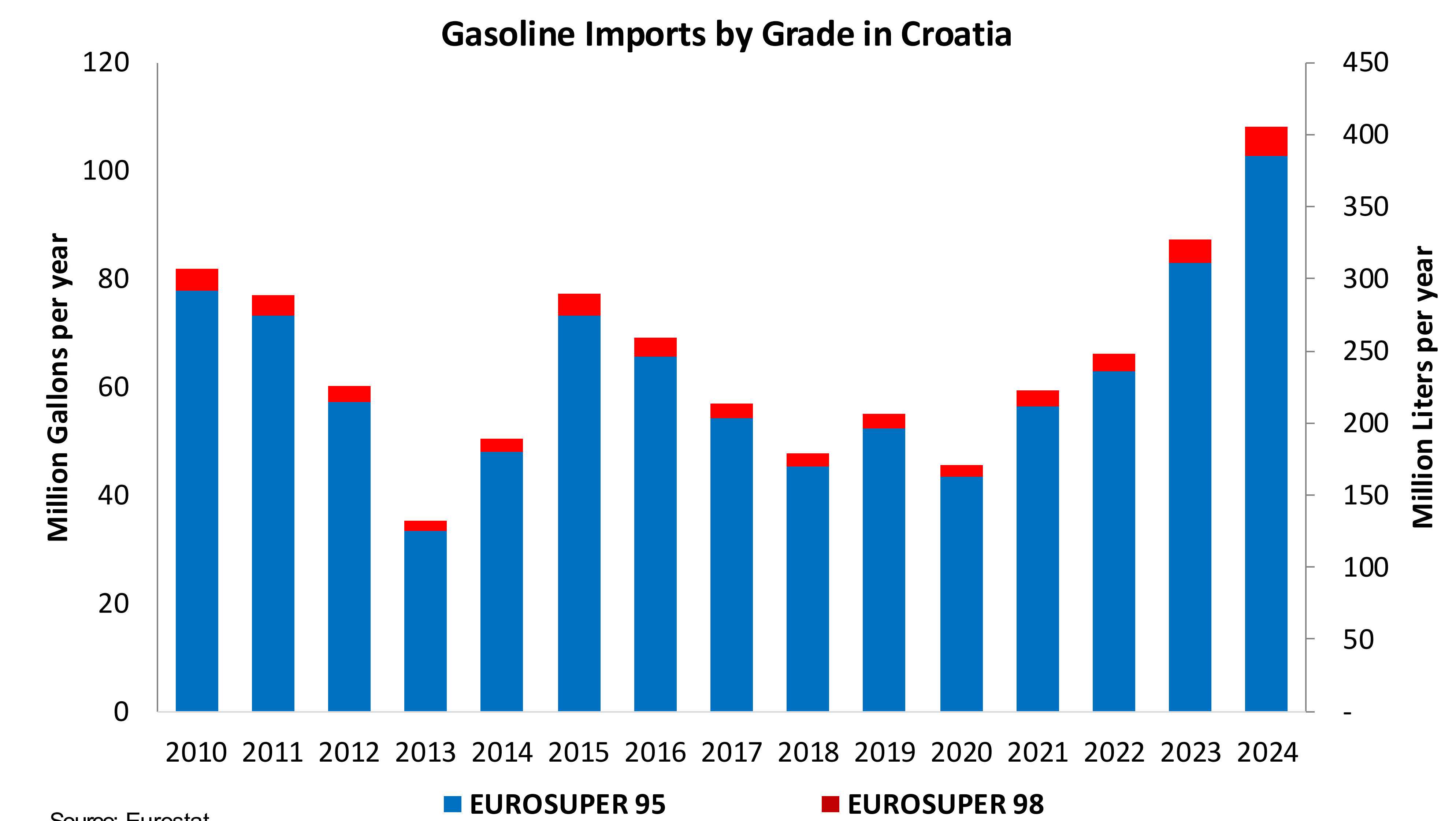
Source: Eurostat

Gasoline Production in Croatia



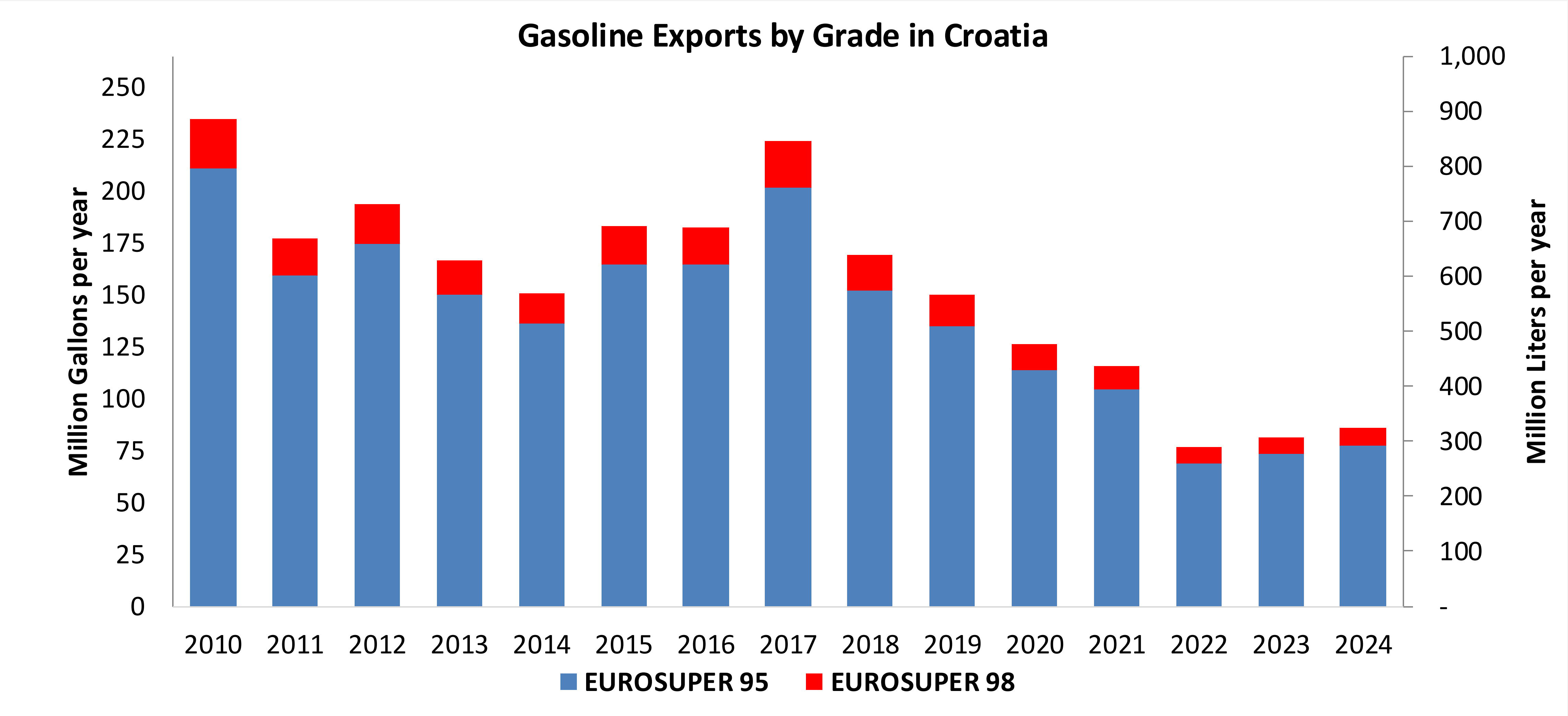
Source: Eurostat

Gasoline Imports in Croatia



Source: Eurostat

Gasoline Exports in Croatia



Source: Eurostat

Ethanol Balance in Croatia

Croatia does not produce or consumes bioethanol in a commercial level. There is a small production of generic ethanol that is exported in minimal quantities to Italy, Slovenia, Austria and Bosnia.

Source: The Global Economy, WITS/Comtrade, Total Croatia News, Axens

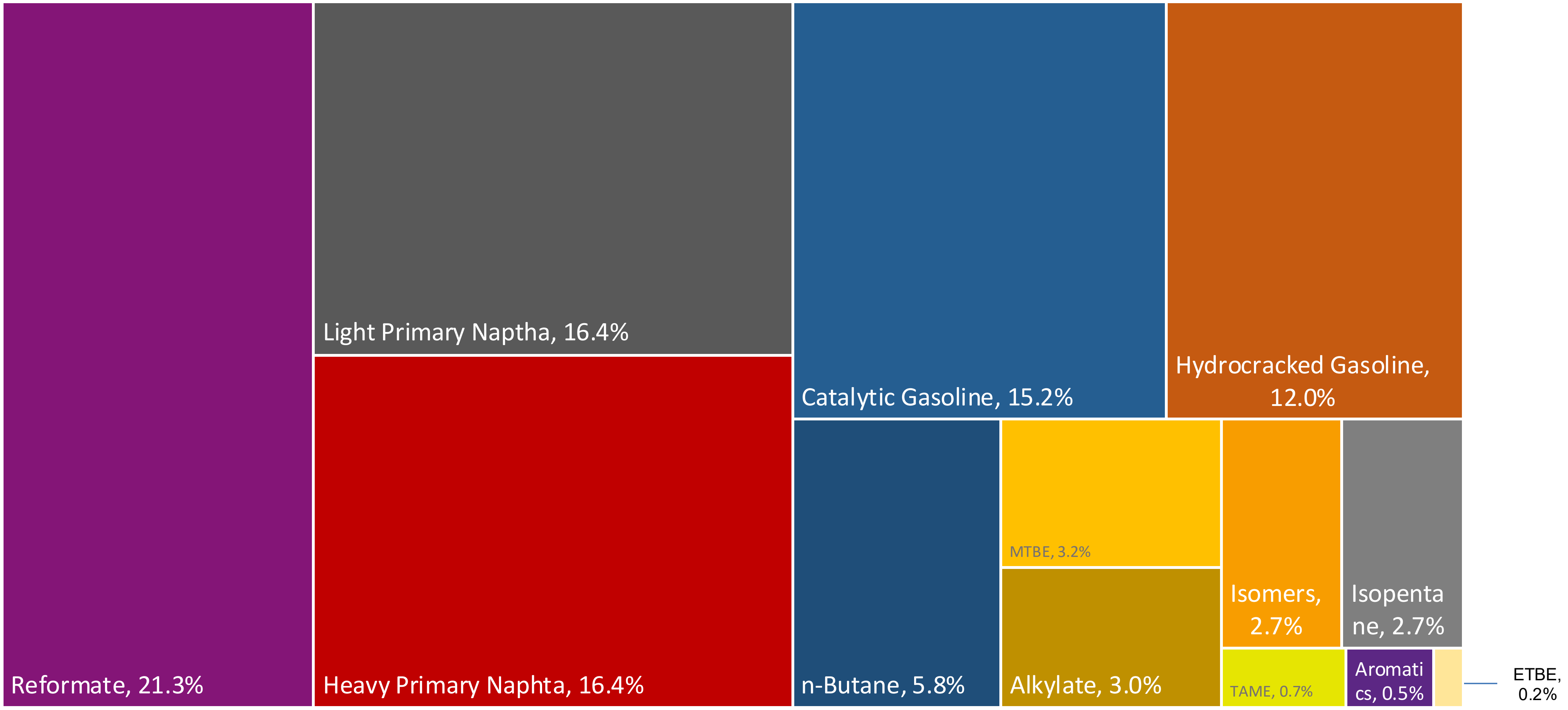
Gasoline Quality in Croatia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	8.8							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	1.0							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

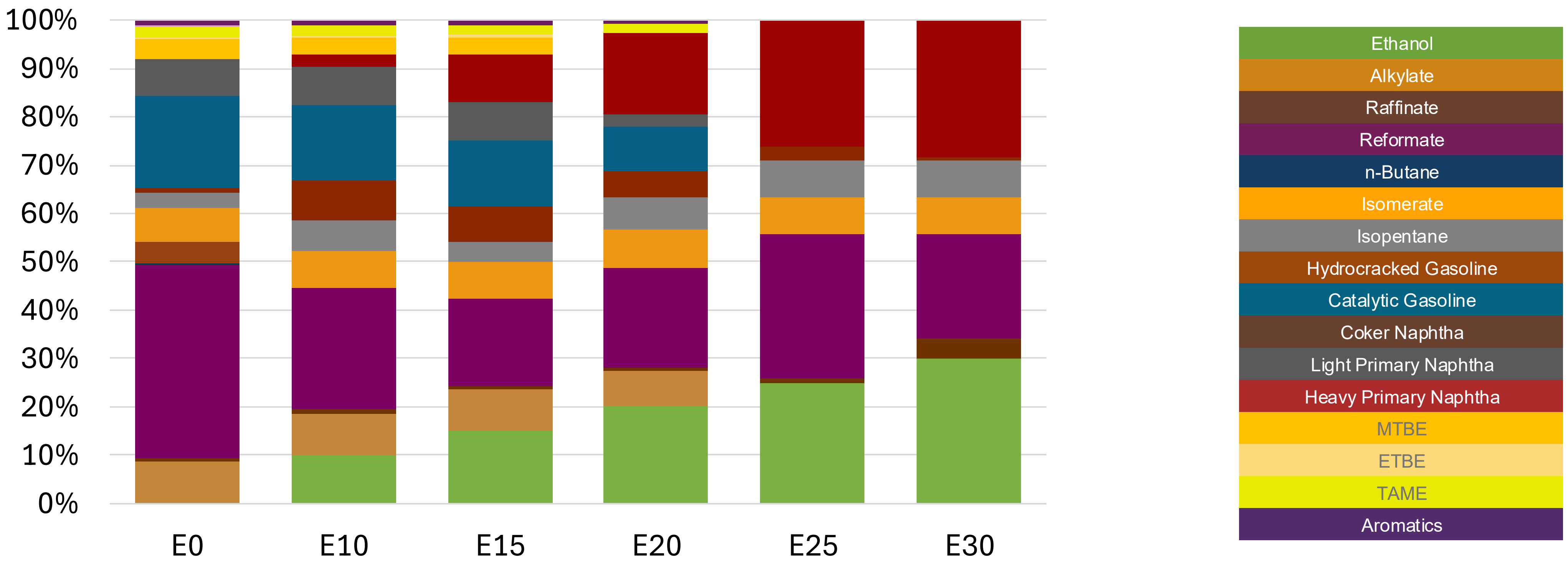
Source: European Commision

Croatia Available Blending Components



Source: Faro90

Croatia – Gasoline 95 – Constant Octane

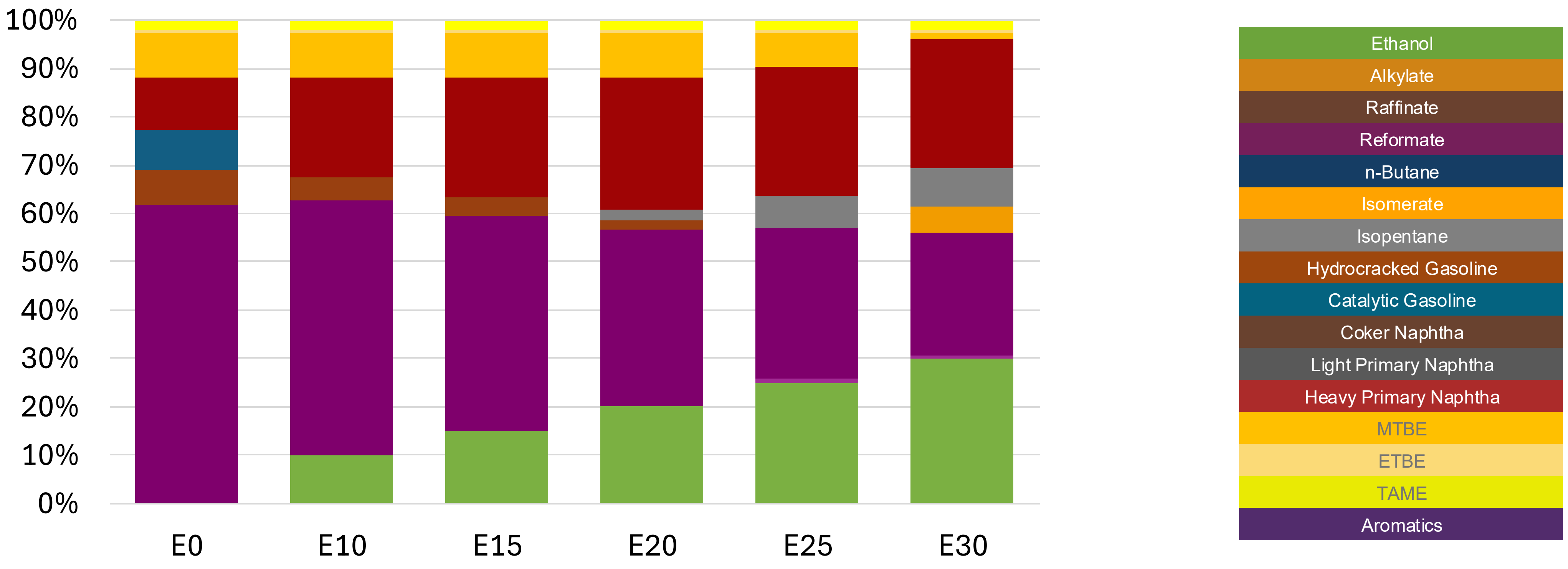


Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US$/gal)	$ 2.258	$ 2.145	$ 2.062	$ 1.986	$ 1.891	$ 1.817

Average CIF 2024 Prices

Source: Faro90

Croatia - Gasoline 98 - Constant Octane

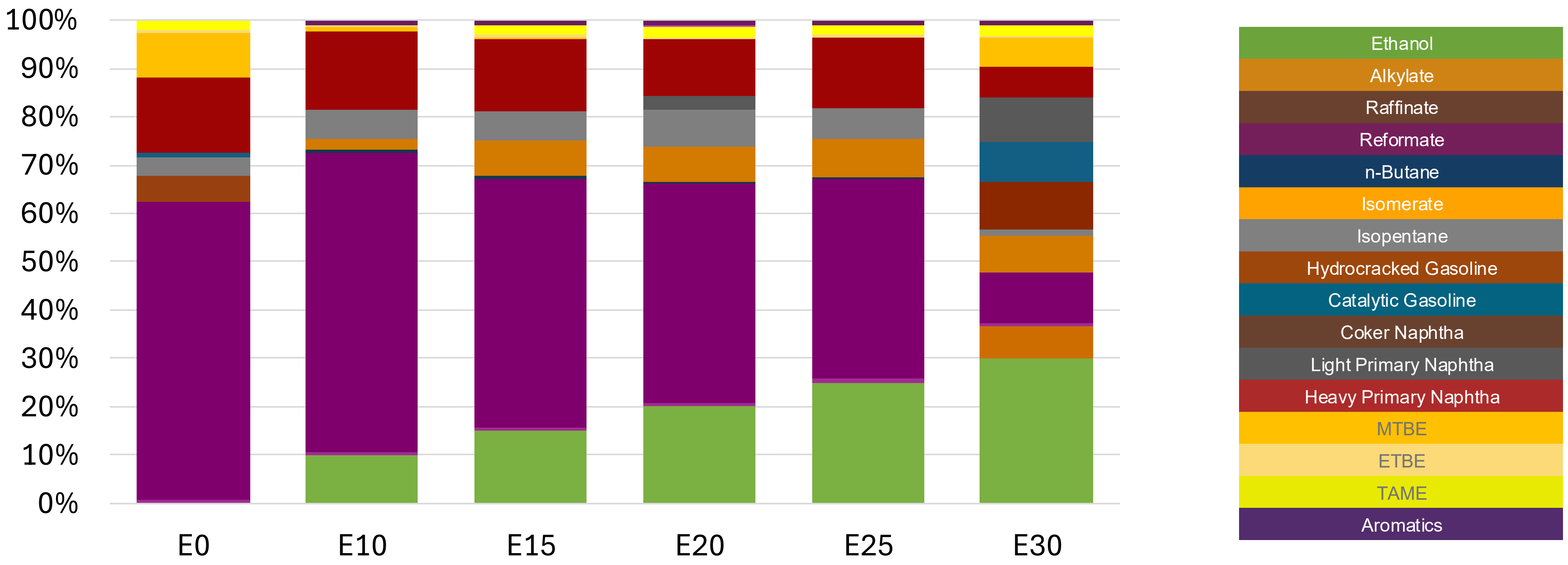


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.307	$ 2.188	$ 2.119	$ 2.049	$ 1.979	$ 1.926

Average CIF 2024 Prices

Source: Faro90

Croatia – Gasoline 95 – Increased Octane

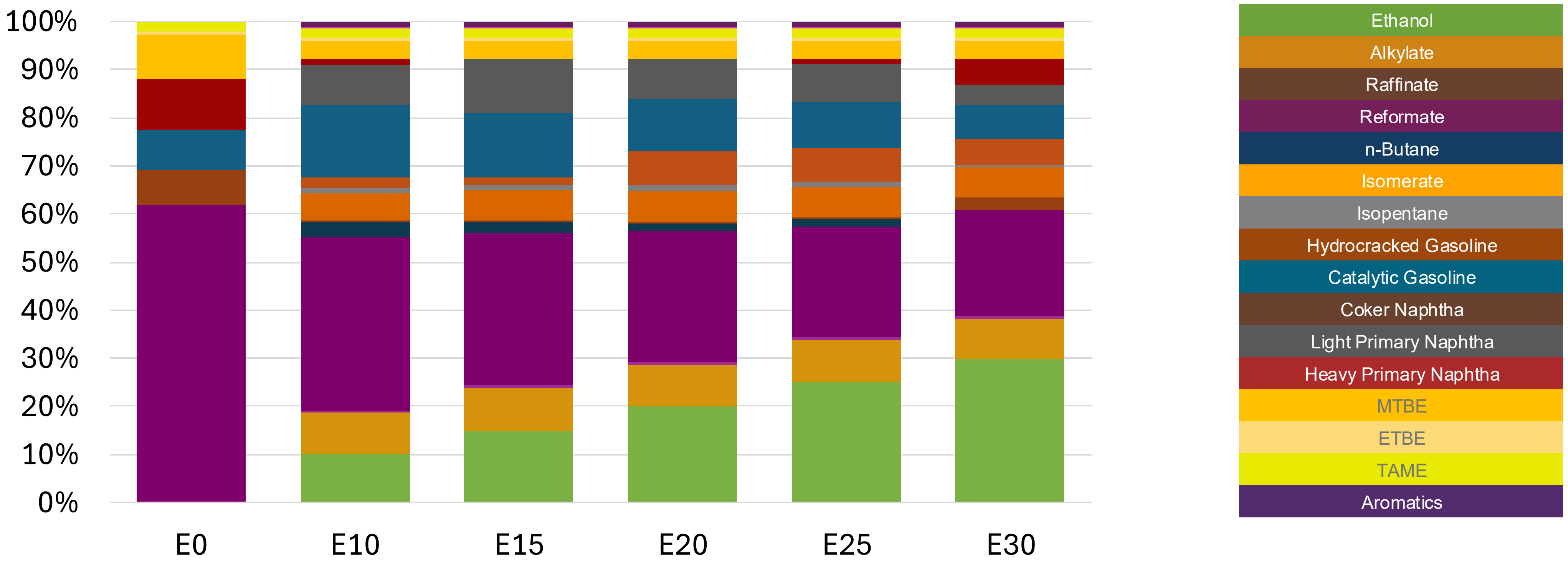


Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US$/gal)	$ 2.177	$ 2.177	$ 2.177	$ 2.177	$ 2.177	$ 2.177

Average CIF 2024 Prices

Source: Faro90

Croatia – Gasoline 98 – Increased Octane



Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US$/gal)	$ 2.307	$ 2.307	$ 2.307	$ 2.307	$ 2.307	$ 2.307

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Croatia

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	634
		Single FI	3-Ways		>8 years	25,598
		Multi FI	3-Ways		>8 years	175,214
			Euro III		>8 years	
			Euro IV		>8 years	
			Euro V		>8 years	40,031
			Euro 6		>8 years	65,949
					4-8 years	120,875
					<4 years	45,411
			Hybrid		All	20,269
		Electric			All	3,283
	Medium	Carburator	3-Ways	PCV Tank	>8 years	63
		Single FI	3-Ways		>8 years	727
		Multi FI	3-Ways		>8 years	26,413
			Euro III		>8 years	42,880
			Euro IV		>8 years	89,459
			Euro V		>8 years	12,243
			Euro 6		>8 years	21,983
					4-8 years	55,443
					<4 years	35,708
			Hybrid		All	17,057
		Electric			All	2,352
Sml Engines	Light	4-Cycle	None	PCV	>8 years	30,933
			Euro III		>8 years	37,503
			Euro IV		>8 years	197,625
			Euro V		>8 years	104,828
			Euro 6		>8 years	
					4-8 years	
					<4 years	183,162
			Hybrid		All	30,036
		Electric			All	4,535
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	1,072
		Fuel-Injection	3-Ways		>8 years	8,237
			Euro III		>8 years	1,951
			Euro IV		>8 years	2,816
			Euro V		>8 years	363
			Euro 6		>8 years	1,494
					4-8 years	1,728
					<4 years	511
			Hybrid		All	2,114
		Electric			All	2,176

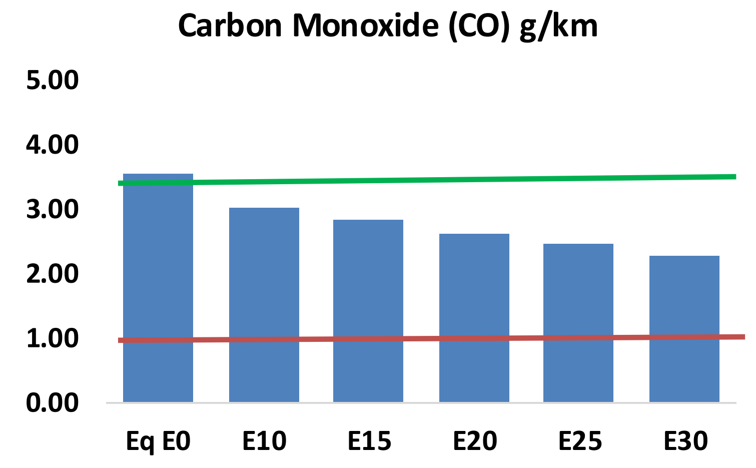
0 20.000 40.000 60.000 80.000 100.000 120.000 140.000 160.000 180.000 200.000

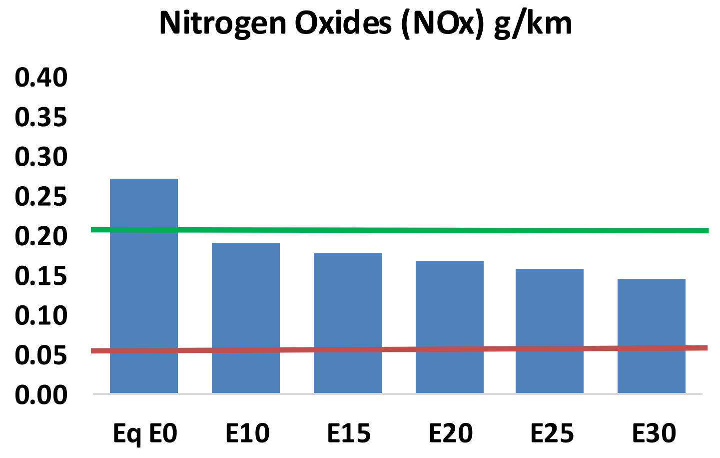
Vehicle Fleet: 2,358,574
Gasoline Vehicle Fleet: 923,614

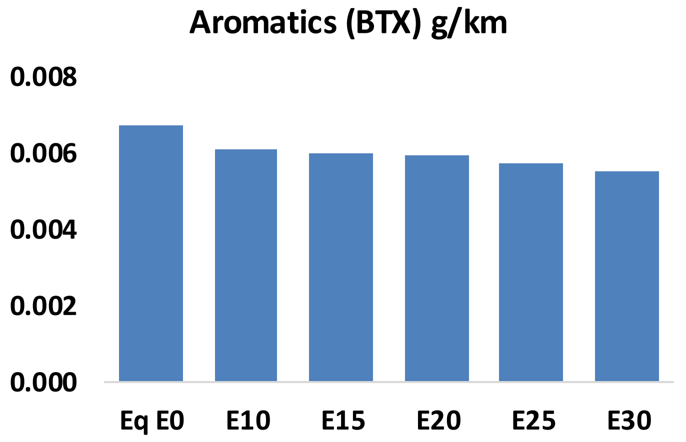
Source: UNECE, Eurostat, ACEA

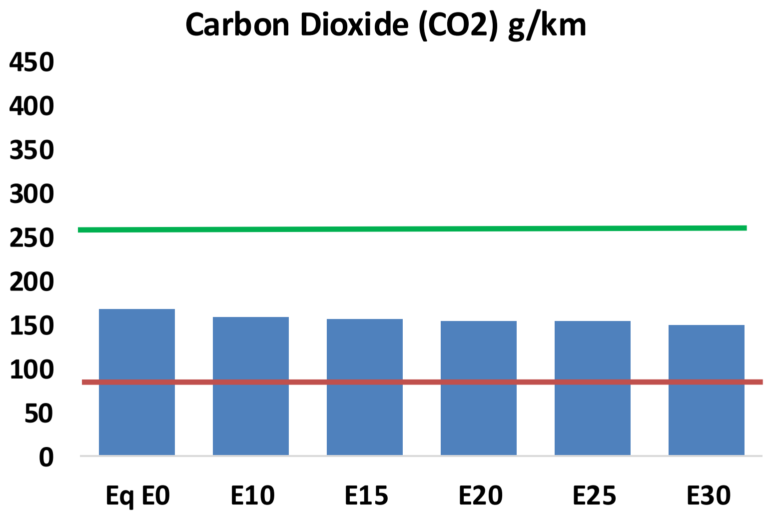
Croatia – Vehicular Emissions

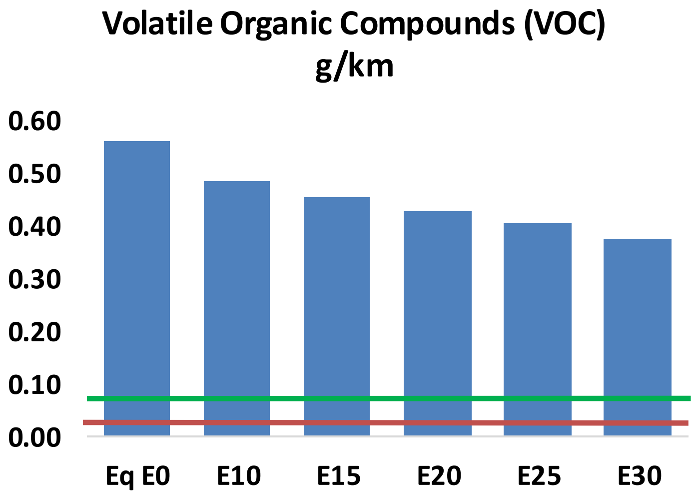
TIER III BIN 125 United States
Euro 6 Standard

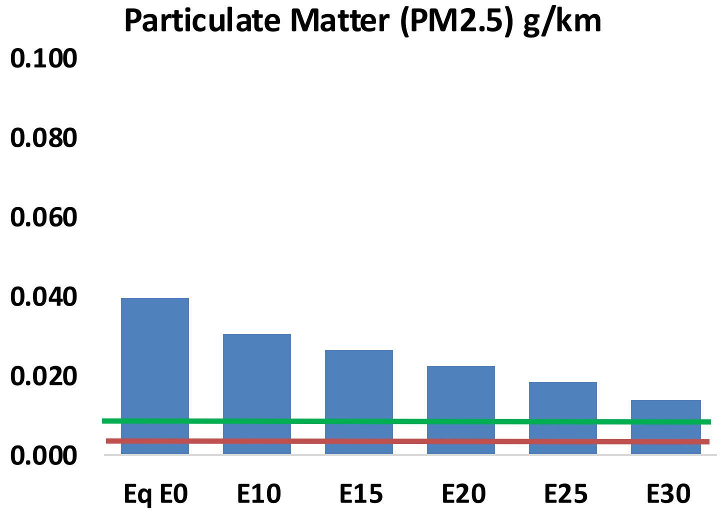












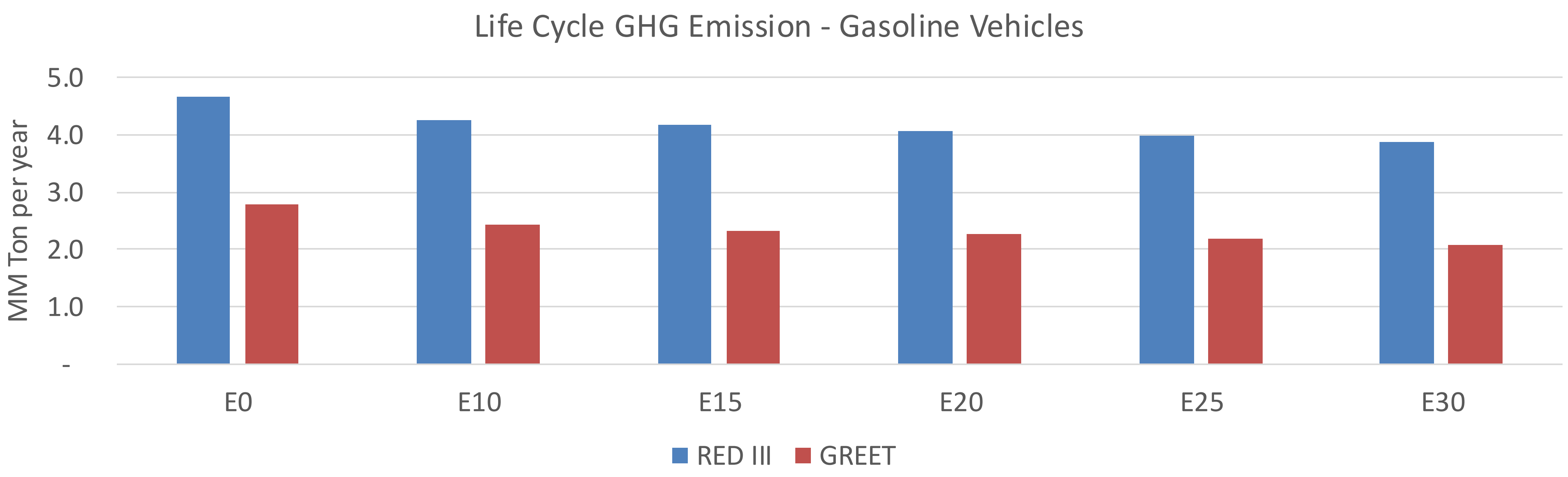
Source: Faro90

Croatia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	91,726	84,984	78,241	72,777	67,305	63,106	58,353	-15%	-27%
CO2	4,317,386	4,206,367	4,101,516	4,019,054	3,978,249	3,955,342	3,882,433	-5%	-8%
VOC	14,448	13,483	12,518	11,755	11,001	10,425	9,643	-13%	-24%
NOx	7,031	5,273	4,922	4,639	4,375	4,084	3,776	-30%	-38%
SOx	48	45	41	38	35	32	28	-15%	-28%
NH3	1,642	1,626	1,610	1,617	1,635	1,648	1,659	-2%	0%
N2O	59	59	59	59	60	61	62	-1%	2%
CH4	3,492	3,492	3,492	3,545	3,622	3,688	3,751	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	101	96	92	89	86	85	82	-9%	-14%
Acetaldehyde	293	336	493	751	1,023	1,169	1,381	68%	249%
Formaldehyde	1,099	1,069	1,246	1,463	1,533	1,696	1,846	13%	39%
Benzene	173	167	158	155	154	147	143	-9%	-11%
PM2.5	293	260	228	198	168	136	103	-22%	-43%
PM10	722	641	560	487	413	335	254	-22%	-43%

Source: Faro90

Croatia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	17.9
Target @2035 (MMT/year)	10.1
Delta	7.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.3%	16.6%	19.4%	22.1%	25.6%
RED II/III	0.0%	8.5%	10.8%	12.7%	14.5%	16.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.7%	5.9%	6.9%	7.9%	9.1%
RED II/III	0.0%	5.1%	6.4%	7.6%	8.6%	10.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

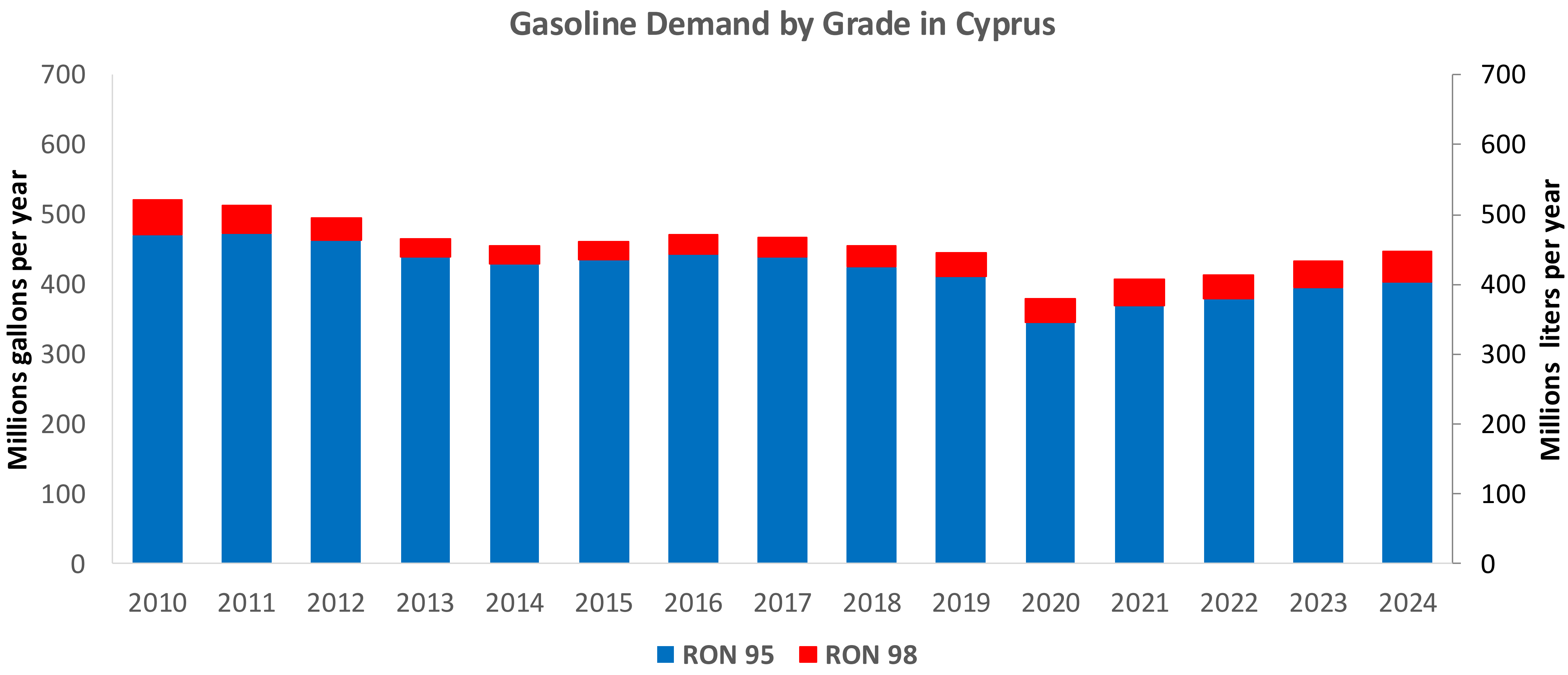
Ethanol Gasoline Blending in Cyprus

In 2024, gasoline consumption in Cyprus was 446 million liters (118 million gallons) and growth rate was 0.1%. Gasoline production and net imports were and 446 million liters (and 118 million gallons) correspondingly. Cyprus complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

Cyprus has no specific ethanol mandate and a target of 14% biofuels in the transport sector by 2030. Ethanol demand was 17.4 million liters in 2024 (4.6 million gallons) equivalent to 3.9% of gasoline demand. Ethanol is supplied by imports.

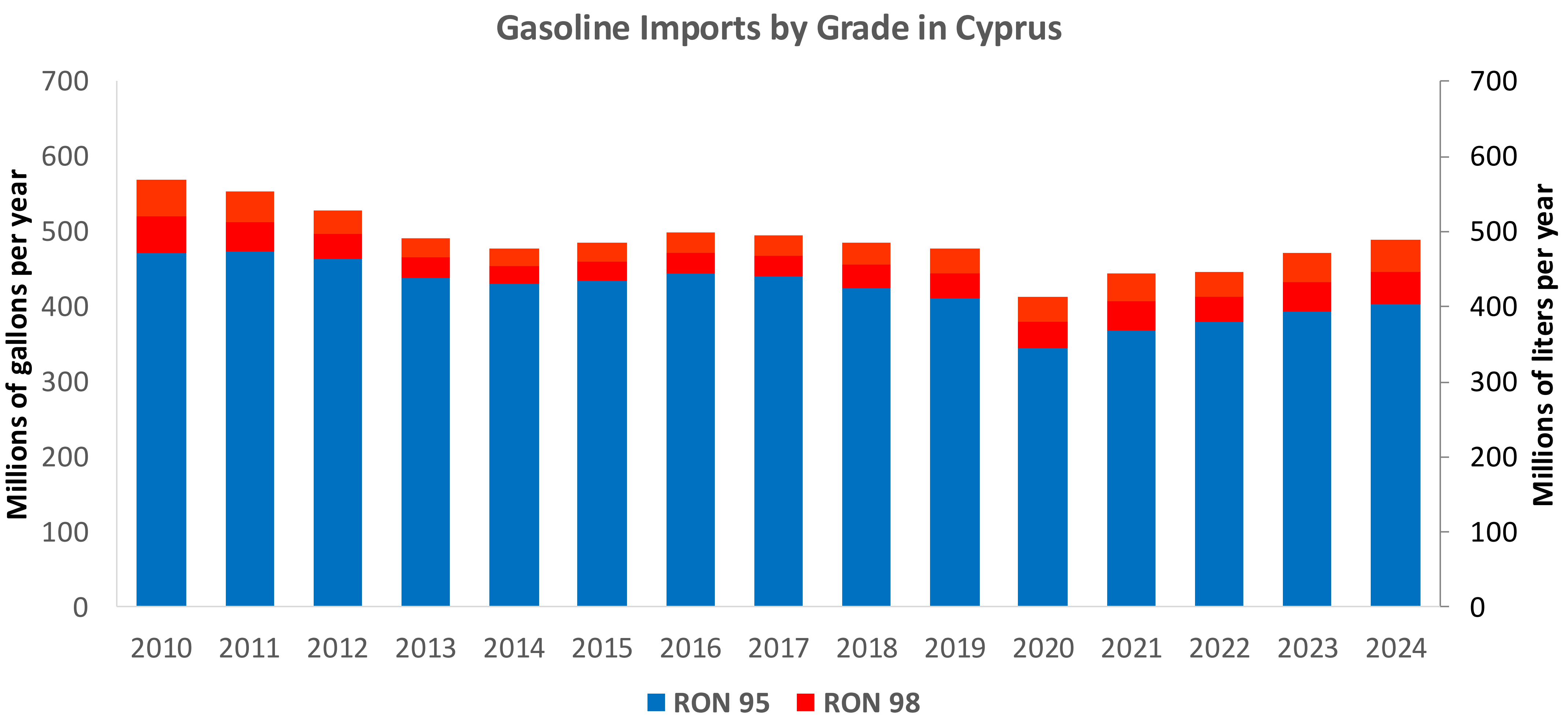
Source: Eurostat, European Commision

Gasoline Demand in Cyprus



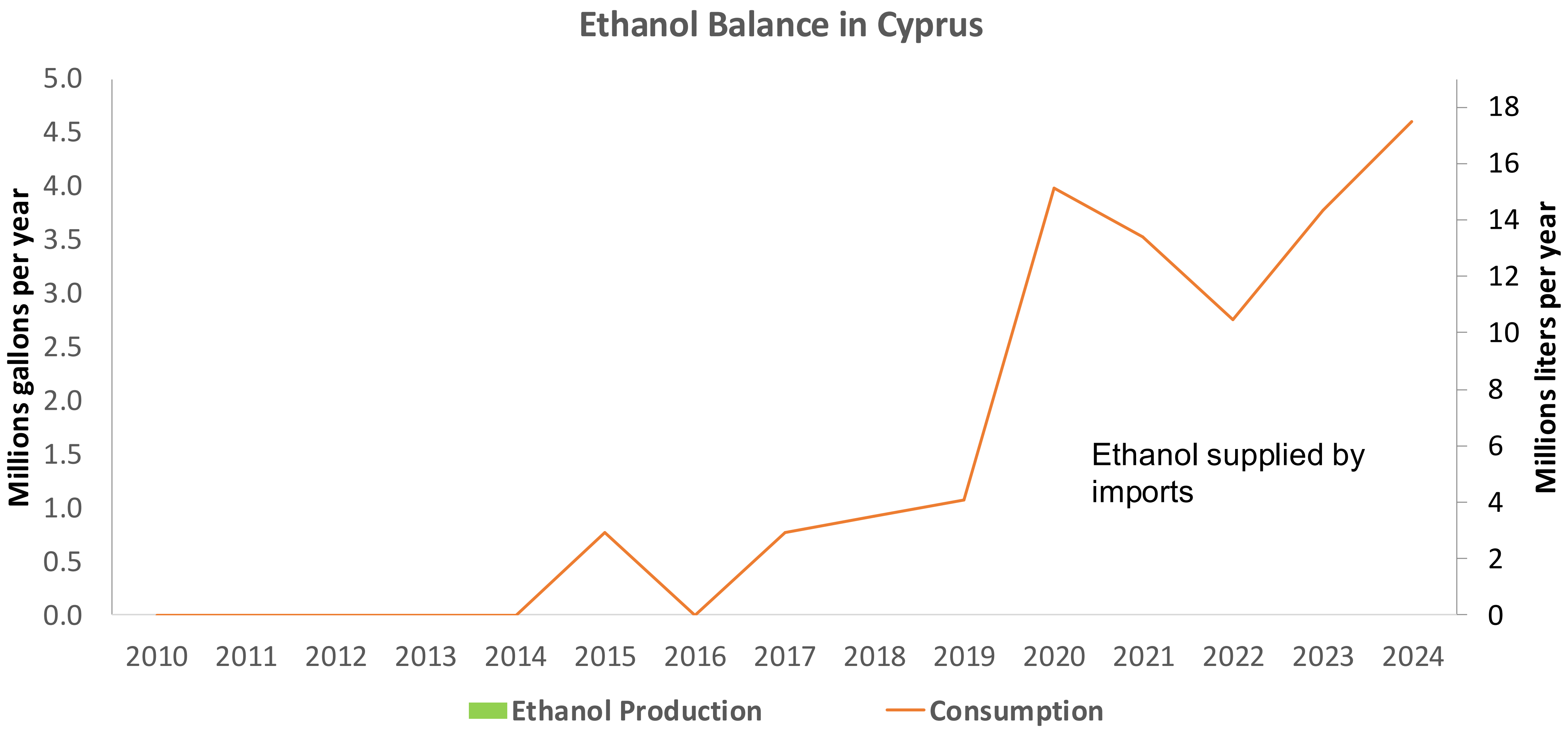
Source: Eurostat

Gasoline Imports in Cyprus



Source: Eurostat

Ethanol Balance in Cyprus



Source: Eurostat, Cyprus Statistics, Global Economy Report

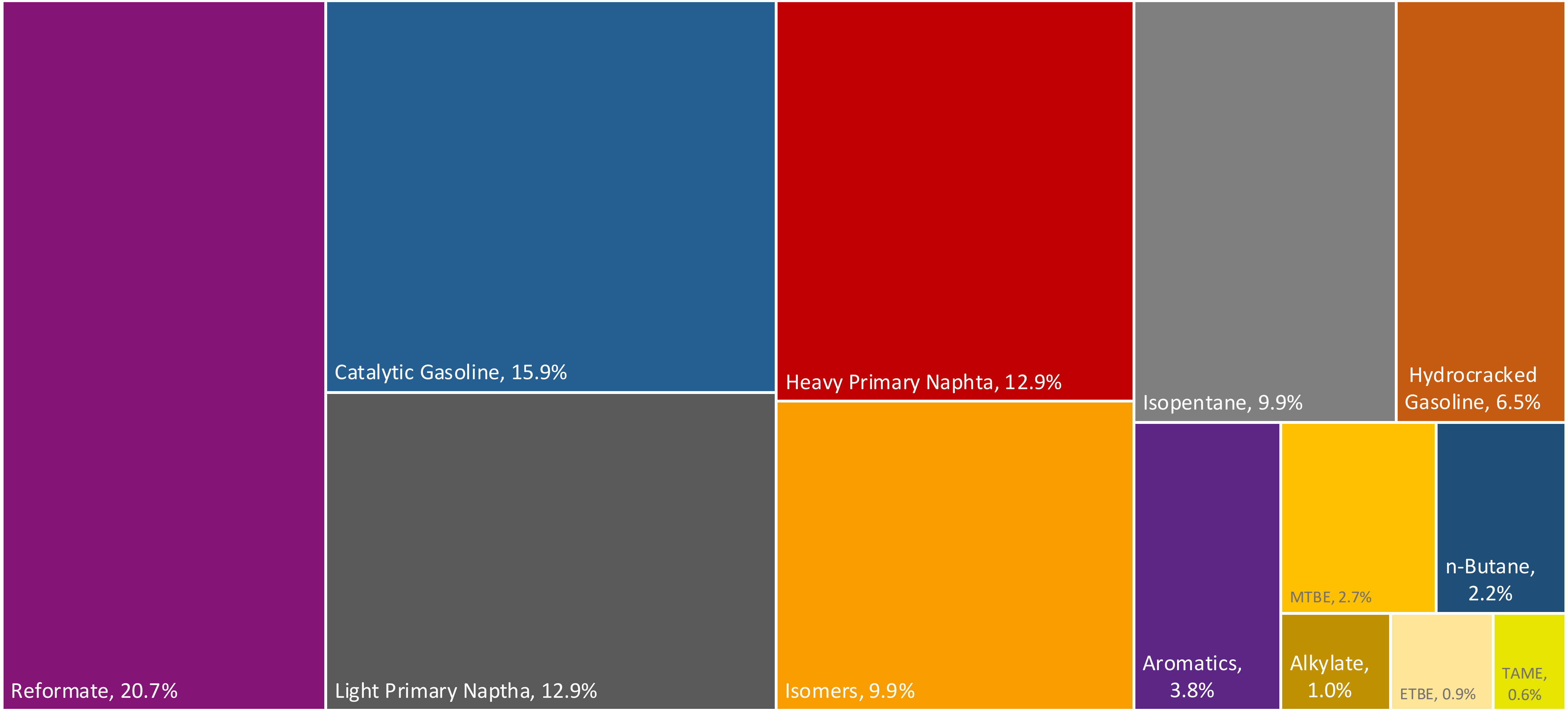
Gasoline Quality in Cyprus

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

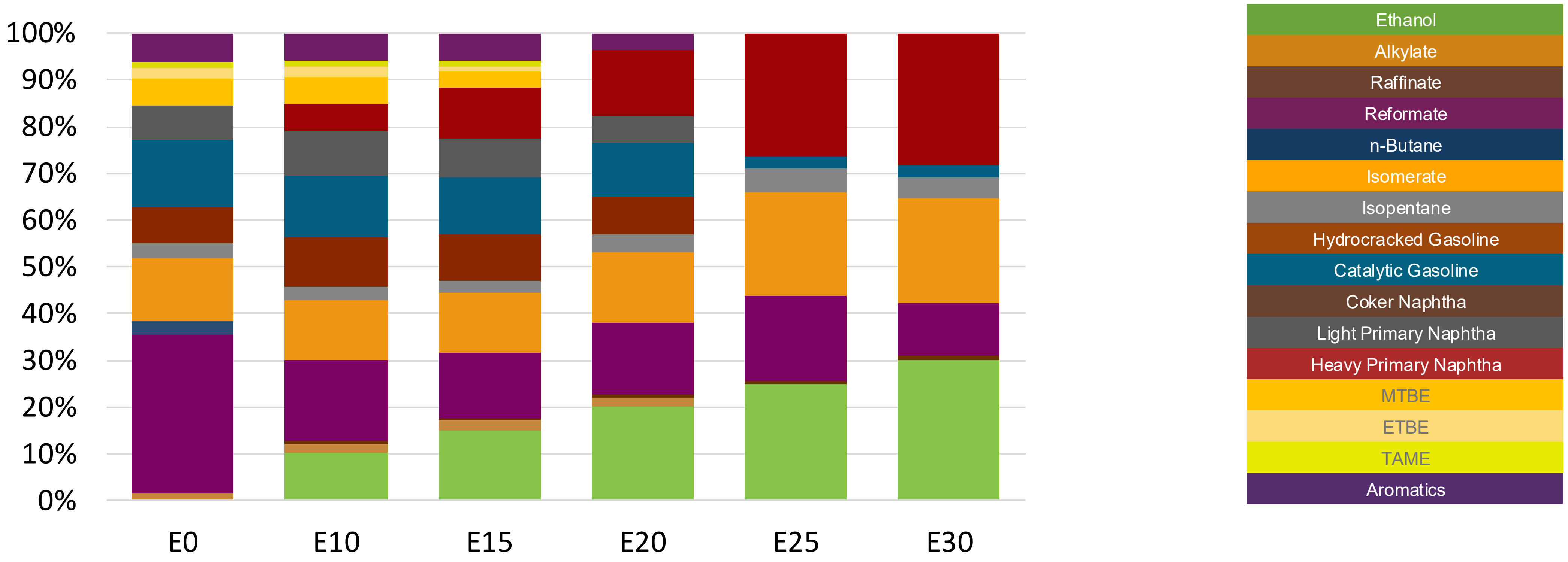
Source: European Commision

Cyprus Available Blending Components



Source: Faro90

Cyprus – Gasoline 95 – Constant Octane

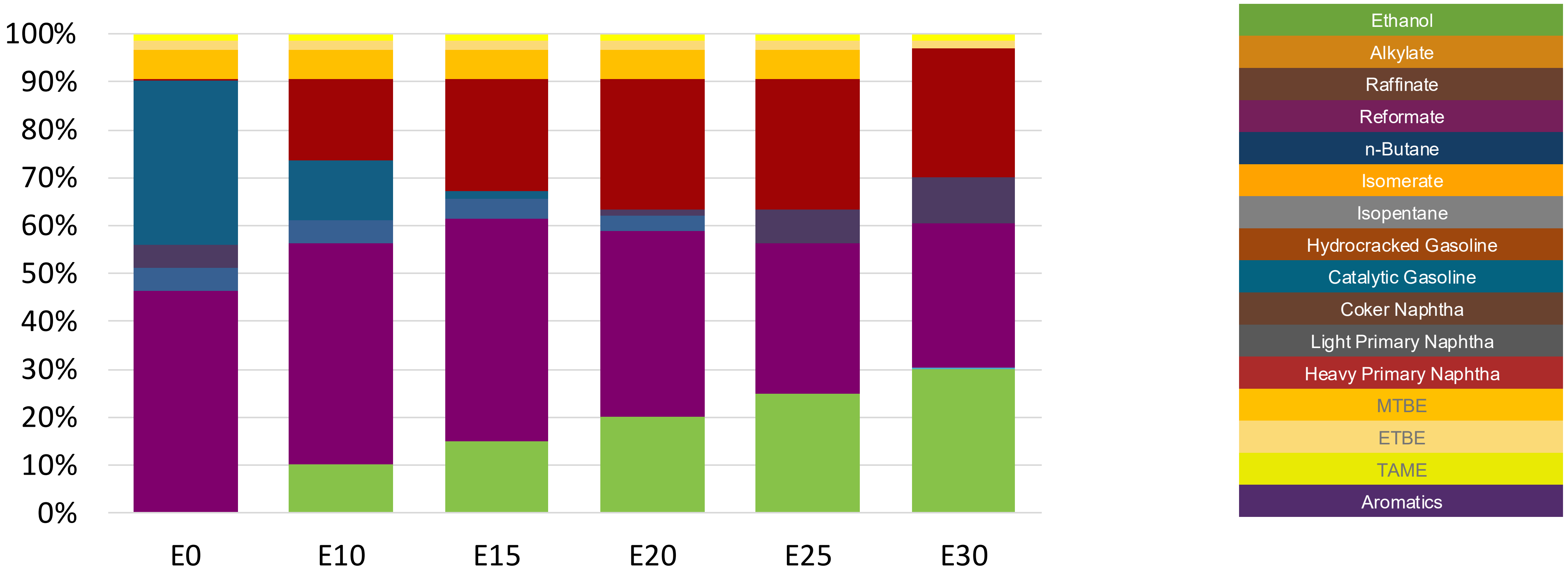


Octane (RON)	95.0	95.0	95.0	95.0	95.0	96.0
Price (US$/gal)	$ 2.258	$ 2.135	$ 2.069	$ 2.002	$ 1.881	$ 1.845

Average CIF 2024 Prices

Source: Faro90

Cyprus - Gasoline 98 - Constant Octane

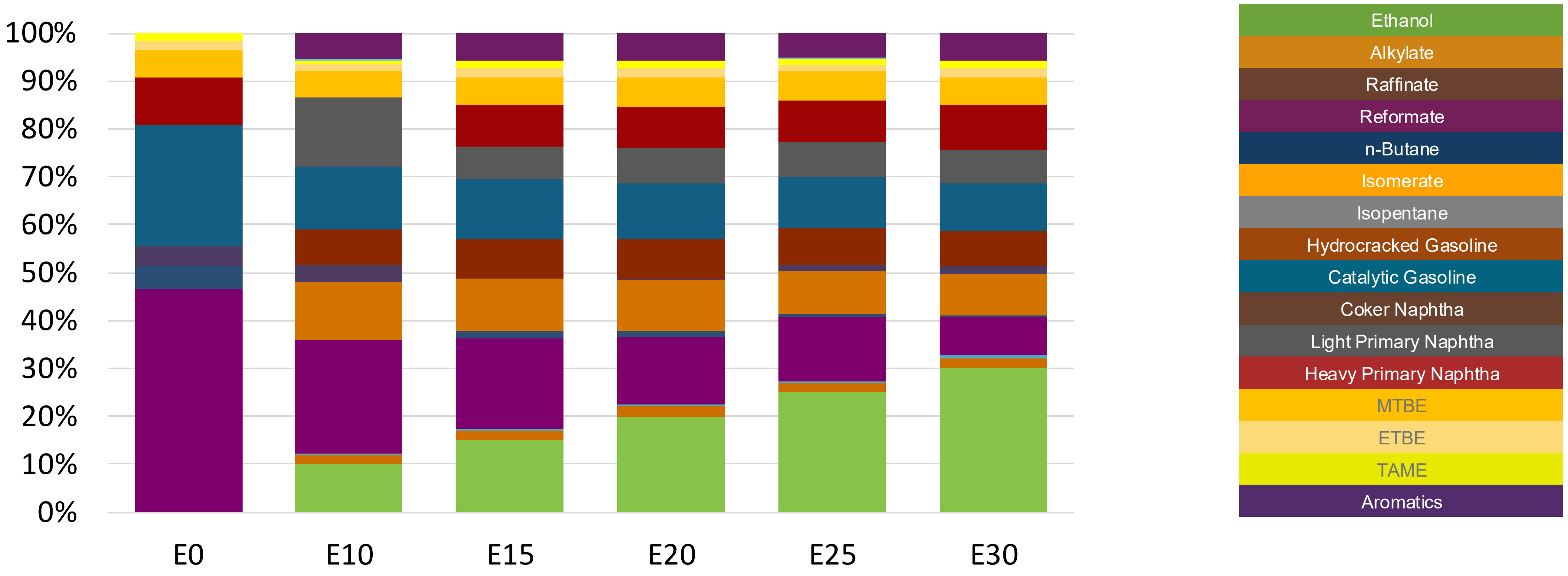


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.330	$ 2.204	$ 2.132	$ 2.057	$ 1.980	$ 1.924

Average CIF 2024 Prices

Source: Faro90

Cyprus – Gasoline 95 – Increased Octane

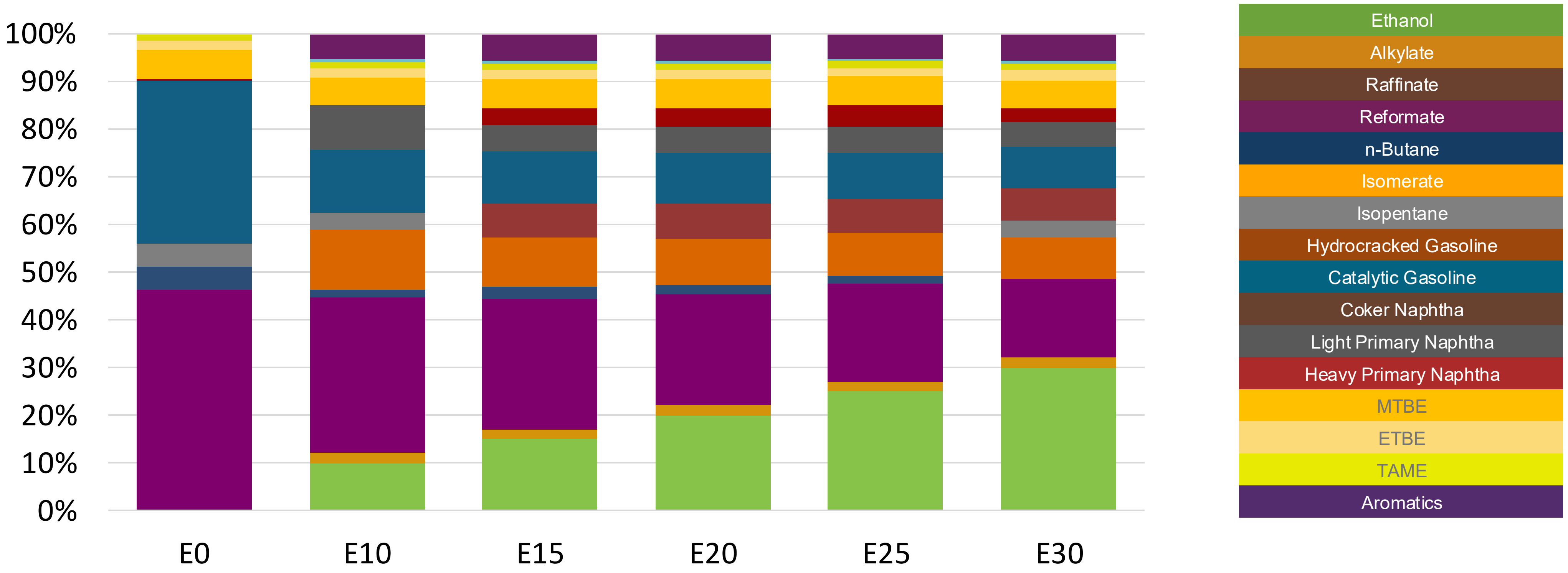


Octane (RON)	95.0	96.2	97.9	99.4	100.8	102.2
Price (US$/gal)	$ 2.196	$ 2.196	$ 2.196	$ 2.196	$ 2.196	$ 2.196

Average CIF 2024 Prices

Source: Faro90

Cyprus – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.4	100.7	102.0	103.3	104.5
Price (US$/gal)	$ 2.330	$ 2.330	$ 2.330	$ 2.330	$ 2.330	$ 2.330

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Cyprus

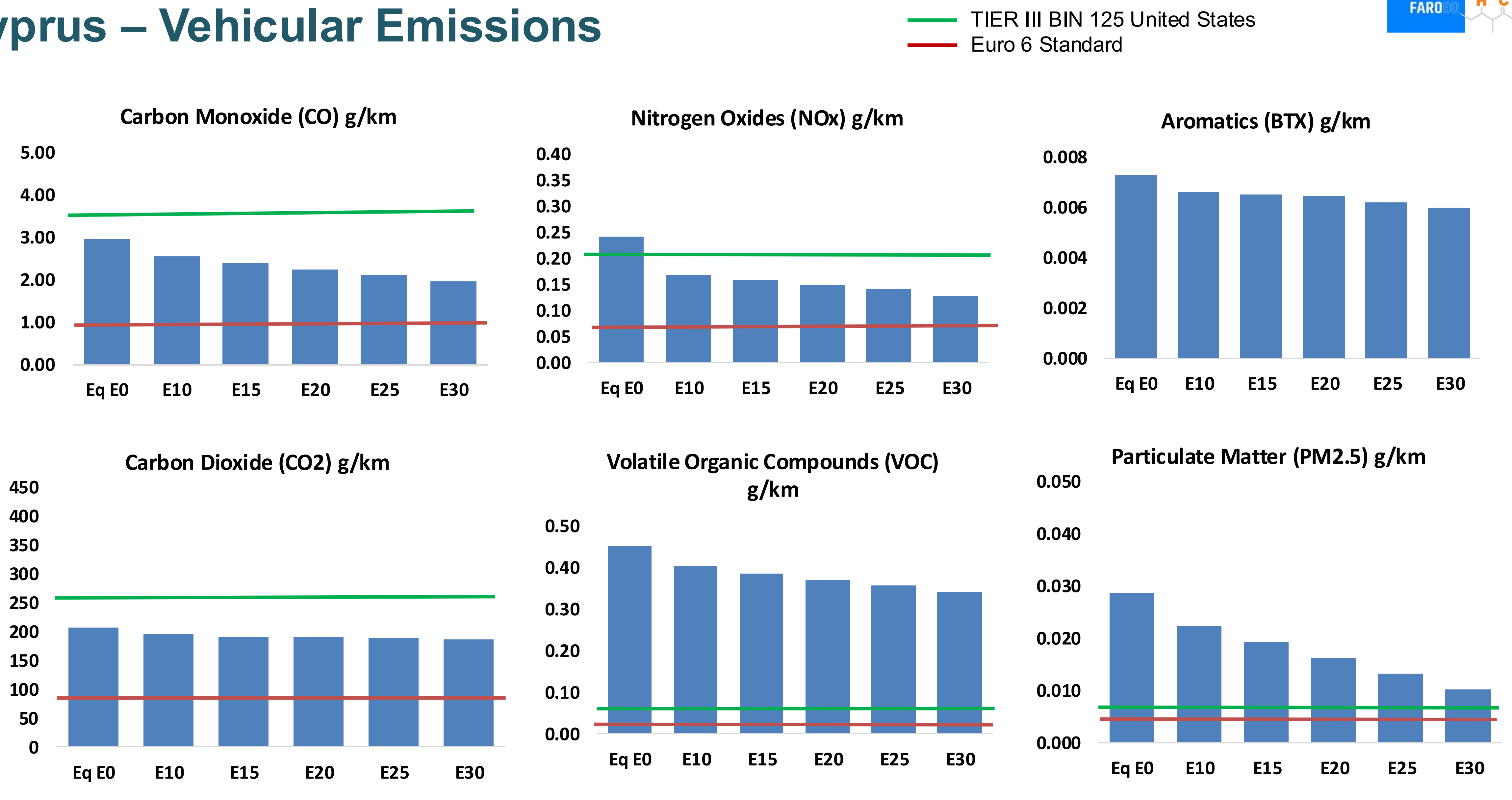
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	556
		Single FI	3-Ways		>8 years	9,050
		Multi FI	3-Ways		>8 years	17,869
			Euro III		>8 years	17,498
			Euro IV		>8 years	87,408
			Euro V		>8 years	47,572
			Euro 6		>8 years	81,464
					4-8 years	144,674
					<4 years	128,128
			Hybrid		All	28,915
		Electric			All	2,691
	Medium	Carburator	3-Ways	PCV Tank	>8 years	1,076
		Single FI	3-Ways		>8 years	14,221
		Multi FI	3-Ways		>8 years	20,318
			Euro III		>8 years	15,556
			Euro IV		>8 years	29,450
			Euro V		>8 years	5,823
			Euro 6		>8 years	7,405
					4-8 years	17,291
					<4 years	15,356
			Hybrid		All	6,847
		Electric			All	637
Sml Engines	Light	4-Cycle	None	PCV	>8 years	27,475
			Euro III		>8 years	17,498
			Euro IV		>8 years	27,516
			Euro V		>8 years	5,212
			Euro 6		>8 years	8,232
					4-8 years	11,453
					<4 years	14,636
			Hybrid		All	3
		Electric			All	1,881
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	545
		Fuel-Injection	3-Ways		>8 years	19,169
			Euro III		>8 years	9,841
			Euro IV		>8 years	18,088
			Euro V		>8 years	3,905
			Euro 6		>8 years	4,082
					4-8 years	7,133
					<4 years	8,392
			Hybrid		All	10,423
		Electric			All	151

0 20,000 40,000 60,000 80,000 100,000 120,000 140,000 160,00

Vehicle Fleet: 895,440
Gasoline Vehicle Fleet: 6685,744

Source: Eurostat, ACEA

Cyprus – Vehicular Emissions



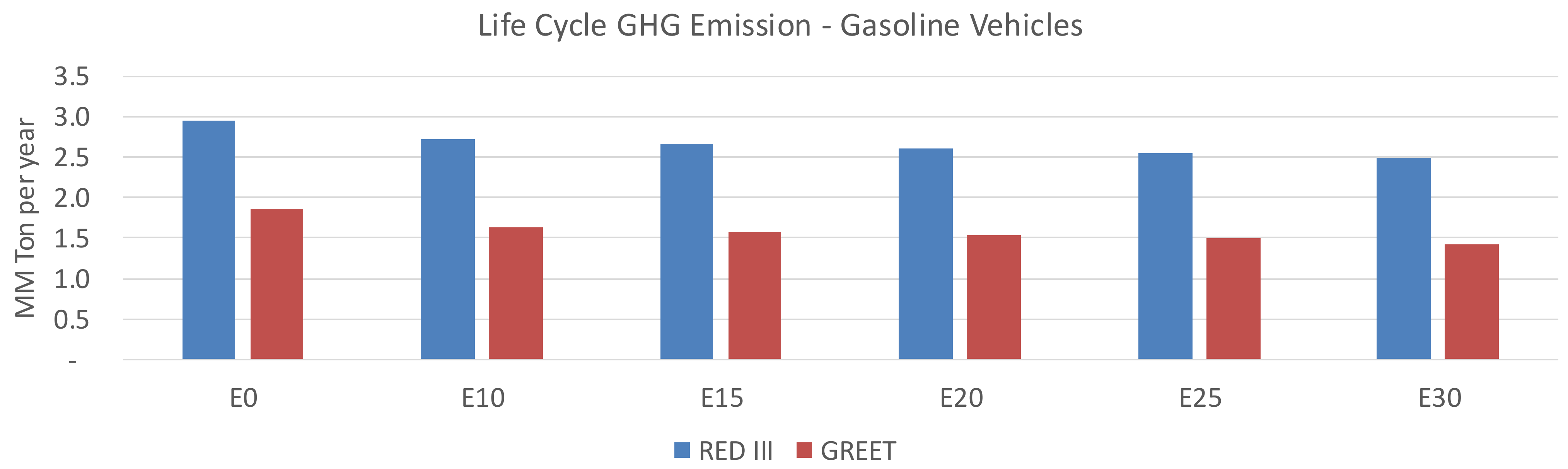
Source: Faro90

Cyprus – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	46,762	43,557	40,352	37,802	35,271	33,335	31,171	-14%	-25%
CO2	3,244,976	3,161,534	3,082,728	3,020,749	2,990,079	2,975,062	2,920,223	-5%	-8%
VOC	7,110	6,737	6,364	6,092	5,834	5,640	5,365	-10%	-18%
NOx	3,799	2,849	2,659	2,507	2,364	2,207	2,040	-30%	-38%
SOx	37	34	31	29	27	24	22	-15%	-28%
NH3	1,157	1,146	1,135	1,140	1,153	1,162	1,169	-2%	0%
N2O	43	43	43	43	44	44	45	-1%	2%
CH4	1,608	1,608	1,608	1,632	1,668	1,698	1,727	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	40	38	36	34	33	32	31	-10%	-17%
Acetaldehyde	103	118	173	263	359	410	485	68%	249%
Formaldehyde	372	362	422	496	519	574	625	13%	39%
Benzene	115	110	105	103	102	98	94	-9%	-11%
PM2.5	204	181	158	137	117	95	72	-22%	-43%
PM10	246	218	191	166	141	114	86	-22%	-43%

Source: Faro90

Cyprus – Gasoline Vehicle Fleet Life Cycle GHG Emissions



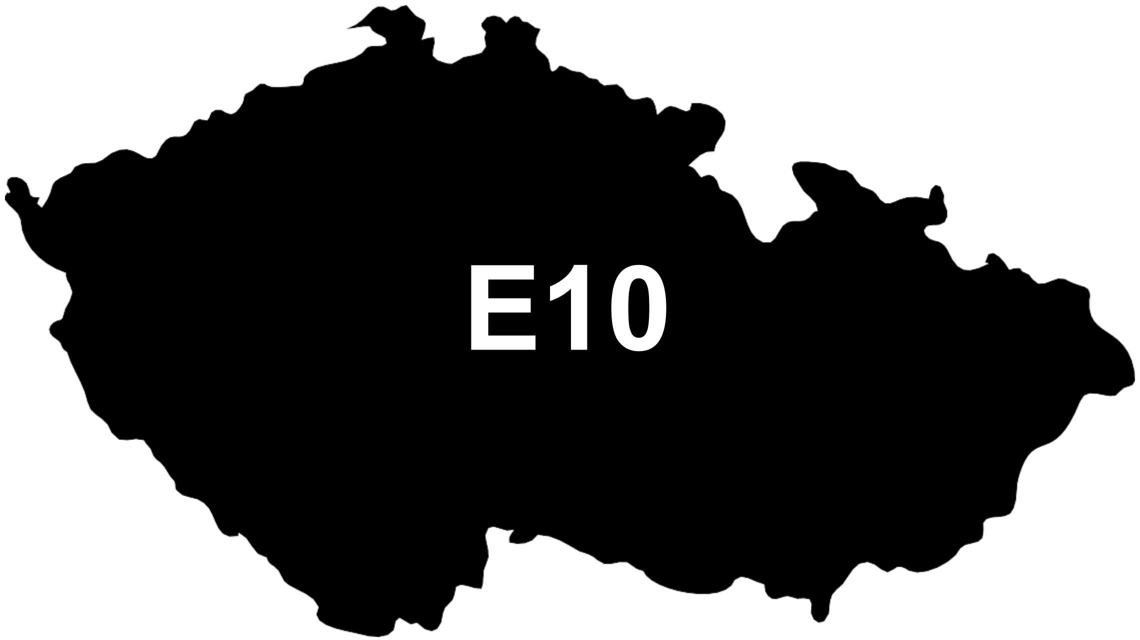
Country Emissions 2024 (MMT/año)	8.3
Target @2035 (MMT/year)	4.7
Delta	3.6

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.8%	15.0%	17.5%	19.8%	23.0%
RED II/III	0.0%	7.9%	10.1%	11.8%	13.5%	15.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.1%	7.7%	9.0%	10.2%	11.9%
RED II/III	0.0%	6.5%	8.2%	9.7%	11.0%	12.7%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Czechia

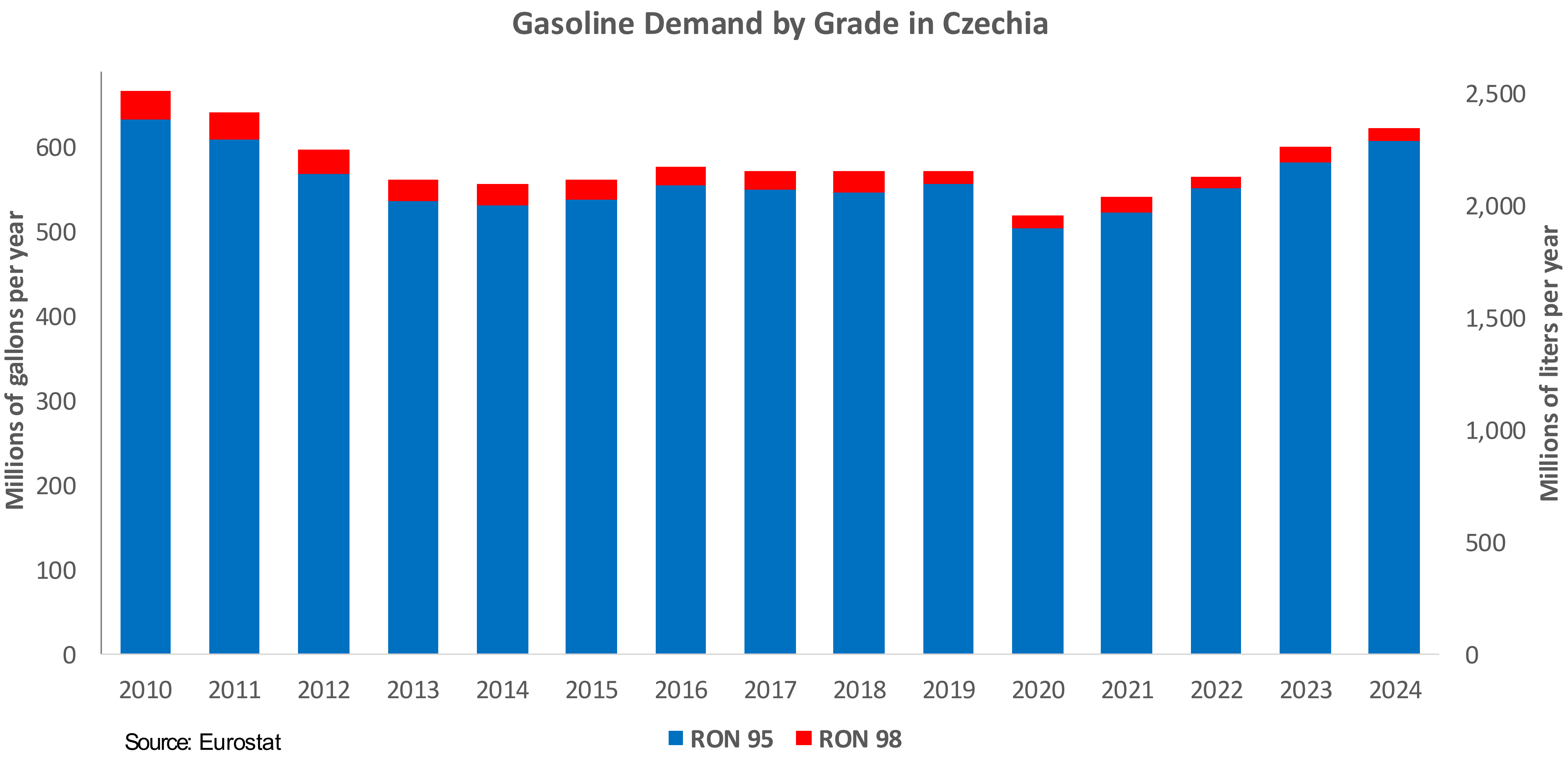


In 2024, gasoline consumption in Czechia was 2,347 million liters (620 million gallons) and growth rate was 1.7%. Gasoline production and net imports were 2,328 and -936 million liters (615 and -247 million gallons) correspondingly. Czechia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.1 RON.

In 2024, ethanol consumption in Czechia was 96 million liters (25 million gallons) representing 4.1% of gasoline demand. Ethanol production and Net Imports were 76 and 21 million liters (25 and 5 million gallons) correspondingly. Ethanol sources are Corn 50% and Sugar Beet 50%.

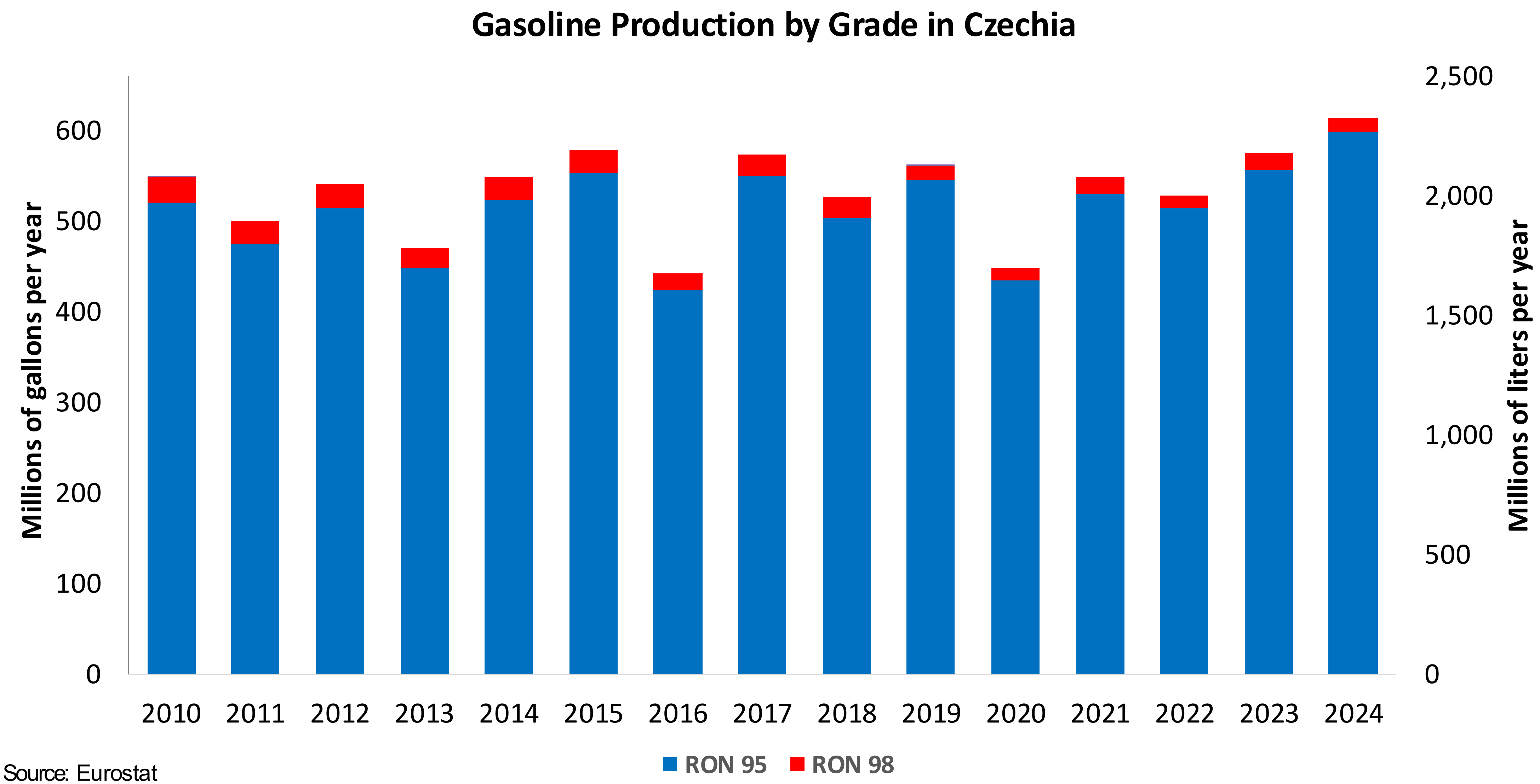
Source: Czechia Statistics, Eurostat

Gasoline Demand in Czechia



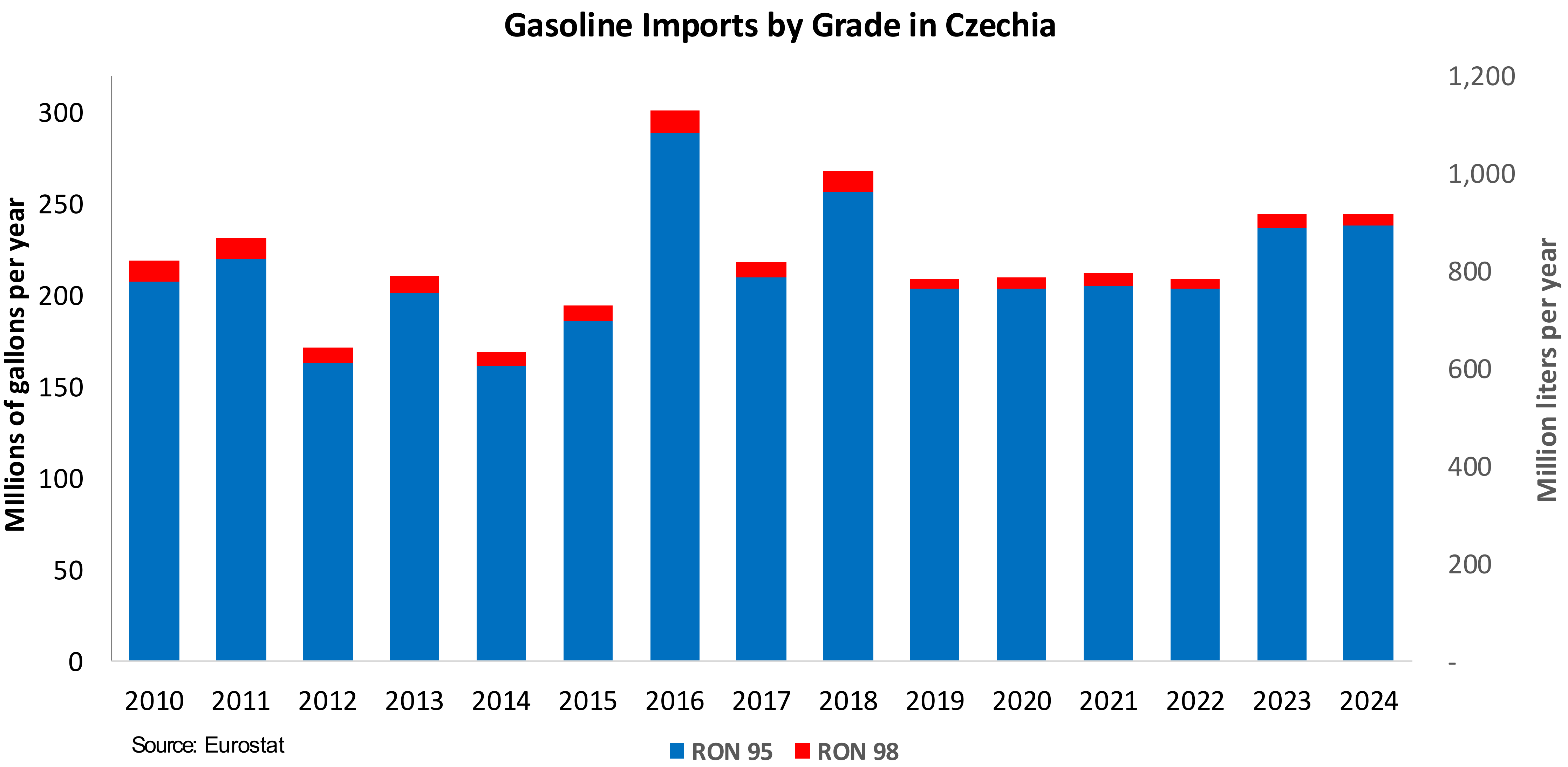
Source: Eurostat

Gasoline Production in Czechia



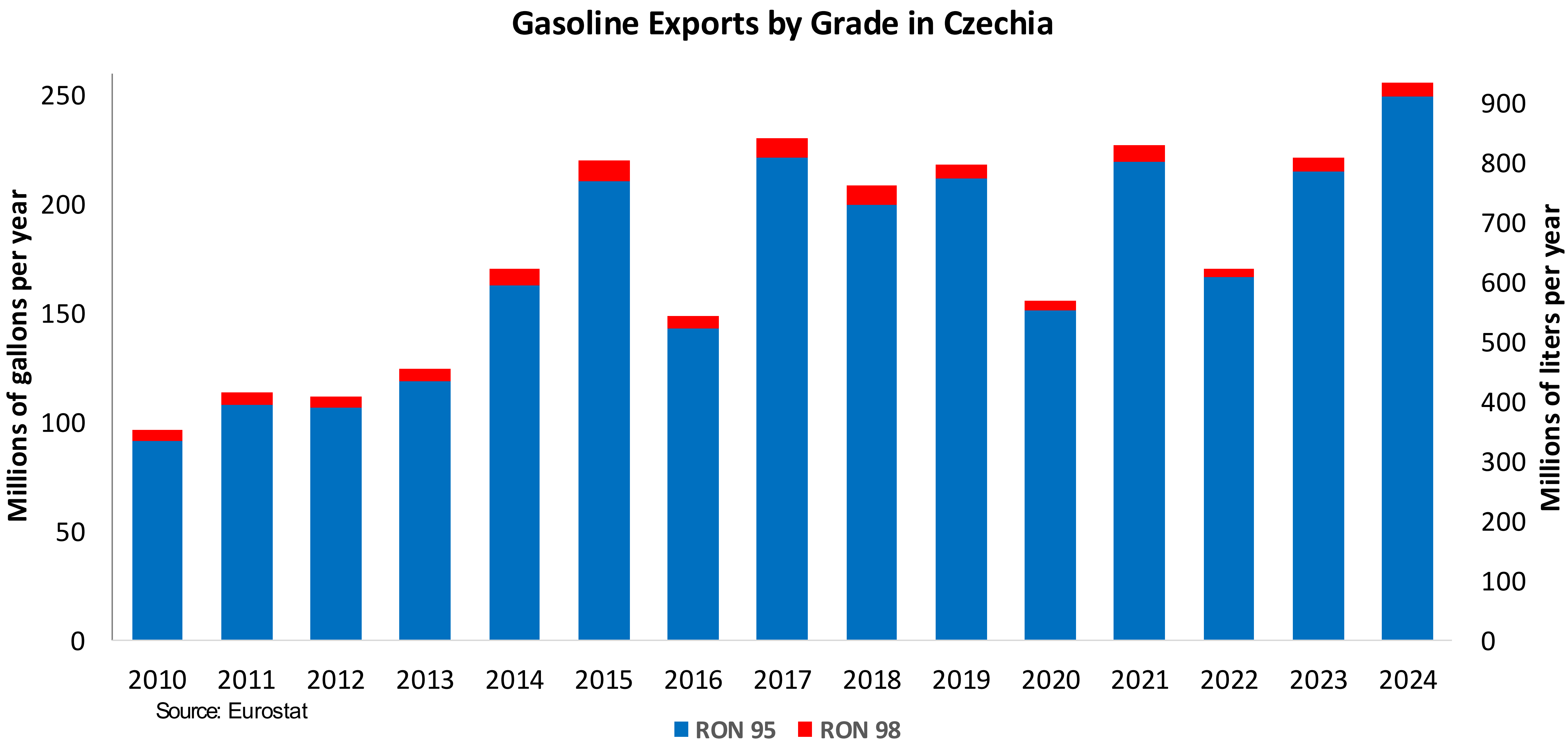
Source: Eurostat

Gasoline Imports in Czechia

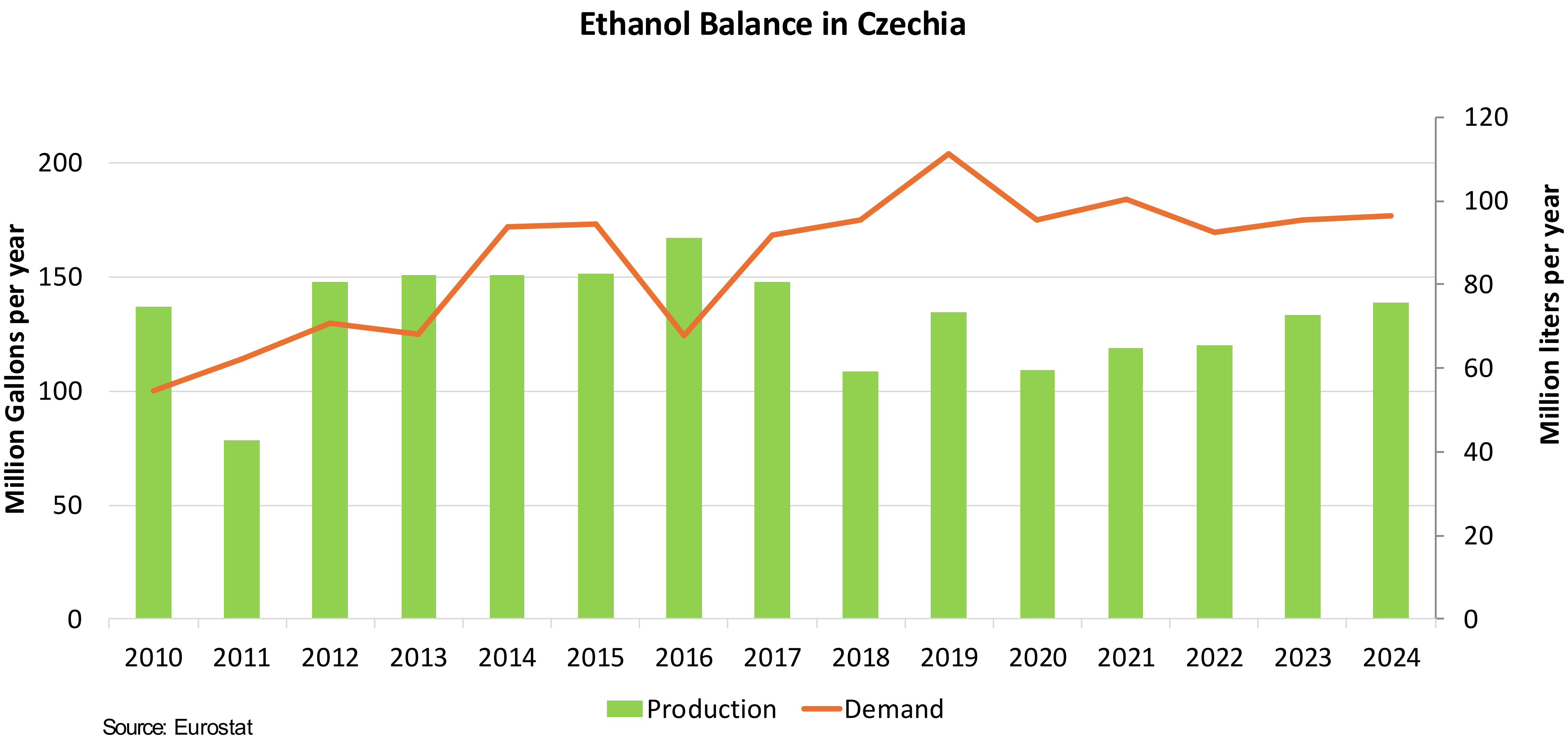


Source: Eurostat

Gasoline Exports in Czechia



Ethanol Balance in Czechia



Source: Eurostat

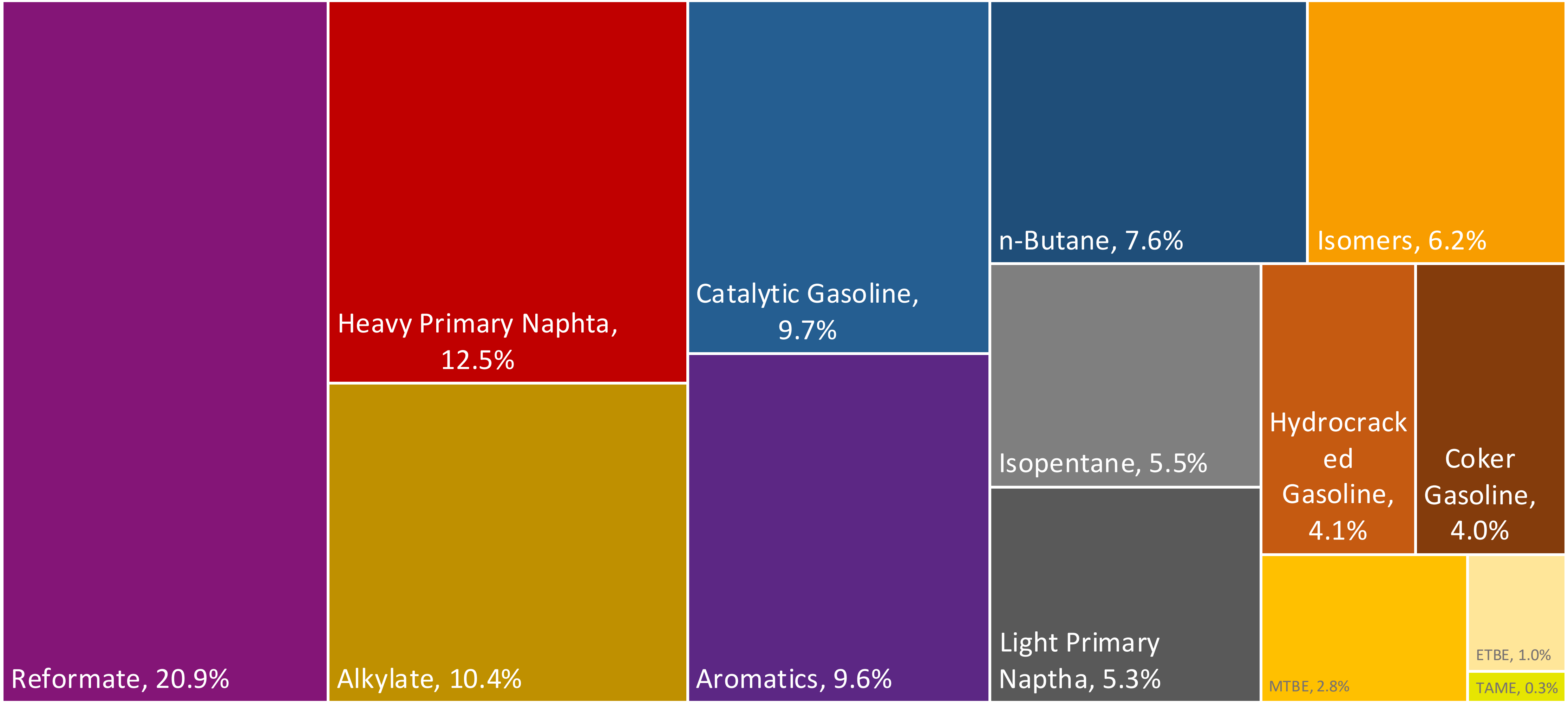
Gasoline Quality in Czechia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.1							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

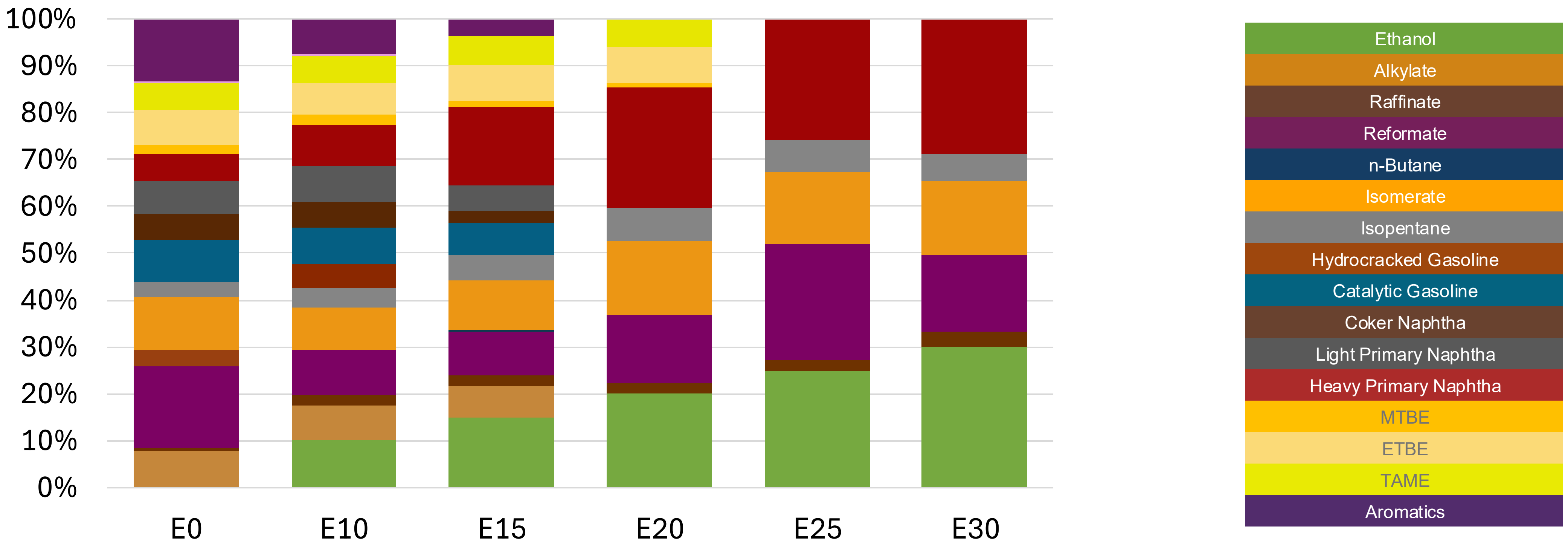
Source: European Commision

Czechia Available Blending Components



Source: Faro90

Czechia – Gasoline 95 – Constant Octane

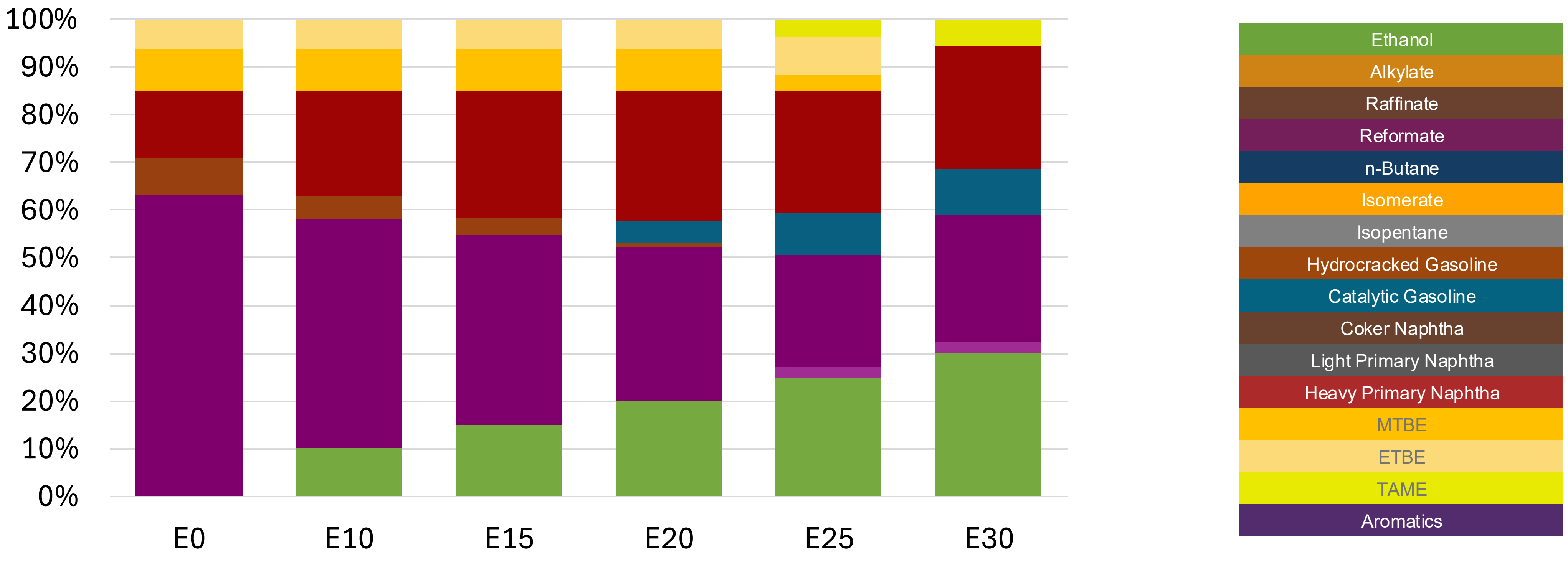


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.314	$ 2.176	$ 2.076	$ 1.961	$ 1.915	$ 1.862

Average CIF 2024 Prices

Source: Faro90

Czechia - Gasoline 98 - Constant Octane

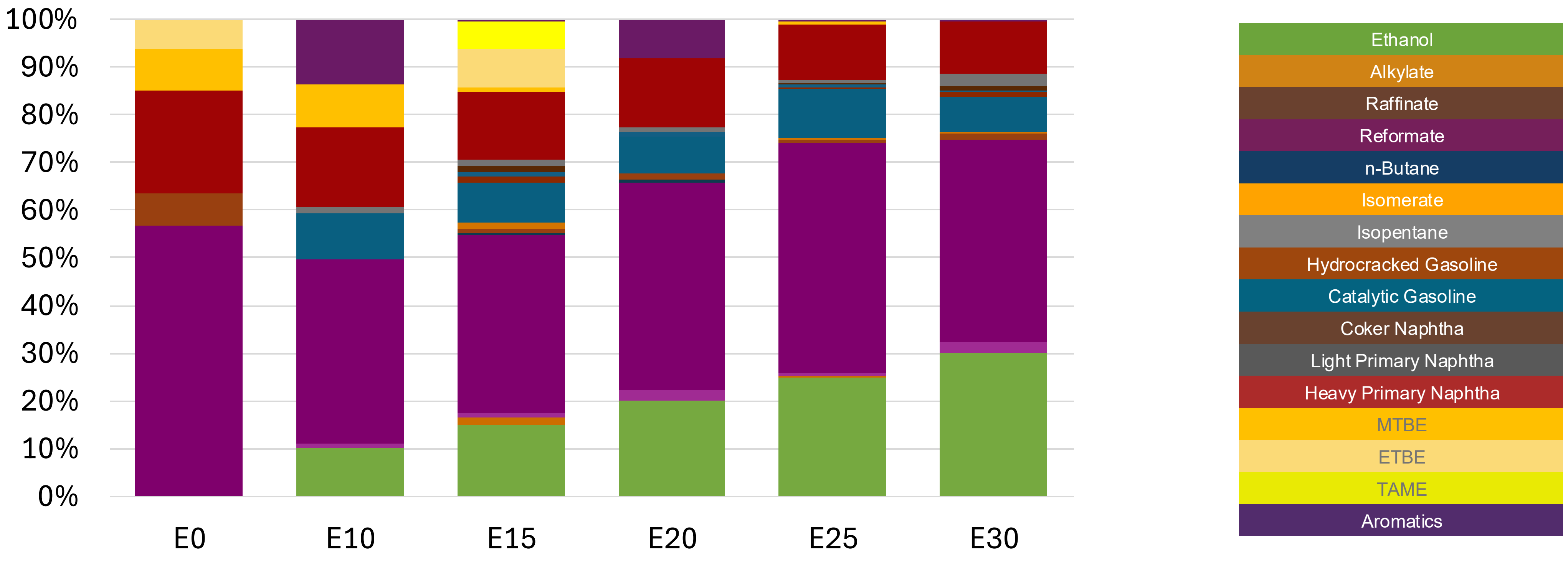


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.334	$ 2.217	$ 2.148	$ 2.076	$ 2.011	$ 1.961

Average CIF 2024 Prices

Source: Faro90

Czechia – Gasoline 95 – Increased Octane

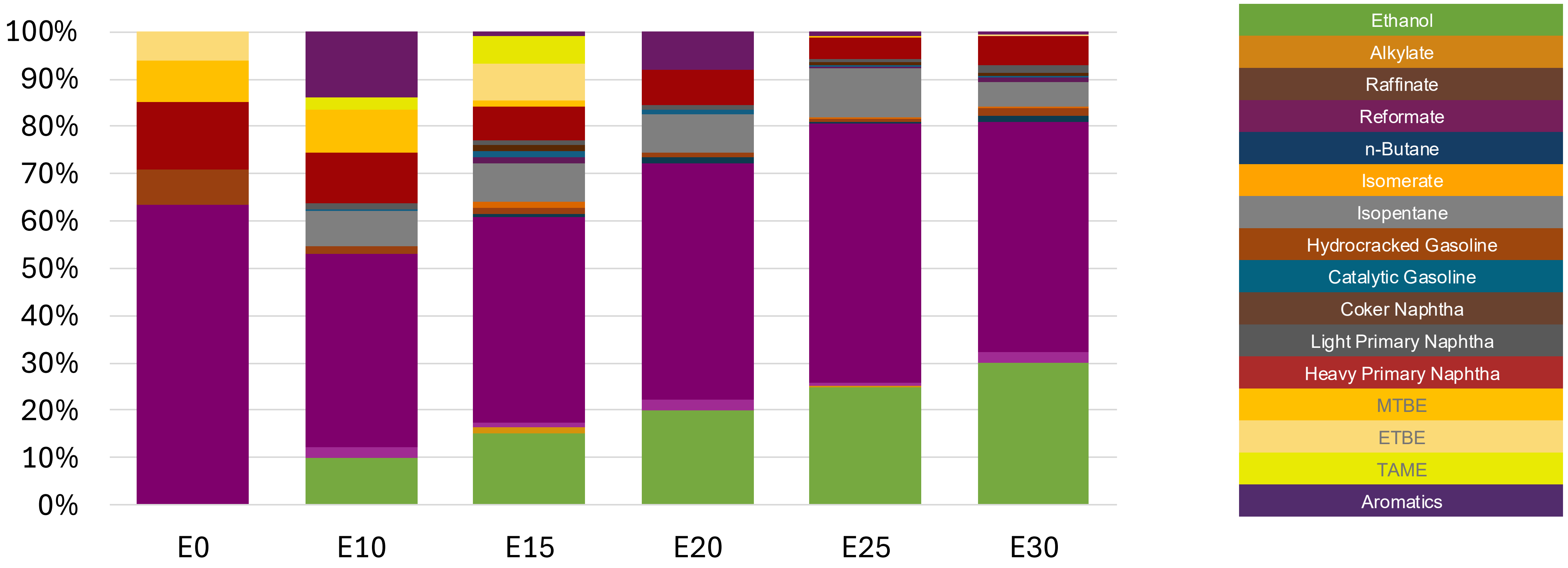


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.202	$ 2.202	$ 2.202	$ 2.202	$ 2.202	$ 2.202

Average CIF 2024 Prices

Source: Faro90

Czechia – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.334	$ 2.334	$ 2.334	$ 2.334	$ 2.334	$ 2.334

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Czechia

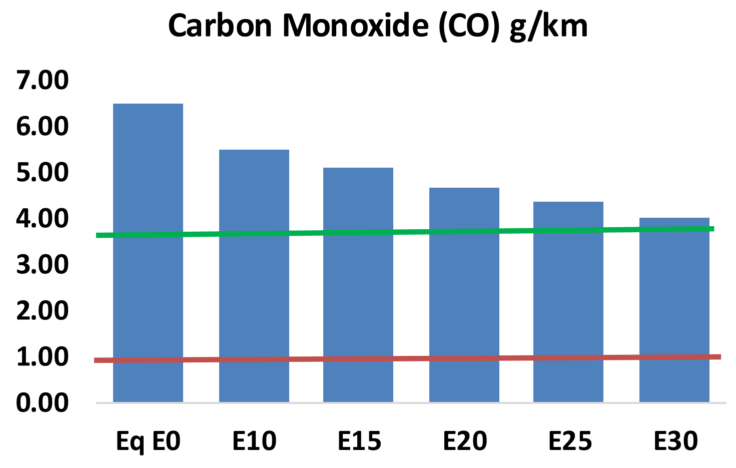
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	79,074
		Single FI	3-Ways		>8 years	496,911
		Multi FI	3-Ways		>8 years	577,810
			Euro III		>8 years	387,206
			Euro IV		>8 years	539,299
			Euro V		>8 years	200,096
			Euro 6		>8 years	444,886
					4-8 years	598,016
					<4 years	712,246
			Hybrid		All	27,035
		Electric			All	73,149
	Medium	Carburator	3-Ways	PCV Tank	>8 years	48,378
		Single FI	3-Ways		>8 years	304,010
		Multi FI	3-Ways		>8 years	353,503
			Euro III		>8 years	236,892
			Euro IV		>8 years	329,942
			Euro V		>8 years	122,419
			Euro 6		>8 years	272,181
					4-8 years	365,866
					<4 years	435,751
			Hybrid		All	16,540
		Electric			All	73,149
Sml Engines	Light	4-Cycle	None	PCV	>8 years	380,304
			Euro III		>8 years	127,130
			Euro IV		>8 years	177,066
			Euro V		>8 years	65,697
			Euro 6		>8 years	146,068
					4-8 years	196,345
					<4 years	233,849
			Hybrid		All	8,662
		Electric			All	23,436
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	207,008
		Fuel-Injection	3-Ways		>8 years	196,071
			Euro III		>8 years	127,335
			Euro IV		>8 years	172,946
			Euro V		>8 years	63,028
			Euro 6		>8 years	138,369
					4-8 years	182,578
					<4 years	225,872
			Hybrid		All	-
		Electric			All	742

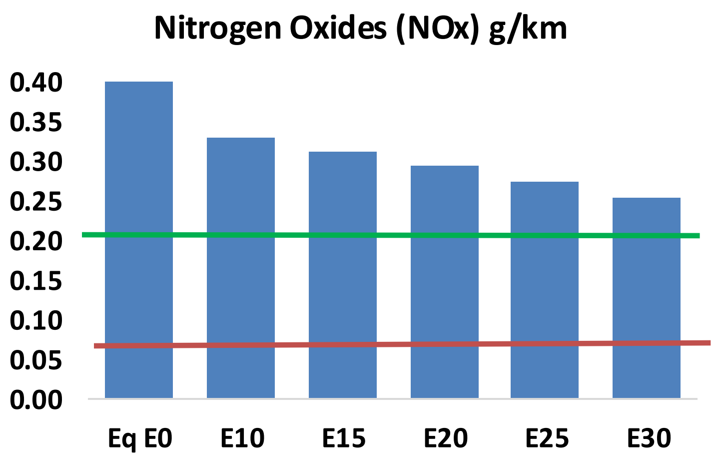
Vehicle Fleet: 9,366,865
Gasoline Vehicle Fleet: 3,468,160

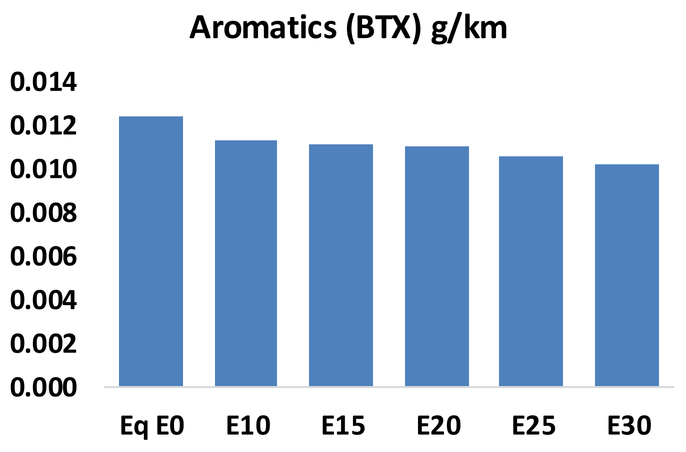
Source: Eurostat, ACEA

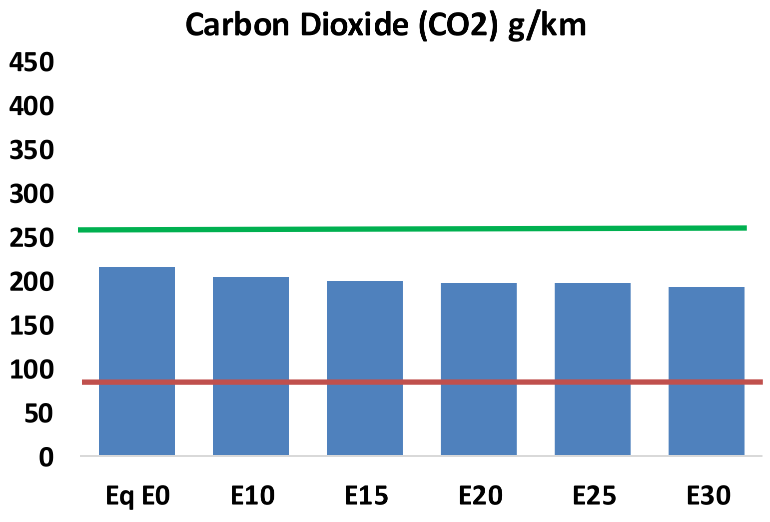
Czechia – Vehicular Emissions

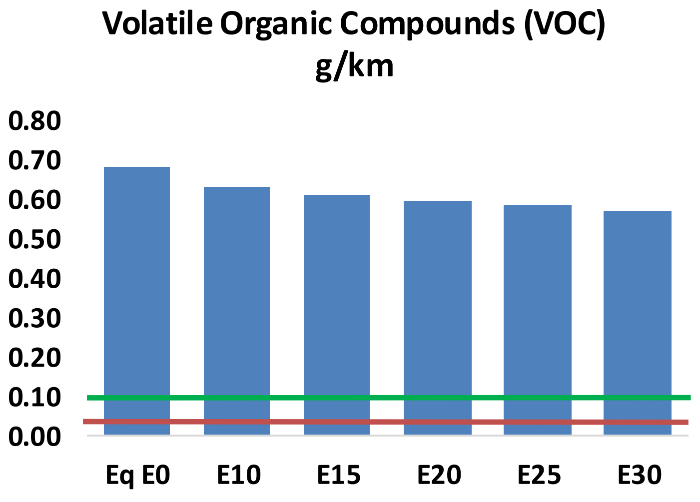
TIER III BIN 125 United States
Euro 6 Standard

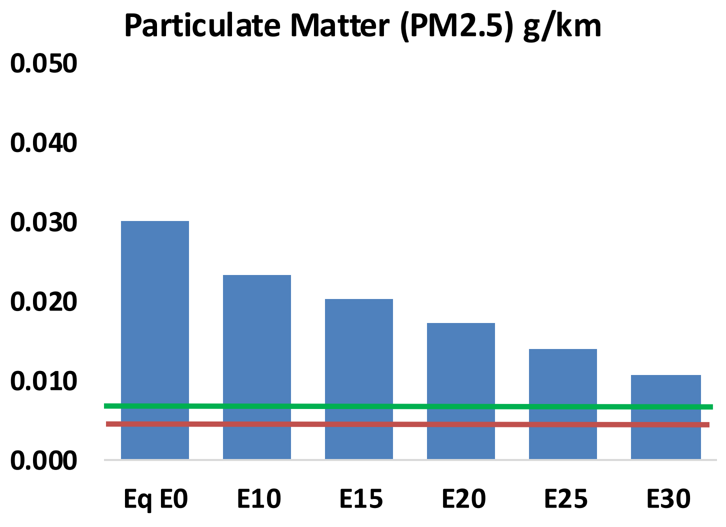












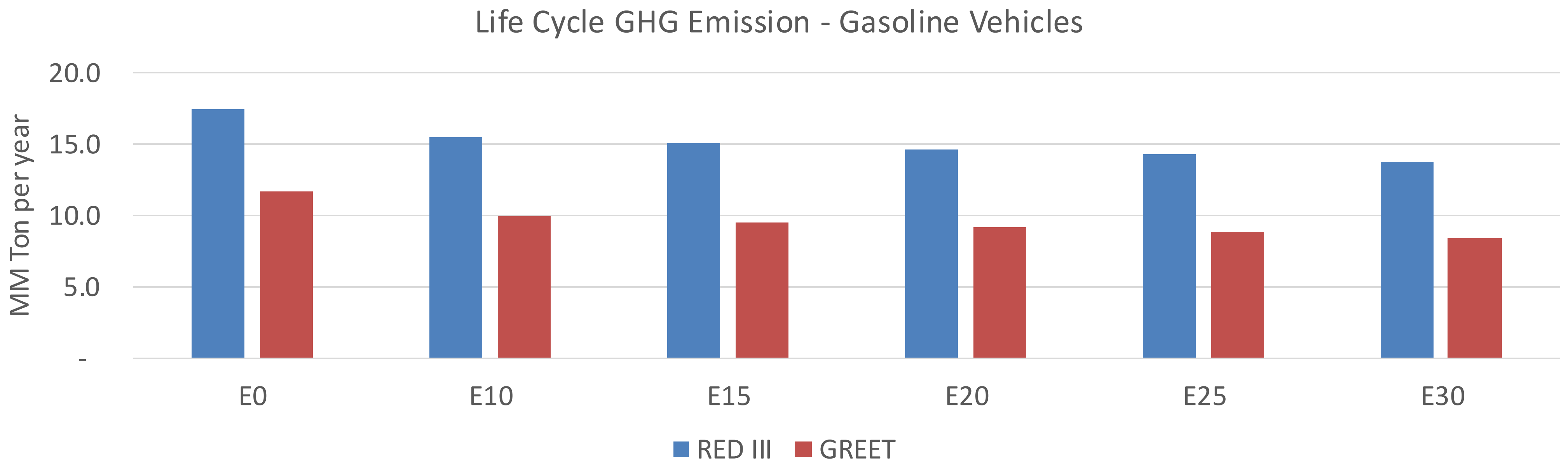
Source: Faro90

Czechia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	514,816	475,239	435,661	403,229	370,591	345,495	316,625	-15%	-28%
CO2	17,030,957	16,593,018	16,179,409	15,854,118	15,693,151	15,618,037	15,330,151	-5%	-8%
VOC	54,073	51,964	49,856	48,528	47,375	46,544	45,240	-8%	-12%
NOx	37,565	28,173	26,295	24,783	23,373	21,817	20,172	-30%	-38%
SOx	196	181	166	154	142	129	115	-15%	-28%
NH3	5,868	5,810	5,752	5,779	5,845	5,890	5,928	-2%	0%
N2O	308	306	305	306	313	316	320	-1%	2%
CH4	11,875	11,875	11,875	12,053	12,314	12,540	12,754	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	361	343	324	310	297	288	274	-10%	-18%
Acetaldehyde	771	885	1,299	1,977	2,694	3,079	3,638	68%	249%
Formaldehyde	2,546	2,475	2,885	3,388	3,549	3,926	4,274	13%	39%
Benzene	987	948	898	884	877	839	812	-9%	-11%
PM2.5	1,023	909	794	689	585	475	360	-22%	-43%
PM10	1,353	1,202	1,050	912	774	628	476	-22%	-43%

Source: Faro90

Czechia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	108.0
Target @2035 (MMT/year)	60.8
Delta	47.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	15.1%	18.6%	21.5%	24.3%	27.9%
RED II/III	0.0%	11.1%	13.8%	16.2%	18.4%	21.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.8%	4.6%	5.3%	6.0%	6.9%
RED II/III	0.0%	4.1%	5.1%	6.0%	6.8%	7.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

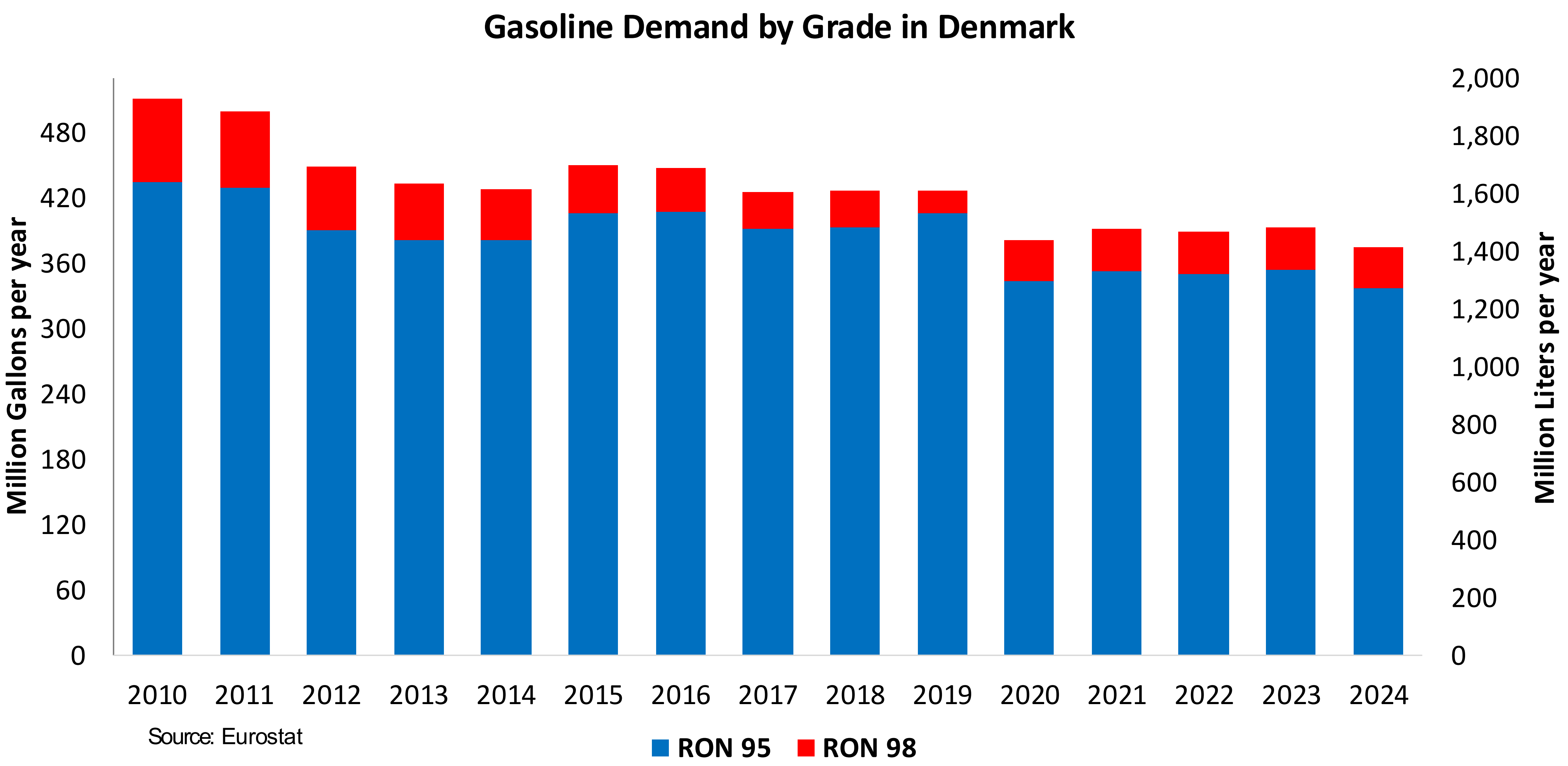
Ethanol Gasoline Blending in Denmark

In 2024, gasoline consumption in Denmark was 1,415 million liters (374 million gallons) and growth rate was -2.6%. Gasoline production and net imports were 2,745 and -1,072 million liters (725 and -283 million gallons) correspondingly. Denmark complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Denmark was 168 million liters (44 million gallons) representing 11.9% of gasoline demand. Ethanol production and Net Imports were 160 and 8 million liters (44 and 2 million gallons) correspondingly.

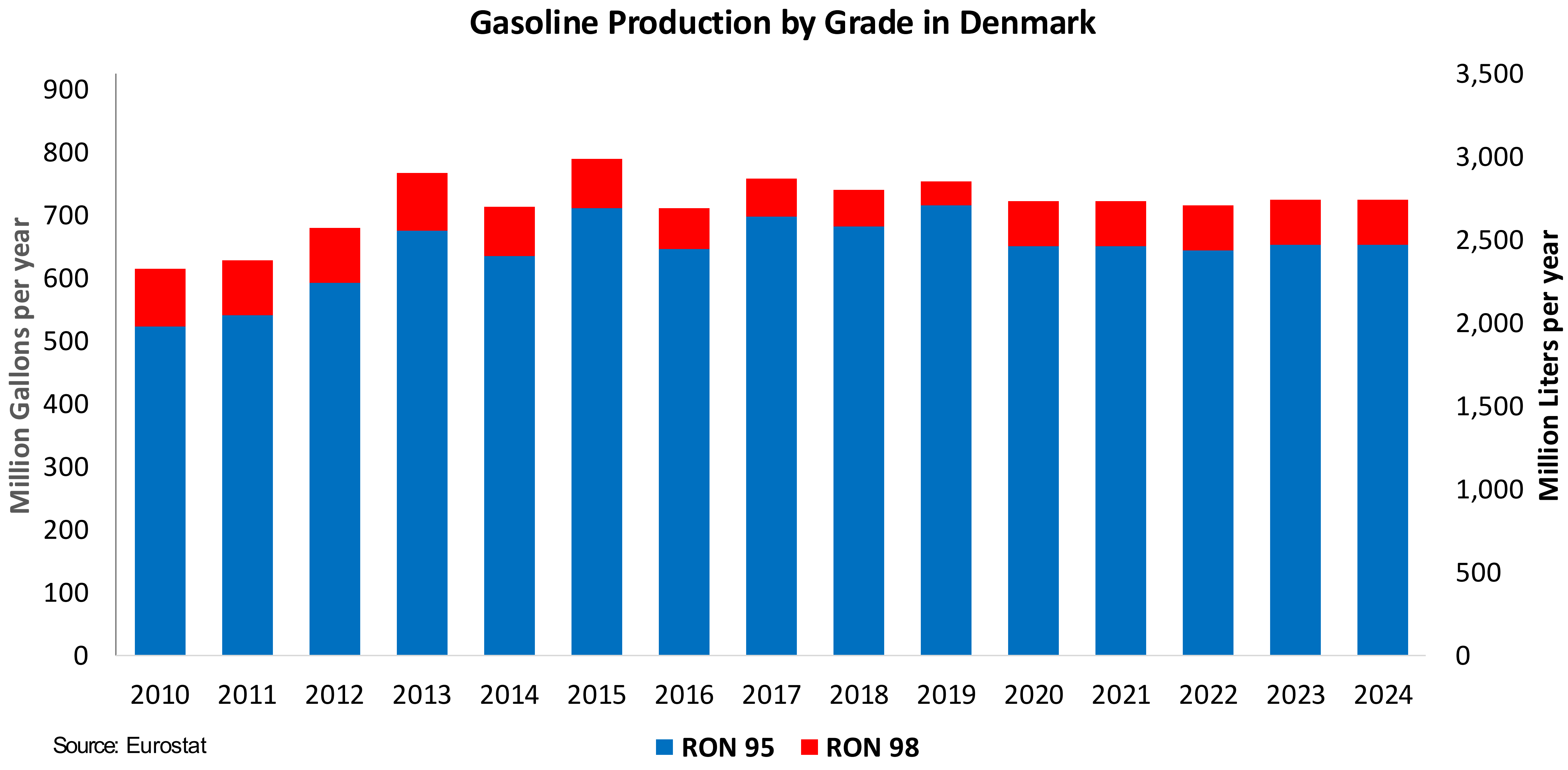
Source: Eurostat, European Commision

Gasoline Demand in Denmark



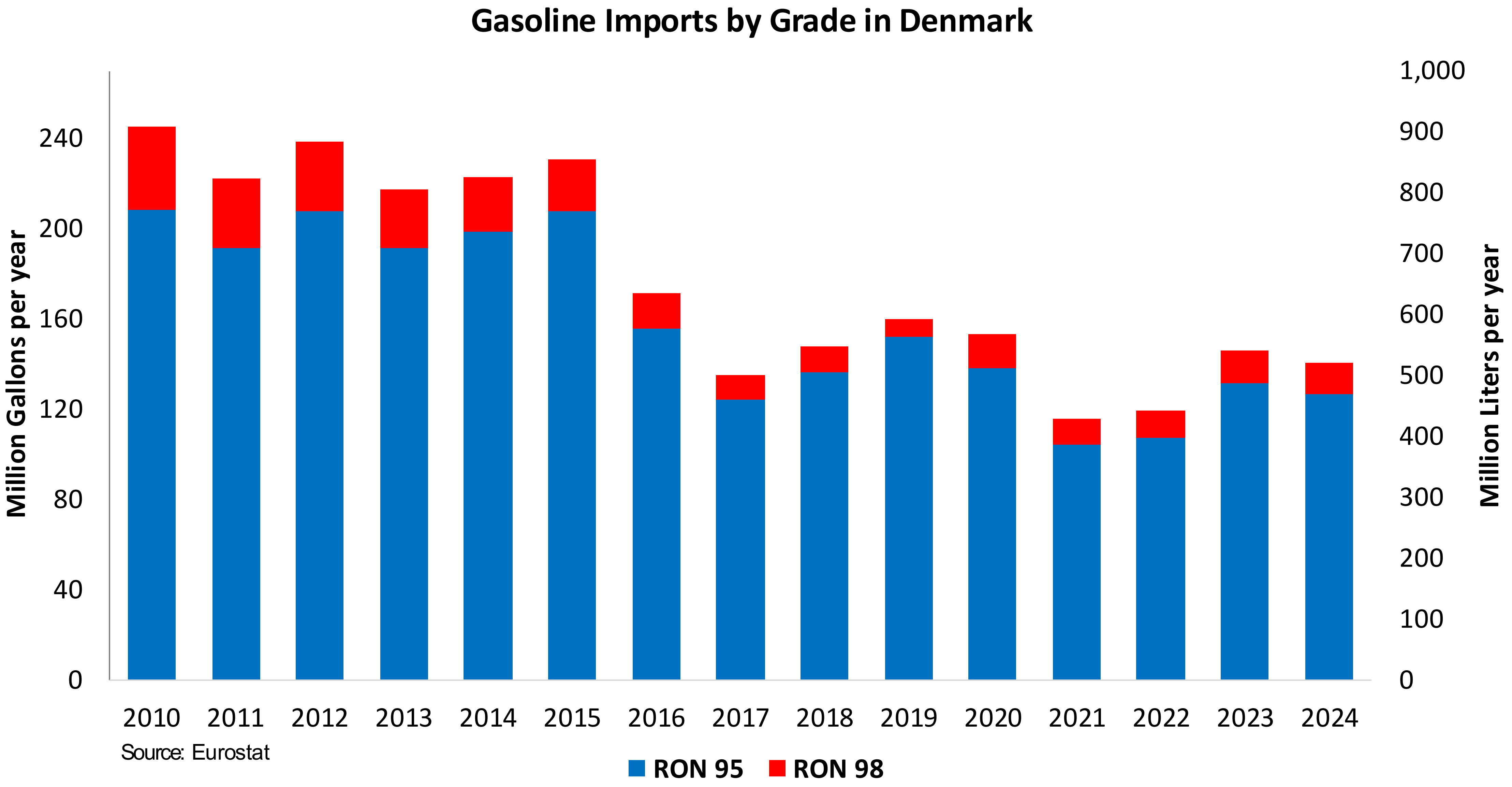
Source: Eurostat

Gasoline Production in Denmark



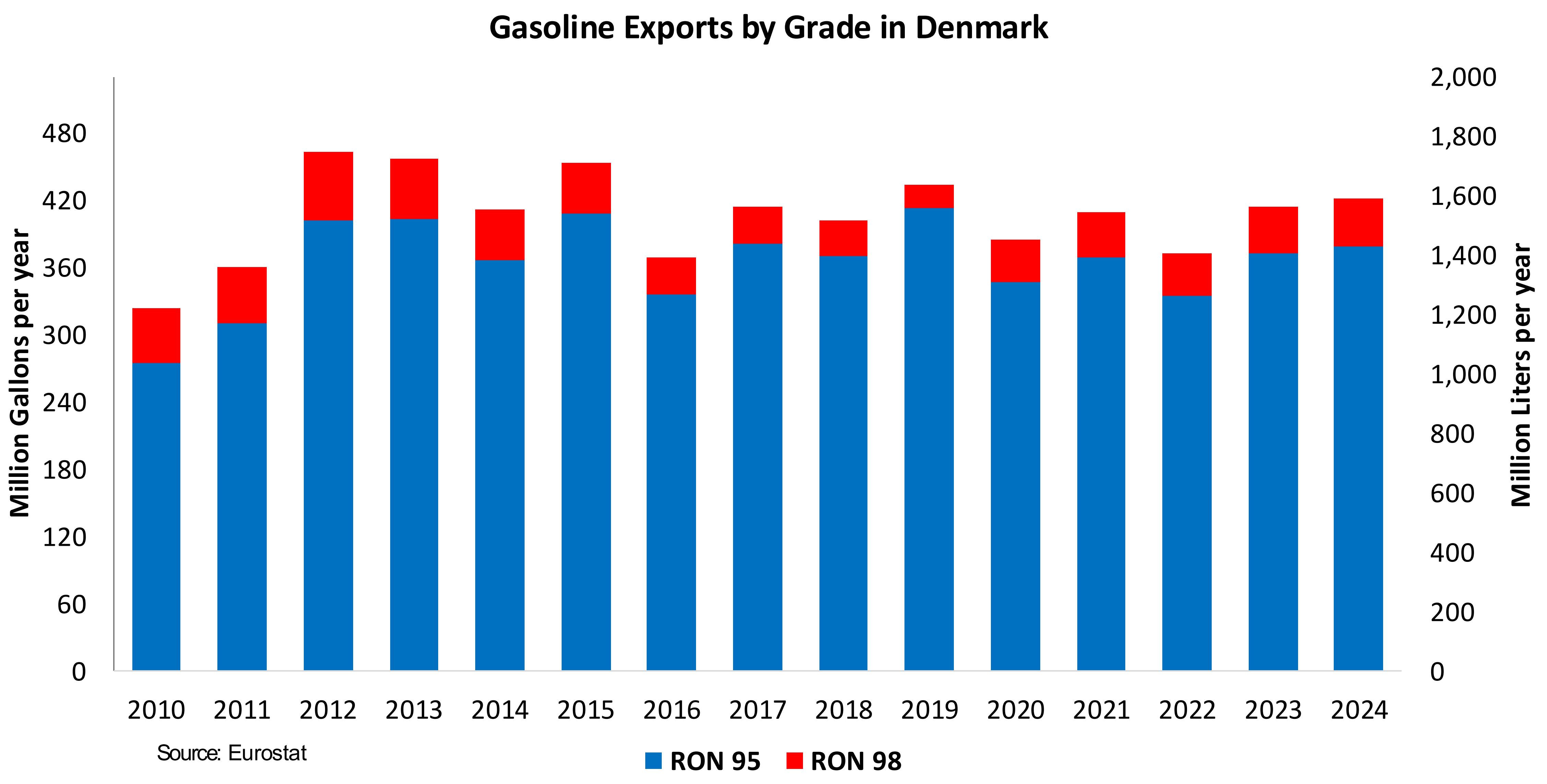
Source: Eurostat

Gasoline Imports in Denmark



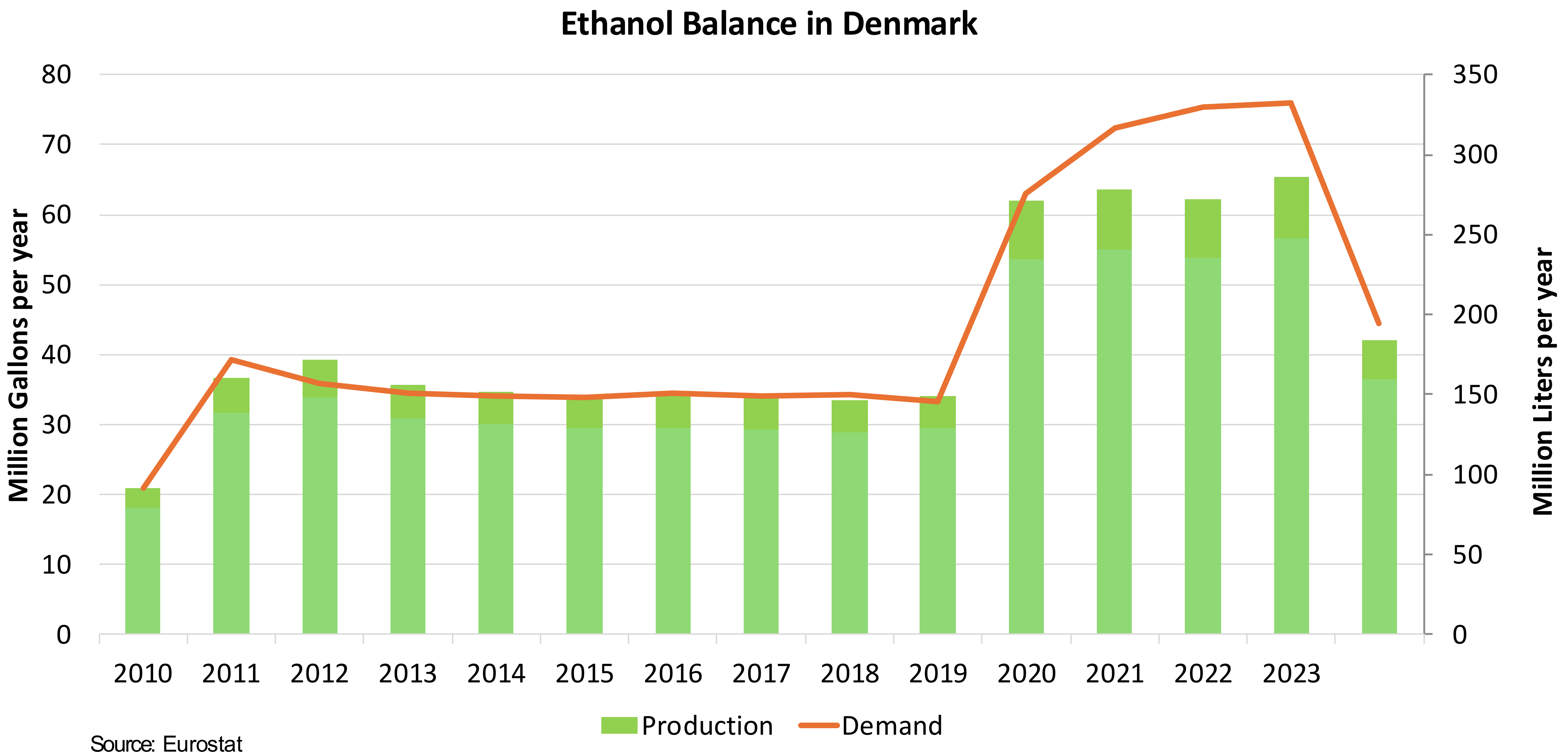
Source: Eurostat

Gasoline Exports in Denmark



Source: Eurostat

Ethanol Balance in Denmark



Source: Eurostat

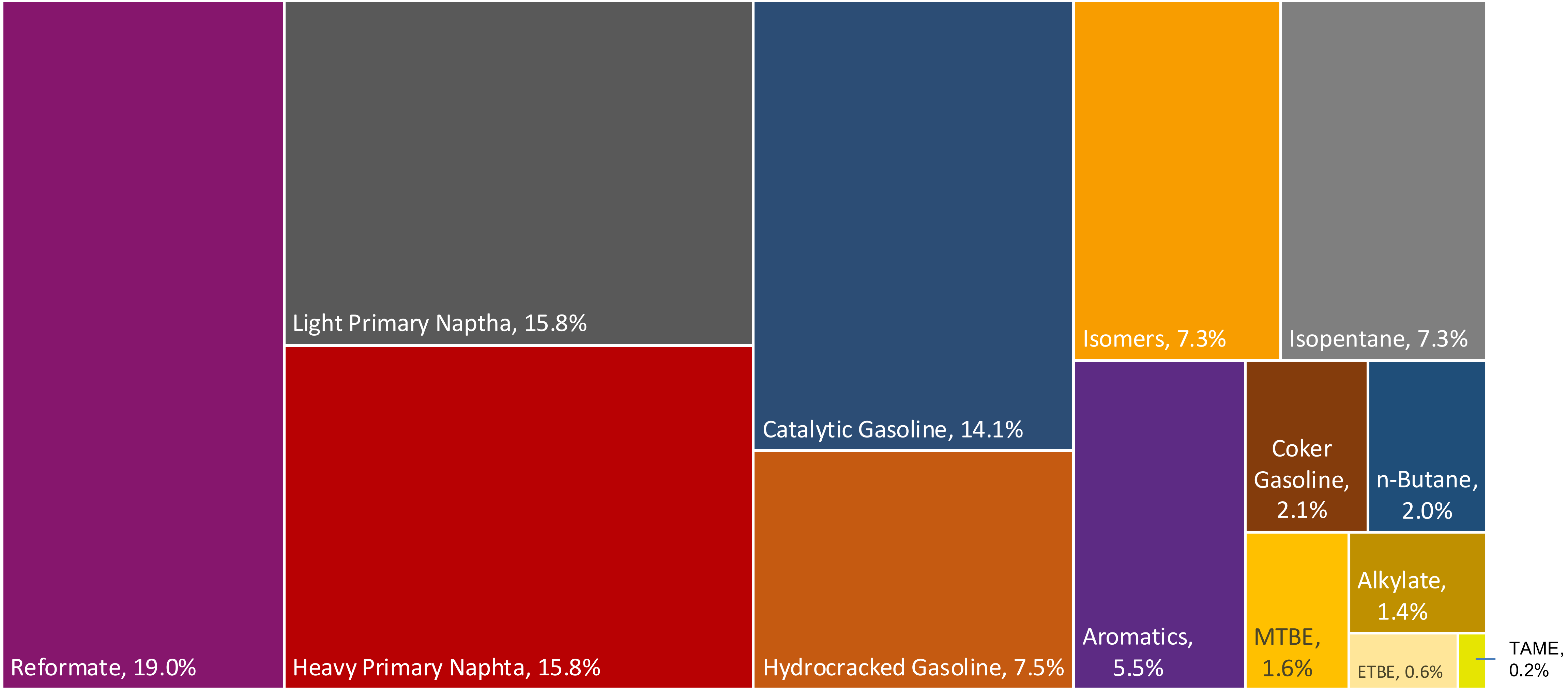
Gasoline Quality in Denmark

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	5.2							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

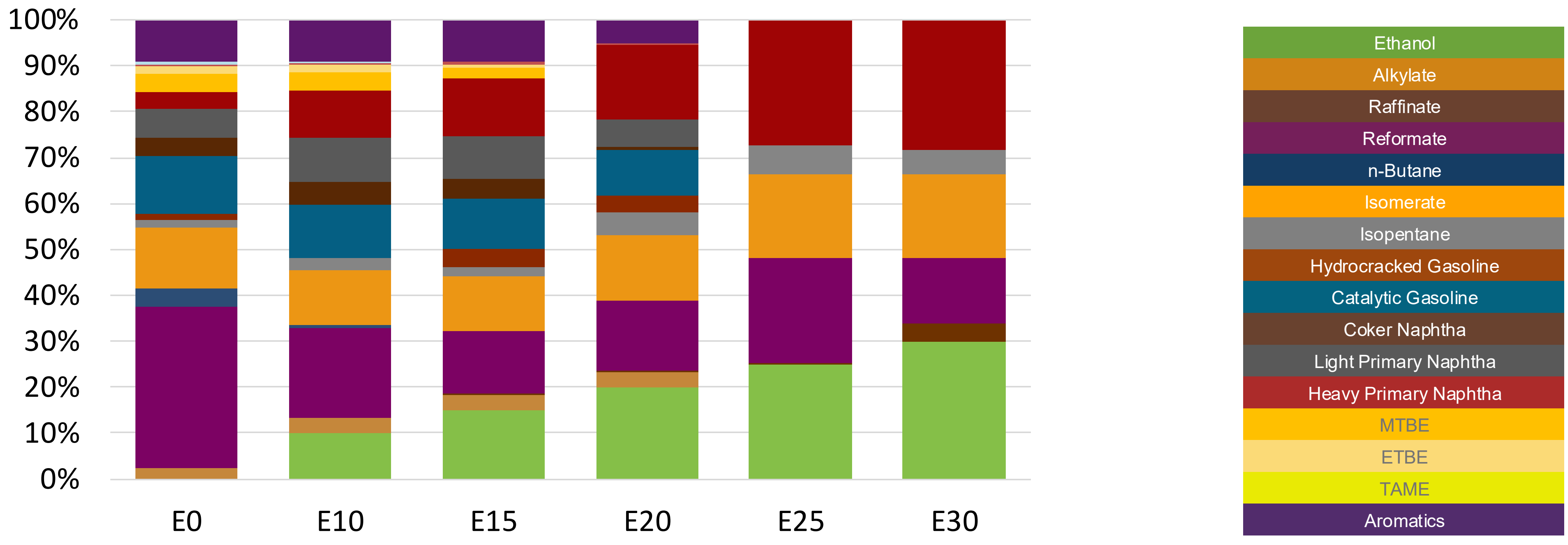
Source: European Commision

Denmark Available Blending Components



Source: Faro90

Denmark – Gasoline 95 – Constant Octane

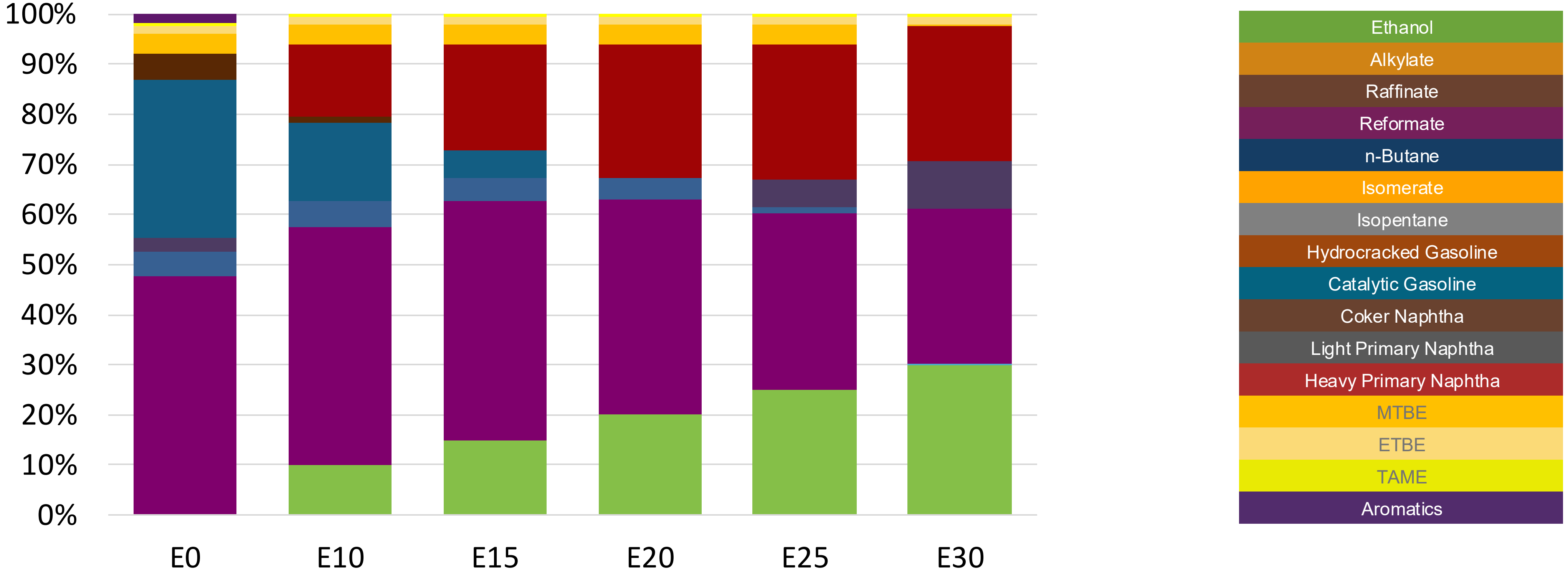


Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US$/gal)	$ 2.272	$ 2.144	$ 2.083	$ 2.002	$ 1.880	$ 1.814

Average CIF 2024 Prices

Source: Faro90

Denmark - Gasoline 98 - Constant Octane

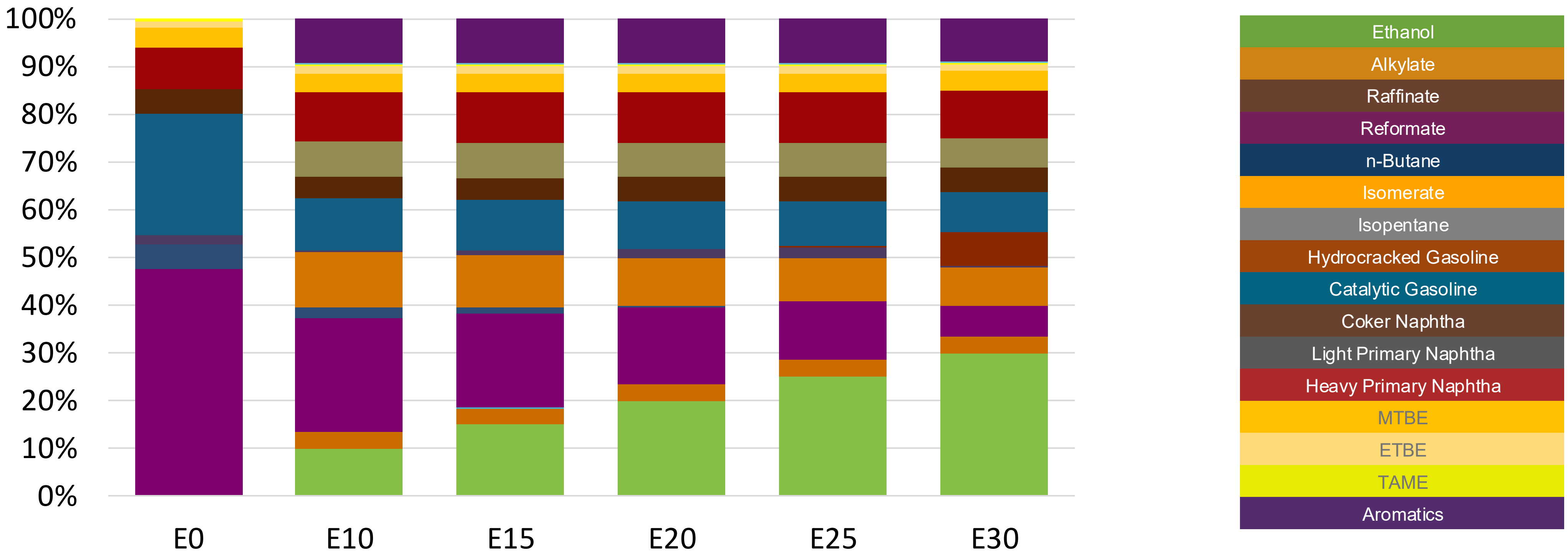


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.347	$ 2.218	$ 2.146	$ 2.069	$ 1.992	$ 1.925

Average CIF 2024 Prices

Source: Faro90

Denmark – Gasoline 95 – Increased Octane

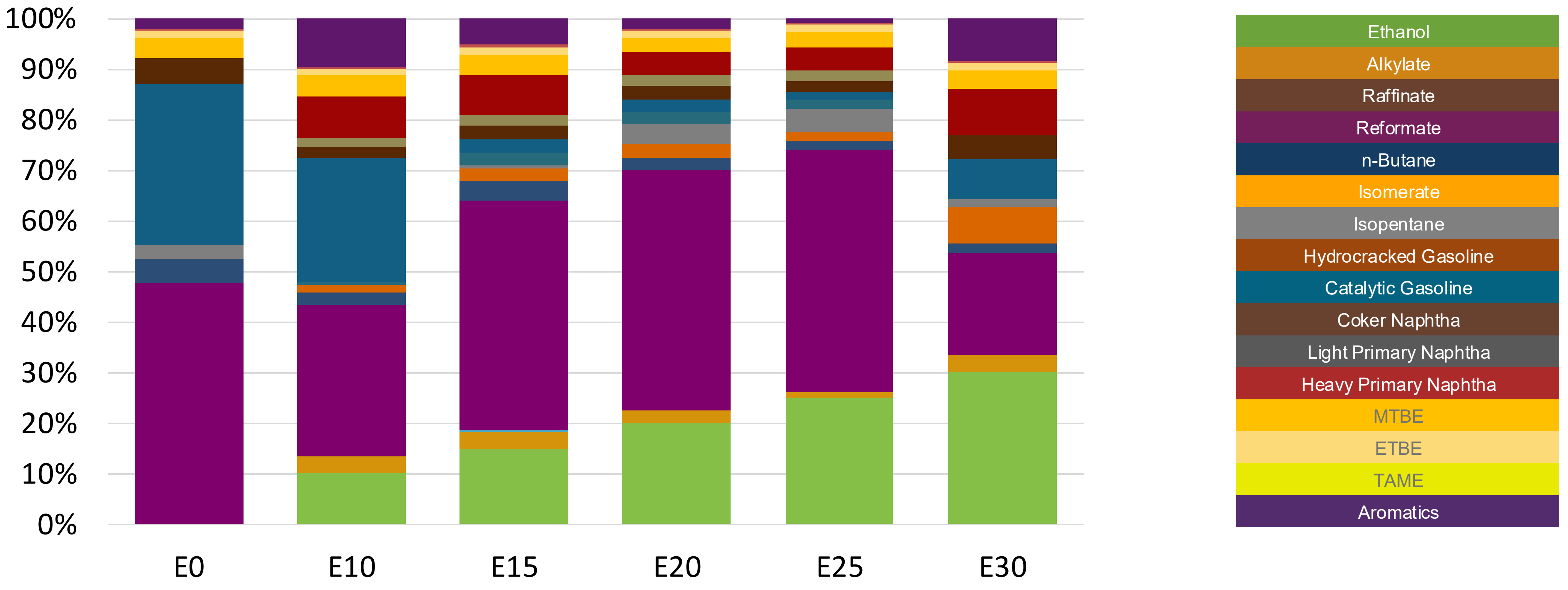


Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US$/gal)	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209

Average CIF 2024 Prices

Source: Faro90

Denmark – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US$/gal)	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Denmark

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	100
		Single FI	3-Ways		>8 years	1,348
		Multi FI	3-Ways		>8 years	227,575
			Euro III		>8 years	64,118
			Euro IV		>8 years	66,181
			Euro V		>8 years	24,646
			Euro 6		>8 years	224,099
					4-8 years	175,229
					<4 years	61,415
			Hybrid		All	50,496
		Electric			All	68,475
	Medium	Carburator	3-Ways	PCV Tank	>8 years	10
		Single FI	3-Ways		>8 years	7
		Multi FI	3-Ways		>8 years	15,531
			Euro III		>8 years	11,541
			Euro IV		>8 years	25,726
			Euro V		>8 years	7,847
			Euro 6		>8 years	74,700
					4-8 years	79,048
					<4 years	47,573
			Hybrid		All	40,116
		Electric			All	51,053
Sml Engines	Light	4-Cycle	None	PCV	>8 years	38,982
			Euro III		>8 years	39,852
			Euro IV		>8 years	231,446
			Euro V		>8 years	138,584
			Euro 6		>8 years	304,572
					4-8 years	234,694
					<4 years	611,357
			Hybrid		All	14,253
		Electric			All	19,328
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	28,176
		Fuel-Injection	3-Ways		>8 years	318,431
			Euro III		>8 years	178,588
			Euro IV		>8 years	205,863
			Euro V		>8 years	57,629
			Euro 6		>8 years	79,560
					4-8 years	37,509
					<4 years	4,873
			Hybrid		All	6,598
		Electric			All	8,396

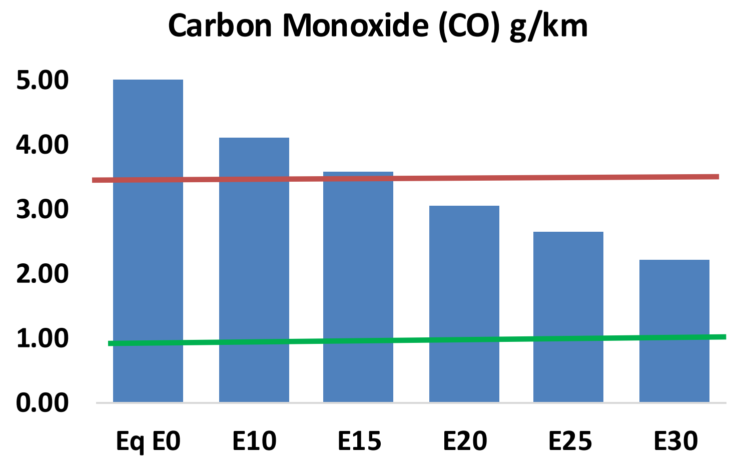
0 100,000 200,000 300,000 400,000 500,000 600,000 700,000

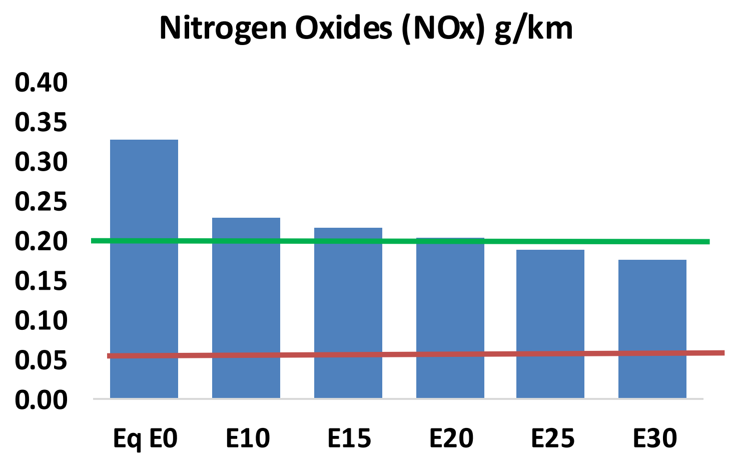
Vehicle Fleet: 3,875,526
Gasoline Vehicle Fleet: 2,055,566

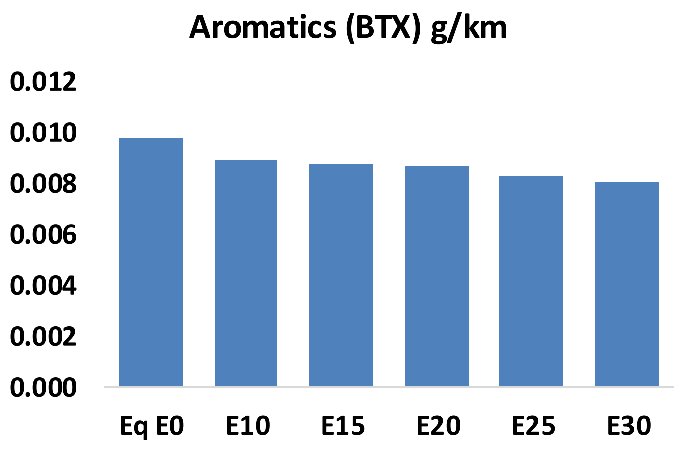
Source: Eurostat, ACEA

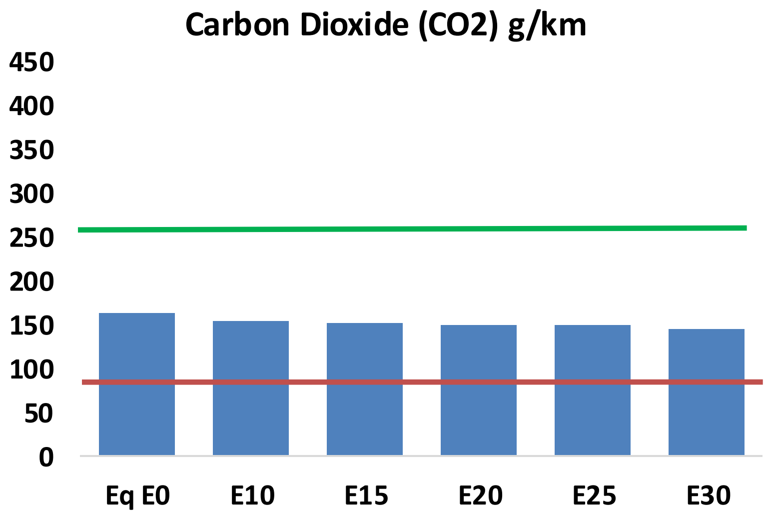
Denmark – Vehicular Emissions

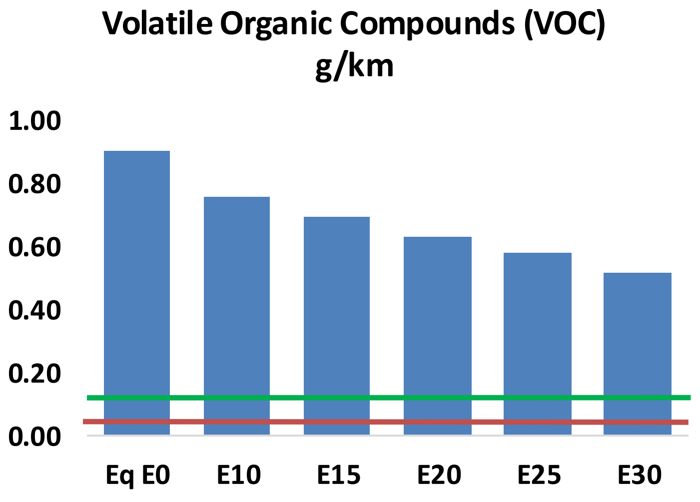
TIER III BIN 125 United States
Euro 6 Standard

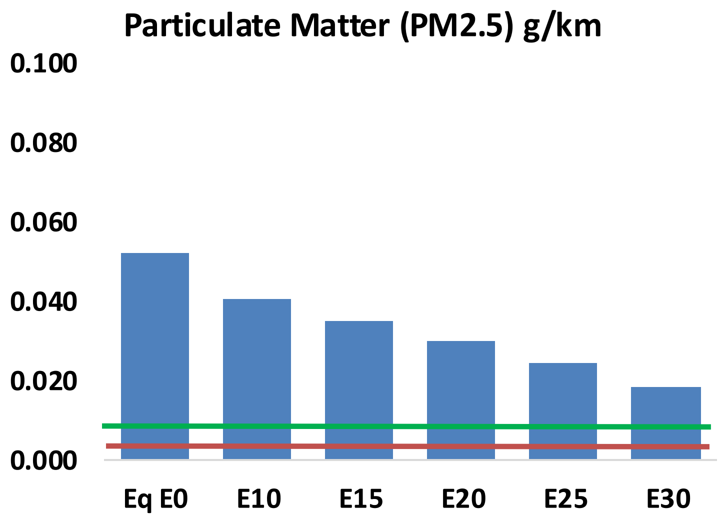












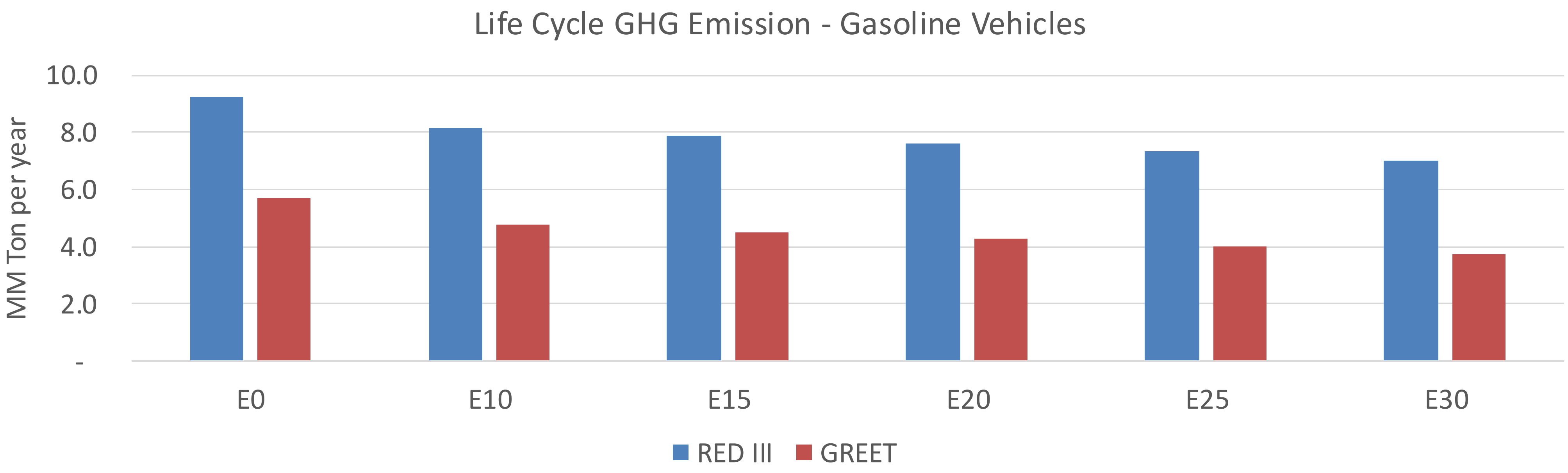
Source: Faro90

Denmark – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	268,045	238,243	208,441	182,226	155,019	133,849	112,318	-22%	-42%
CO2	8,251,751	8,039,563	7,839,163	7,681,555	7,603,564	7,562,067	7,422,676	-5%	-8%
VOC	46,038	42,219	38,400	35,216	31,986	29,495	26,198	-17%	-31%
NOx	16,648	12,486	11,654	10,984	10,359	9,669	8,940	-30%	-38%
SOx	97	90	82	76	70	64	57	-15%	-28%
NH3	3,339	3,306	3,272	3,288	3,325	3,351	3,373	-2%	0%
N2O	140	140	139	140	143	144	146	-1%	2%
CH4	10,288	10,288	10,288	10,442	10,669	10,864	11,049	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	269	252	236	223	210	200	187	-12%	-22%
Acetaldehyde	843	967	1,419	2,161	2,944	3,364	3,975	68%	249%
Formaldehyde	3,275	3,184	3,712	4,358	4,566	5,051	5,499	13%	39%
Benzene	498	478	453	446	442	423	409	-9%	-11%
PM2.5	557	494	432	375	318	258	196	-22%	-43%
PM10	2,084	1,850	1,617	1,404	1,192	967	733	-22%	-43%

Source: Faro90

Denmark – Gasoline Vehicle Fleet Life Cycle GHG Emissions



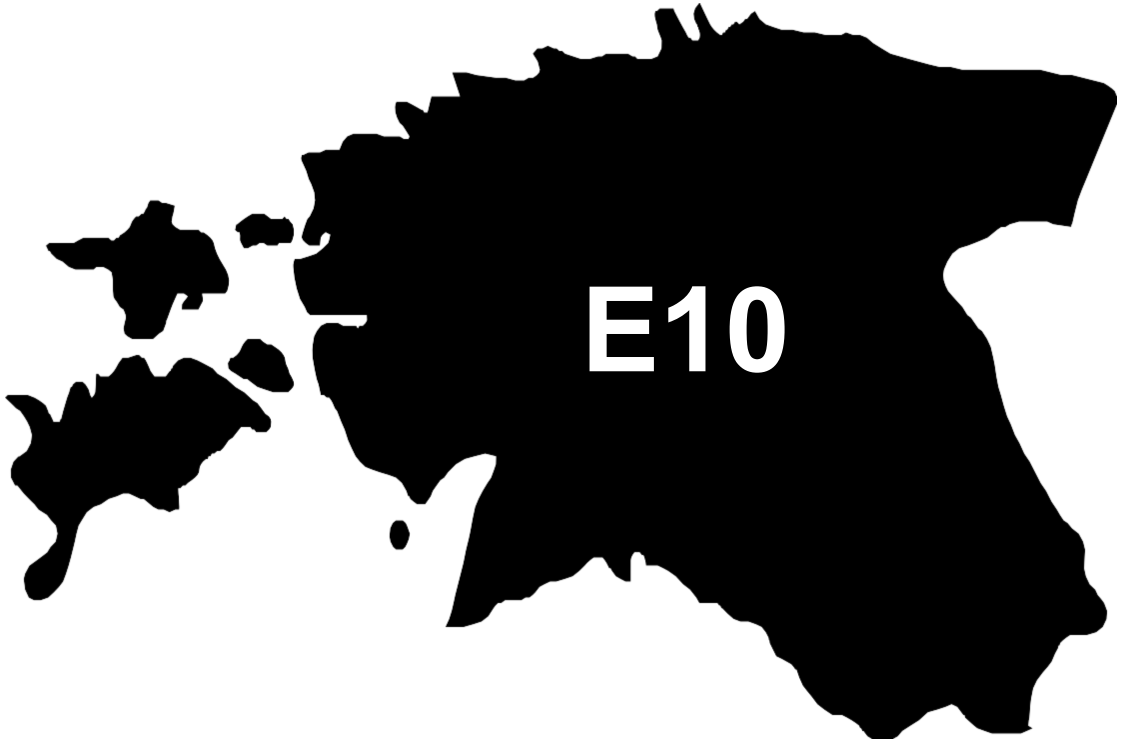
Country Emissions 2024 (MMT/año)	45.5
Target @2035 (MMT/year)	25.6
Delta	19.9

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	16.6%	21.2%	25.2%	29.3%	34.1%
RED II/III	0.0%	11.4%	14.7%	17.6%	20.5%	23.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.8%	6.1%	7.2%	8.4%	9.8%
RED II/III	0.0%	5.3%	6.8%	8.2%	9.5%	11.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Estonia

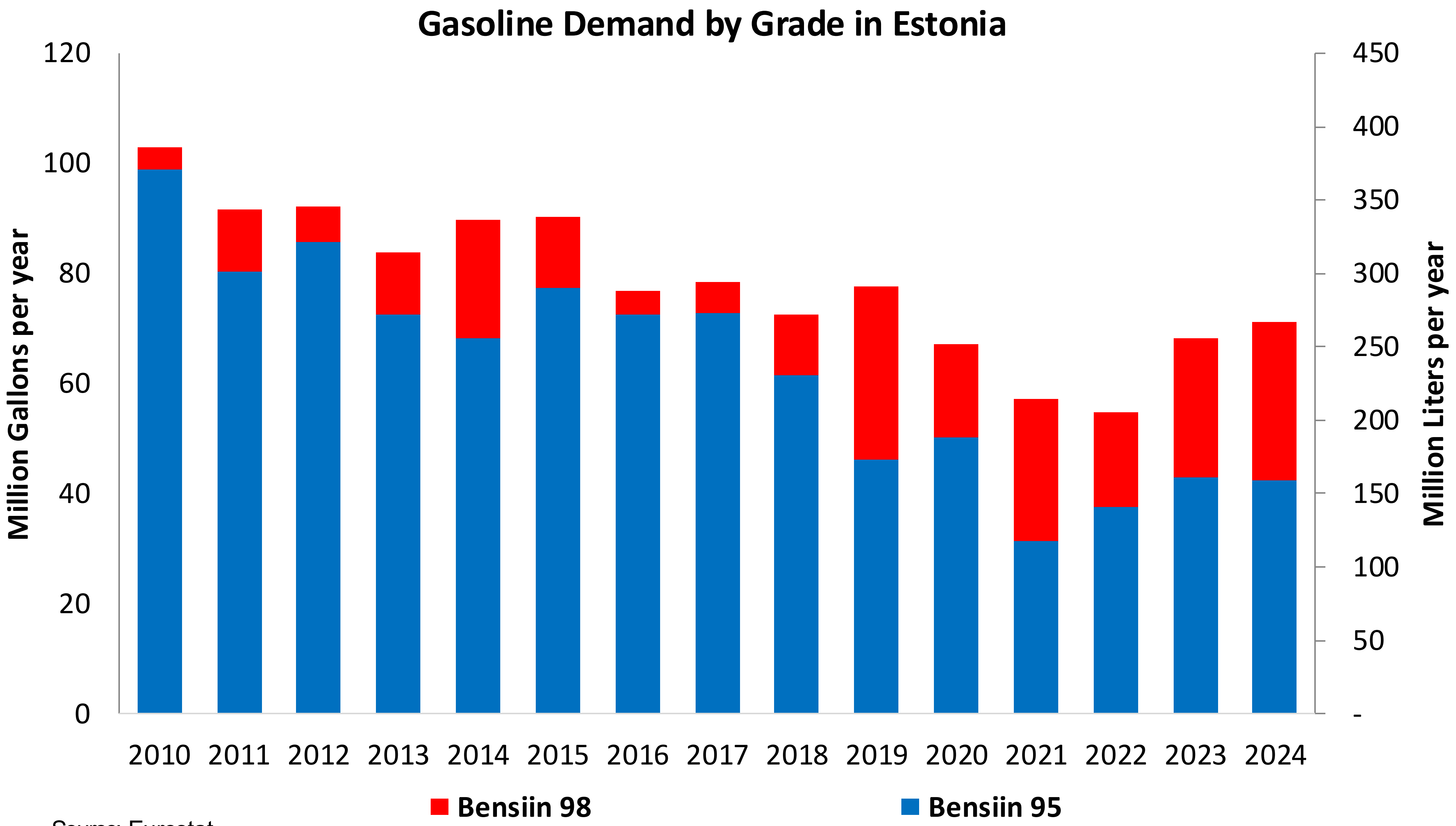


In 2024, gasoline consumption in Estonia was 267 million liters (71 million gallons) and growth rate was -1.8%. There is no gasoline production. Gasoline net imports were 267 million liters (70 million gallons). Estonia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 96.2 RON.

In 2024, ethanol consumption in Estonia was 5 million liters (1 million gallons) representing 1.8% of gasoline demand. Ethanol imports were 5 million liters (1 million gallons). Ethanol sources is Cereals 100%.

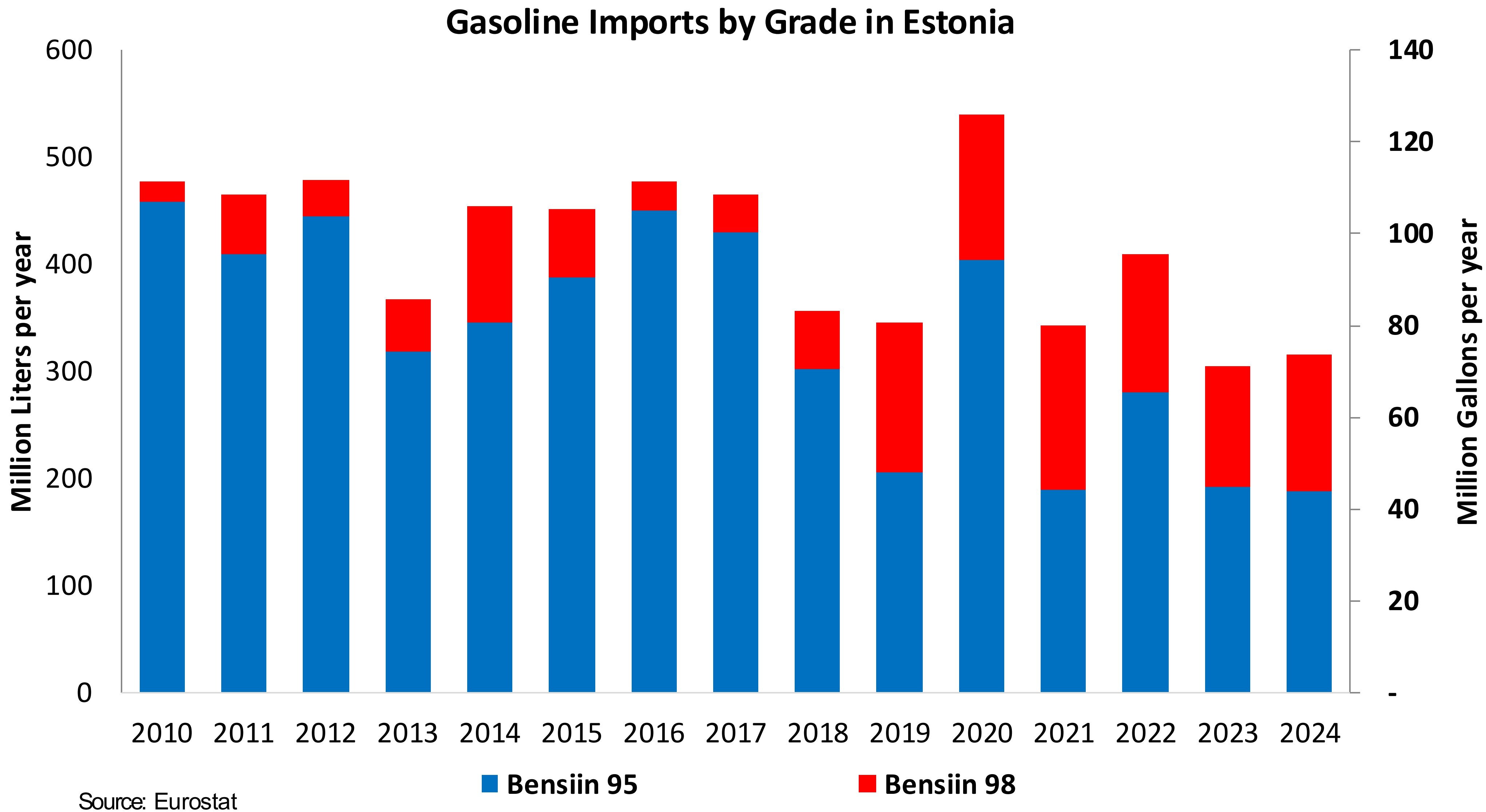
Source: Eurostat, Eurpean Commision

Gasoline Demand in Estonia



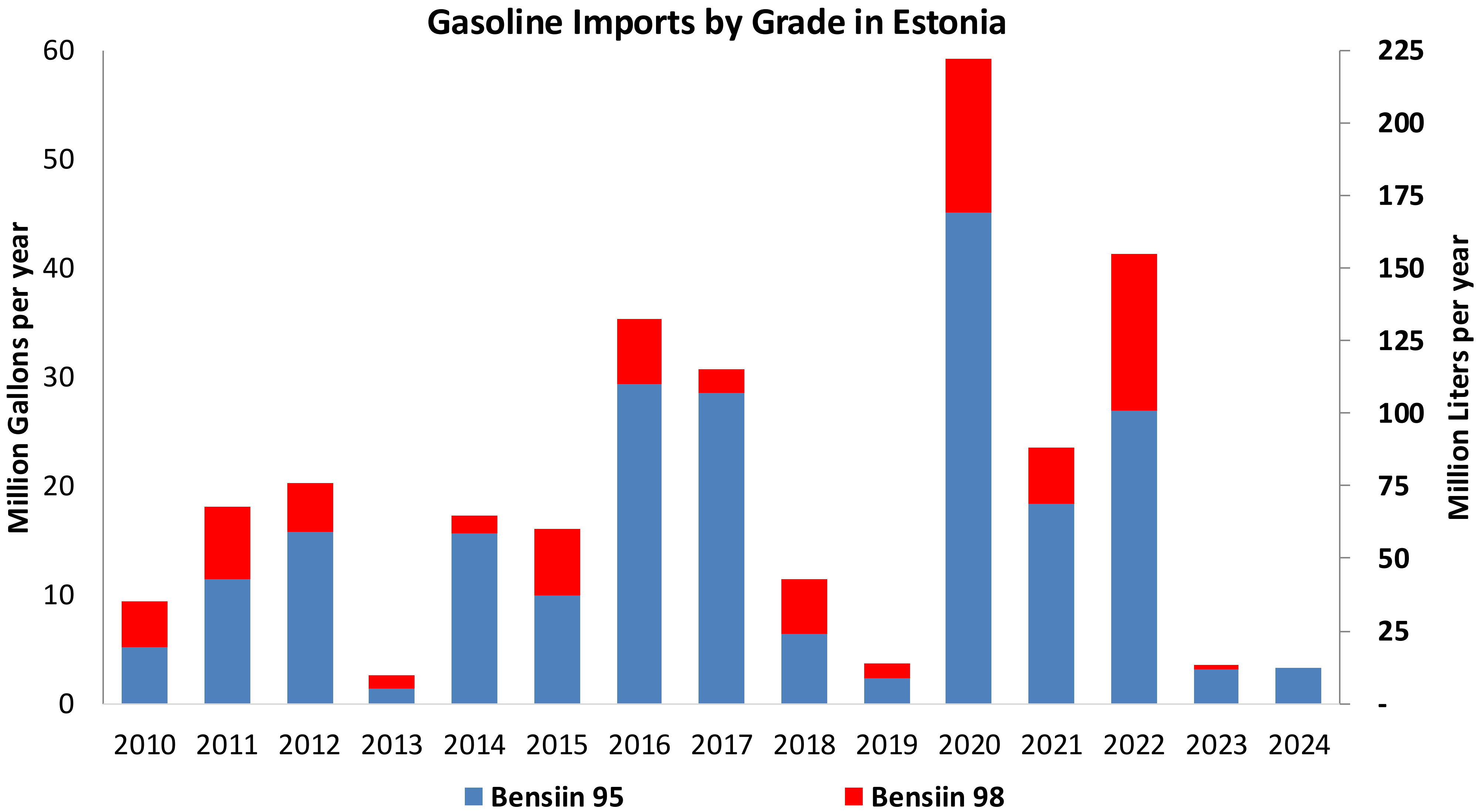
Source: Eurostat

Gasoline Imports in Estonia



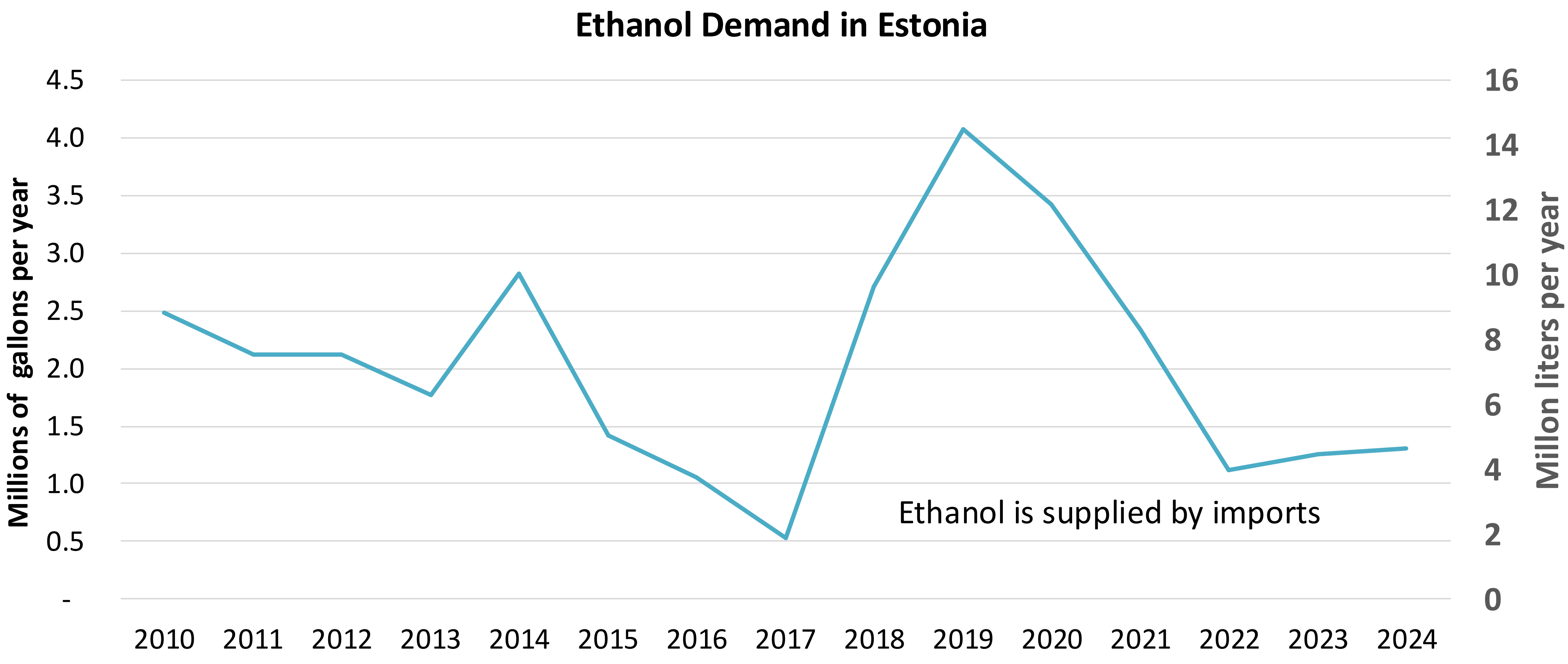
Source: Eurostat

Gasoline Exports in Estonia



Source: Eurostat

Ethanol Balance in Estonia



Source: Eurostat, Statistics Estonia

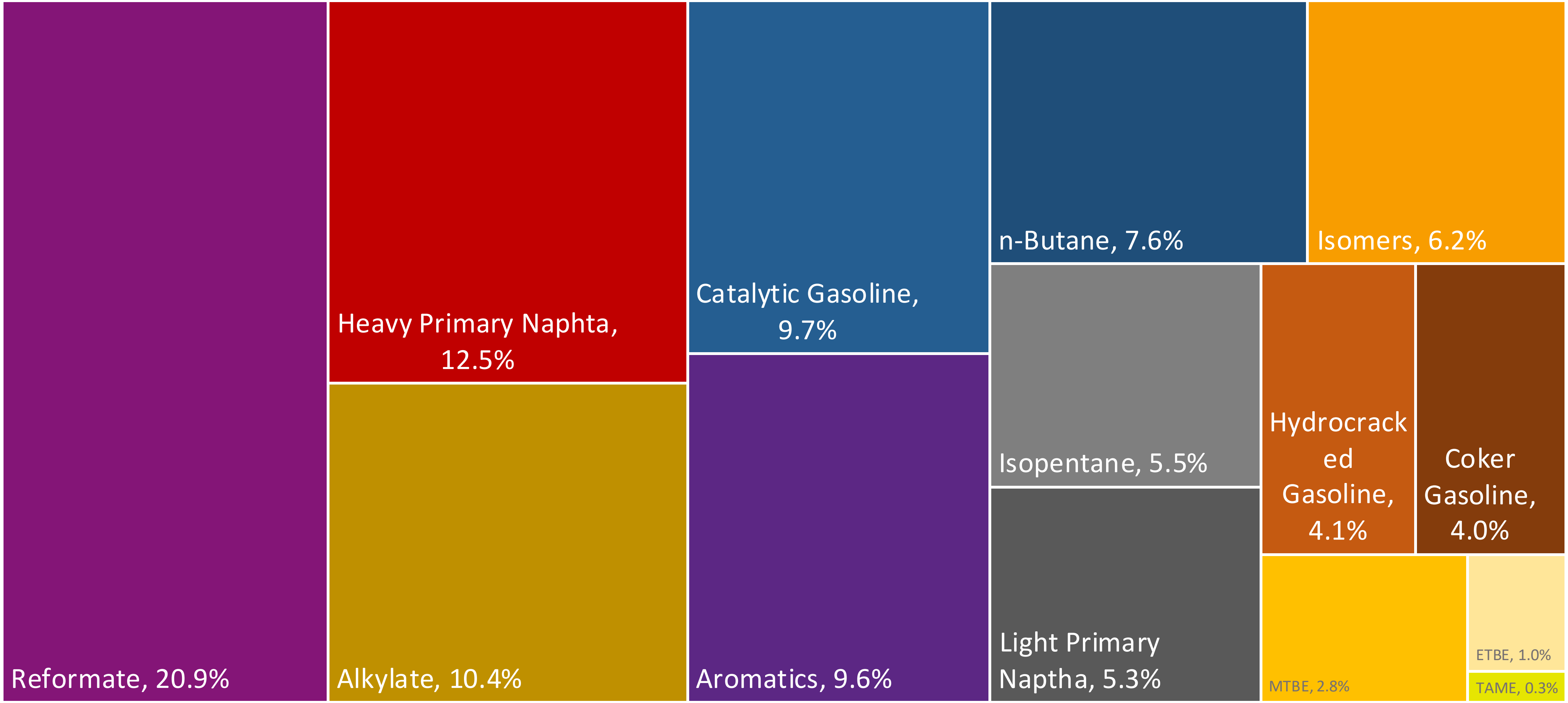
Gasoline Quality in Estonia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	7.5							
Advanced Biofuels (%cal)	0.5							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

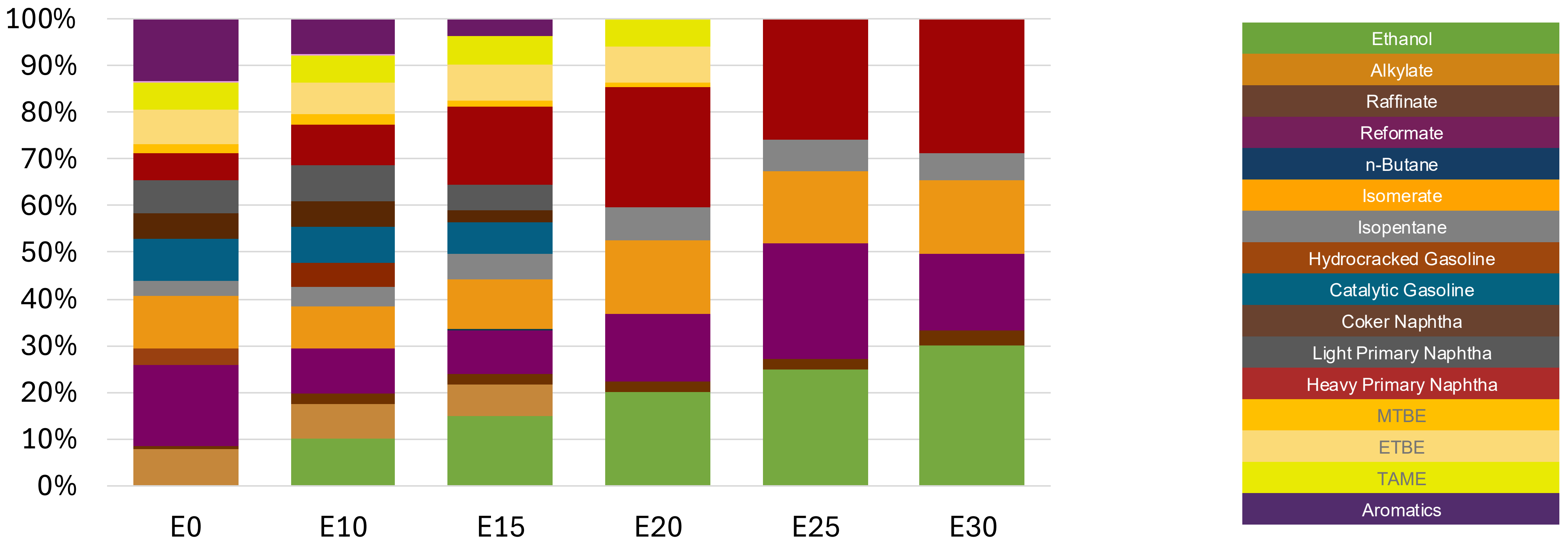
Source: European Commision

Estonia Available Blending Components



Source: Faro90

Estonia – Gasoline 95 – Constant Octane

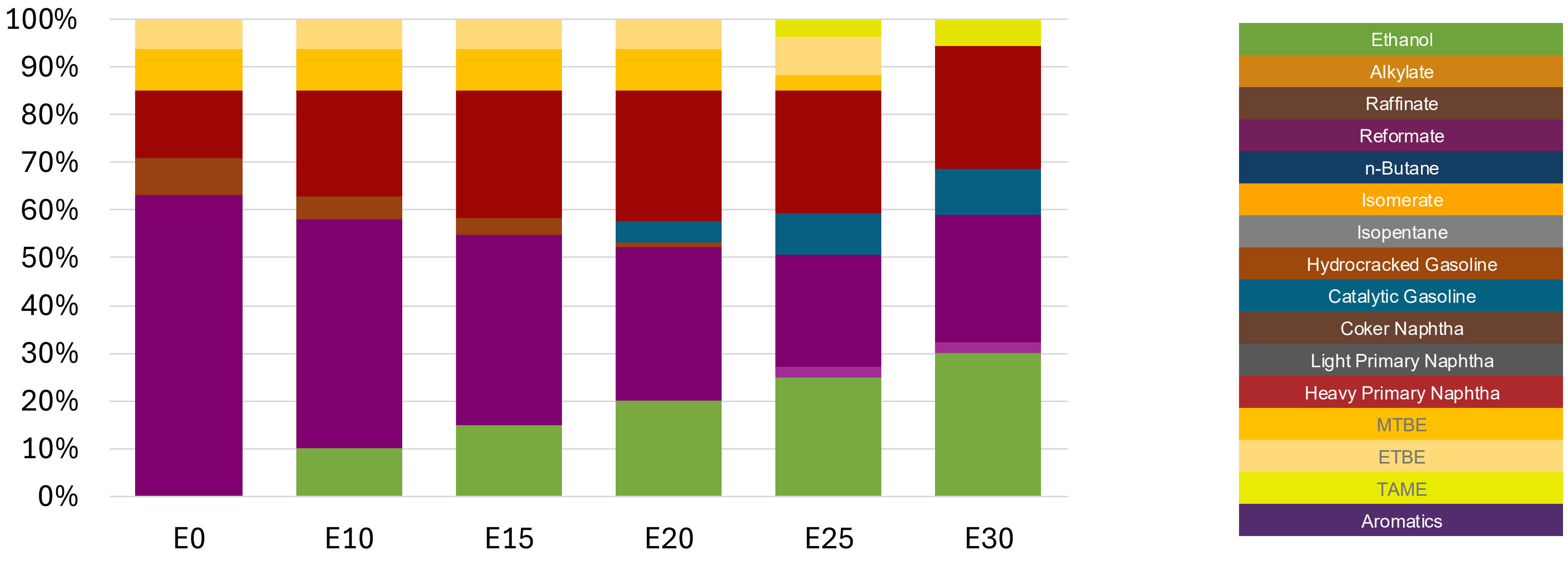


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.274	$ 2.136	$ 2.036	$ 1.921	$ 1.875	$ 1.822

Average CIF 2024 Prices

Source: Faro90

Estonia - Gasoline 98 - Constant Octane

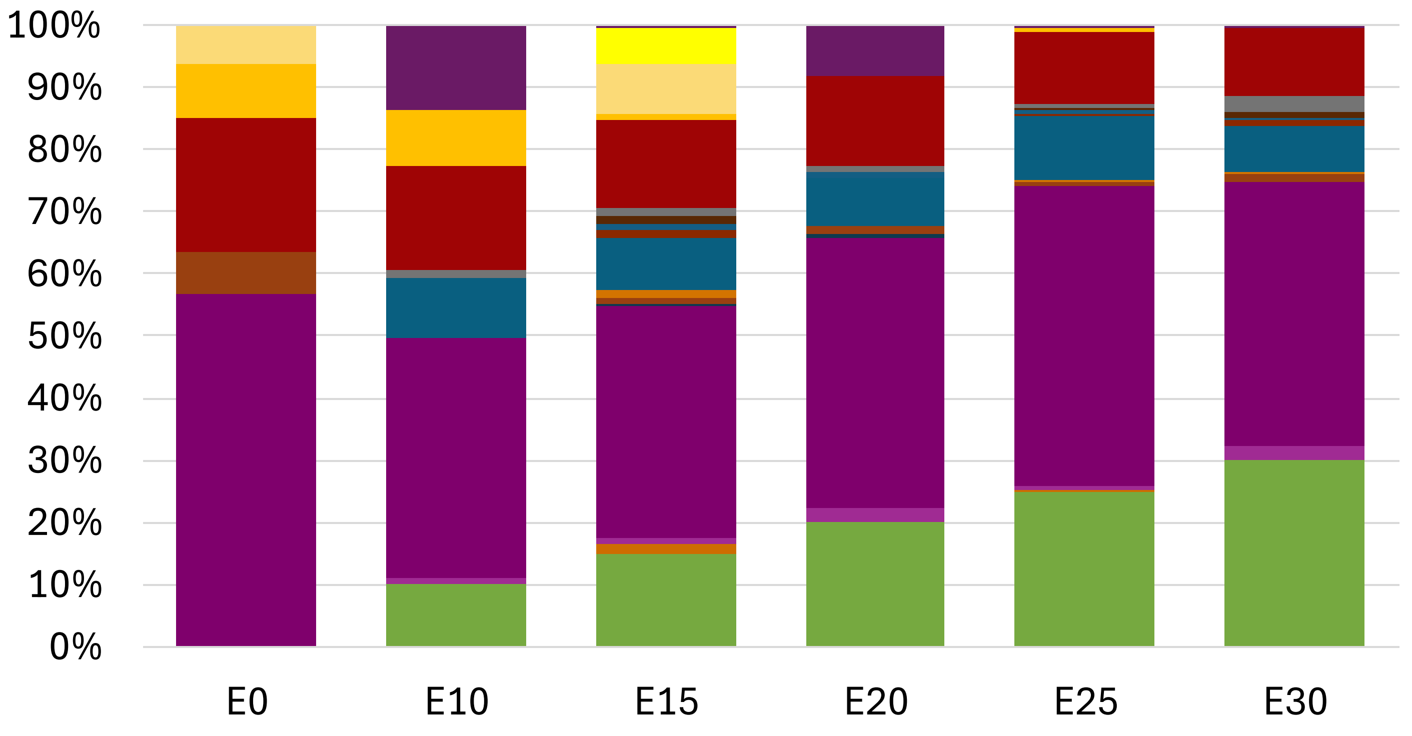


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.294	$ 2.177	$ 2.108	$ 2.036	$ 1.971	$ 1.921

Average CIF 2024 Prices

Source: Faro90

Estonia – Gasoline 95 – Increased Octane

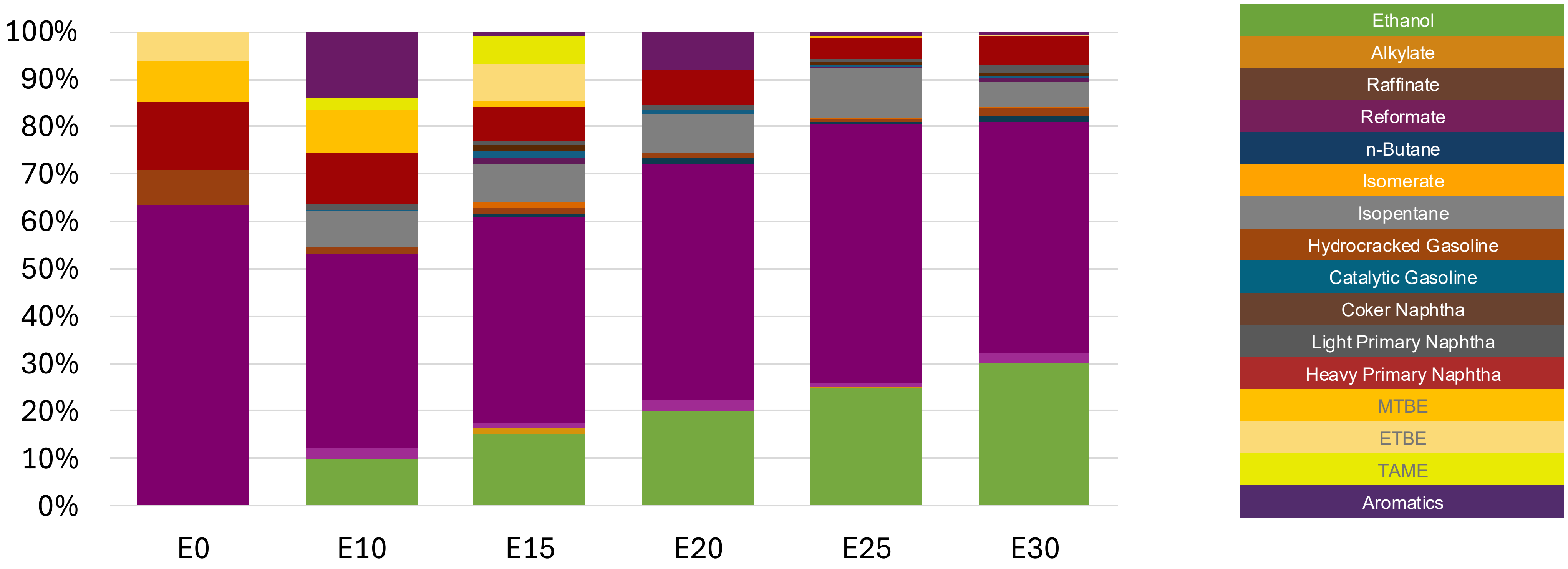


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162

Average CIF 2024 Prices

Source: Faro90

Estonia – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294

Average CIF 2024 Prices

Source: Faro90

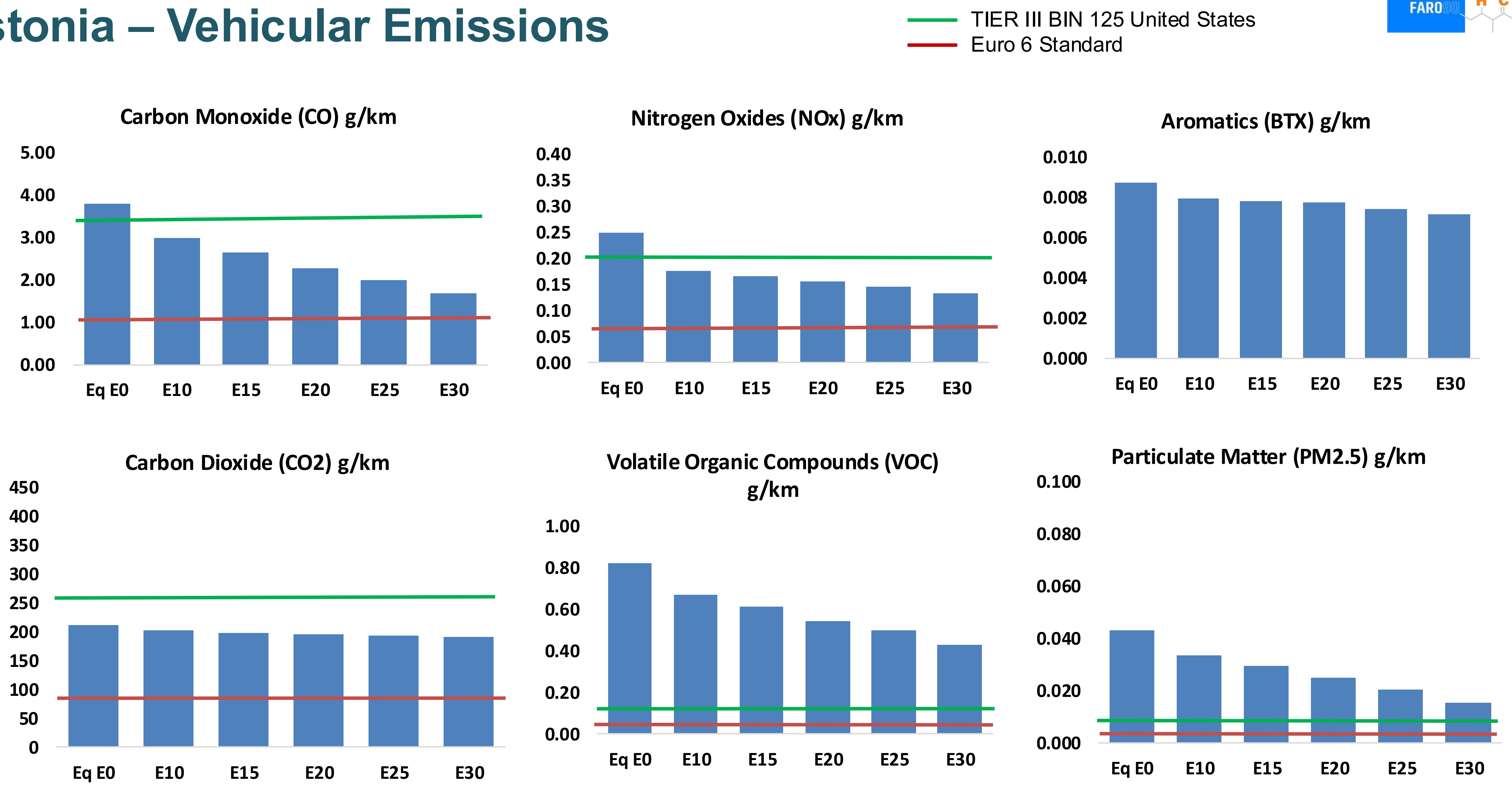
Gasoline Vehicle Fleet - Estonia

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	202
		Single FI	3-Ways		>8 years	6,340
		Multi FI	3-Ways		>8 years	64,777
			Euro III		>8 years	85,307
			Euro IV		>8 years	186,431
			Euro V		>8 years	70,437
			Euro 6		>8 years	138,584
					4-8 years	120,678
					<4 years	39,805
			Hybrid		All	81,342
		Electric			All	14,722
	Medium	Carburator	3-Ways	PCV Tank	>8 years	20
		Single FI	3-Ways		>8 years	169
		Multi FI	3-Ways		>8 years	9,814
			Euro III		>8 years	17,916
			Euro IV		>8 years	65,469
			Euro V		>8 years	21,688
			Euro 6		>8 years	46,195
					4-8 years	54,738
					<4 years	30,615
			Hybrid		All	58,156
		Electric			All	10,387
Sml Engines	Light	4-Cycle	None	PCV	>8 years	91
			Euro III		>8 years	186
			Euro IV		>8 years	2,376
			Euro V		>8 years	952
			Euro 6		>8 years	3,432
					4-8 years	2,982
					<4 years	6,112
			Hybrid		All	2,346
		Electric			All	422
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	1,113
		Fuel-Injection	3-Ways		>8 years	8,013
			Euro III		>8 years	4,753
			Euro IV		>8 years	23,826
			Euro V		>8 years	8,719
			Euro 6		>8 years	24,367
					4-8 years	29,213
					<4 years	32,909
			Hybrid		All	100
		Electric			All	623

Vehicle Fleet: 1,276,323
Gasoline Vehicle Fleet: 616,792

Source: Eurostat, ACEA

Estonia – Vehicular Emissions



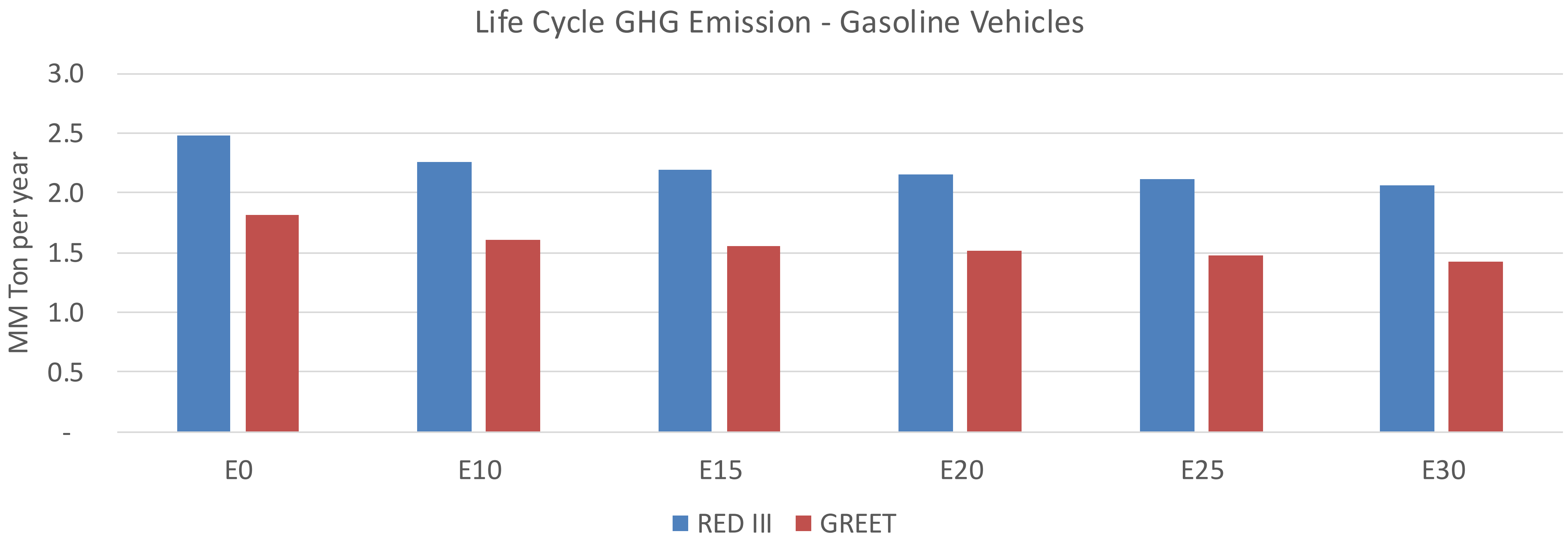
Source: Faro90

Estonia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	59,676	53,345	47,014	41,486	35,767	31,322	26,777	-21%	-40%
CO2	3,336,894	3,251,088	3,170,049	3,106,315	3,074,776	3,056,519	3,000,178	-5%	-8%
VOC	12,895	11,734	10,573	9,589	8,583	7,805	6,784	-18%	-33%
NOx	3,952	2,964	2,766	2,607	2,459	2,295	2,122	-30%	-38%
SOx	40	37	34	31	29	26	24	-15%	-28%
NH3	1,269	1,257	1,244	1,250	1,264	1,274	1,282	-2%	0%
N2O	53	53	53	53	54	55	55	-1%	2%
CH4	2,578	2,578	2,578	2,616	2,673	2,722	2,769	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	57	54	51	49	47	45	43	-10%	-18%
Acetaldehyde	199	228	335	510	695	794	938	68%	249%
Formaldehyde	758	737	859	1,009	1,057	1,169	1,273	13%	39%
Benzene	137	132	125	123	122	117	113	-9%	-11%
PM2.5	207	184	161	139	118	96	73	-22%	-43%
PM10	474	421	368	319	271	220	167	-22%	-43%

Source: Faro90

Estonia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	15.6
Target @2035 (MMT/year)	8.8
Delta	6.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.7%	14.5%	16.7%	18.7%	21.5%
RED II/III	0.0%	8.9%	11.1%	12.8%	14.4%	16.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.1%	3.9%	4.4%	5.0%	5.7%
RED II/III	0.0%	3.2%	4.0%	4.7%	5.2%	6.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

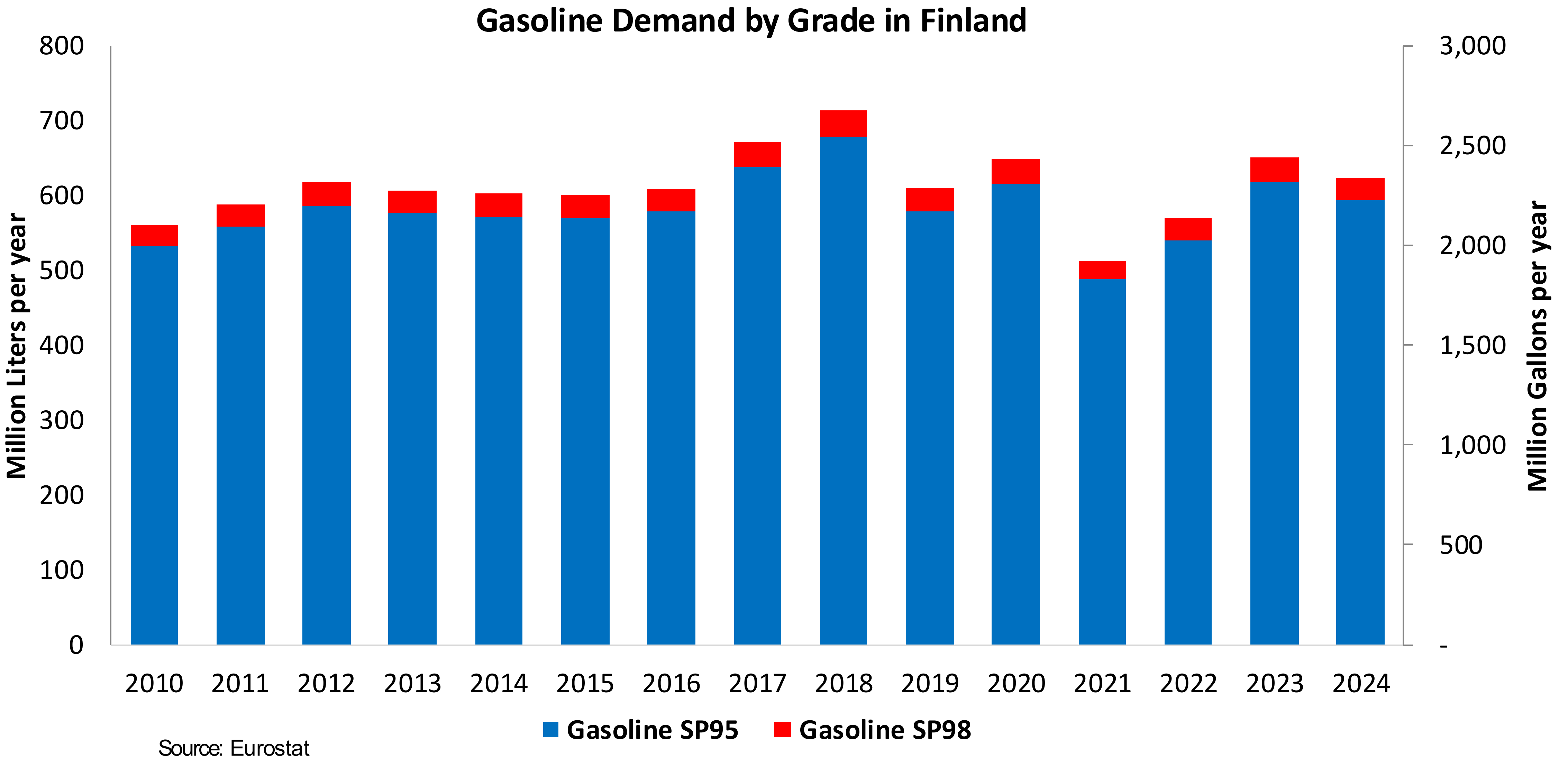
Ethanol Gasoline Blending in Finland

In 2024, gasoline consumption in Finland was 2,341 million liters (618 million gallons) and growth rate was 0.5%. Gasoline production and net imports were 5,450 and -3,109 million liters (1,440 and -821 million gallons) correspondingly. Finland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Finland was 299 million liters (79 million gallons) representing 12.8% of gasoline demand. Ethanol production and Net Imports were 67 and 233 million liters (79 and 61 million gallons) correspondingly. Ethanol sources are Corn 5%, Cereals 85% and 2G lignocelulosic Ethanol 10%.

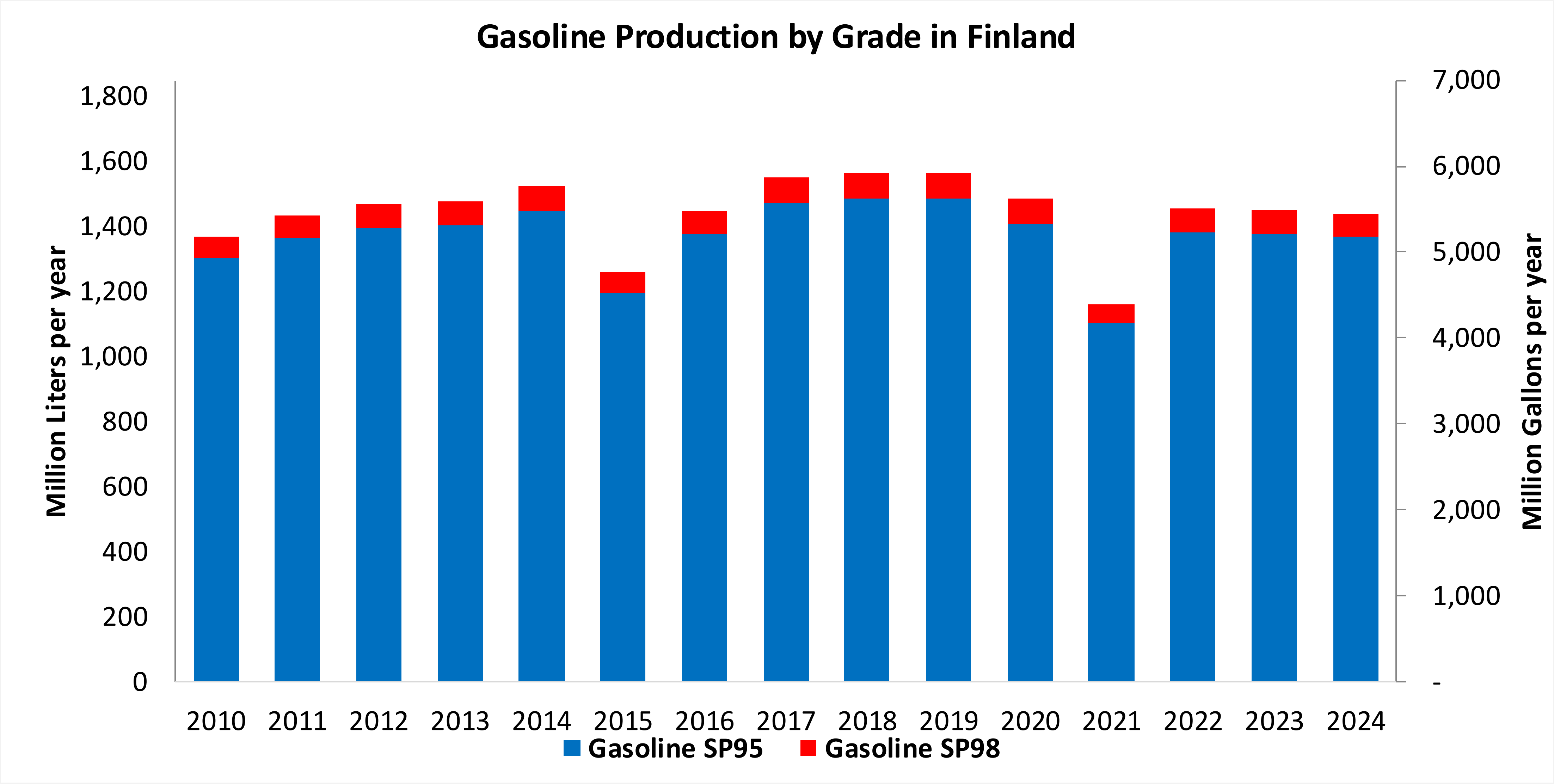
Source: Eurostat, European Commision

Gasoline Demand in Finland



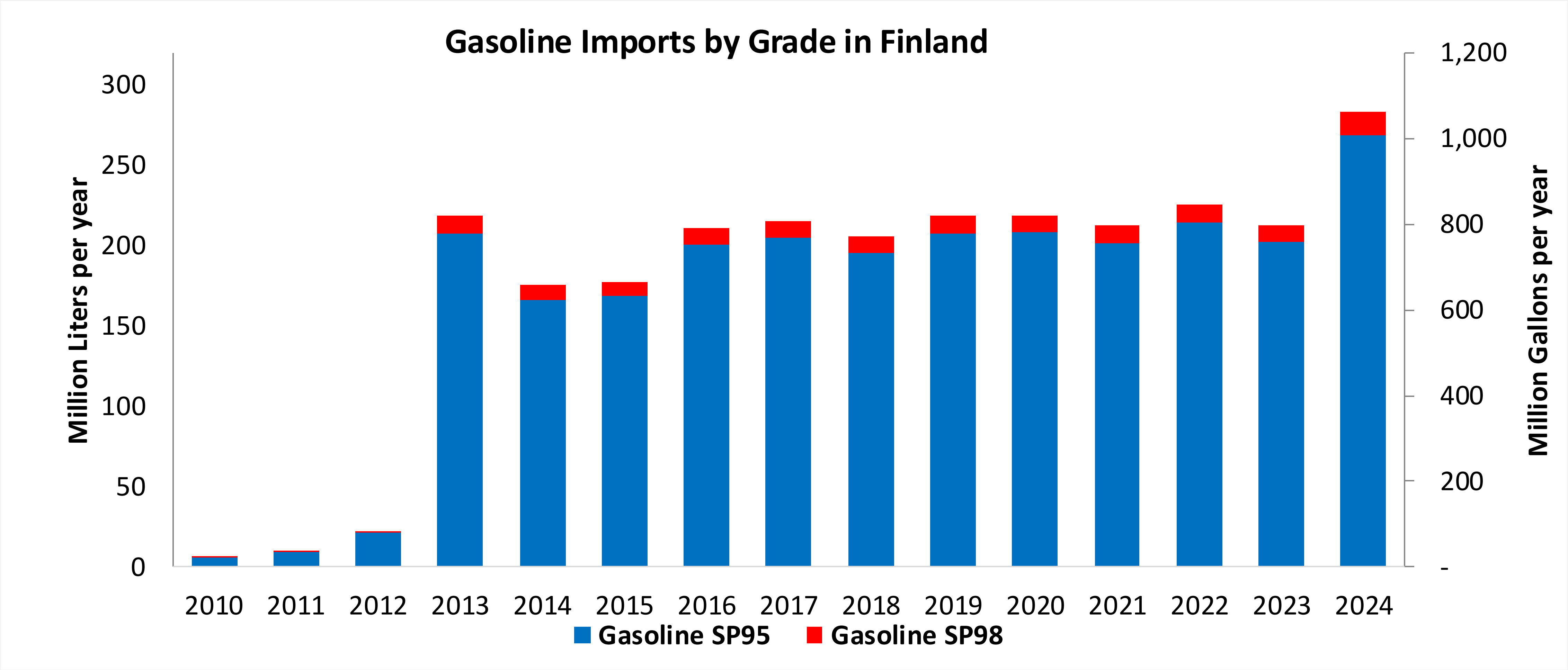
Source: Eurostat

Gasoline Production in Finland



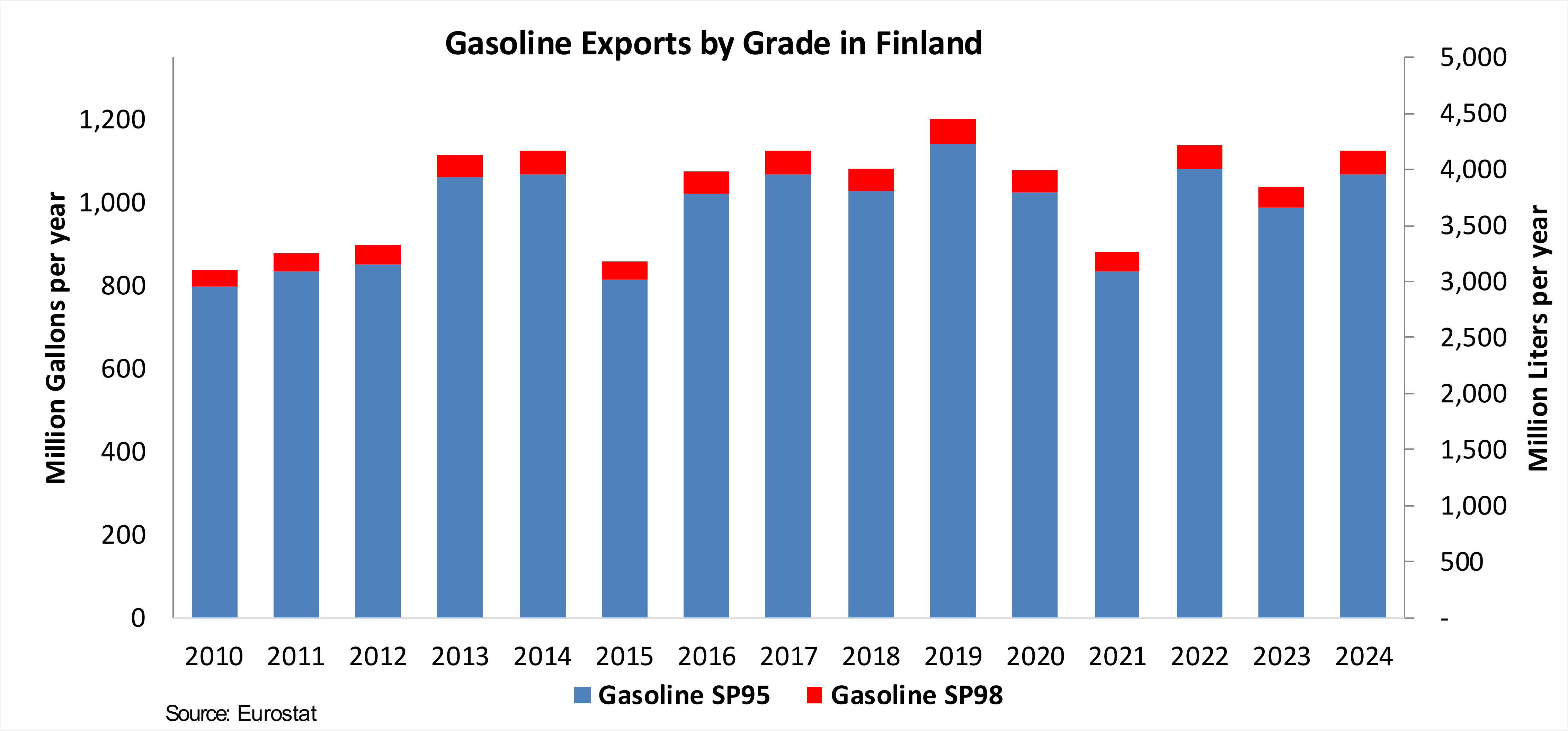
Source: Eurostat

Gasoline Imports in Finland

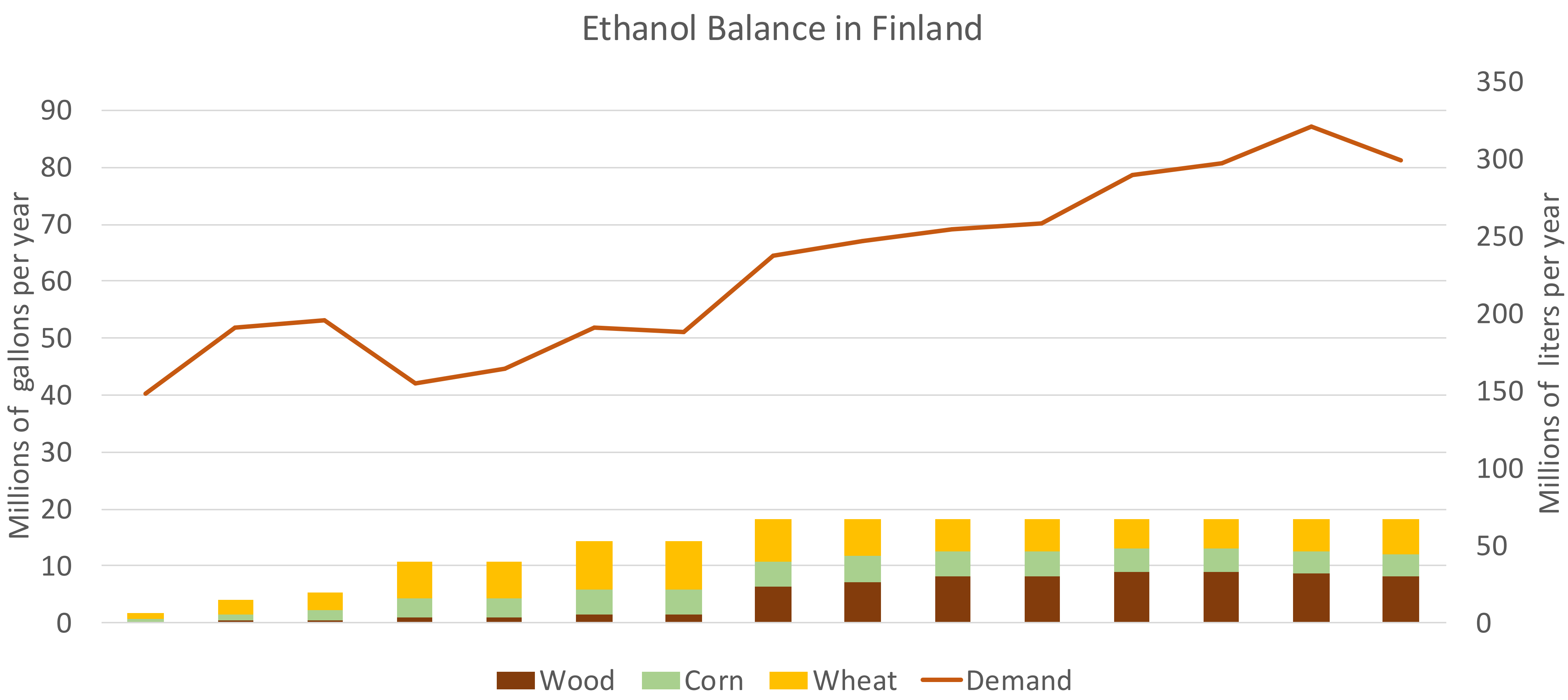


Source: Eurostat

Gasoline Exports in Finland



Ethanol Balance in Finland



Source: Eurostat

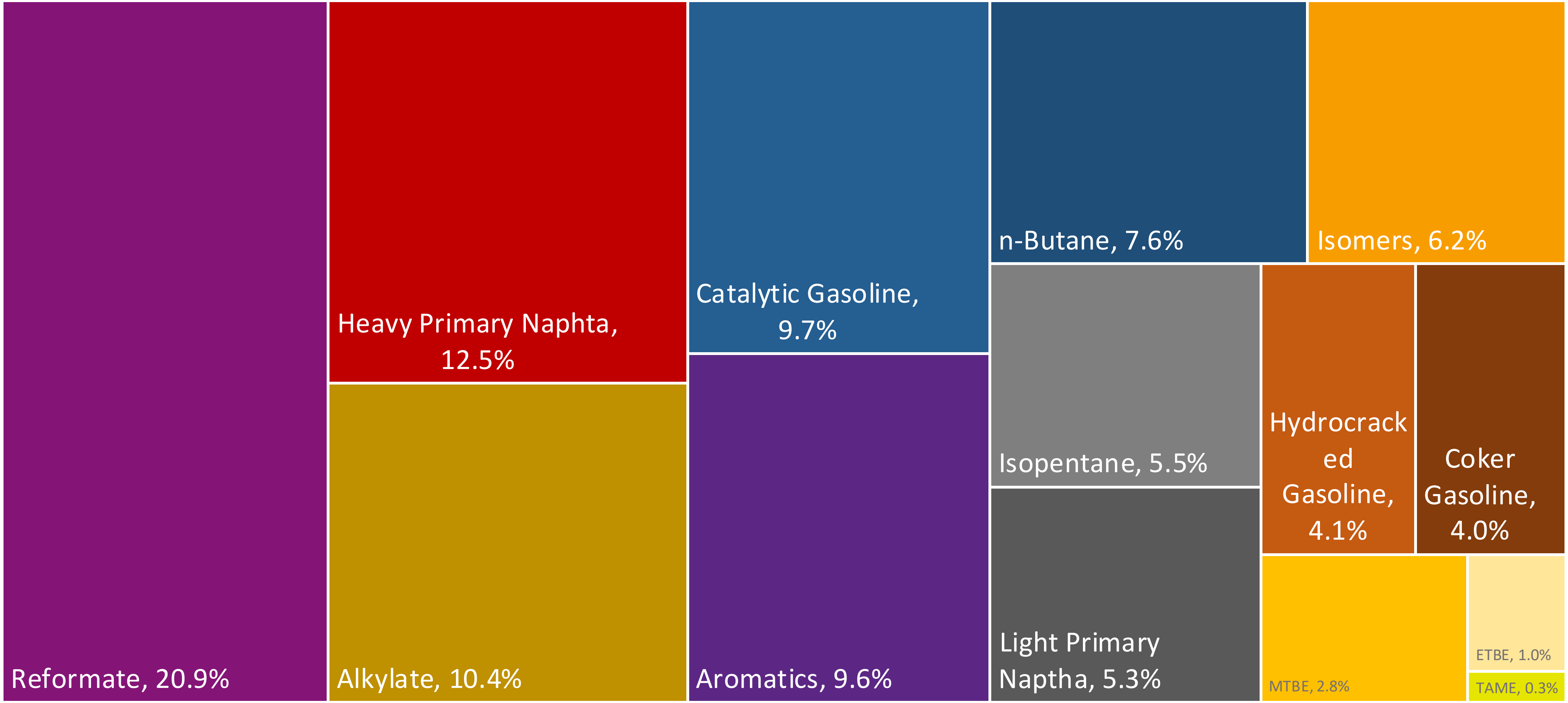
Gasoline Quality in Finland

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	16.5							
Advanced Biofuels (%cal)	4.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

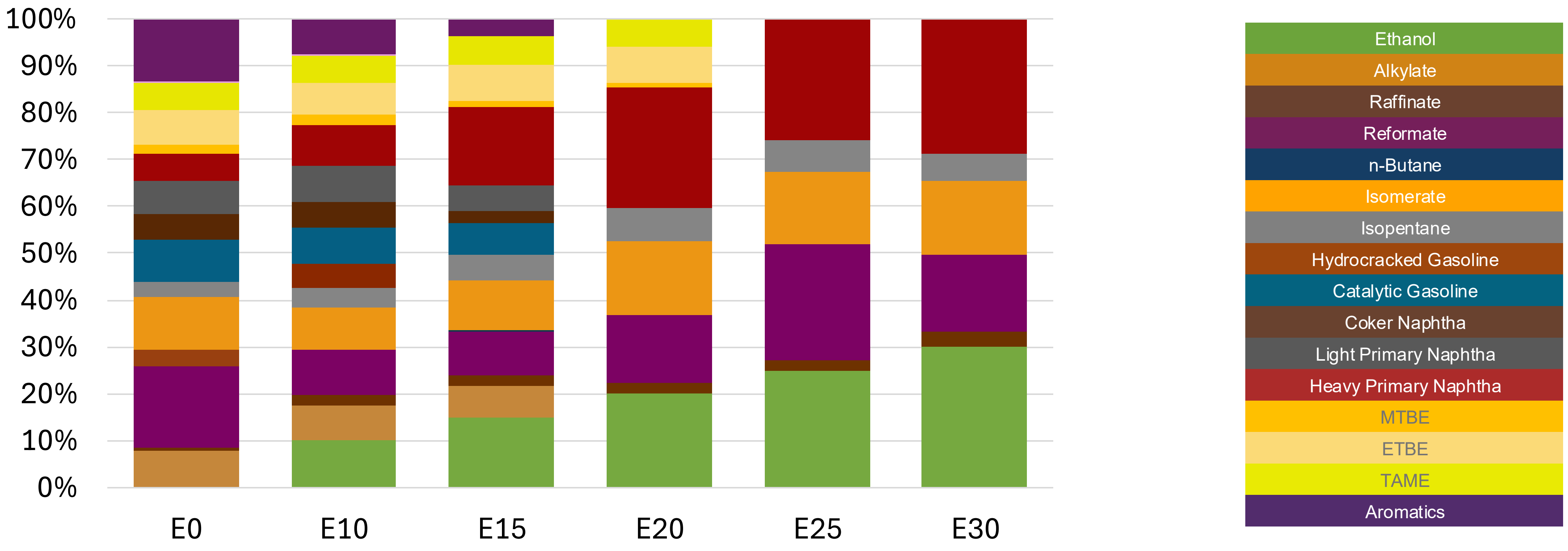
Source: European Commision

Finland Available Blending Components



Source: Faro90

Finland – Gasoline 95 – Constant Octane

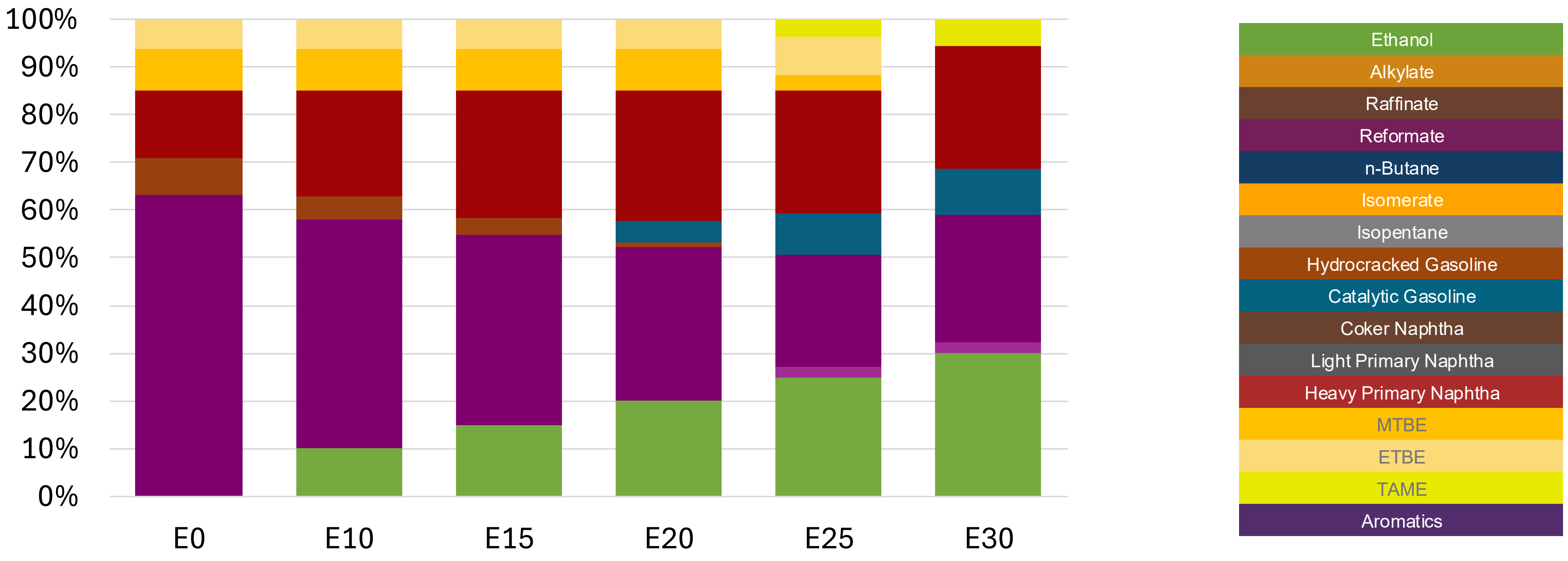


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.294	$ 2.156	$ 2.056	$ 1.941	$ 1.895	$ 1.842

Average CIF 2024 Prices

Source: Faro90

Finland - Gasoline 98 - Constant Octane

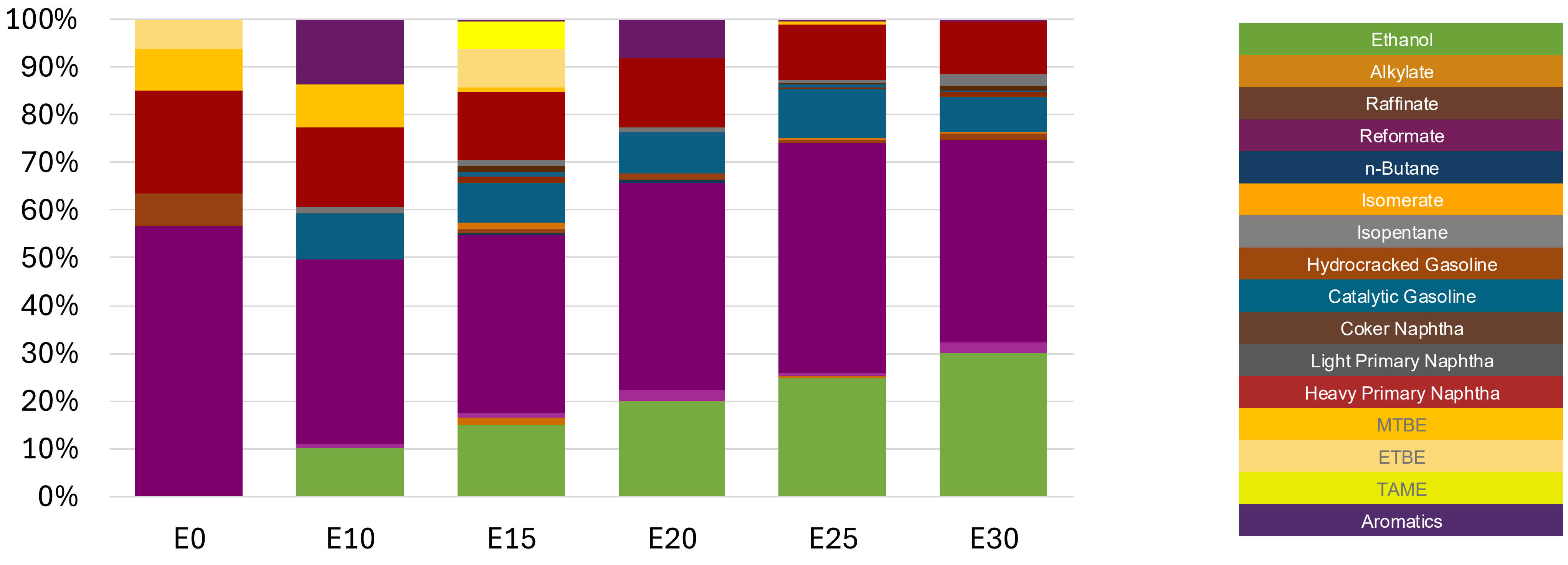


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.314	$ 2.197	$ 2.128	$ 2.056	$ 1.991	$ 1.941

Average CIF 2024 Prices

Source: Faro90

Finland – Gasoline 95 – Increased Octane

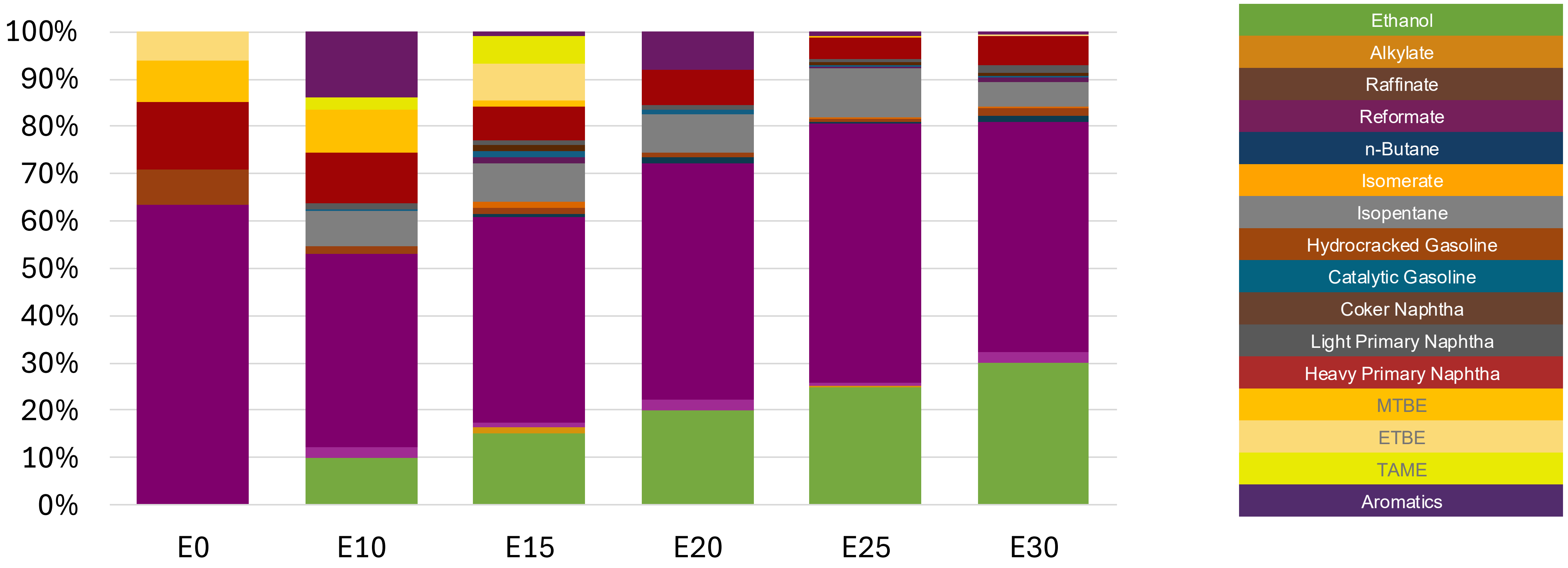


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.182	$ 2.182	$ 2.182	$ 2.182	$ 2.182	$ 2.182

Average CIF 2024 Prices

Source: Faro90

Finland – Gasoline 98 – Increased Octane

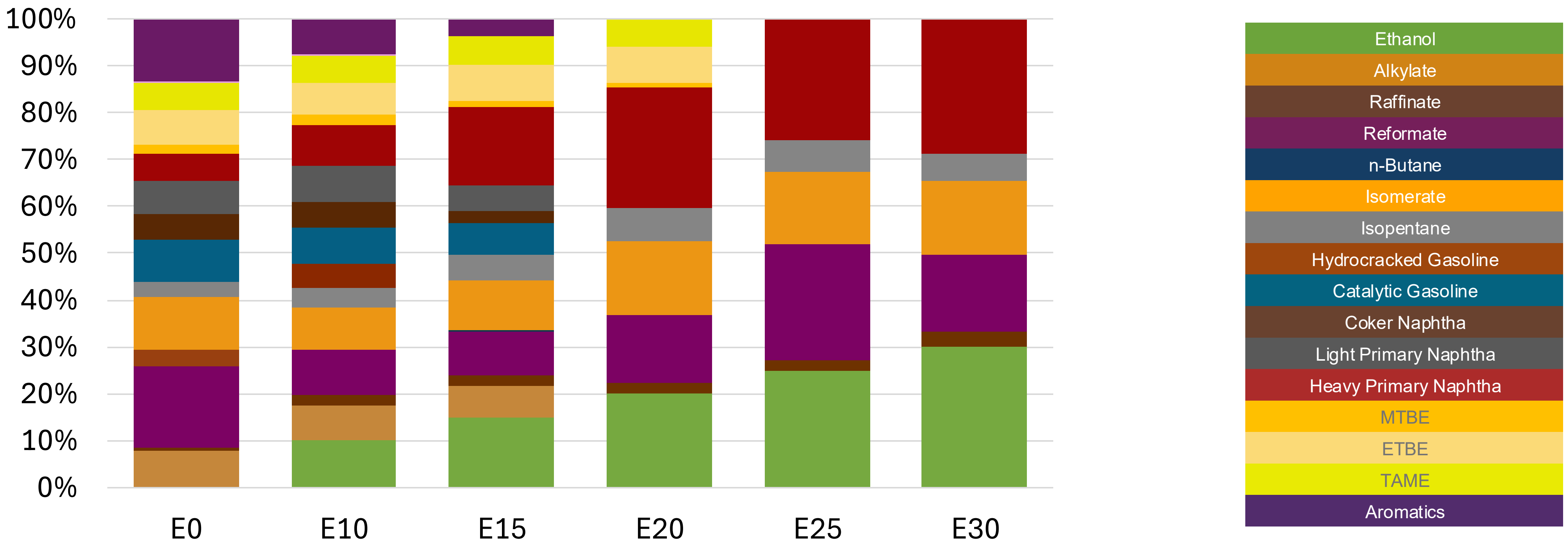


Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.314	$ 2.314	$ 2.314	$ 2.314	$ 2.314	$ 2.314

Average CIF 2024 Prices

Source: Faro90

Finland – Gasoline 95 – Constant Octane

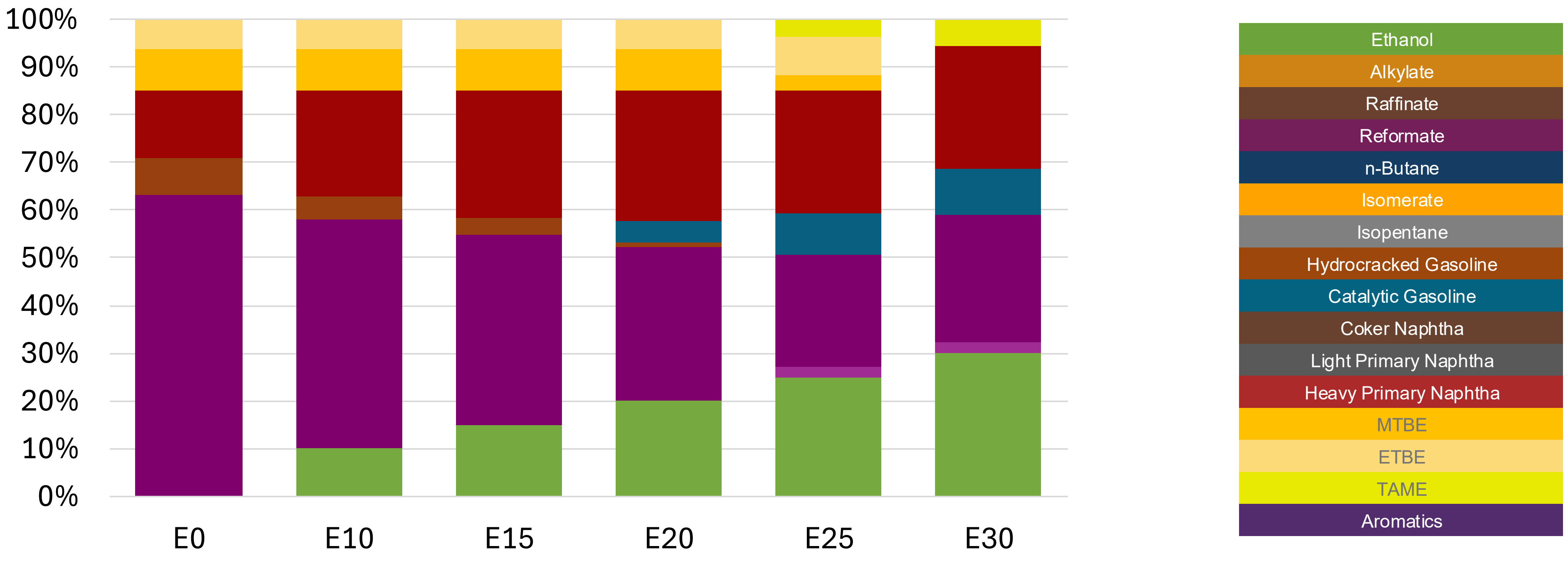


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.274	$ 2.136	$ 2.036	$ 1.921	$ 1.875	$ 1.822

Average CIF 2024 Prices

Source: Faro90

Finland - Gasoline 98 - Constant Octane

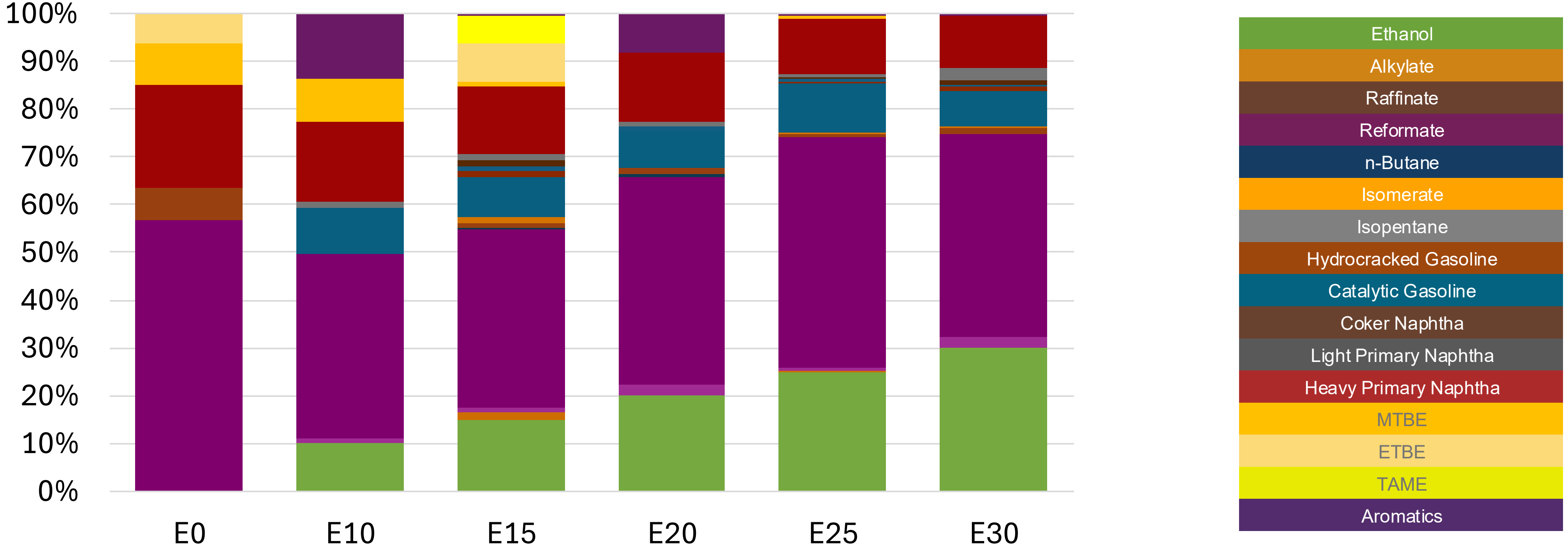


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.294	$ 2.177	$ 2.108	$ 2.036	$ 1.971	$ 1.921

Average CIF 2024 Prices

Source: Faro90

Finland – Gasoline 95 – Increased Octane

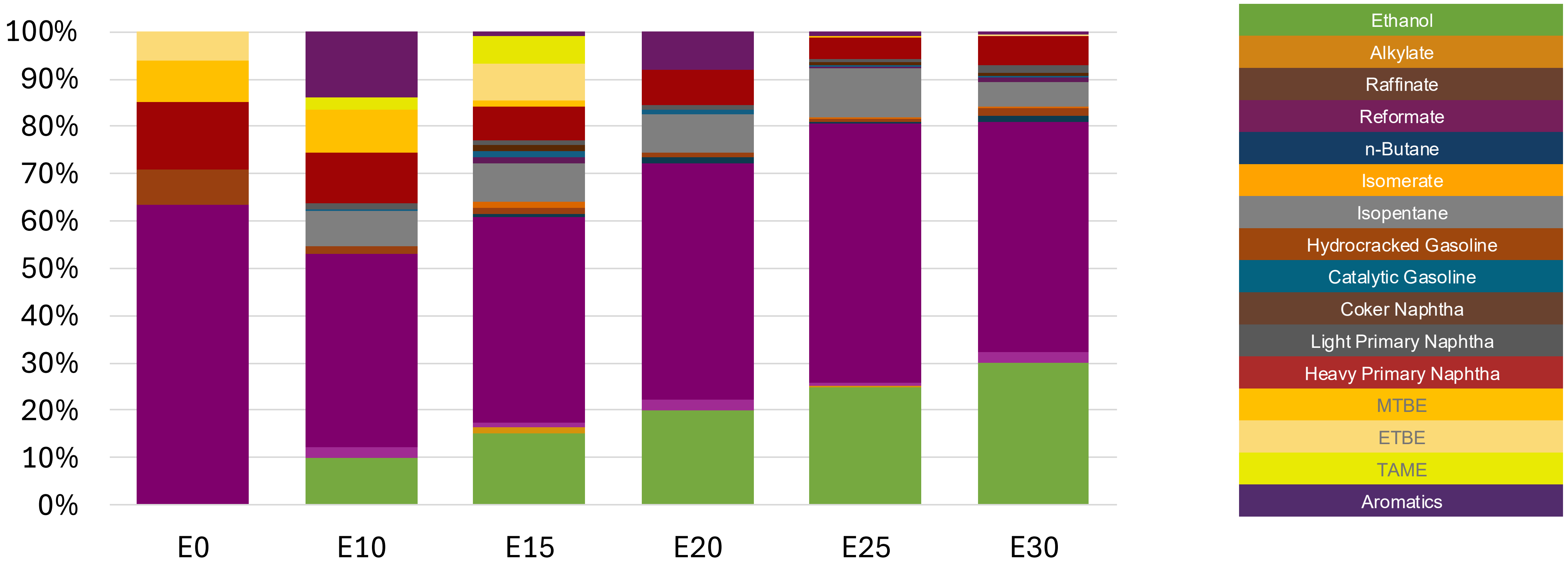


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162

Average CIF 2024 Prices

Source: Faro90

Finland – Gasoline 98 – Increased Octane



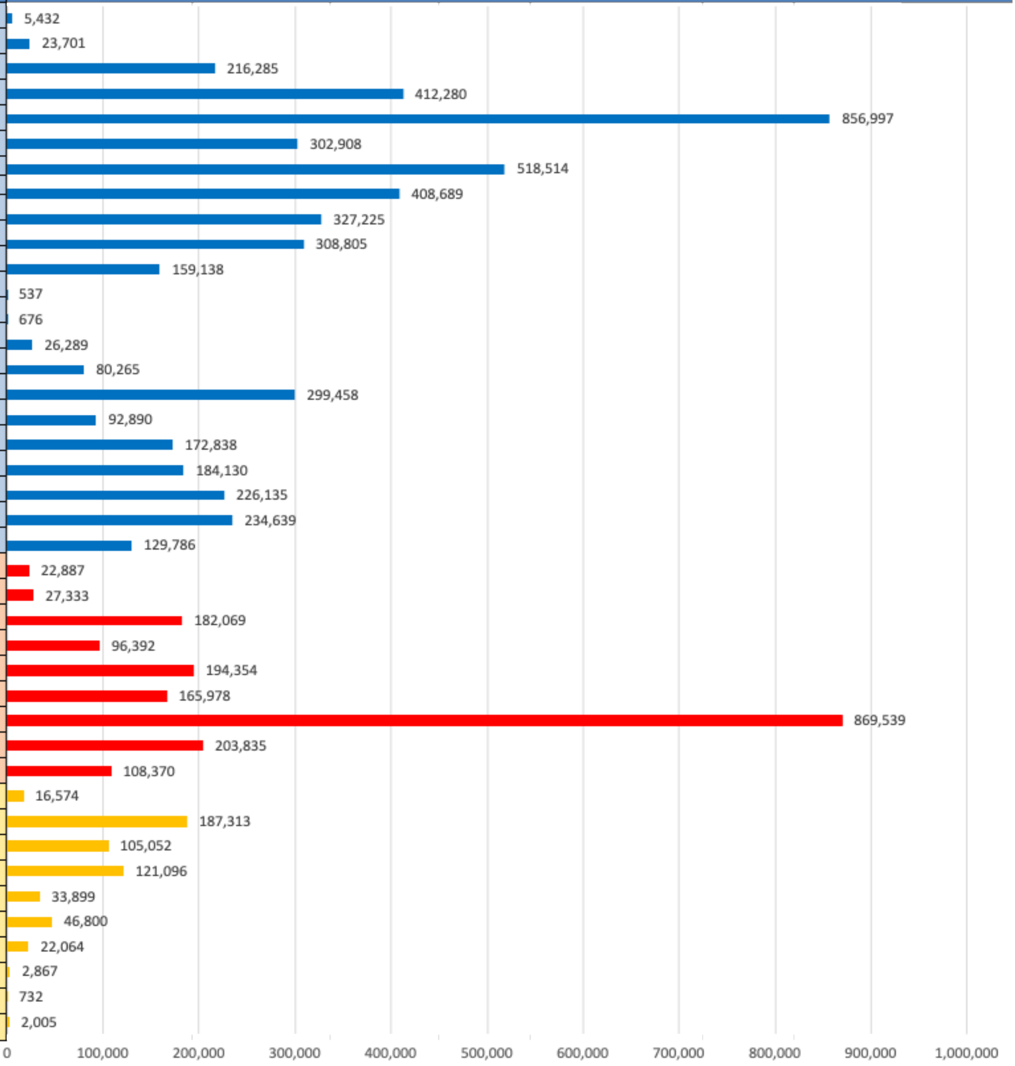
Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Finland

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	5,432
Auto/Sml Truck	Light	Single FI	3-Ways	PCV Tank	>8 years	23,701
Auto/Sml Truck	Light	Multi FI	3-Ways	PCV Tank	>8 years	216,285
Auto/Sml Truck	Light	Multi FI	Euro III	PCV Tank	>8 years	412,280
Auto/Sml Truck	Light	Multi FI	Euro IV	PCV Tank	>8 years	856,997
Auto/Sml Truck	Light	Multi FI	Euro V	PCV Tank	>8 years	302,908
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	>8 years	518,514
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	4-8 years	408,689
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	<4 years	327,225
Auto/Sml Truck	Light	Multi FI	Hybrid	PCV Tank	All	308,805
Auto/Sml Truck	Light	Electric			All	159,138
Auto/Sml Truck	Medium	Carburator	3-Ways	PCV Tank	>8 years	537
Auto/Sml Truck	Medium	Single FI	3-Ways	PCV Tank	>8 years	676
Auto/Sml Truck	Medium	Multi FI	3-Ways	PCV Tank	>8 years	26,289
Auto/Sml Truck	Medium	Multi FI	Euro III	PCV Tank	>8 years	80,265
Auto/Sml Truck	Medium	Multi FI	Euro IV	PCV Tank	>8 years	299,458
Auto/Sml Truck	Medium	Multi FI	Euro V	PCV Tank	>8 years	92,890
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	>8 years	172,838
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	4-8 years	184,130
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	<4 years	226,135
Auto/Sml Truck	Medium	Multi FI	Hybrid	PCV Tank	All	234,639
Auto/Sml Truck	Medium	Electric			All	129,786
Sml Engines	Light	4-Cycle	None	PCV	>8 years	22,887
Sml Engines	Light	4-Cycle	Euro III	PCV	>8 years	27,333
Sml Engines	Light	4-Cycle	Euro IV	PCV	>8 years	182,069
Sml Engines	Light	4-Cycle	Euro V	PCV	>8 years	96,392
Sml Engines	Light	4-Cycle	Euro 6	PCV	>8 years	194,354
Sml Engines	Light	4-Cycle	Euro 6	PCV	4-8 years	165,978
Sml Engines	Light	4-Cycle	Euro 6	PCV	<4 years	869,539
Sml Engines	Light	4-Cycle	Hybrid	PCV	All	203,835
Sml Engines	Light	Electric			All	108,370
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	16,574
Truck	Heavy	Fuel-Injection	3-Ways	PCV	>8 years	187,313
Truck	Heavy	Fuel-Injection	Euro III	PCV	>8 years	105,052
Truck	Heavy	Fuel-Injection	Euro IV	PCV	>8 years	121,096
Truck	Heavy	Fuel-Injection	Euro V	PCV	>8 years	33,899
Truck	Heavy	Fuel-Injection	Euro 6	PCV	>8 years	46,800
Truck	Heavy	Fuel-Injection	Euro 6	PCV	4-8 years	22,064
Truck	Heavy	Fuel-Injection	Euro 6	PCV	<4 years	2,867
Truck	Heavy	Fuel-Injection	Hybrid	PCV	All	732
Truck	Heavy	Electric			All	2,005

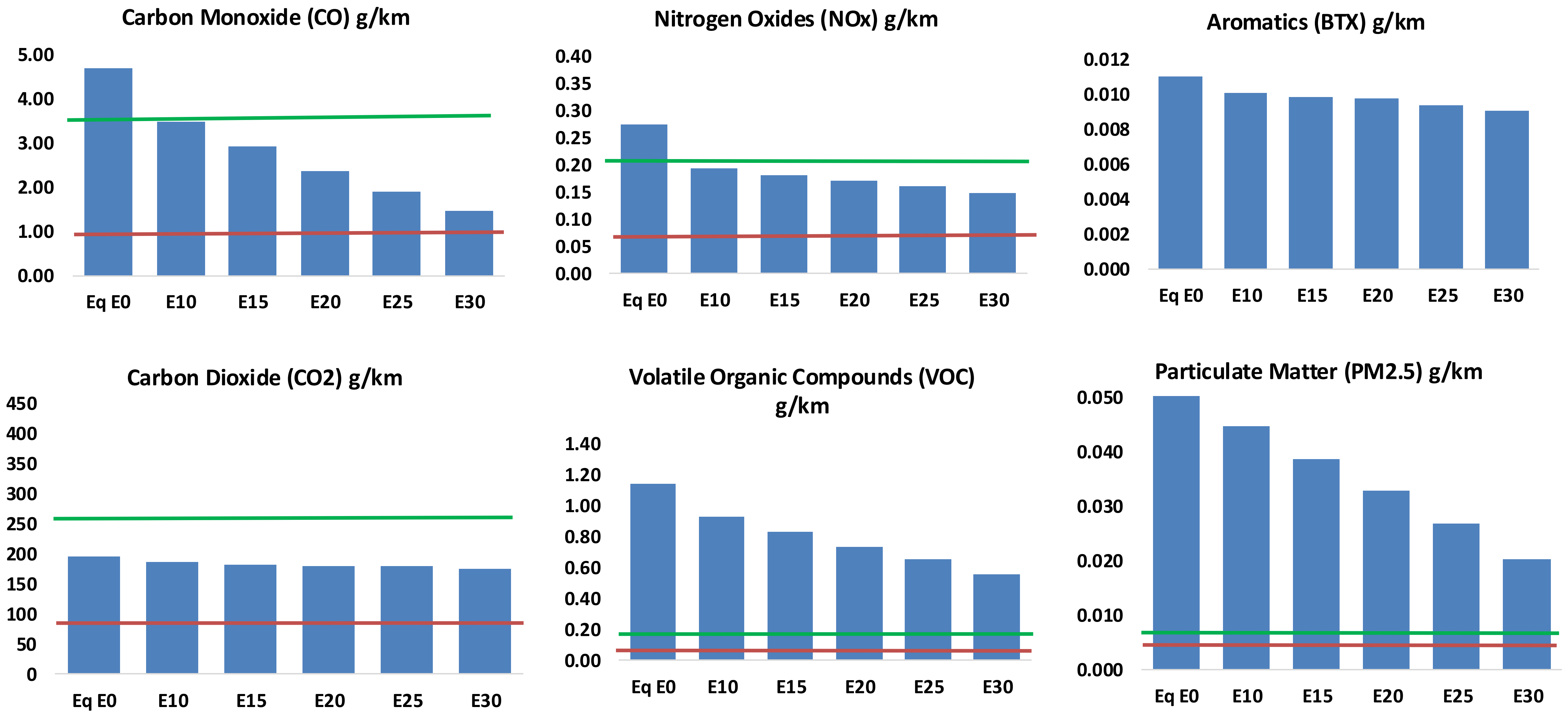


Vehicle Fleet: 7,396,775
Gasoline Vehicle Fleet: 3,837,401

Source: Eurostat, ACEA

Finland – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



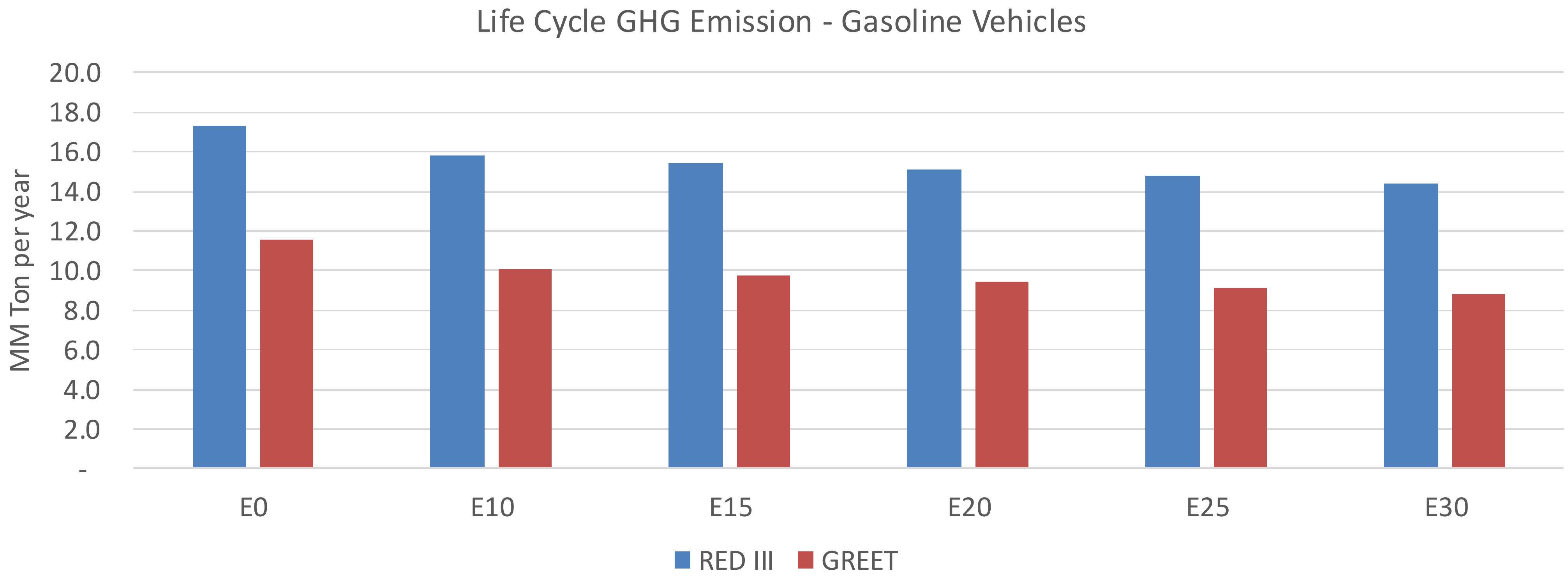
Source: Faro90

Finland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	469,869	408,719	347,570	292,577	234,989	190,028	145,320	-26%	-50%
CO2	19,452,404	18,952,199	18,479,784	18,108,243	17,924,390	17,818,561	17,490,113	-5%	-8%
VOC	114,075	103,064	92,053	82,593	72,871	65,335	55,493	-19%	-36%
NOx	27,517	20,638	19,262	18,154	17,121	15,982	14,777	-30%	-38%
SOx	221	204	188	174	160	145	130	-15%	-28%
NH3	7,866	7,788	7,710	7,747	7,834	7,895	7,946	-2%	0%
N2O	324	322	321	323	329	333	337	-1%	2%
CH4	22,977	22,977	22,977	23,322	23,828	24,264	24,678	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	467	434	400	373	347	326	299	-14%	-26%
Acetaldehyde	1,791	2,056	3,017	4,594	6,259	7,152	8,451	68%	249%
Formaldehyde	7,151	6,952	8,104	9,516	9,969	11,029	12,006	13%	39%
Benzene	1,102	1,059	1,003	987	979	936	906	-9%	-11%
PM2.5	1,230	1,092	954	828	703	571	433	-22%	-43%
PM10	4,504	3,999	3,494	3,035	2,577	2,091	1,585	-22%	-43%

Source: Faro90

Finland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



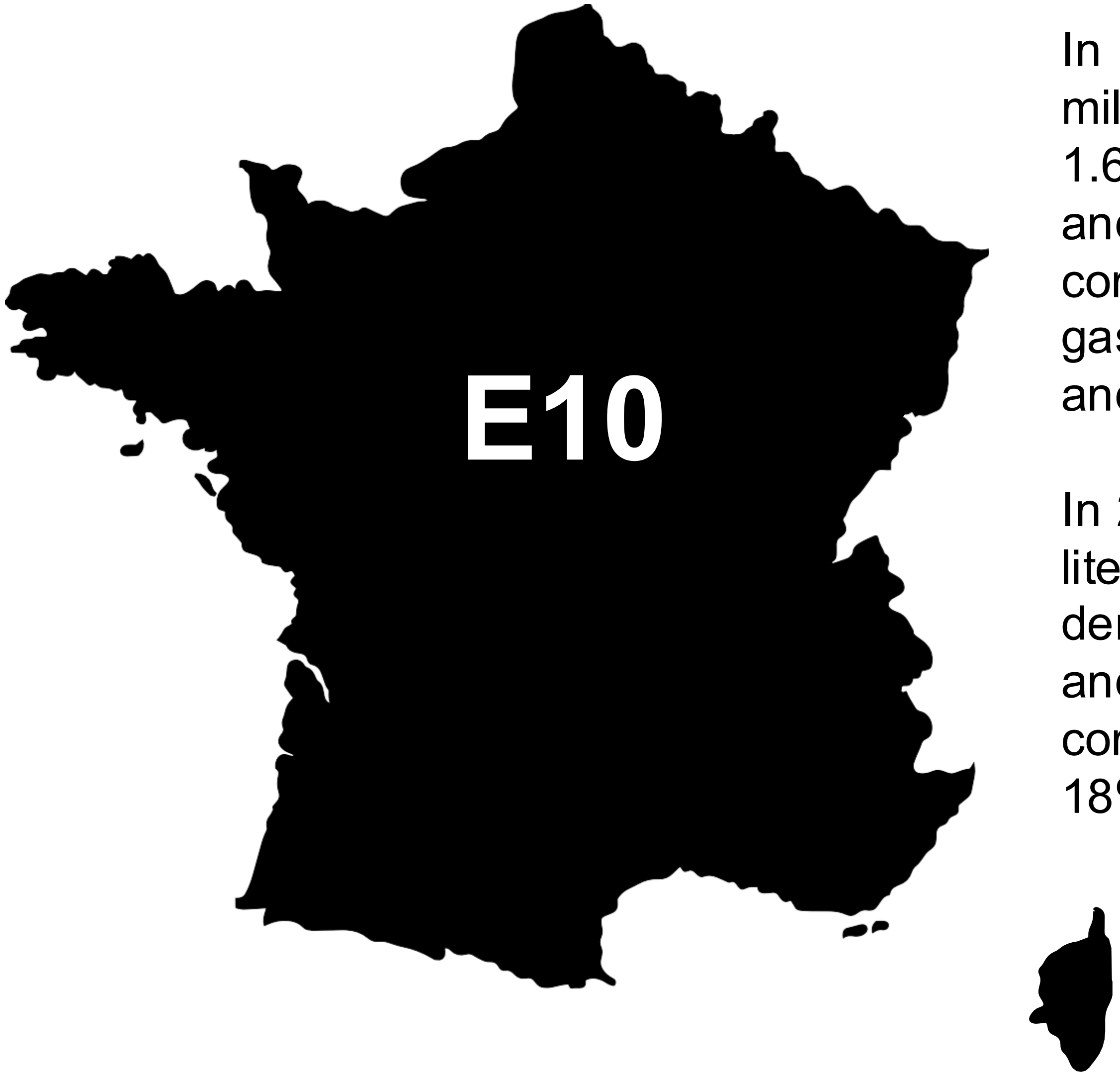
Country Emissions 2024 (MMT/año)	64.9
Target @2035 (MMT/year)	36.6
Delta	28.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.7%	15.9%	18.5%	21.0%	24.2%
RED II/III	0.0%	8.9%	11.3%	13.1%	14.9%	17.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.2%	6.5%	7.6%	8.6%	9.9%
RED II/III	0.0%	5.5%	6.9%	8.0%	9.1%	10.5%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in France

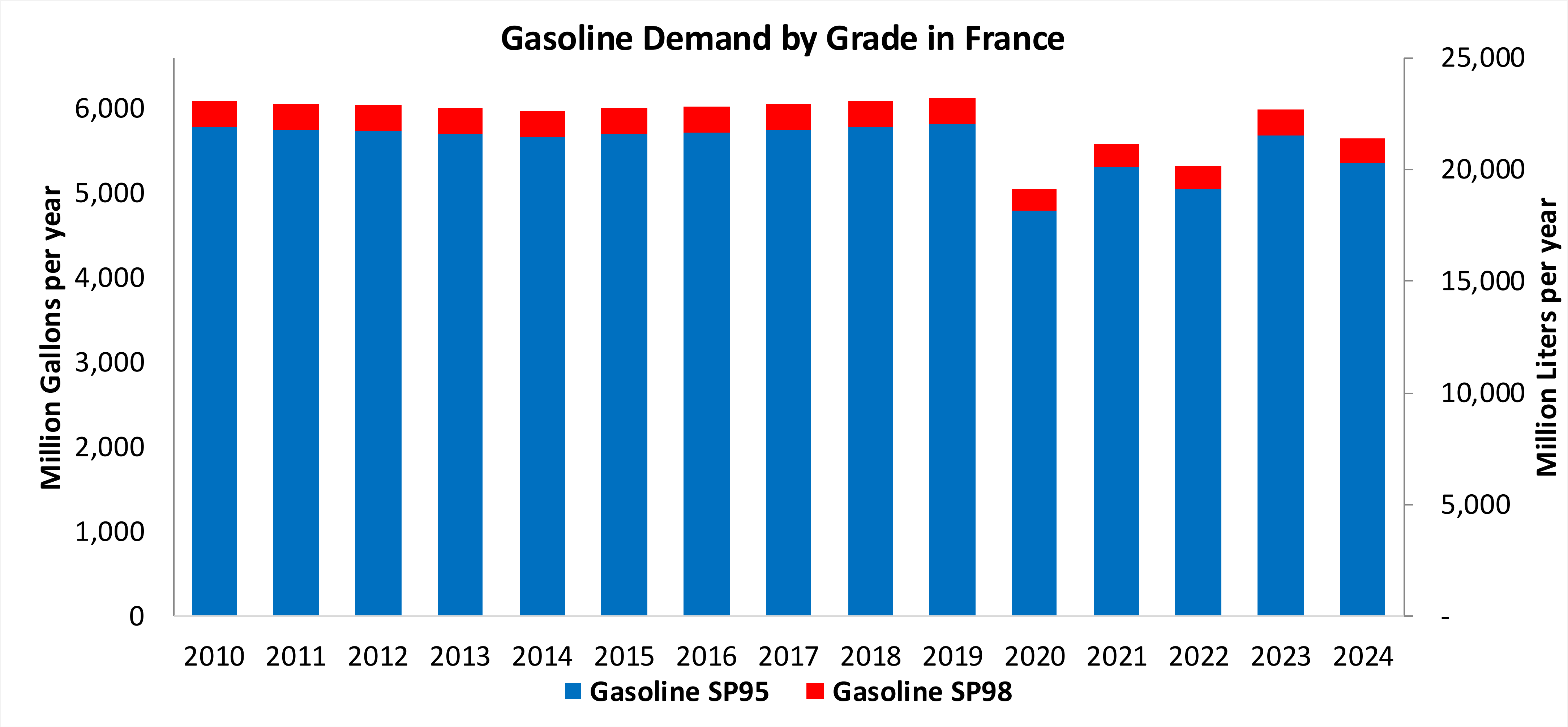


In 2024, gasoline consumption in France was 21,379 million liters (5,648 million gallons) and growth rate was -1.6%. Gasoline production and net imports were 10,333 and 11,046 million liters (2,730 and 2,918 million gallons) correspondingly. France complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in France was 2,622 million liters (693 million gallons) representing 12.3% of gasoline demand. Ethanol production and Net Imports were 2,154 and 468 million liters (693 and 124 million gallons) correspondingly. Ethanol sources are Corn 27%, Cereals 18% and Sugar Beet 37%.

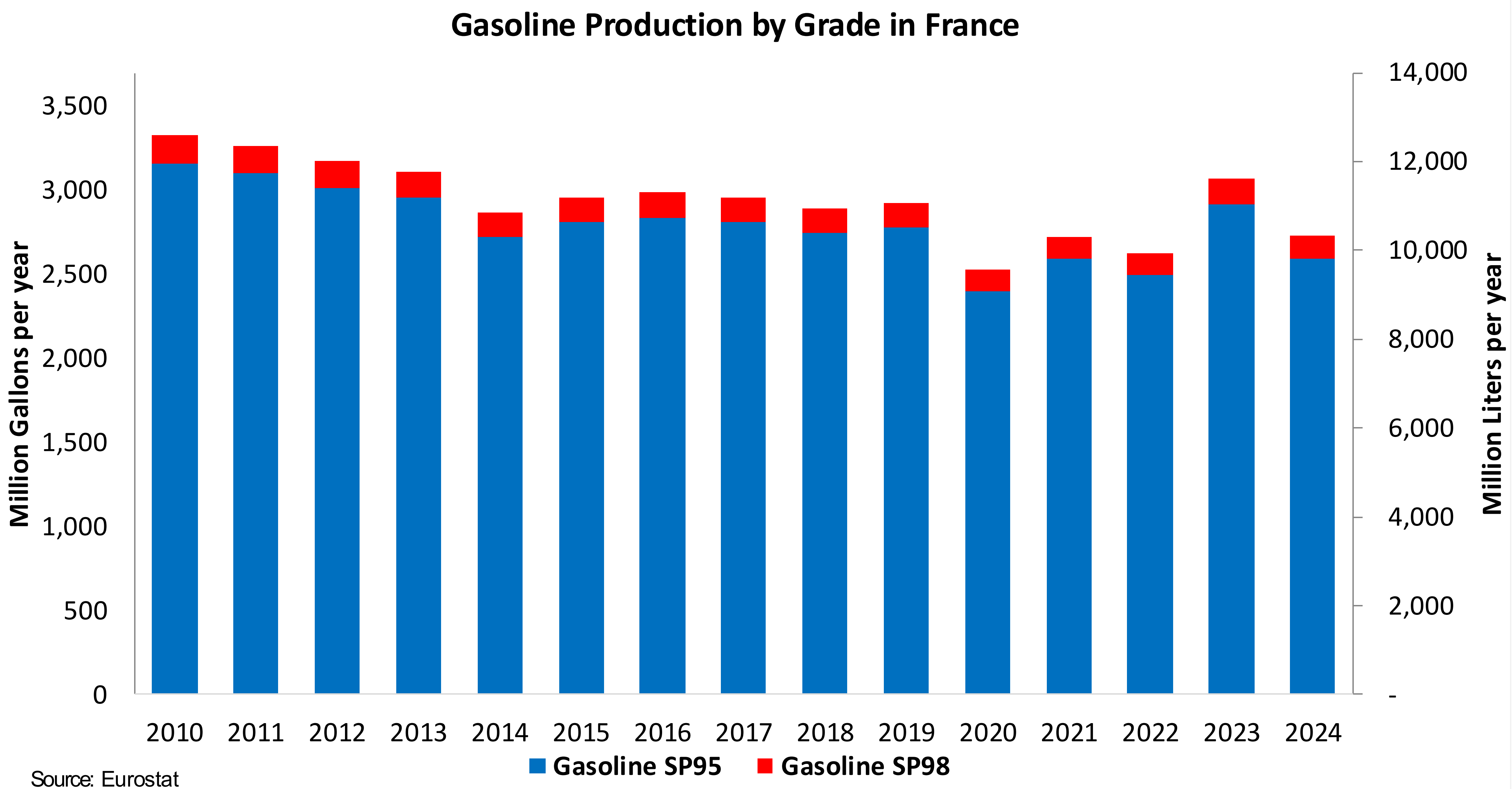
Source: Eurostat, European Commision

Gasoline Demand in France



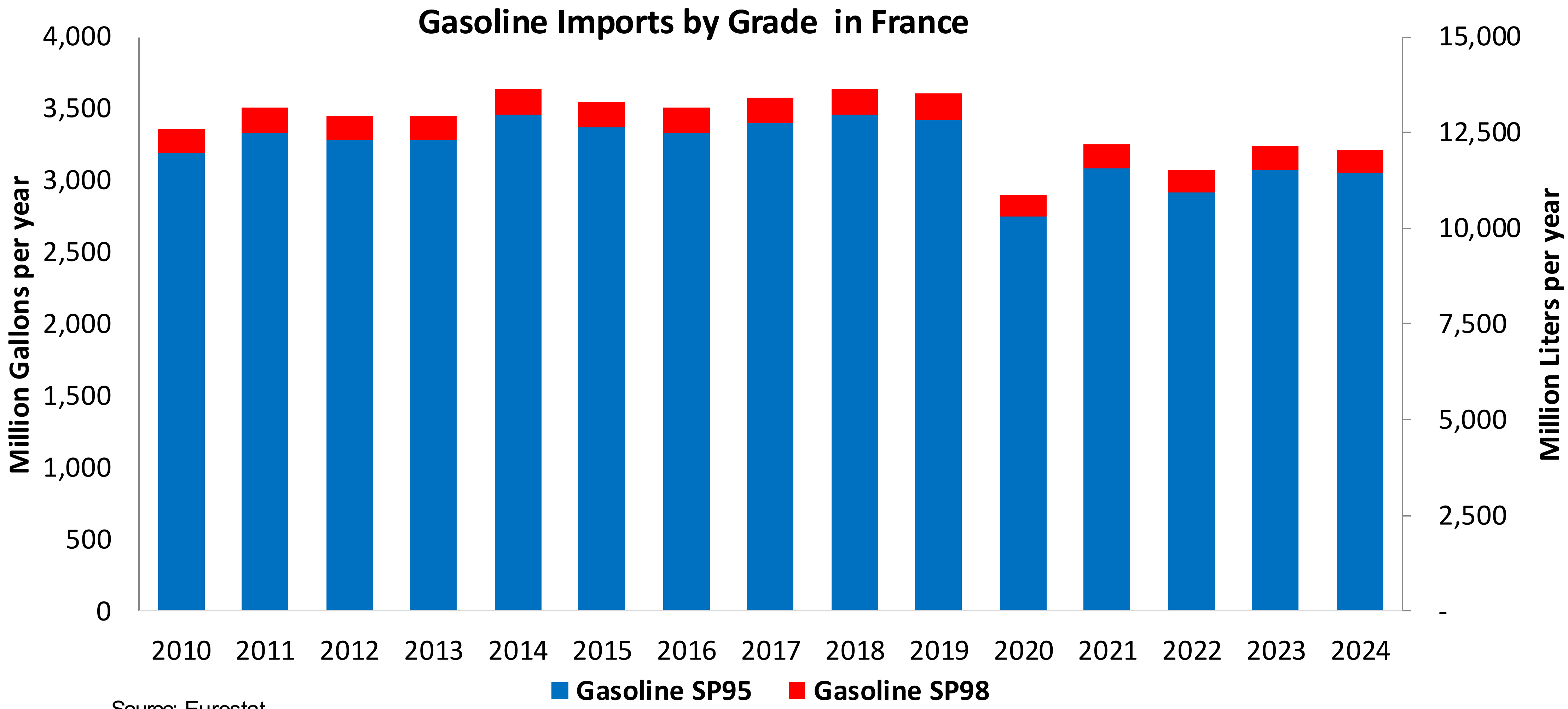
Source: Eurostat

Gasoline Production in France



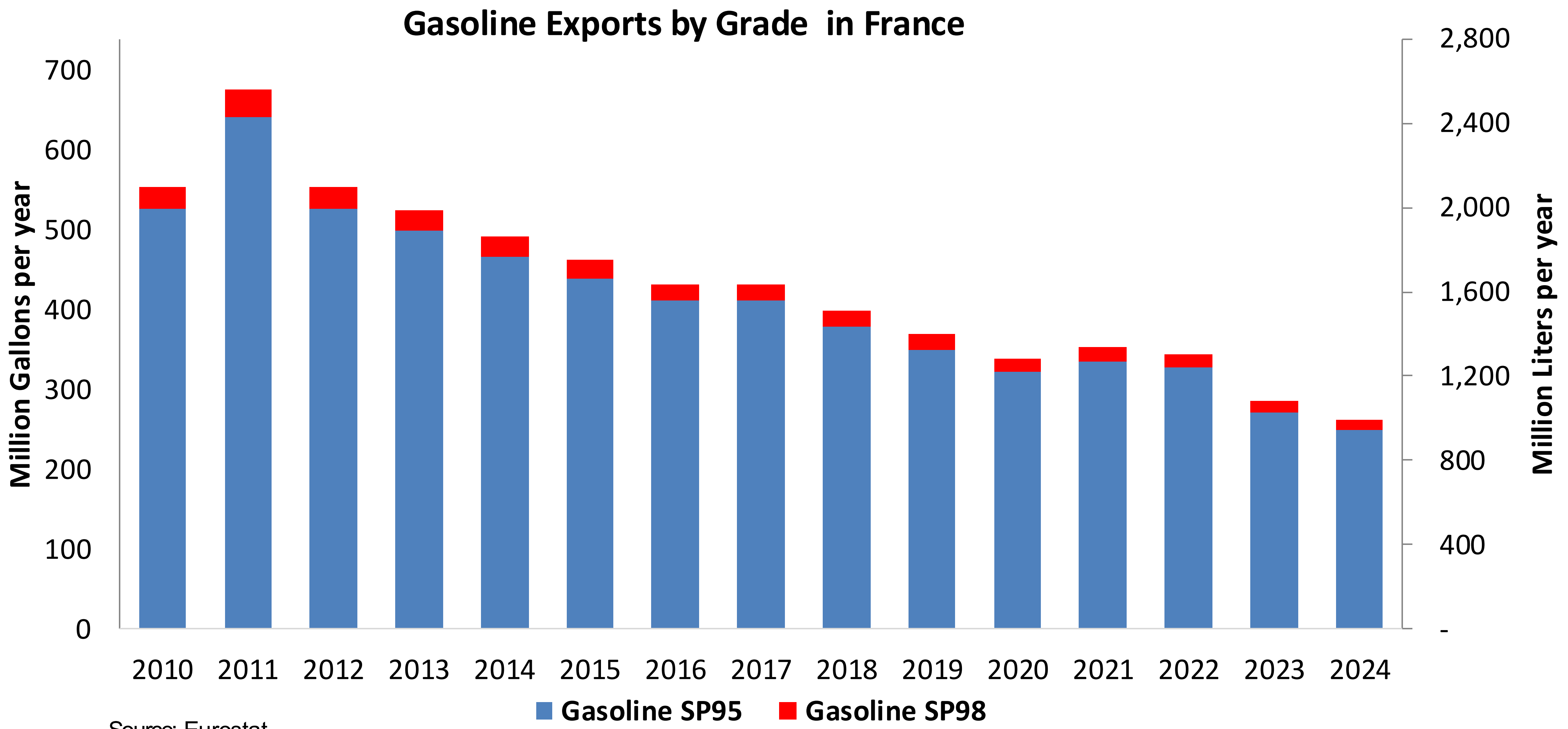
Source: Eurostat

Gasoline Imports in France



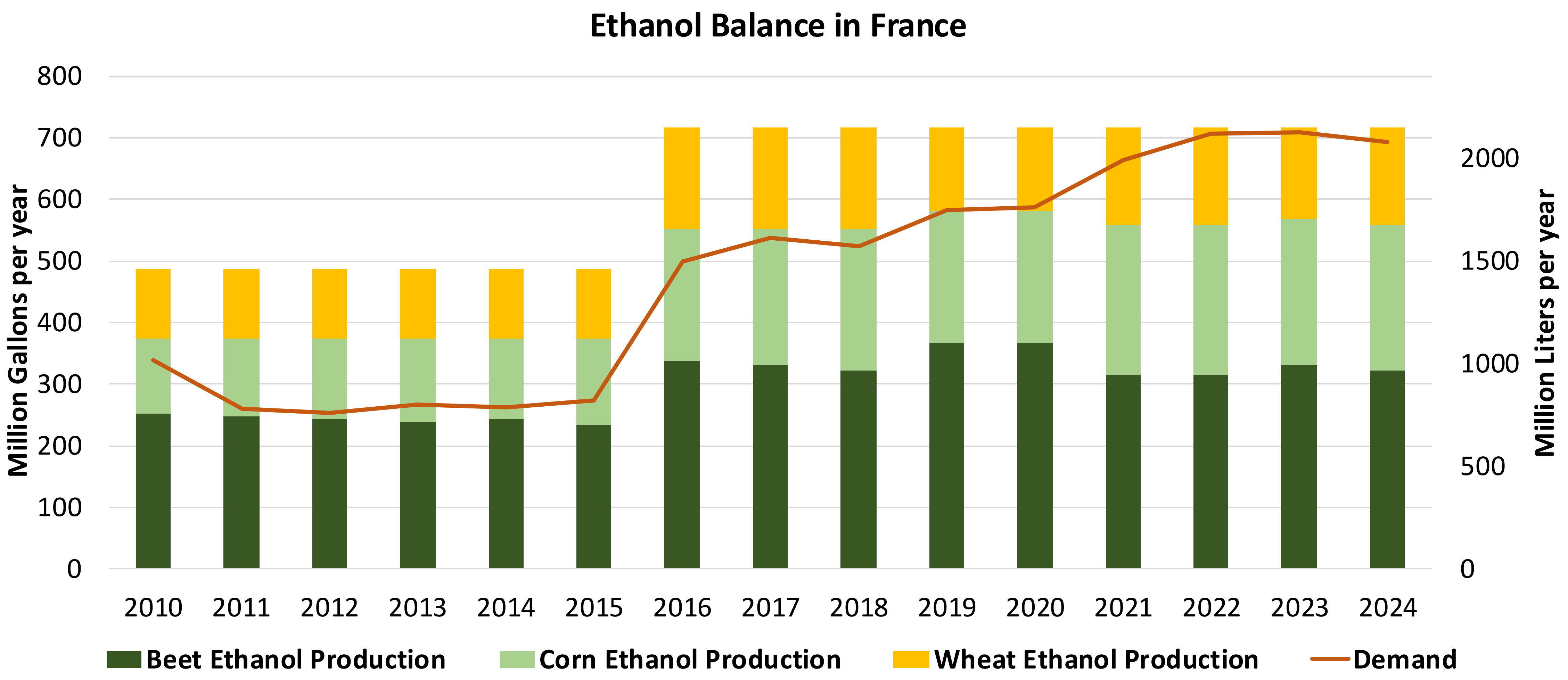
Source: Eurostat

Gasoline Exports in France



Source: Eurostat

Ethanol Balance in France



Source: Eurostat

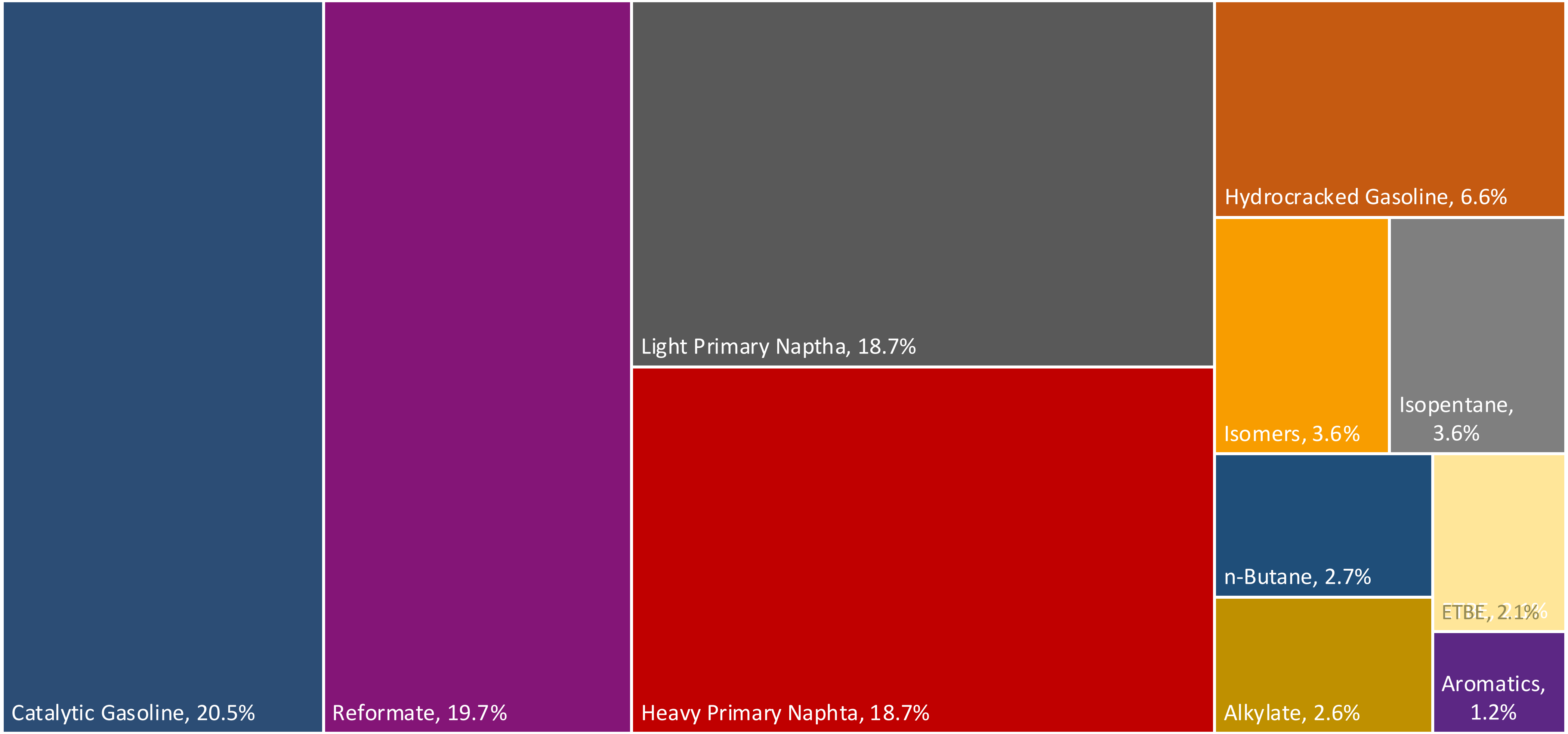
Gasoline Quality in France

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.2							
Bioethanol in Gasoline Sales (%cal)	8.6							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

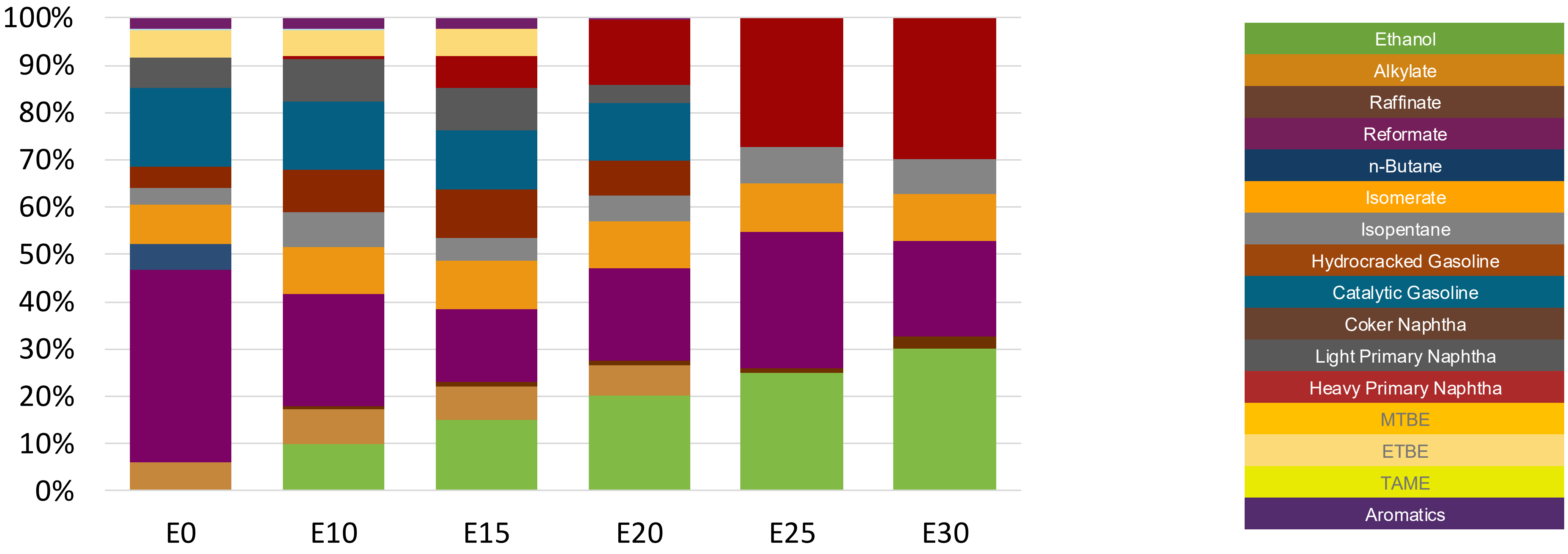
Source: European Commision

France Available Blending Components



Source: Faro90

France – Gasoline 95 – Constant Octane

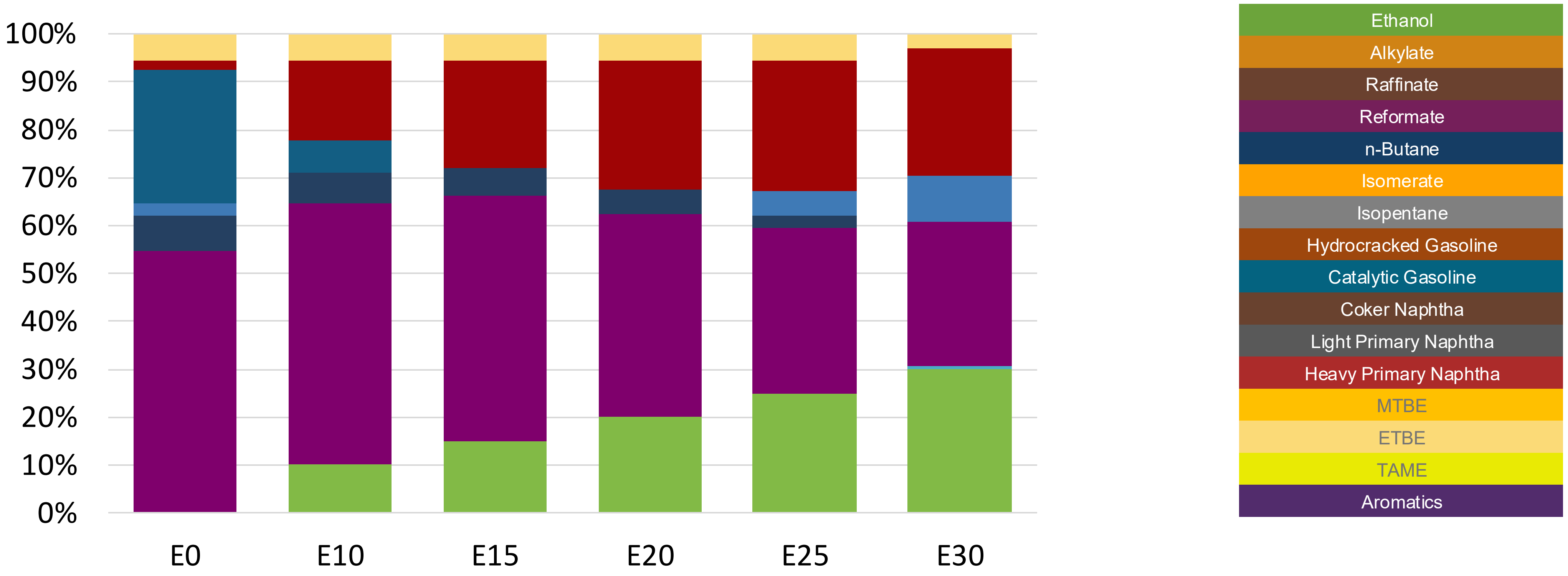


Octane (RON)	95.0	95.0	95.0	95.0	95.2	95.7
Price (US$/gal)	$ 2.268	$ 2.158	$ 2.079	$ 1.996	$ 1.882	$ 1.810

Average CIF 2024 Prices

Source: Faro90

France - Gasoline 98 - Constant Octane

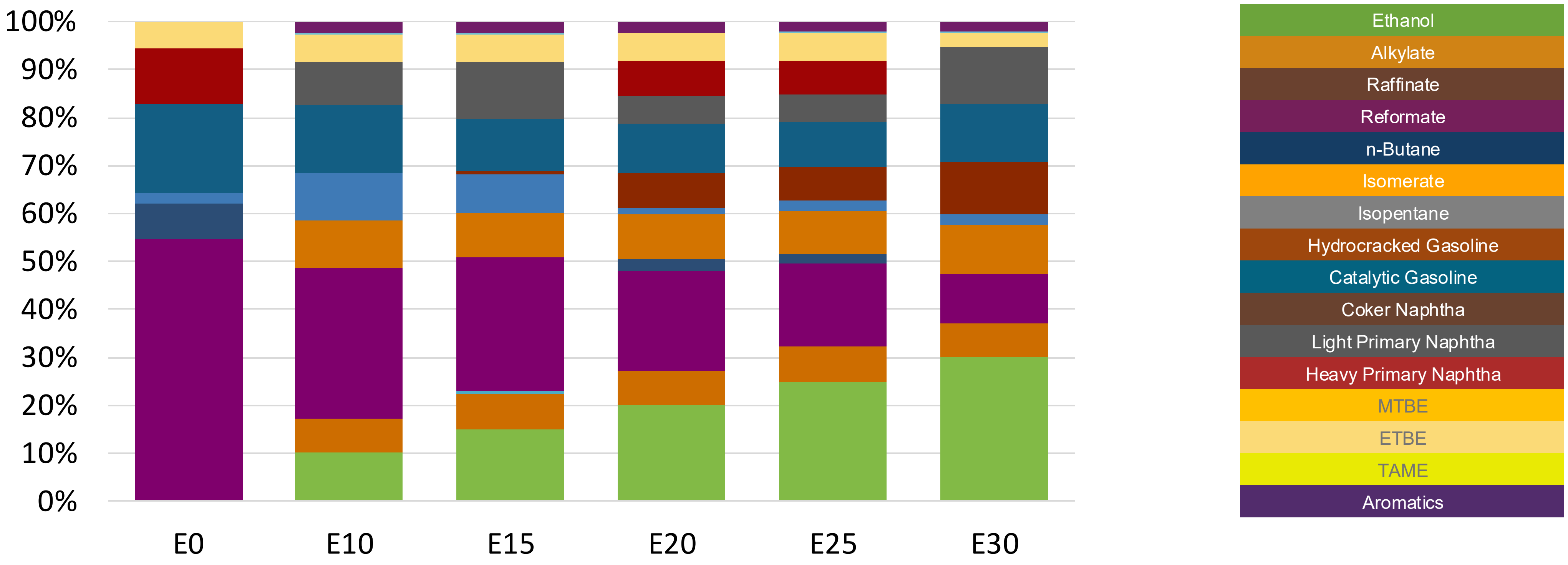


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.346	$ 2.222	$ 2.151	$ 2.076	$ 1.998	$ 1.923

Average CIF 2024 Prices

Source: Faro90

France – Gasoline 95 – Increased Octane

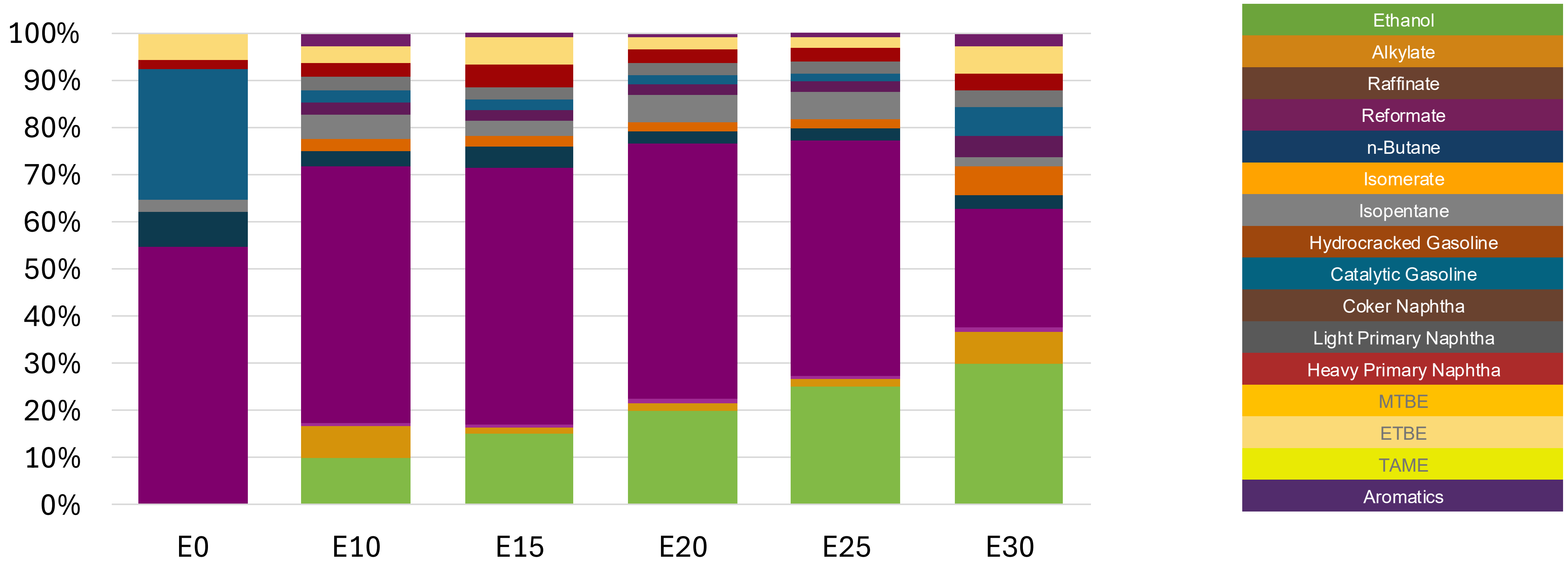


Octane (RON)	95.0	96.5	97.9	99.4	100.8	101.7
Price (US$/gal)	$ 2.212	$ 2.212	$ 2.212	$ 2.212	$ 2.212	$ 2.212

Average CIF 2024 Prices

Source: Faro90

France – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.3	101.4	102.4	103.7	104.6
Price (US$/gal)	$ 2.346	$ 2.346	$ 2.346	$ 2.346	$ 2.346	$ 2.346

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - France

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	94,264
		Single FI	3-Ways		>8 years	153,505
		Multi FI	3-Ways		>8 years	2,060,781
			Euro III		>8 years	5,256,949
			Euro IV		>8 years	8,656,854
			Euro V		>8 years	2,367,571
			Euro 6		>8 years	8,242,962
					4-8 years	3,239,948
					<4 years	1,412,932
			Hybrid		All	1,072,977
		Electric			All	1,848,366
	Medium	Carburator	3-Ways	PCV Tank	>8 years	9,323
		Single FI	3-Ways		>8 years	4,554
		Multi FI	3-Ways		>8 years	308,227
			Euro III		>8 years	998,432
			Euro IV		>8 years	3,021,161
			Euro V		>8 years	723,620
			Euro 6		>8 years	2,747,654
					4-8 years	1,450,623
					<4 years	1,056,596
			Hybrid		All	891,093
		Electric			All	1,544,140
Sml Engines	Light	4-Cycle	None	PCV	>8 years	26,320
			Euro III		>8 years	29,544
			Euro IV		>8 years	186,487
			Euro V		>8 years	66,231
			Euro 6		>8 years	150,474
					4-8 years	189,826
					<4 years	428,210
			Hybrid		All	50,602
		Electric			All	87,405
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	19,060
		Fuel-Injection	3-Ways		>8 years	215,409
			Euro III		>8 years	120,809
			Euro IV		>8 years	139,261
			Euro V		>8 years	38,984
			Euro 6		>8 years	53,820
					4-8 years	25,374
					<4 years	3,297
			Hybrid		All	732
		Electric			All	2,005

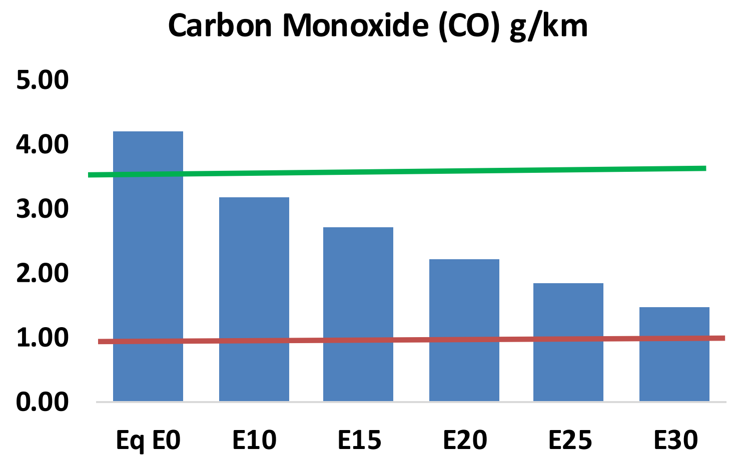
0 1,000,000 2,000,000 3,000,000 4,000,000 5,000,000 6,000,000 7,000,000 8,000,000 9,000,000 10,000,000

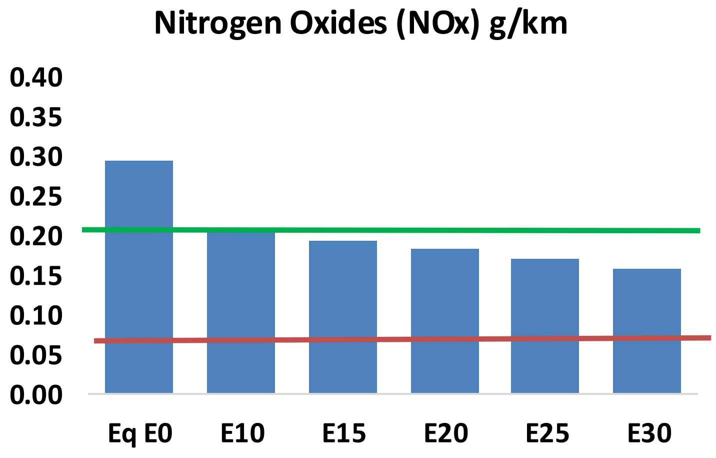
Vehicle Fleet: 48,964,974
Gasoline Vehicle Fleet: 24,096,952

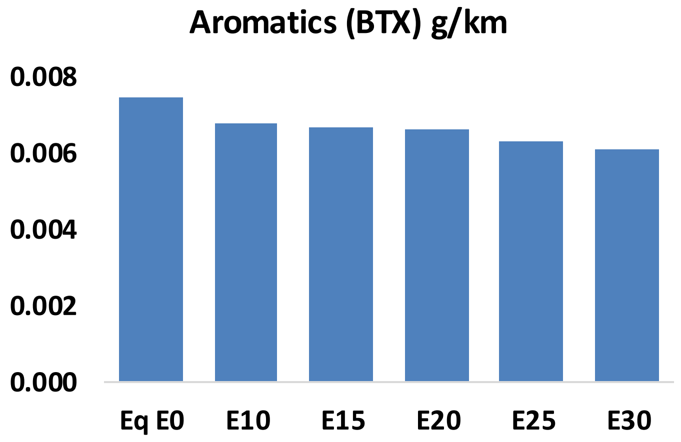
Source: Eurostat, ACEA

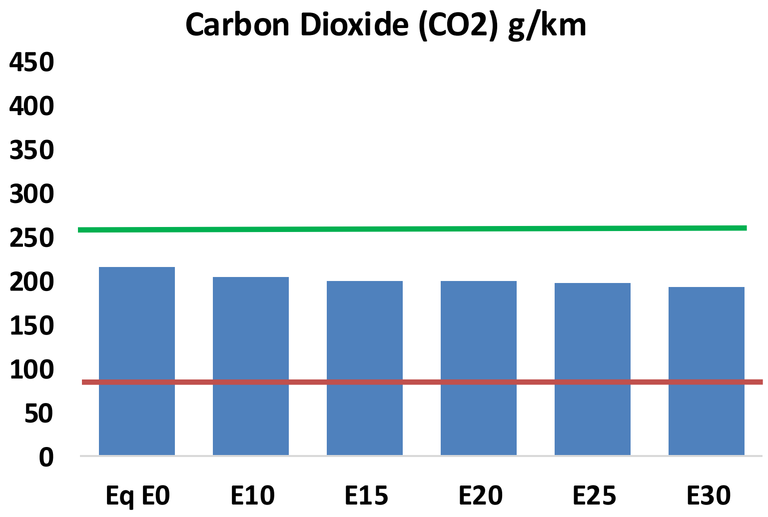
France – Vehicular Emissions

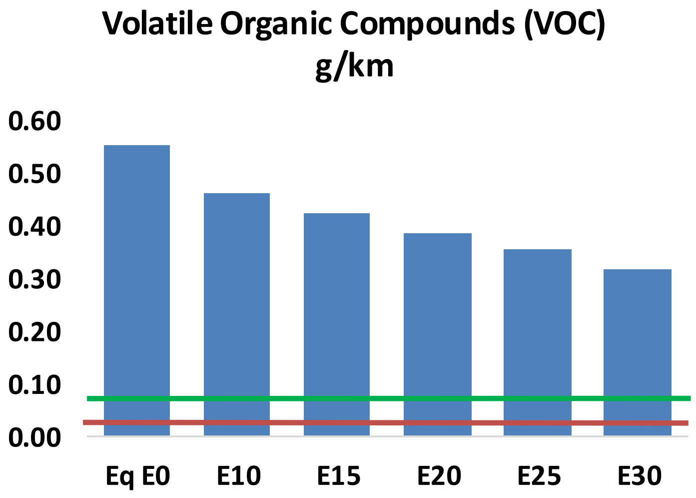
TIER III BIN 125 United States
Euro 6 Standard

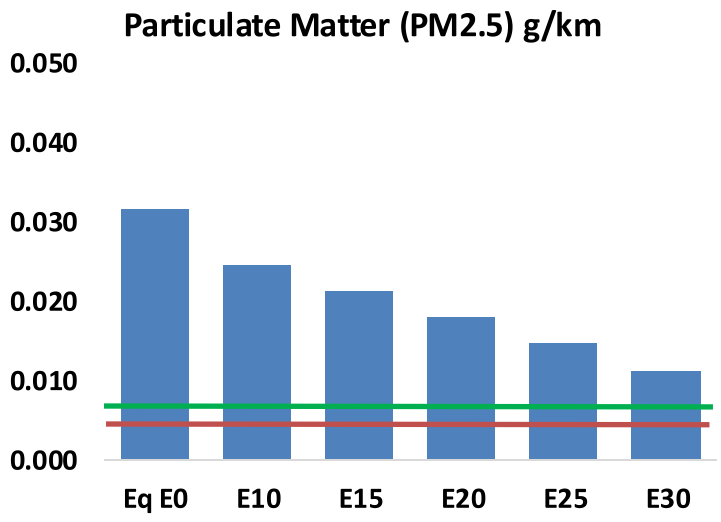












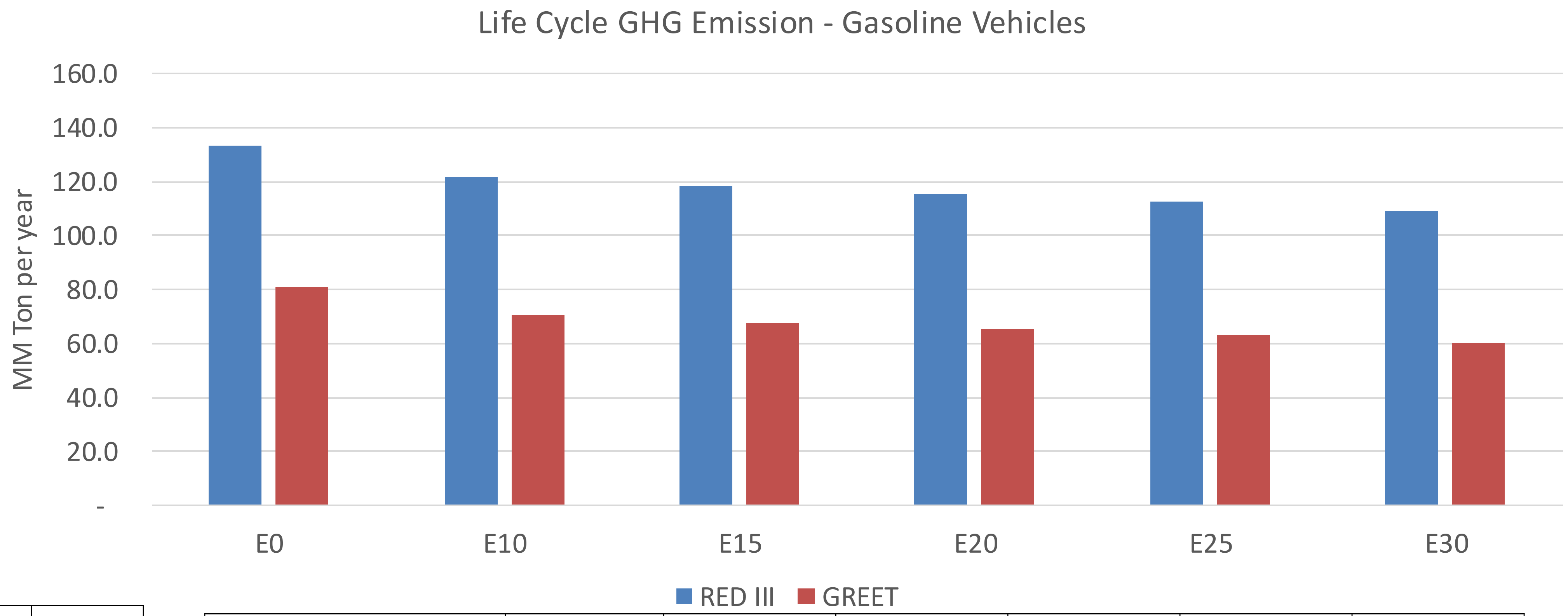
Source: Faro90

France – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,605,439	2,284,795	1,964,150	1,677,912	1,379,054	1,145,988	912,581	-25%	-47%
CO2	133,470,121	130,038,032	126,796,615	124,247,335	122,985,852	122,244,413	119,991,097	-5%	-8%
VOC	343,295	314,878	286,460	262,778	238,764	220,244	195,723	-17%	-30%
NOx	182,003	136,502	127,402	120,076	113,242	105,707	97,736	-30%	-38%
SOx	1,536	1,420	1,304	1,206	1,112	1,010	902	-15%	-28%
NH3	47,099	46,633	46,166	46,385	46,907	47,274	47,579	-2%	0%
N2O	1,983	1,972	1,964	1,974	2,014	2,037	2,060	-1%	2%
CH4	70,260	70,260	70,260	71,314	72,859	74,194	75,459	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,334	2,199	2,065	1,963	1,864	1,790	1,686	-12%	-20%
Acetaldehyde	5,934	6,810	9,993	15,218	20,733	23,691	27,994	68%	249%
Formaldehyde	20,829	20,251	23,606	27,719	29,040	32,126	34,973	13%	39%
Benzene	4,604	4,424	4,192	4,126	4,092	3,913	3,787	-9%	-11%
PM2.5	8,142	7,229	6,317	5,486	4,658	3,780	2,864	-22%	-43%
PM10	11,339	10,069	8,798	7,640	6,488	5,265	3,989	-22%	-43%

Source: Faro90

France – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	331.1
Target @2035 (MMT/year)	186.4
Delta	144.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.9%	16.4%	19.2%	22.0%	25.6%
RED II/III	0.0%	8.8%	11.4%	13.6%	15.6%	18.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.2%	9.2%	10.8%	12.3%	14.3%
RED II/III	0.0%	8.1%	10.5%	12.5%	14.4%	16.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

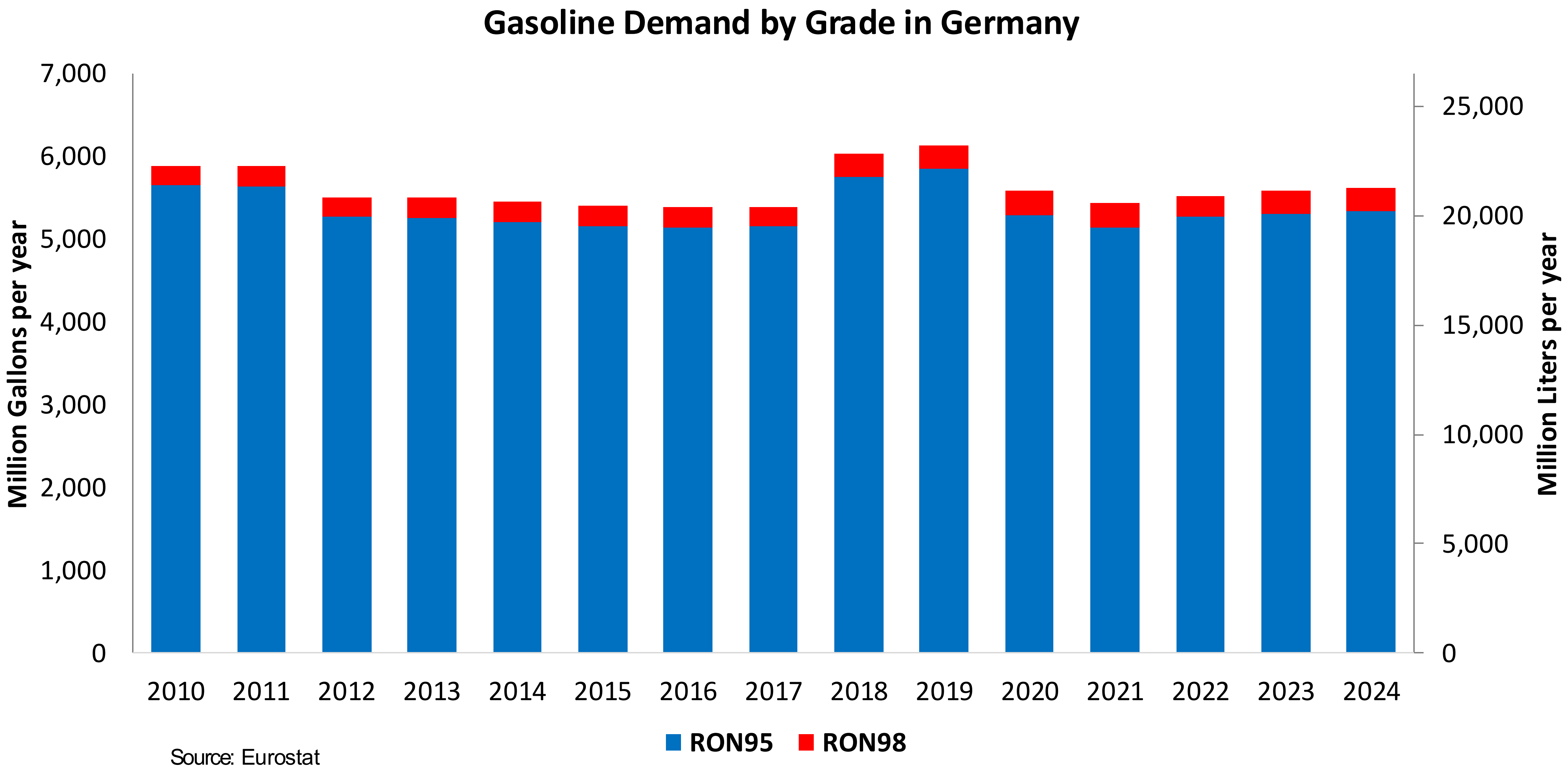
Ethanol Gasoline Blending in Germany

In 2024, gasoline consumption in Germany was 21,292 million liters (5,625 million gallons) and growth rate was -1.7%. Gasoline production and net imports were 22,284 and -3,611 million liters (5,887 and -954 million gallons) correspondingly. Germany complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Germany was 2,262 million liters (598 million gallons) representing 10.6% of gasoline demand. Ethanol production and Net Imports were 1,333 and 929 million liters (598 and 245 million gallons) correspondingly. Ethanol sources are Cereals 64% and Sugar Beet 33%.

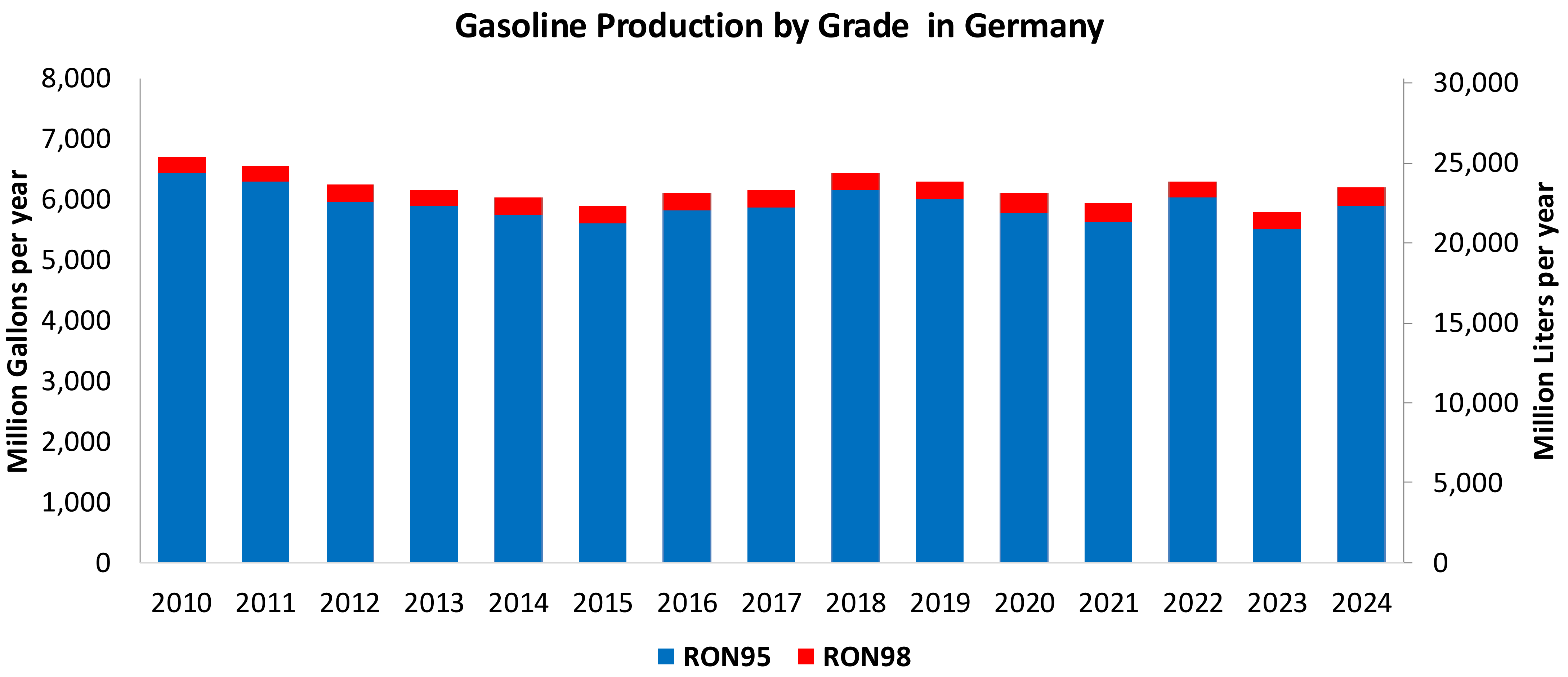
Source: Eurostat, European Commision

Gasoline Demand in Germany



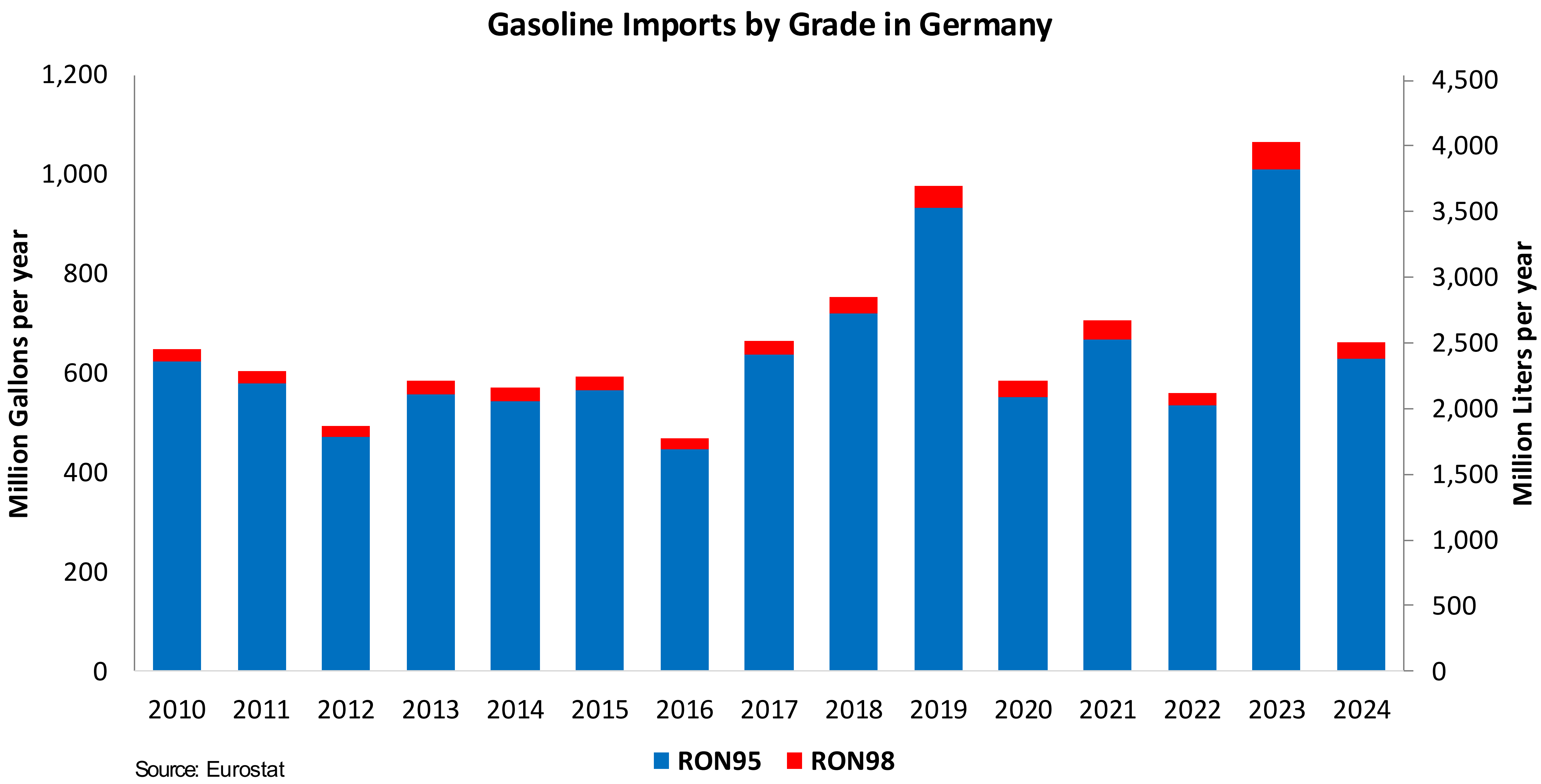
Source: Eurostat

Gasoline Production in Germany



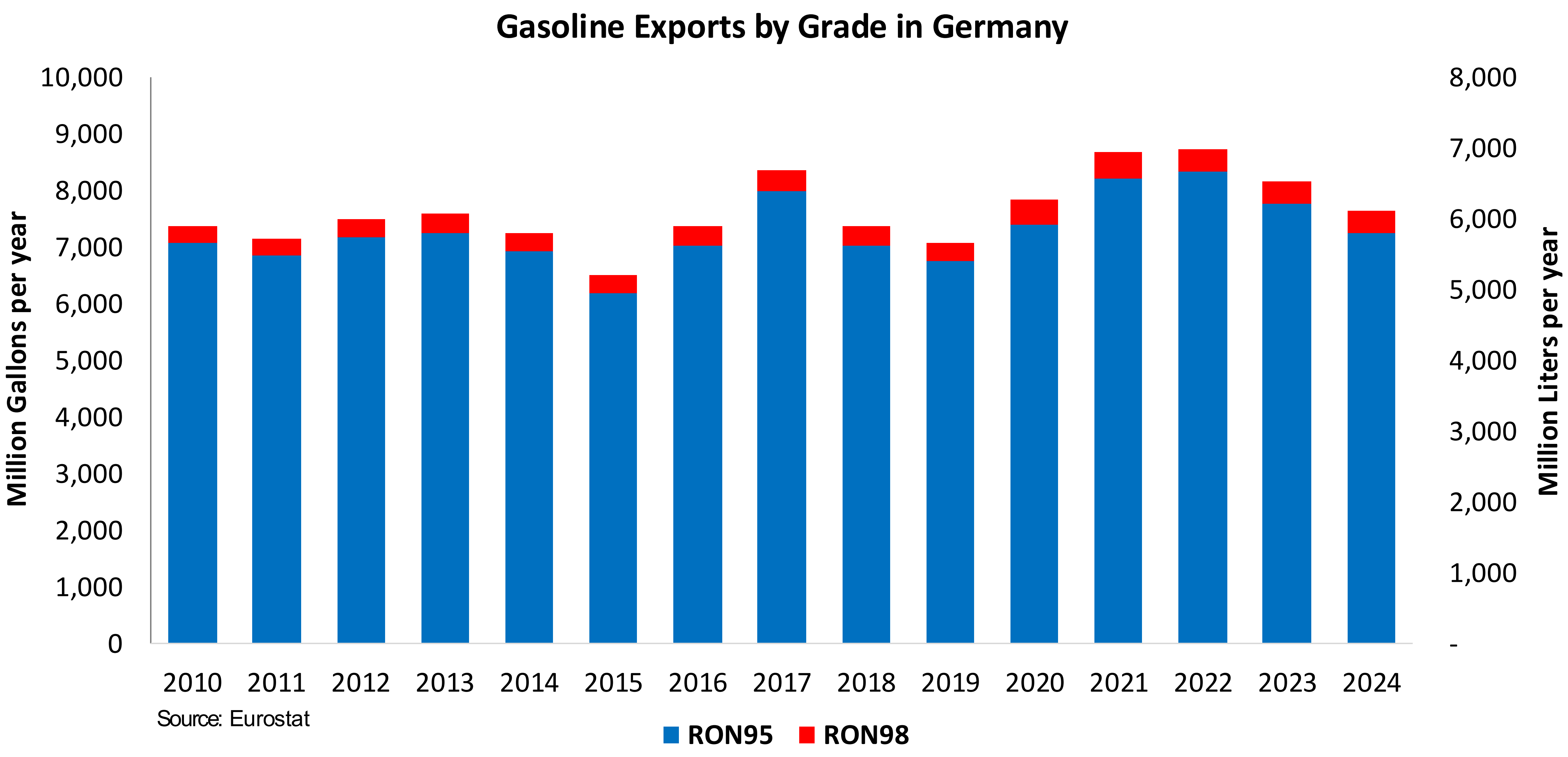
Source: Eurostat

Gasoline Imports in Germany



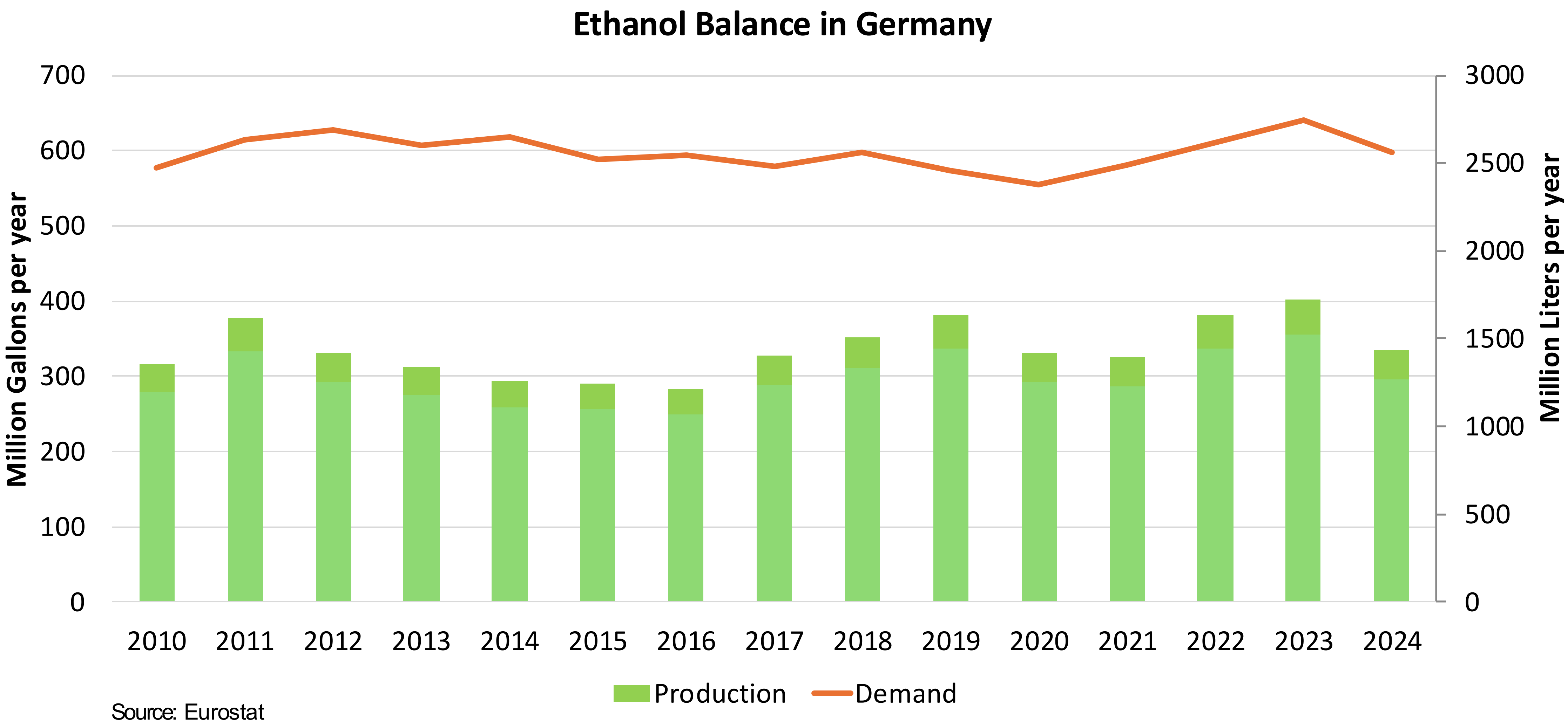
Source: Eurostat

Gasoline Exports in Germany



Source: Eurostat

Ethanol Balance in Germany



Source: Eurostat

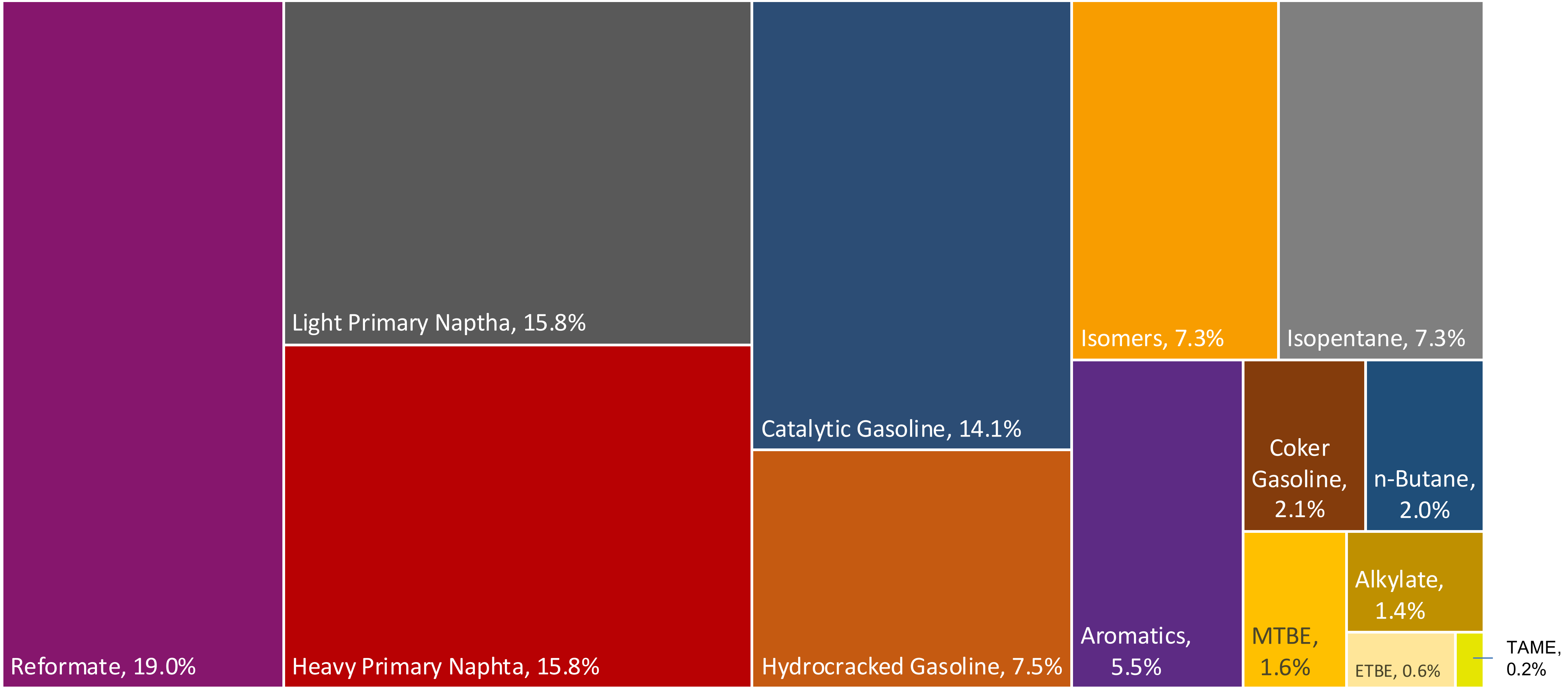
Gasoline Quality in Germany

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.7							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	10.5							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

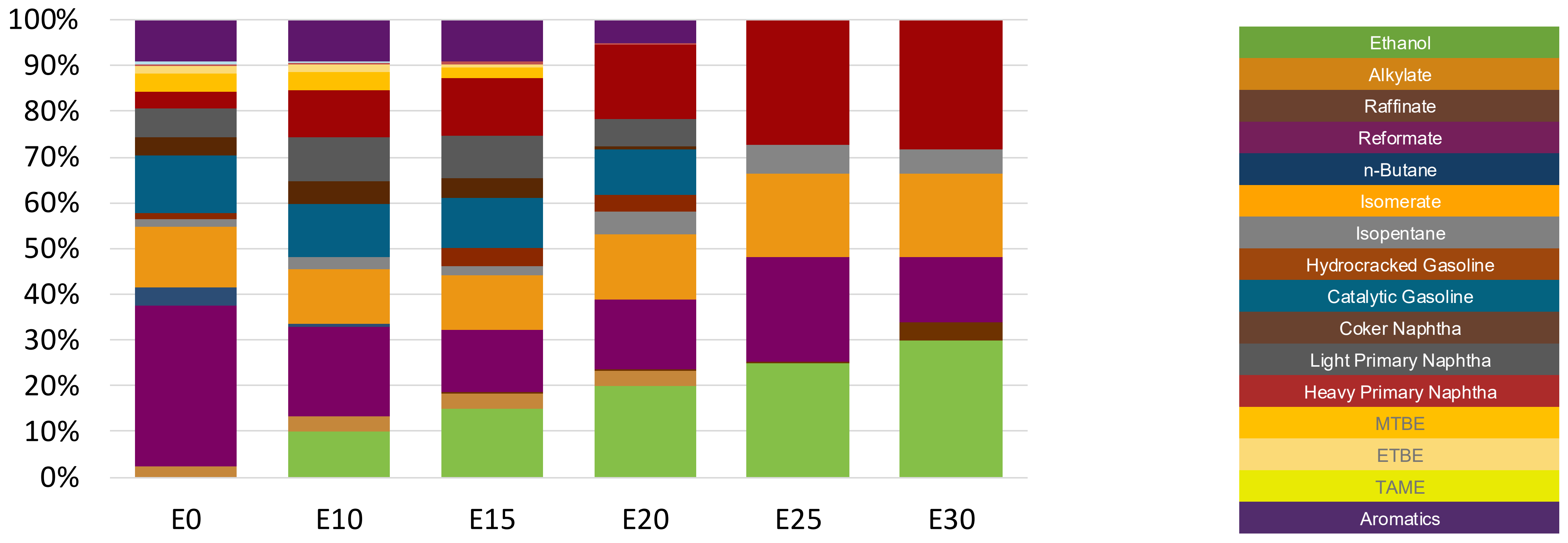
Source: European Commision

Germany Available Blending Components



Source: Faro90

Germany – Gasoline 95 – Constant Octane

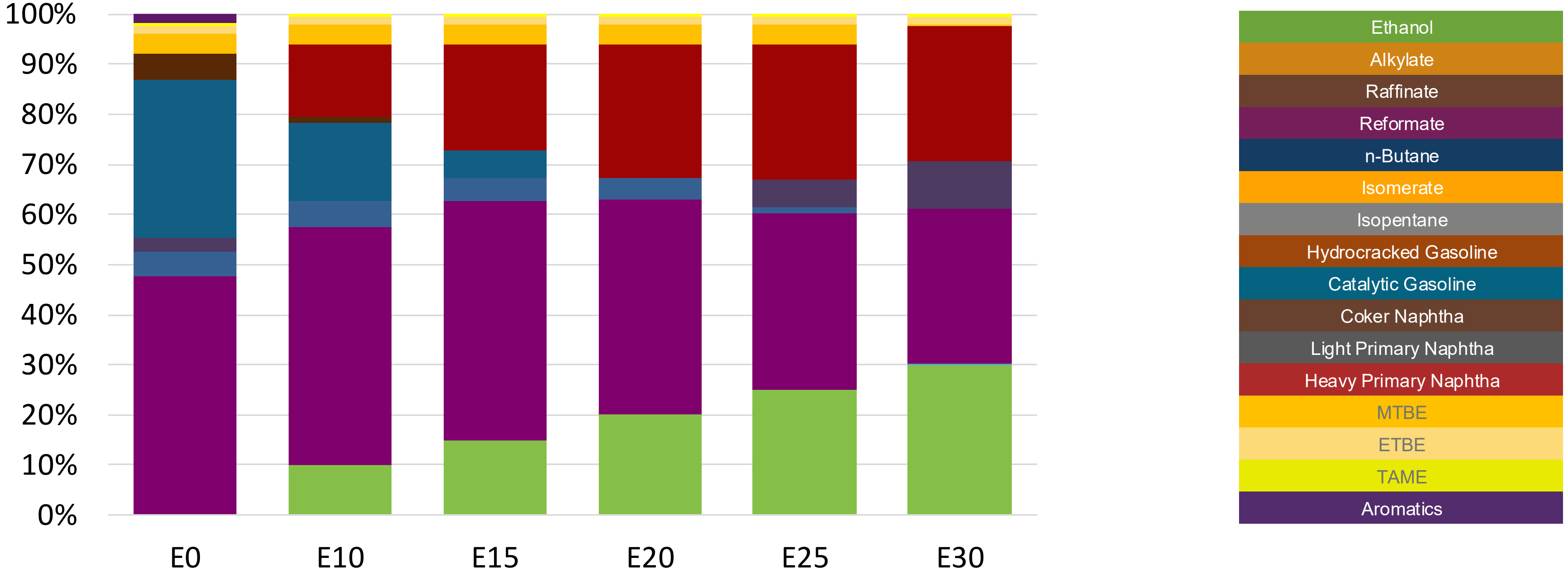


Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US$/gal)	$ 2.272	$ 2.144	$ 2.083	$ 2.002	$ 1.880	$ 1.814

Average CIF 2024 Prices

Source: Faro90

Germany - Gasoline 98 - Constant Octane

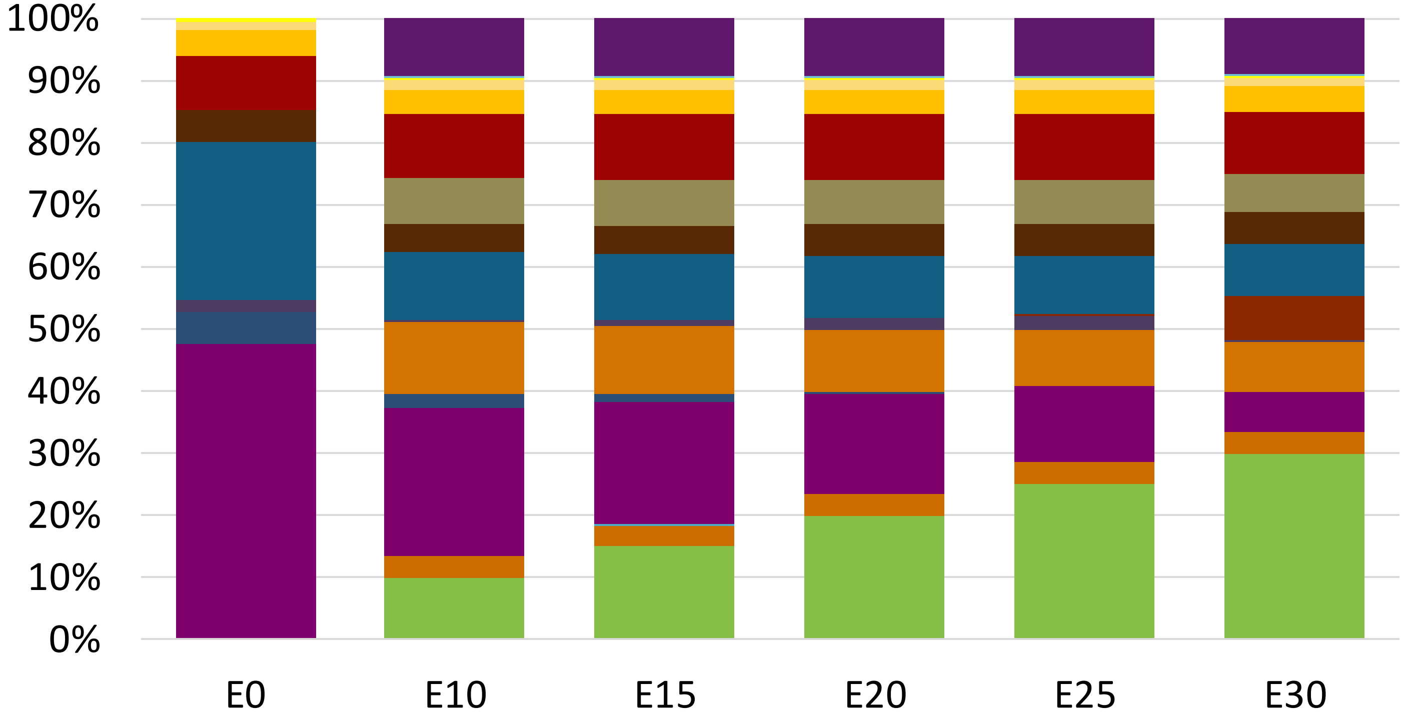


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.347	$ 2.218	$ 2.146	$ 2.069	$ 1.992	$ 1.925

Average CIF 2024 Prices

Source: Faro90

Germany – Gasoline 95 – Increased Octane

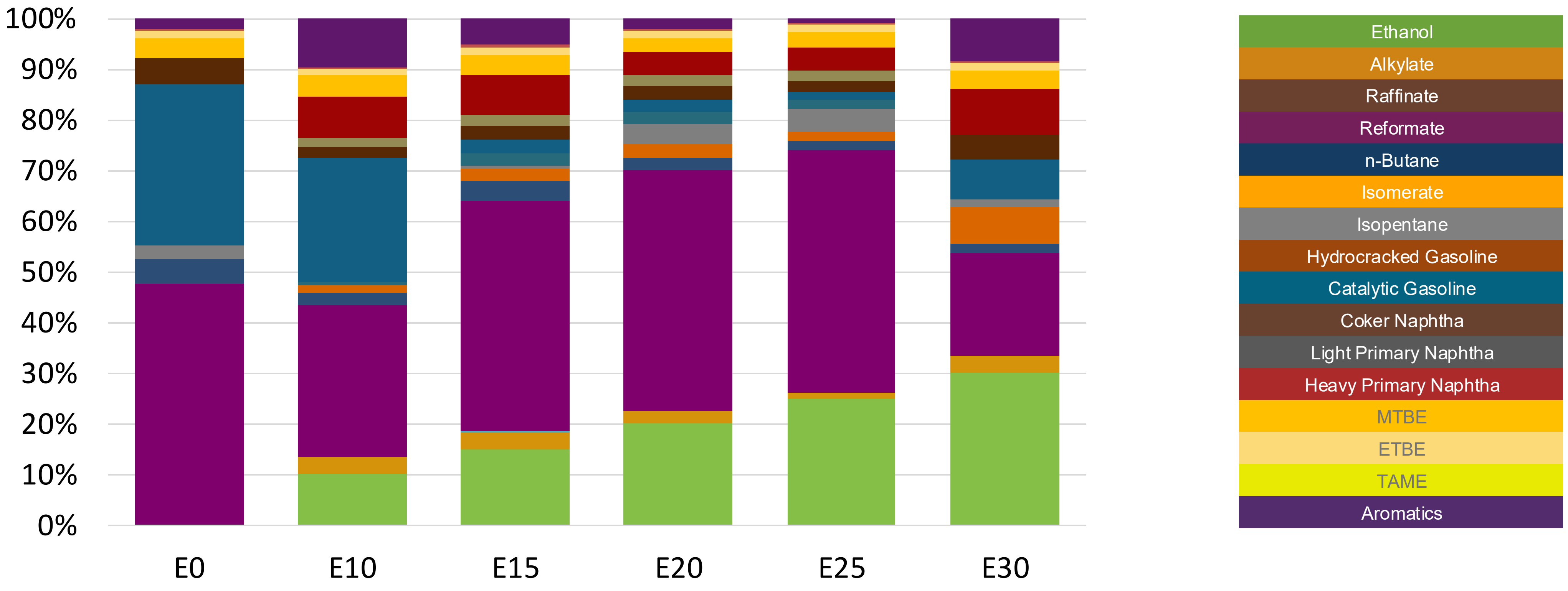


Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US$/gal)	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209

Average CIF 2024 Prices

Source: Faro90

Germany – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US$/gal)	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Germany

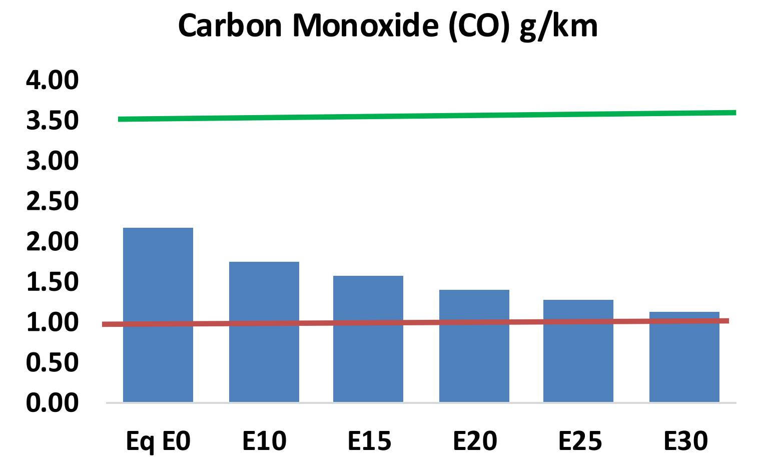
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	19,353
		Single FI	3-Ways		>8 years	174,176
		Multi FI	3-Ways		>8 years	1,655,439
			Euro III		>8 years	1,603,283
			Euro IV		>8 years	5,474,789
			Euro V		>8 years	2,555,744
			Euro 6		>8 years	4,974,662
					4-8 years	10,980,947
					<4 years	16,791,954
			Hybrid		All	2,100,972
		Electric			All	814,320
	Medium	Carburator	3-Ways	PCV Tank	>8 years	1,163
		Single FI	3-Ways		>8 years	10,469
		Multi FI	3-Ways		>8 years	86,505
			Euro III		>8 years	21,876
			Euro IV		>8 years	36,990
			Euro V		>8 years	13,890
			Euro 6		>8 years	42,183
					4-8 years	67,255
					<4 years	106,528
			Hybrid		All	1,757
		Electric			All	2,010
Sml Engines	Light	4-Cycle	None	PCV	>8 years	1,256,474
			Euro III		>8 years	604,481
			Euro IV		>8 years	806,968
			Euro V		>8 years	263,309
			Euro 6		>8 years	482,272
					4-8 years	616,089
					<4 years	1,061,737
			Hybrid		All	1,010
		Electric			All	40,510
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	40,285
		Fuel-Injection	3-Ways		>8 years	3,198,889
			Euro III		>8 years	268,027
			Euro IV		>8 years	494,603
			Euro V		>8 years	207,082
			Euro 6		>8 years	828,974
					4-8 years	1,383,193
					<4 years	2,009,814
			Hybrid		All	788
		Electric			All	691

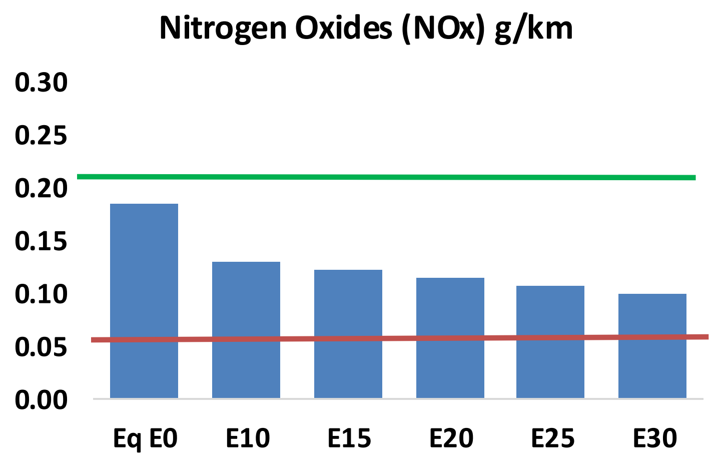
Vehicle Fleet: 61,101,463
Gasoline Vehicle Fleet: 32,331,793

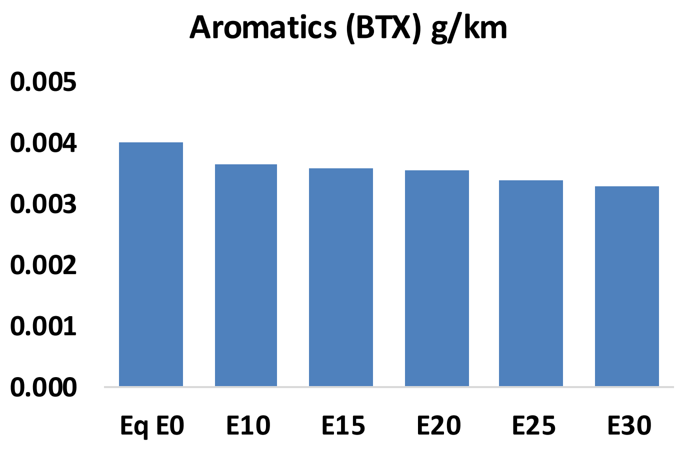
Source: Eurostat, ACEA

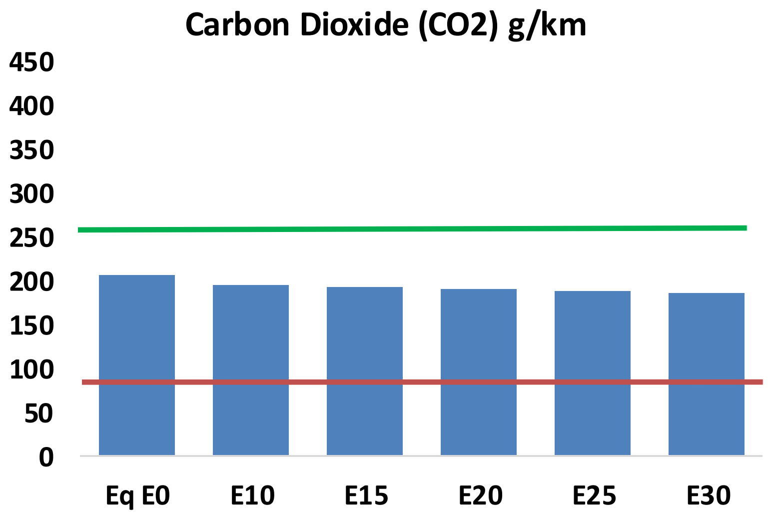
Germany – Vehicular Emissions

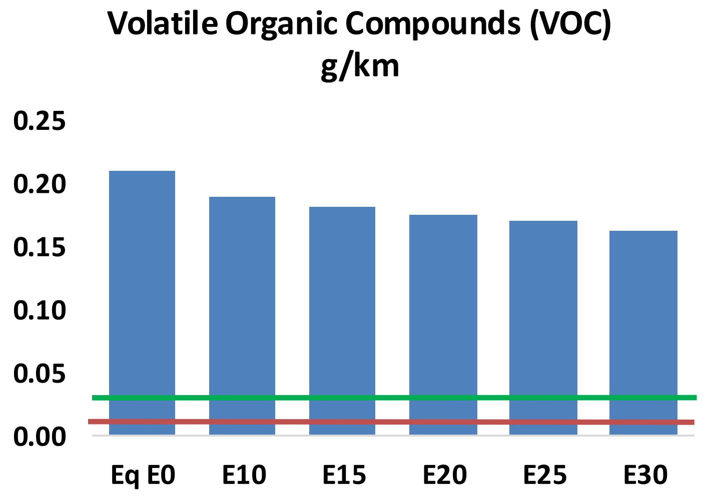
TIER III BIN 125 United States
Euro 6 Standard

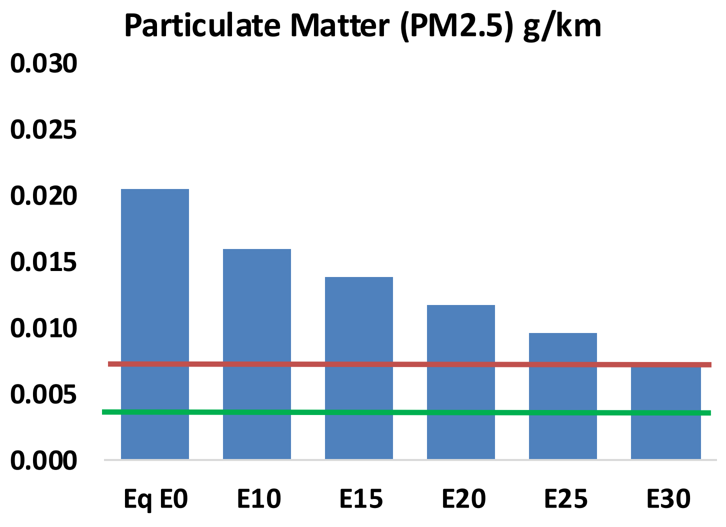












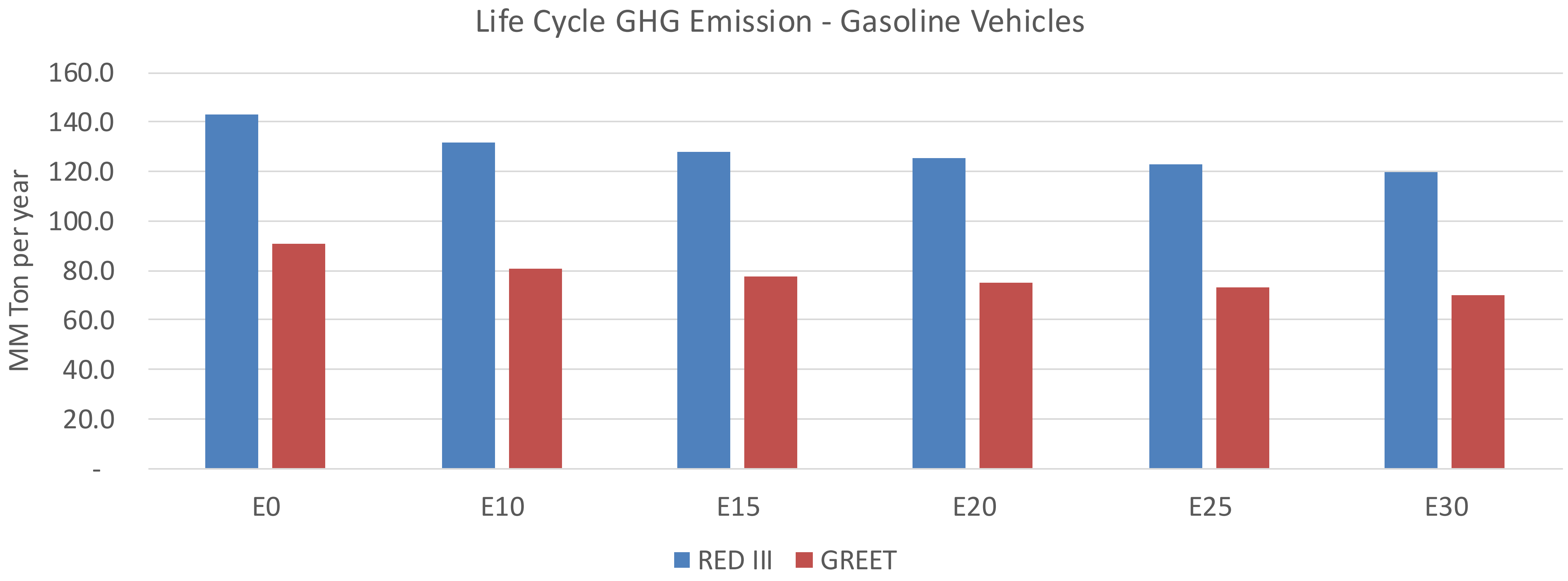
Source: Faro90

Germany – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,772,395	1,605,976	1,439,558	1,297,299	1,151,430	1,038,448	921,561	-19%	-35%
CO2	169,687,313	165,323,925	161,202,948	157,961,920	156,358,133	155,431,613	152,566,562	-5%	-8%
VOC	172,340	163,809	155,277	149,207	143,521	139,277	133,163	-10%	-17%
NOx	152,188	114,141	106,531	100,406	94,691	88,391	81,725	-30%	-38%
SOx	1,977	1,827	1,678	1,551	1,430	1,299	1,161	-15%	-28%
NH3	58,379	57,801	57,223	57,495	58,141	58,596	58,974	-2%	0%
N2O	1,820	1,810	1,803	1,811	1,849	1,869	1,891	-1%	2%
CH4	41,275	41,275	41,275	41,894	42,802	43,586	44,329	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,662	1,583	1,505	1,451	1,401	1,363	1,309	-9%	-16%
Acetaldehyde	3,372	3,870	5,679	8,648	11,783	13,464	15,909	68%	249%
Formaldehyde	10,885	10,583	12,337	14,486	15,176	16,789	18,277	13%	39%
Benzene	3,279	3,151	2,986	2,939	2,914	2,787	2,697	-9%	-11%
PM2.5	10,785	9,576	8,368	7,267	6,171	5,007	3,795	-22%	-43%
PM10	6,036	5,360	4,683	4,067	3,453	2,802	2,124	-22%	-43%

Source: Faro90

Germany – Gasoline Vehicle Fleet Life Cycle GHG Emissions



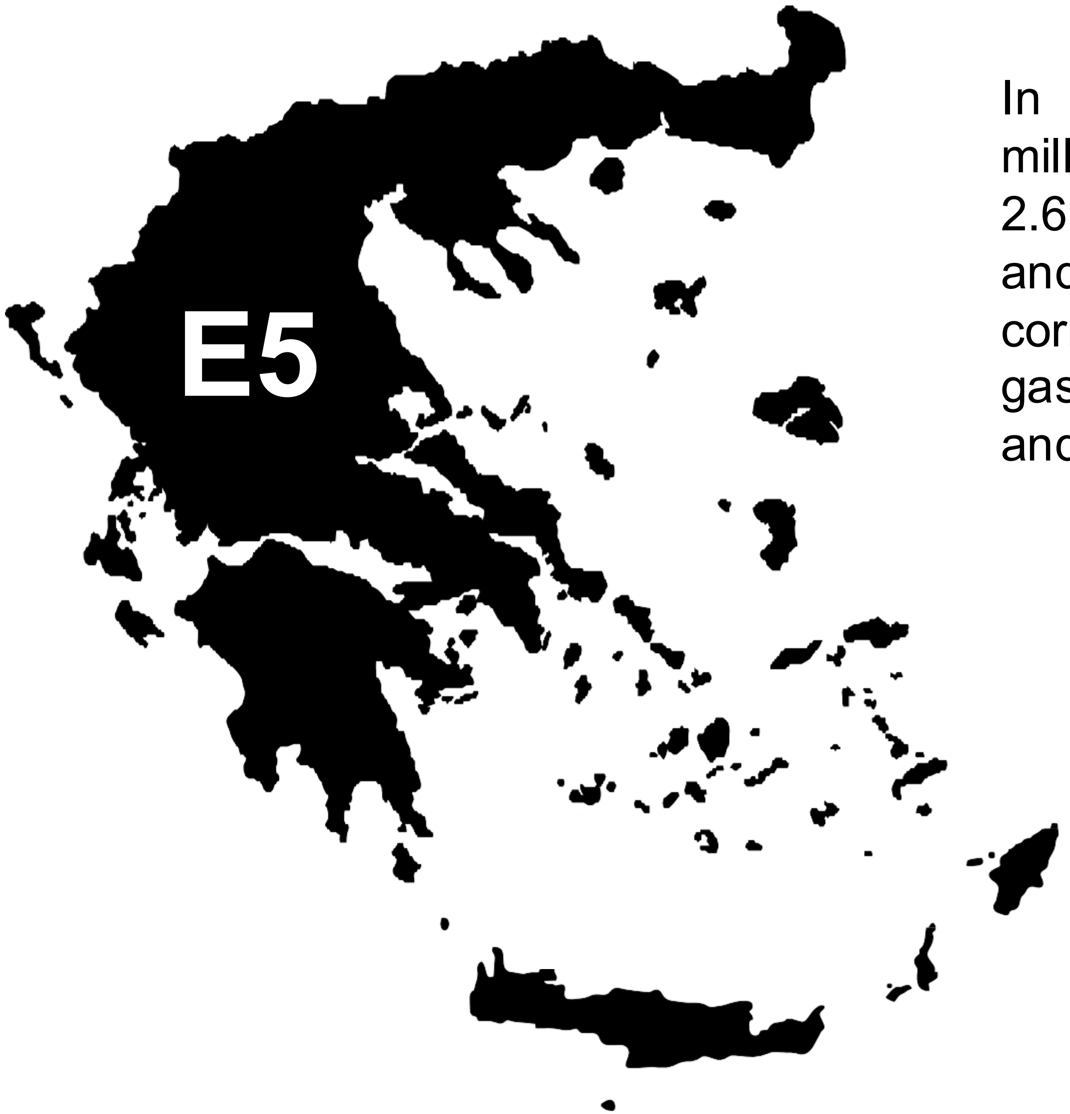
Country Emissions 2024 (MMT/año)	717.1
Target @2035 (MMT/year)	403.6
Delta	313.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.2%	14.4%	16.9%	19.3%	22.6%
RED II/III	0.0%	7.9%	10.3%	12.2%	14.1%	16.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.2%	4.2%	4.9%	5.6%	6.5%
RED II/III	0.0%	3.6%	4.7%	5.6%	6.4%	7.5%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Greece

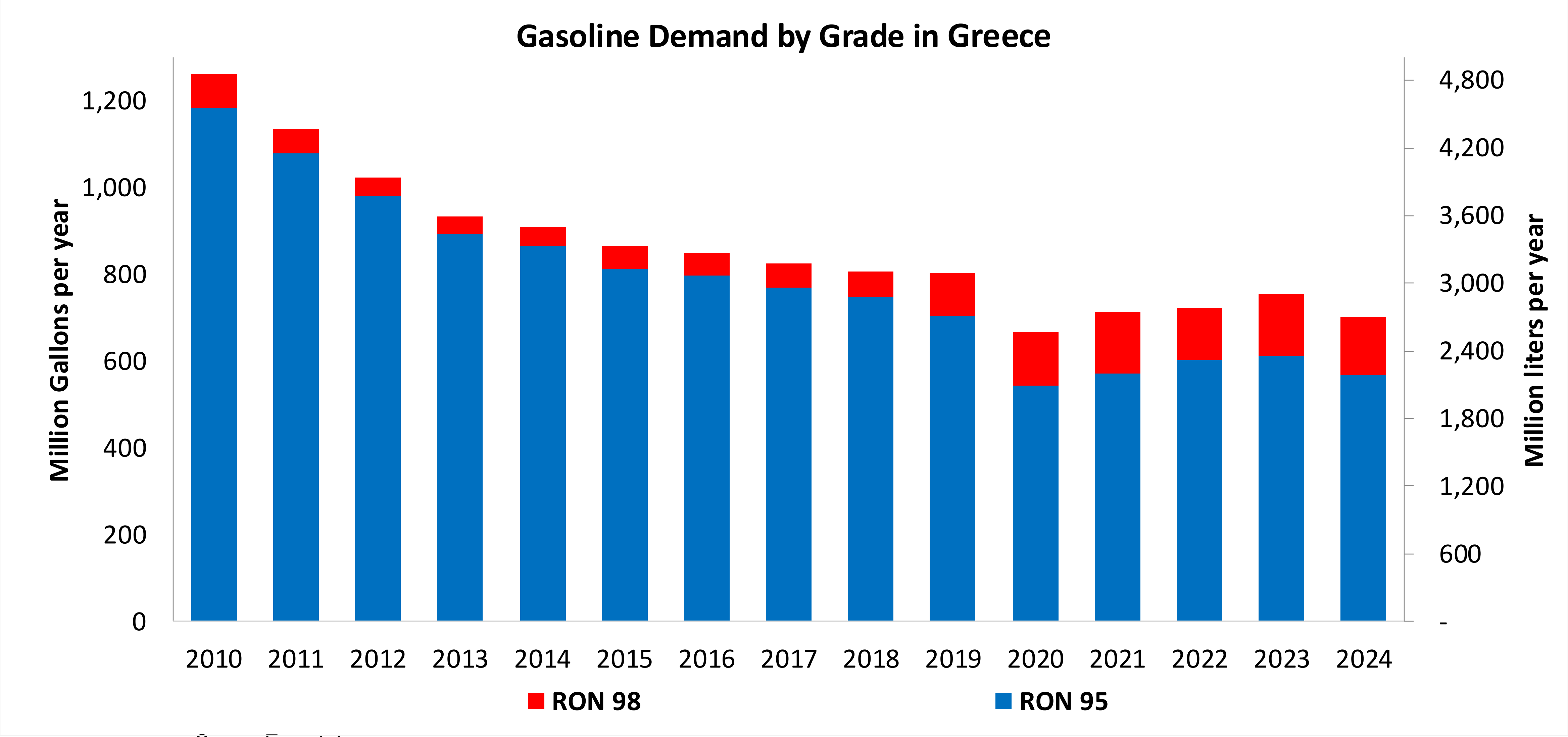


In 2024, gasoline consumption in Greece was 2,660 million liters (703 million gallons) and growth rate was -2.6%. Gasoline production and net imports were 2,811 and -151 million liters (743 and -40 million gallons) correspondingly. Greece complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

Ethanol consumption in Greece is determined by 3.3% bioethanol content mandate (by energy). Demand was 134 million liters (35 million gallons) representing 5.0% of gasoline demand. Ethanol is supplied by imports.

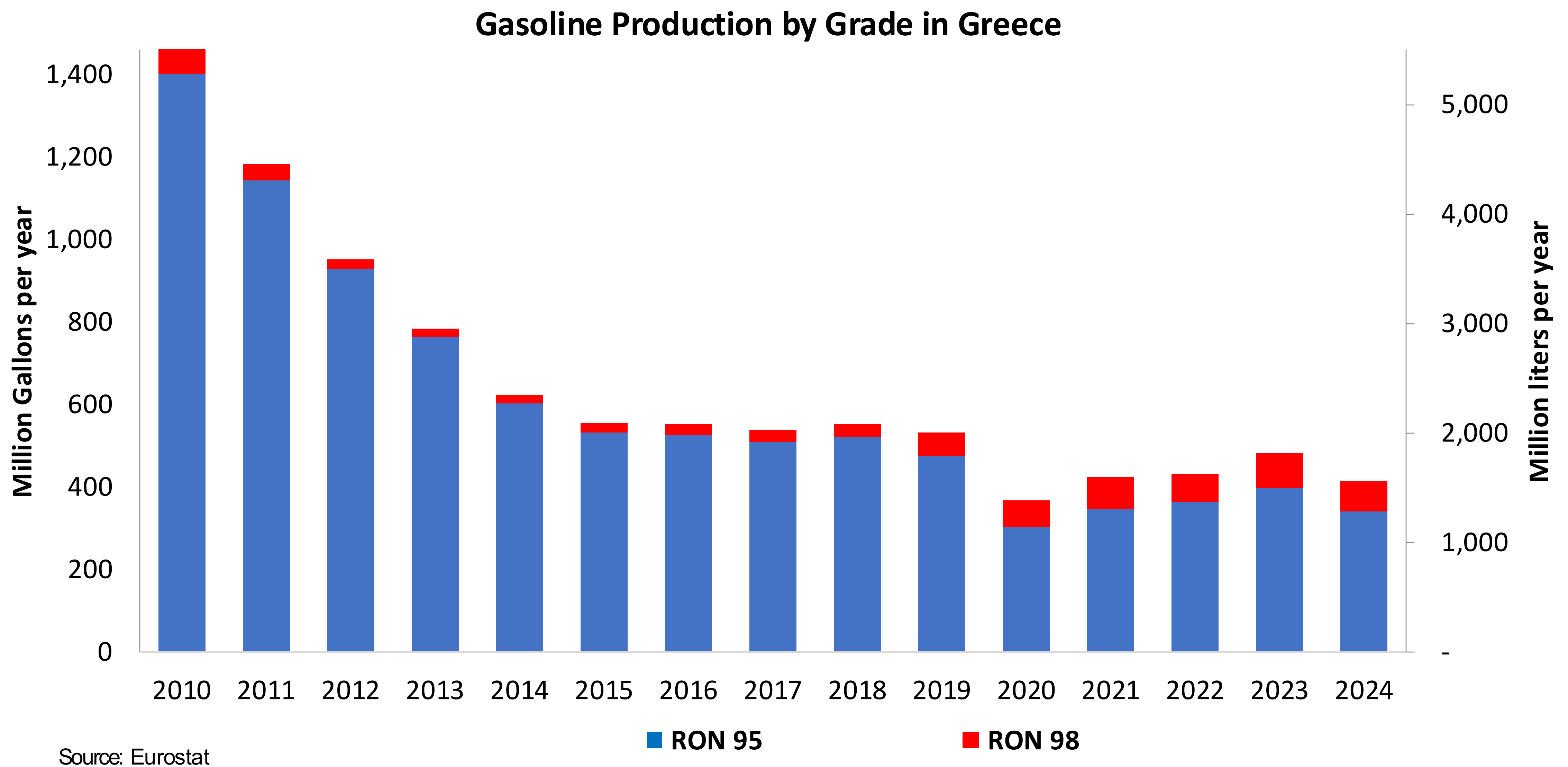
Source: Eurostat, European Commision, USDA

Gasoline Demand in Greece



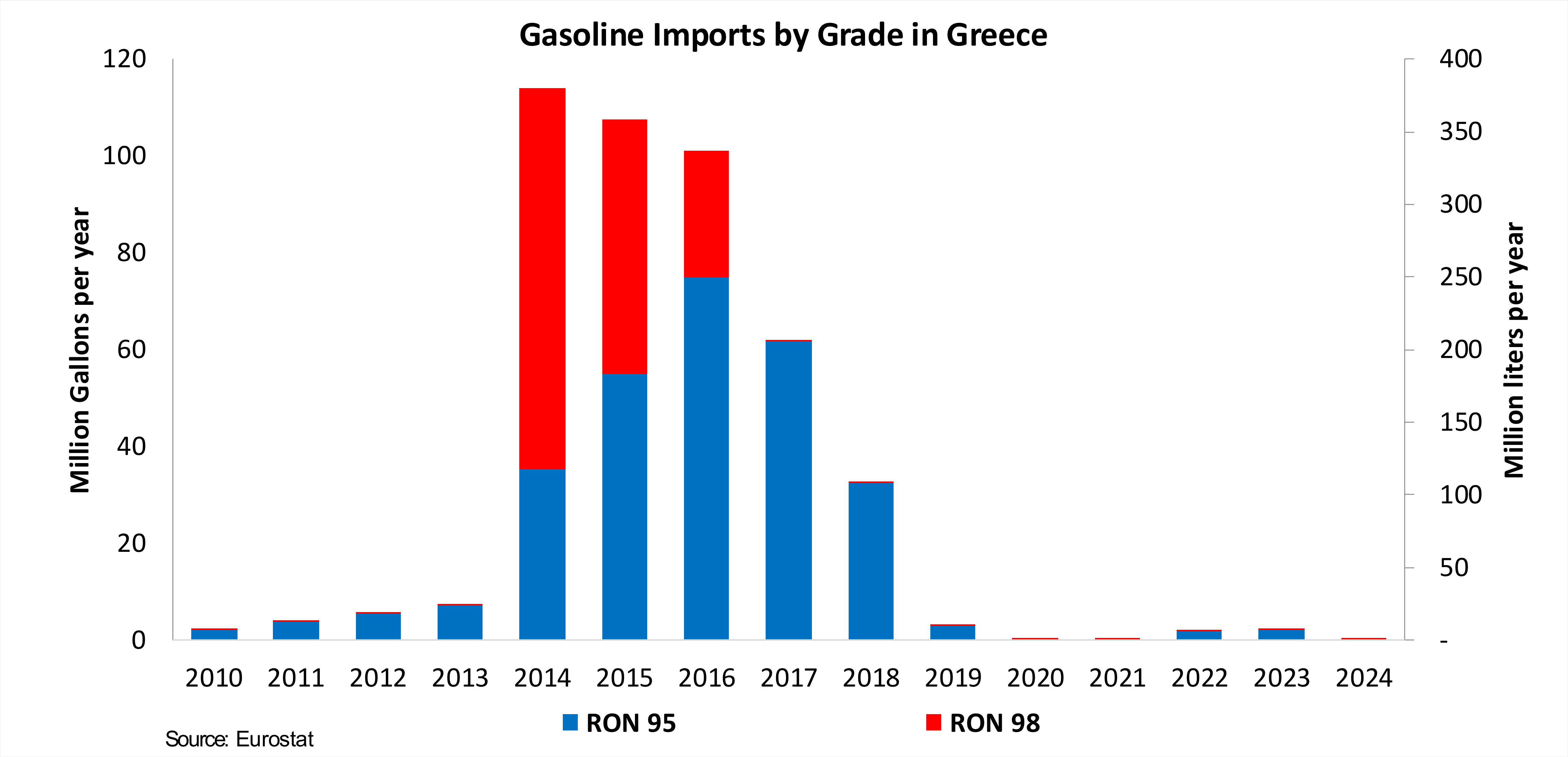
Source: Eurostat

Gasoline Production in Greece

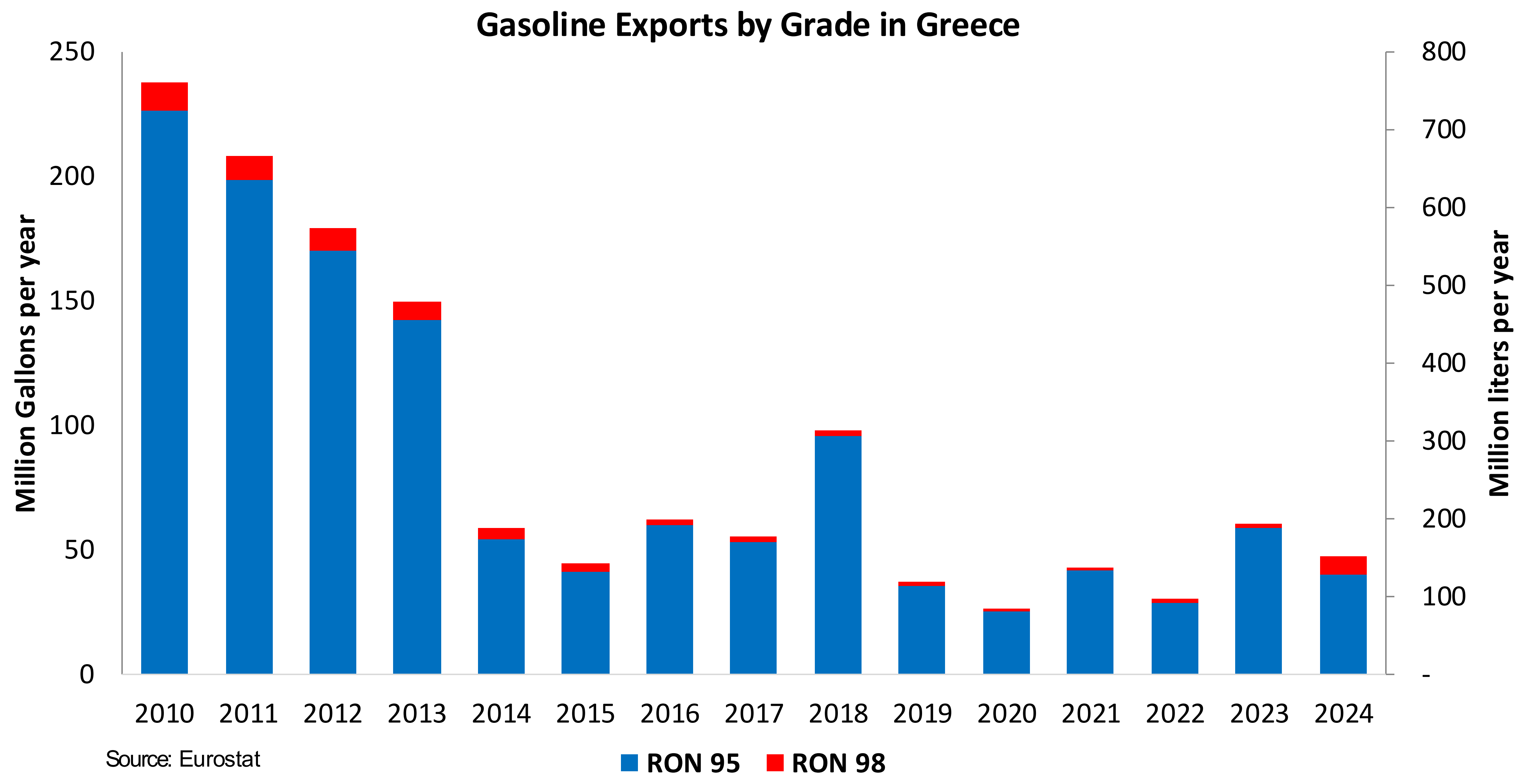


Source: Eurostat

Gasoline Imports in Greece

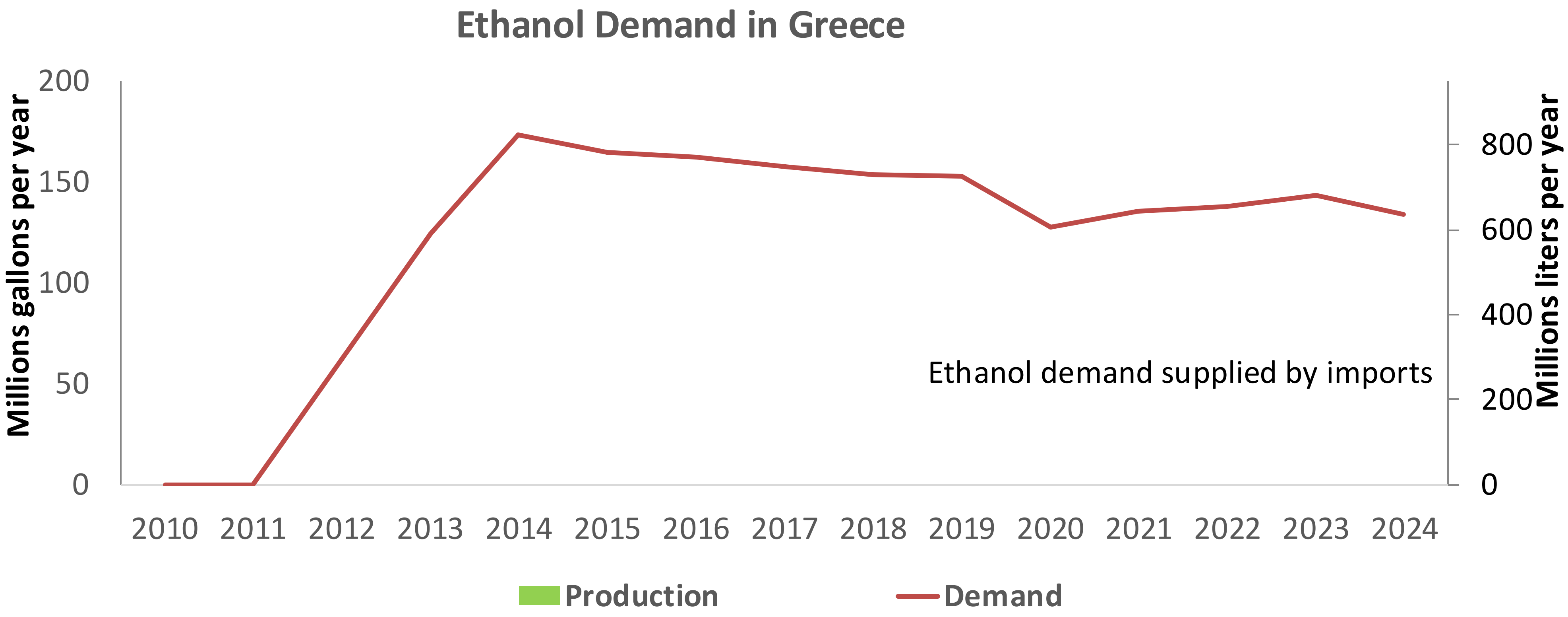


Gasoline Exports in Greece



Source: Eurostat

Ethanol Balance in Greece



Source: Eurostat, USDA – Fuel Mandates in EU

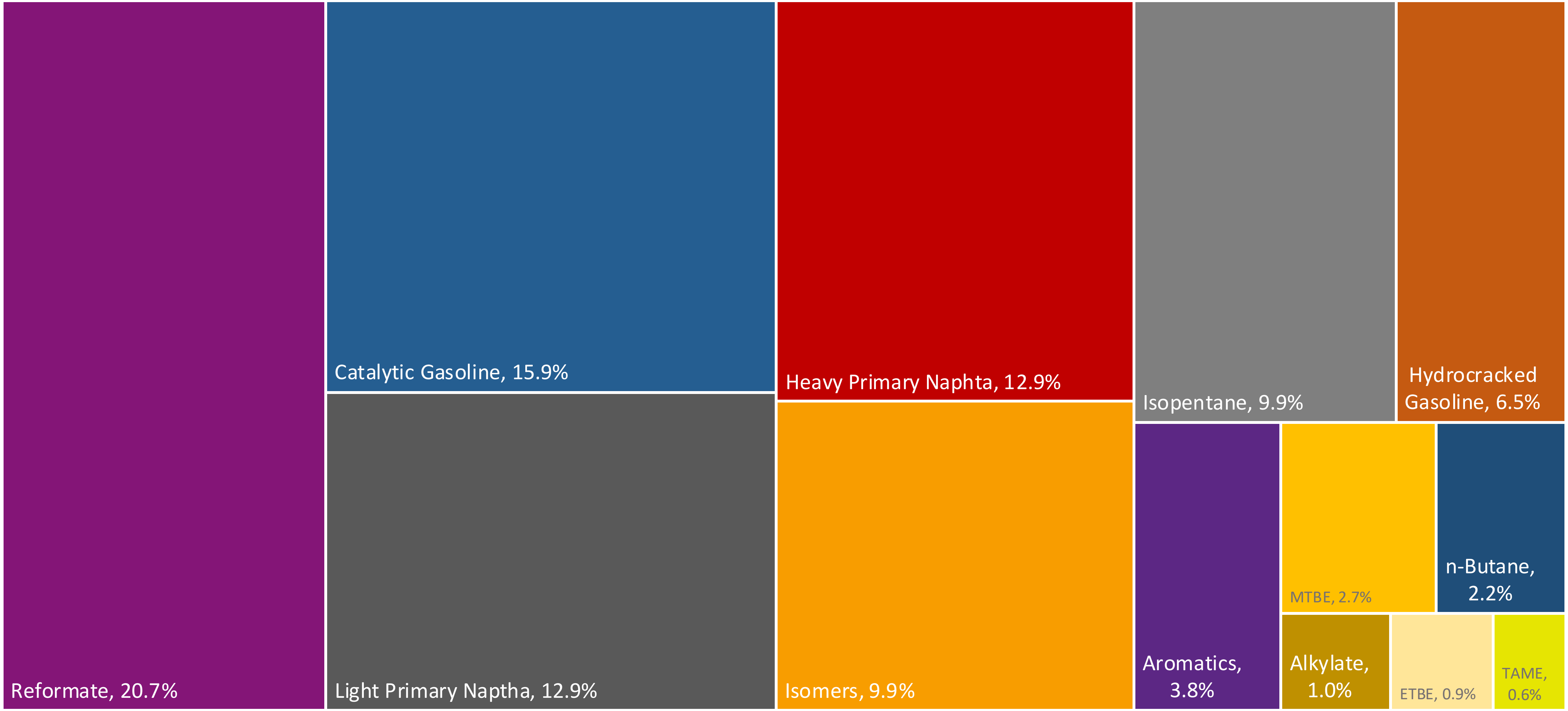
Gasoline Quality in Greece

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.0							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	3.3							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

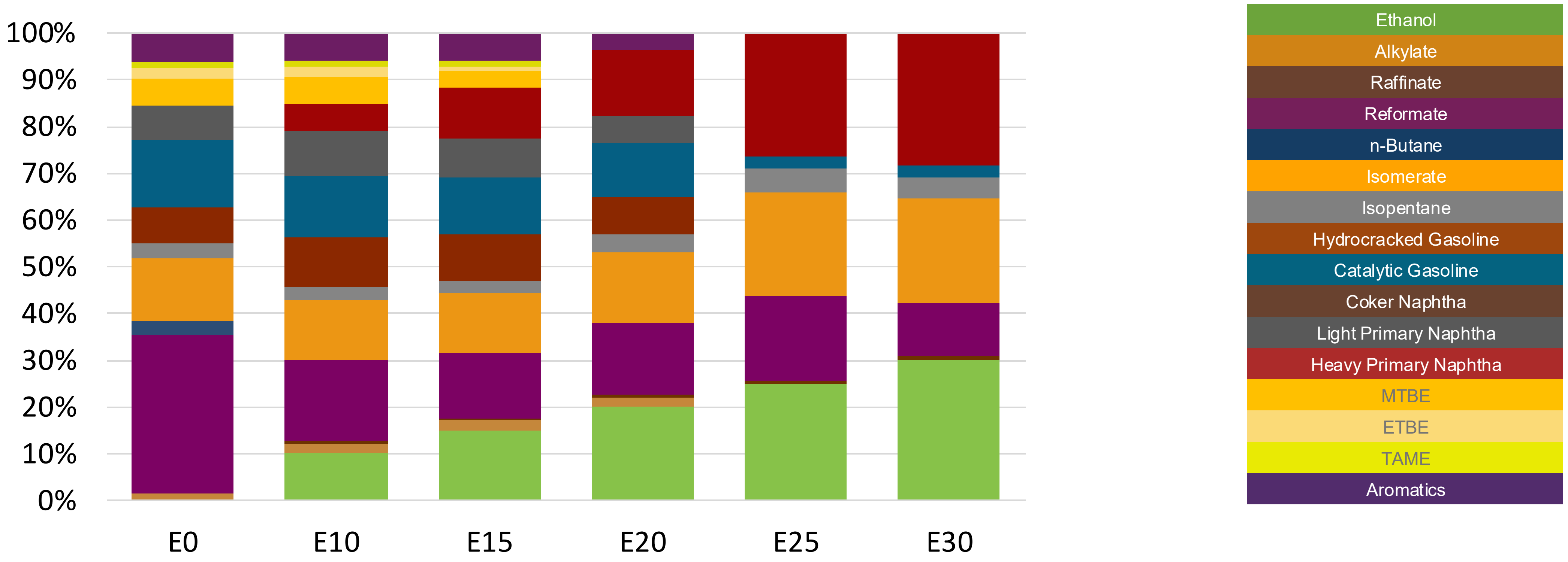
Source: European Commision

Greece Available Blending Components



Source: Faro90

Greece – Gasoline 95 – Constant Octane

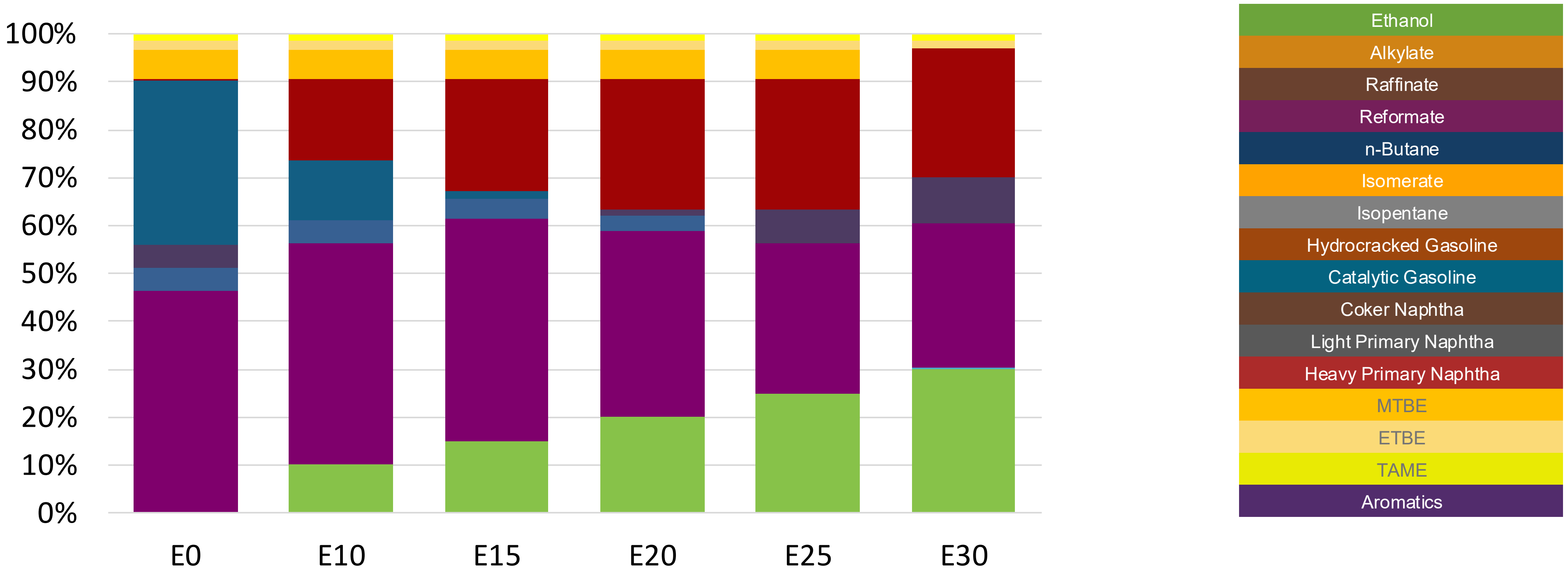


Octane (RON)	95.0	95.0	95.0	95.0	95.0	96.0
Price (US$/gal)	$ 2.258	$ 2.135	$ 2.069	$ 2.002	$ 1.881	$ 1.845

Average CIF 2024 Prices

Source: Faro90

Greece - Gasoline 98 - Constant Octane

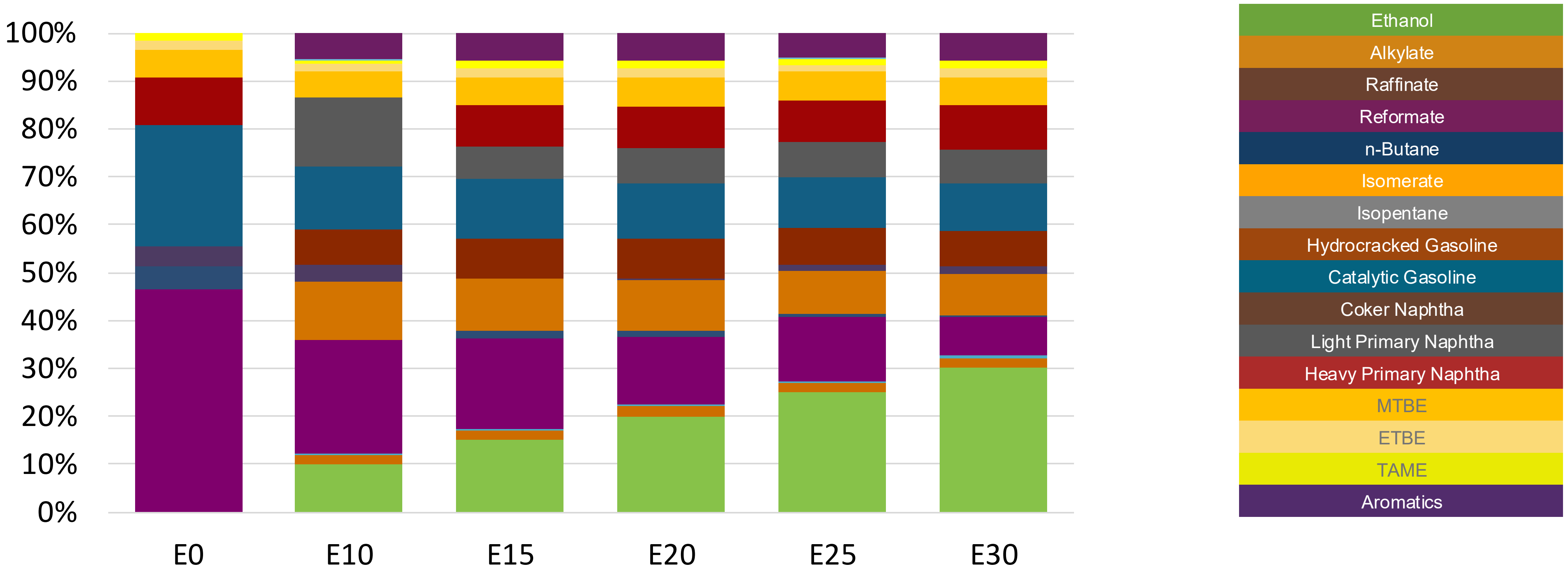


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.330	$ 2.204	$ 2.132	$ 2.057	$ 1.980	$ 1.924

Average CIF 2024 Prices

Source: Faro90

Greece – Gasoline 95 – Increased Octane

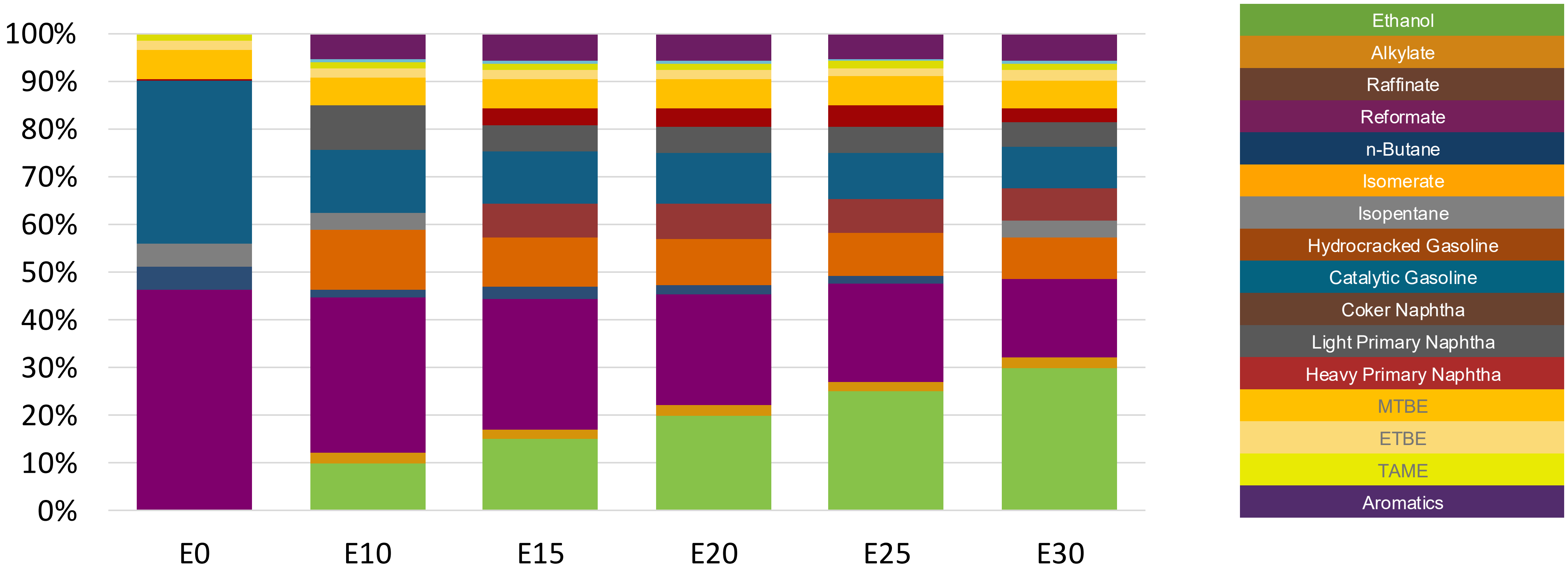


Octane (RON)	95.0	96.2	97.9	99.4	100.8	102.2
Price (US$/gal)	$ 2.196	$ 2.196	$ 2.196	$ 2.196	$ 2.196	$ 2.196

Average CIF 2024 Prices

Source: Faro90

Greece – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.4	100.7	102.0	103.3	104.5
Price (US$/gal)	$ 2.330	$ 2.330	$ 2.330	$ 2.330	$ 2.330	$ 2.330

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Greece

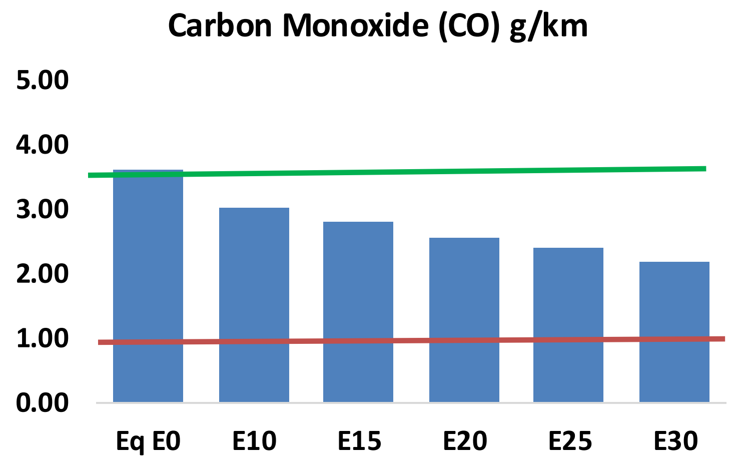
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	-
		Single FI	3-Ways		>8 years	70,969
		Multi FI	3-Ways		>8 years	353,479
			Euro III		>8 years	830,998
			Euro IV		>8 years	1,080,821
			Euro V		>8 years	127,791
			Euro 6		>8 years	240,045
					4-8 years	262,415
					<4 years	103,556
			Hybrid		All	132,272
		Electric			All	20,177
	Medium	Carburator	3-Ways	PCV Tank	>8 years	-
		Single FI	3-Ways		>8 years	2,050
		Multi FI	3-Ways		>8 years	54,901
			Euro III		>8 years	154,007
			Euro IV		>8 years	378,495
			Euro V		>8 years	38,985
			Euro 6		>8 years	80,015
					4-8 years	119,271
					<4 years	85,597
			Hybrid		All	104,759
		Electric			All	16,634
Sml Engines	Light	4-Cycle	None	PCV	>8 years	27,633
			Euro III		>8 years	85,380
			Euro IV		>8 years	299,742
			Euro V		>8 years	62,400
			Euro 6		>8 years	125,785
					4-8 years	129,891
					<4 years	229,319
			Hybrid		All	57,134
		Electric			All	8,873
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	-
		Fuel-Injection	3-Ways		>8 years	166,369
			Euro III		>8 years	145,539
			Euro IV		>8 years	456,964
			Euro V		>8 years	113,516
			Euro 6		>8 years	218,570
					4-8 years	126,348
					<4 years	75,940
			Hybrid		All	115
		Electric			All	1,321

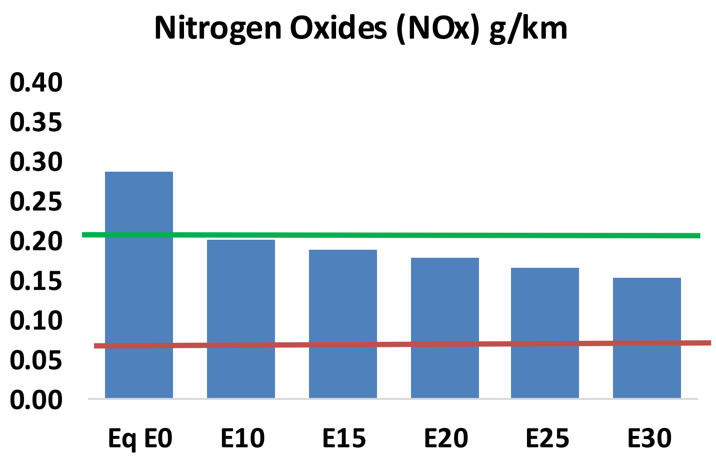
Vehicle Fleet: 6,588,079
Gasoline Vehicle Fleet: 4,489,264

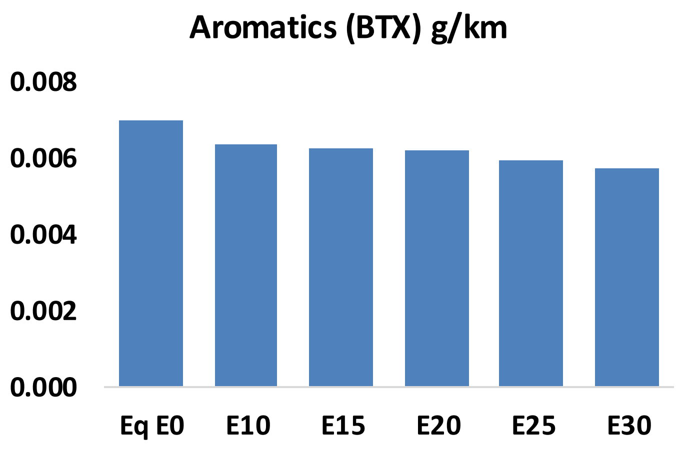
Source: Eurostat, ACEA

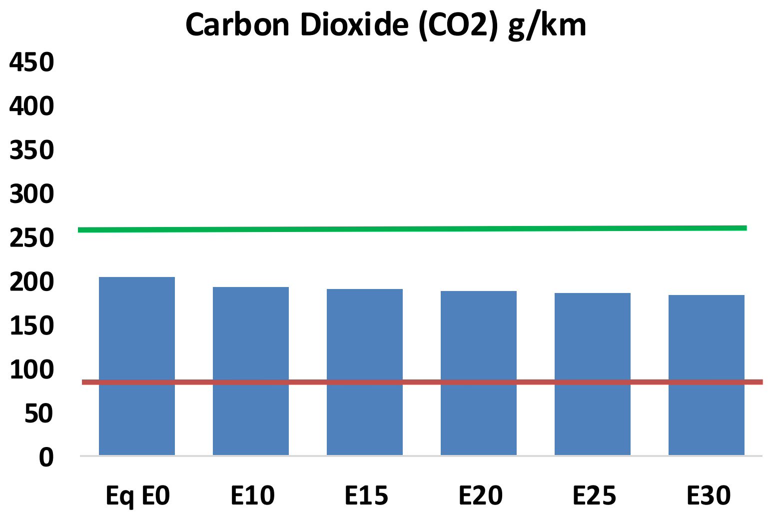
Greece – Vehicular Emissions

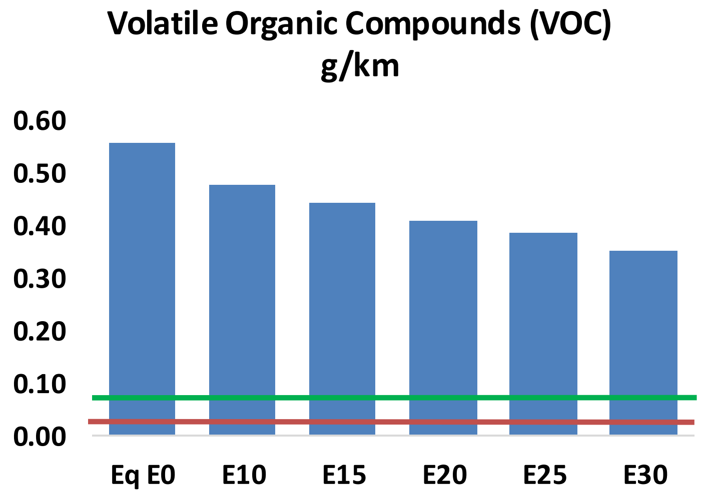
TIER III BIN 125 United States
Euro 6 Standard

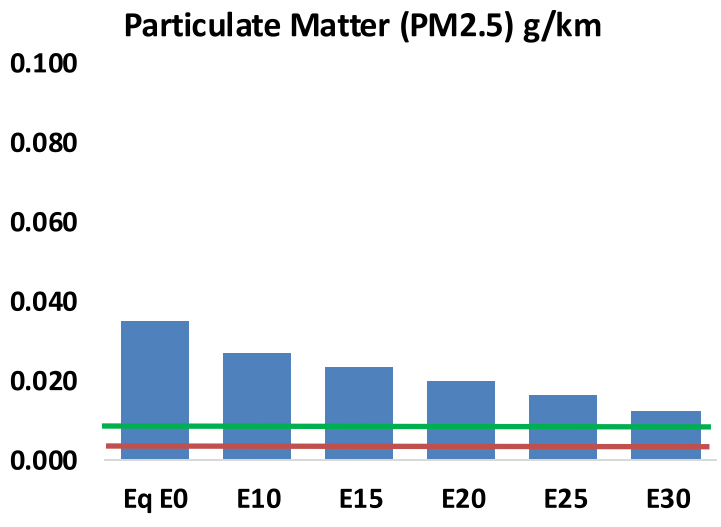












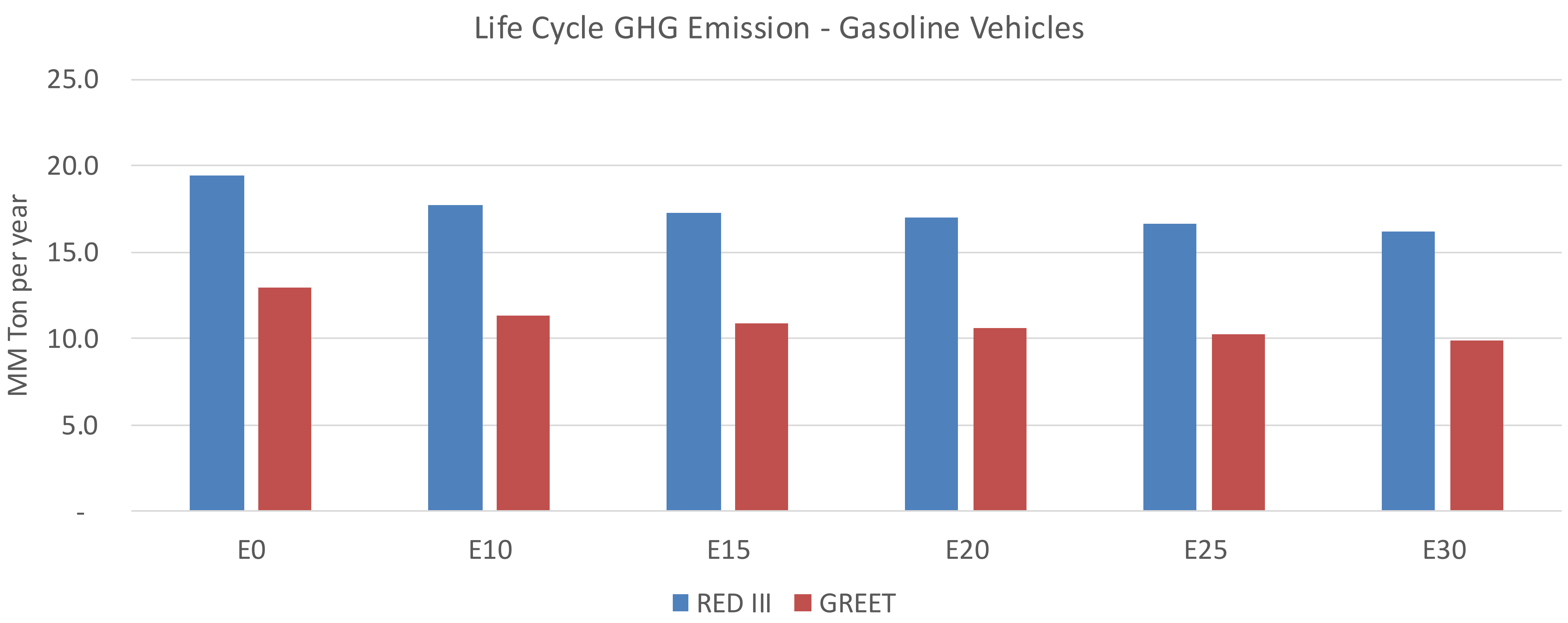
Source: Faro90

Greece – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	385,776	355,555	325,335	300,460	275,378	256,076	234,621	-16%	-29%
CO2	21,880,668	21,318,022	20,786,634	20,368,713	20,161,910	20,045,323	19,675,830	-5%	-8%
VOC	59,679	55,319	50,959	47,431	43,901	41,193	37,557	-15%	-26%
NOx	30,711	23,034	21,498	20,262	19,109	17,837	16,492	-30%	-38%
SOx	260	240	221	204	188	171	153	-15%	-28%
NH3	7,990	7,911	7,832	7,869	7,958	8,020	8,072	-2%	0%
N2O	301	299	298	299	305	309	312	-1%	2%
CH4	12,854	12,854	12,854	13,047	13,330	13,574	13,806	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	391	376	360	351	342	336	326	-8%	-13%
Acetaldehyde	1,097	1,259	1,847	2,812	3,831	4,378	5,173	68%	249%
Formaldehyde	3,922	3,813	4,445	5,220	5,469	6,050	6,586	13%	39%
Benzene	749	719	682	671	665	636	616	-9%	-11%
PM2.5	1,379	1,225	1,070	929	789	640	485	-22%	-43%
PM10	2,338	2,076	1,814	1,575	1,337	1,085	822	-22%	-43%

Source: Faro90

Greece – Gasoline Vehicle Fleet Life Cycle GHG Emissions



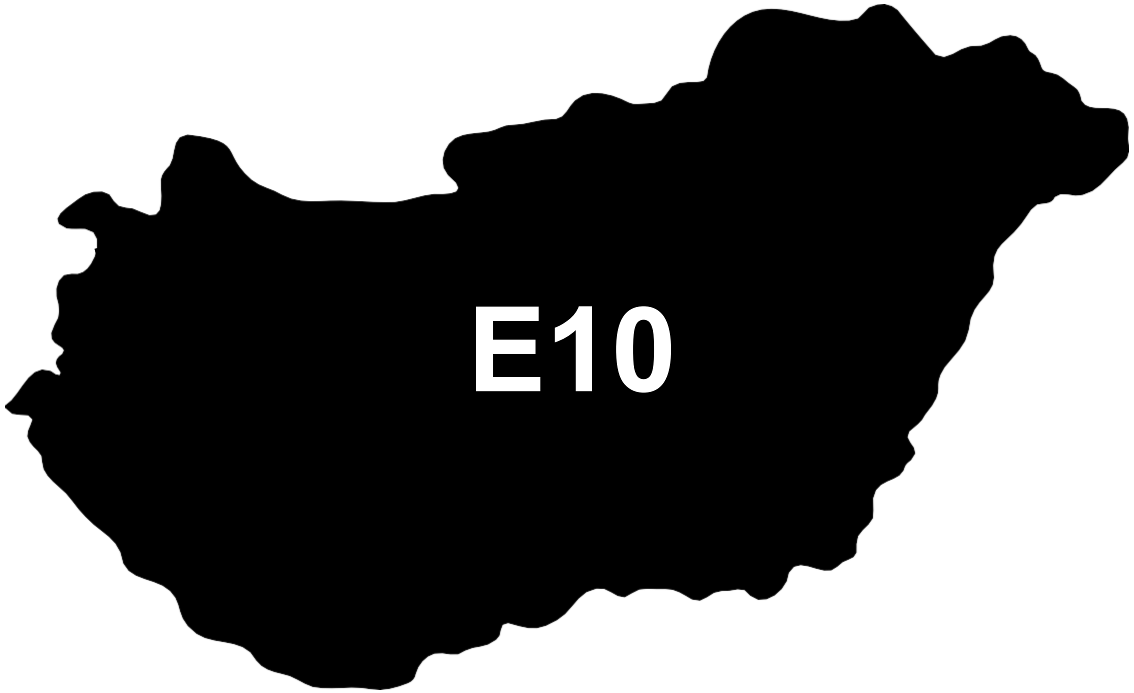
Country Emissions 2024 (MMT/año)	74.3
Target @2035 (MMT/year)	41.8
Delta	32.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.6%	15.7%	18.1%	20.5%	23.7%
RED II/III	0.0%	8.8%	11.0%	12.8%	14.5%	16.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.0%	6.2%	7.2%	8.2%	9.4%
RED II/III	0.0%	5.2%	6.6%	7.7%	8.7%	10.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Hungary

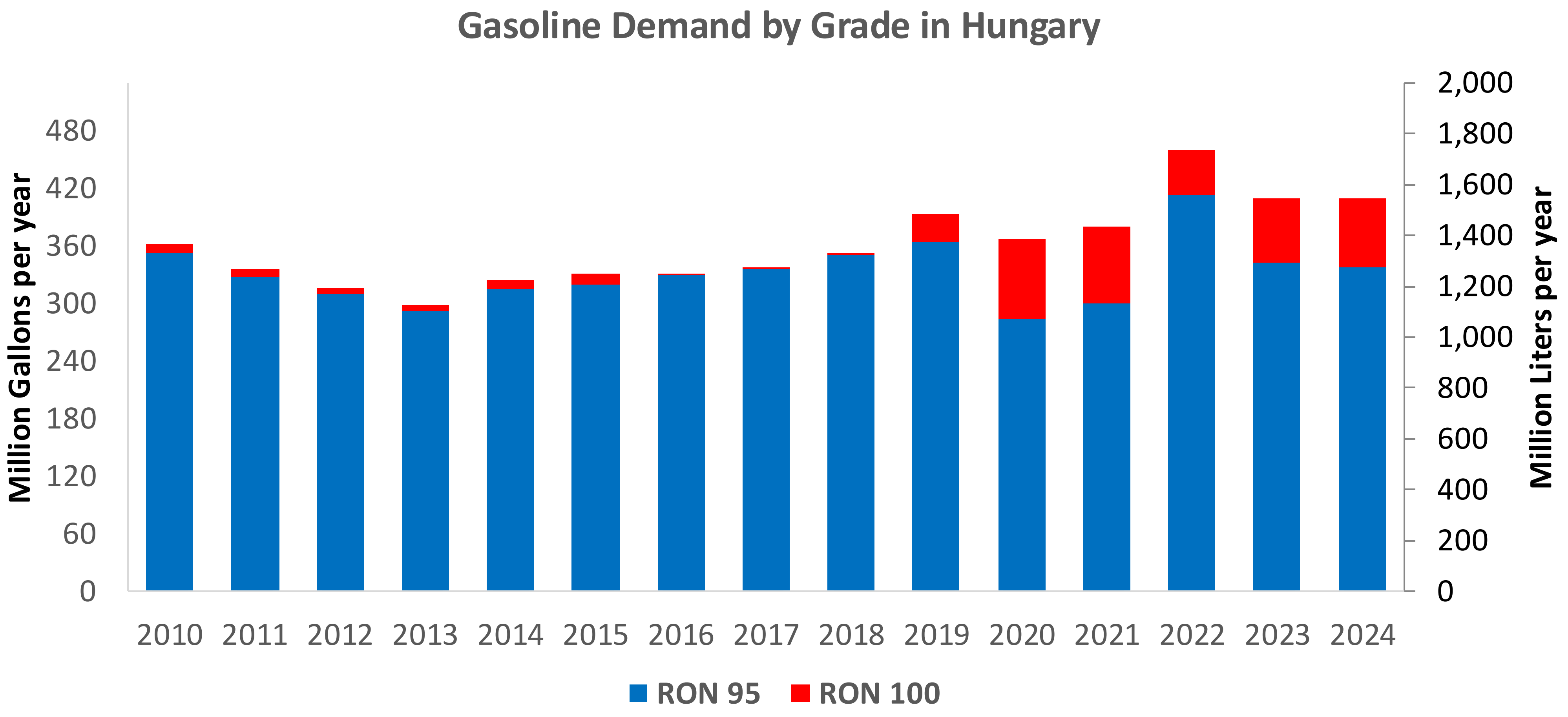


In 2024, gasoline consumption in Hungary was 1,544 million liters (408 million gallons) and growth rate was 0.8%. Gasoline production and net imports were 1,701 and 79 million liters (449 and 21 million gallons) correspondingly. Hungary complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.9 RON.

In 2024, ethanol consumption in Hungary was 152 million liters (40 million gallons) representing 9.8% of gasoline demand. Ethanol production and Net Imports were 609 and -457 million liters (40 and -121 million gallons) correspondingly. Ethanol source is Corn 100%.

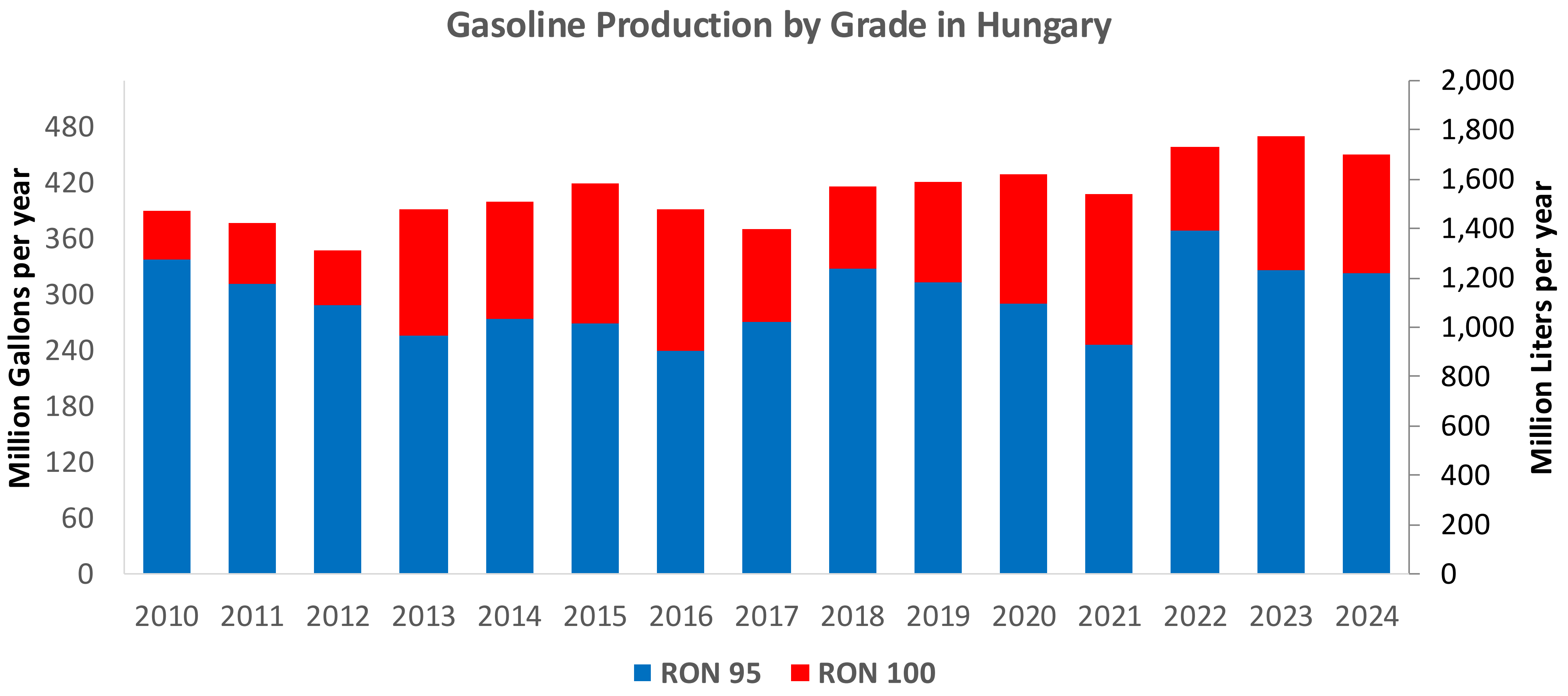
Source: Eurostat, European Commision

Gasoline Demand in Hungary



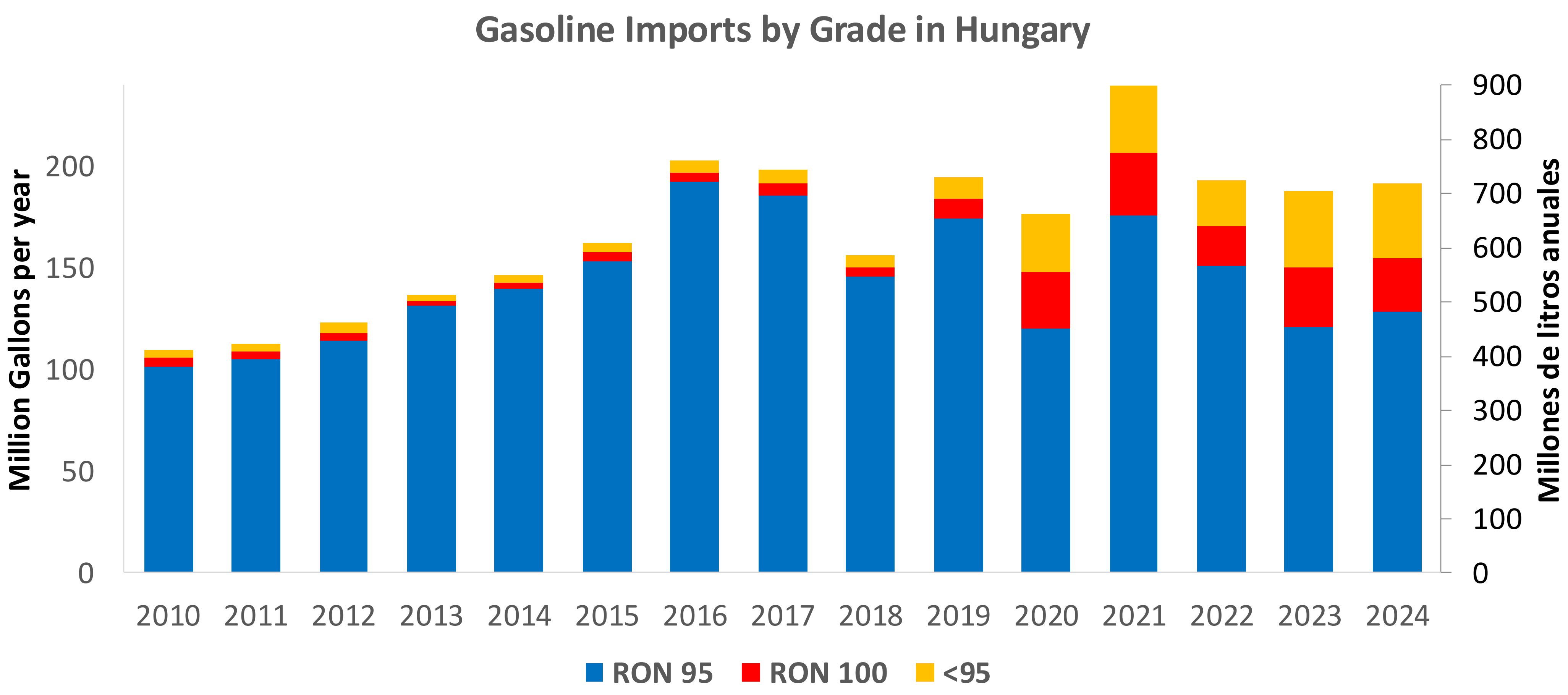
Source: Eurostat

Gasoline Production in Hungary



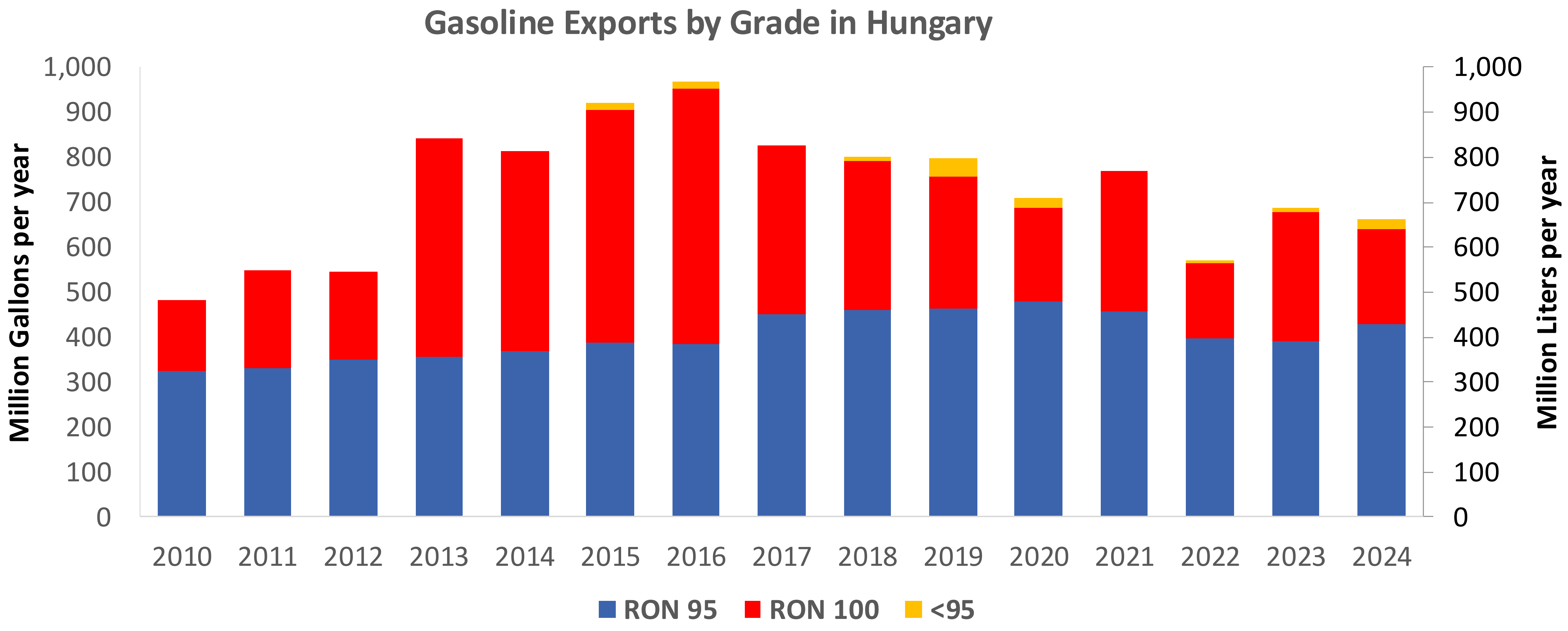
Source: Eurostat

Gasoline Imports in Hungary



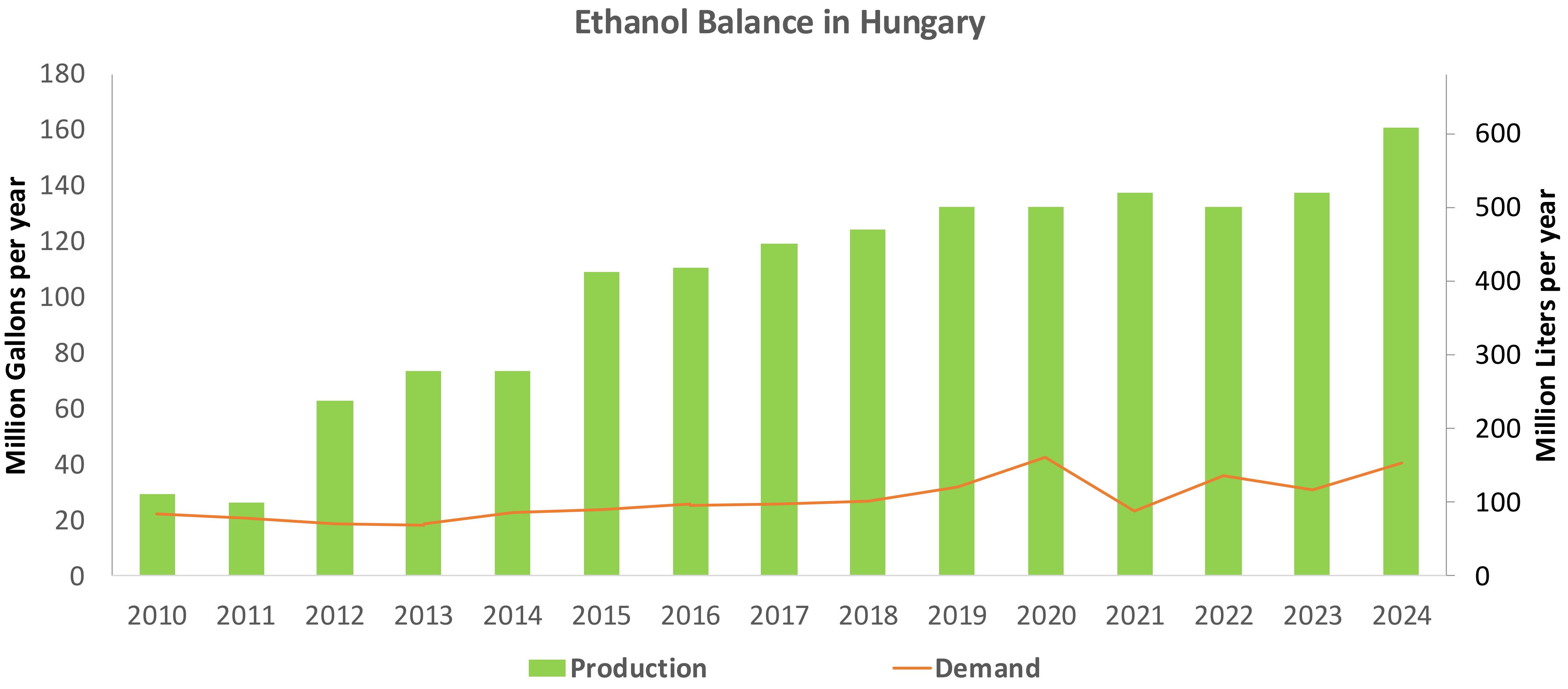
Source: Eurostat

Gasoline Exports in Hungary



Source: Eurostat

Ethanol Balance in Hungary



Source: Eurostat

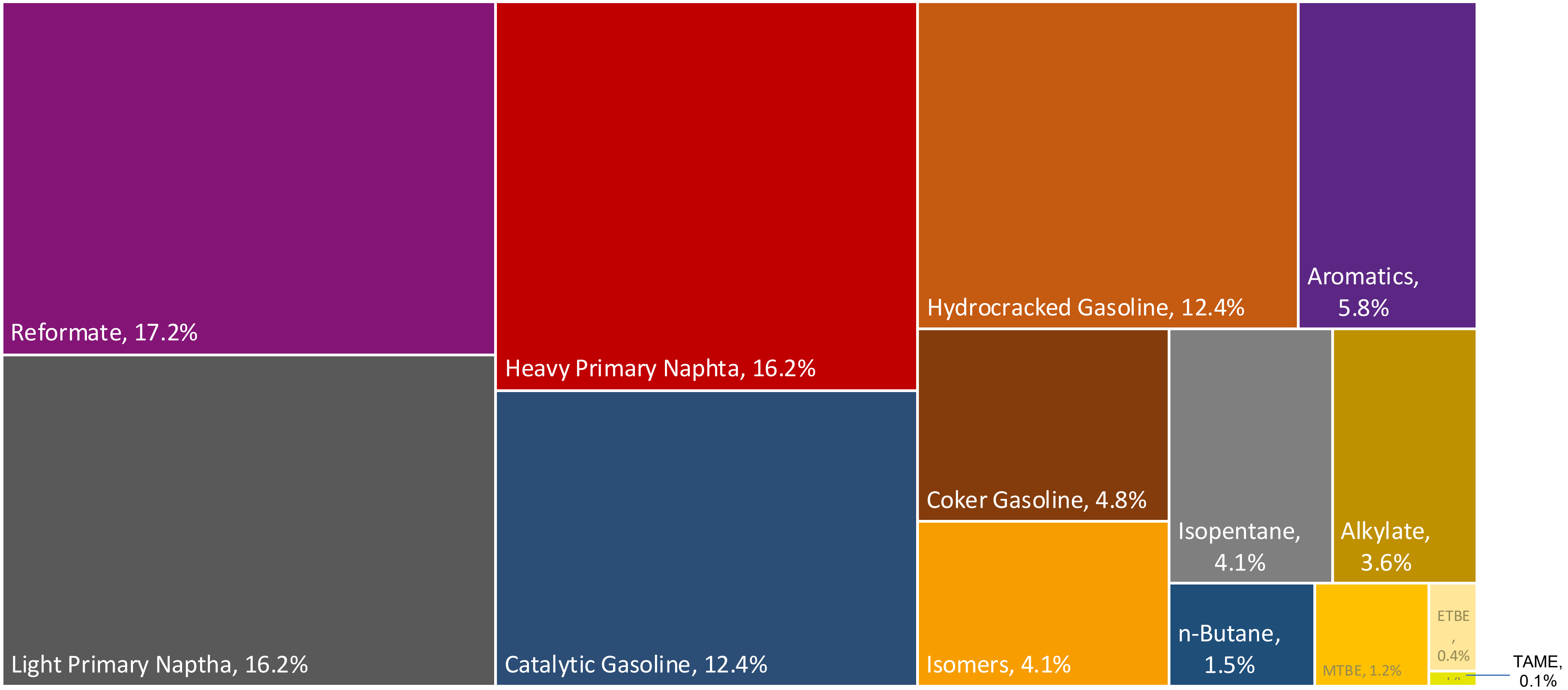
Gasoline Quality in Hungary

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.5							
Bioethanol in Gasoline Sales (%cal)	6.1							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

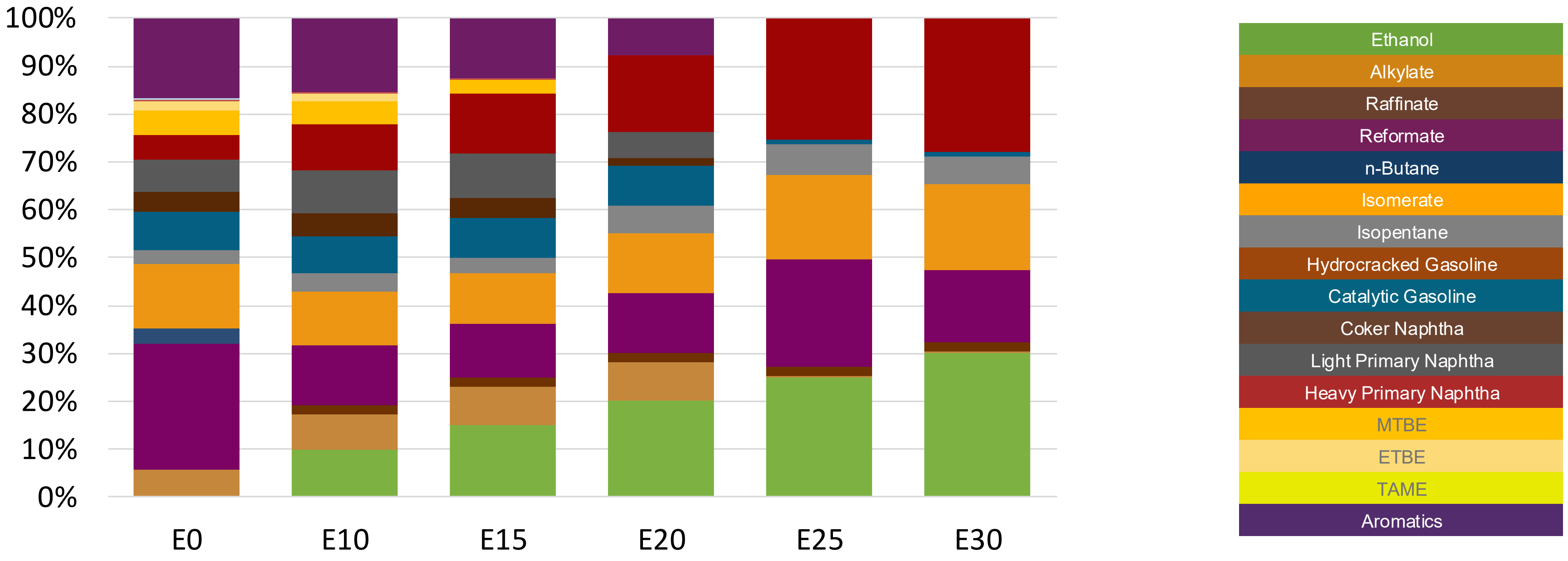
Source: European Commision

Hungary Available Blending Components



Source: Faro90

Hungary – Gasoline 95 – Constant Octane

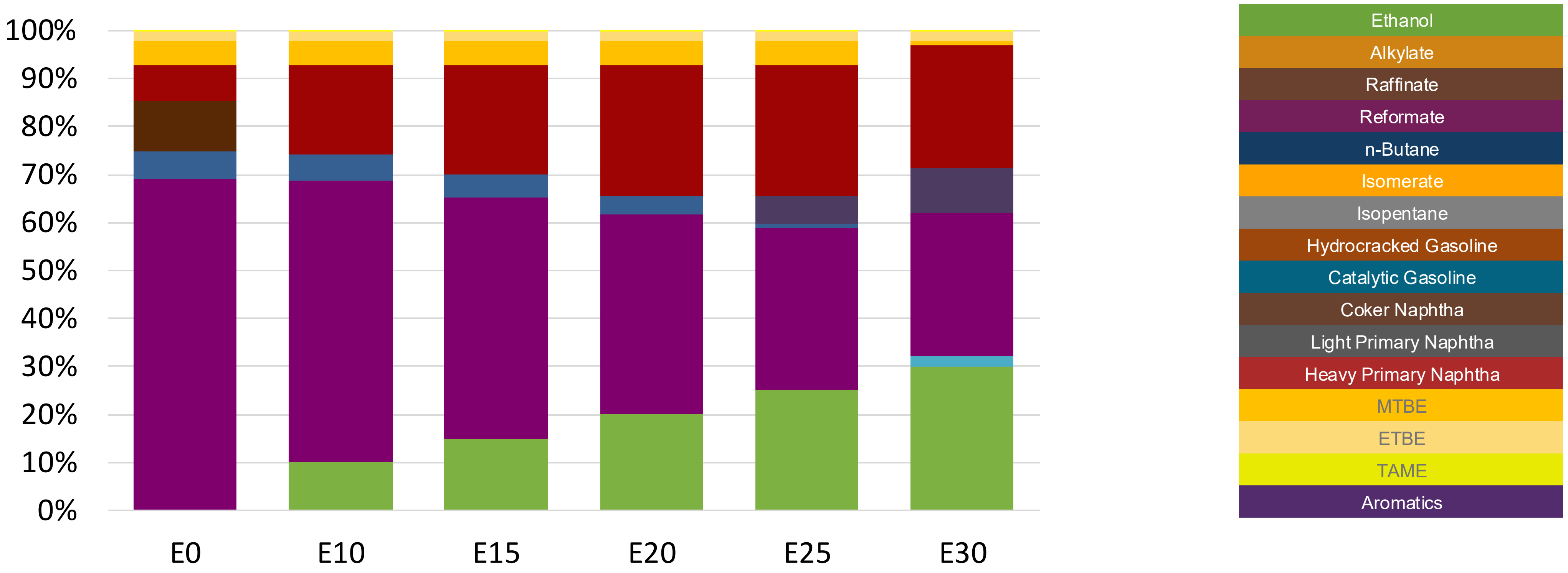


Octane (RON)	95.0	95.0	95.0	95.0	95.0	96.0
Price (US$/gal)	$ 2.297	$ 2.168	$ 2.099	$ 2.021	$ 1.878	$ 1.841

Average CIF 2024 Prices

Source: Faro90

Hungary - Gasoline 98 - Constant Octane

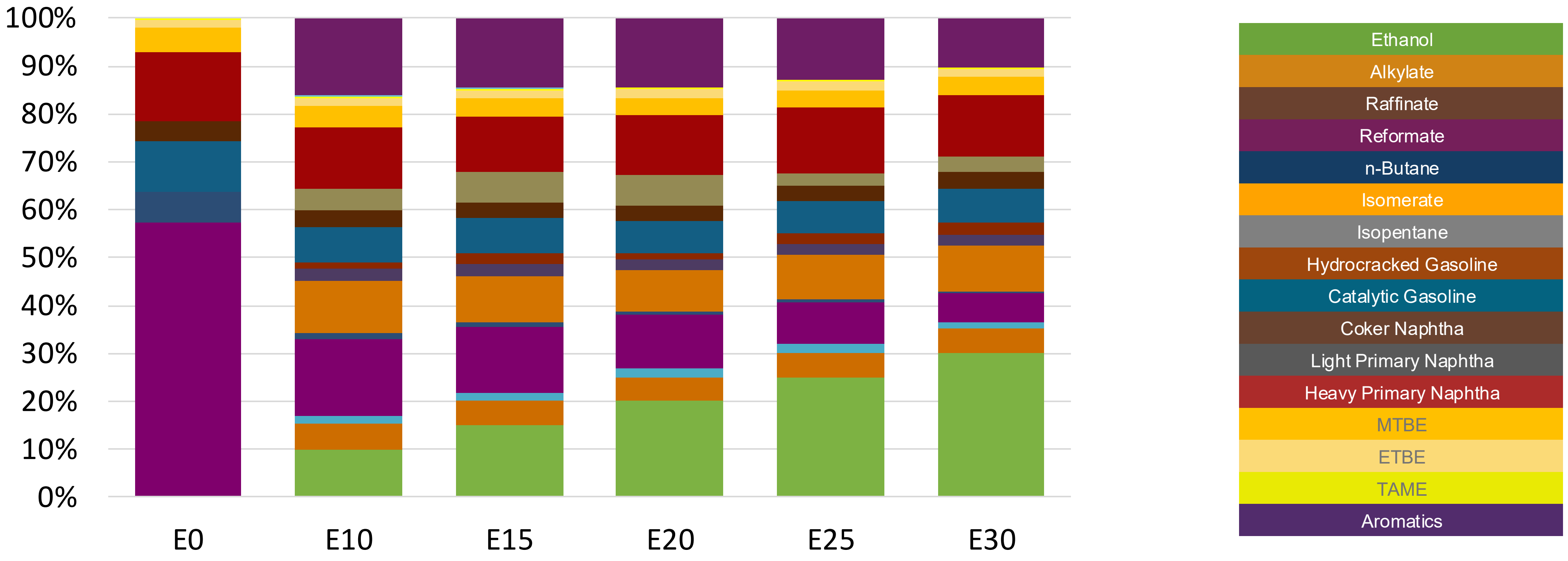


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.329	$ 2.208	$ 2.139	$ 2.064	$ 1.987	$ 1.923

Average CIF 2024 Prices

Source: Faro90

Hungary – Gasoline 95 – Increased Octane

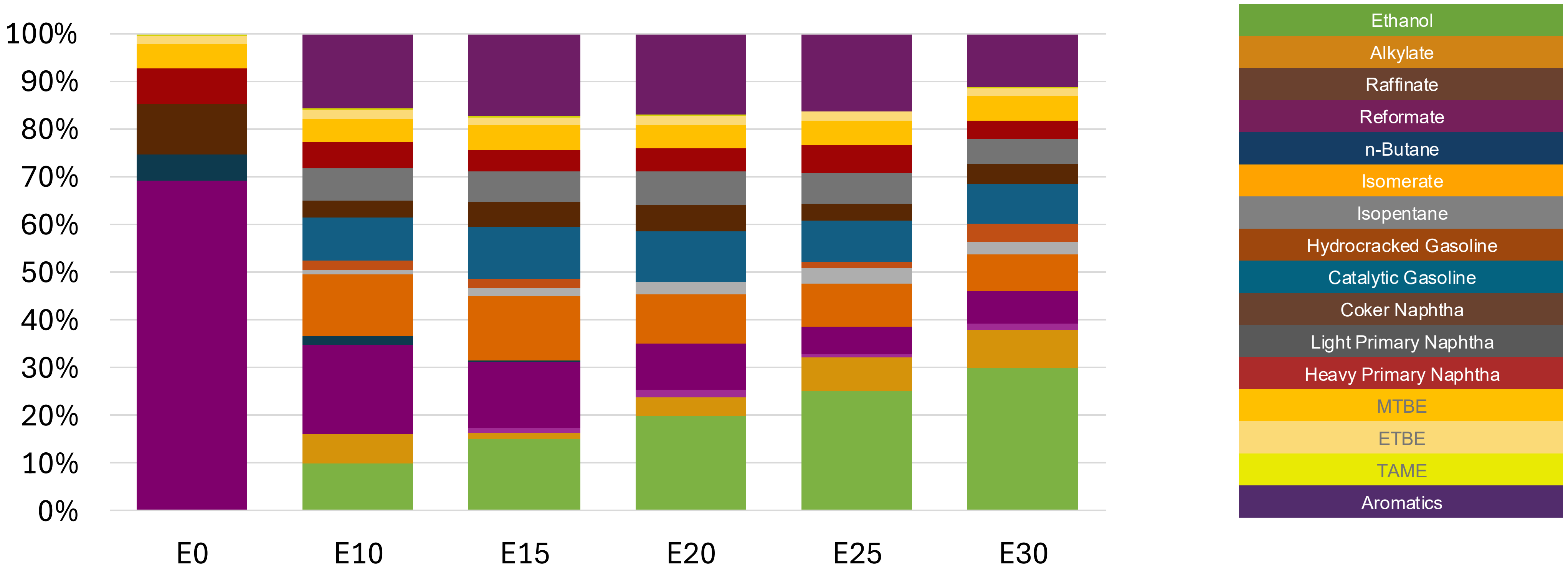


Octane (RON)	95.0	95.7	97.2	98.8	100.3	101.8
Price (US$/gal)	$ 2.198	$ 2.198	$ 2.198	$ 2.198	$ 2.198	$ 2.198

Average CIF 2024 Prices

Source: Faro90

Hungary – Gasoline 98 – Increased Octane



Octane (RON)	98.0	98.5	99.9	101.2	102.5	104.0
Price (US$/gal)	$ 2.329	$ 2.329	$ 2.329	$ 2.329	$ 2.329	$ 2.329

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Hungary

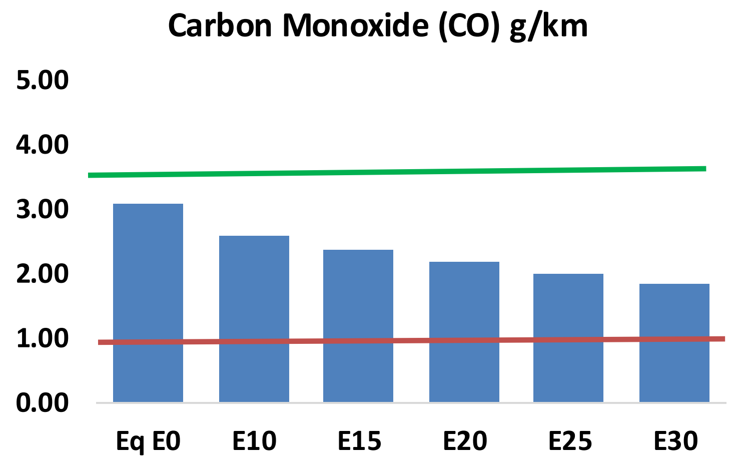
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	231
		Single FI	3-Ways		>8 years	33,605
		Multi FI	3-Ways		>8 years	213,752
			Euro III		>8 years	570,834
			Euro IV		>8 years	738,941
			Euro V		>8 years	101,691
			Euro 6		>8 years	310,820
					4-8 years	556,204
					<4 years	477,469
			Hybrid		All	16,801
		Electric			All	59,501
	Medium	Carburator	3-Ways	PCV Tank	>8 years	0
		Single FI	3-Ways		>8 years	178
		Multi FI	3-Ways		>8 years	4,076
			Euro III		>8 years	13,202
			Euro IV		>8 years	42,335
			Euro V		>8 years	22,847
			Euro 6		>8 years	62,736
					4-8 years	104,694
					<4 years	256,031
			Hybrid		All	27,354
		Electric			All	5,948
Sml Engines	Light	4-Cycle	None	PCV	>8 years	1,544
			Euro III		>8 years	6,707
			Euro IV		>8 years	21,442
			Euro V		>8 years	2,884
			Euro 6		>8 years	7,501
					4-8 years	13,975
					<4 years	24,355
			Hybrid		All	0
		Electric			All	697
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	604
		Fuel-Injection	3-Ways		>8 years	45,371
			Euro III		>8 years	68,035
			Euro IV		>8 years	111,858
			Euro V		>8 years	30,788
			Euro 6		>8 years	93,492
					4-8 years	121,368
					<4 years	123,655
			Hybrid		All	27
		Electric			All	4,937

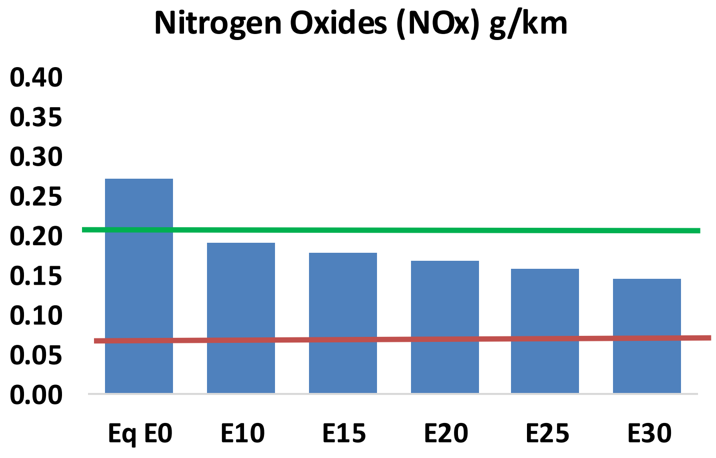
Vehicle Fleet: 4,298,491
Gasoline Vehicle Fleet: 2,317,855

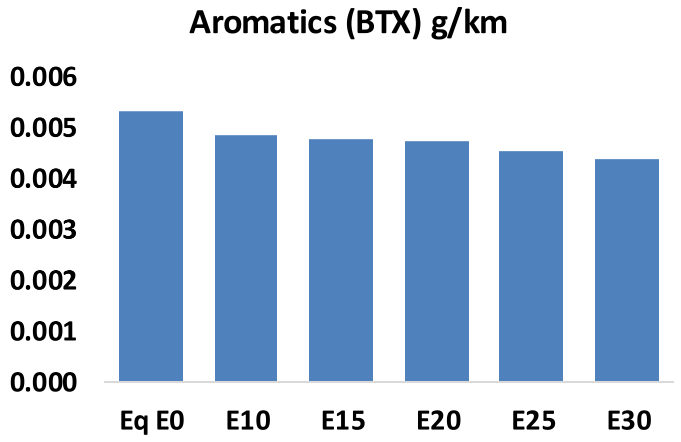
Source: Eurostat, ACEA

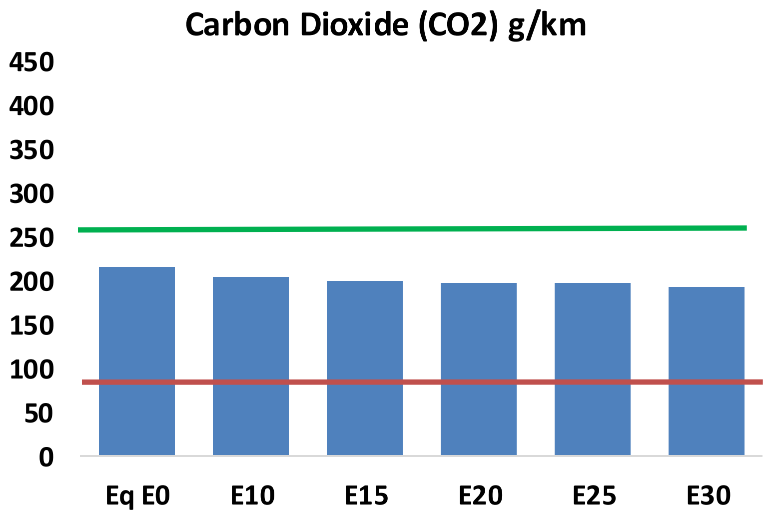
Hungary – Vehicular Emissions

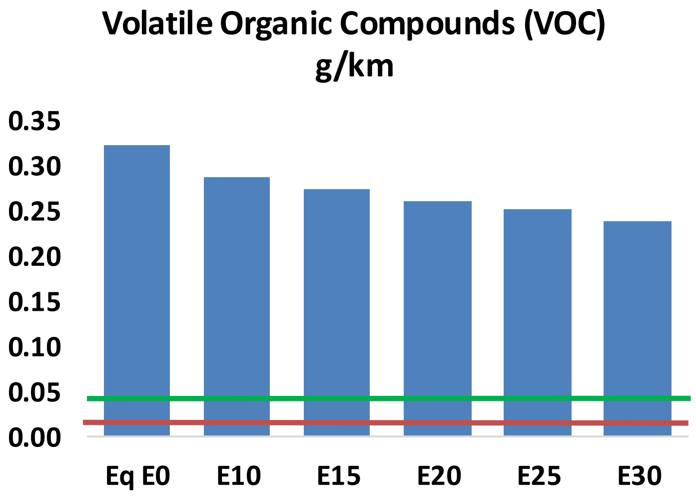
TIER III BIN 125 United States
Euro 6 Standard

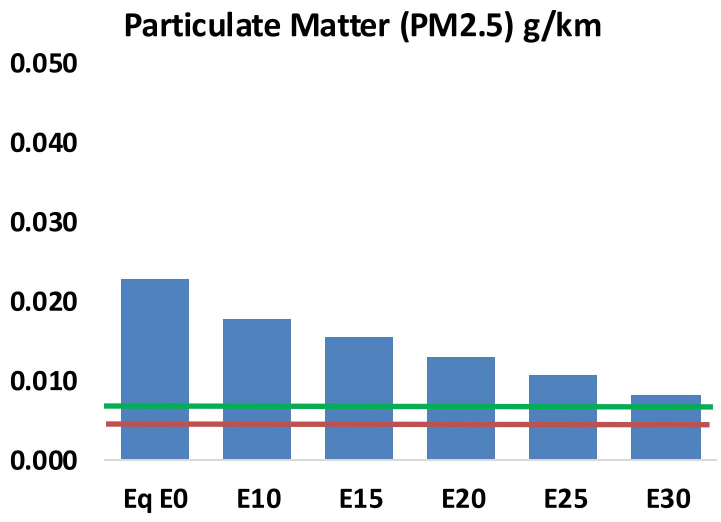












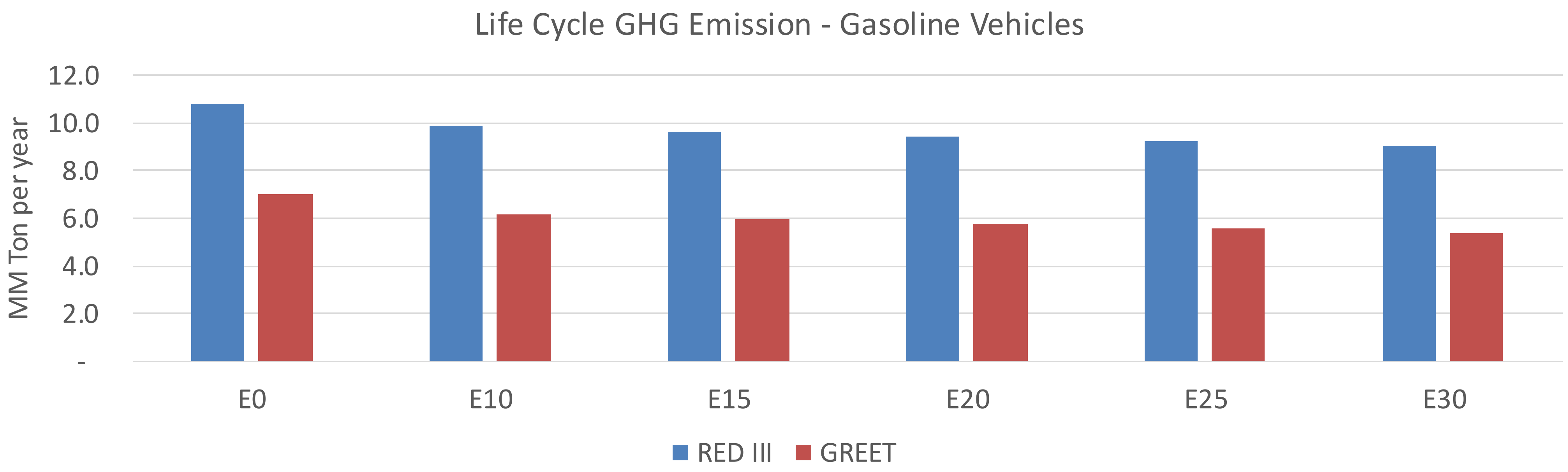
Source: Faro90

Hungary – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	174,918	160,652	146,385	134,534	122,535	113,286	103,266	-16%	-30%
CO2	12,154,422	11,841,879	11,546,701	11,314,551	11,199,675	11,138,457	10,933,143	-5%	-8%
VOC	18,282	17,261	16,241	15,479	14,748	14,196	13,422	-11%	-19%
NOx	15,402	11,552	10,782	10,162	9,583	8,946	8,271	-30%	-38%
SOx	138	127	117	108	100	90	81	-15%	-28%
NH3	4,122	4,081	4,041	4,060	4,105	4,138	4,164	-2%	0%
N2O	188	187	186	187	191	193	195	-1%	2%
CH4	3,983	3,983	3,983	4,043	4,131	4,206	4,278	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	159	153	148	145	142	141	138	-7%	-10%
Acetaldehyde	342	393	576	877	1,195	1,366	1,614	68%	249%
Formaldehyde	1,074	1,044	1,217	1,429	1,497	1,657	1,803	13%	39%
Benzene	302	290	275	270	268	256	248	-9%	-11%
PM2.5	750	666	582	506	429	348	264	-22%	-43%
PM10	537	477	417	362	307	249	189	-22%	-43%

Source: Faro90

Hungary – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	64.6
Target @2035 (MMT/year)	36.4
Delta	28.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.2%	15.3%	17.8%	20.1%	23.3%
RED II/III	0.0%	8.4%	10.7%	12.5%	14.2%	16.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.0%	3.8%	4.4%	5.0%	5.8%
RED II/III	0.0%	3.2%	4.1%	4.8%	5.4%	6.3%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

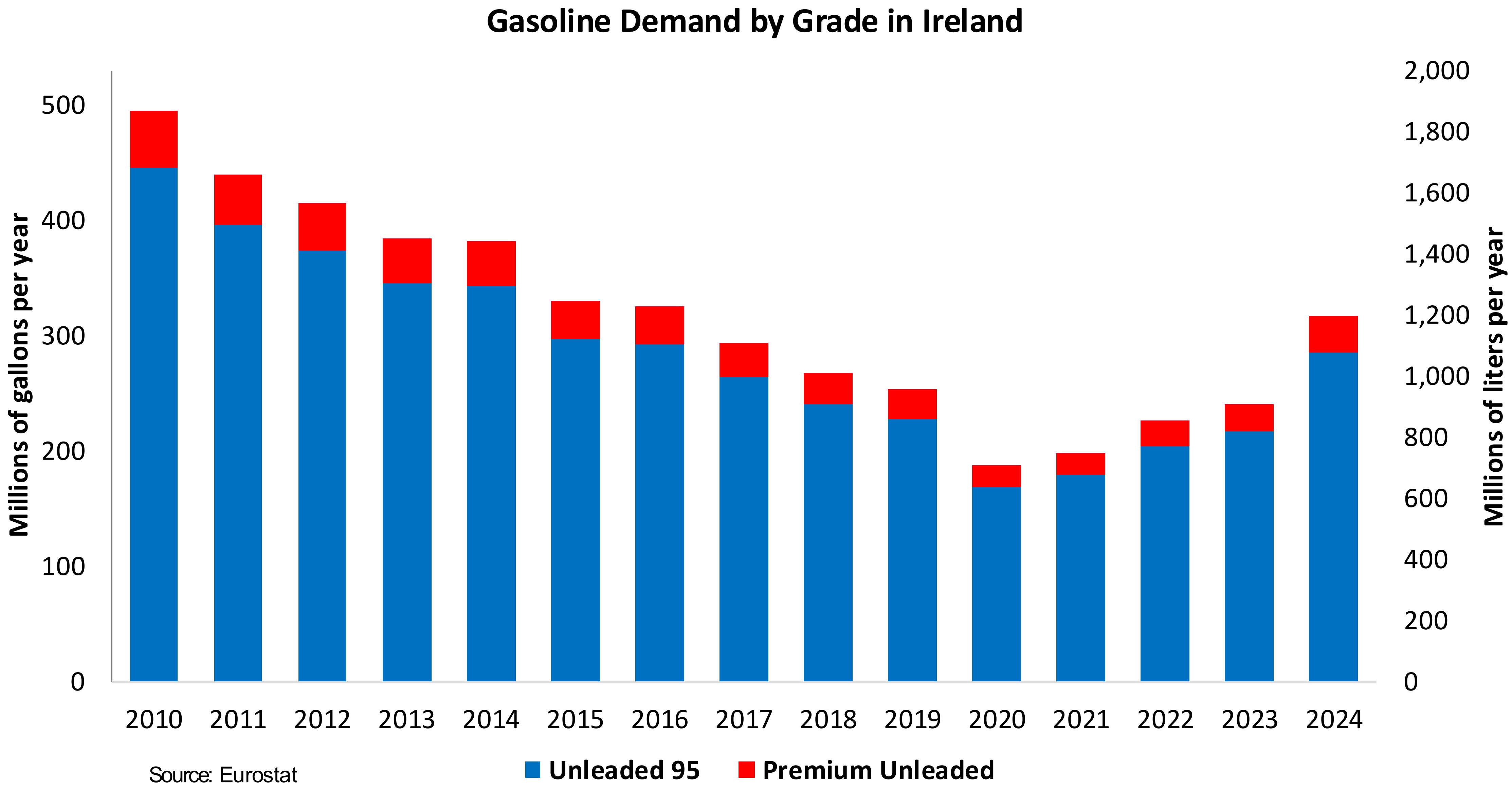
Ethanol Gasoline Blending in Ireland

In 2024, gasoline consumption in Ireland was 1,197 million liters (316 million gallons) and growth rate was 4.6%. Gasoline production and net imports were 676 and 521 million liters (179 and 138 million gallons) correspondingly. Ireland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Ireland was 60 million liters (16 million gallons) representing 5.0% of gasoline demand. Ethanol production and Net Imports were 22 and 39 million liters (16 and 10 million gallons) correspondingly. Ethanol source is Cereals 100%.

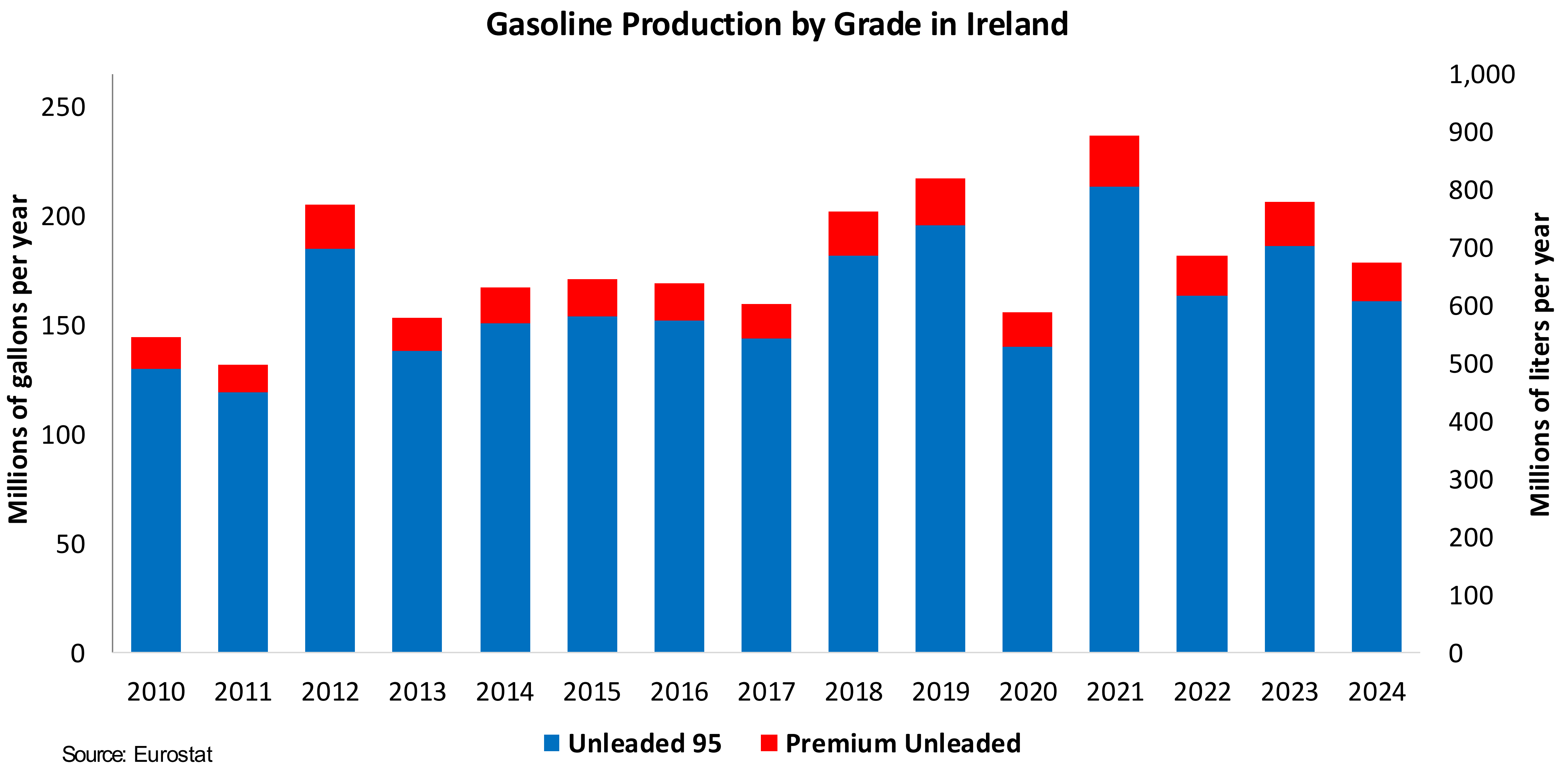
Source: Eurostat, European Commision

Gasoline Demand in Ireland



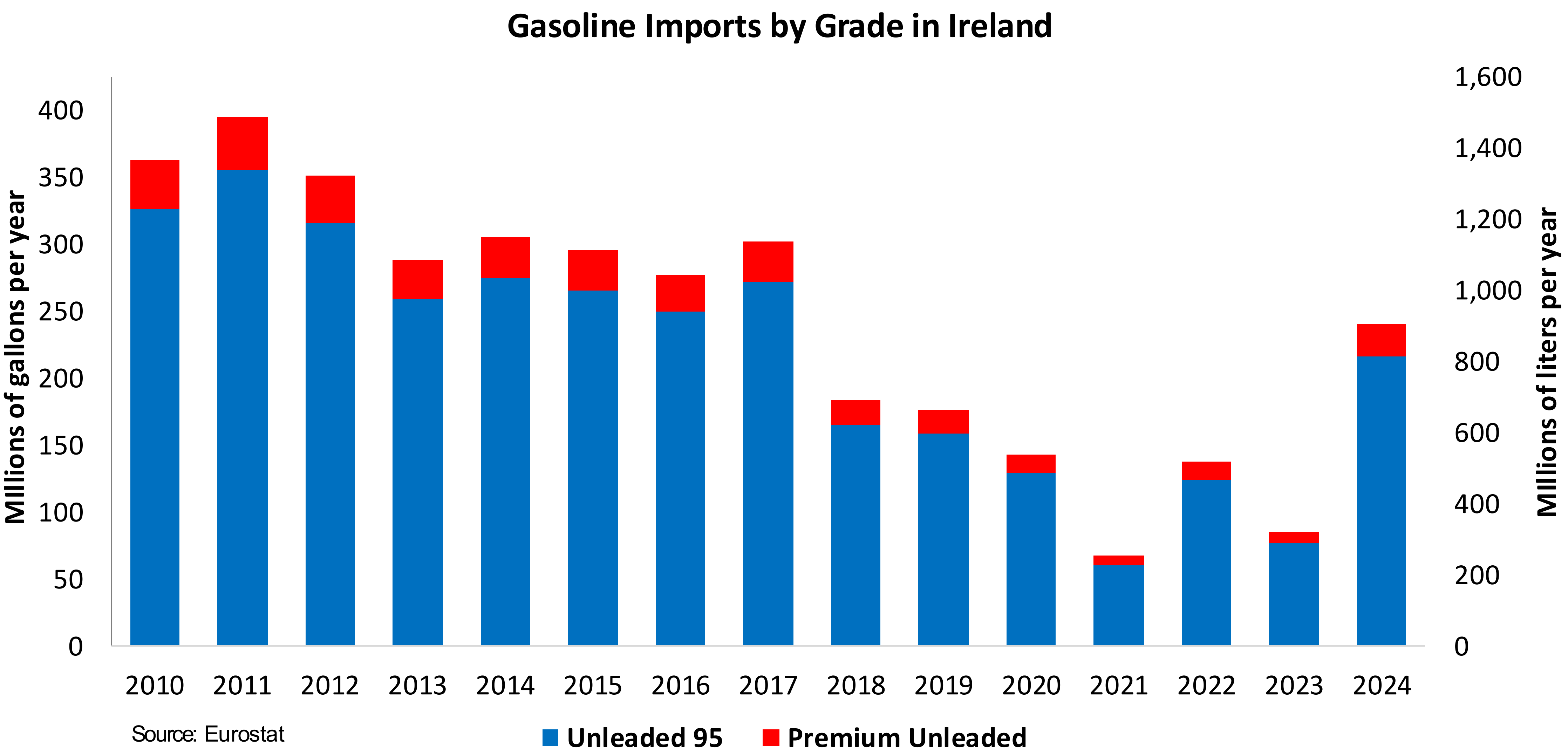
Source: Eurostat

Gasoline Production in Ireland



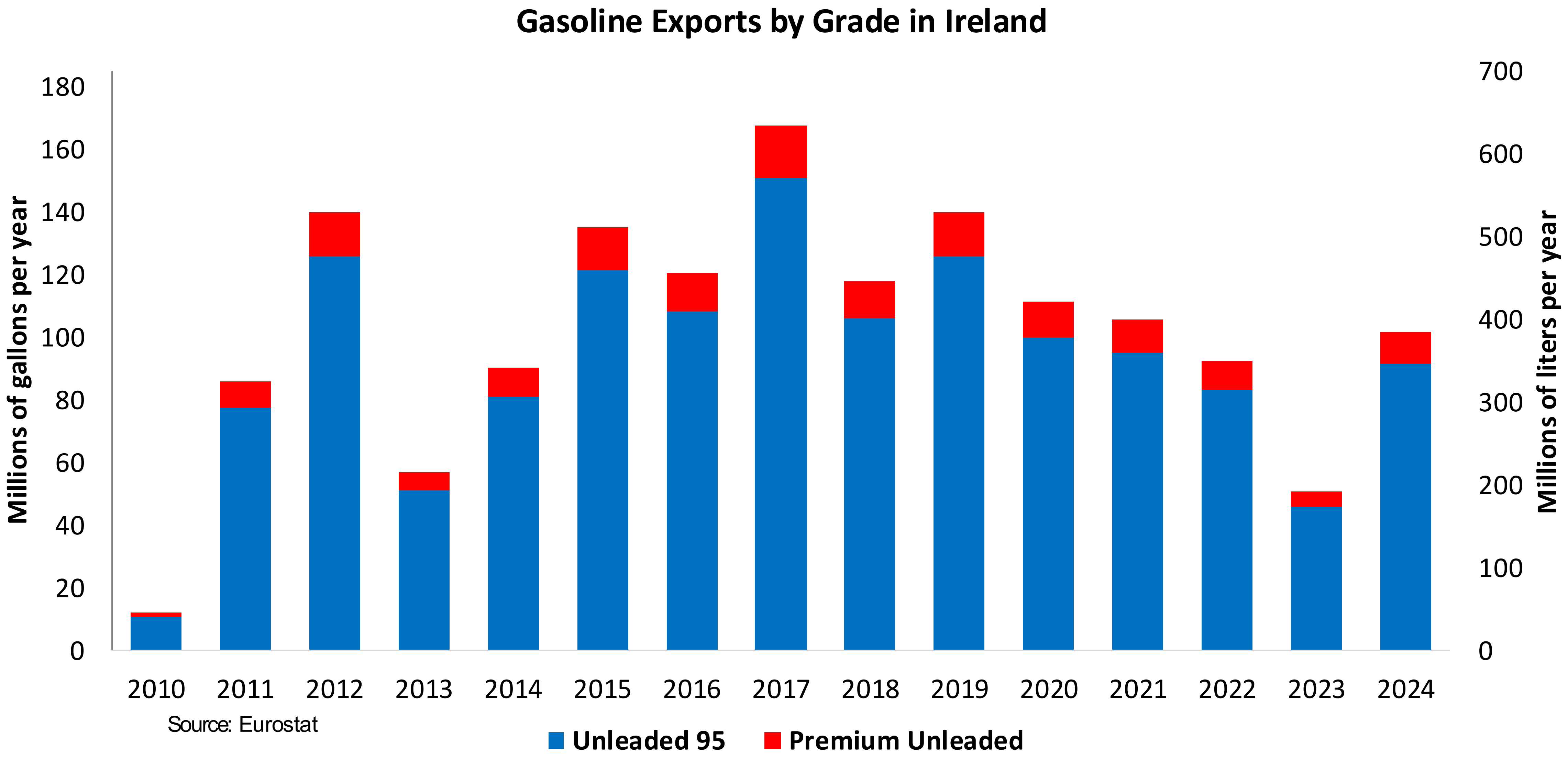
Source: Eurostat

Gasoline Imports in Ireland



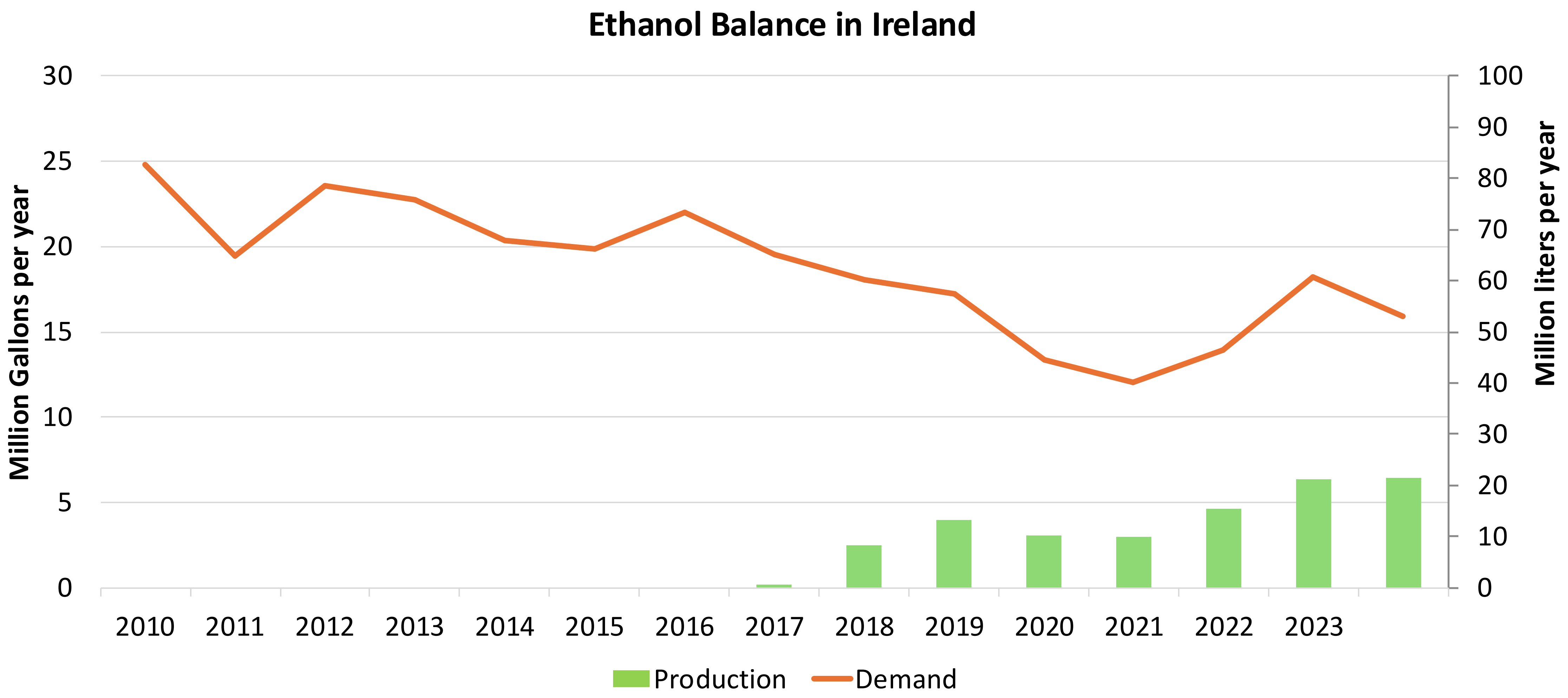
Source: Eurostat

Gasoline Exports in Ireland



Source: Eurostat

Ethanol Balance in Ireland



Source: Eurostat

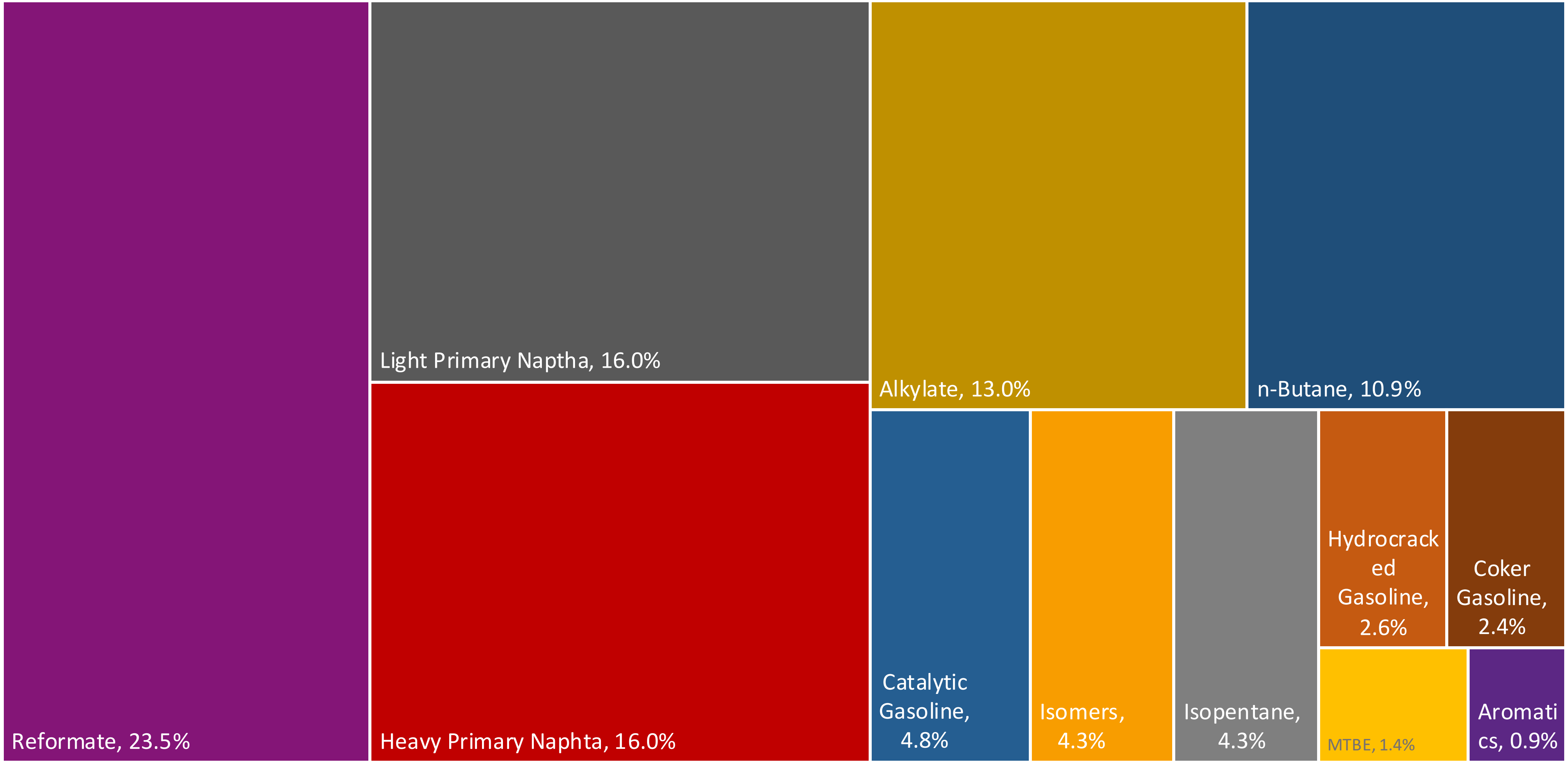
Gasoline Quality in Ireland

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	21.0							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

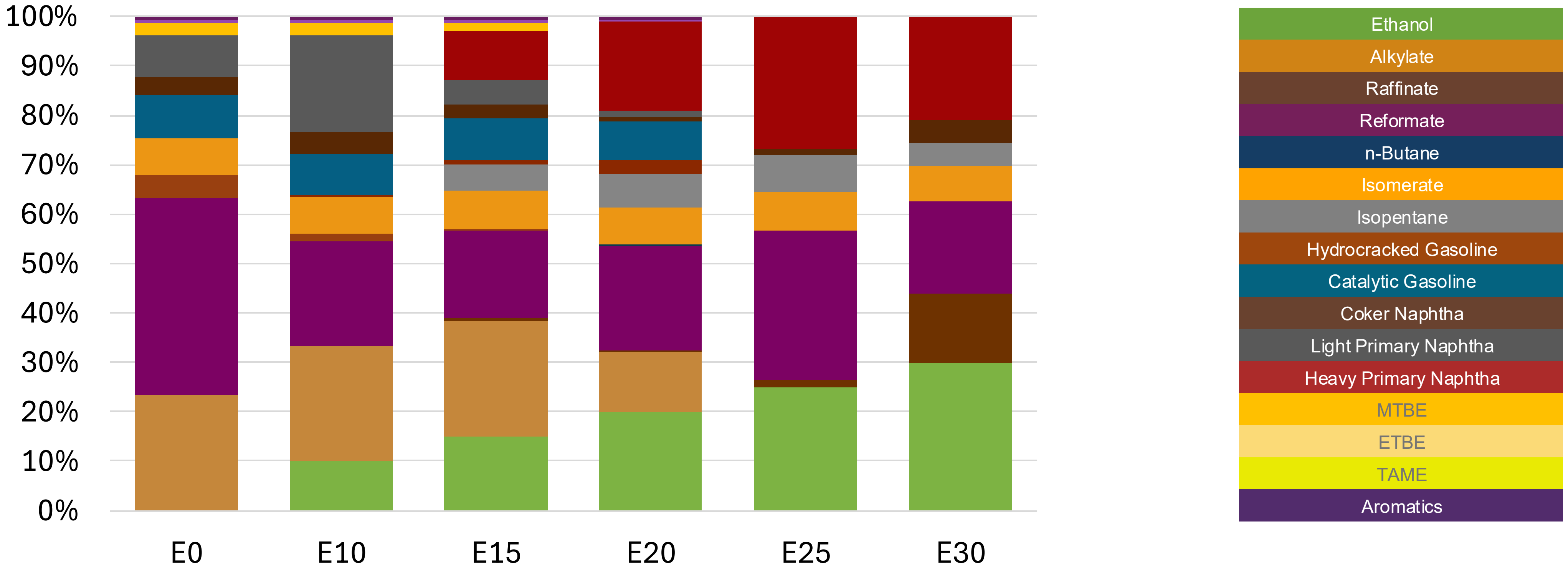
Source: European Commision

Ireland Available Blending Components



Source: Faro90

Ireland – Gasoline 95 – Constant Octane

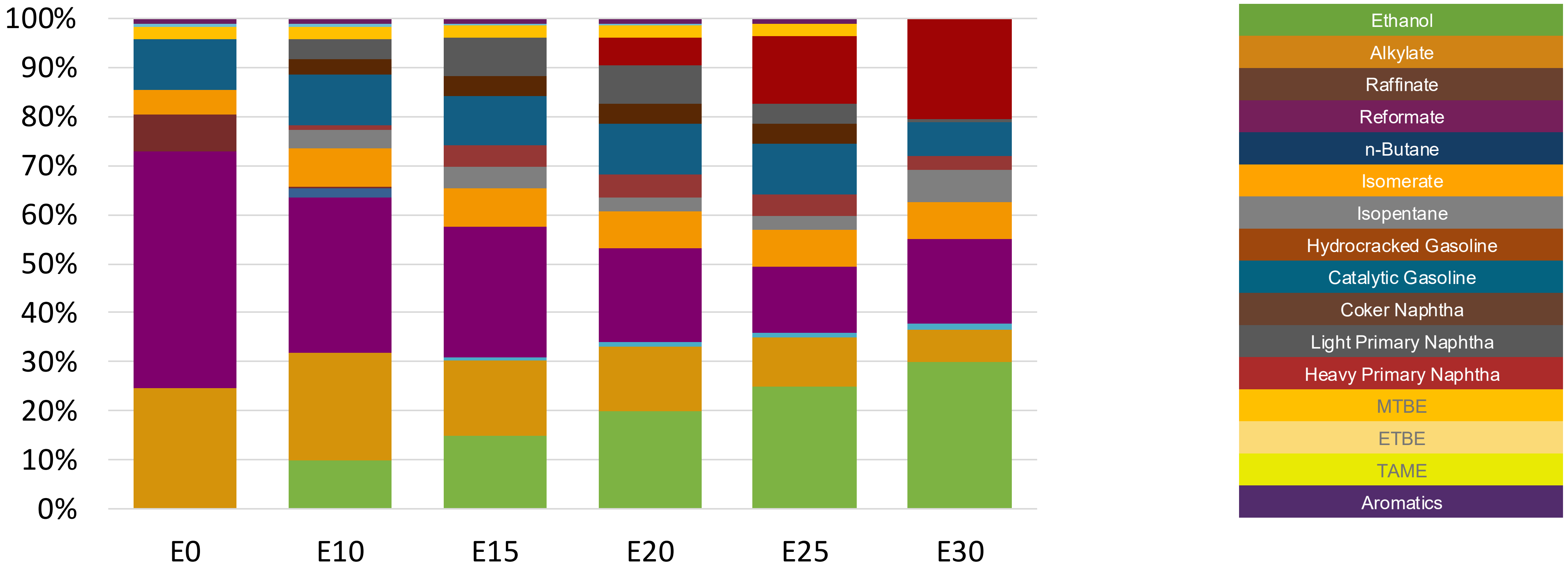


Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US$/gal)	$ 2.317	$ 2.187	$ 2.105	$ 1.996	$ 1.886	$ 1.821

Average CIF 2024 Prices

Source: Faro90

Ireland - Gasoline 98 - Constant Octane

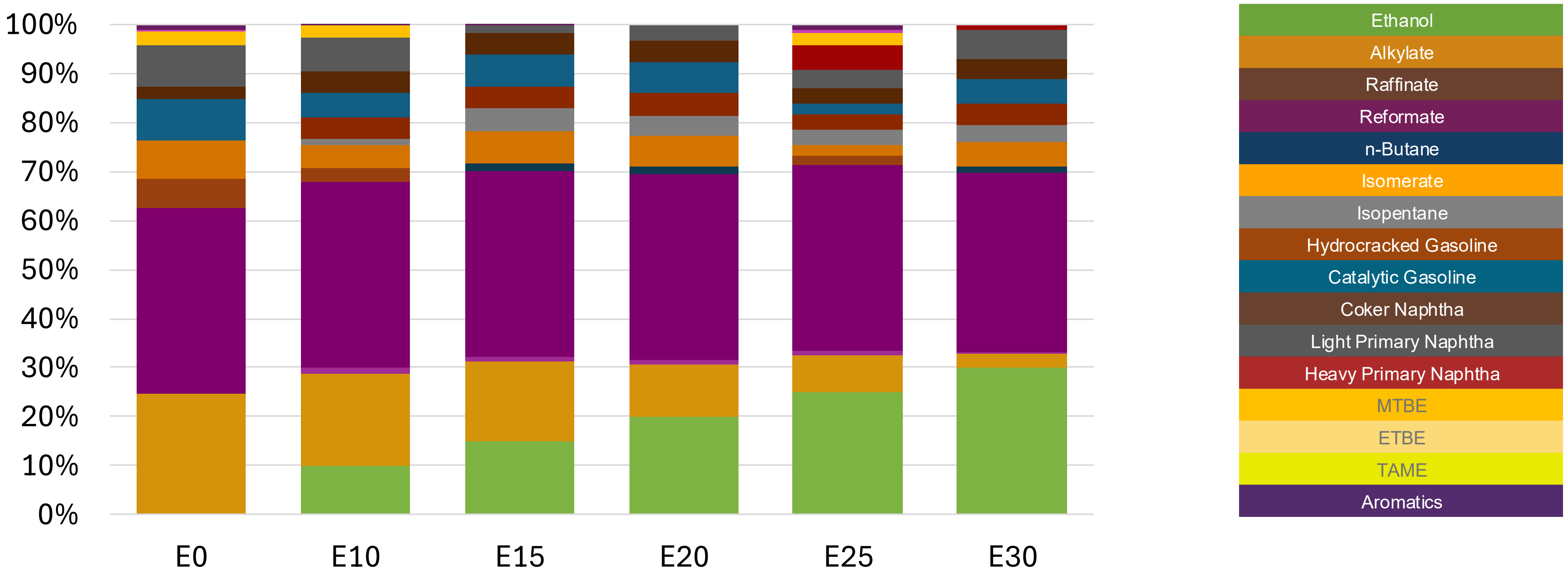


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.440	$ 2.317	$ 2.240	$ 2.157	$ 2.062	$ 1.967

Average CIF 2024 Prices

Source: Faro90

Ireland – Gasoline 95 – Increased Octane

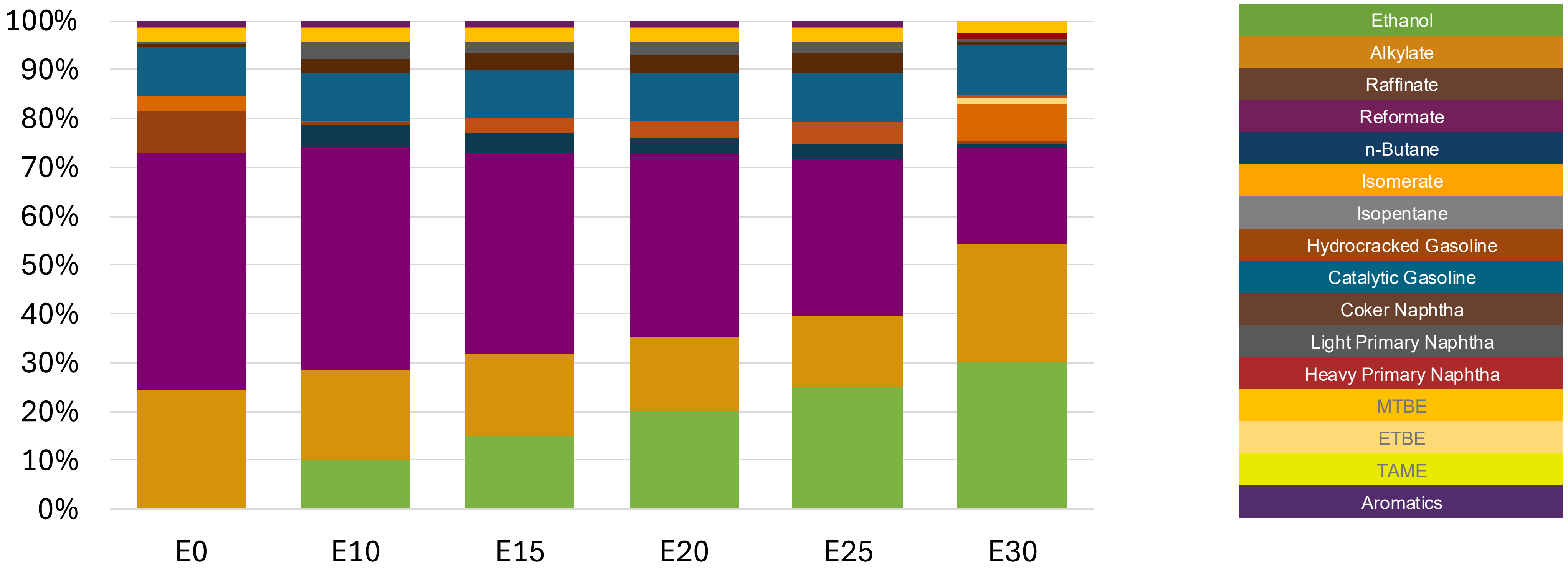


Octane (RON)	95.0	97.9	99.3	100.9	102.6	103.9
Price (US$/gal)	$ 2.301	$ 2.301	$ 2.301	$ 2.301	$ 2.301	$ 2.301

Average CIF 2024 Prices

Source: Faro90

Ireland – Gasoline 98 – Increased Octane



Octane (RON)	98.0	100.7	102.0	103.3	104.5	105.0
Price (US$/gal)	$ 2.425	$ 2.425	$ 2.425	$ 2.425	$ 2.425	$ 2.425

Average CIF 2024 Prices

Source: Faro90

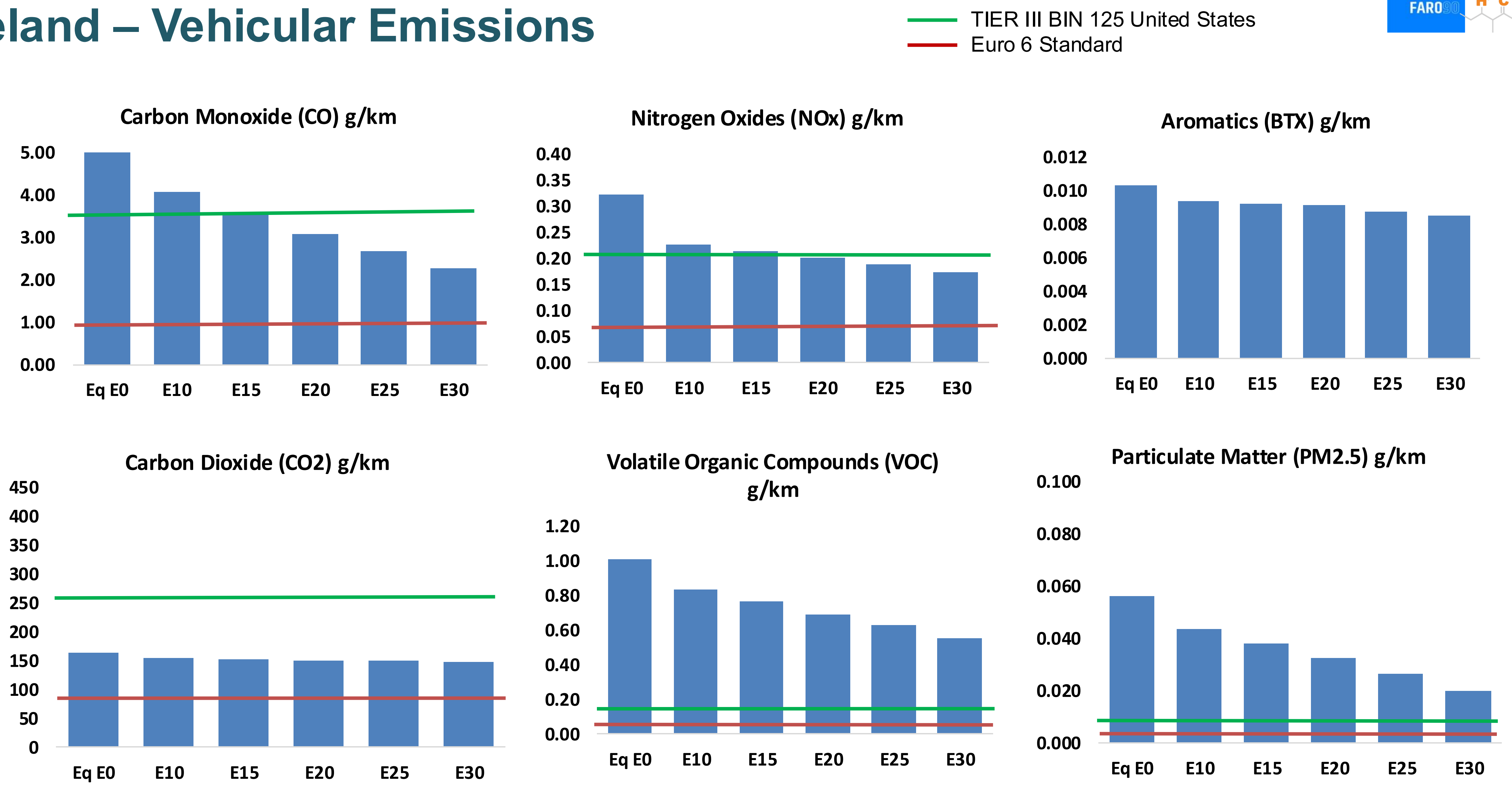
Gasoline Vehicle Fleet - Ireland

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	665
Auto/Sml Truck	Light	Single FI	3-Ways	PCV Tank	>8 years	4,144
Auto/Sml Truck	Light	Multi FI	3-Ways	PCV Tank	>8 years	182,350
Auto/Sml Truck	Light	Multi FI	Euro III	PCV Tank	>8 years	74,376
Auto/Sml Truck	Light	Multi FI	Euro IV	PCV Tank	>8 years	126,900
Auto/Sml Truck	Light	Multi FI	Euro V	PCV Tank	>8 years	45,876
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	>8 years	110,643
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	4-8 years	73,652
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	<4 years	84,390
Auto/Sml Truck	Light	Multi FI	Hybrid	PCV Tank	All	67,895
Auto/Sml Truck	Light	Electric			All	42,814
Auto/Sml Truck	Medium	Carburator	3-Ways	PCV Tank	>8 years	66
Auto/Sml Truck	Medium	Single FI	3-Ways	PCV Tank	>8 years	36
Auto/Sml Truck	Medium	Multi FI	3-Ways	PCV Tank	>8 years	14,554
Auto/Sml Truck	Medium	Multi FI	Euro III	PCV Tank	>8 years	12,905
Auto/Sml Truck	Medium	Multi FI	Euro IV	PCV Tank	>8 years	42,313
Auto/Sml Truck	Medium	Multi FI	Euro V	PCV Tank	>8 years	14,119
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	>8 years	36,881
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	4-8 years	32,793
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	<4 years	66,776
Auto/Sml Truck	Medium	Multi FI	Hybrid	PCV Tank	All	47,405
Auto/Sml Truck	Medium	Electric			All	32,969
Sml Engines	Light	4-Cycle	None	PCV	>8 years	32,042
Sml Engines	Light	4-Cycle	Euro III	PCV	>8 years	32,589
Sml Engines	Light	4-Cycle	Euro IV	PCV	>8 years	189,233
Sml Engines	Light	4-Cycle	Euro V	PCV	>8 years	113,307
Sml Engines	Light	4-Cycle	Euro 6	PCV	>8 years	237,737
Sml Engines	Light	4-Cycle	Euro 6	PCV	4-8 years	190,758
Sml Engines	Light	4-Cycle	Euro 6	PCV	<4 years	502,310
Sml Engines	Light	4-Cycle	Hybrid	PCV	All	19,308
Sml Engines	Light	Electric			All	12,175
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	23,204
Truck	Heavy	Fuel-Injection	3-Ways	PCV	>8 years	262,238
Truck	Heavy	Fuel-Injection	Euro III	PCV	>8 years	147,072
Truck	Heavy	Fuel-Injection	Euro IV	PCV	>8 years	169,535
Truck	Heavy	Fuel-Injection	Euro V	PCV	>8 years	47,459
Truck	Heavy	Fuel-Injection	Euro 6	PCV	>8 years	65,520
Truck	Heavy	Fuel-Injection	Euro 6	PCV	4-8 years	30,890
Truck	Heavy	Fuel-Injection	Euro 6	PCV	<4 years	4,013
Truck	Heavy	Fuel-Injection	Hybrid	PCV	All	7,855
Truck	Heavy	Electric			All	5,463

Vehicle Fleet: 3,207,228
Gasoline Vehicle Fleet: 1,641,194

Source: Eurostat, ACEA

Ireland – Vehicular Emissions



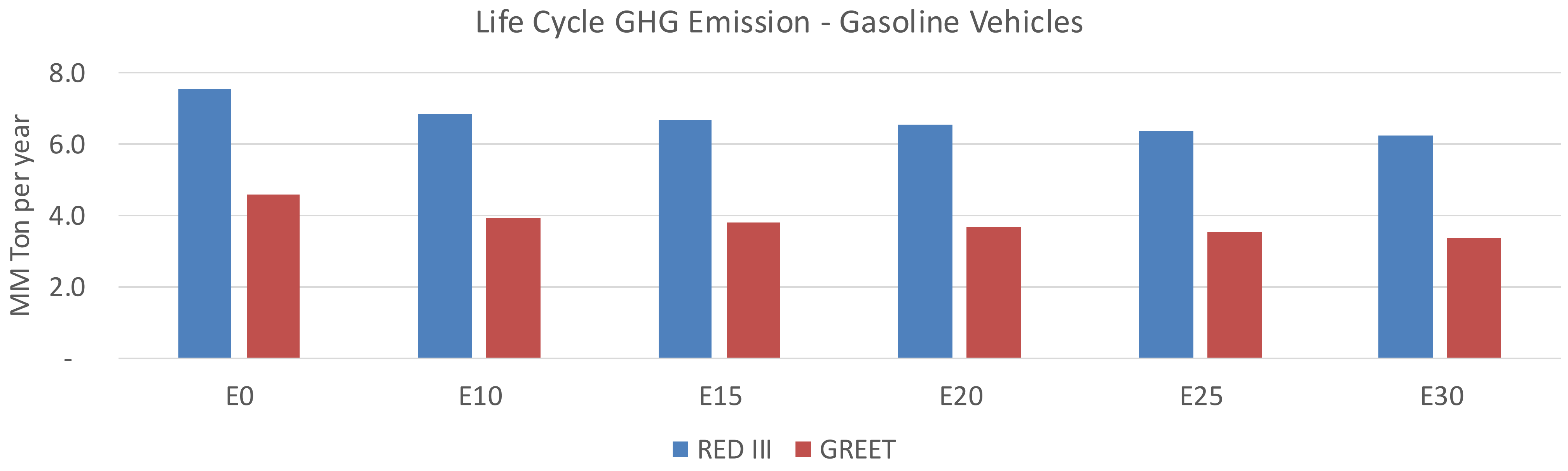
Source: Faro90

Ireland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	211,328	188,438	165,548	145,494	124,716	108,559	92,049	-22%	-41%
CO2	6,629,738	6,459,259	6,298,251	6,171,623	6,108,963	6,075,101	5,963,120	-5%	-8%
VOC	40,883	37,376	33,869	30,924	27,928	25,614	22,561	-17%	-32%
NOx	13,148	9,861	9,204	8,675	8,181	7,637	7,061	-30%	-38%
SOx	74	68	63	58	53	49	43	-15%	-28%
NH3	2,749	2,722	2,695	2,708	2,738	2,760	2,777	-2%	0%
N2O	118	117	116	117	119	121	122	-1%	2%
CH4	8,968	8,968	8,968	9,103	9,300	9,471	9,632	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	216	203	190	180	170	162	152	-12%	-22%
Acetaldehyde	725	832	1,221	1,860	2,534	2,895	3,421	68%	249%
Formaldehyde	2,850	2,771	3,230	3,793	3,973	4,396	4,785	13%	39%
Benzene	420	404	383	377	374	357	346	-9%	-11%
PM2.5	447	397	347	301	256	208	157	-22%	-43%
PM10	1,837	1,631	1,425	1,238	1,051	853	646	-22%	-43%

Source: Faro90

Ireland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	70.0
Target @ 2035 (MMT/year)	39.4
Delta	30.6

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.1%	17.5%	20.3%	23.1%	26.6%
RED II/III	0.0%	9.1%	11.4%	13.3%	15.1%	17.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.1%	2.6%	3.1%	3.5%	4.0%
RED II/III	0.0%	2.2%	2.8%	3.3%	3.7%	4.3%

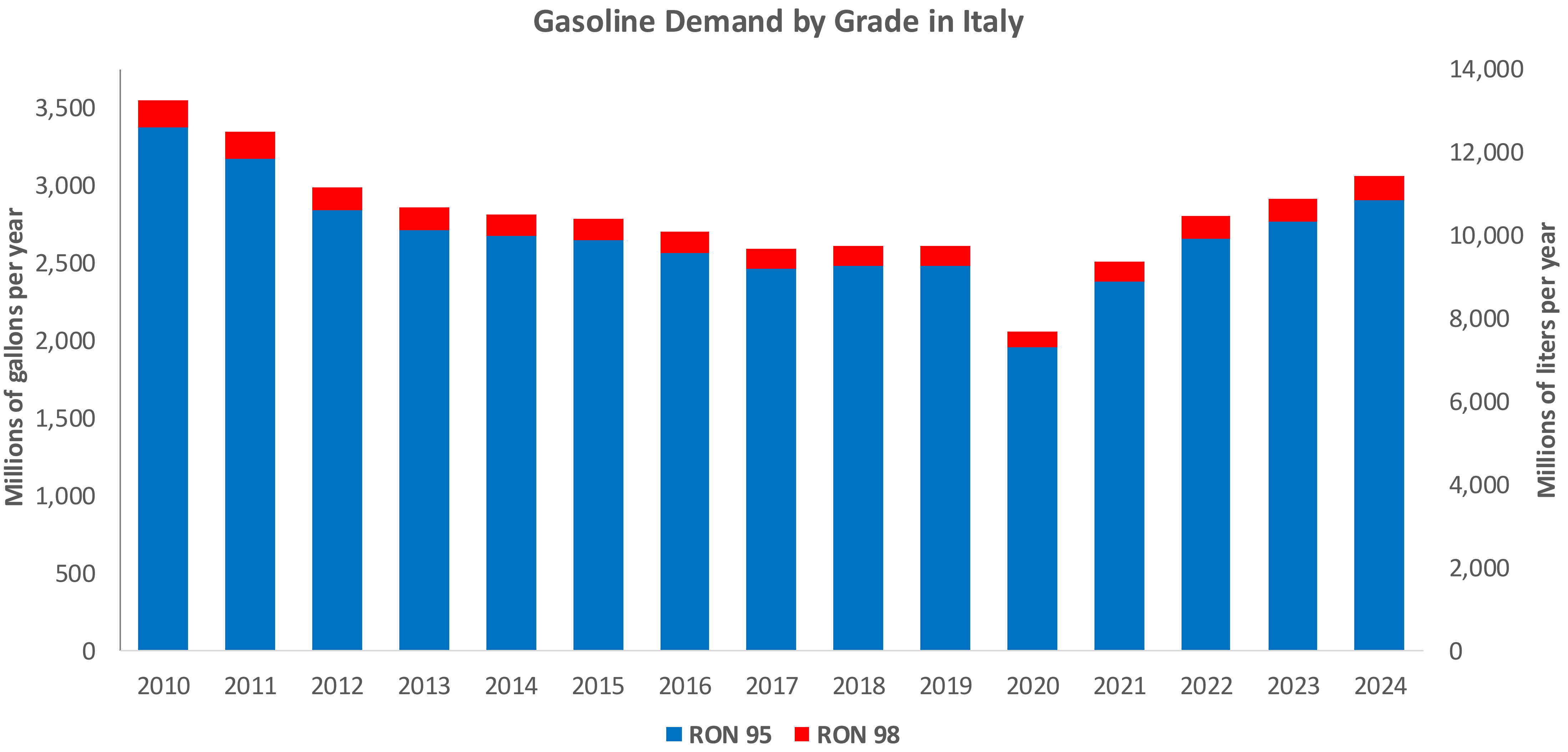
Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Italy

In 2024, gasoline consumption in Italy was 11,426 million liters (3,018 million gallons) and growth rate was 3.2%. Gasoline production and net imports were 17,947 and -6,521 million liters (4,741 and -1,723 million gallons) correspondingly. Italy complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

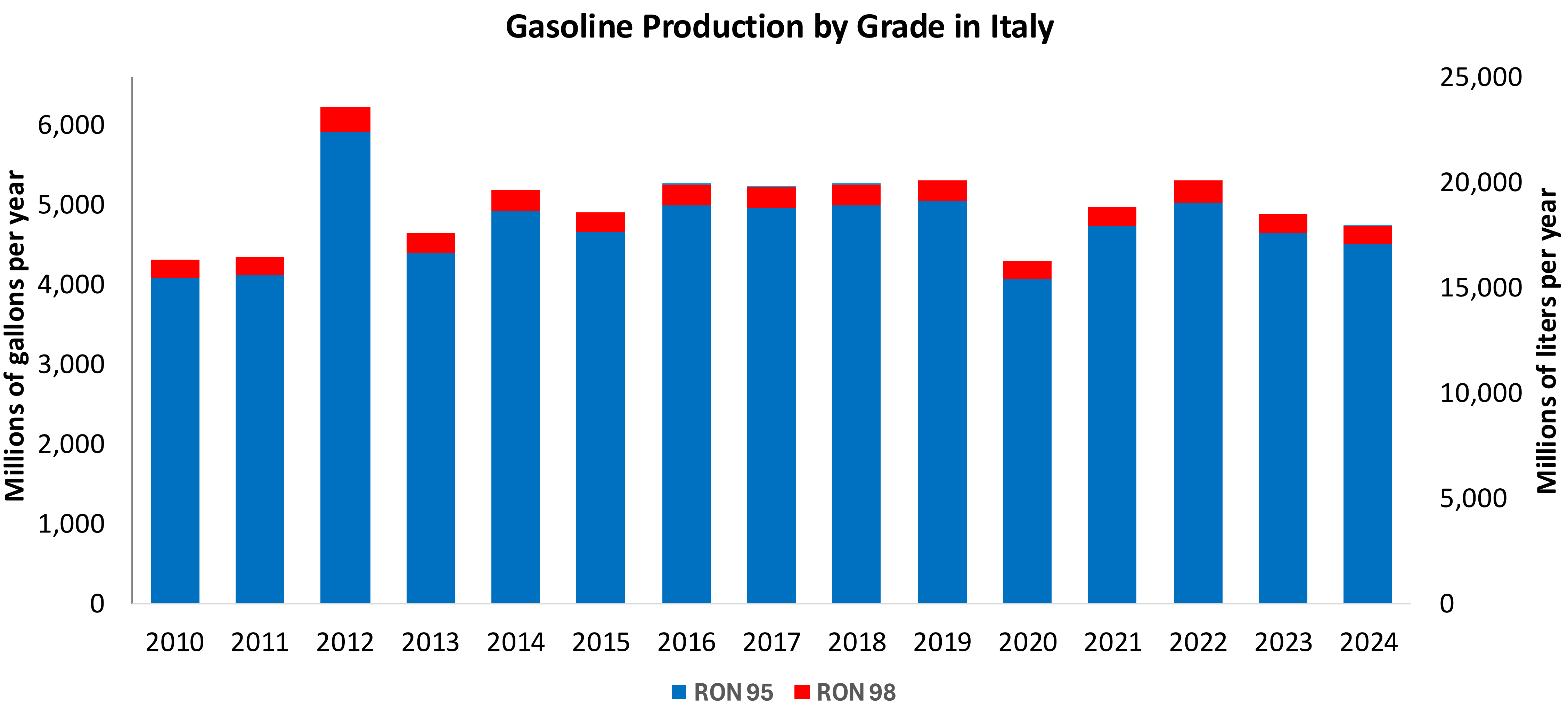
In 2024, ethanol consumption in Italy was 640 million liters (169 million gallons) representing 5.6% of gasoline demand. Ethanol production and Net Imports were 700 and -60 million liters (169 and -16 million gallons) correspondingly. Ethanol sources are Wine and Second Generation Lignocelulosic ethanol.

Gasoline Demand in Italy



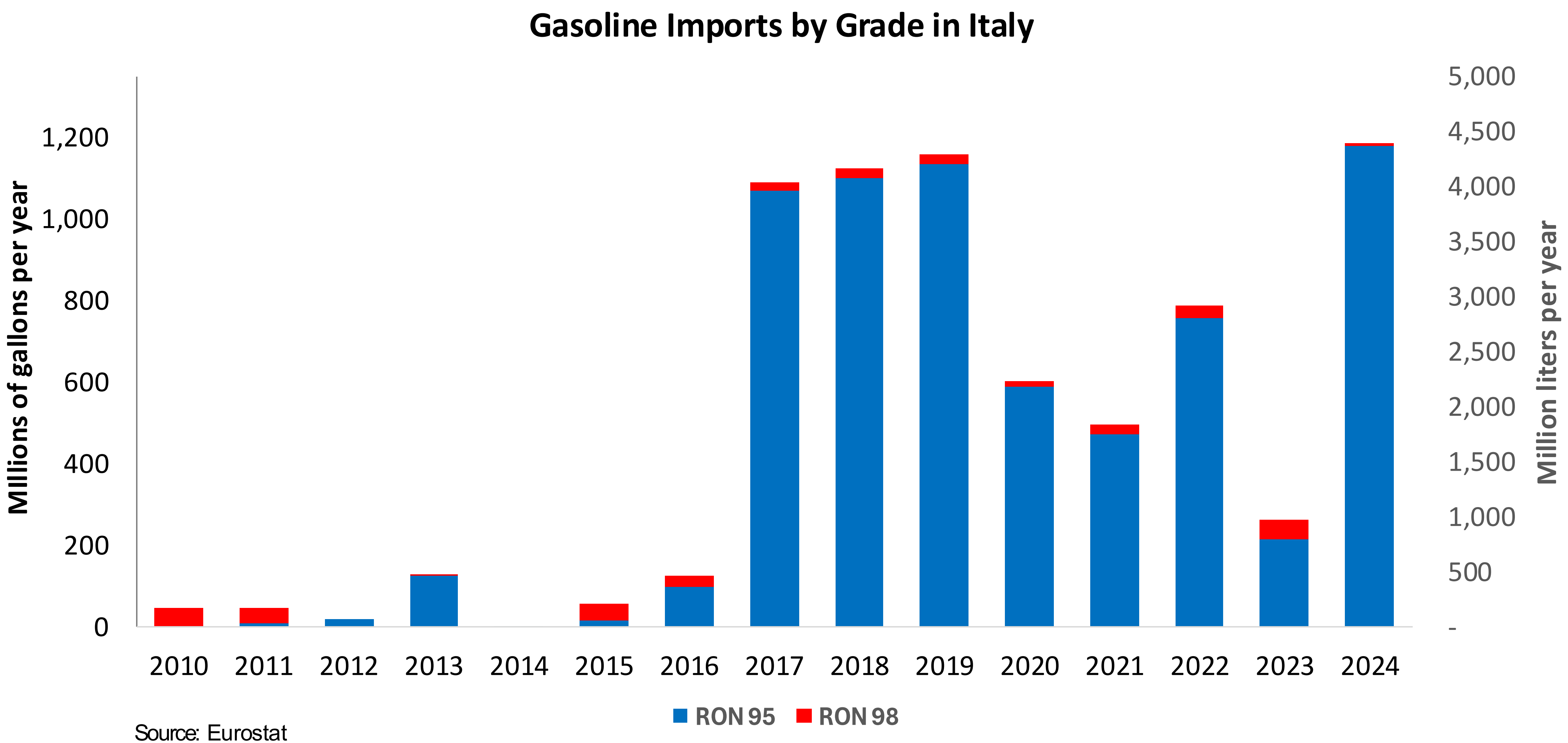
Source: Eurostat

Gasoline Production in Italy



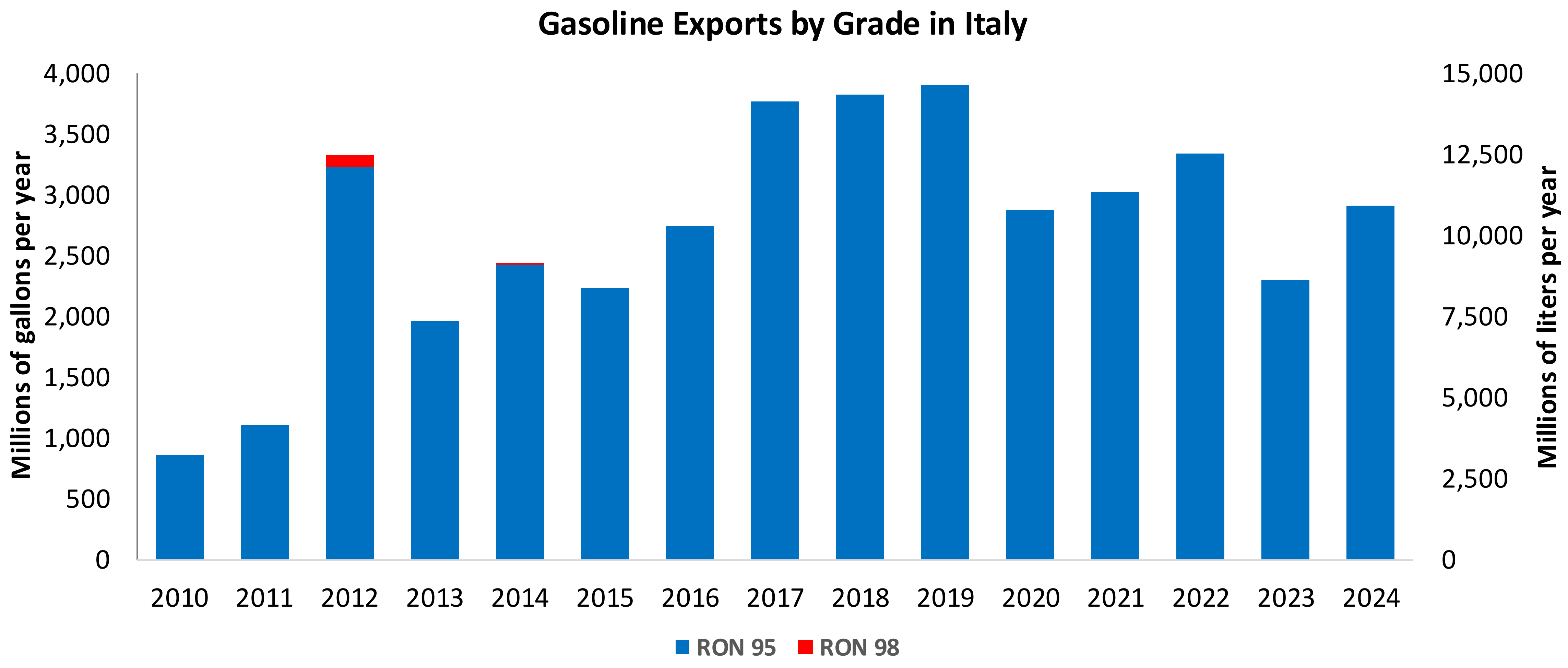
Source: Eurostat

Gasoline Imports in Italy



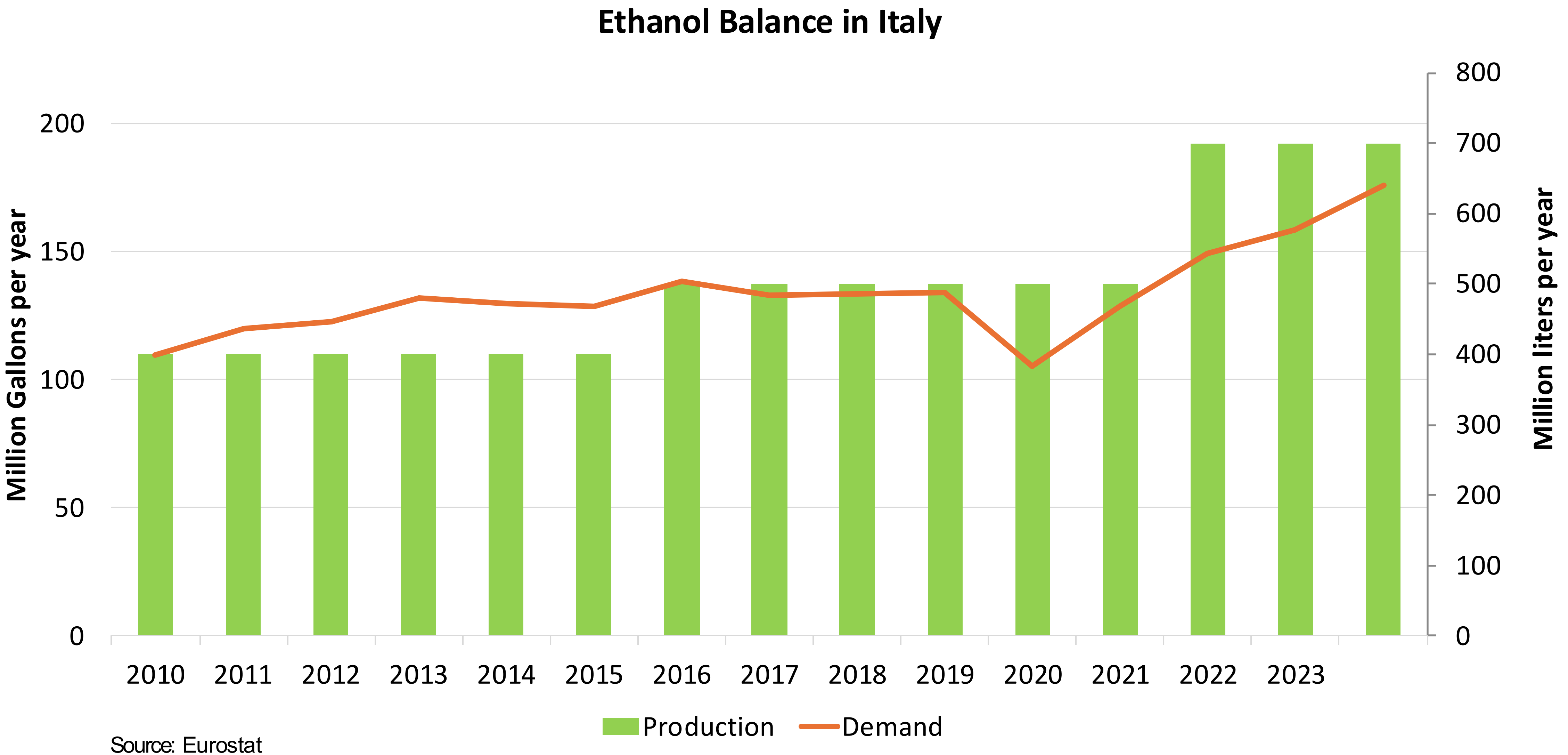
Source: Eurostat

Gasoline Exports in Italy



Source: Eurostat

Ethanol Balance in Italy



Source: Eurostat

Gasoline Quality in Italy

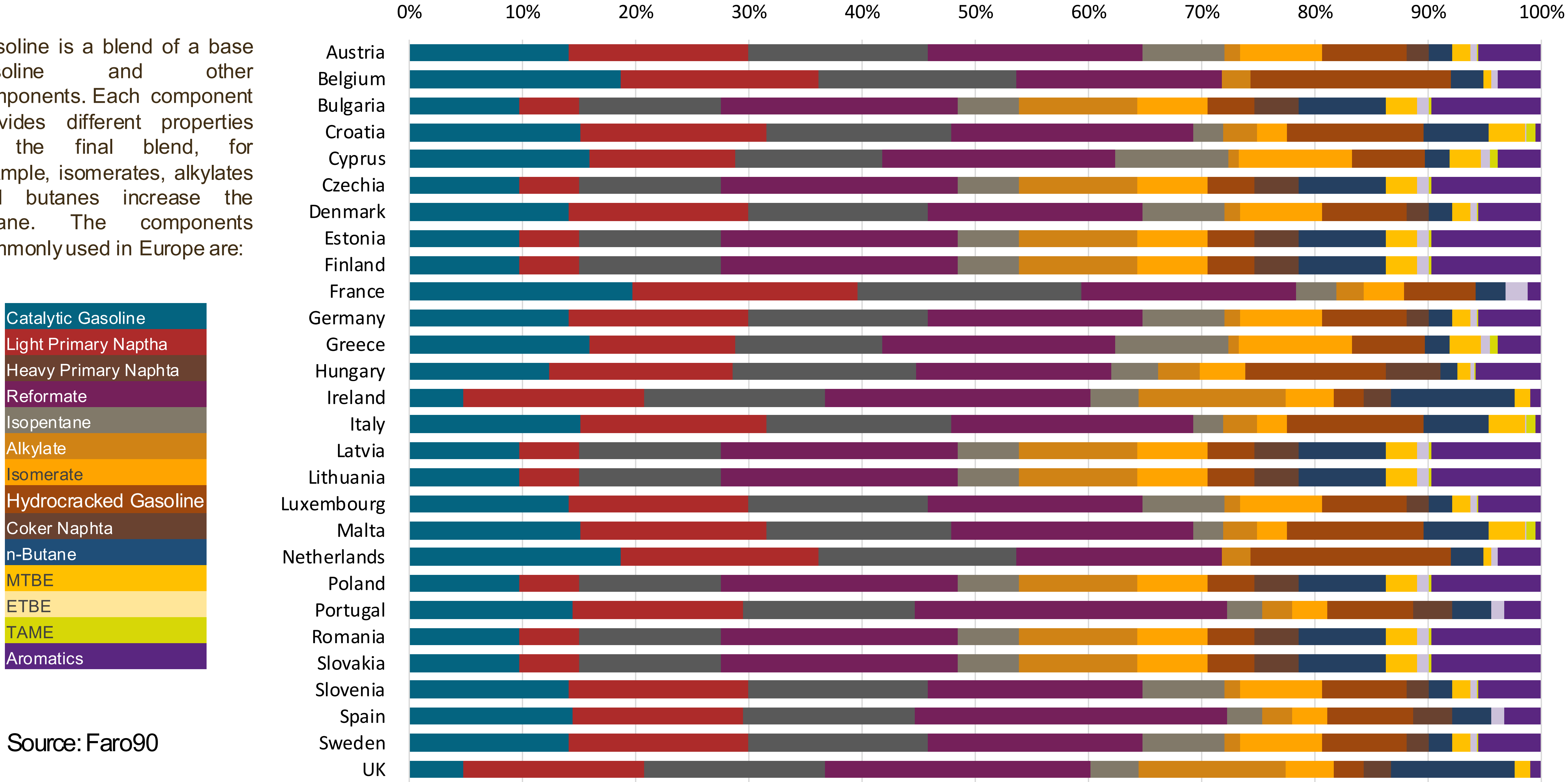
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.7							
Advanced Biofuels (%cal)	4.9							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

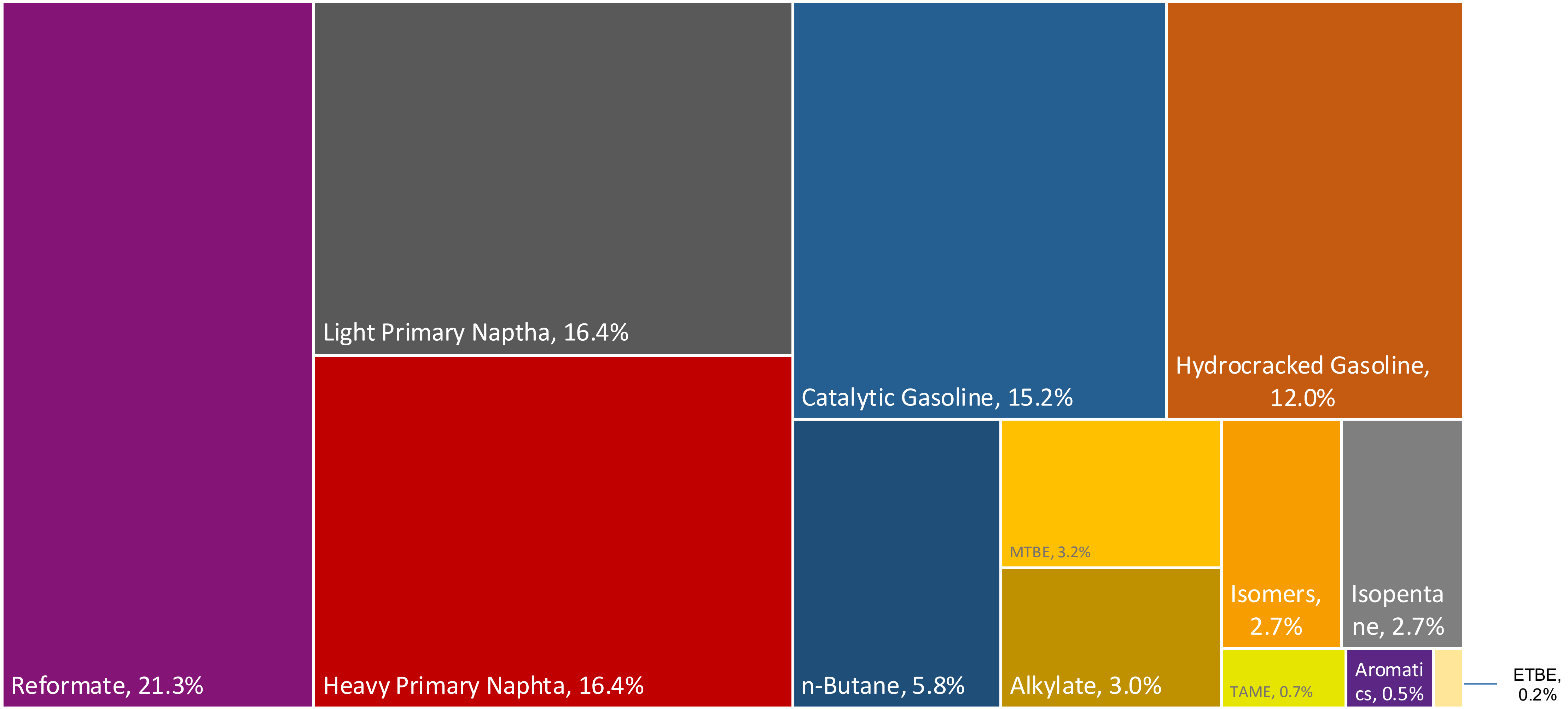
Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



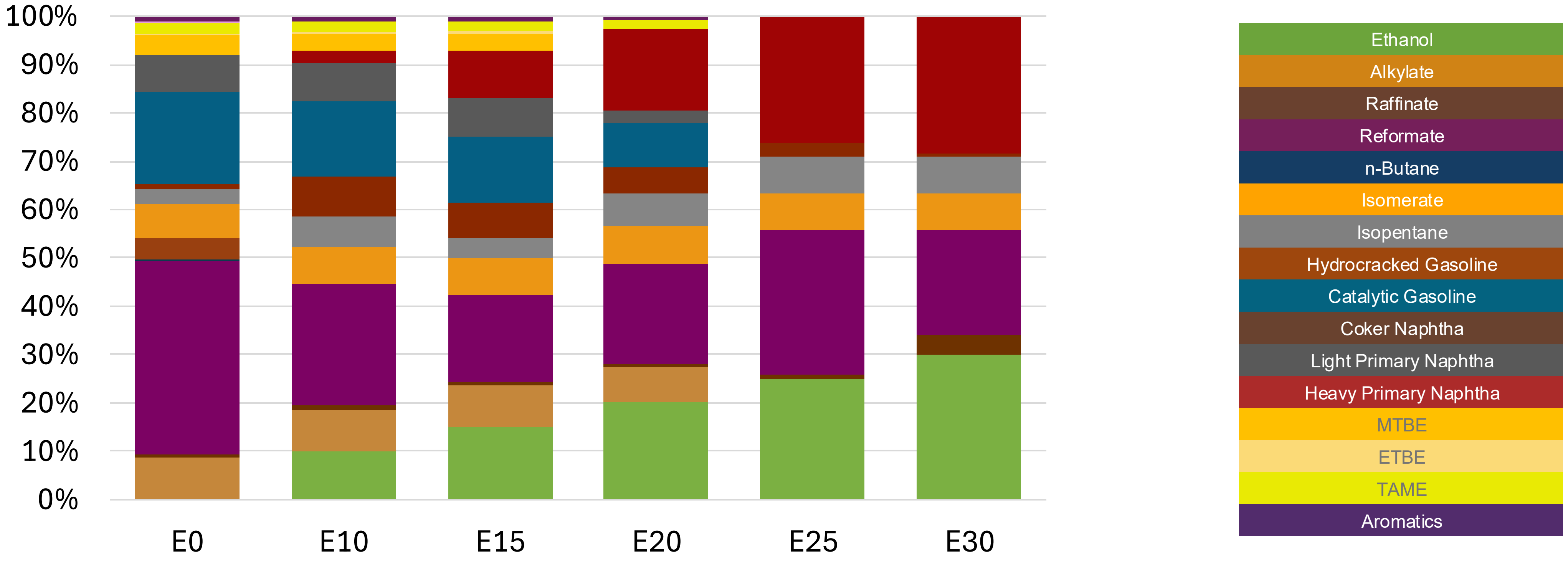
Source: Faro90

Italy Available Blending Components



Source: Faro90

Italy – Gasoline 95 – Constant Octane

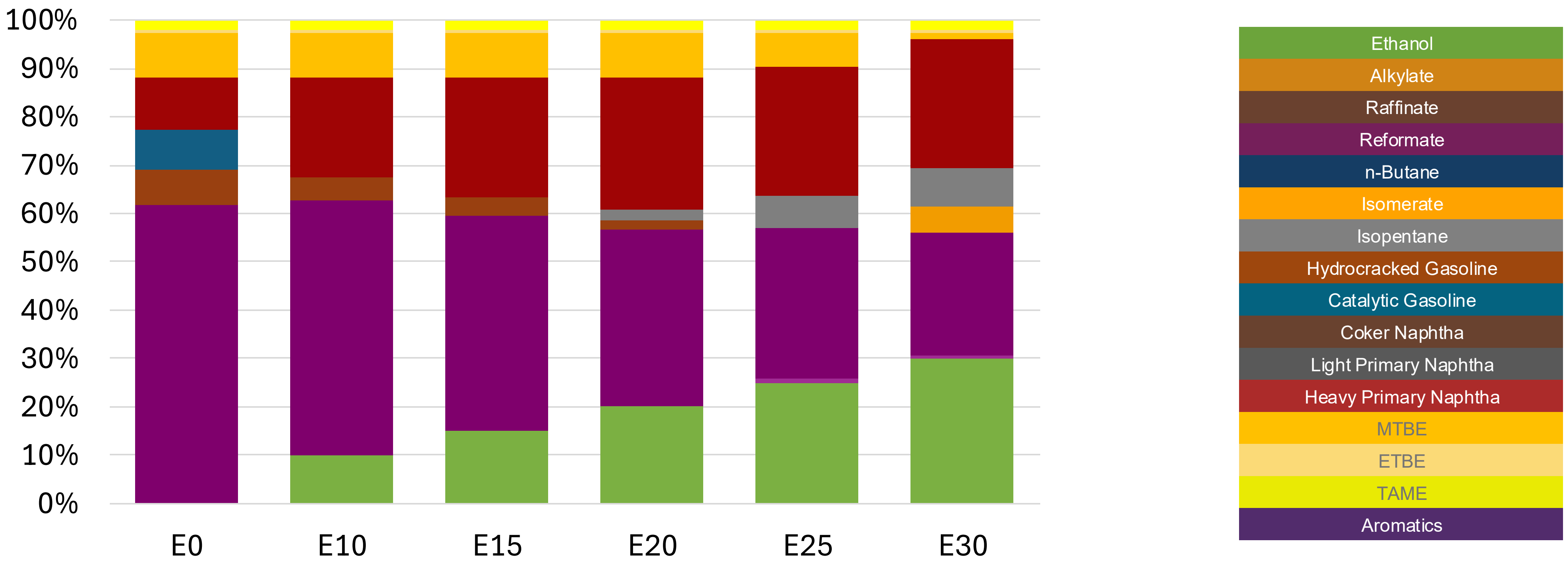


Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US$/gal)	$ 2.258	$ 2.145	$ 2.062	$ 1.986	$ 1.891	$ 1.817

Average CIF 2024 Prices

Source: Faro90

Italy - Gasoline 98 - Constant Octane

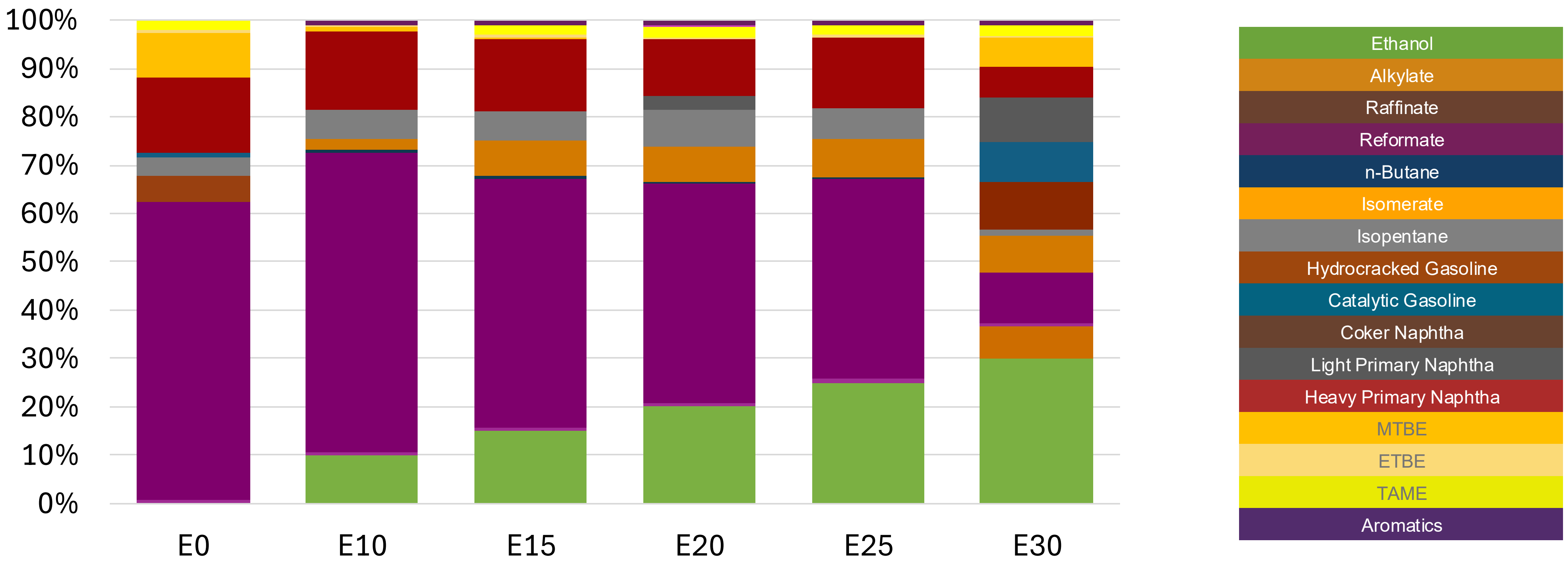


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.307	$ 2.188	$ 2.119	$ 2.049	$ 1.979	$ 1.926

Average CIF 2024 Prices

Source: Faro90

Italy – Gasoline 95 – Increased Octane

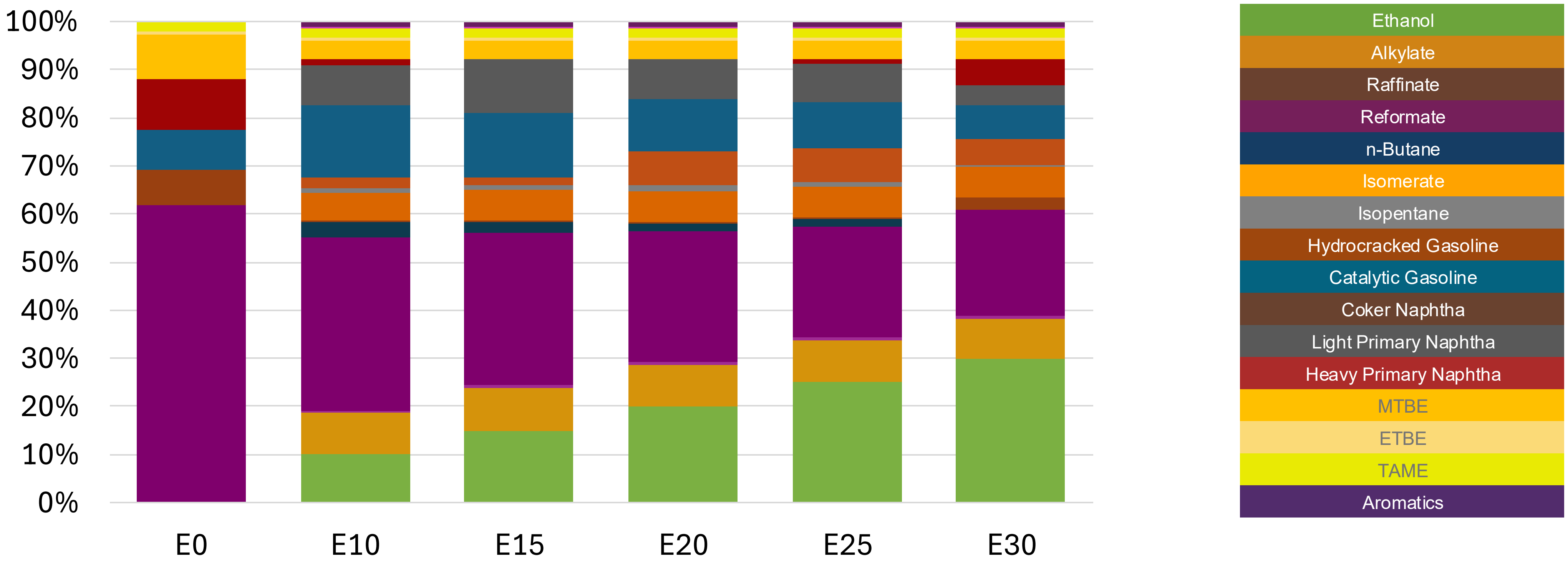


Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US$/gal)	$ 2.177	$ 2.177	$ 2.177	$ 2.177	$ 2.177	$ 2.177

Average CIF 2024 Prices

Source: Faro90

Italy – Gasoline 98 – Increased Octane



Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US$/gal)	$ 2.307	$ 2.307	$ 2.307	$ 2.307	$ 2.307	$ 2.307

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Italy

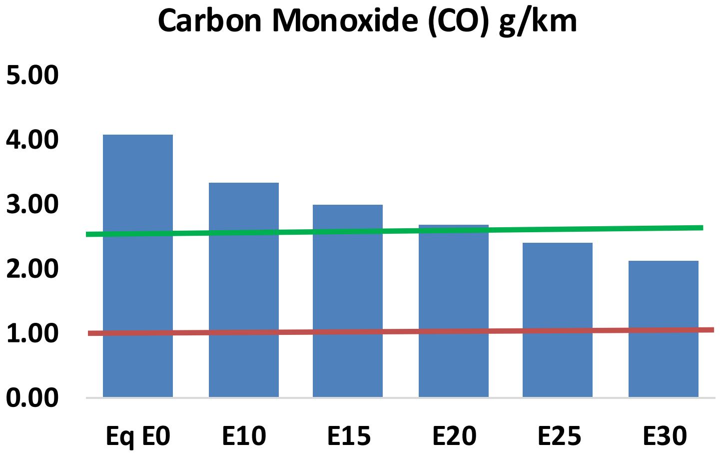
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	999,433
		Single FI	3-Ways		>8 years	870,534
		Multi FI	3-Ways		>8 years	2,211,158
			Euro III		>8 years	2,427,825
			Euro IV		>8 years	5,601,663
			Euro V		>8 years	1,672,037
			Euro 6		>8 years	4,047,100
					4-8 years	4,888,395
					<4 years	4,201,859
			Hybrid		All	1,209
		Electric			All	1,564
	Medium	Carburator	3-Ways	PCV Tank	>8 years	538,156
		Single FI	3-Ways		>8 years	468,749
		Multi FI	3-Ways		>8 years	1,190,624
			Euro III		>8 years	1,307,291
			Euro IV		>8 years	3,016,280
			Euro V		>8 years	900,328
			Euro 6		>8 years	2,179,208
					4-8 years	2,632,213
					<4 years	2,262,539
			Hybrid		All	651
		Electric			All	842
Sml Engines	Light	4-Cycle	None	PCV	>8 years	2,287,812
			Euro III		>8 years	1,000,017
			Euro IV		>8 years	1,590,335
			Euro V		>8 years	333,345
			Euro 6		>8 years	730,725
					4-8 years	853,794
					<4 years	1,227,742
			Hybrid		All	368
		Electric			All	57,168
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	685,119
		Fuel-Injection	3-Ways		>8 years	728,143
			Euro III		>8 years	689,622
			Euro IV		>8 years	991,361
			Euro V		>8 years	255,001
			Euro 6		>8 years	670,803
					4-8 years	882,382
					<4 years	786,697
			Hybrid		All	45,846
		Electric			All	25,755

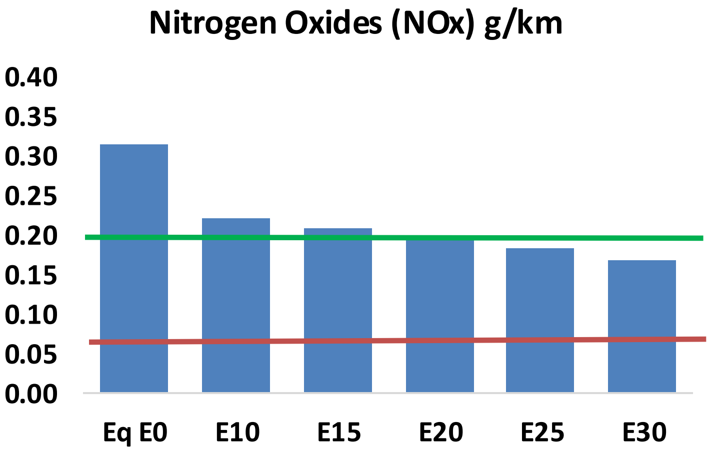
Vehicle Fleet: 55,261,692
Gasoline Vehicle Fleet: 19,966,496

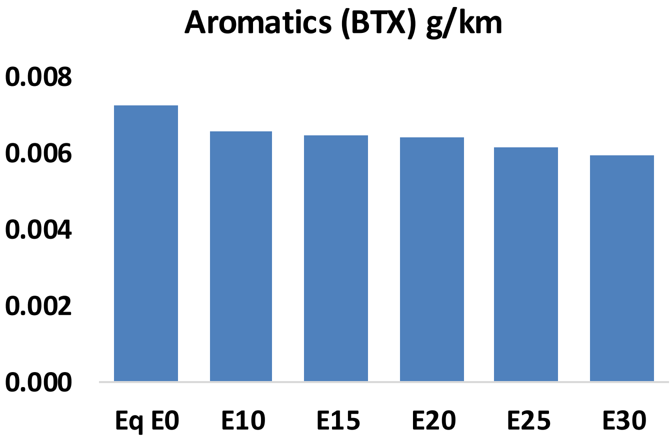
Source: Eurostat, ACEA

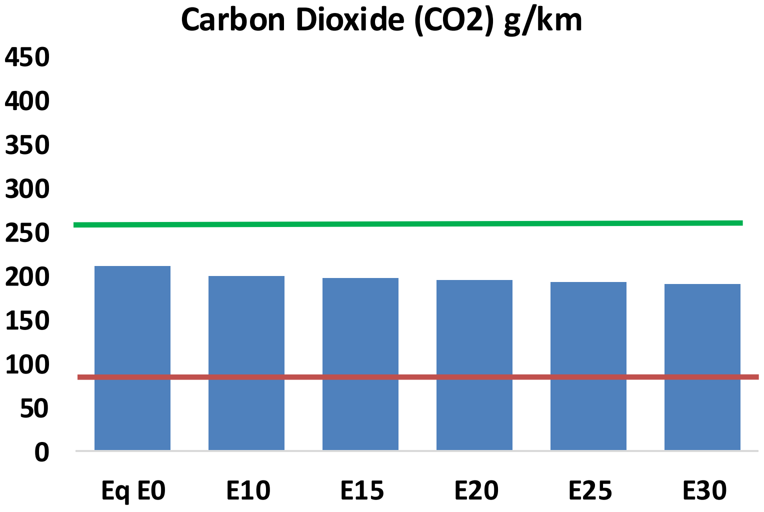
Italy – Vehicular Emissions

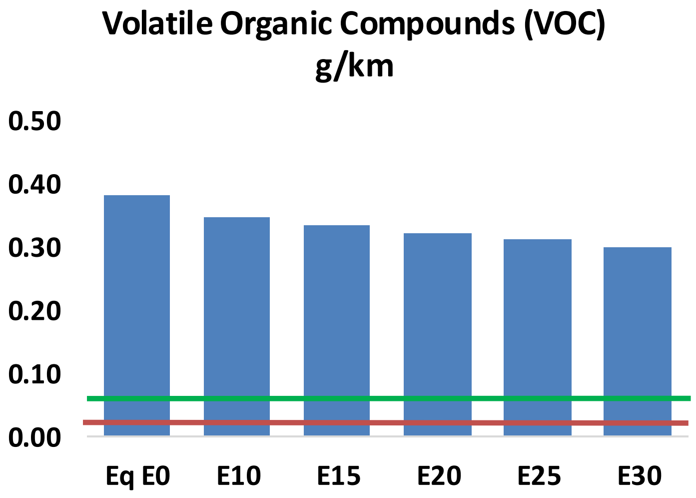
TIER III BIN 125 United States
Euro 6 Standard

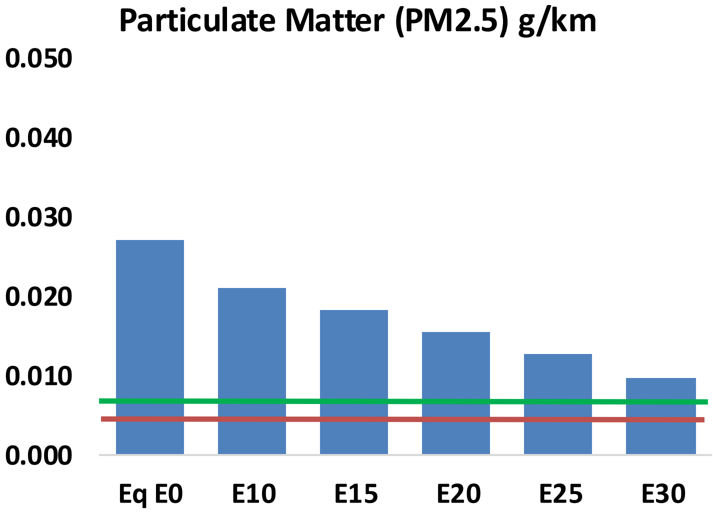












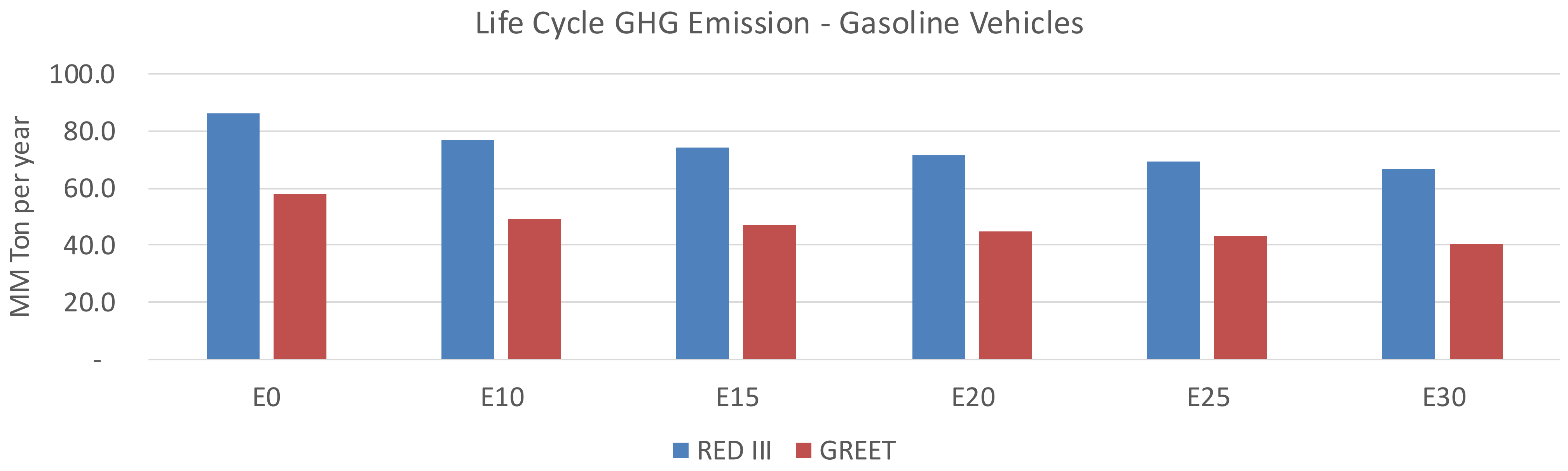
Source: Faro90

Italy – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,868,067	1,694,065	1,520,062	1,371,545	1,219,358	1,101,511	973,684	-19%	-35%
CO2	96,686,516	94,200,291	91,852,190	90,005,478	89,091,652	88,644,245	87,010,277	-5%	-8%
VOC	175,091	166,801	158,512	152,728	147,372	143,394	137,590	-9%	-16%
NOx	144,409	108,307	101,087	95,274	89,852	83,873	77,548	-30%	-38%
SOx	1,112	1,028	944	873	805	731	653	-15%	-28%
NH3	33,122	32,794	32,466	32,620	32,987	33,245	33,460	-2%	0%
N2O	1,578	1,570	1,563	1,571	1,603	1,621	1,640	-1%	2%
CH4	40,511	40,511	40,511	41,118	42,010	42,779	43,509	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,540	1,462	1,384	1,328	1,275	1,236	1,179	-10%	-17%
Acetaldehyde	3,291	3,777	5,542	8,440	11,498	13,139	15,525	68%	249%
Formaldehyde	10,962	10,658	12,424	14,588	15,283	16,907	18,406	13%	39%
Benzene	3,310	3,181	3,014	2,967	2,942	2,814	2,723	-9%	-11%
PM2.5	5,909	5,246	4,584	3,981	3,381	2,743	2,079	-22%	-43%
PM10	6,477	5,751	5,025	4,364	3,706	3,007	2,279	-22%	-43%

Source: Faro90

Italy – Gasoline Vehicle Fleet Life Cycle GHG Emissions



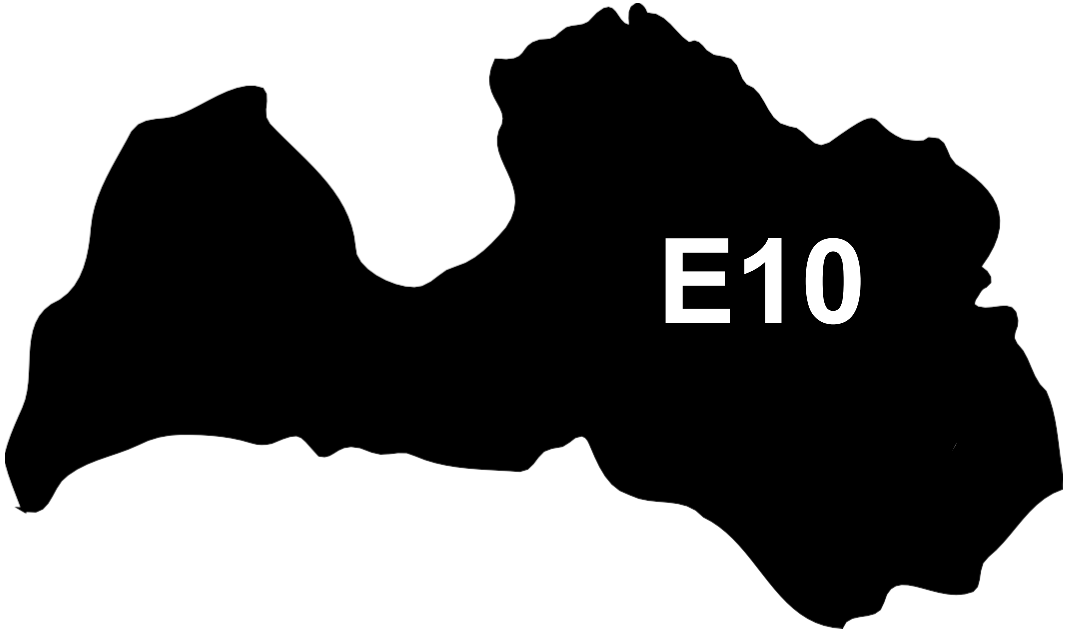
Country Emissions 2024 (MMT/año)	357.1
Target @2035 (MMT/year)	201.0
Delta	156.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.7%	18.8%	22.3%	25.7%	30.0%
RED II/III	0.0%	10.8%	14.0%	16.7%	19.3%	22.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.5%	7.0%	8.3%	9.5%	11.1%
RED II/III	0.0%	6.0%	7.7%	9.2%	10.6%	12.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Latvia

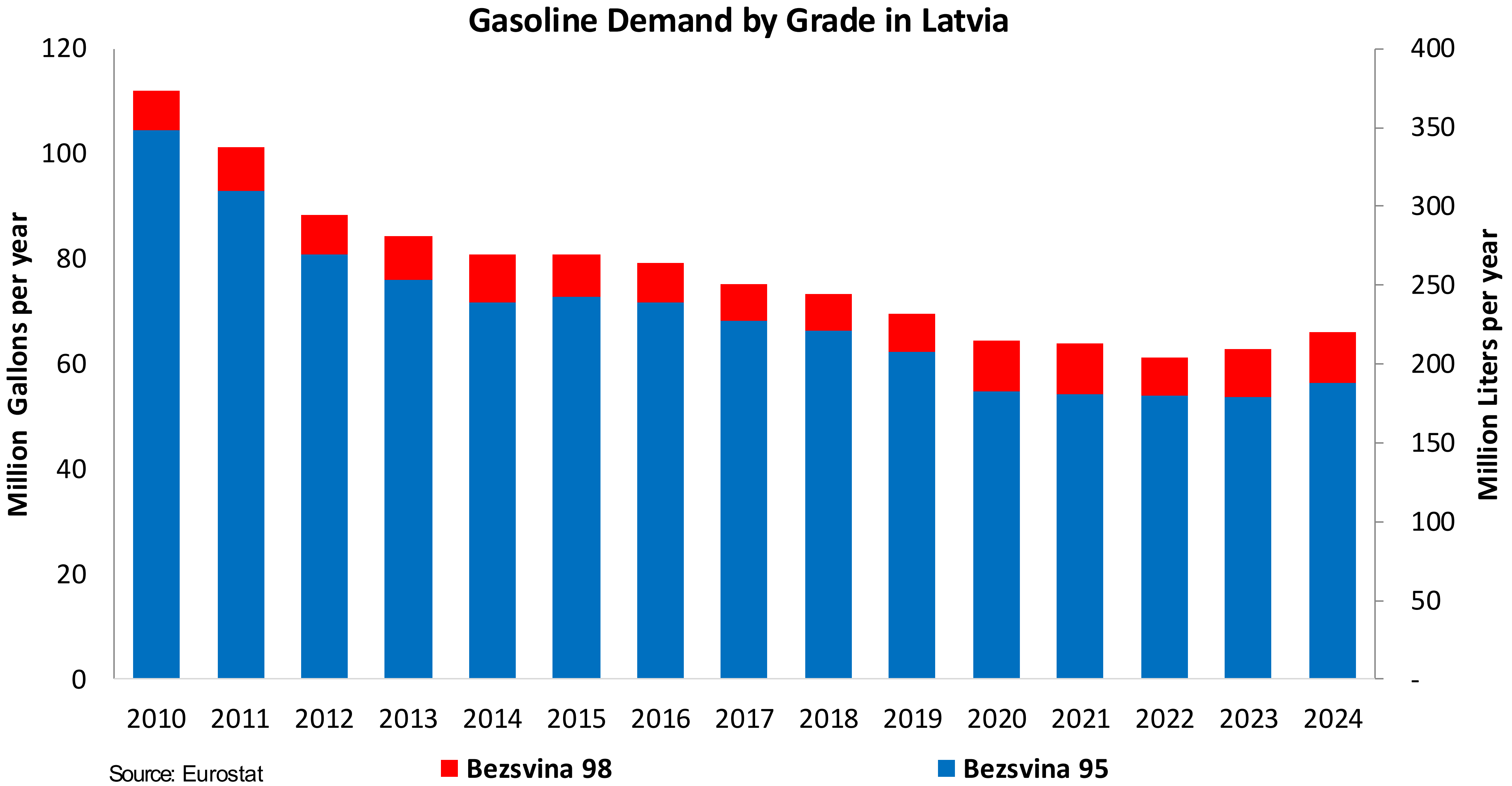


In 2024, gasoline consumption in Latvia was 220 million liters (58 million gallons) and growth rate was -1.1%. There is no gasoline production. Gasoline net imports were 200 million liters (53 million gallons). Latvia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.4 RON.

In 2024, ethanol consumption in Latvia was 23 million liters (6 million gallons) representing 10.4% of gasoline demand. Ethanol production and Net Imports were 25 and -3 million liters (6 and -1 million gallons) correspondingly. Ethanol source Cereals 100%.

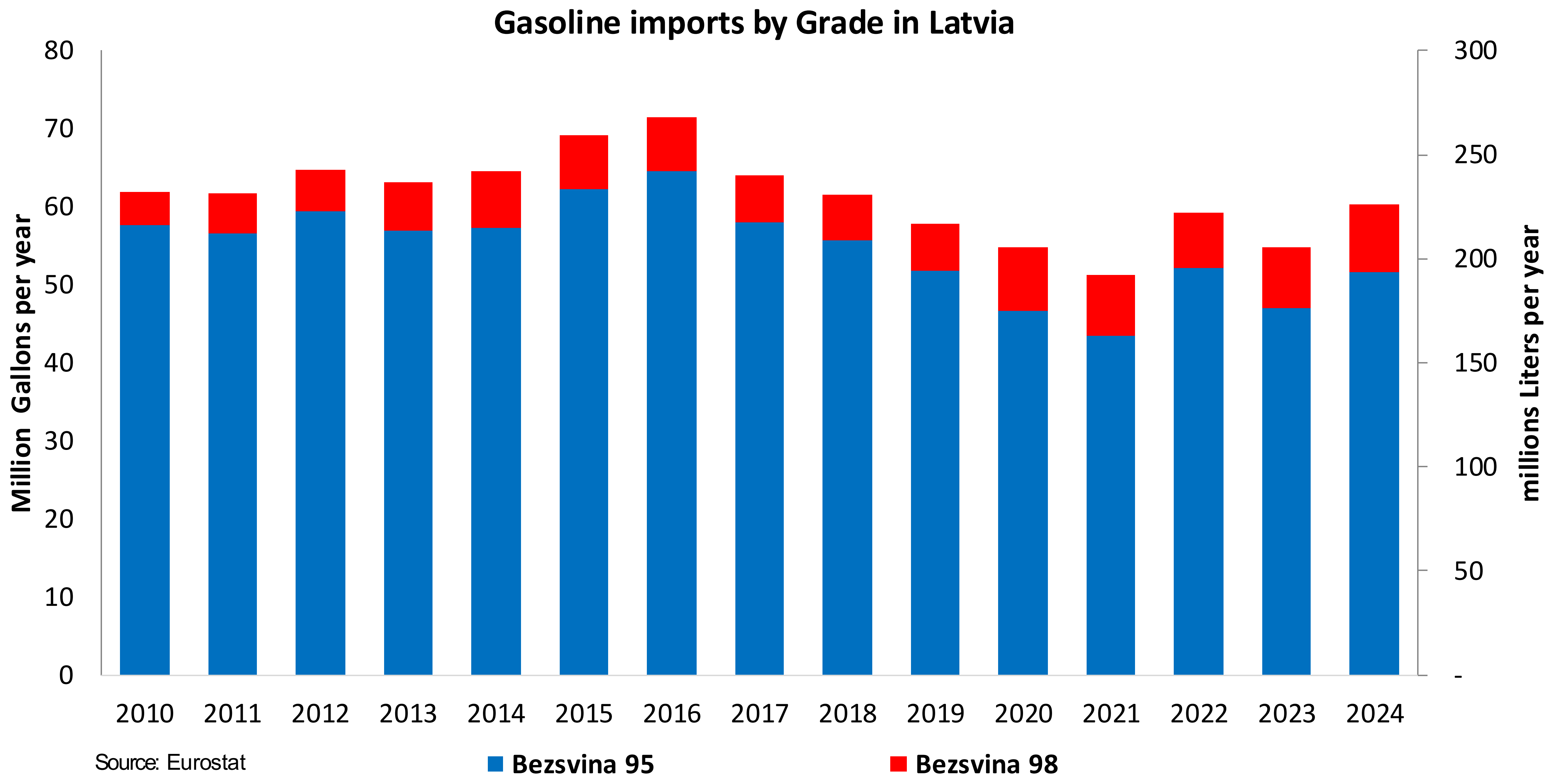
Source: Eurostat, European Commision

Gasoline Demand in Latvia



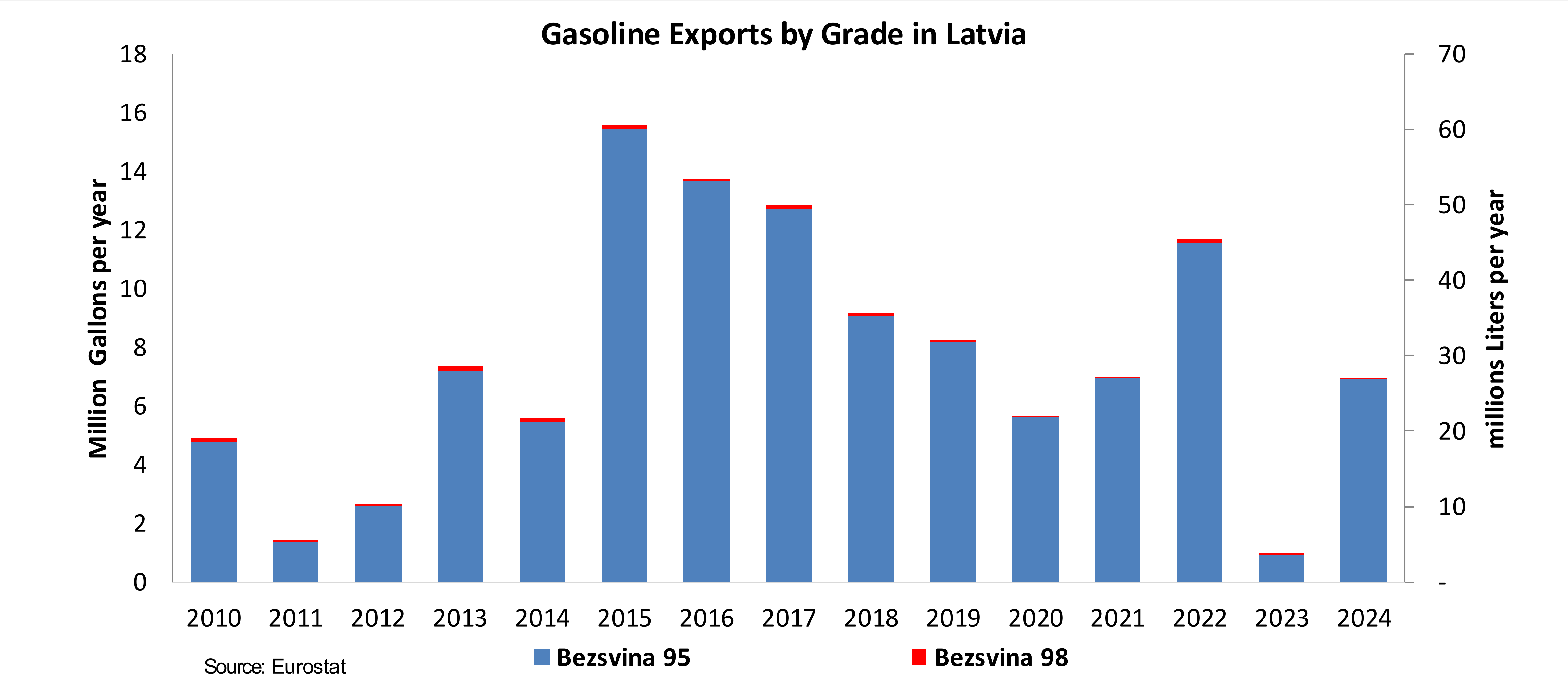
Source: Eurostat

Gasoline Imports in Latvia

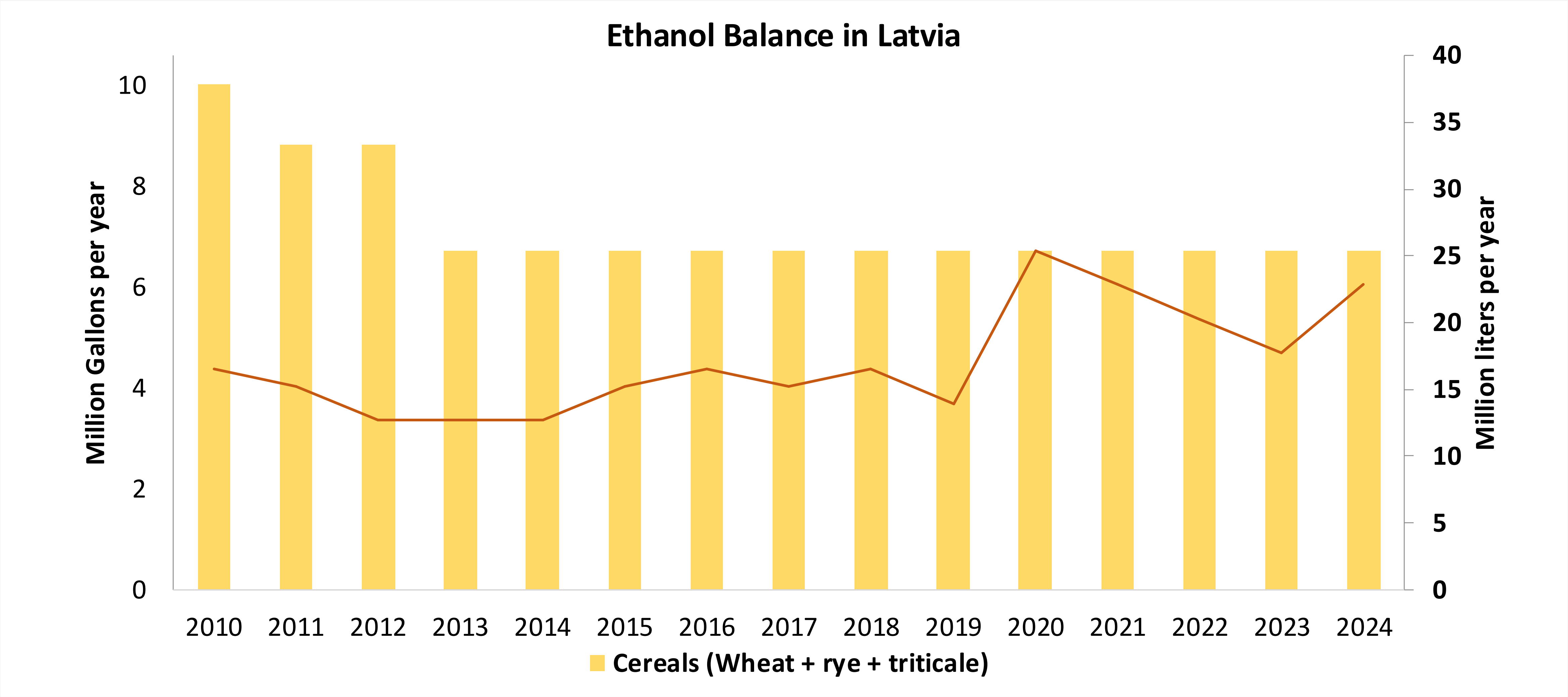


Source: Eurostat

Gasoline Exports in Latvia



Ethanol Balance in Latvia



Source: Eurostat, Latvia Statistics

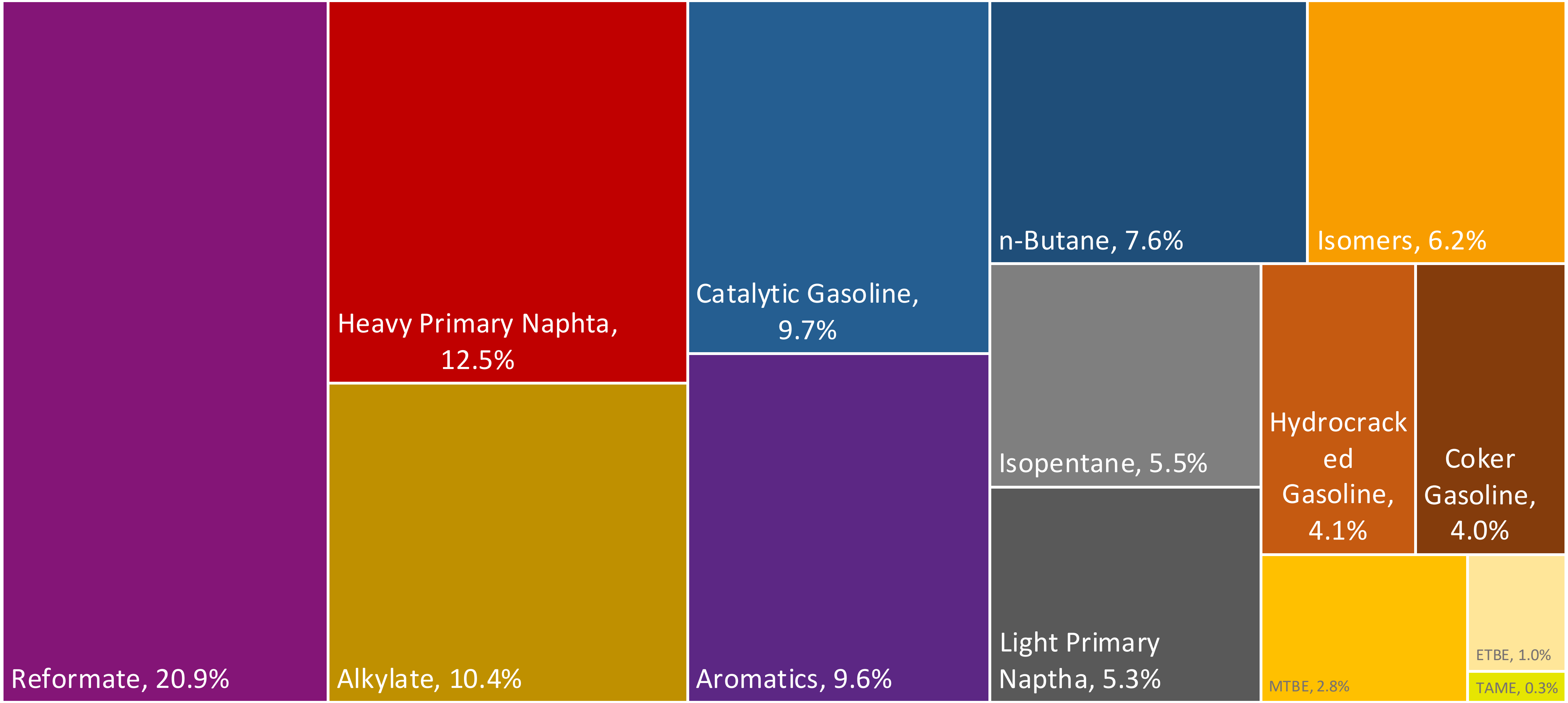
Gasoline Quality in Latvia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

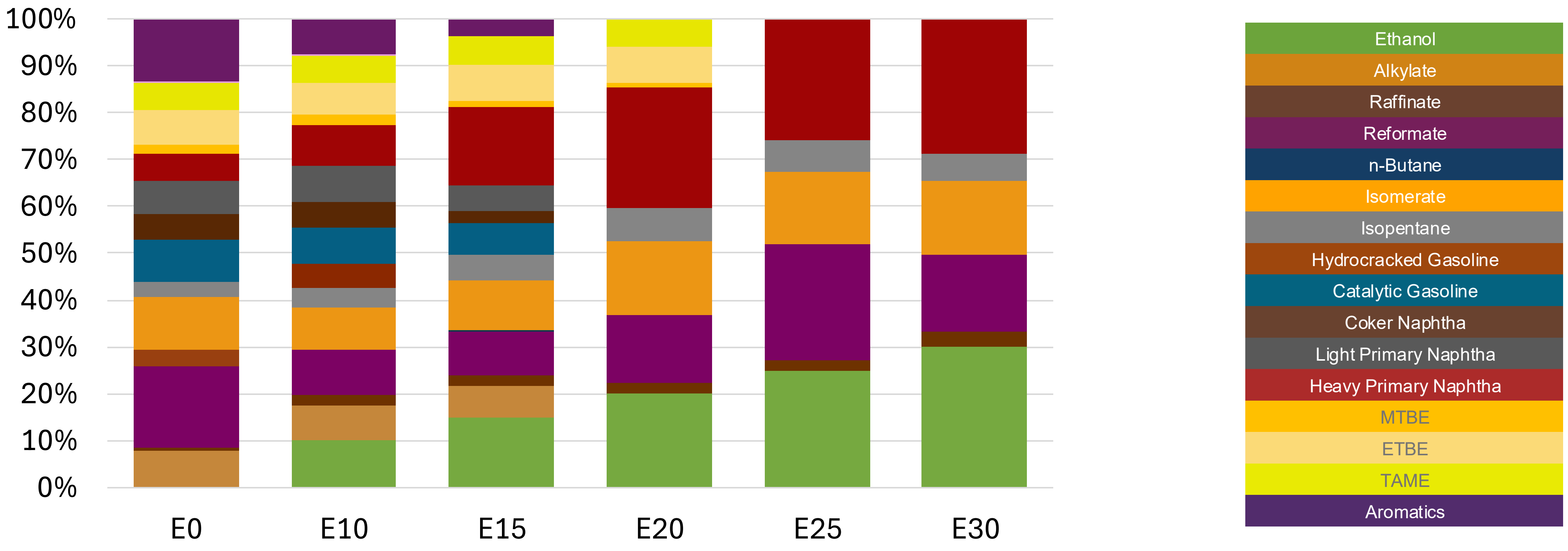
Source: European Commision

Latvia Available Blending Components



Source: Faro90

Latvia – Gasoline 95 – Constant Octane

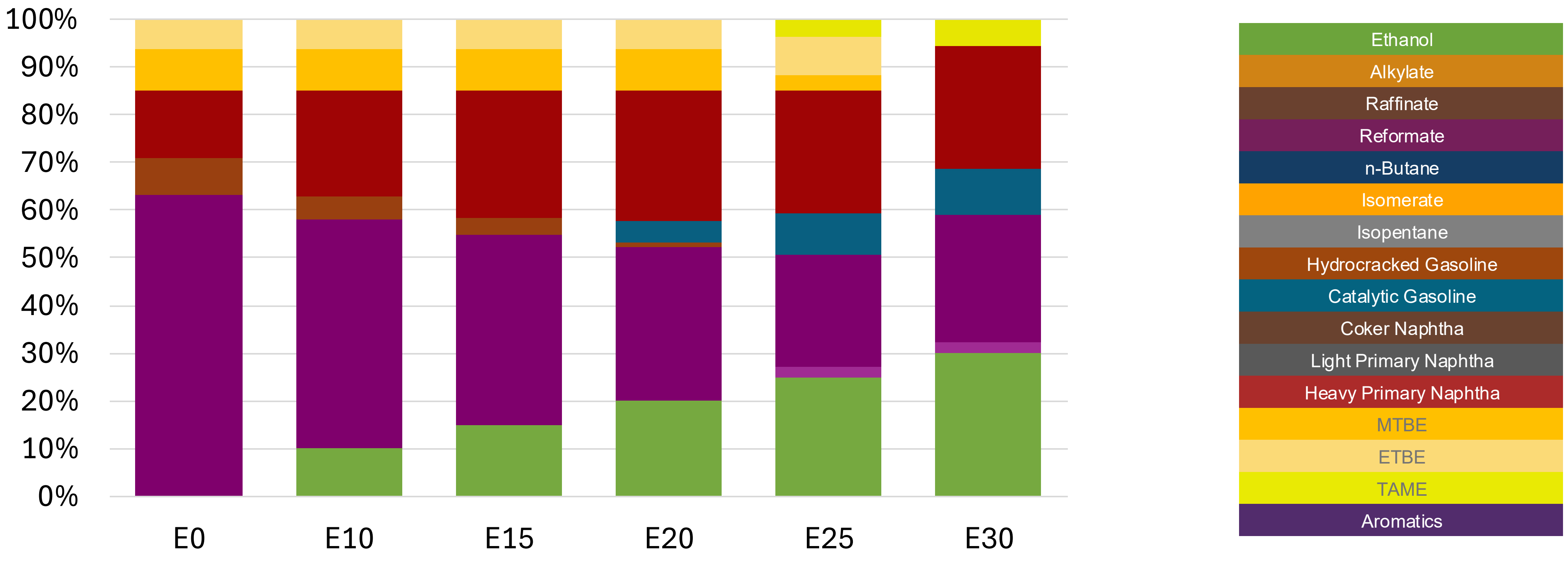


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.274	$ 2.136	$ 2.036	$ 1.921	$ 1.875	$ 1.822

Average CIF 2024 Prices

Source: Faro90

Latvia - Gasoline 98 - Constant Octane

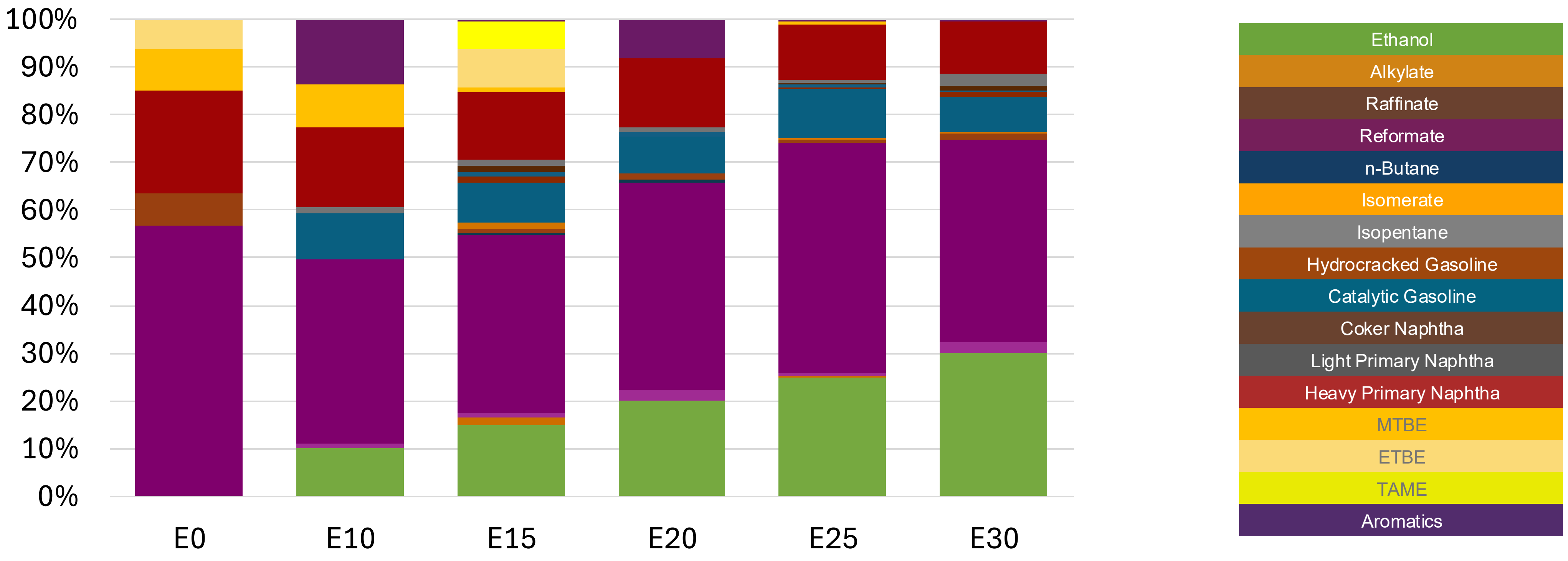


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.294	$ 2.177	$ 2.108	$ 2.036	$ 1.971	$ 1.921

Average CIF 2024 Prices

Source: Faro90

Latvia – Gasoline 95 – Increased Octane

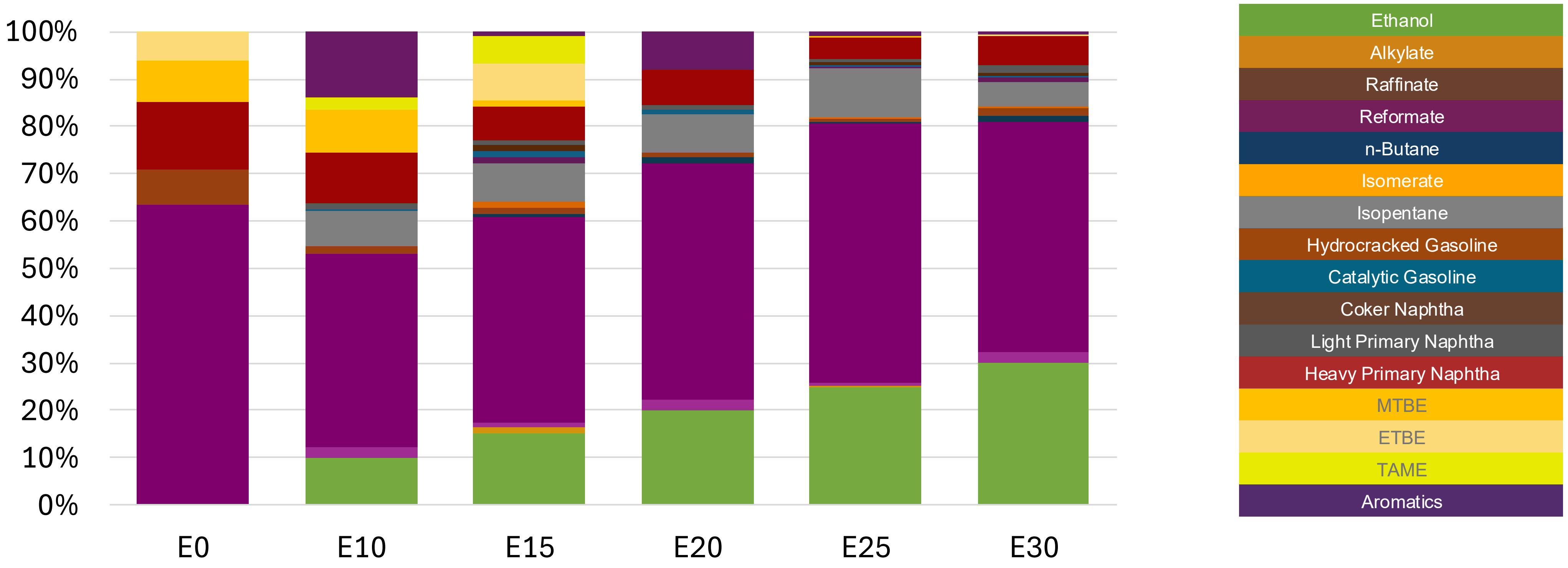


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162

Average CIF 2024 Prices

Source: Faro90

Latvia – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Latvia

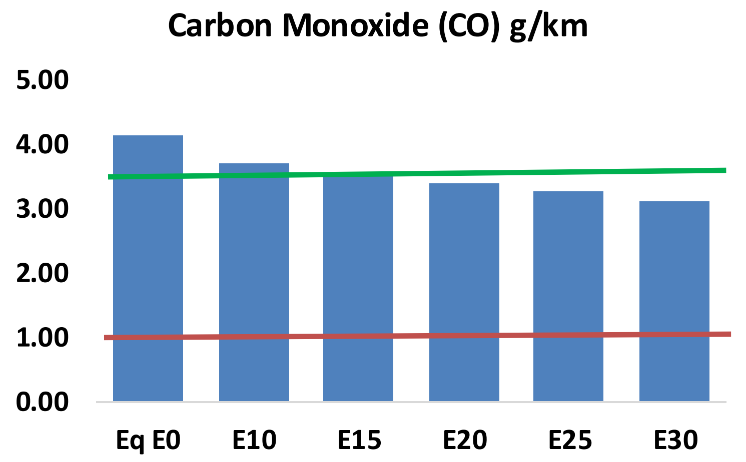
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	-
		Single FI	3-Ways		>8 years	9,291
		Multi FI	3-Ways		>8 years	186,731
			Euro III		>8 years	131,332
			Euro IV		>8 years	216,673
			Euro V		>8 years	69,475
			Euro 6		>8 years	73,328
					4-8 years	43,482
					<4 years	63,509
			Hybrid		All	1,394
		Electric			All	8,366
	Medium	Carburator	3-Ways	PCV Tank	>8 years	-
		Single FI	3-Ways		>8 years	251
		Multi FI	3-Ways		>8 years	19,669
			Euro III		>8 years	25,311
			Euro IV		>8 years	75,052
			Euro V		>8 years	21,387
			Euro 6		>8 years	24,443
					4-8 years	19,668
					<4 years	53,957
			Hybrid		All	1,022
		Electric			All	6,150
Sml Engines	Light	4-Cycle	None	PCV	>8 years	128
			Euro III		>8 years	948
			Euro IV		>8 years	6,679
			Euro V		>8 years	2,223
			Euro 6		>8 years	3,066
					4-8 years	3,297
					<4 years	5,918
			Hybrid		All	52
		Electric			All	313
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	-
		Fuel-Injection	3-Ways		>8 years	13,871
			Euro III		>8 years	24,067
			Euro IV		>8 years	59,352
			Euro V		>8 years	25,855
			Euro 6		>8 years	37,341
					4-8 years	35,094
					<4 years	42,230
			Hybrid		All	10
		Electric			All	469

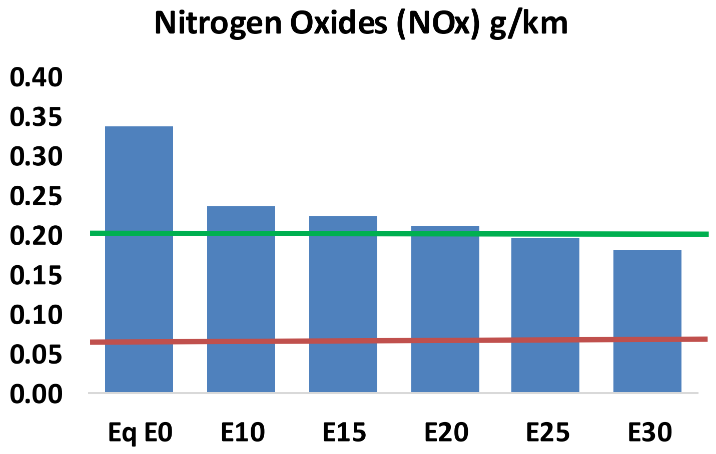
Vehicle Fleet: 1,311,403
Gasoline Vehicle Fleet: 333,918

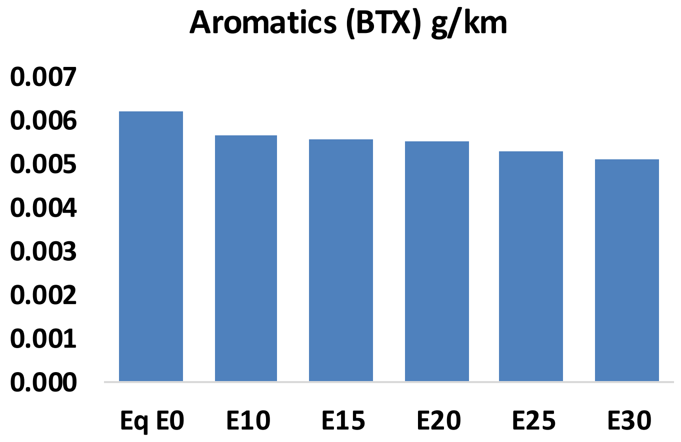
Source: Eurostat, ACEA

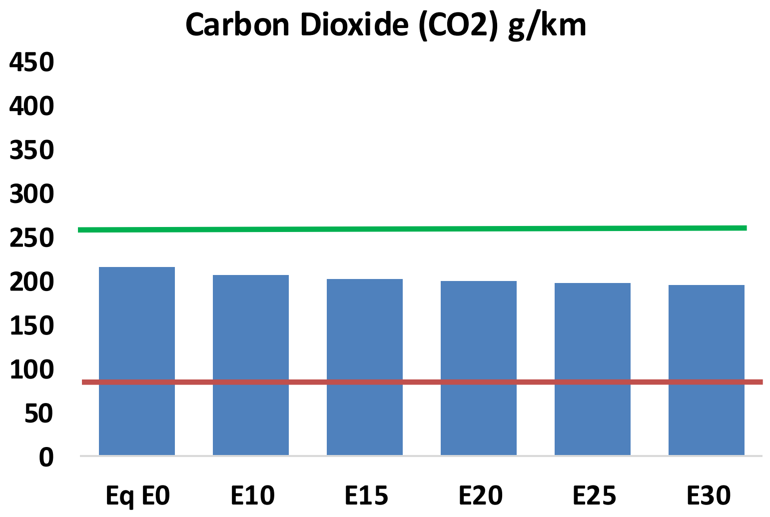
Latvia – Vehicular Emissions

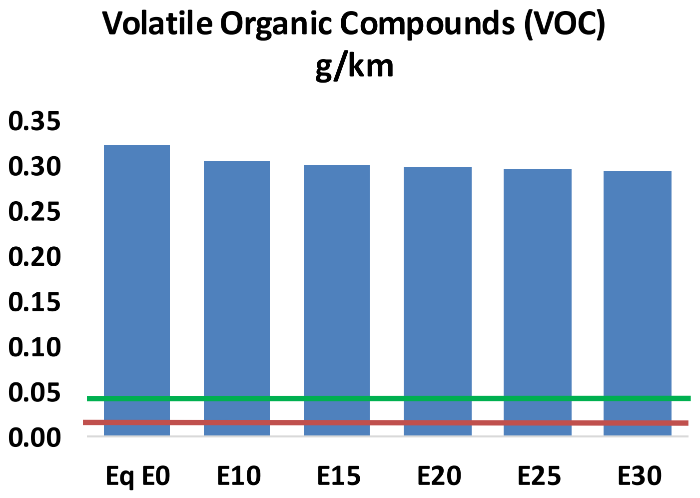
TIER III BIN 125 United States
Euro 6 Standard

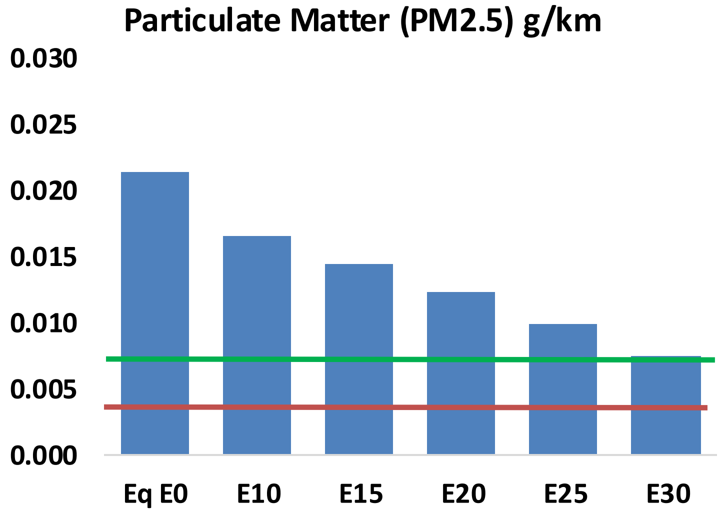












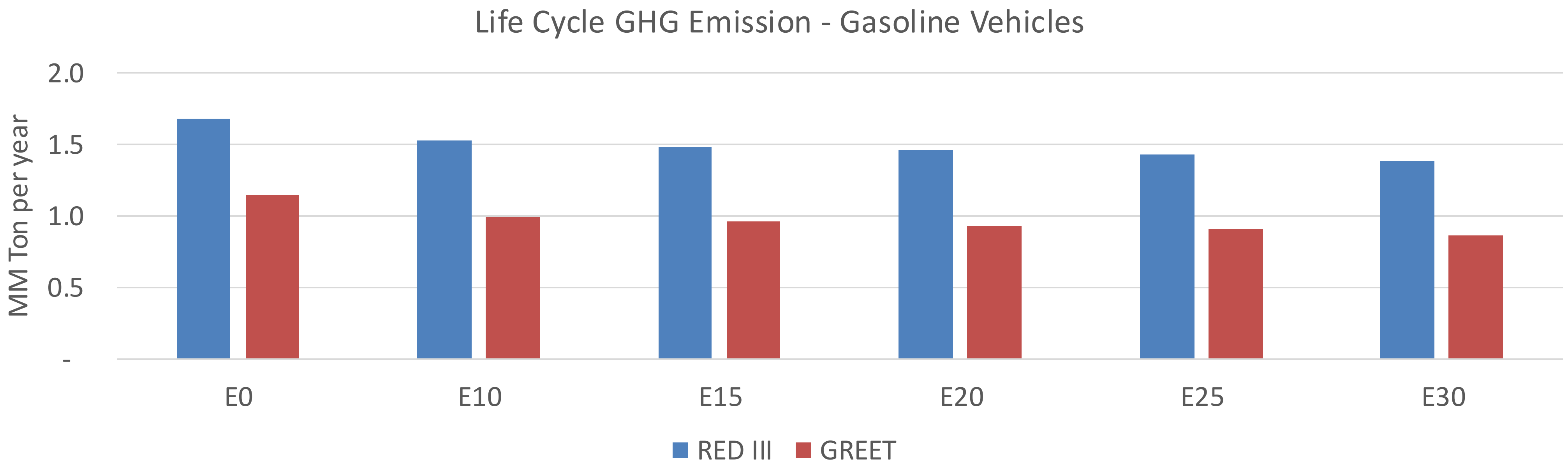
Source: Faro90

Latvia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	36,253	34,335	32,417	31,014	29,681	28,680	27,419	-11%	-18%
CO2	1,902,185	1,853,272	1,807,076	1,770,744	1,752,766	1,741,942	1,709,833	-5%	-8%
VOC	2,835	2,759	2,682	2,646	2,624	2,610	2,578	-5%	-7%
NOx	2,975	2,232	2,083	1,963	1,851	1,728	1,598	-30%	-38%
SOx	23	21	19	18	16	15	13	-15%	-28%
NH3	639	633	626	629	636	641	646	-2%	0%
N2O	29	29	29	29	30	30	30	-1%	2%
CH4	629	629	629	638	652	664	676	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	35	34	34	33	33	33	33	-4%	-4%
Acetaldehyde	68	78	114	174	237	270	319	68%	249%
Formaldehyde	197	191	223	262	274	303	330	13%	39%
Benzene	55	53	50	49	49	46	45	-9%	-11%
PM2.5	117	104	91	79	67	54	41	-22%	-43%
PM10	71	63	55	48	41	33	25	-22%	-43%

Source: Faro90

Latvia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



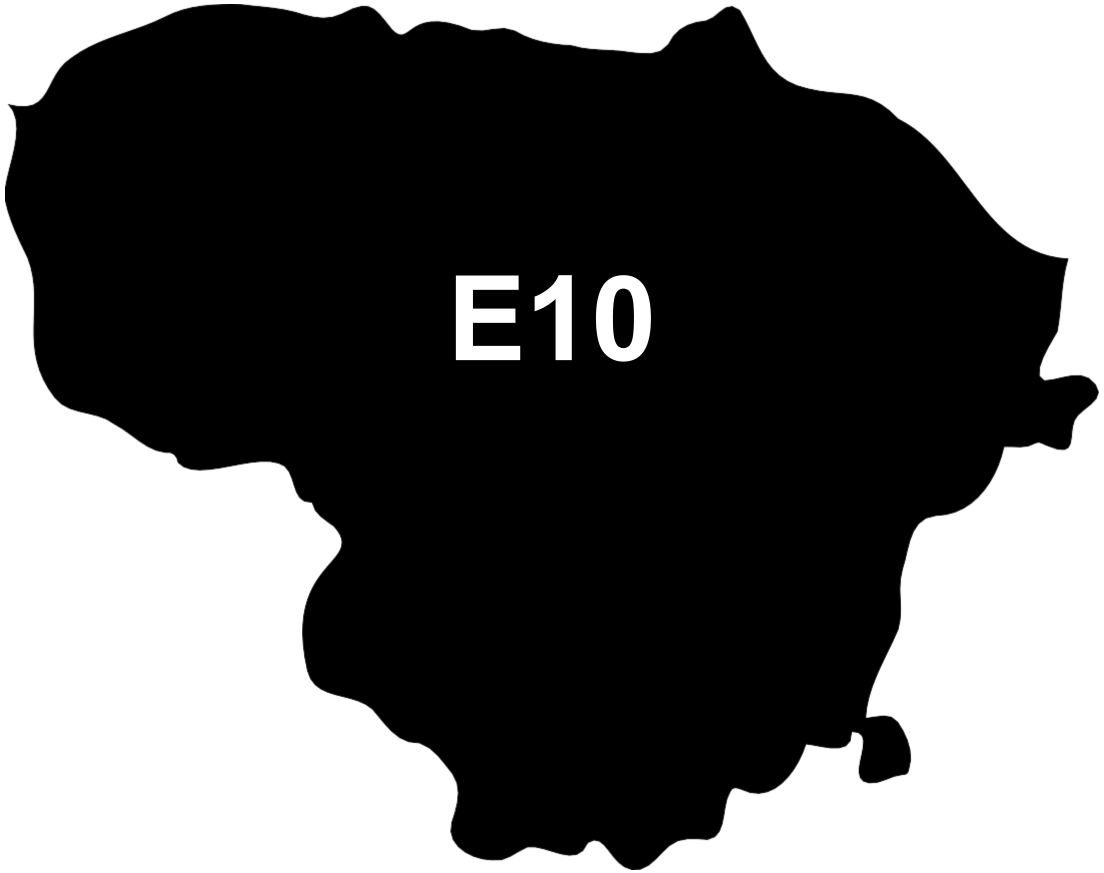
Country Emissions 2024 (MMT/año)	12.1
Target @2035 (MMT/year)	6.8
Delta	5.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.1%	16.2%	18.7%	21.1%	24.3%
RED II/III	0.0%	9.3%	11.6%	13.4%	15.2%	17.5%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.8%	3.5%	4.0%	4.6%	5.3%
RED II/III	0.0%	3.0%	3.7%	4.3%	4.8%	5.6%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Lithuania

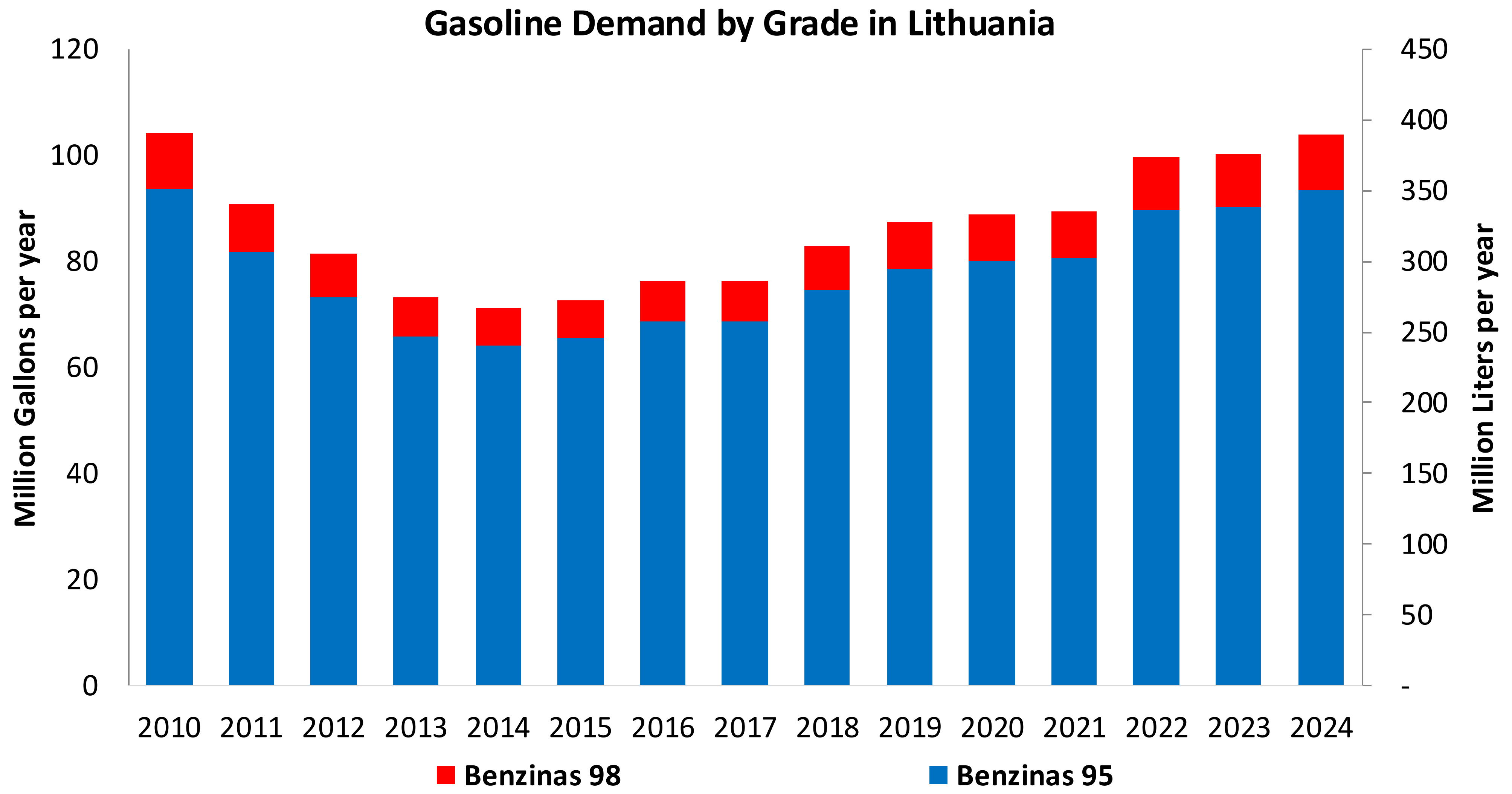


In 2024, gasoline consumption in Lithuania was 390 million liters (103 million gallons) and growth rate was 3.5%. Gasoline production and net imports were 3,762 and -3,361 million liters (994 and -888 million gallons) correspondingly. Lithuania complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Lithuania was 45 million liters (12 million gallons) representing 11.4% of gasoline demand. Ethanol production and Net Imports were 26 and 19 million liters (12 and 5 million gallons) correspondingly. Ethanol sources are Corn 15% and Cereals 85%.

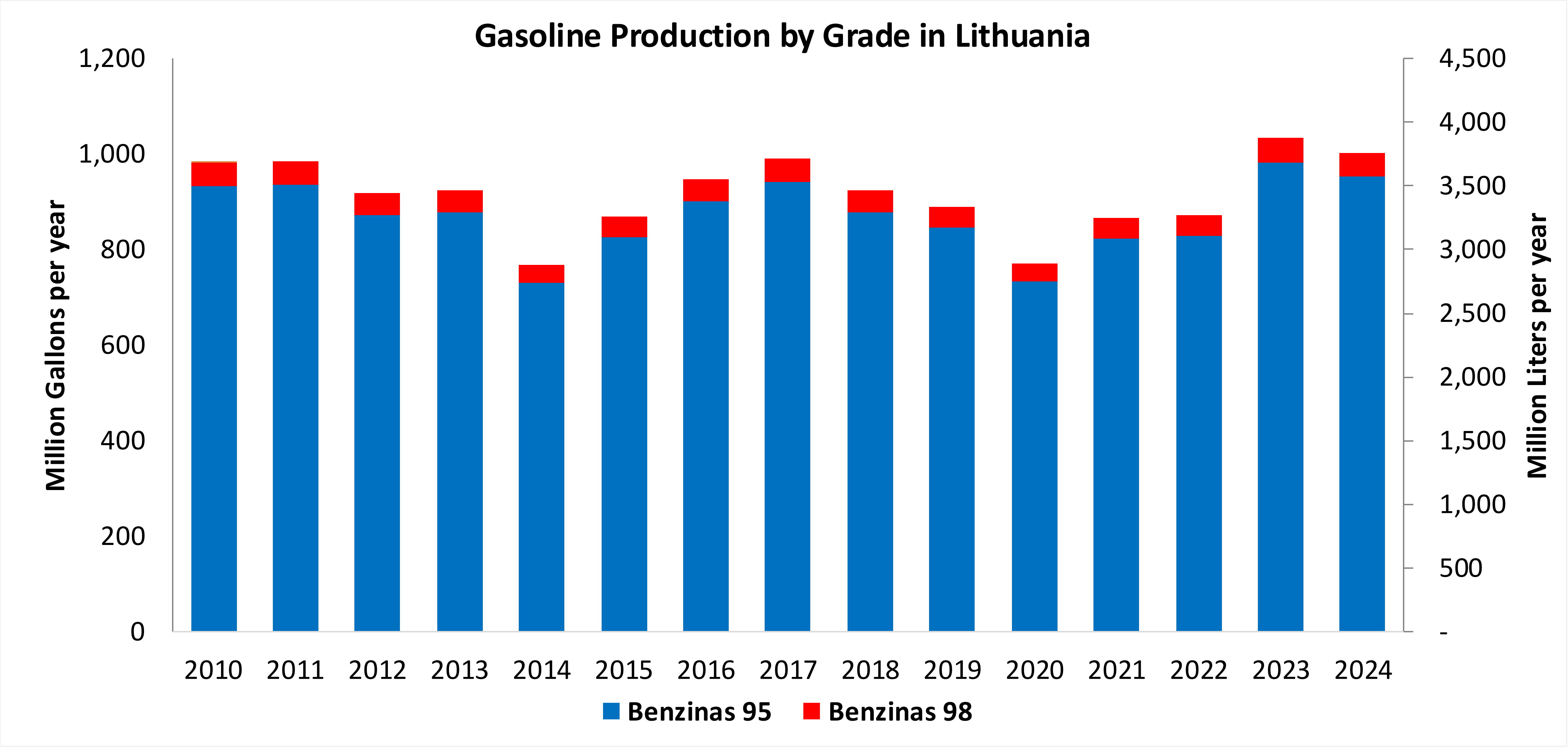
Source: Eurostat, European Commision

Gasoline Demand in Lithuania



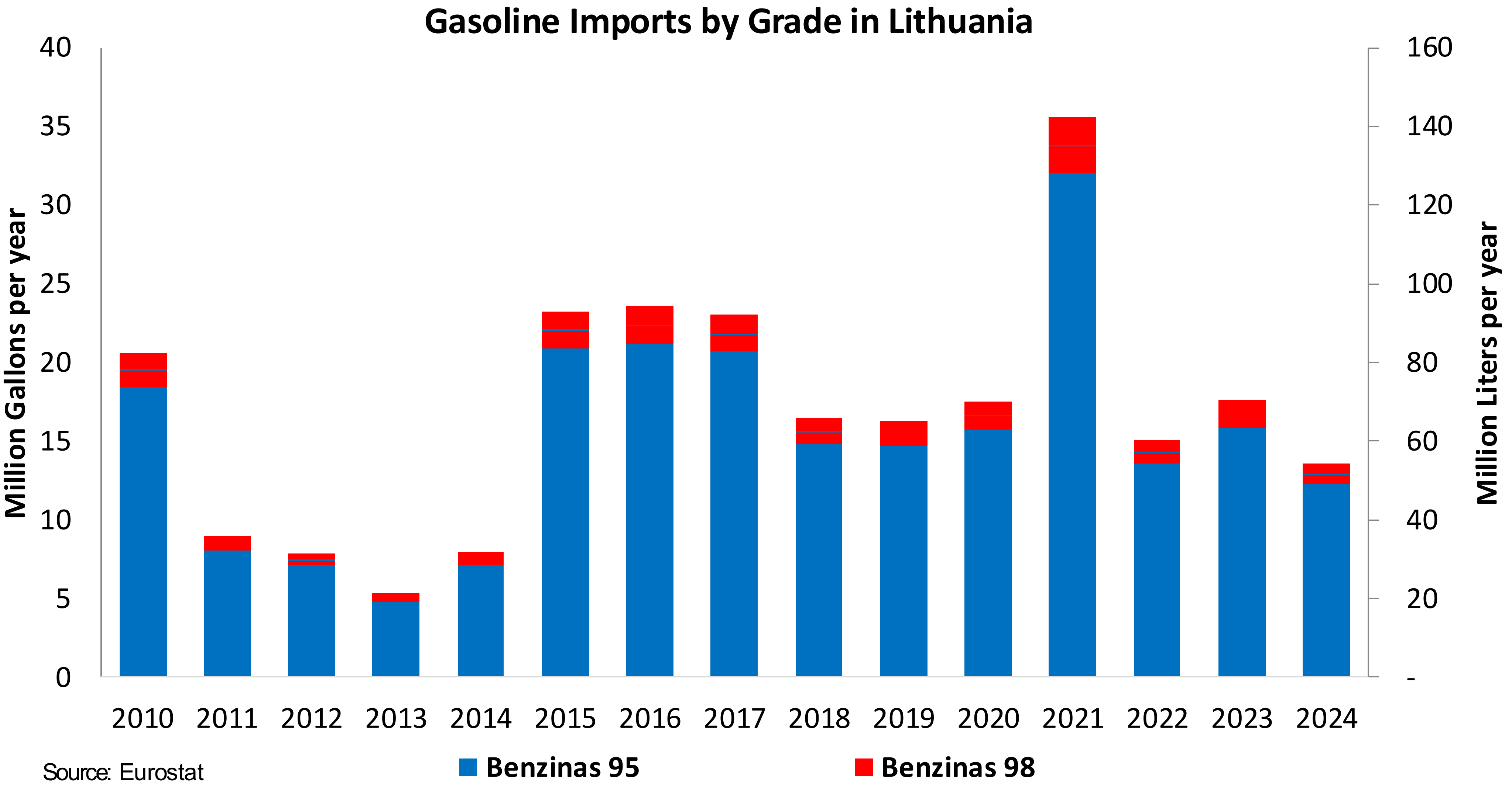
Source: Eurostat

Gasoline Production in Lithuania



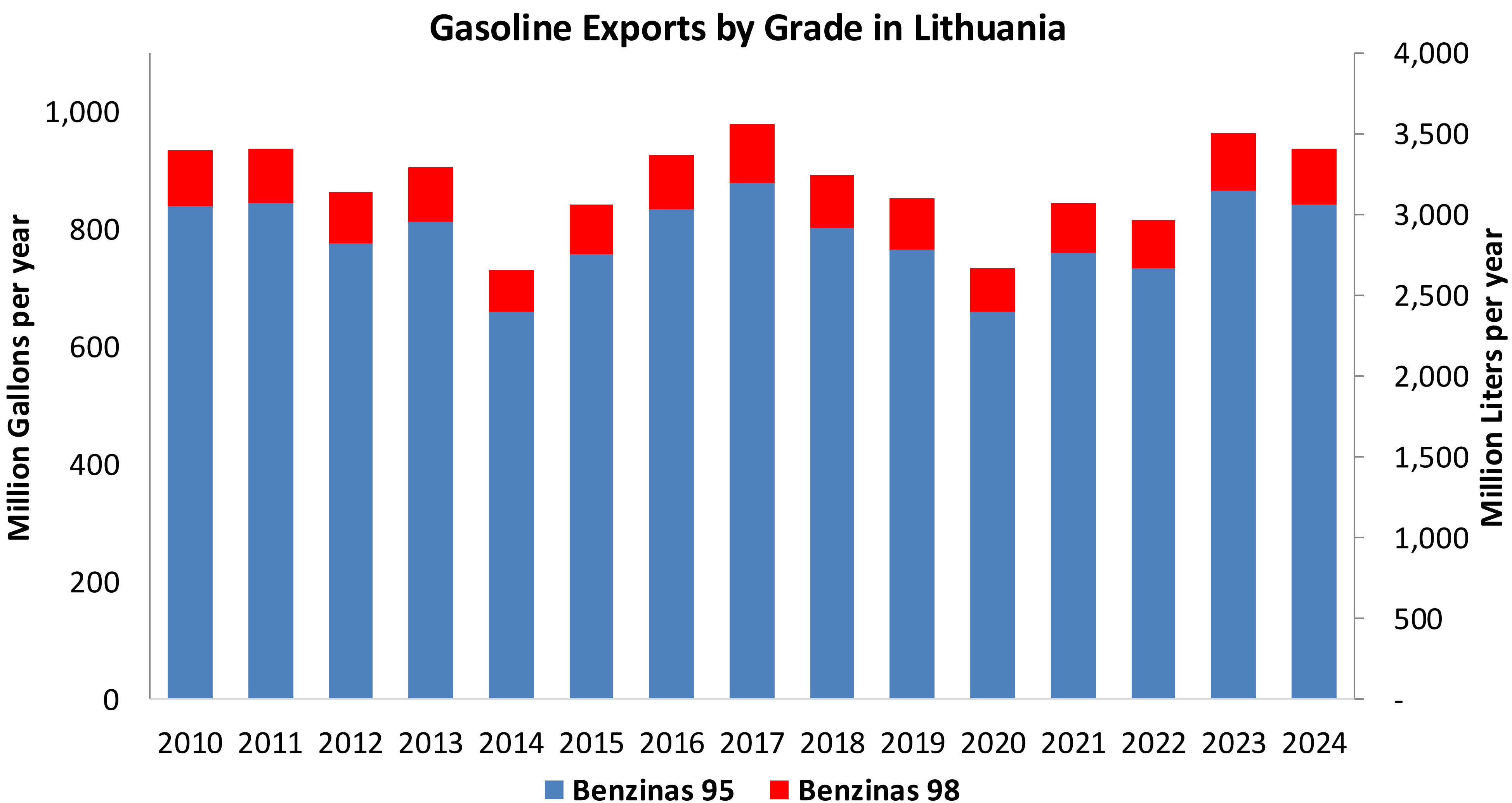
Source: Eurostat

Gasoline Imports in Lithuania



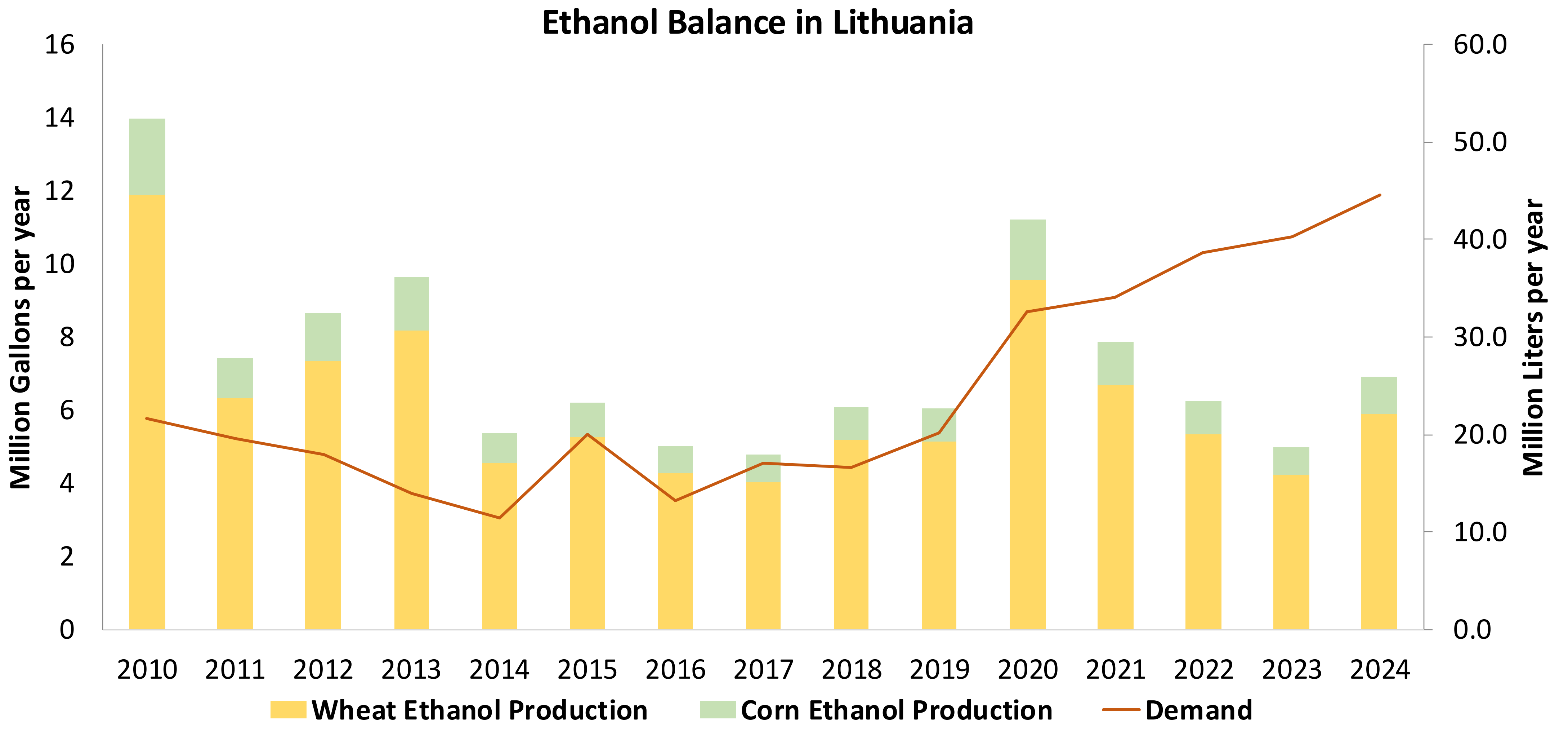
Source: Eurostat

Gasoline Exports in Lithuania



Source: Eurostat

Ethanol Balance in Lithuania



Source: Eurostat

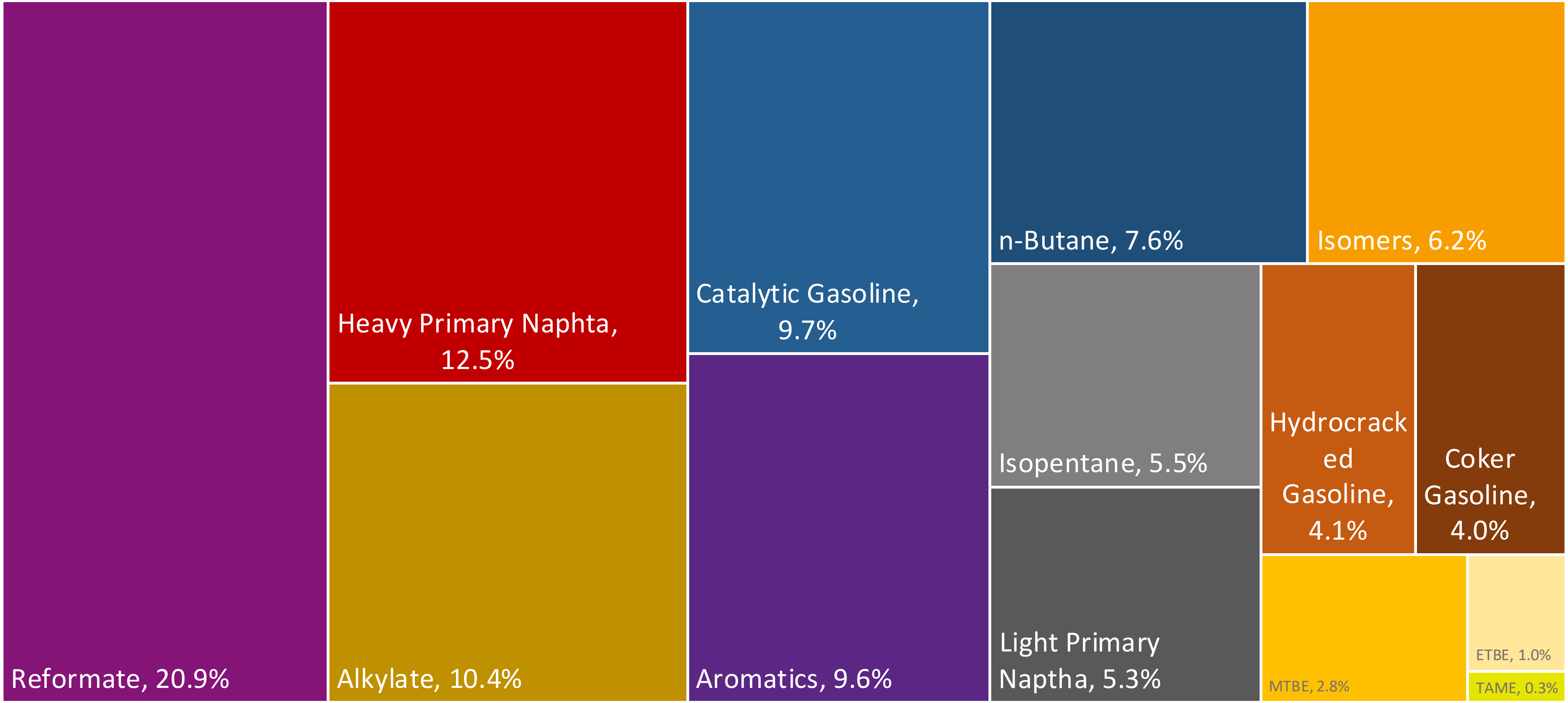
Gasoline Quality in Lithuania

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	8.6							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

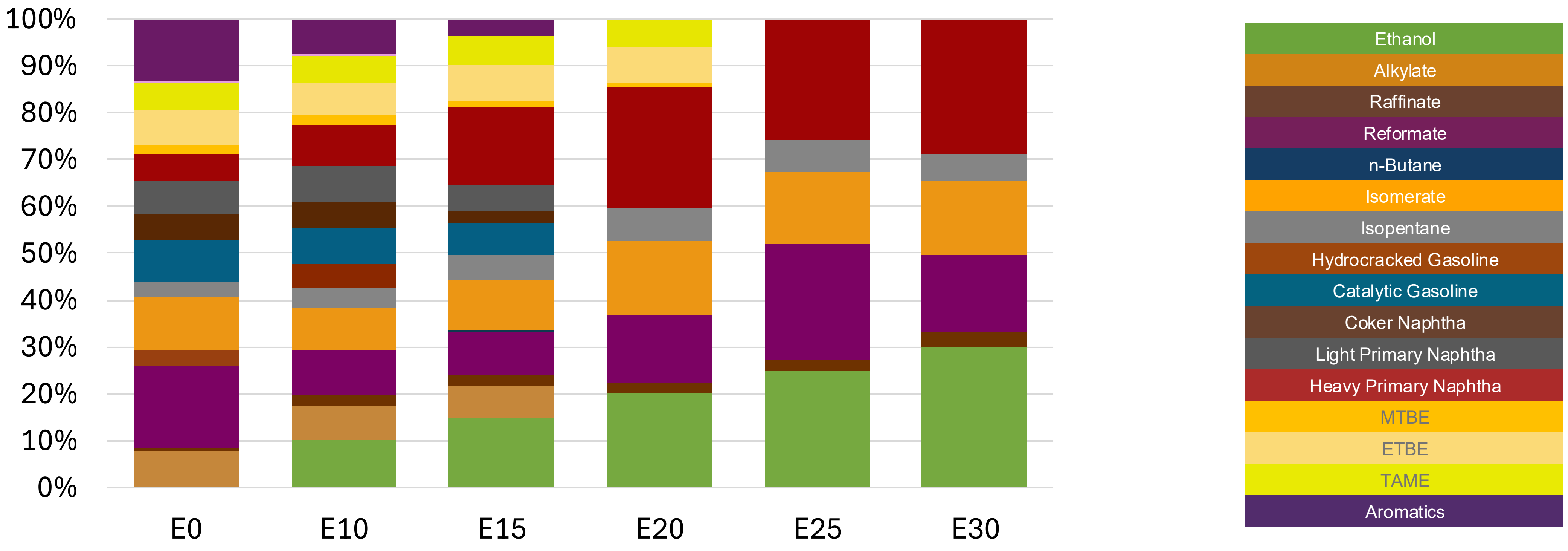
Source: European Commision

Lithuania Available Blending Components



Source: Faro90

Lithuania – Gasoline 95 – Constant Octane

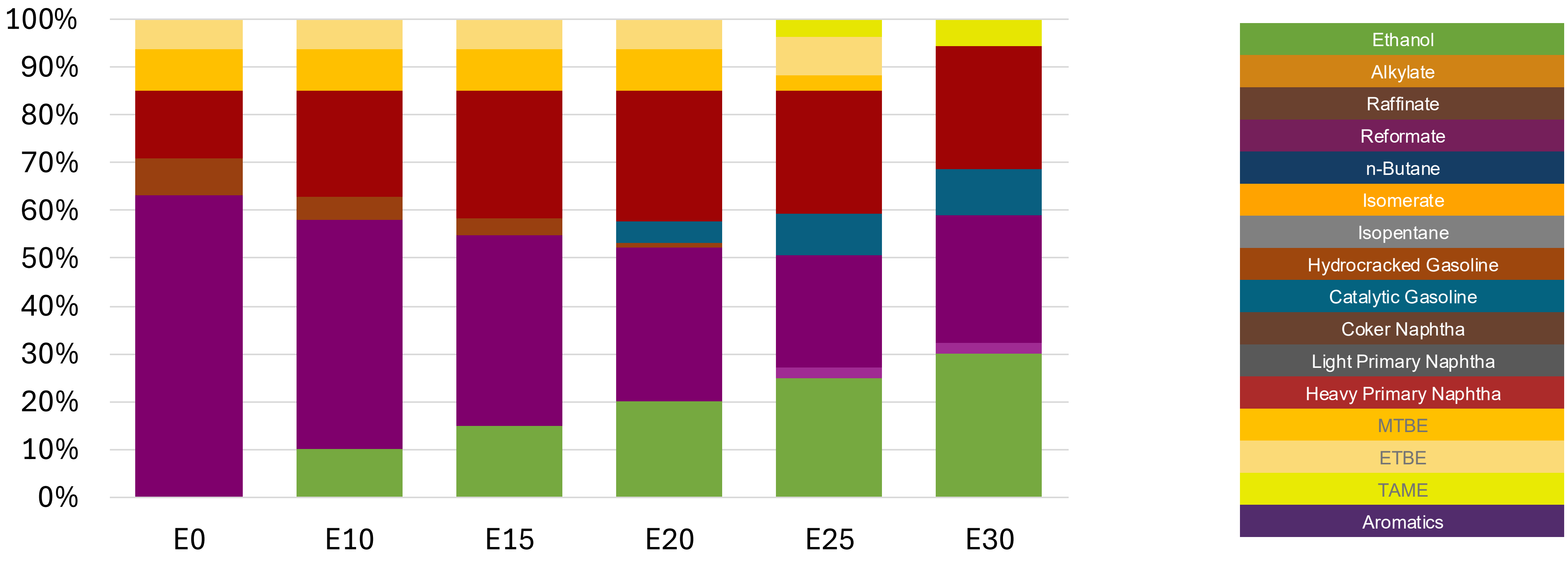


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.274	$ 2.136	$ 2.036	$ 1.921	$ 1.875	$ 1.822

Average CIF 2024 Prices

Source: Faro90

Lithuania - Gasoline 98 - Constant Octane

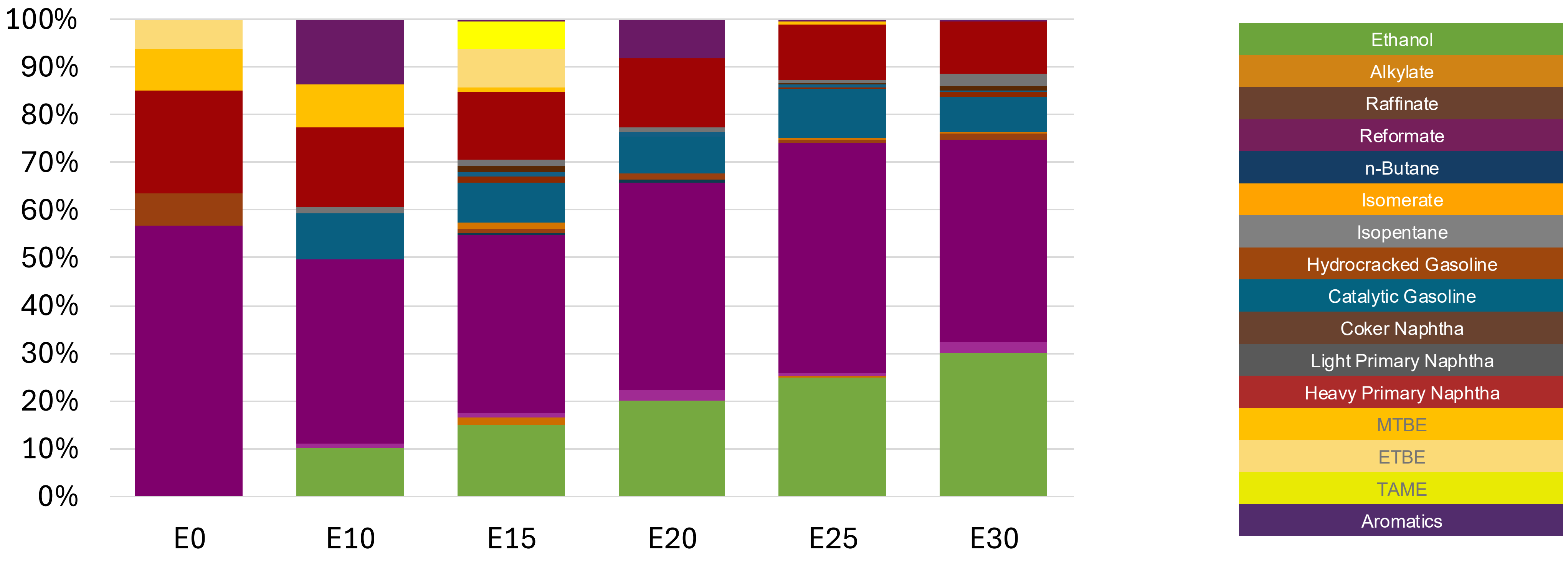


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.294	$ 2.177	$ 2.108	$ 2.036	$ 1.971	$ 1.921

Average CIF 2024 Prices

Source: Faro90

Lithuania – Gasoline 95 – Increased Octane

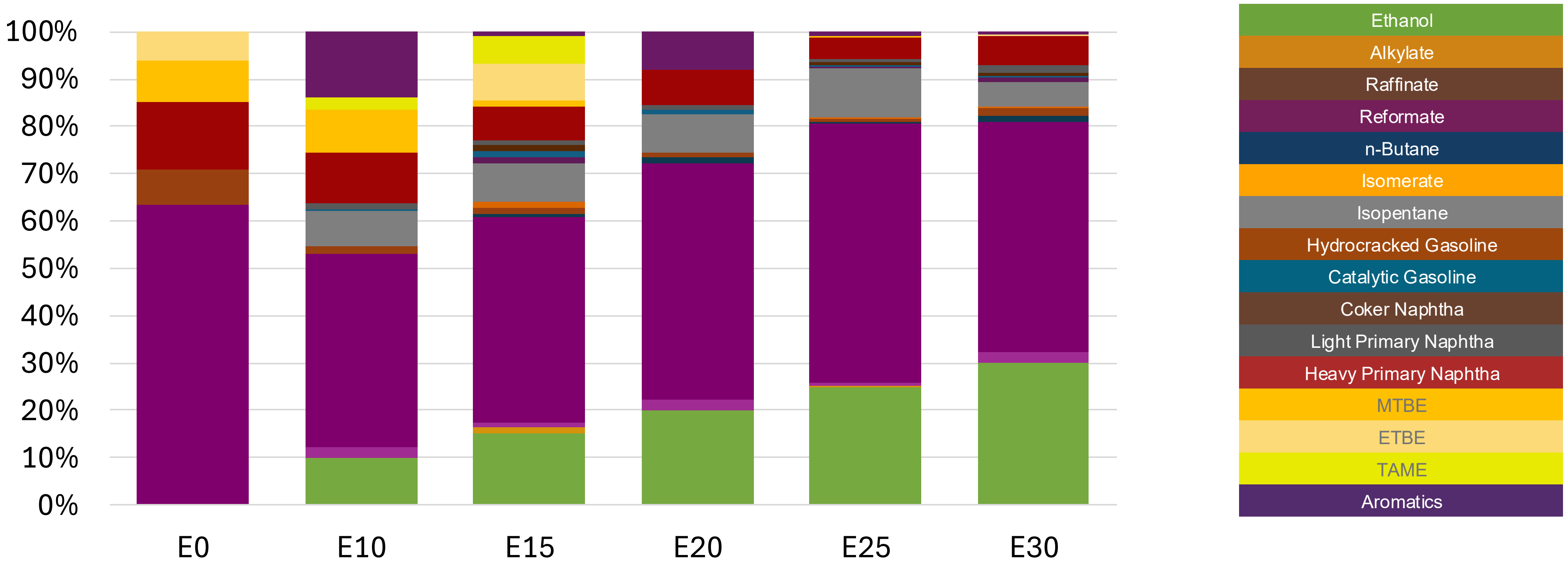


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162

Average CIF 2024 Prices

Source: Faro90

Lithuania – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Lithuania

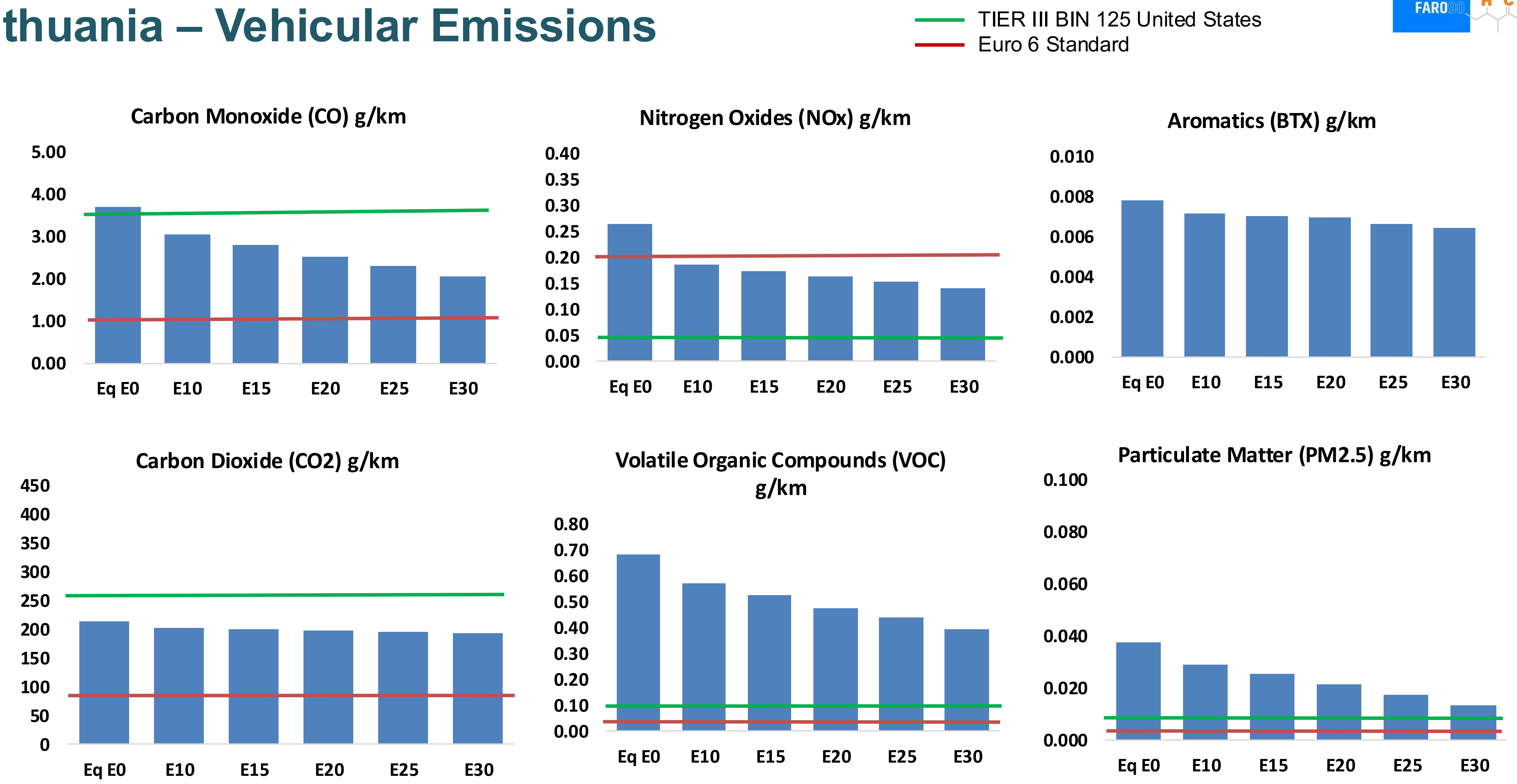
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	790
		Single FI	3-Ways		>8 years	18,498
		Multi FI	3-Ways		>8 years	277,557
			Euro III		>8 years	160,441
			Euro IV		>8 years	374,388
			Euro V		>8 years	131,925
			Euro 6		>8 years	388,917
					4-8 years	404,463
					<4 years	187,055
			Hybrid		All	166,600
		Electric			All	24,698
	Medium	Carburator	3-Ways	PCV Tank	>8 years	16
		Single FI	3-Ways		>8 years	500
		Multi FI	3-Ways		>8 years	31,495
			Euro III		>8 years	32,274
			Euro IV		>8 years	128,169
			Euro V		>8 years	40,697
			Euro 6		>8 years	129,639
					4-8 years	183,715
					<4 years	146,617
			Hybrid		All	116,948
		Electric			All	19,121
Sml Engines	Light	4-Cycle	None	PCV	>8 years	99
			Euro III		>8 years	869
			Euro IV		>8 years	11,393
			Euro V		>8 years	5,193
			Euro 6		>8 years	13,375
					4-8 years	10,222
					<4 years	6,158
			Hybrid		All	5,088
		Electric			All	786
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	15,344
		Fuel-Injection	3-Ways		>8 years	101,246
			Euro III		>8 years	34,226
			Euro IV		>8 years	106,050
			Euro V		>8 years	42,548
			Euro 6		>8 years	72,206
					4-8 years	79,325
					<4 years	78,889
			Hybrid		All	19
		Electric			All	907

0 50,000 100,000 150,000 200,000 250,000 300,000 350,000 400,000 450,000

Vehicle Fleet: 3,548,467
Gasoline Vehicle Fleet 745,407

Source: Eurostat, ACEA

Lithuania – Vehicular Emissions



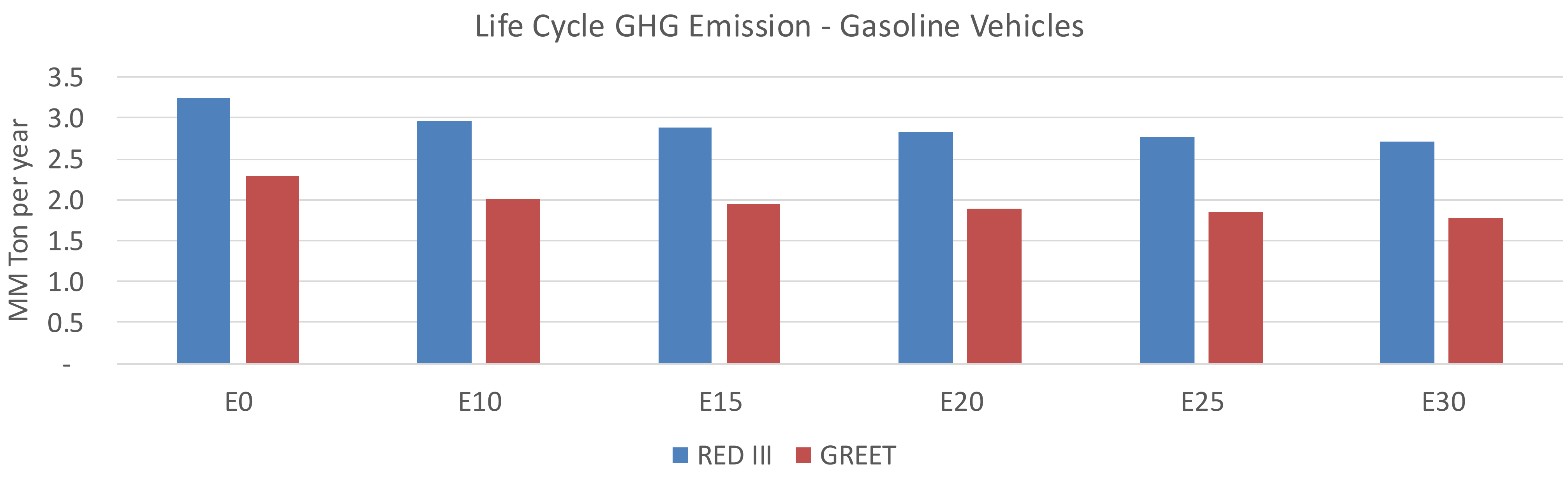
Source: Faro90

Lithuania – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	71,126	64,920	58,715	53,486	48,158	44,041	39,682	-17%	-32%
CO2	4,091,103	3,985,903	3,886,547	3,808,407	3,769,741	3,747,441	3,678,365	-5%	-8%
VOC	13,034	11,965	10,896	10,007	9,106	8,411	7,490	-16%	-30%
NOx	5,080	3,810	3,556	3,352	3,161	2,950	2,728	-30%	-38%
SOx	49	45	42	39	35	32	29	-15%	-28%
NH3	1,503	1,488	1,473	1,480	1,497	1,508	1,518	-2%	0%
N2O	69	68	68	68	70	70	71	-1%	2%
CH4	2,644	2,644	2,644	2,684	2,742	2,792	2,840	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	67	64	62	60	58	57	55	-8%	-14%
Acetaldehyde	210	242	354	540	735	840	993	68%	249%
Formaldehyde	773	752	876	1,029	1,078	1,193	1,298	13%	39%
Benzene	151	145	137	135	134	128	124	-9%	-11%
PM2.5	252	224	196	170	144	117	89	-22%	-43%
PM10	463	411	359	312	265	215	163	-22%	-43%

Source: Faro90

Lithuania – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	25.4
Target @2035 (MMT/year)	14.3
Delta	11.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.9%	14.9%	17.2%	19.3%	22.3%
RED II/III	0.0%	8.8%	11.0%	12.8%	14.4%	16.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.5%	3.1%	3.5%	4.0%	4.6%
RED II/III	0.0%	2.6%	3.2%	3.7%	4.2%	4.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

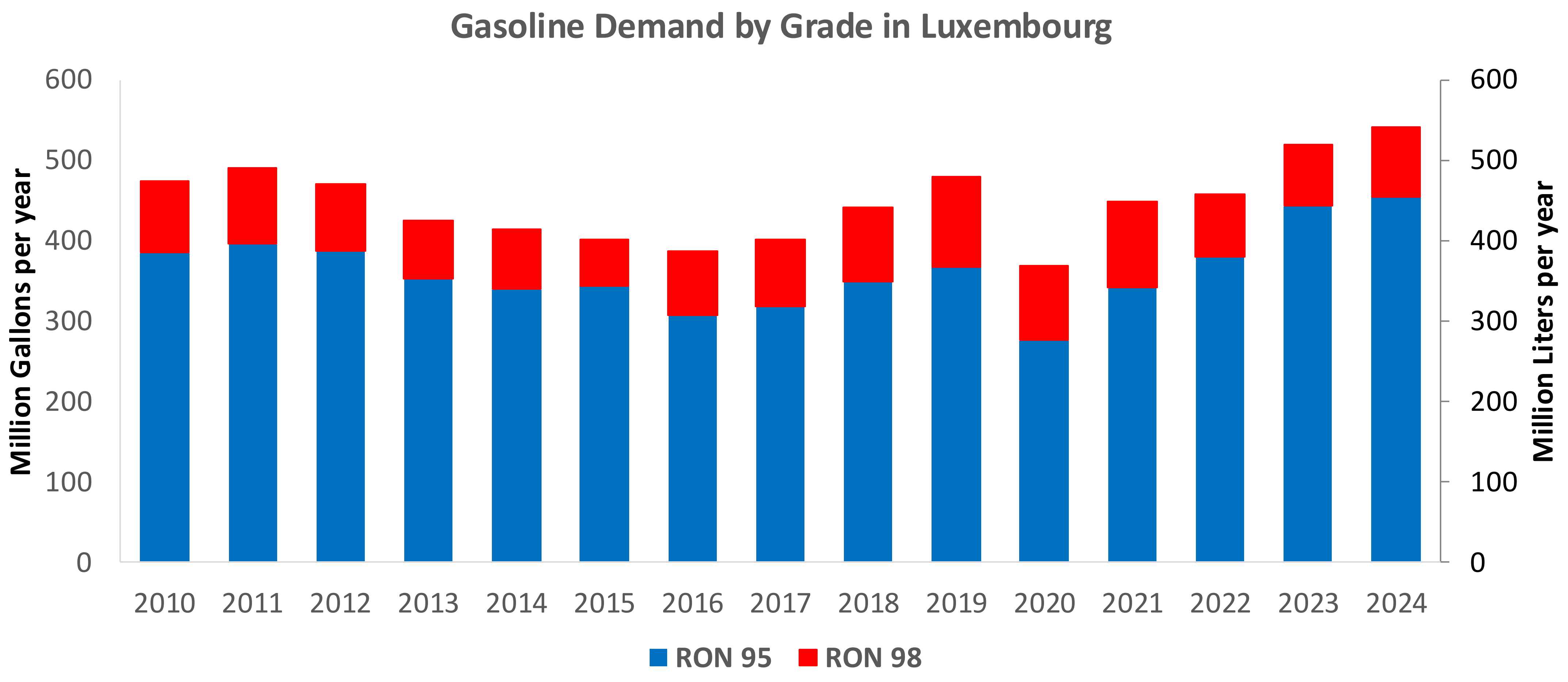
Ethanol Gasoline Blending in Luxembourg

In 2024, gasoline consumption in Luxembourg was 540 million liters (143 million gallons) and growth rate was 2.4%. There is no gasoline production. Gasoline net imports were 540 million liters (143 million gallons) correspondingly. Luxembourg complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

In 2024, ethanol consumption in Luxembourg was 29 million liters (8 million gallons) representing 5.3% of gasoline demand. Net Imports were 29 million liters (8 million gallons). Ethanol sources are Sugar Cane 80% and Corn 20%.

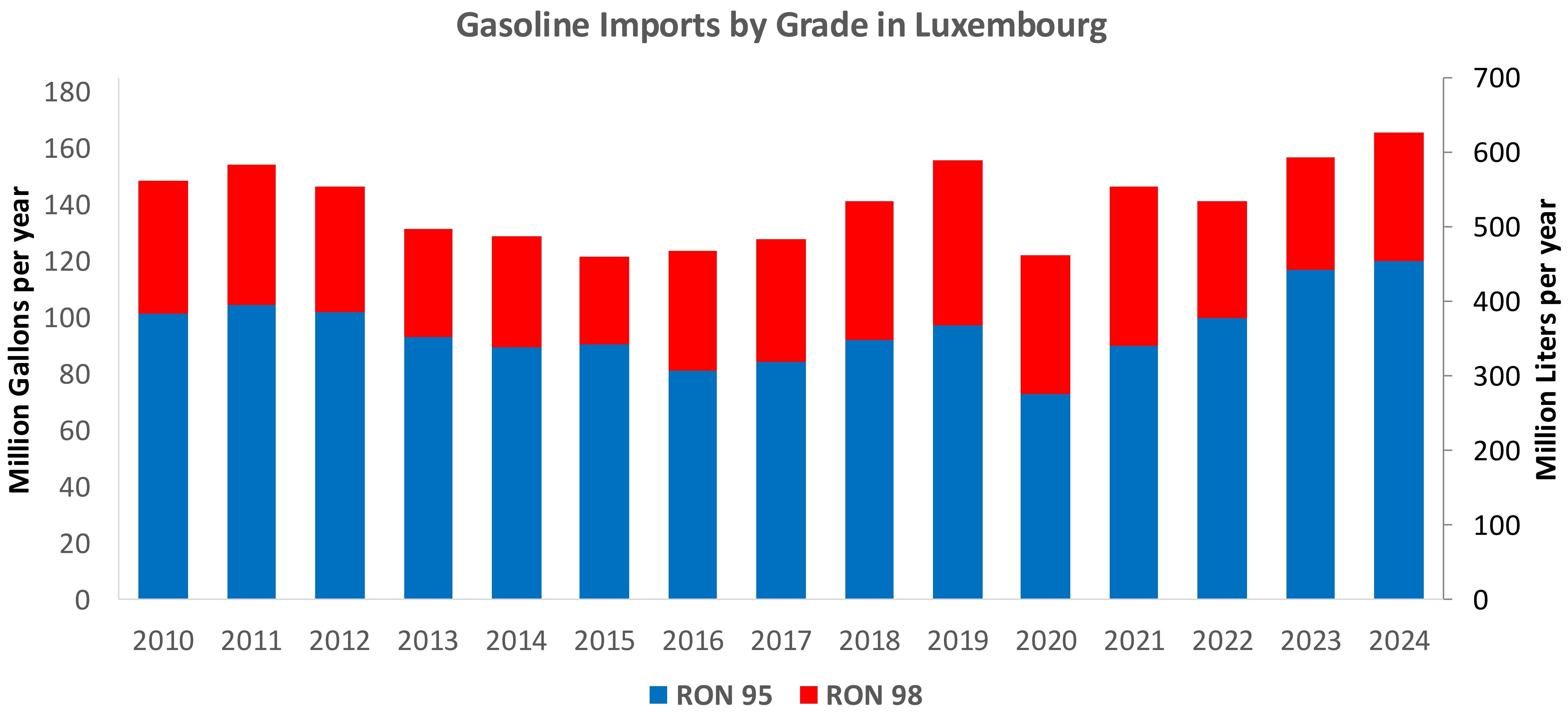
Source: Eurostat, European Commision

Gasoline Demand in Luxembourg



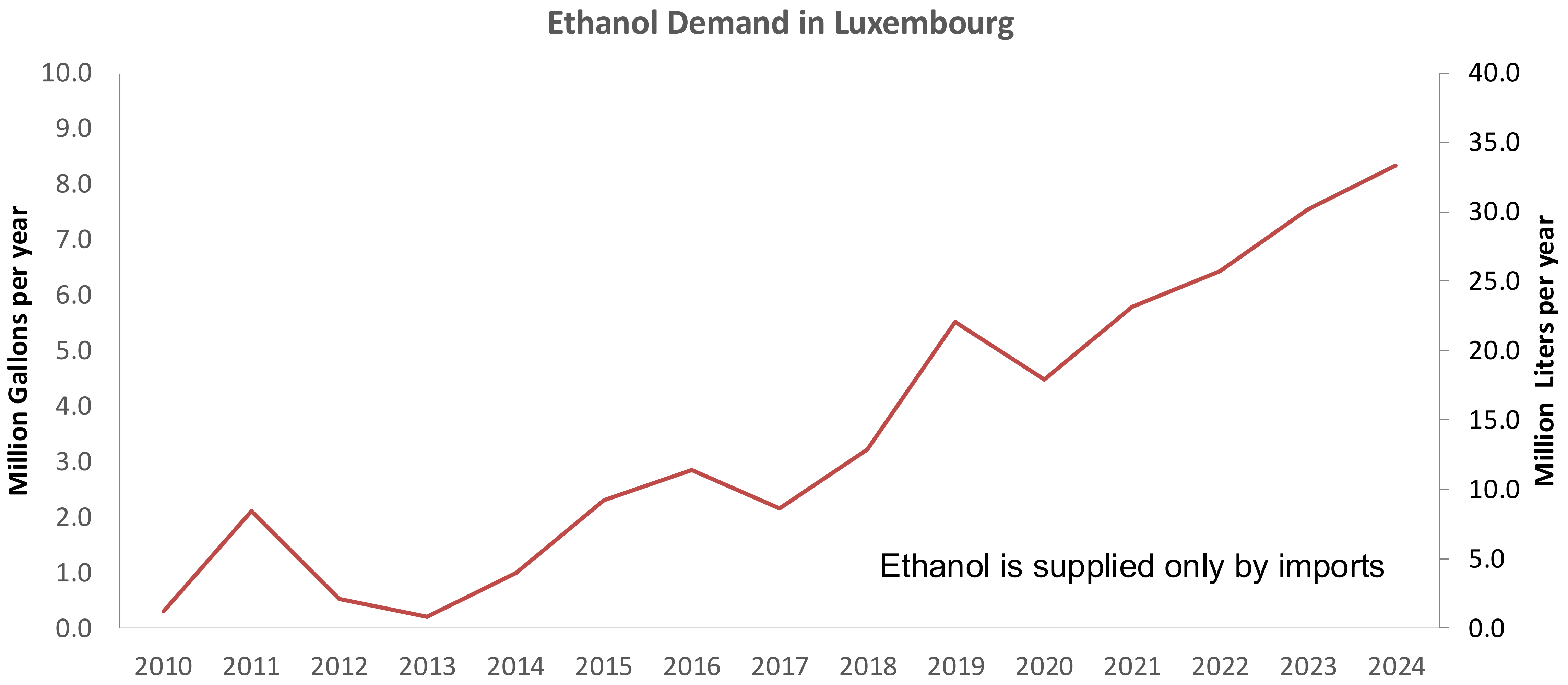
Source: Eurostat

Gasoline Imports in Luxembourg



Source: Eurostat

Ethanol Balance in Luxembourg



Source: Eurostat

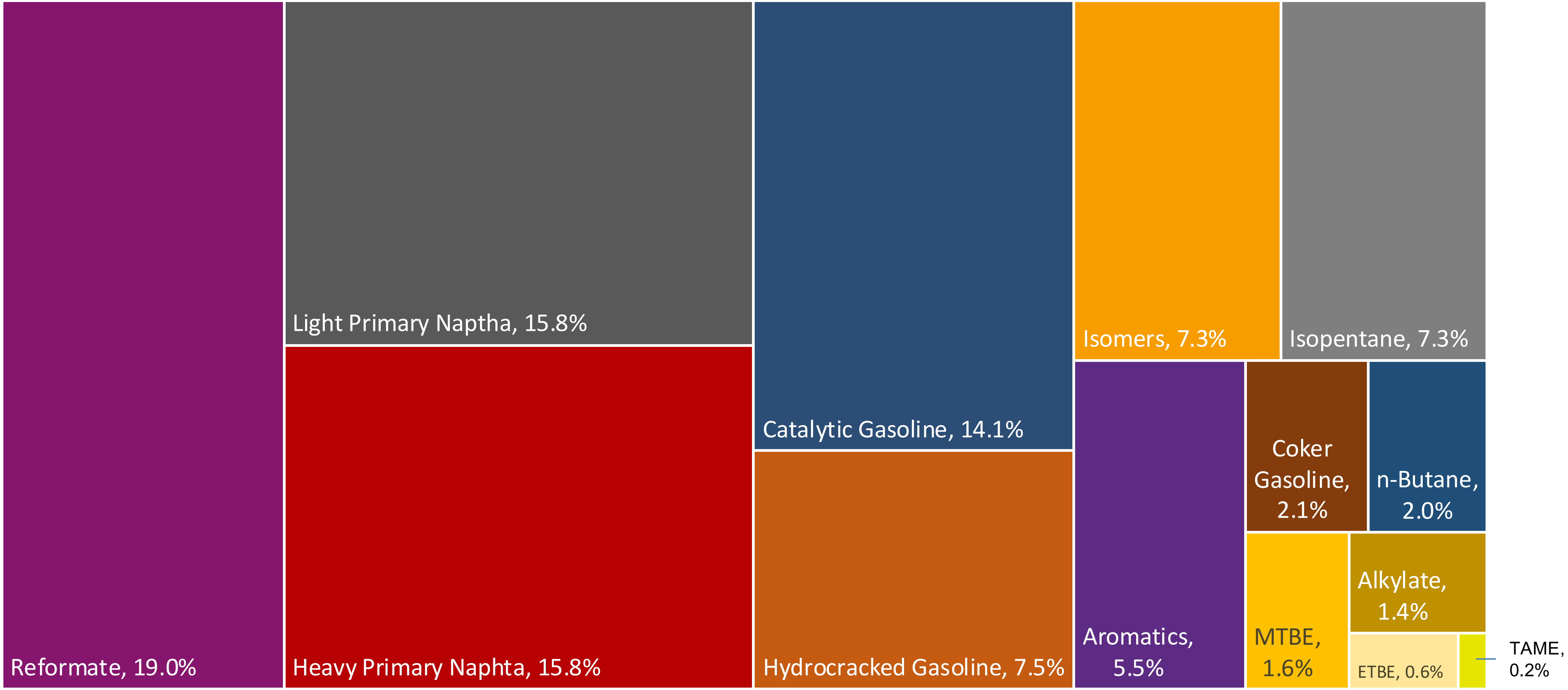
Gasoline Quality in Luxembourg

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	7.7							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

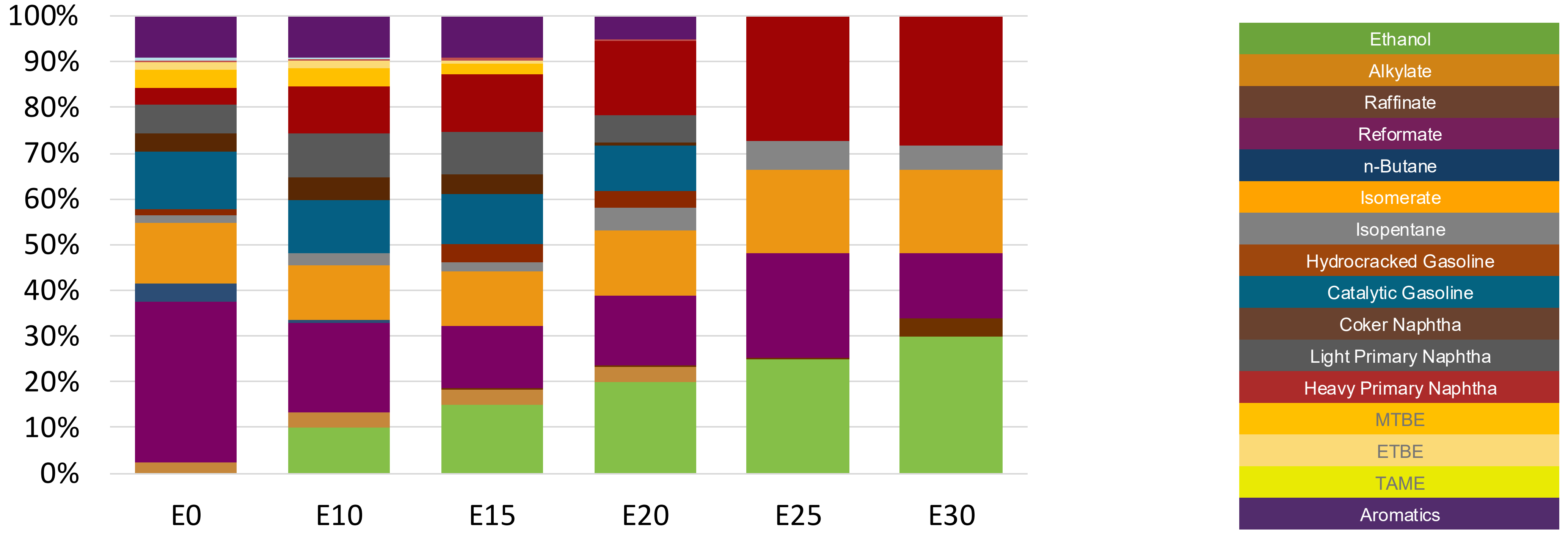
Source: European Commision

Luxembourg Available Blending Components



Source: Faro90

Luxembourg – Gasoline 95 – Constant Octane

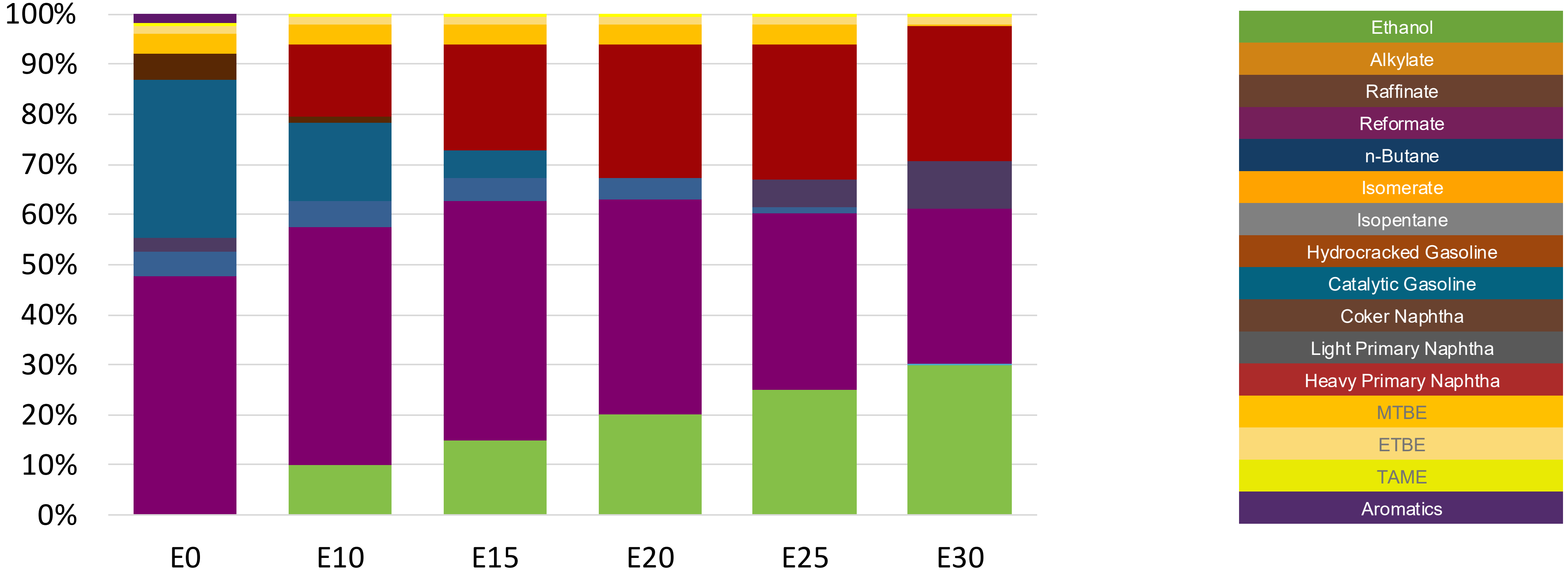


Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US$/gal)	$ 2.272	$ 2.144	$ 2.083	$ 2.002	$ 1.880	$ 1.814

Average CIF 2024 Prices

Source: Faro90

Luxembourg - Gasoline 98 - Constant Octane

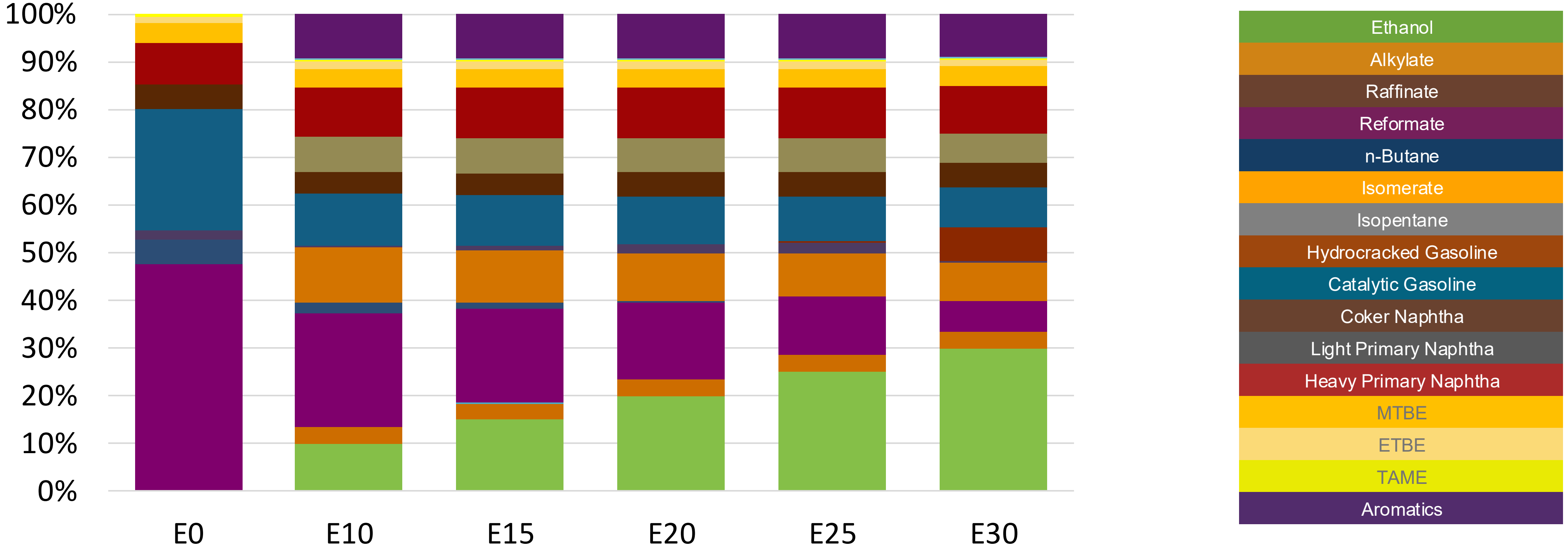


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.347	$ 2.218	$ 2.146	$ 2.069	$ 1.992	$ 1.925

Average CIF 2024 Prices

Source: Faro90

Luxembourg – Gasoline 95 – Increased Octane

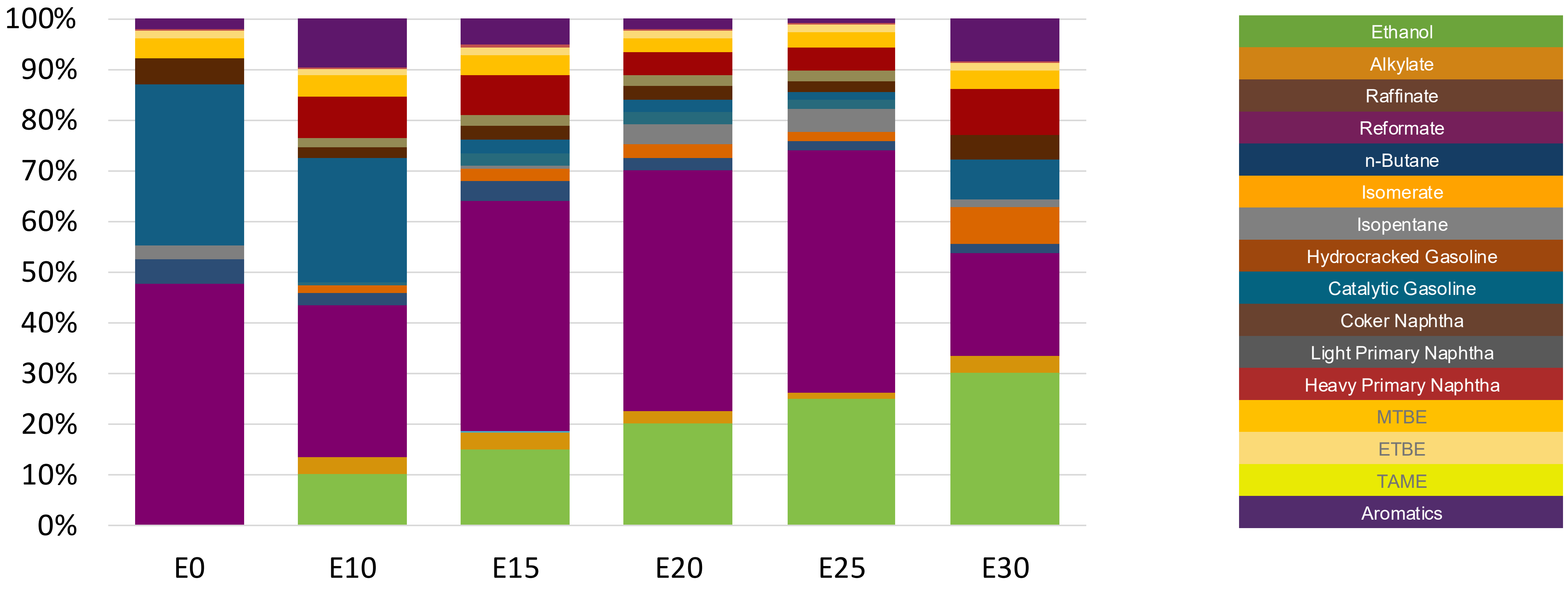


Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US$/gal)	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209	$ 2.209

Average CIF 2024 Prices

Source: Faro90

Luxembourg – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US$/gal)	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347	$ 2.347

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Luxembourg

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	1
		Single FI	3-Ways		>8 years	79
		Multi FI	3-Ways		>8 years	2,258
			Euro III		>8 years	12,158
			Euro IV		>8 years	94,410
			Euro V		>8 years	74,496
			Euro 6		>8 years	202,467
					4-8 years	256,114
					<4 years	144,161
			Hybrid		All	22,891
		Electric			All	20,545
	Medium	Carburator	3-Ways	PCV Tank	>8 years	0
		Single FI	3-Ways		>8 years	6
		Multi FI	3-Ways		>8 years	153
			Euro III		>8 years	797
			Euro IV		>8 years	5,974
			Euro V		>8 years	4,989
			Euro 6		>8 years	15,681
					4-8 years	25,748
					<4 years	25,672
			Hybrid		All	0
		Electric			All	474
Sml Engines	Light	4-Cycle	None	PCV	>8 years	114
			Euro III		>8 years	595
			Euro IV		>8 years	4,558
			Euro V		>8 years	3,628
			Euro 6		>8 years	11,903
					4-8 years	13,472
					<4 years	16,083
			Hybrid		All	0
		Electric			All	233
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	0
		Fuel-Injection	3-Ways		>8 years	36
			Euro III		>8 years	358
			Euro IV		>8 years	5,344
			Euro V		>8 years	5,786
			Euro 6		>8 years	16,314
					4-8 years	24,361
					<4 years	26,467
			Hybrid		All	6
		Electric			All	420

0 50.000 100.000 150.000 200.000 250.000 300.000

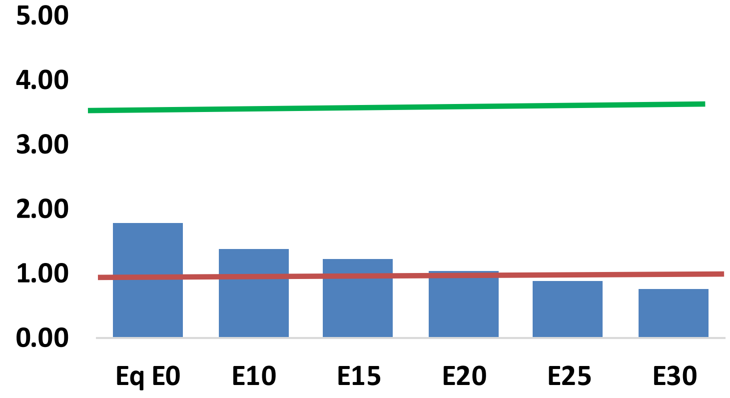
Vehicle Fleet: 1,038,755
Gasoline Vehicle Fleet: 694,397

Source: Eurostat, ACEA

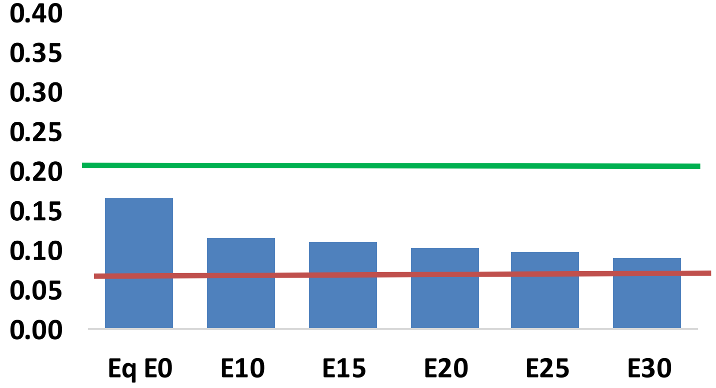
Luxembourg – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard

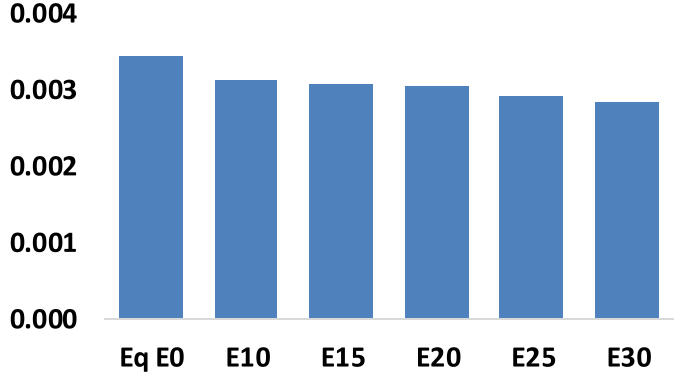
Carbon Monoxide (CO) g/km



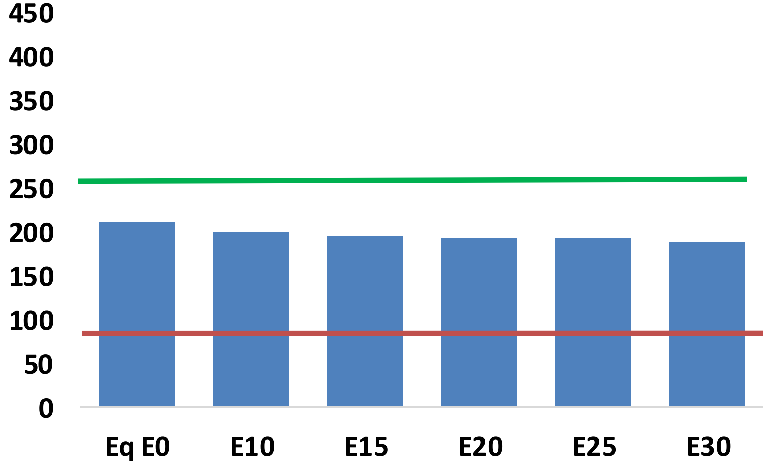
Nitrogen Oxides (NOx) g/km



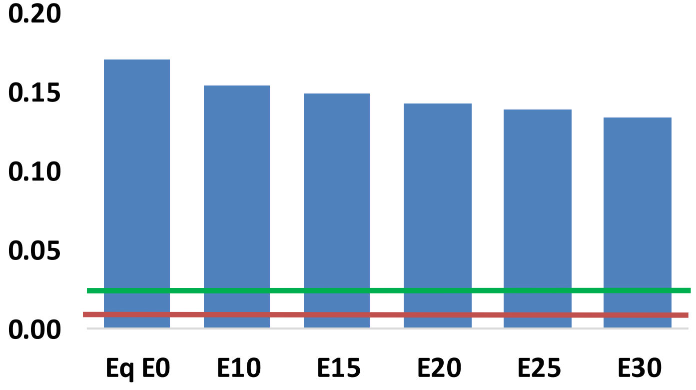
Aromatics (BTX) g/km



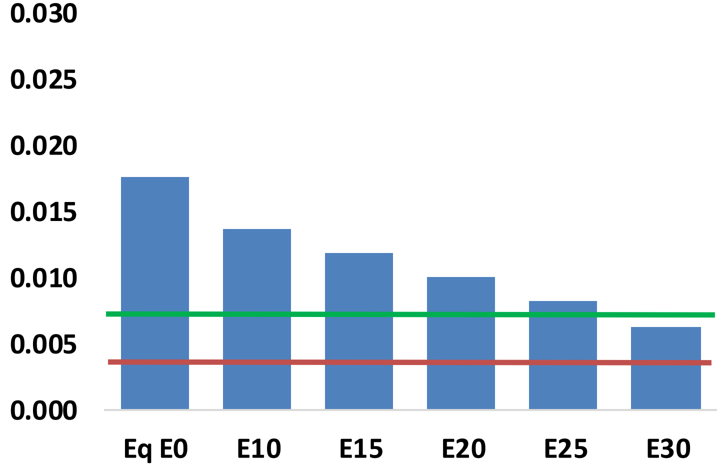
Carbon Dioxide (CO2) g/km



Volatile Organic Compounds (VOC) g/km



Particulate Matter (PM2.5) g/km



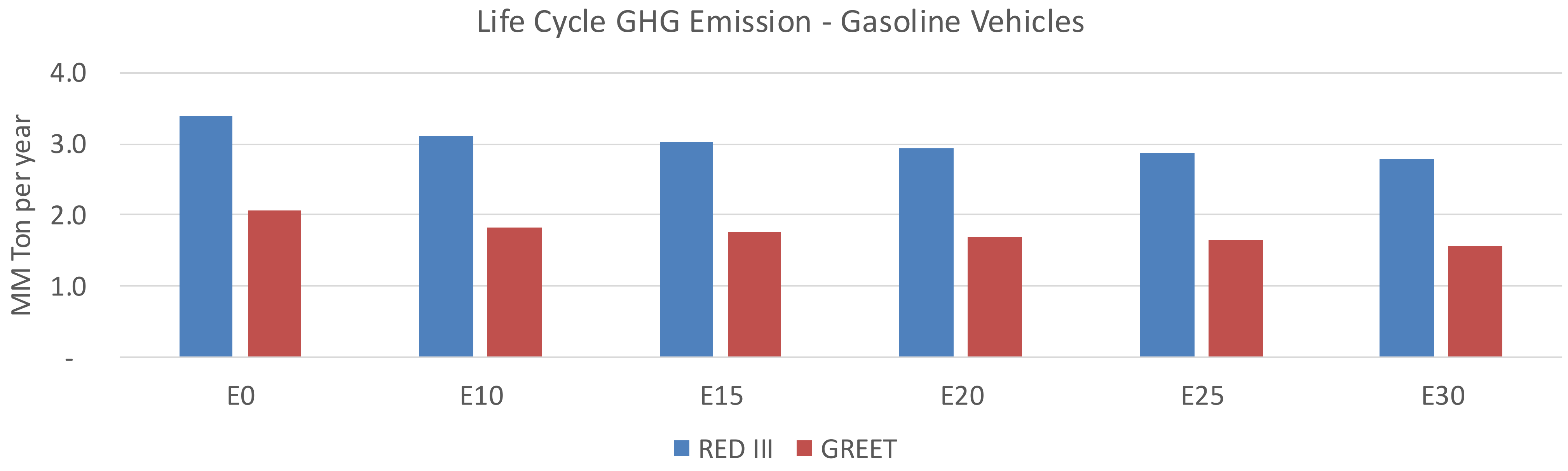
Source: Faro90

Luxembourg – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	32,931	29,249	25,567	22,326	18,961	16,342	13,726	-22%	-42%
CO2	3,882,918	3,783,071	3,688,772	3,614,608	3,577,909	3,556,182	3,490,631	-5%	-8%
VOC	3,142	2,992	2,843	2,738	2,642	2,570	2,465	-10%	-16%
NOx	3,081	2,311	2,157	2,033	1,917	1,790	1,655	-30%	-38%
SOx	44	41	37	34	32	29	26	-15%	-28%
NH3	1,305	1,292	1,279	1,285	1,299	1,309	1,318	-2%	0%
N2O	48	47	47	47	48	49	49	-1%	2%
CH4	747	747	747	758	775	789	802	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	31	29	28	27	27	26	25	-8%	-14%
Acetaldehyde	55	64	93	142	194	221	261	68%	249%
Formaldehyde	162	157	184	216	226	250	272	13%	39%
Benzene	63	61	58	57	56	54	52	-9%	-11%
PM2.5	244	216	189	164	139	113	86	-22%	-43%
PM10	81	72	62	54	46	37	28	-22%	-43%

Source: Faro90

Luxembourg – Gasoline Vehicle Fleet Life Cycle GHG Emissions



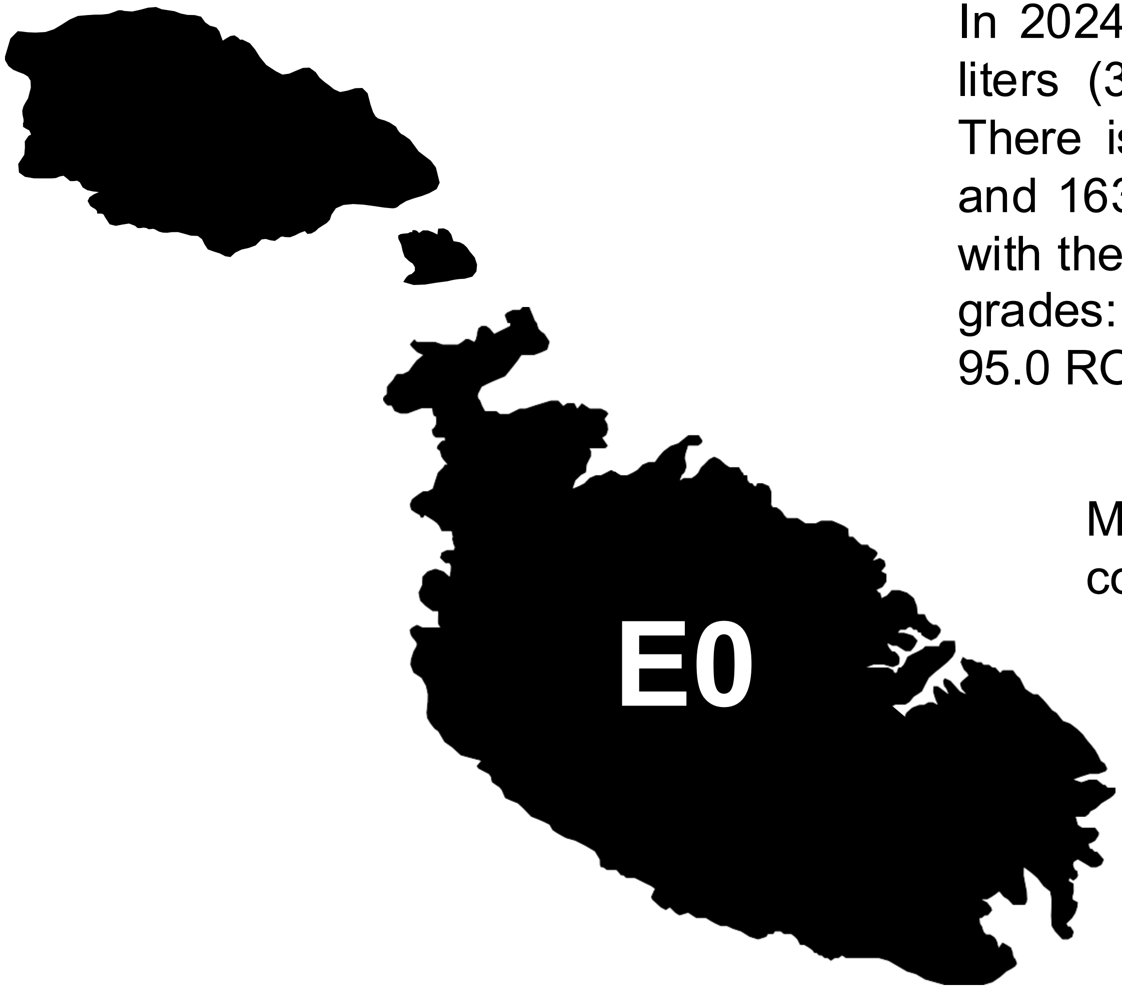
Country Emissions 2024 (MMT/año)	9.0
Target @2035 (MMT/year)	5.1
Delta	3.9

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.2%	14.7%	17.5%	20.2%	23.8%
RED II/III	0.0%	8.1%	10.8%	13.0%	15.2%	17.8%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.9%	7.7%	9.2%	10.6%	12.5%
RED II/III	0.0%	7.0%	9.3%	11.2%	13.1%	15.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Malta

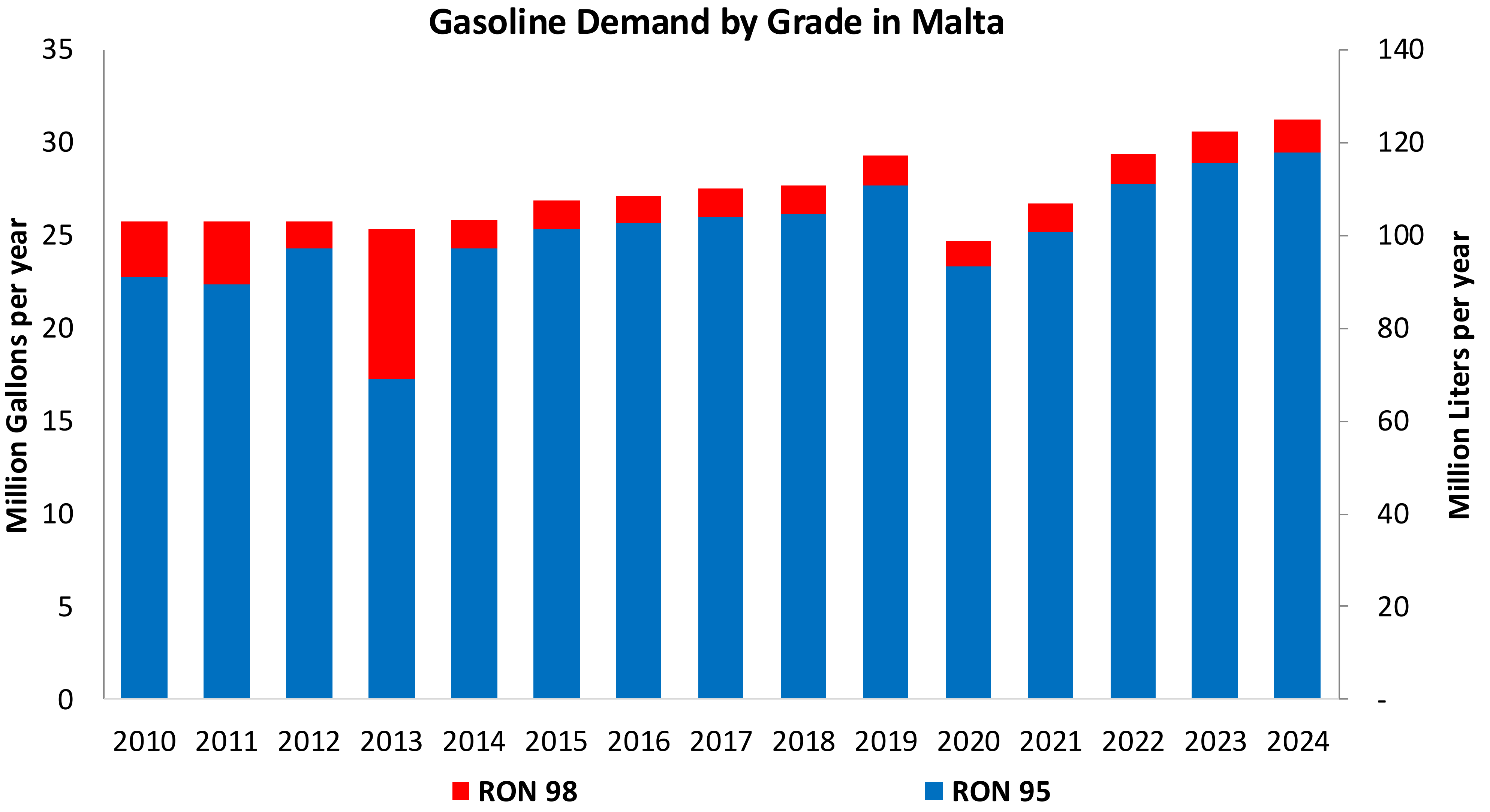


In 2024, gasoline consumption in Malta was 118 million liters (31 million gallons) and growth rate was 1.2%. There is no gasoline production. Gasoline imports were and 163 million liters (43 million gallons). Malta complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.0 RON.

Malta does not produce or consume bioethanol at a commercial level

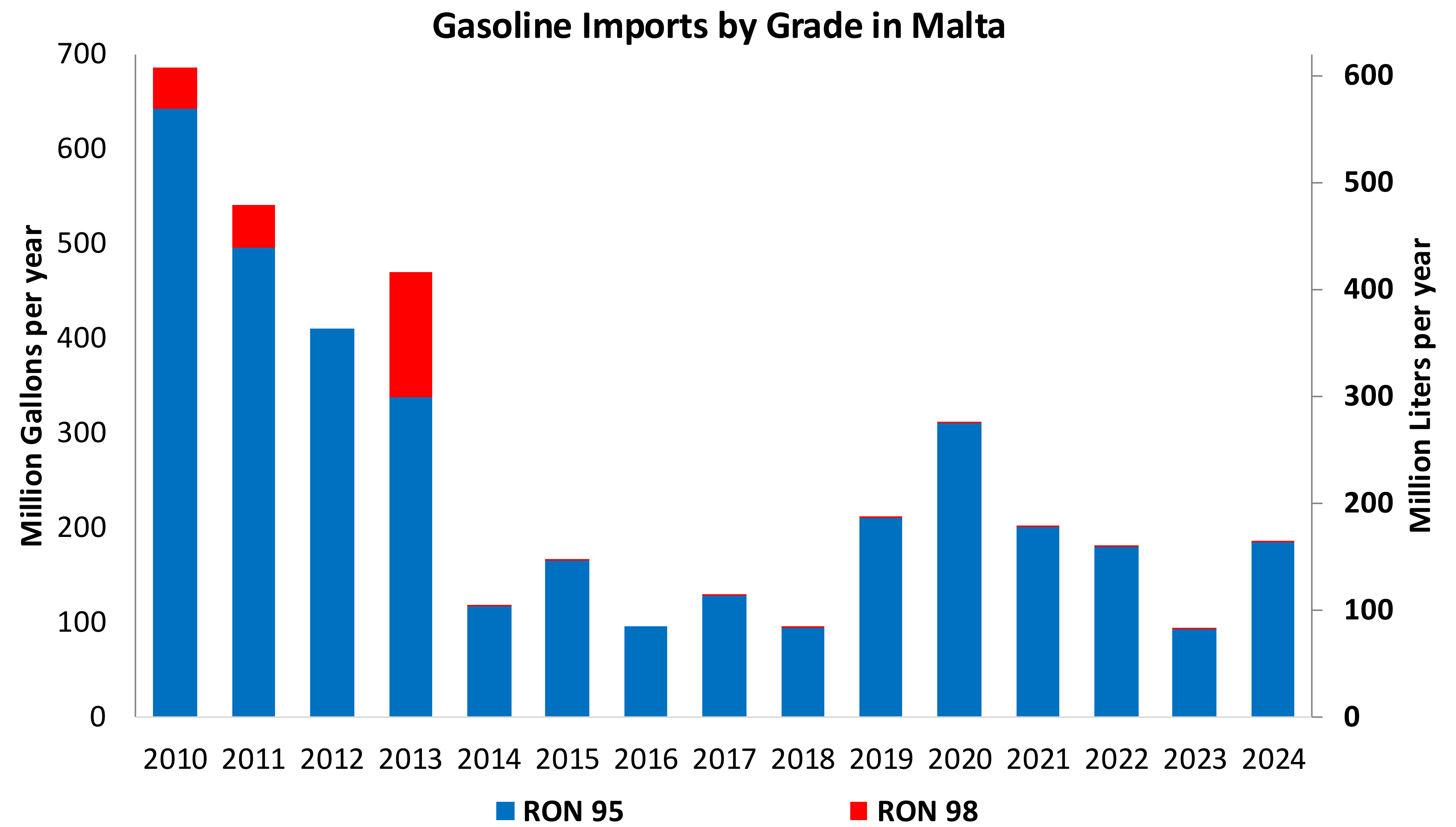
Source: European Commision, Eurostat

Gasoline Demand in Malta



Source: Eurostat

Gasoline Imports in Malta



Source: Eurostat

Ethanol Balance in Malta

Malta does not produce or consume bioethanol at a commercial level

Source: Eurostat

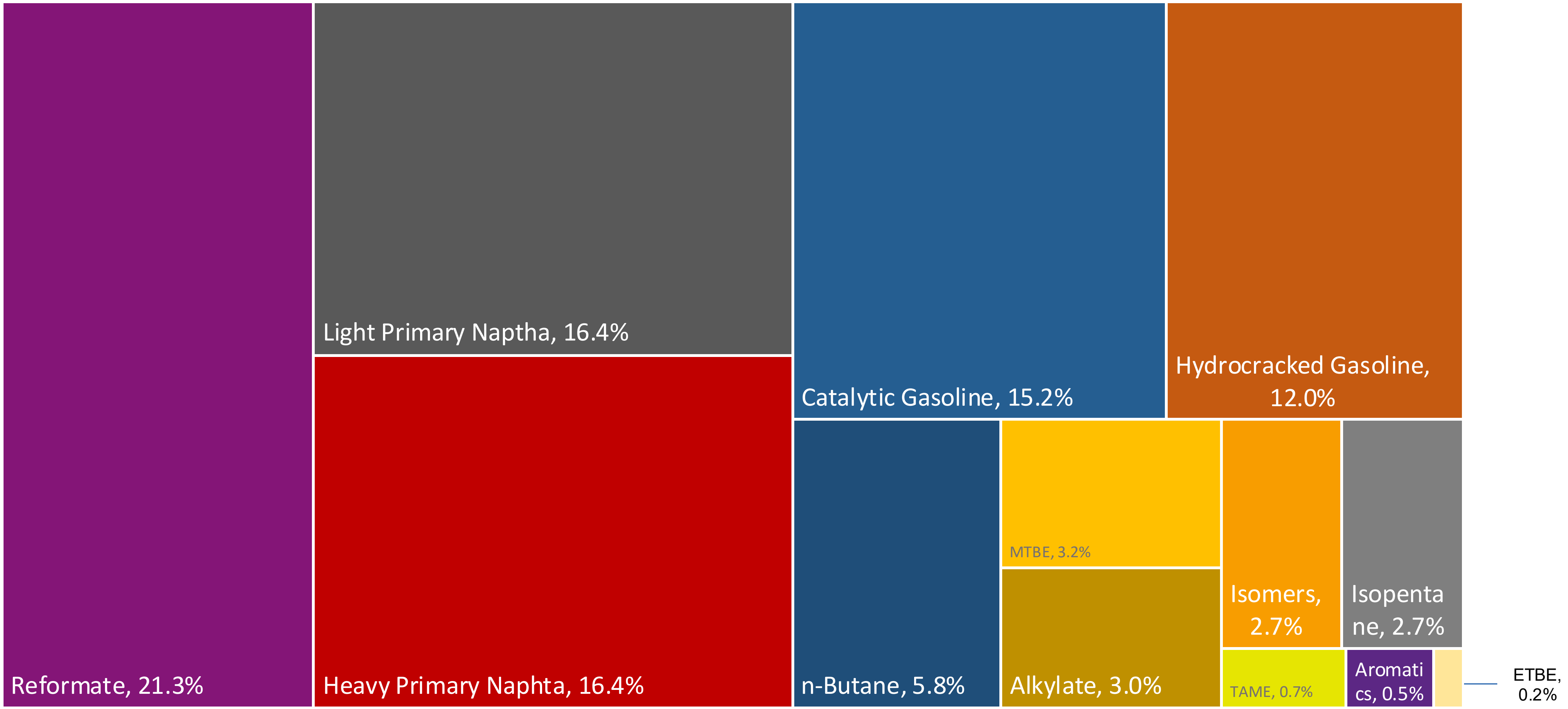
Gasoline Quality in Malta

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

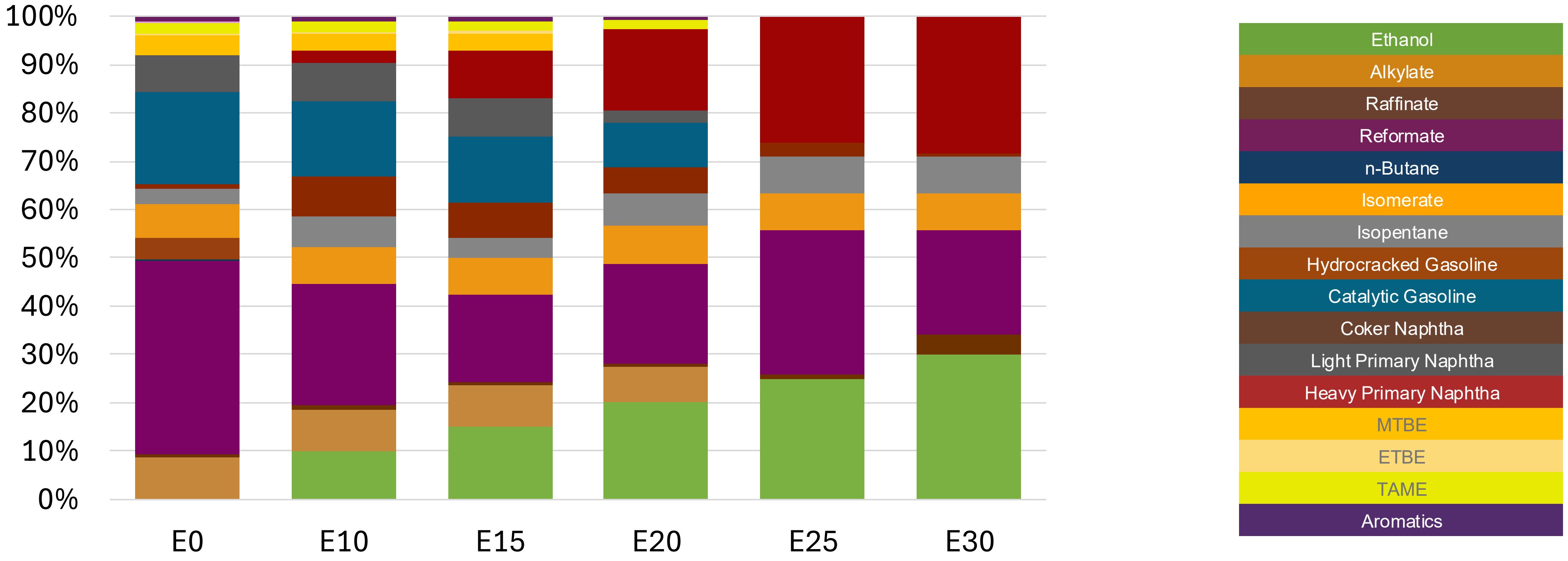
Source: European Commision

Malta Available Blending Components



Source: Faro90

Malta – Gasoline 95 – Constant Octane

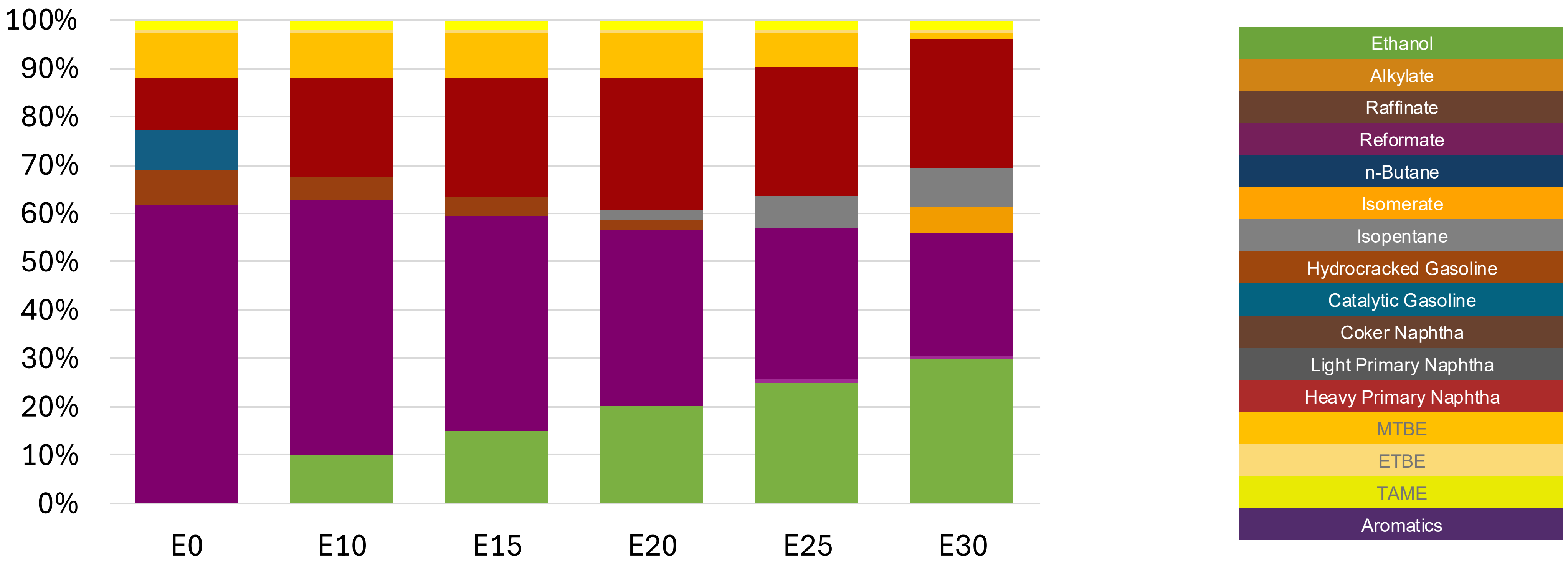


Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US$/gal)	$ 2.258	$ 2.145	$ 2.062	$ 1.986	$ 1.891	$ 1.817

Average CIF 2024 Prices

Source: Faro90

Malta - Gasoline 98 - Constant Octane

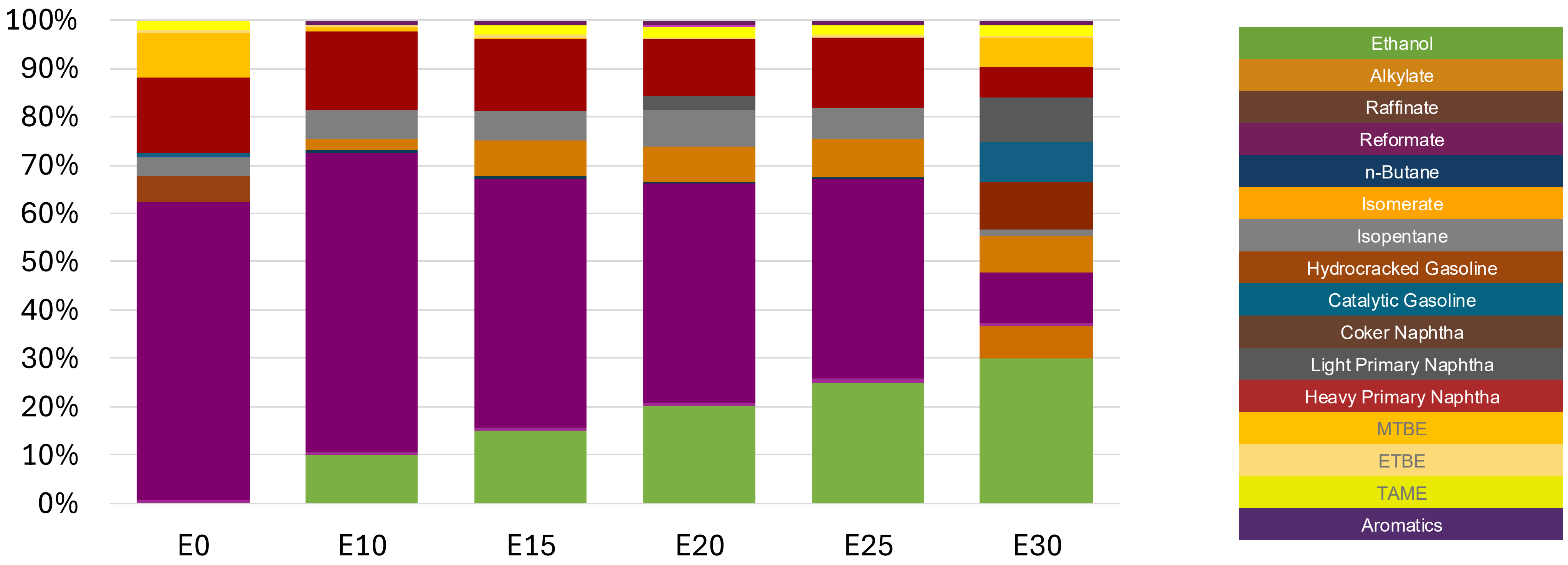


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.307	$ 2.188	$ 2.119	$ 2.049	$ 1.979	$ 1.926

Average CIF 2024 Prices

Source: Faro90

Malta – Gasoline 95 – Increased Octane

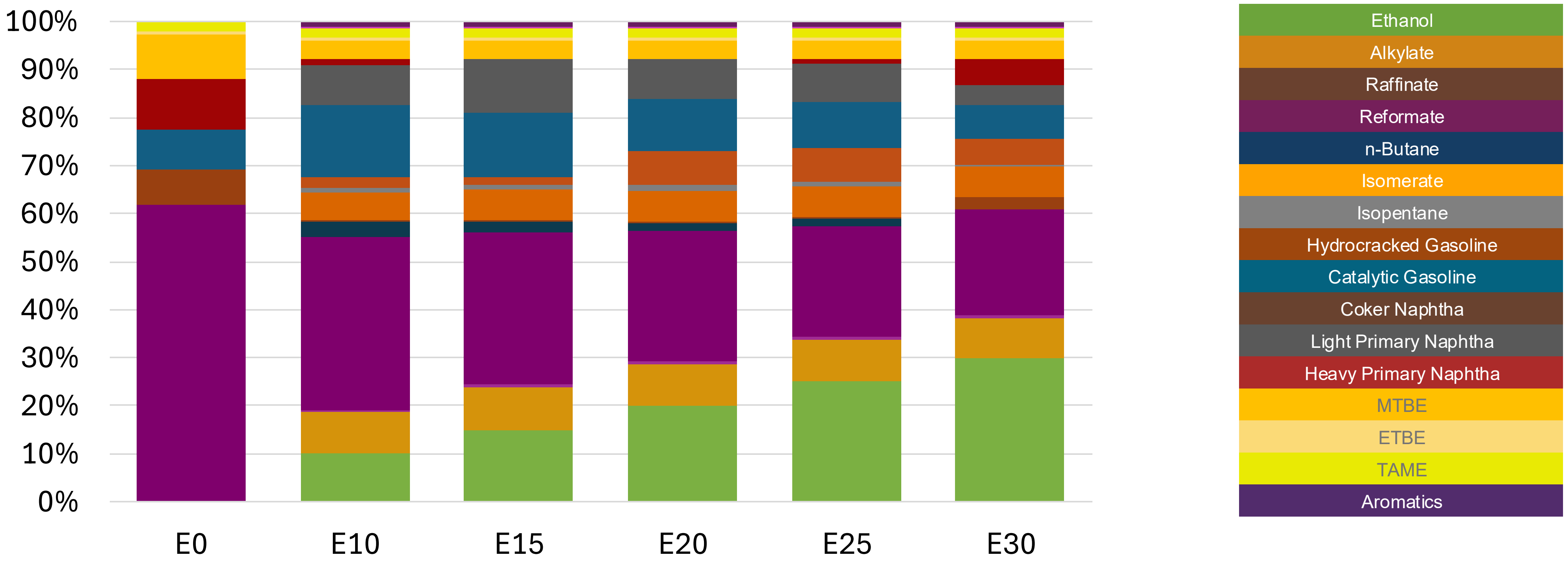


Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US$/gal)	$ 2.177	$ 2.177	$ 2.177	$ 2.177	$ 2.177	$ 2.177

Average CIF 2024 Prices

Source: Faro90

Malta – Gasoline 98 – Increased Octane



Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US$/gal)	$ 2.307	$ 2.307	$ 2.307	$ 2.307	$ 2.307	$ 2.307

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Malta

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	91
		Single FI	3-Ways		>8 years	2,132
		Multi FI	3-Ways		>8 years	24,208
			Euro III		>8 years	30,464
			Euro IV		>8 years	46,373
			Euro V		>8 years	21,413
			Euro 6		>8 years	37,579
					4-8 years	13,870
					<4 years	992
			Hybrid		All	17,942
		Electric			All	1,276
	Medium	Carburator	3-Ways	PCV Tank	>8 years	9
		Single FI	3-Ways		>8 years	63
		Multi FI	3-Ways		>8 years	3,358
			Euro III		>8 years	5,565
			Euro IV		>8 years	16,212
			Euro V		>8 years	6,561
			Euro 6		>8 years	12,526
					4-8 years	6,143
					<4 years	693
			Hybrid		All	12,628
		Electric			All	991
Sml Engines	Light	4-Cycle	None	PCV	>8 years	299
			Euro III		>8 years	835
			Euro IV		>8 years	2,980
			Euro V		>8 years	1,805
			Euro 6		>8 years	7,860
					4-8 years	8,911
					<4 years	14,140
			Hybrid		All	1,842
		Electric			All	368
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	-
		Fuel-Injection	3-Ways		>8 years	5,059
			Euro III		>8 years	2,655
			Euro IV		>8 years	6,689
			Euro V		>8 years	2,757
			Euro 6		>8 years	5,752
					4-8 years	3,941
					<4 years	7,101
			Hybrid		All	14,838
		Electric			All	15,735

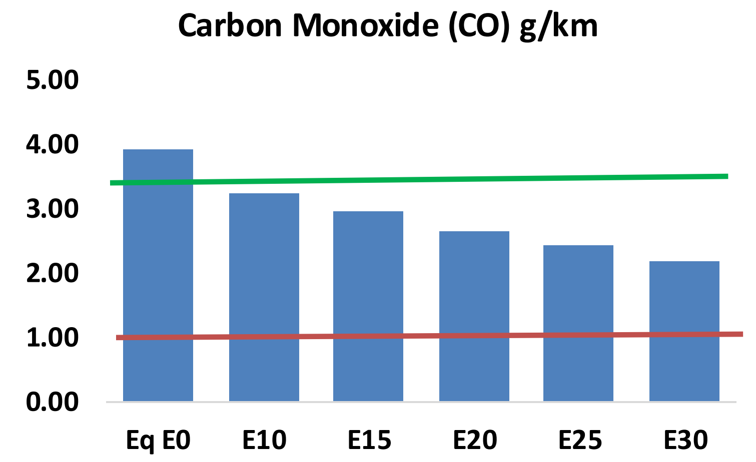
0 5,000 10,000 15,000 20,000 25,000 30,000 35,000 40,000 45,000 50,000

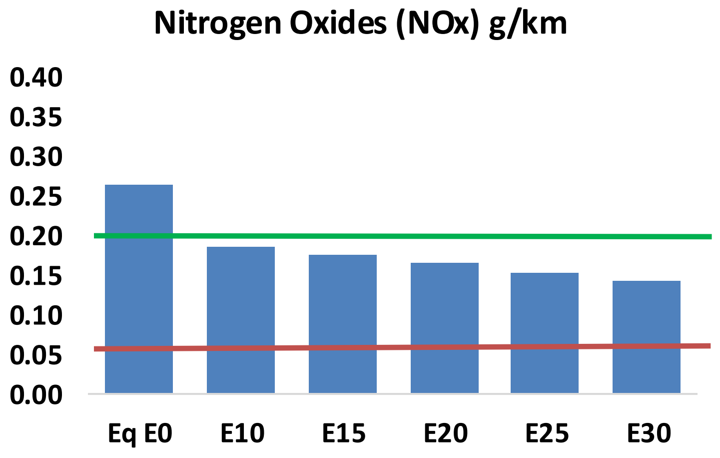
Vehicle Fleet: 364,656
Gasoline Vehicle Fleet 207,158

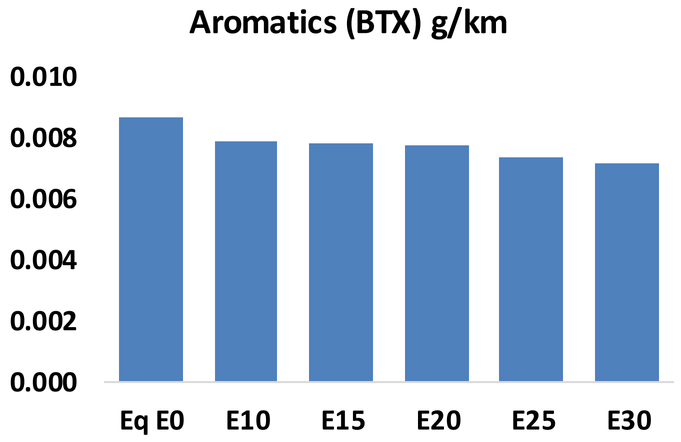
Source: Eurostat, ACEA

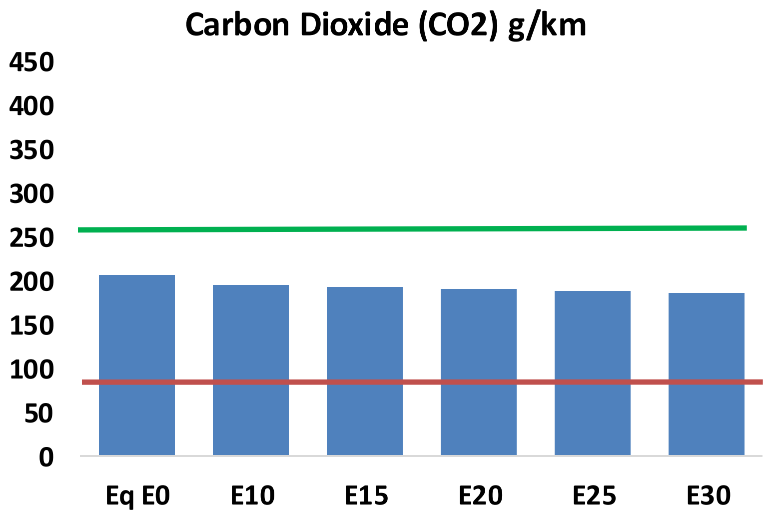
Malta – Vehicular Emissions

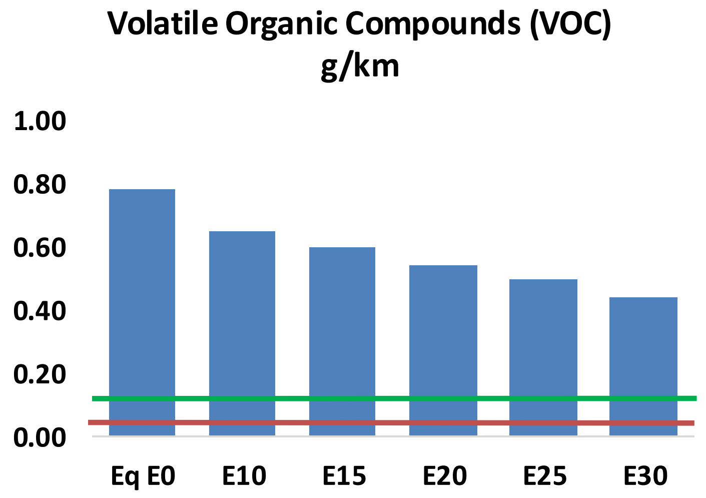
TIER III BIN 125 United States
Euro 6 Standard

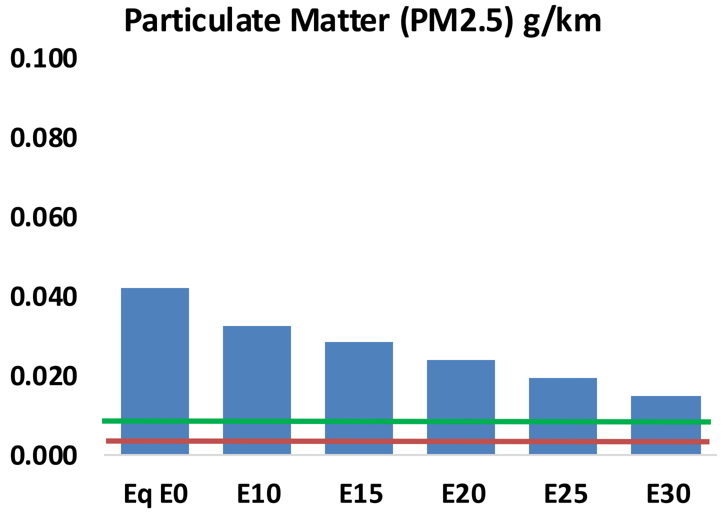












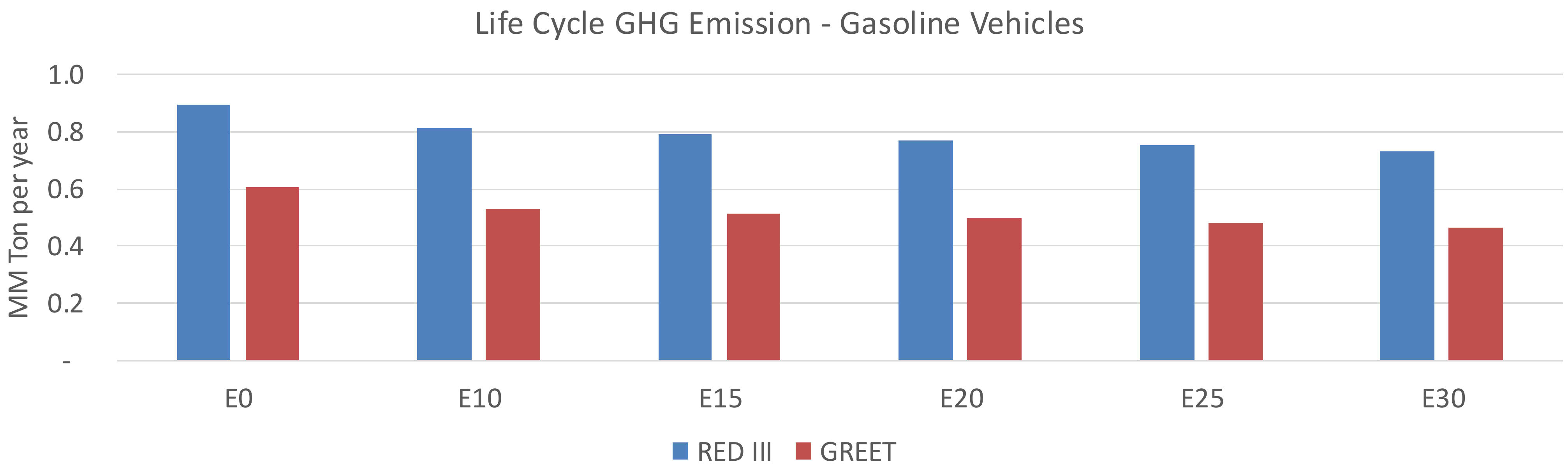
Source: Faro90

Malta – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	19,908	18,177	16,446	14,988	13,502	12,355	11,130	-17%	-32%
CO2	1,052,550	1,025,484	999,922	979,818	969,870	964,236	946,463	-5%	-8%
VOC	3,980	3,647	3,315	3,038	2,756	2,539	2,252	-17%	-31%
NOx	1,355	1,016	948	894	843	787	728	-30%	-38%
SOx	13	12	11	10	9	8	7	-15%	-28%
NH3	404	400	396	398	402	405	408	-2%	0%
N2O	15	15	15	15	16	16	16	-1%	2%
CH4	807	807	807	819	837	852	867	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	20	19	18	18	17	17	17	-8%	-13%
Acetaldehyde	66	76	111	169	230	263	310	68%	249%
Formaldehyde	245	238	277	326	341	378	411	13%	39%
Benzene	44	43	40	40	39	38	36	-9%	-11%
PM2.5	66	59	51	45	38	31	23	-22%	-43%
PM10	148	131	115	99	84	69	52	-22%	-43%

Source: Faro90

Malta – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	2.2
Target @ 2035 (MMT/year)	1.2
Delta	1.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.4%	15.6%	18.1%	20.5%	23.7%
RED II/III	0.0%	9.2%	11.7%	13.7%	15.6%	18.1%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.8%	9.8%	11.4%	12.9%	15.0%
RED II/III	0.0%	8.5%	10.9%	12.8%	14.6%	16.9%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

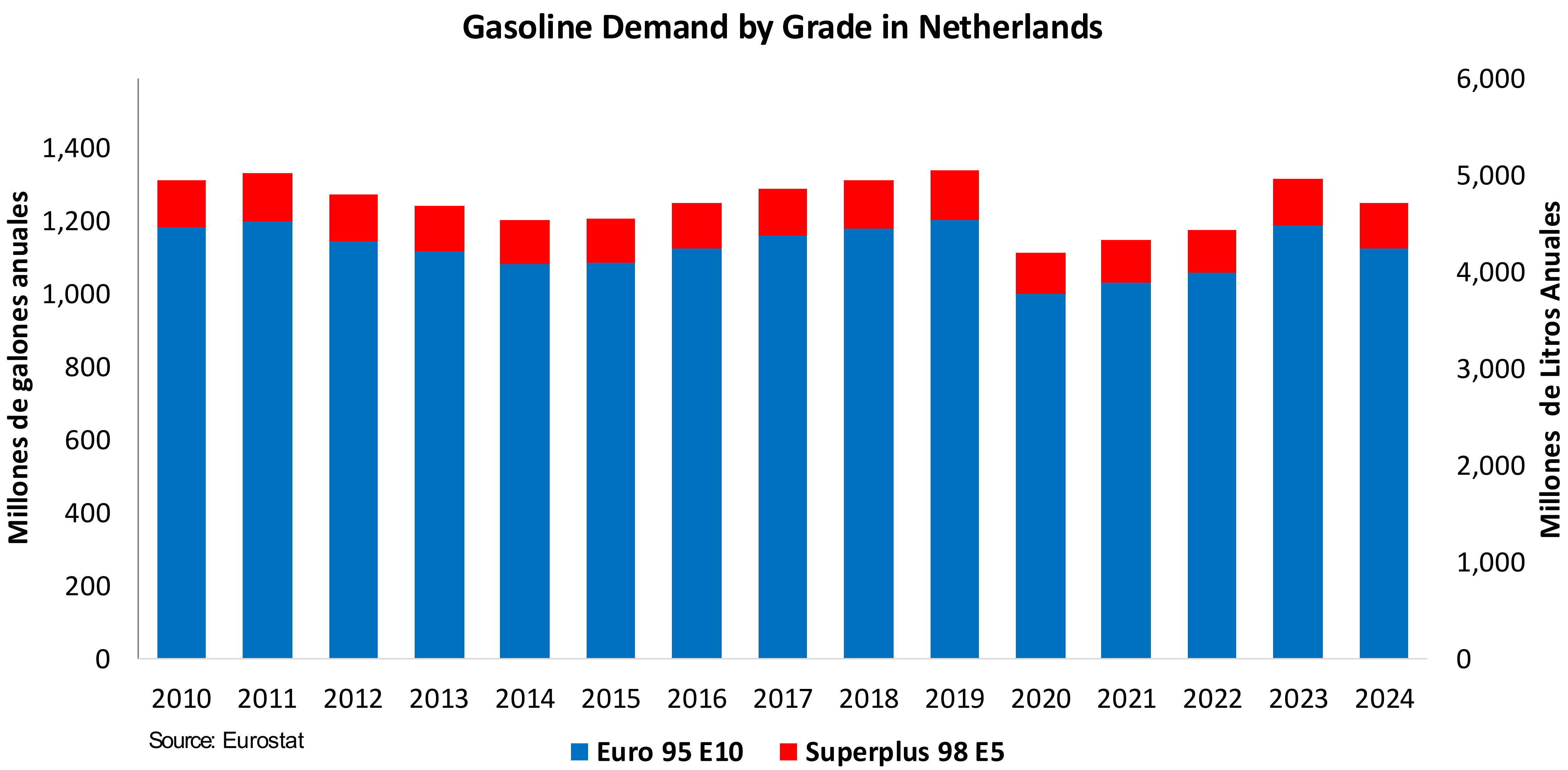
Ethanol Gasoline Blending in Netherlands

In 2024, gasoline consumption in Netherlands was 4,723 million liters (1,248 million gallons) and growth rate was -1.3%. Gasoline production and net imports were 6,295 and -14,373 million liters (1,663 and -3,797 million gallons) correspondingly. Netherlands complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Netherlands was 507 million liters (134 million gallons) representing 10.7% of gasoline demand. Ethanol production and Net Imports were 1,080 and -574 million liters (134 and -152 million gallons) correspondingly. Ethanol sources are Sugar Cane 5% and Cereals 95%.

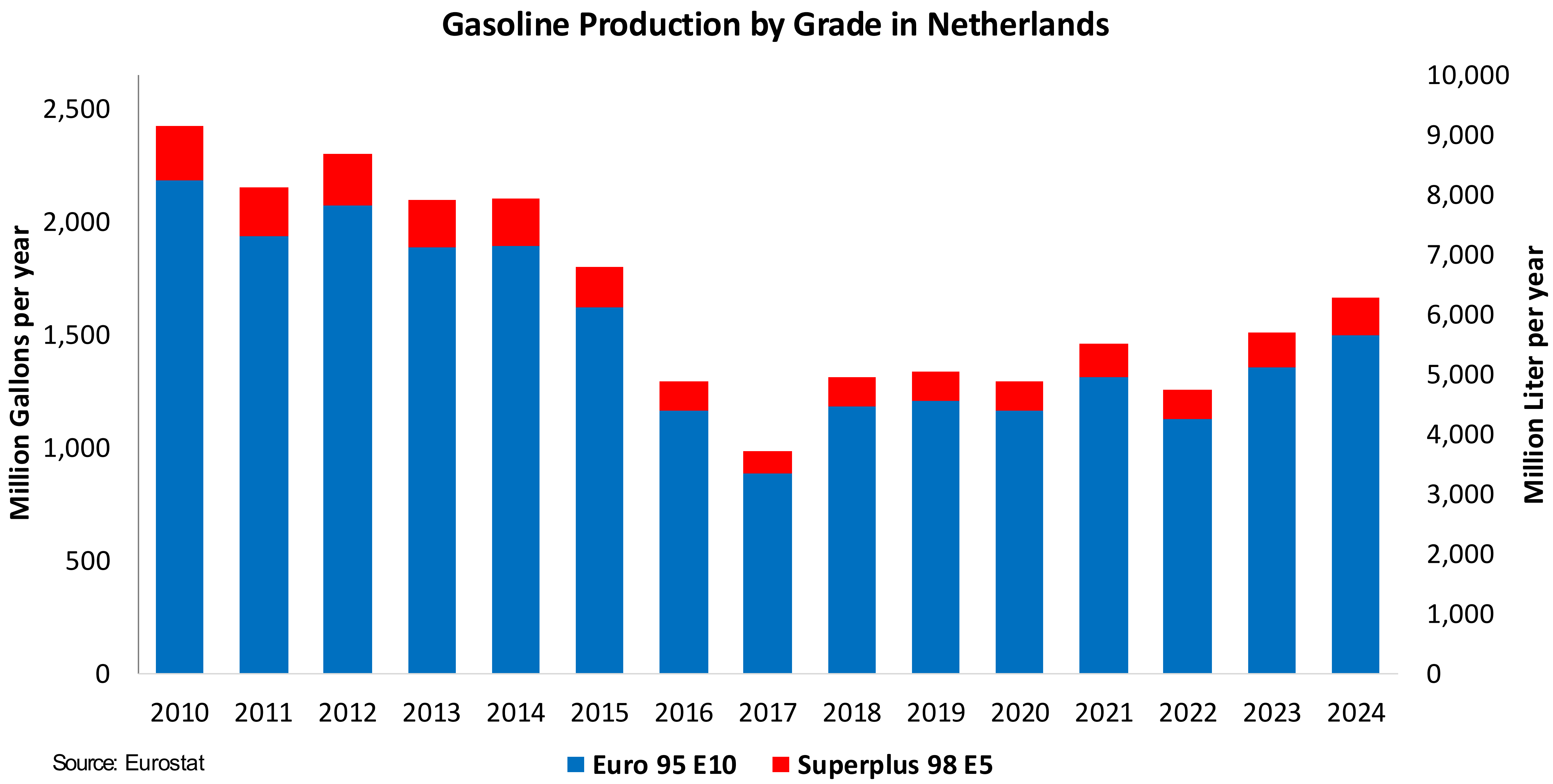
Source: Eurostat, European Commision

Gasoline Demand in Netherlands



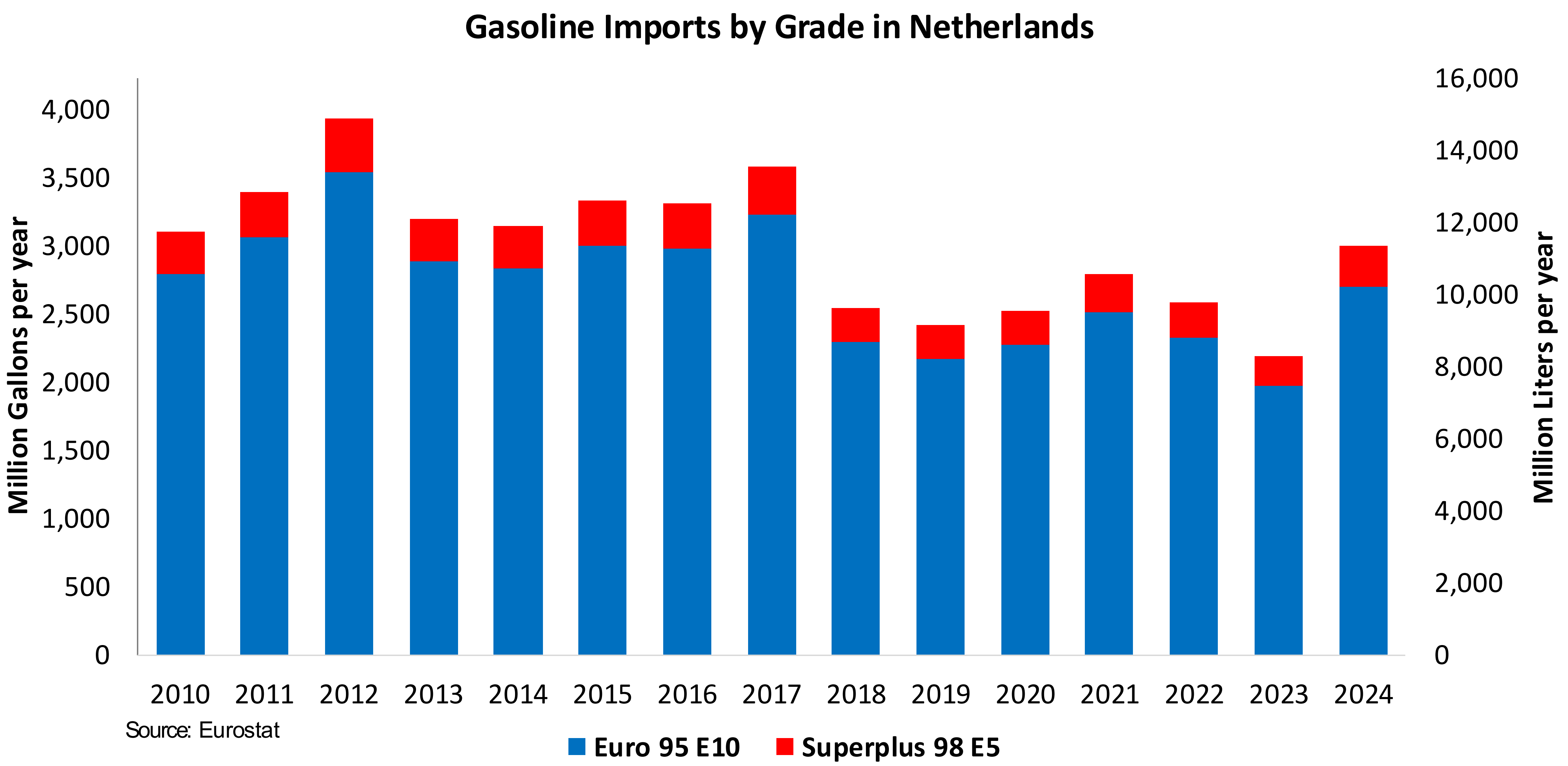
Source: Eurostat

Gasoline Production in Netherlands



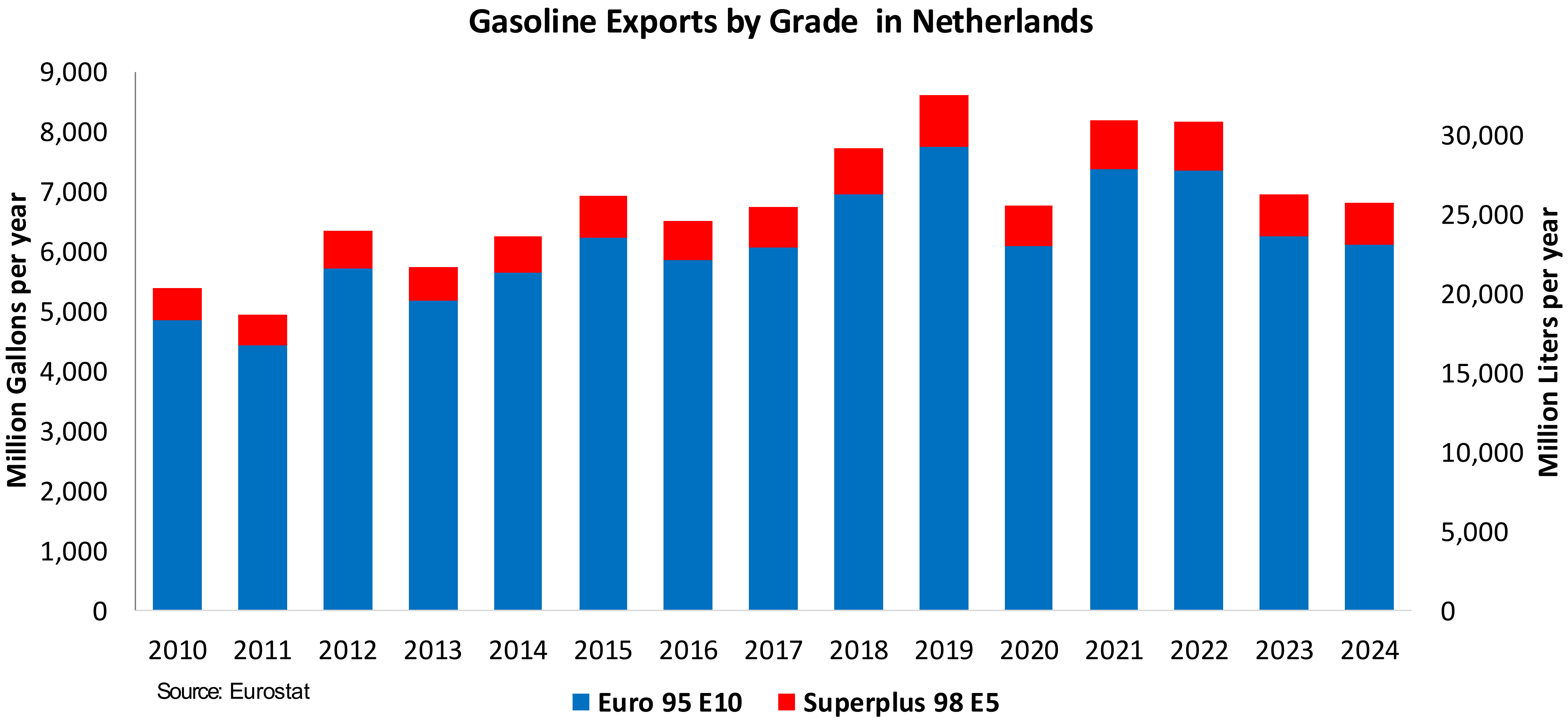
Source: Eurostat

Gasoline Imports in Netherlands



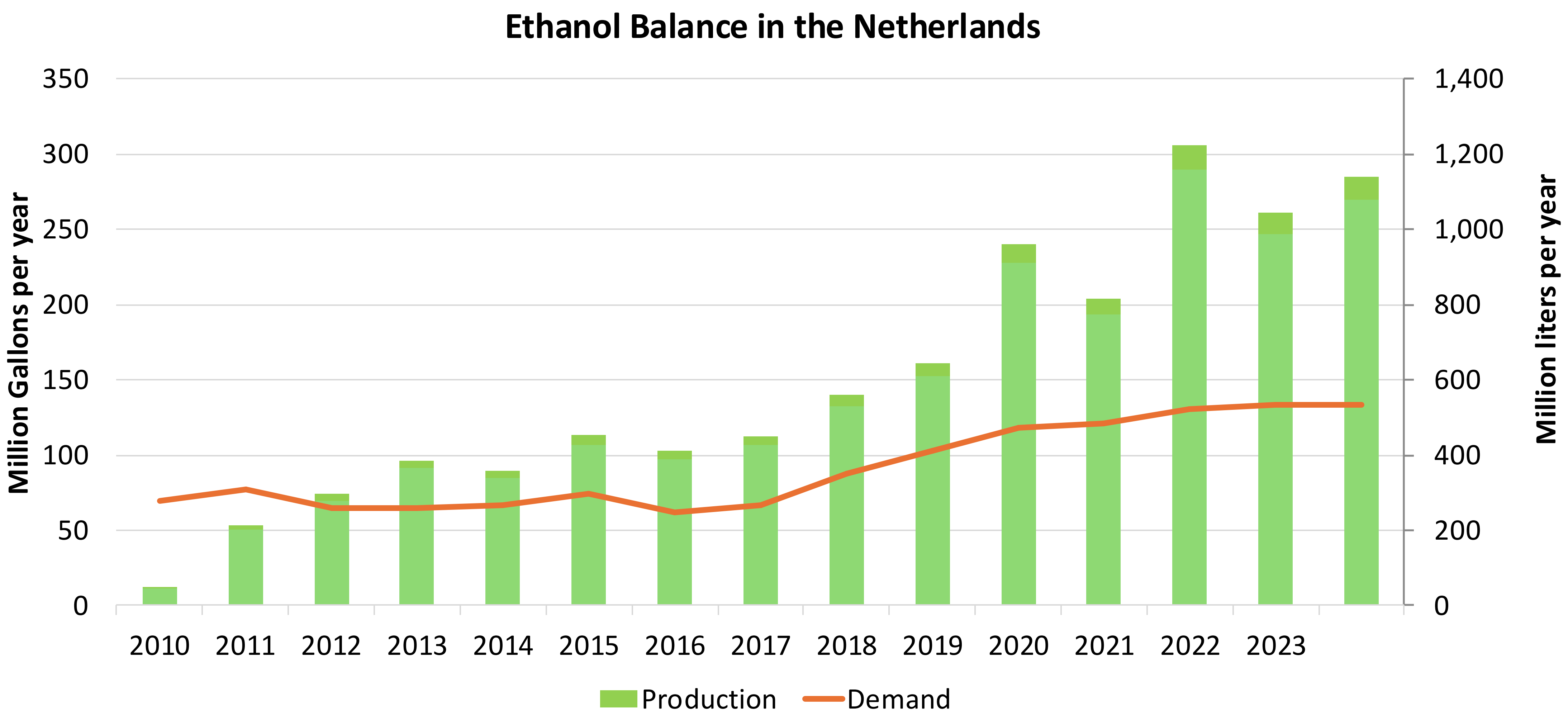
Source: Eurostat

Gasoline Exports in Netherlands



Source: Eurostat

Ethanol Balance in Netherlands



Source: Eurostat, Statistics Netherlands

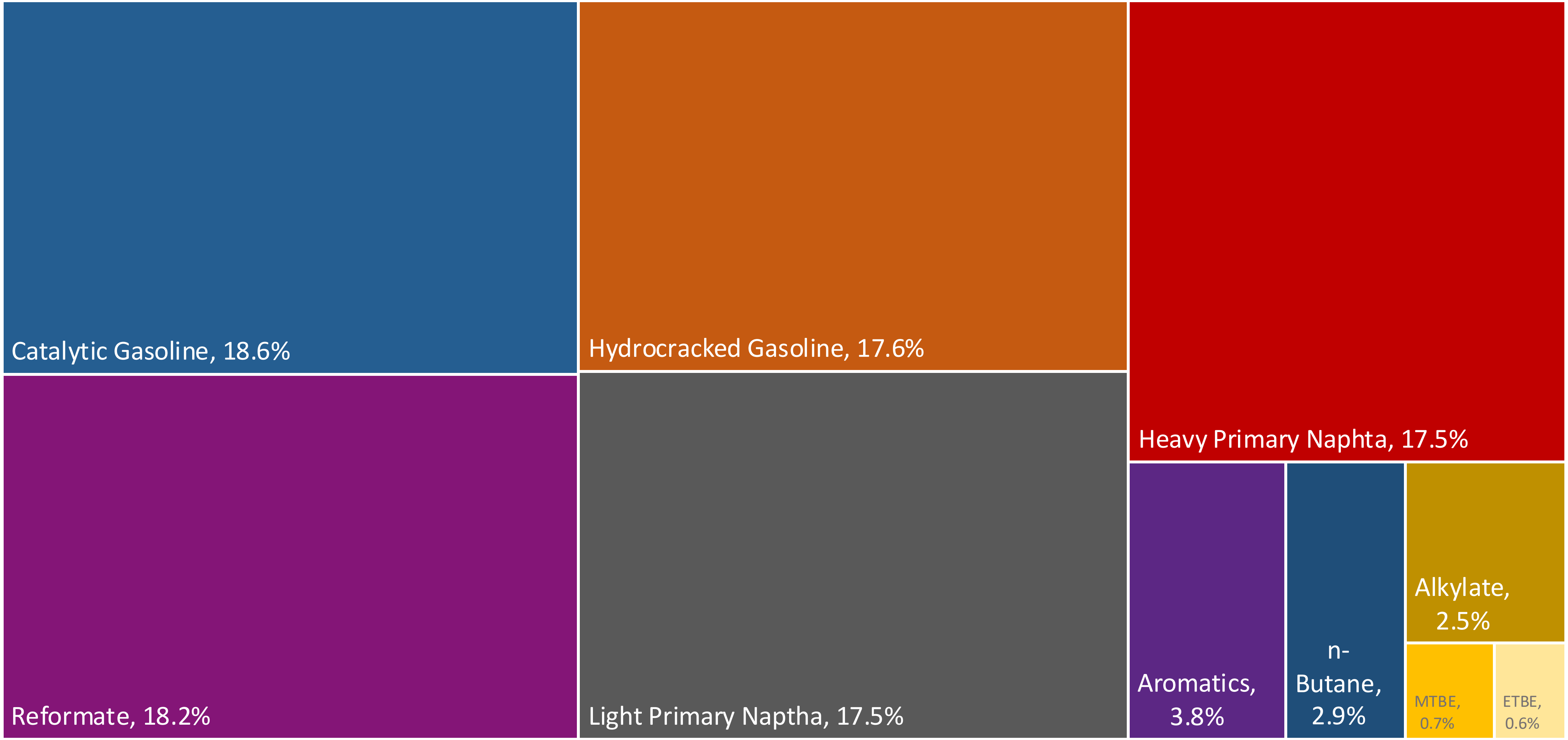
Gasoline Quality in Netherlands

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	28.4							
Advanced Biofuels (%cal)	3.6							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

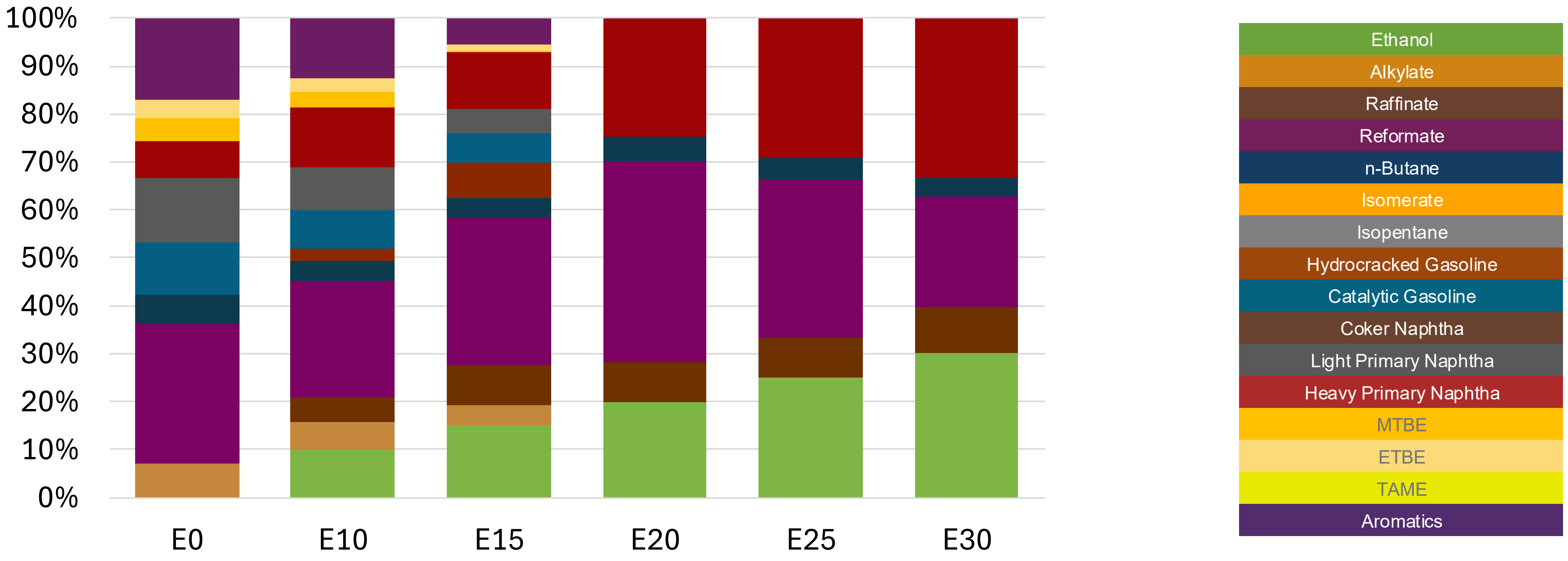
Source: European Commision

Netherlands Available Blending Components



Source: Faro90

Netherlands – Gasoline 95 – Constant Octane

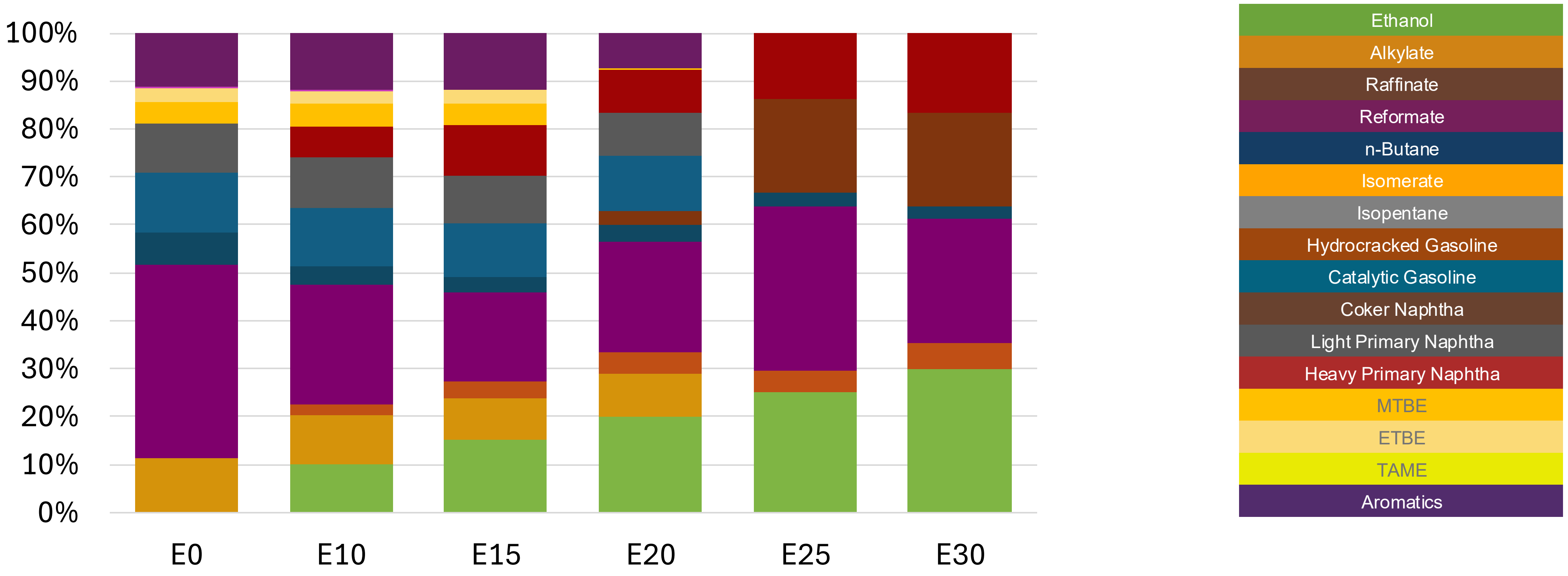


Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US$/gal)	$ 2.281	$ 2.147	$ 2.072	$ 1.951	$ 1.875	$ 1.780

Average CIF 2024 Prices

Source: Faro90

Netherlands - Gasoline 98 - Constant Octane

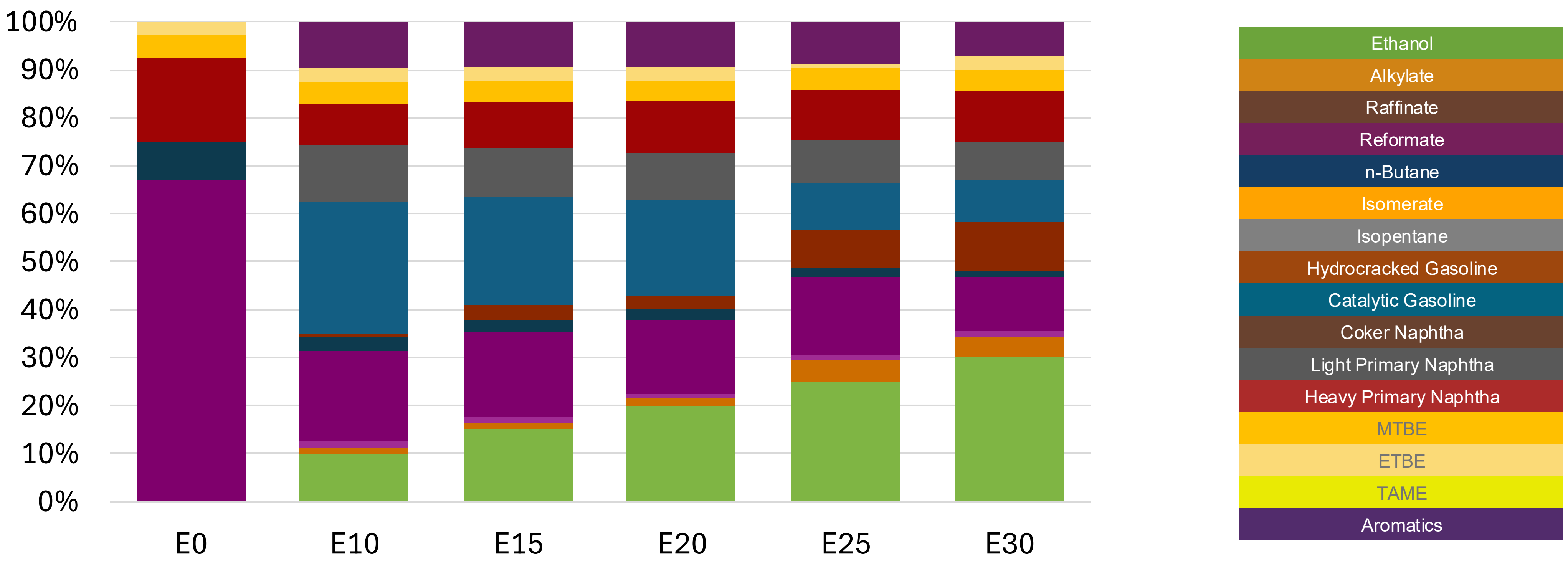


Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US$/gal)	$ 2.409	$ 2.290	$ 2.216	$ 2.164	$ 2.057	$ 1.994

Average CIF 2024 Prices

Source: Faro90

Netherlands – Gasoline 95 – Increased Octane

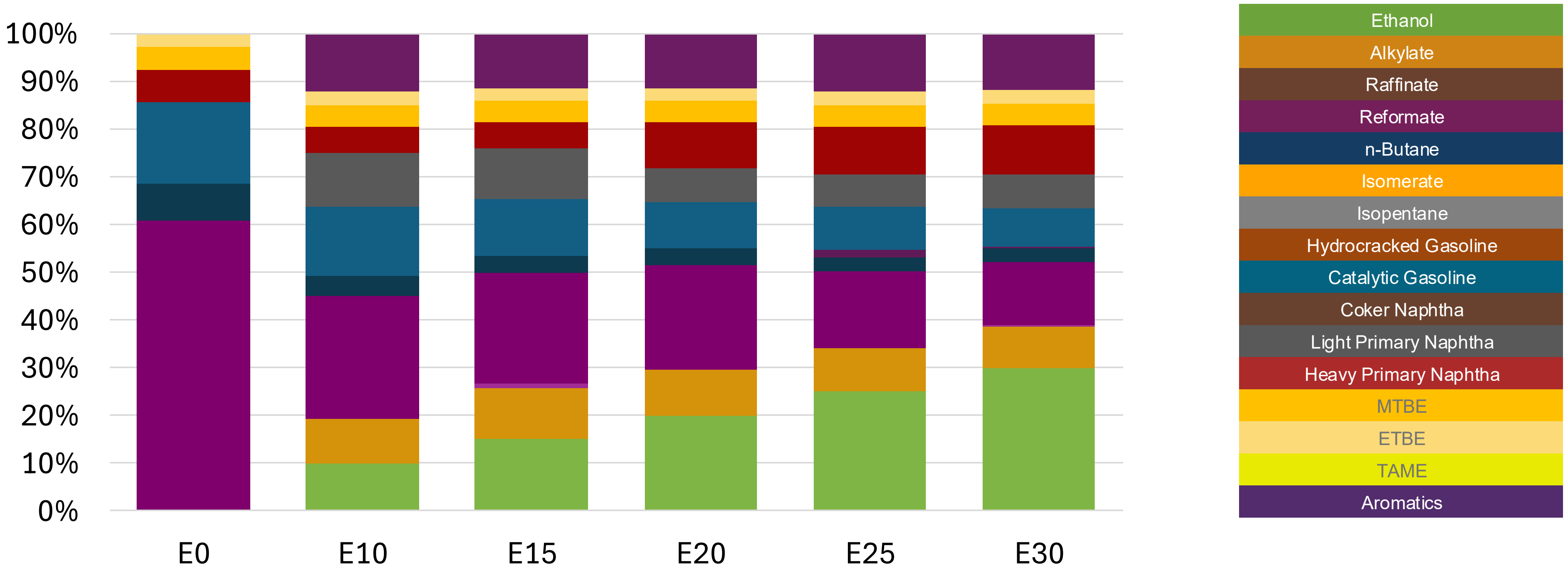


Octane (RON)	95.0	96.5	97.9	99.4	100.5	102.1
Price (US$/gal)	$ 2.193	$ 2.193	$ 2.193	$ 2.193	$ 2.193	$ 2.193

Average CIF 2024 Prices

Source: Faro90

Netherlands – Gasoline 98 – Increased Octane



Octane (RON)	98.0	98.8	100.1	101.6	102.9	104.2
Price (US$/gal)	$ 2.329	$ 2.329	$ 2.329	$ 2.329	$ 2.329	$ 2.329

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Netherlands

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	6,216
		Single FI	3-Ways		>8 years	12,798
		Multi FI	3-Ways		>8 years	793,203
			Euro III		>8 years	96,743
			Euro IV		>8 years	700,482
			Euro V		>8 years	484,680
			Euro 6		>8 years	235,874
					4-8 years	134,841
					<4 years	200,369
			Hybrid		All	161,840
		Electric			All	19,143
	Medium	Carburator	3-Ways	PCV Tank	>8 years	615
		Single FI	3-Ways		>8 years	180
		Multi FI	3-Ways		>8 years	84,622
			Euro III		>8 years	26,441
			Euro IV		>8 years	236,181
			Euro V		>8 years	146,819
			Euro 6		>8 years	78,625
					4-8 years	67,627
					<4 years	158,736
			Hybrid		All	93,586
		Electric			All	11,070
Sml Engines	Light	4-Cycle	None	PCV	>8 years	103,199
			Euro III		>8 years	106,680
			Euro IV		>8 years	603,459
			Euro V		>8 years	358,694
			Euro 6		>8 years	858,679
					4-8 years	653,077
					<4 years	1,634,834
			Hybrid		All	42,874
		Electric			All	5,071
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	74,584
		Fuel-Injection	3-Ways		>8 years	842,906
			Euro III		>8 years	472,733
			Euro IV		>8 years	544,933
			Euro V		>8 years	152,548
			Euro 6		>8 years	210,599
					4-8 years	99,290
					<4 years	12,899
			Hybrid		All	14,445
		Electric			All	1,709

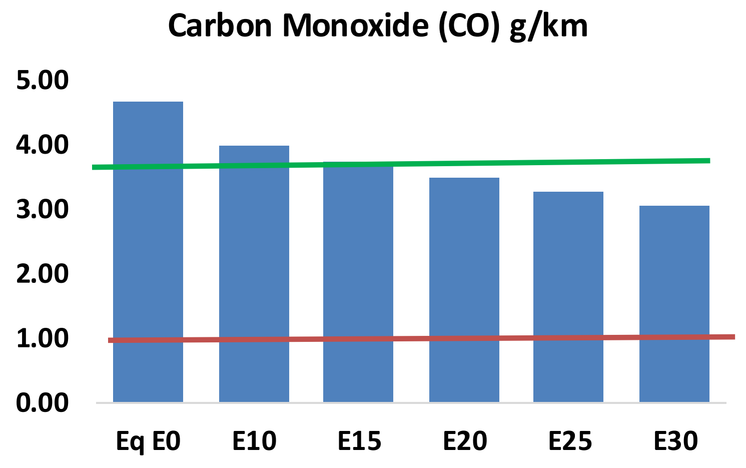
0 200,000 400,000 600,000 800,000 1,000,000 1,200,000 1,400,000 1,600,000 1,800,000

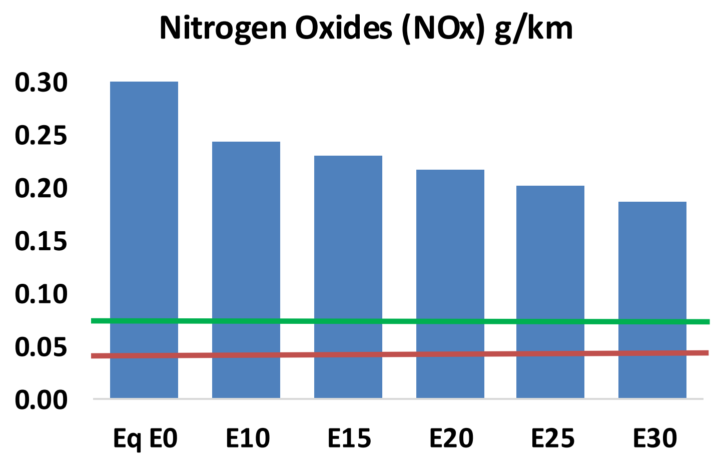
Vehicle Fleet: 10,593,903
Gasoline Vehicle Fleet: 7,721,374

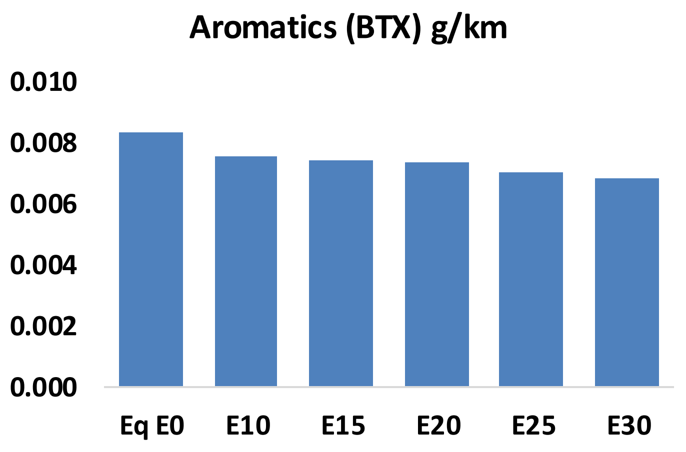
Source: Eurostat, ACEA

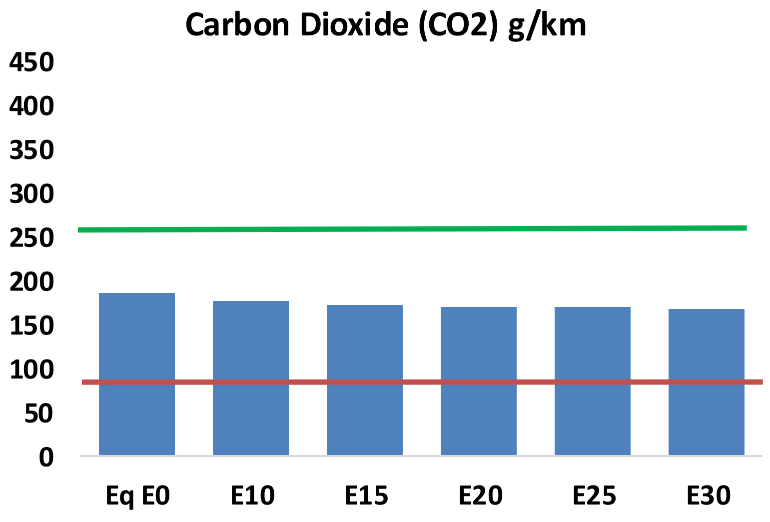
Netherlands – Vehicular Emissions

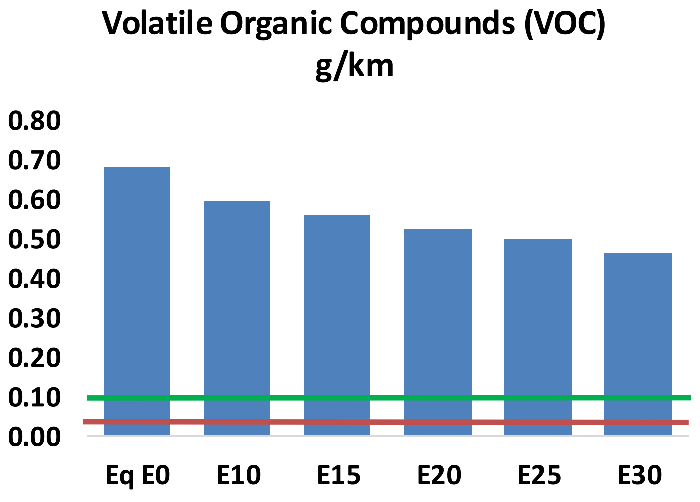
TIER III BIN 125 United States
Euro 6 Standard

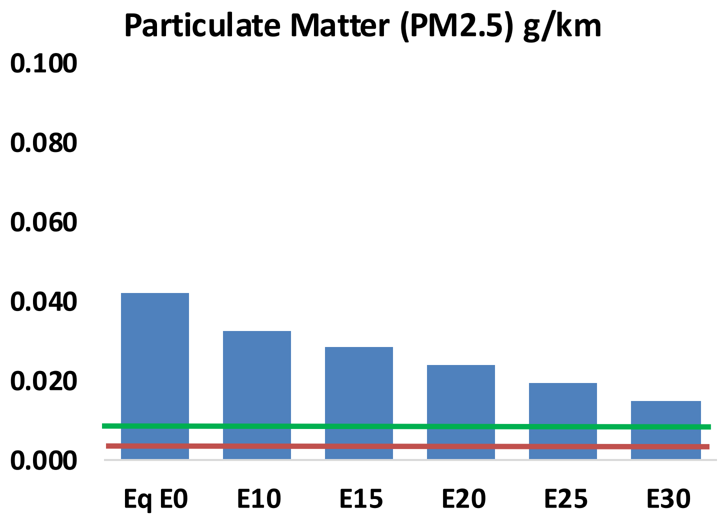












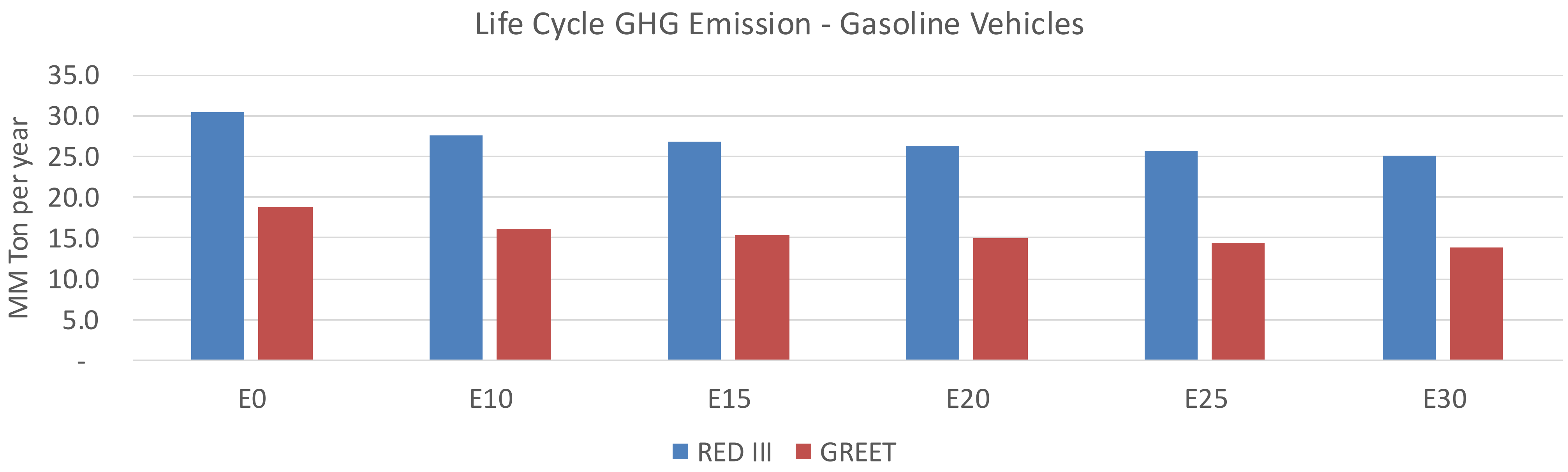
Source: Faro90

Netherlands – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	704,445	654,707	604,969	565,072	525,320	494,872	460,719	-14%	-25%
CO2	28,033,974	27,313,101	26,632,276	26,096,827	25,831,865	25,714,823	25,240,825	-5%	-8%
VOC	103,360	96,697	90,034	84,824	79,696	75,788	70,450	-13%	-23%
NOx	52,857	39,643	37,000	34,873	32,888	30,700	28,384	-30%	-38%
SOx	333	308	283	261	241	219	195	-15%	-28%
NH3	10,778	10,671	10,564	10,615	10,734	10,818	10,888	-2%	0%
N2O	473	470	468	470	480	485	491	-1%	2%
CH4	23,057	23,057	23,057	23,403	23,910	24,348	24,763	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	737	710	682	666	652	642	626	-7%	-12%
Acetaldehyde	2,047	2,349	3,447	5,250	7,152	8,173	9,657	68%	249%
Formaldehyde	7,461	7,253	8,455	9,928	10,401	11,507	12,526	13%	39%
Benzene	1,258	1,209	1,146	1,128	1,118	1,070	1,035	-9%	-11%
PM2.5	1,819	1,615	1,411	1,225	1,041	844	640	-22%	-43%
PM10	4,542	4,033	3,524	3,060	2,599	2,109	1,598	-22%	-43%

Source: Faro90

Netherlands – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	171.1
Target @ 2035 (MMT/year)	96.3
Delta	74.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.0%	18.6%	20.2%	22.9%	26.4%
RED II/III	0.0%	9.2%	11.5%	13.4%	15.3%	17.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.5%	4.7%	5.1%	5.8%	6.6%
RED II/III	0.0%	3.7%	4.7%	5.5%	6.2%	7.1%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

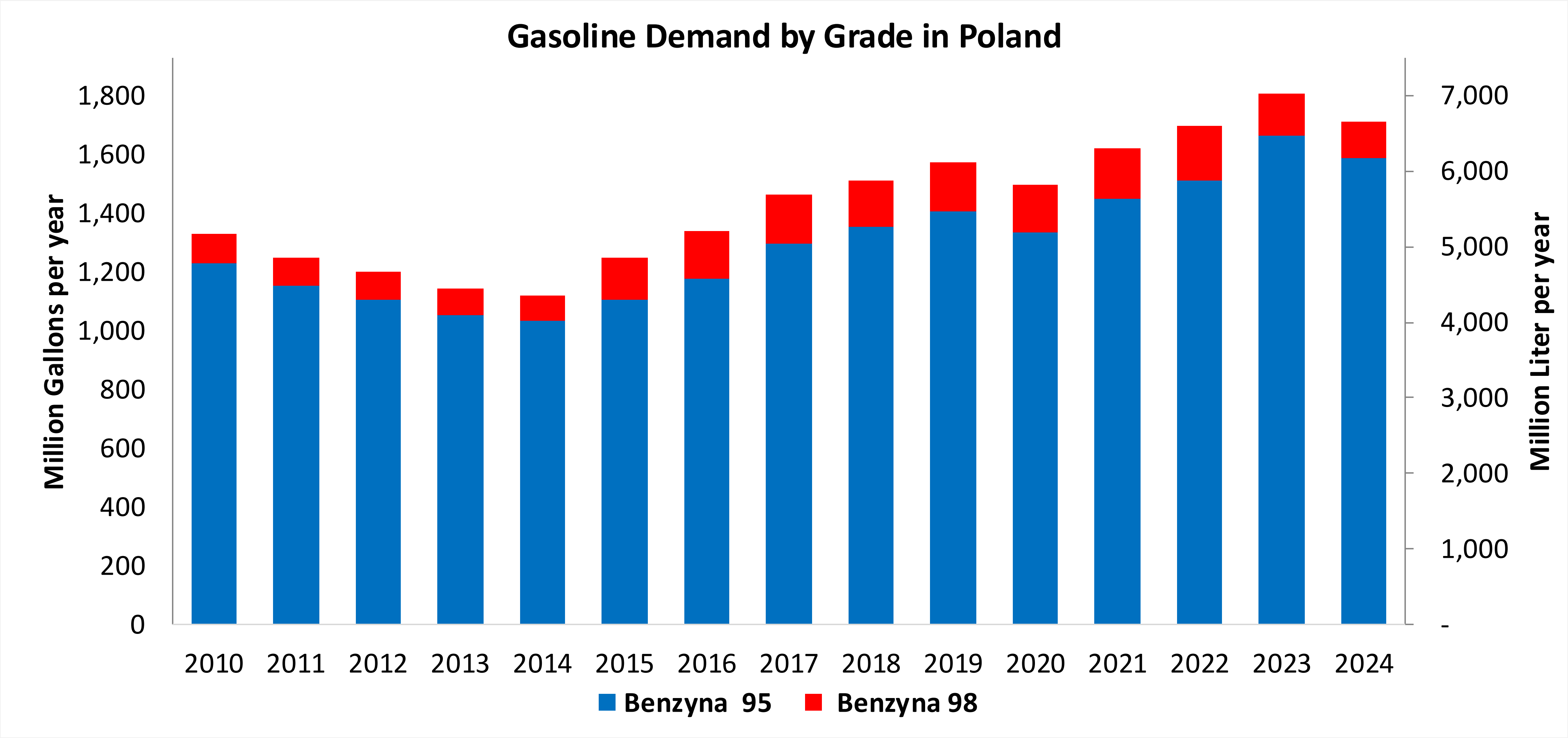
Ethanol Gasoline Blending in Poland

In 2024, gasoline consumption in Poland was 6,493 million liters (1,715 million gallons) and growth rate was 1.7%. Gasoline production and net imports were 5,606 and 913 million liters (1,481 and 241 million gallons) correspondingly. Poland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Poland was 420 million liters (111 million gallons) representing 6.5% of gasoline demand. Ethanol production and Net Imports were 452 and -32 million liters (111 and -9 million gallons) correspondingly. Ethanol sources are Corn 57% and Cereals 43%.

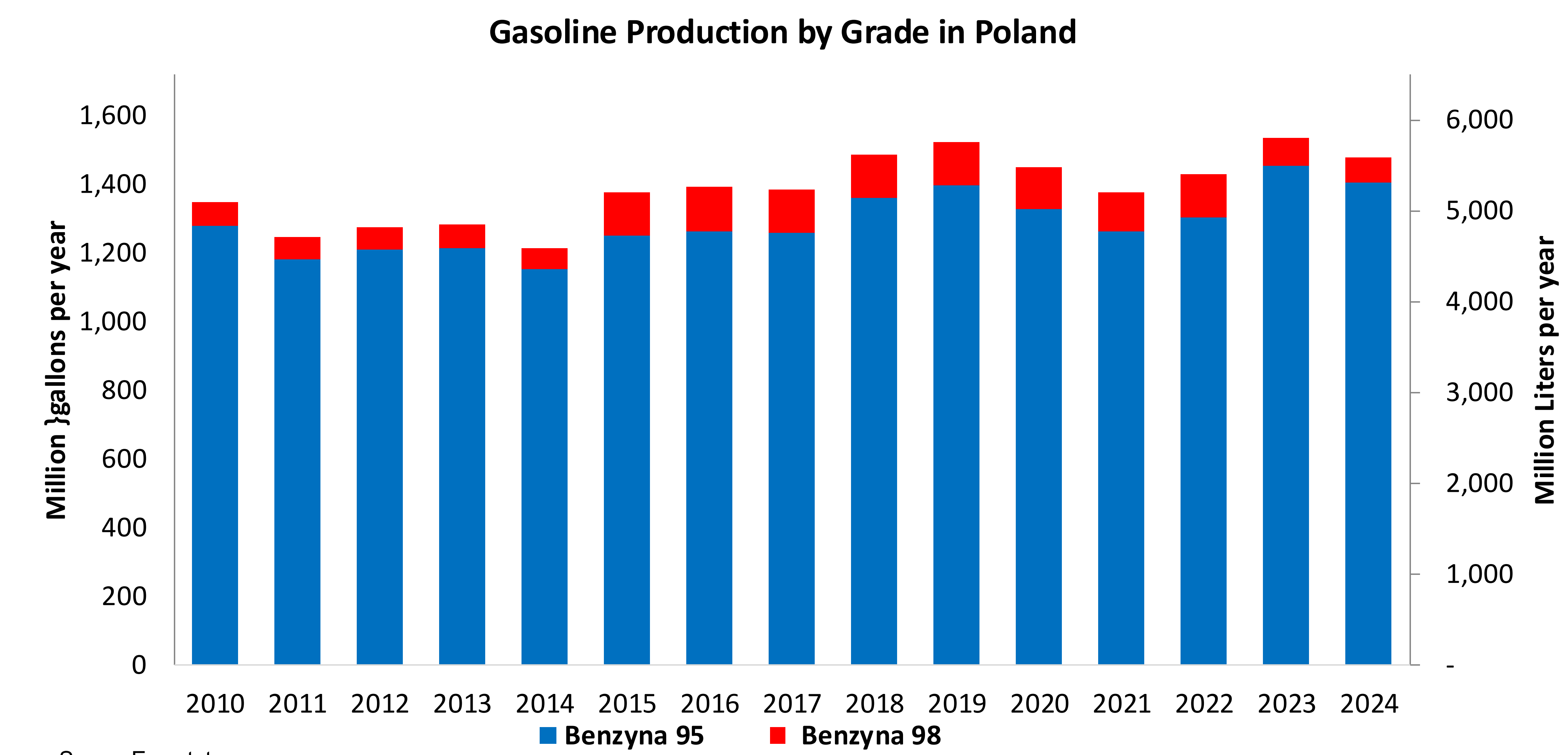
Source: Eurostat, European Commision

Gasoline Demand in Poland



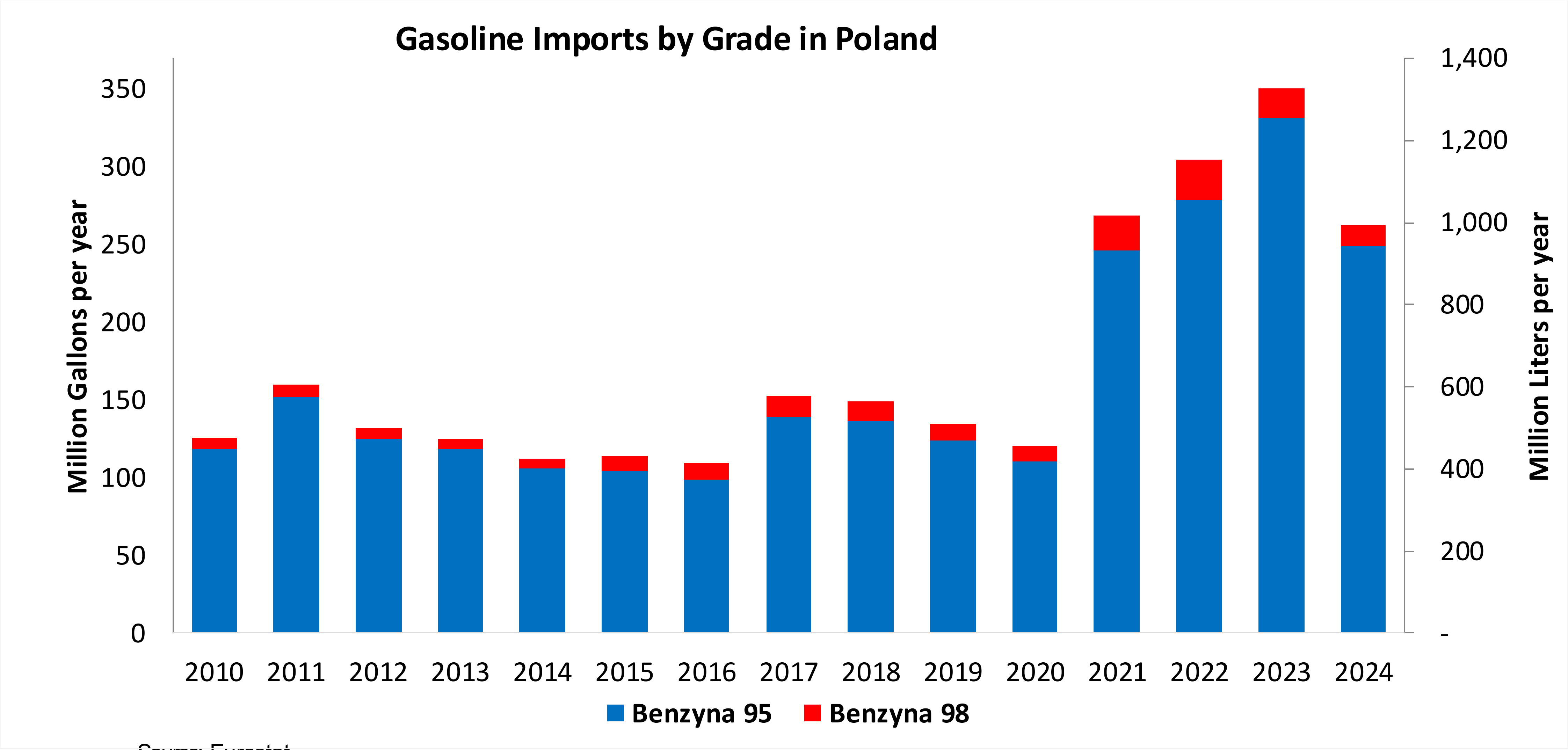
Source: Eurostat

Gasoline Production in Poland



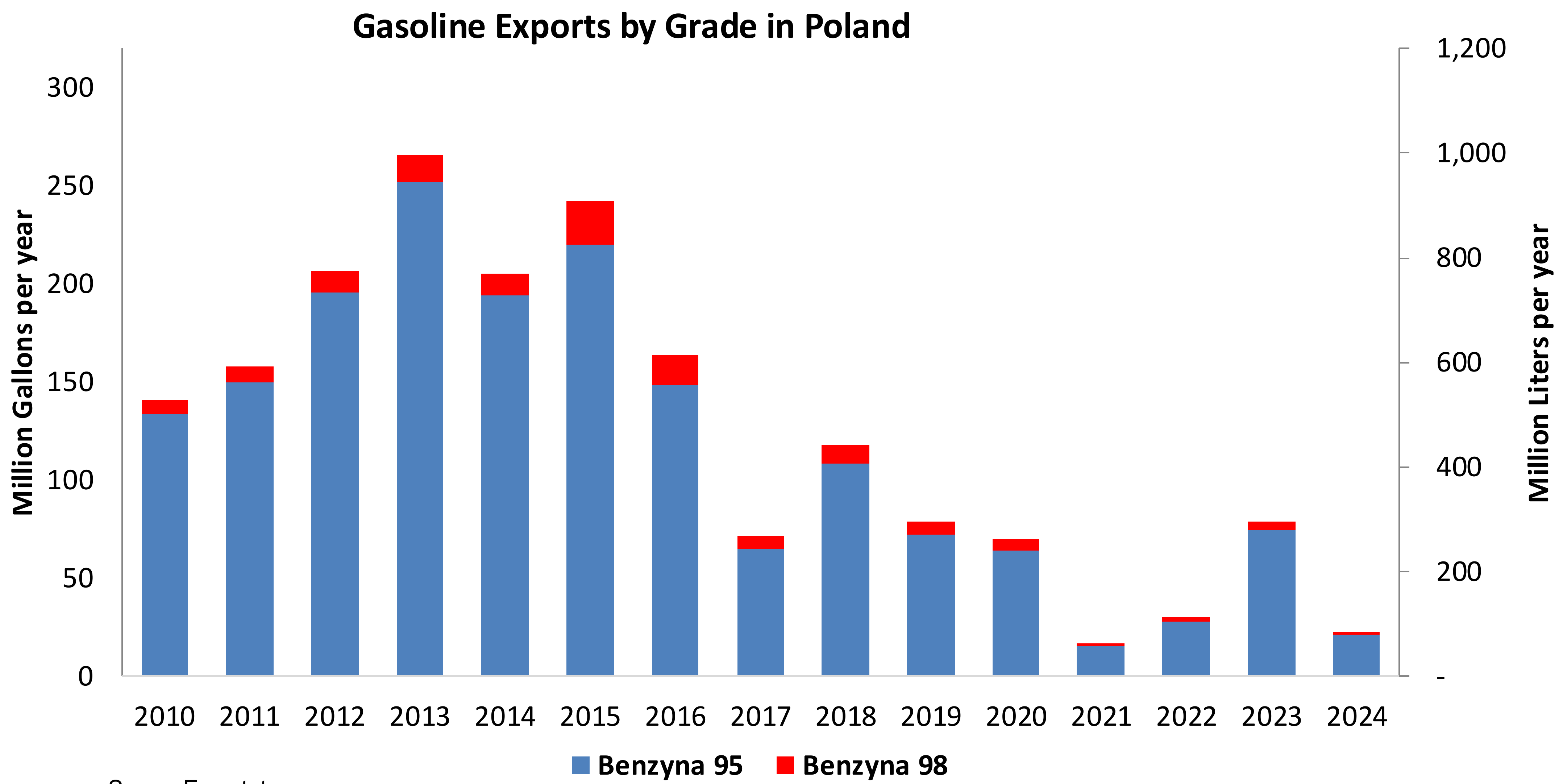
Source: Eurostat

Gasoline Imports in Poland



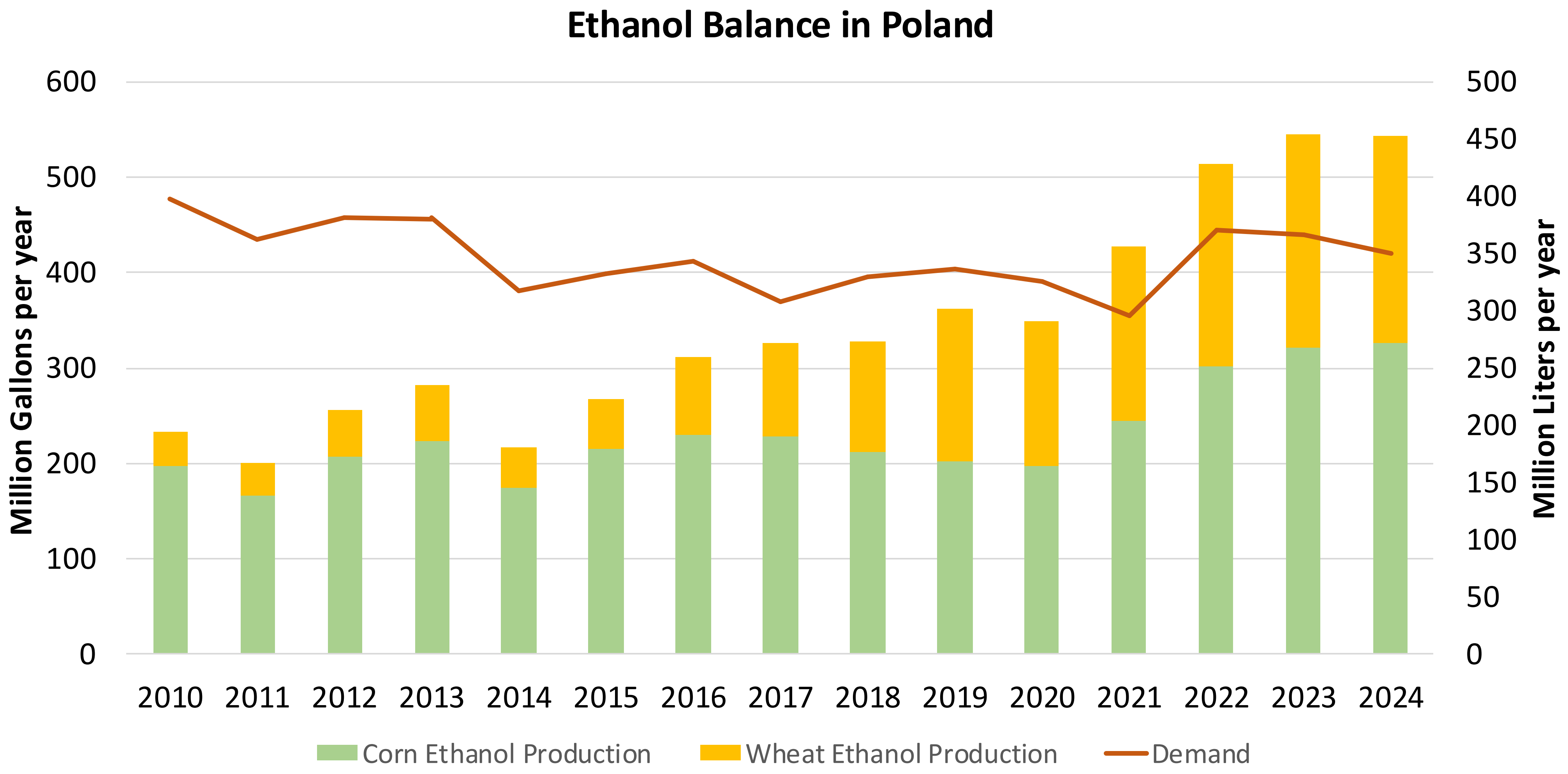
Source: Eurostat

Gasoline Exports in Poland



Source: Eurostat

Ethanol Balance in Poland



Source: Eurostat

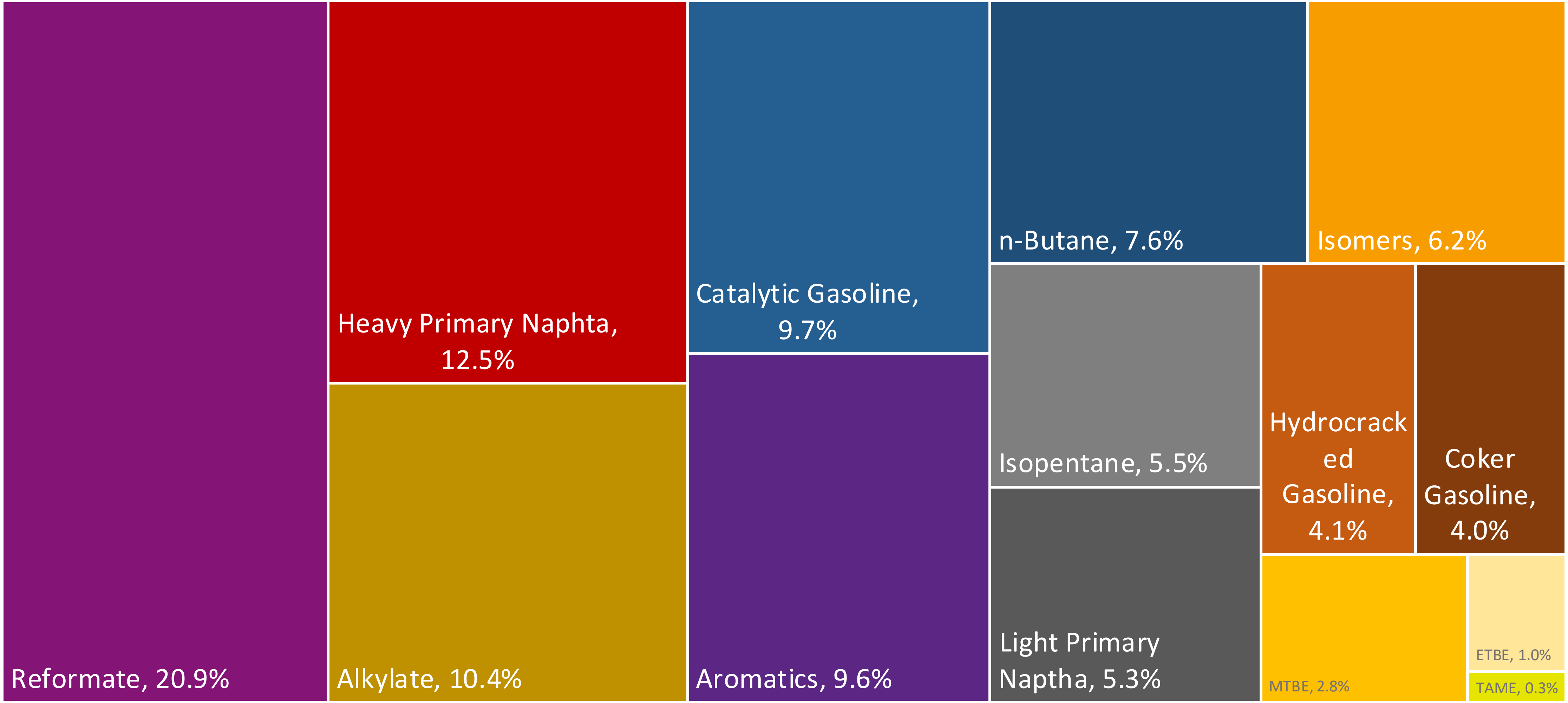
Gasoline Quality in Poland

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	9.2							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	4.6							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

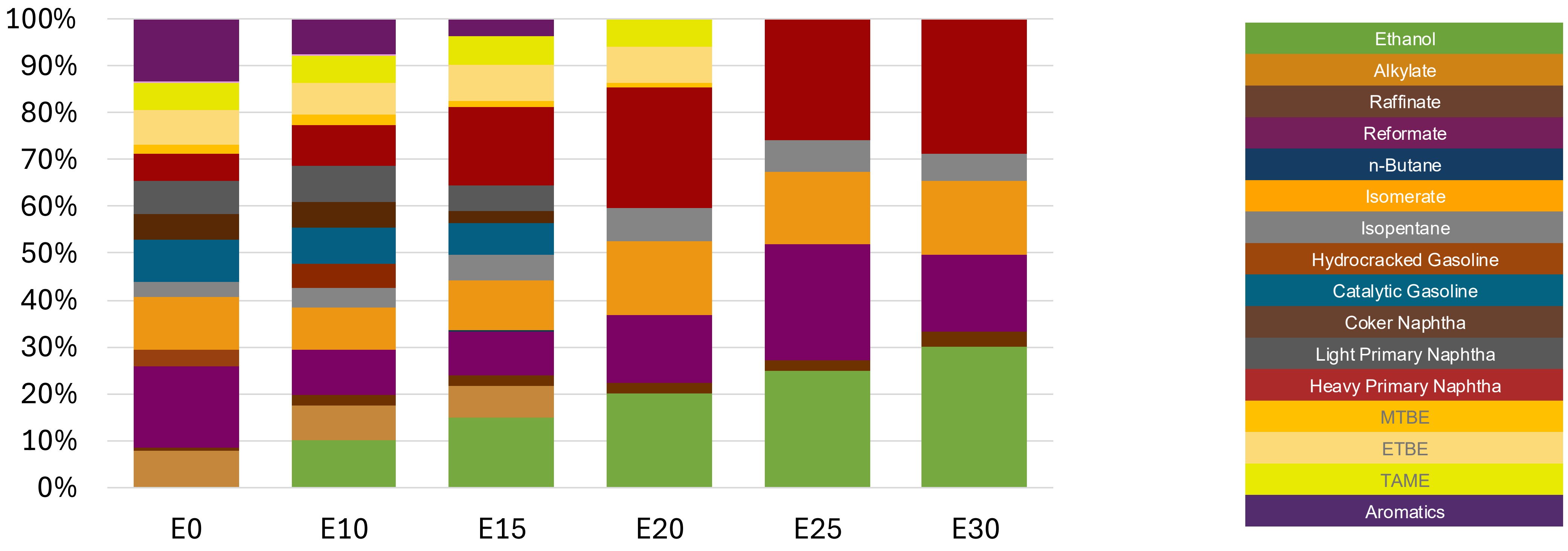
Source: European Commision

Poland Available Blending Components



Source: Faro90

Poland – Gasoline 95 – Constant Octane

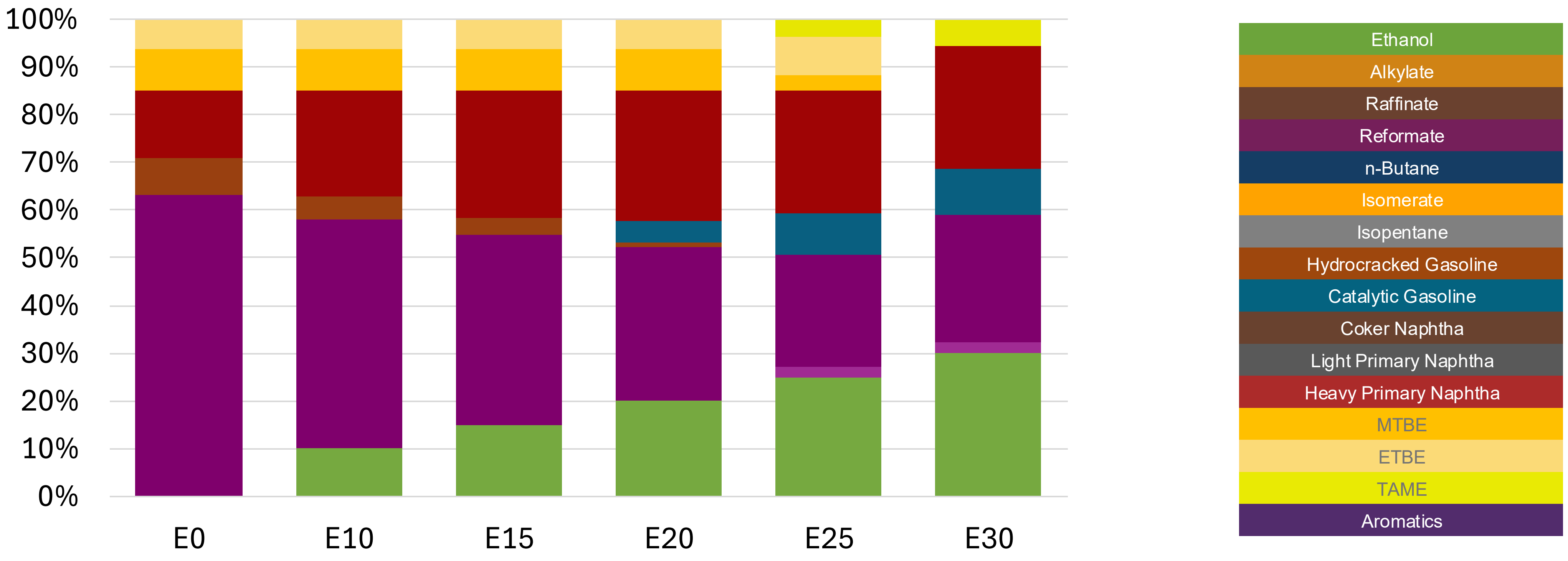


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.284	$ 2.146	$ 2.046	$ 1.931	$ 1.885	$ 1.832

Average CIF 2024 Prices

Source: Faro90

Poland - Gasoline 98 - Constant Octane

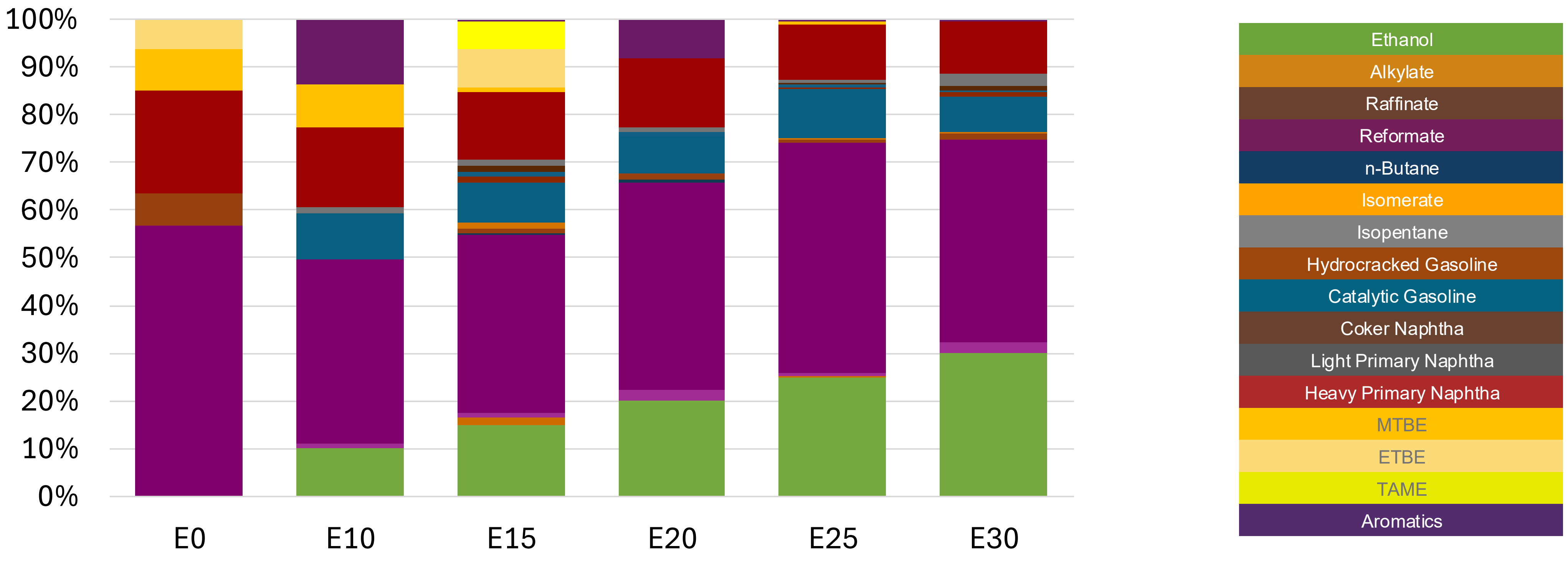


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.304	$ 2.187	$ 2.118	$ 2.046	$ 1.981	$ 1.931

Average CIF 2024 Prices

Source: Faro90

Poland – Gasoline 95 – Increased Octane

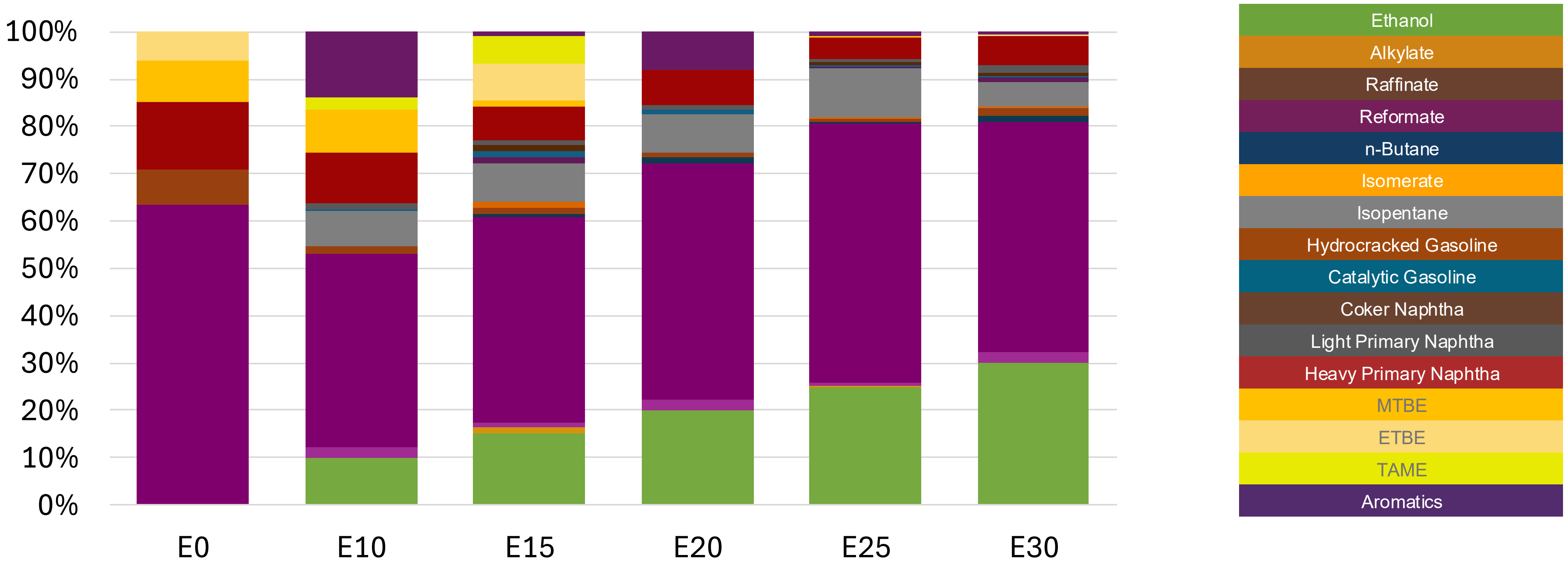


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.172	$ 2.172	$ 2.172	$ 2.172	$ 2.172	$ 2.172

Average CIF 2024 Prices

Source: Faro90

Poland – Gasoline 98 – Increased Octane



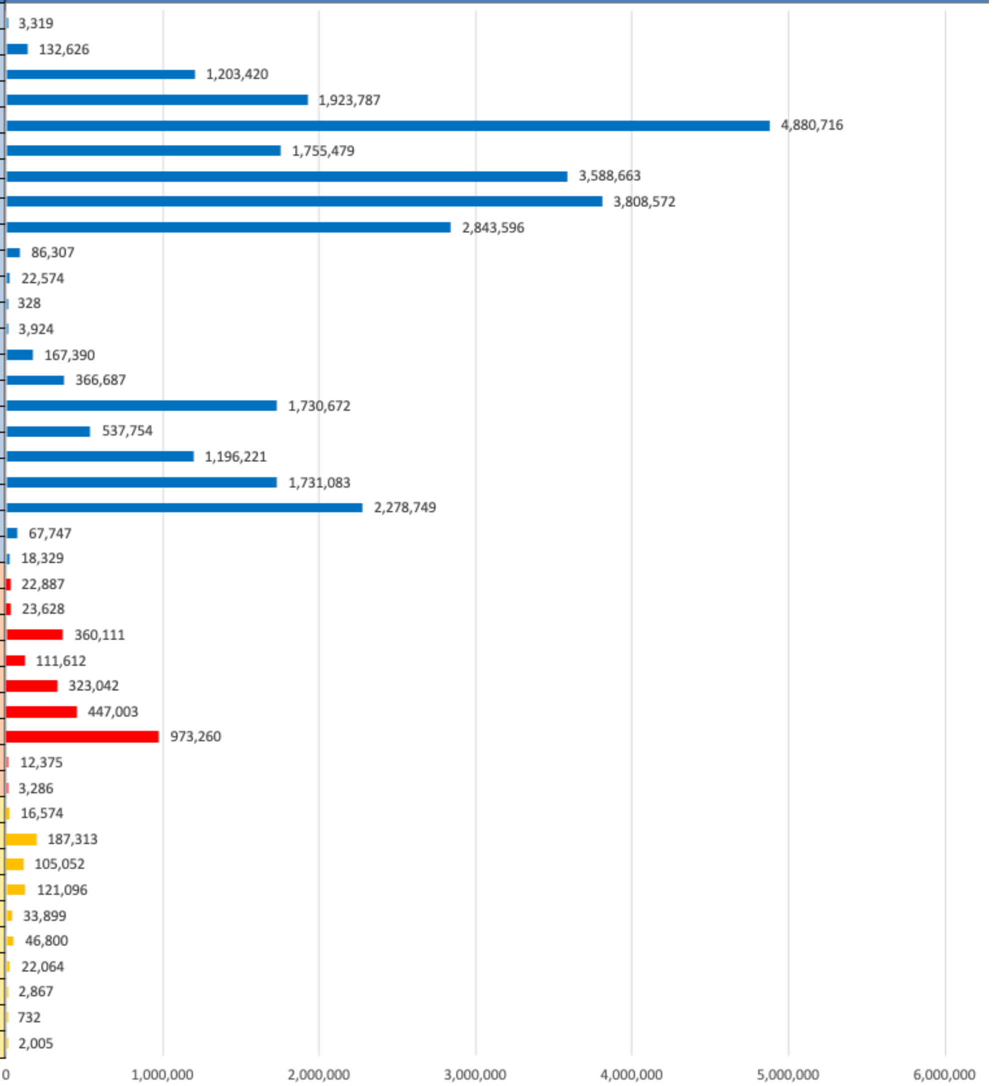
Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.304	$ 2.304	$ 2.304	$ 2.304	$ 2.304	$ 2.304

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Poland

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	3,319
		Single FI	3-Ways		>8 years	132,626
		Multi FI	3-Ways		>8 years	1,203,420
			Euro III		>8 years	1,923,787
			Euro IV		>8 years	4,880,716
			Euro V		>8 years	1,755,479
			Euro 6		>8 years	3,588,663
					4-8 years	3,808,572
					<4 years	2,843,596
			Hybrid		All	86,307
		Electric			All	22,574
	Medium	Carburator	3-Ways	PCV Tank	>8 years	328
		Single FI	3-Ways		>8 years	3,924
		Multi FI	3-Ways		>8 years	167,390
			Euro III		>8 years	366,687
			Euro IV		>8 years	1,730,672
			Euro V		>8 years	537,754
			Euro 6		>8 years	1,196,221
					4-8 years	1,731,083
					<4 years	2,278,749
			Hybrid		All	67,747
		Electric			All	18,329
Sml Engines	Light	4-Cycle	None	PCV	>8 years	22,887
			Euro III		>8 years	23,628
			Euro IV		>8 years	360,111
			Euro V		>8 years	111,612
			Euro 6		>8 years	323,042
					4-8 years	447,003
					<4 years	973,260
			Hybrid		All	12,375
		Electric			All	3,286
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	16,574
		Fuel-Injection	3-Ways		>8 years	187,313
			Euro III		>8 years	105,052
			Euro IV		>8 years	121,096
			Euro V		>8 years	33,899
			Euro 6		>8 years	46,800
					4-8 years	22,064
					<4 years	2,867
			Hybrid		All	732
		Electric			All	2,005

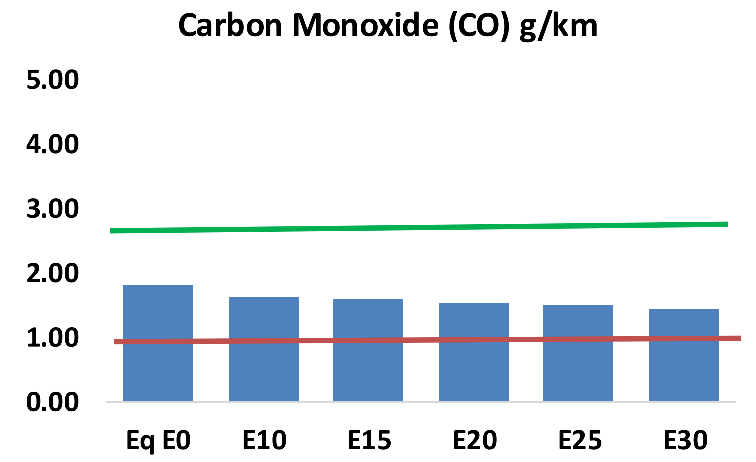


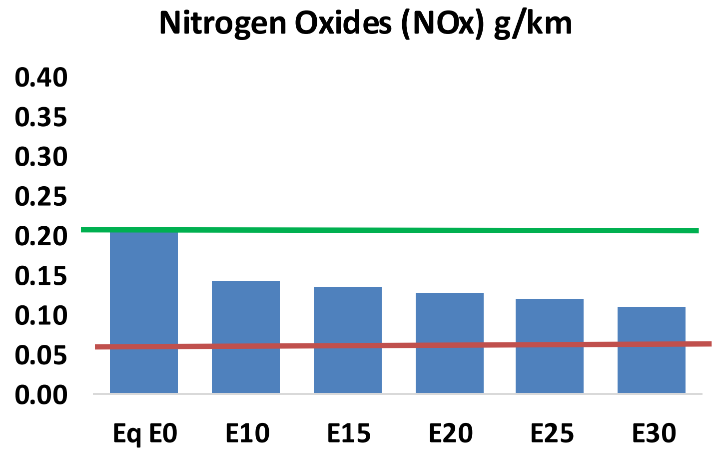
Vehicle Fleet: 31,163,548
Gasoline Vehicle Fleet: 13,217,757

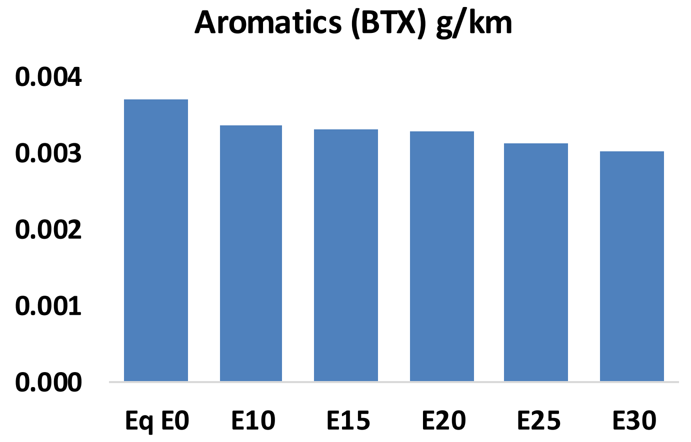
Source: Eurostat, ACEA

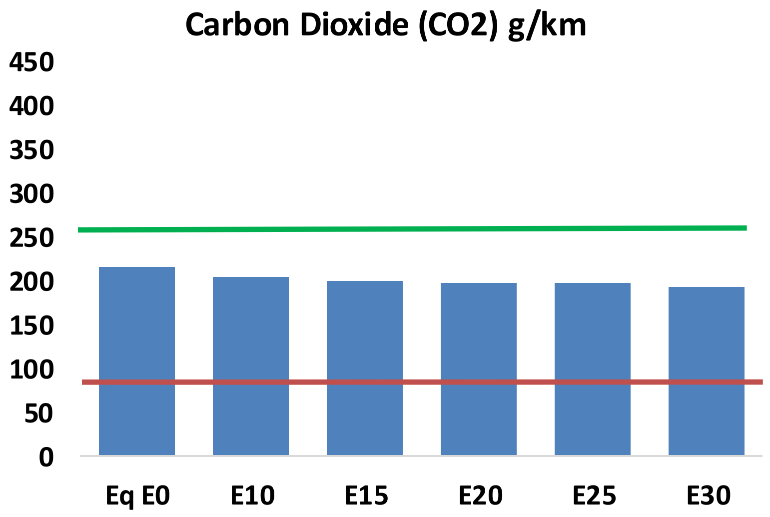
Poland – Vehicular Emissions

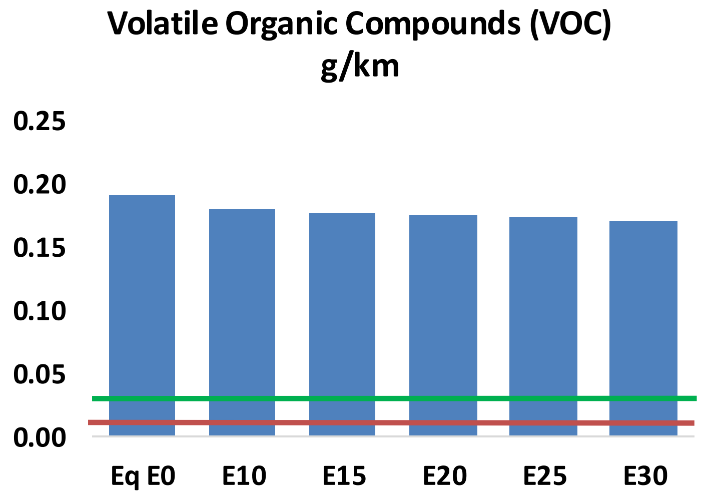
TIER III BIN 125 United States
Euro 6 Standard

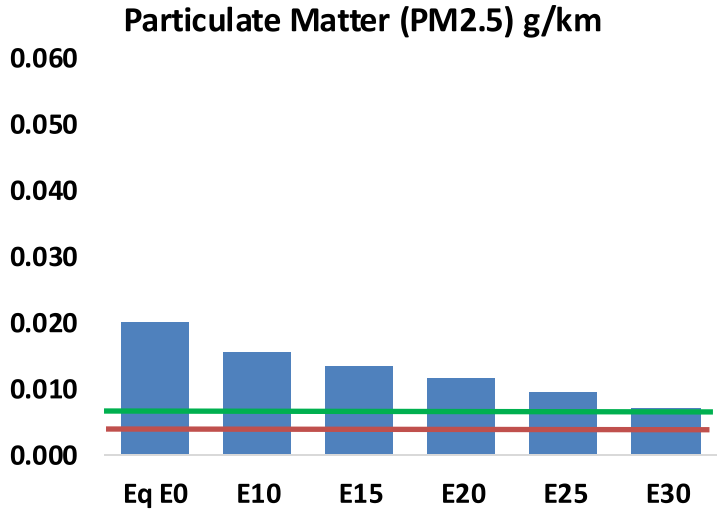












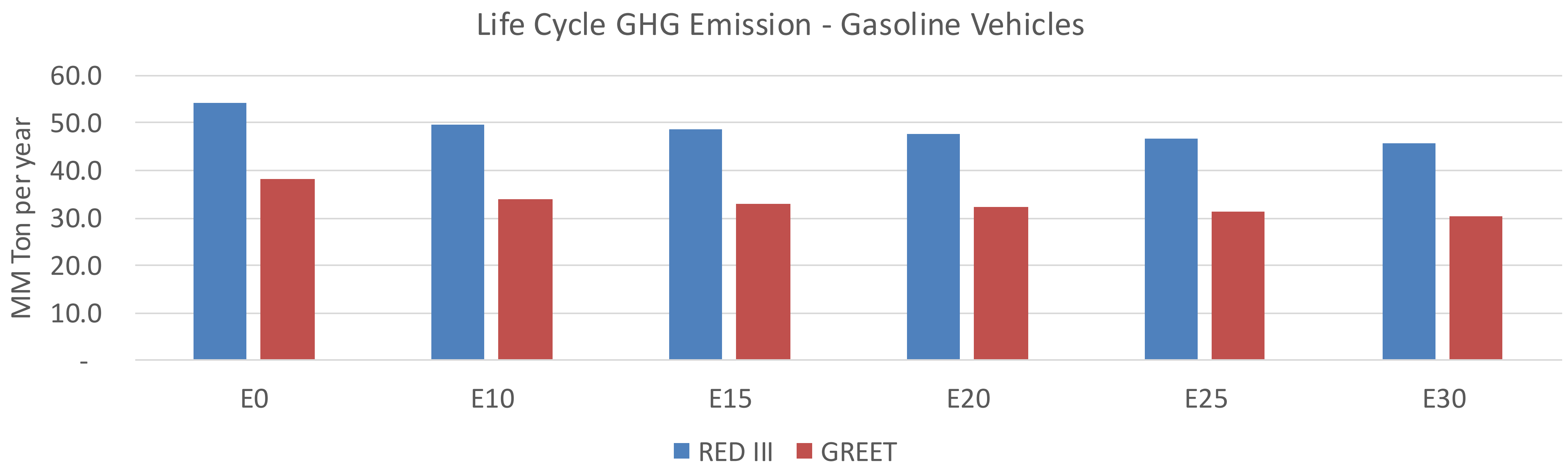
Source: Faro90

Poland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	615,660	588,117	560,575	541,866	524,816	512,241	495,401	-9%	-15%
CO2	73,575,358	71,683,421	69,896,590	68,491,301	67,795,909	67,393,631	66,151,372	-5%	-8%
VOC	65,622	63,617	61,613	60,564	59,781	59,261	58,277	-6%	-9%
NOx	70,624	52,968	49,437	46,594	43,942	41,019	37,925	-30%	-38%
SOx	880	813	747	691	637	578	517	-15%	-28%
NH3	24,677	24,432	24,188	24,303	24,576	24,769	24,928	-2%	0%
N2O	1,147	1,140	1,136	1,141	1,165	1,178	1,191	-1%	2%
CH4	15,774	15,774	15,774	16,011	16,358	16,658	16,942	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	674	662	650	648	650	652	651	-4%	-4%
Acetaldehyde	1,351	1,550	2,275	3,464	4,720	5,393	6,373	68%	249%
Formaldehyde	4,074	3,961	4,617	5,421	5,680	6,283	6,840	13%	39%
Benzene	1,265	1,216	1,152	1,134	1,124	1,075	1,040	-9%	-11%
PM2.5	4,537	4,028	3,520	3,057	2,596	2,106	1,596	-22%	-43%
PM10	2,330	2,069	1,808	1,570	1,333	1,082	820	-22%	-43%

Source: Faro90

Poland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	337.8
Target @2035 (MMT/year)	190.2
Delta	147.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.9%	13.8%	16.0%	18.0%	20.9%
RED II/III	0.0%	8.1%	10.3%	12.0%	13.6%	15.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.8%	3.6%	4.1%	4.7%	5.4%
RED II/III	0.0%	3.0%	3.8%	4.4%	5.0%	5.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

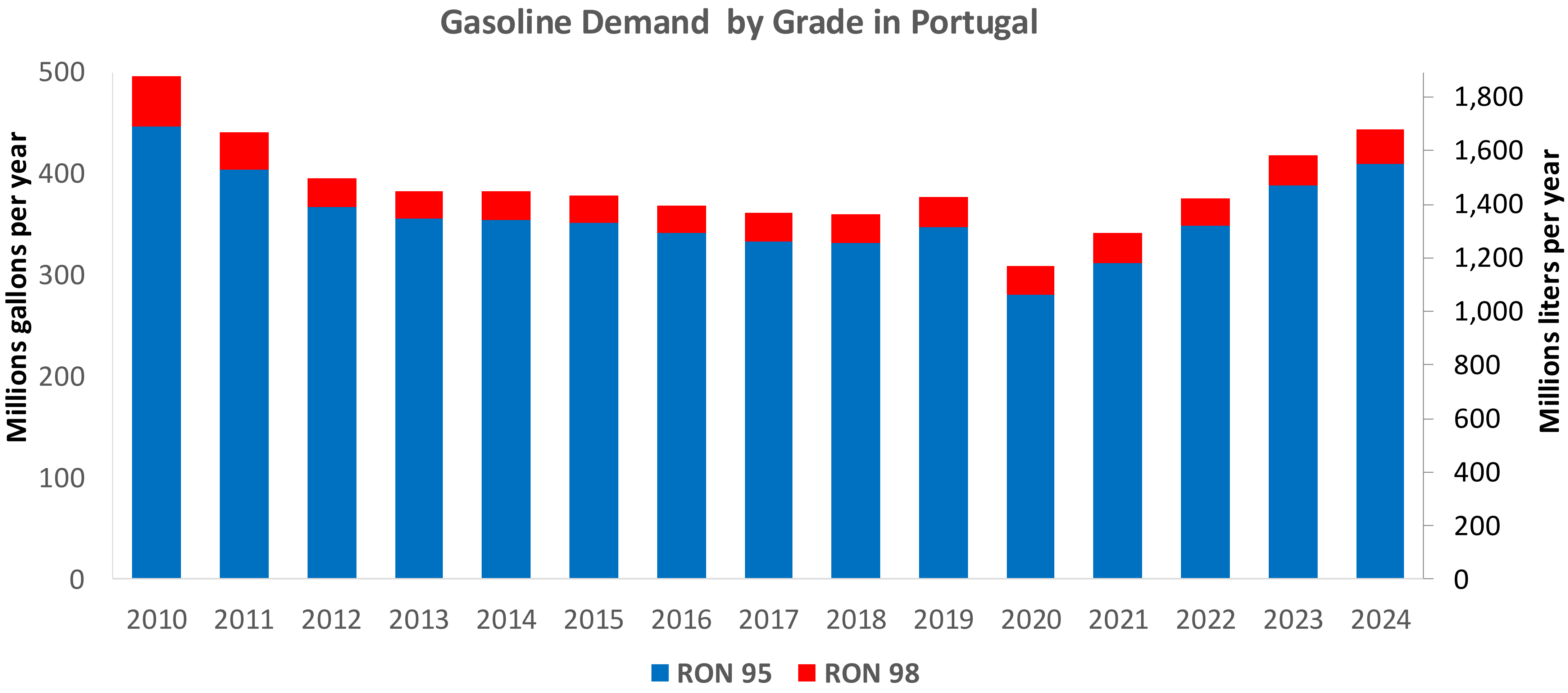
Ethanol Gasoline Blending in Portugal

In 2024, gasoline consumption in Portugal was 1,688 million liters (446 million gallons) and growth rate was 0.2%. Gasoline production and net imports were 2,990 and -1,310 million liters (790 and -346 million gallons) correspondingly. Portugal complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Portugal was 64 million liters (17 million gallons) representing 3.8% of gasoline demand. Portugal does not produce ethanol, supply is done through imports. Local production plants are being built.

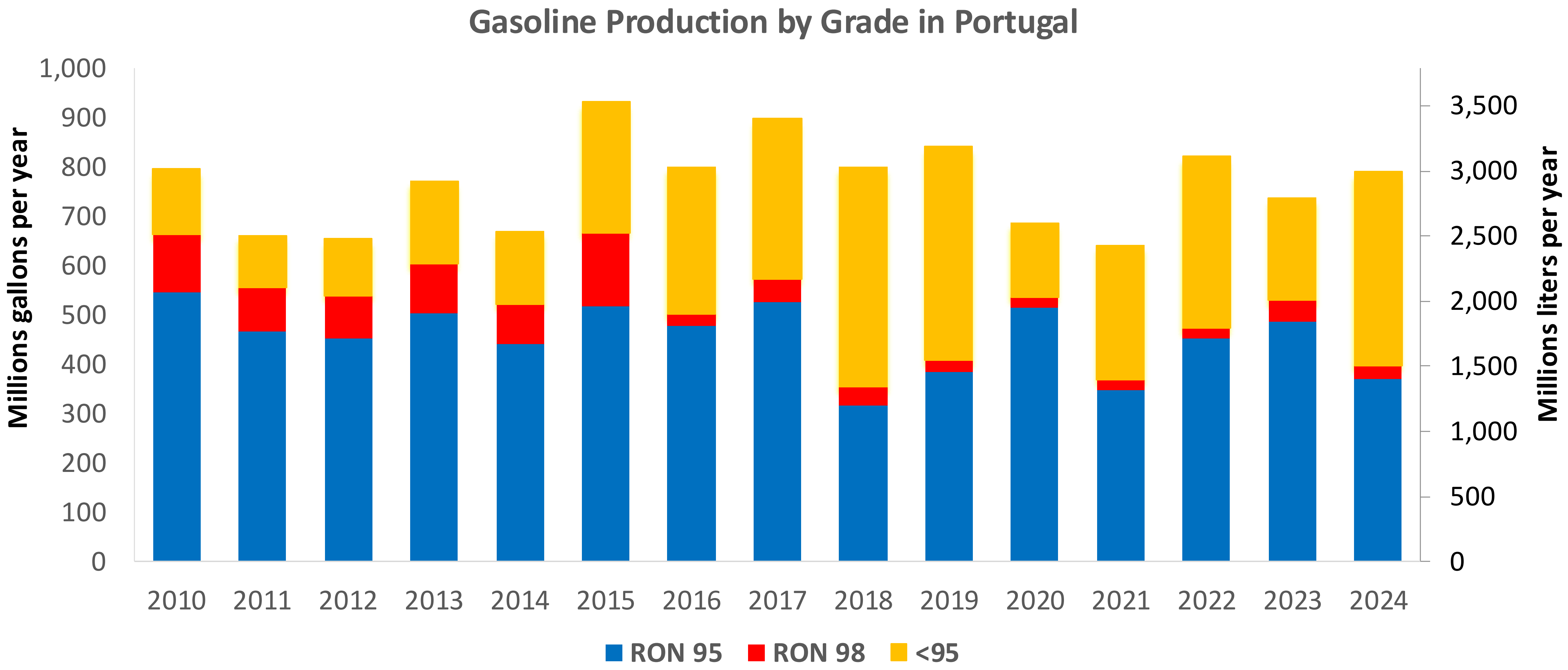
Source: Eurostat, European Commision

Gasoline Demand in Portugal



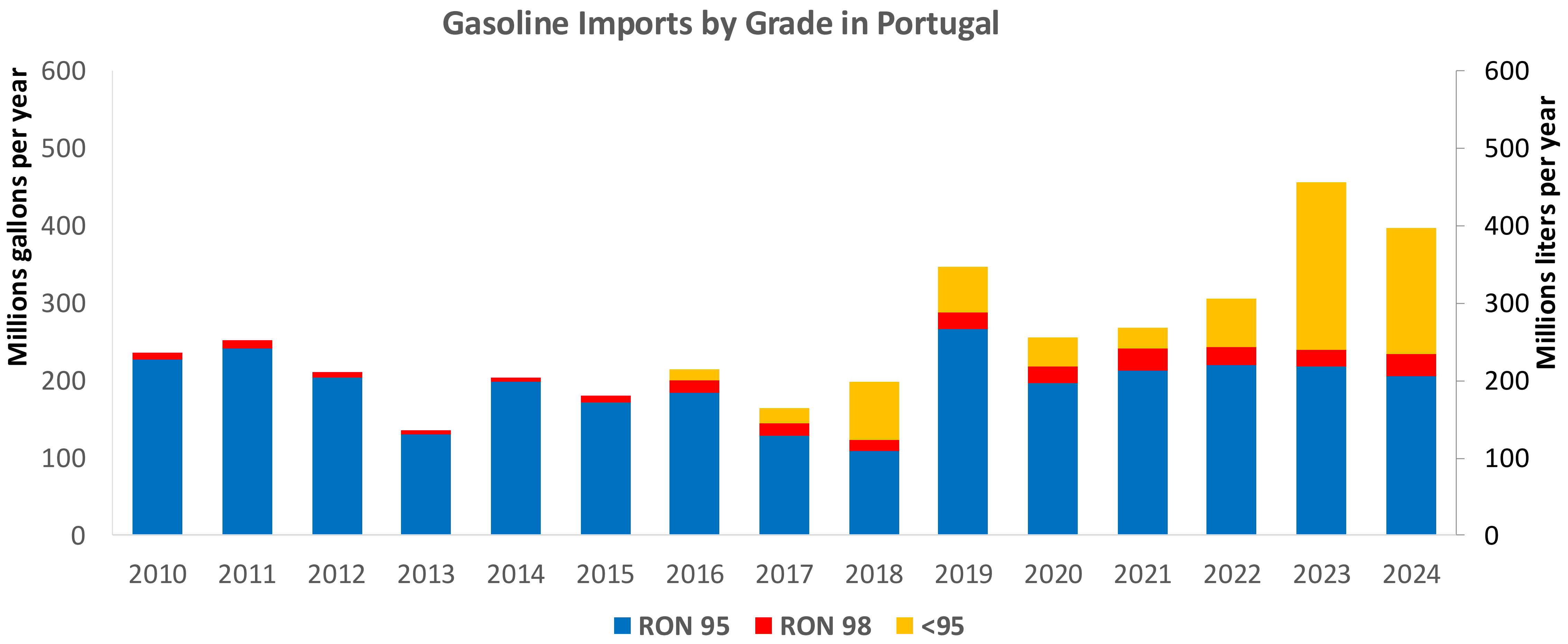
Source: Eurostat

Gasoline Production in Portugal



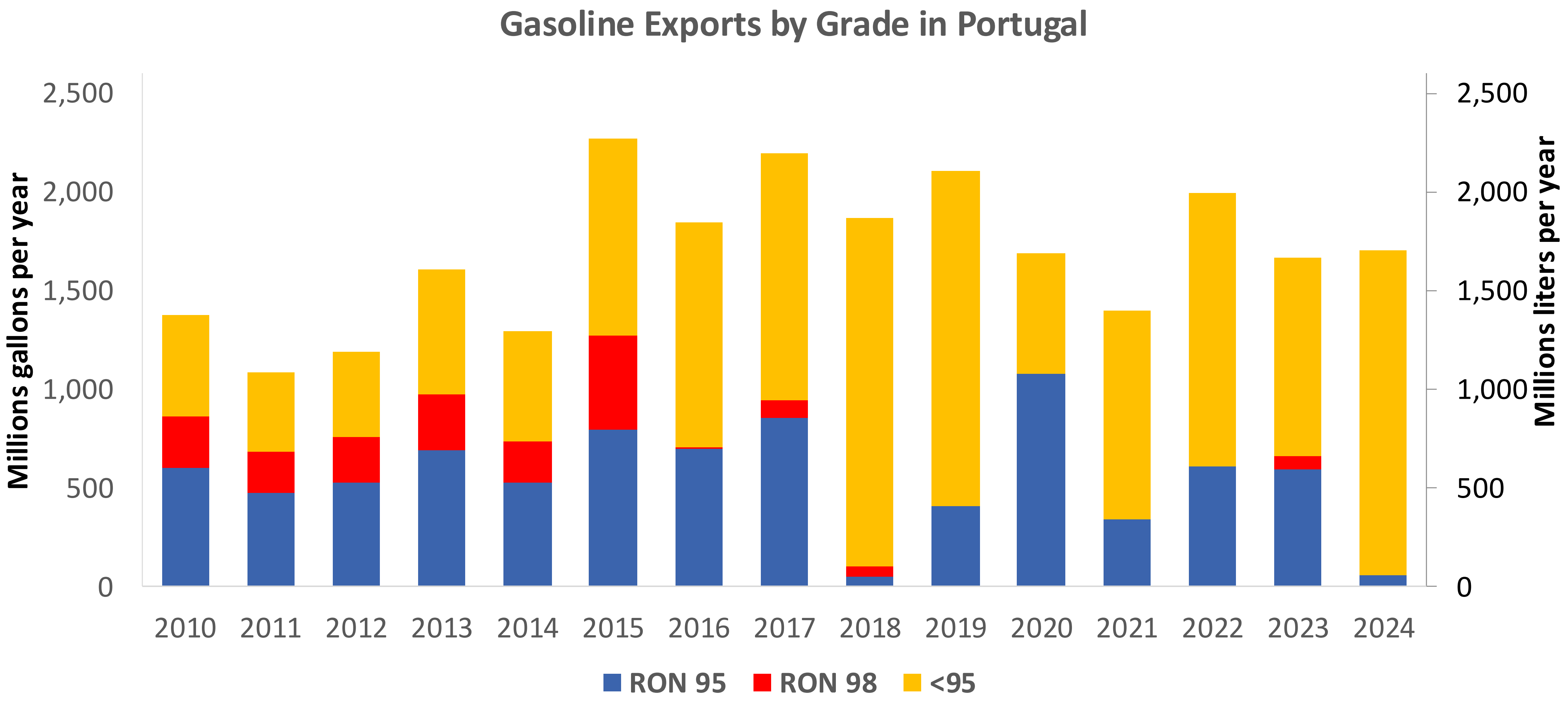
Source: Eurostat

Gasoline Imports in Portugal



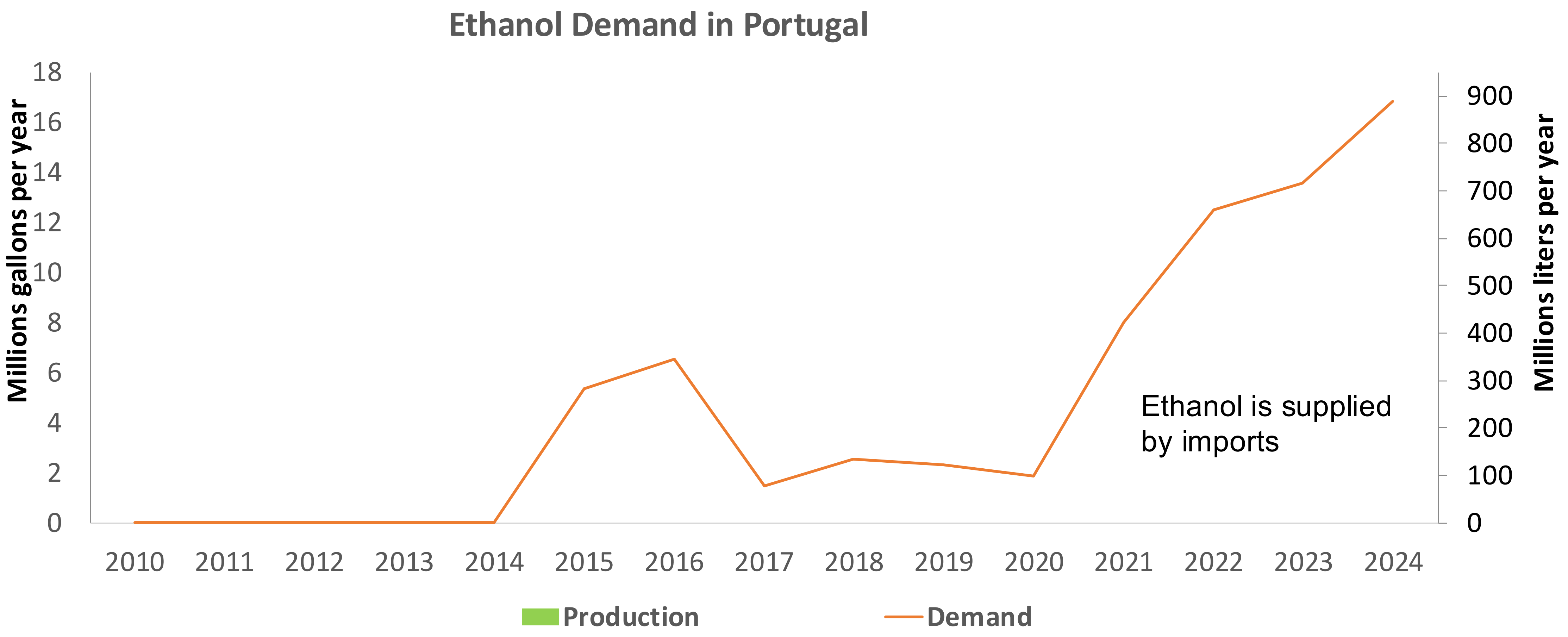
Source: Eurostat

Gasoline Exports in Portugal



Source: Eurostat

Ethanol Balance in Portugal



Source: Eurostat

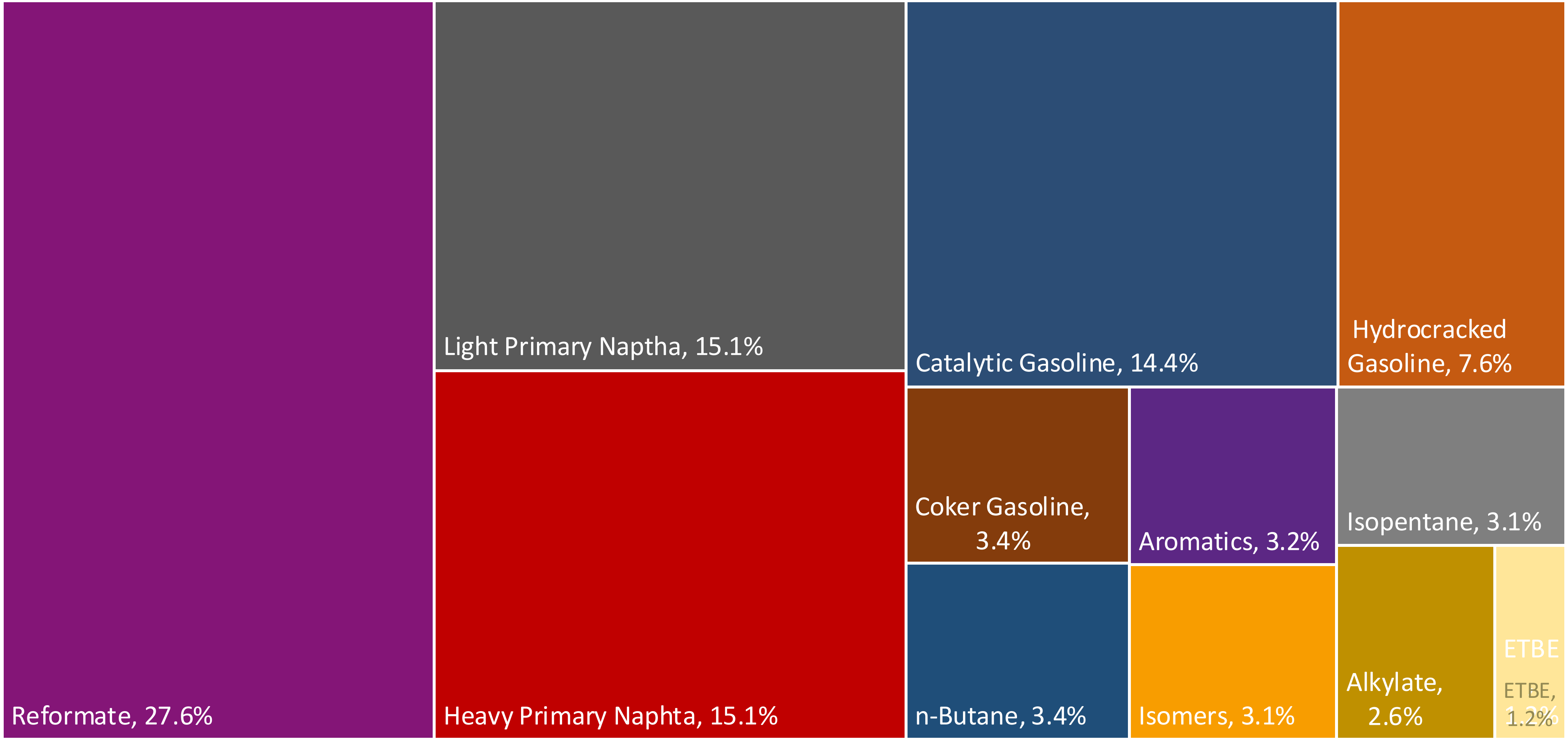
Gasoline Quality in Portugal

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	13.0							
Advanced Biofuels (%cal)	2.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

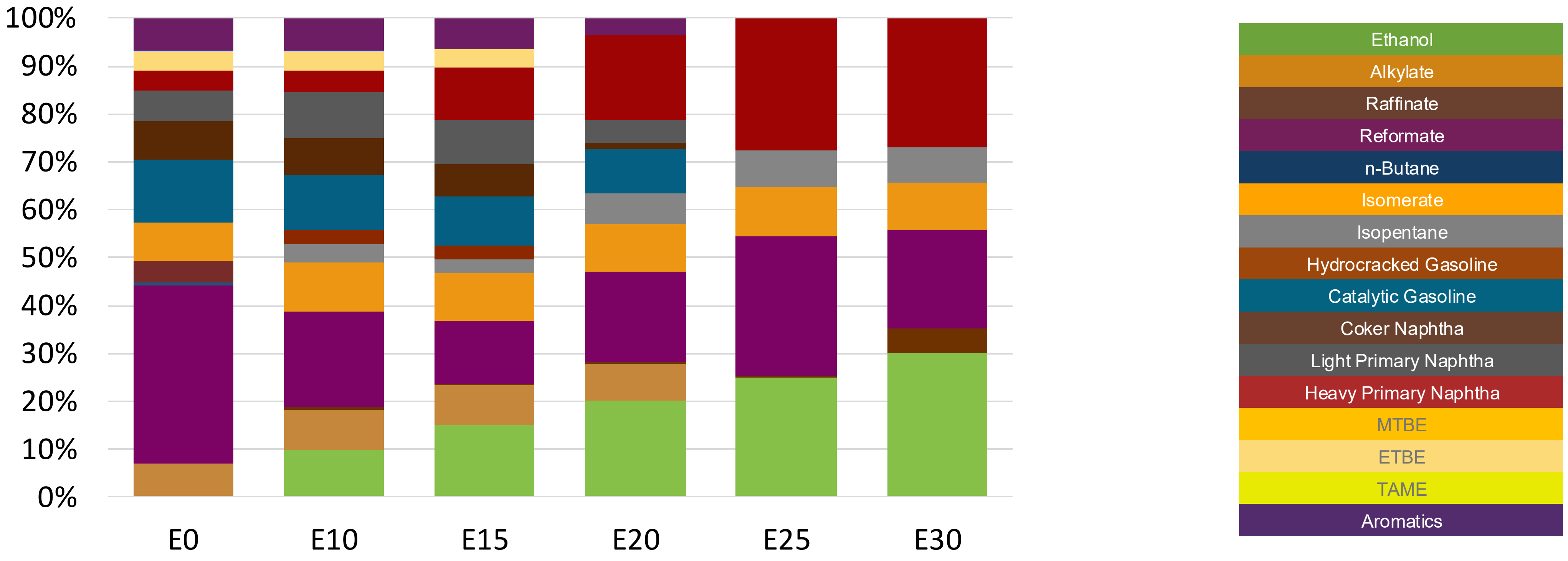
Source: European Commision

Portugal Available Blending Components



Source: Faro90

Portugal – Gasoline 95 – Constant Octane

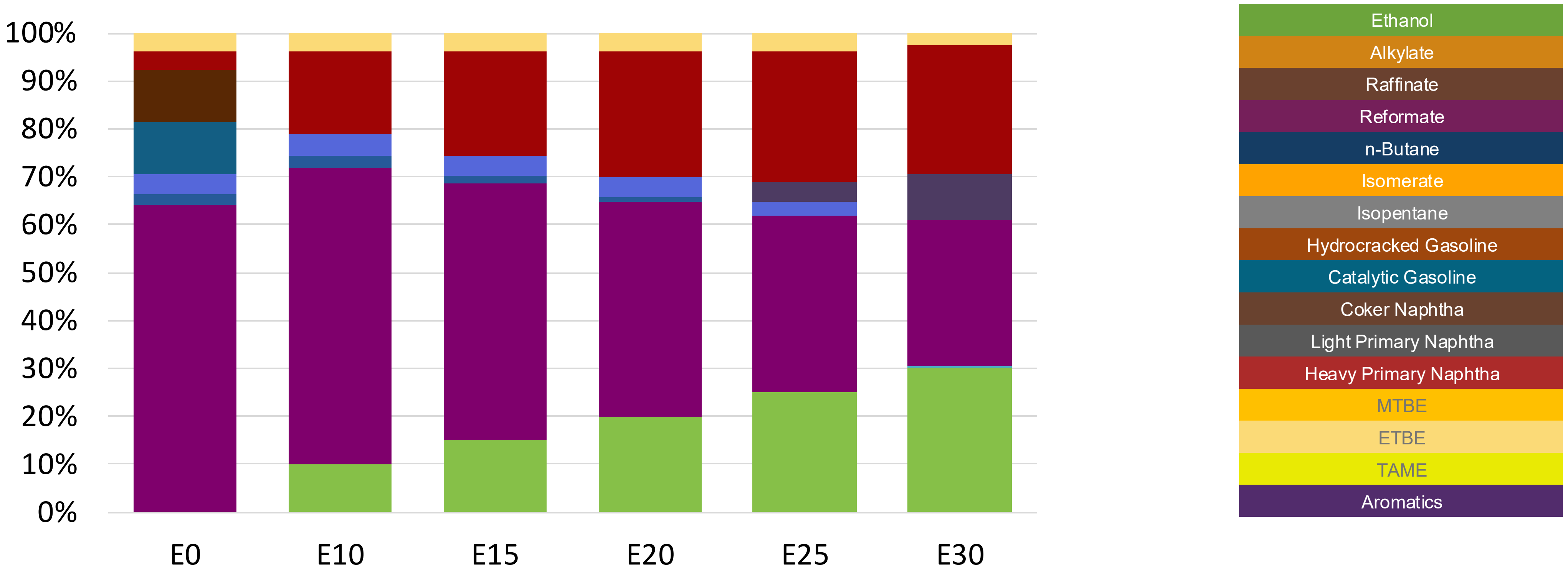


Octane (RON)	95.0	95.0	95.0	95.0	95.3	95.8
Price (US$/gal)	$ 2.290	$ 2.178	$ 2.098	$ 2.008	$ 1.893	$ 1.830

Average CIF 2024 Prices

Source: Faro90

Portugal - Gasoline 98 - Constant Octane

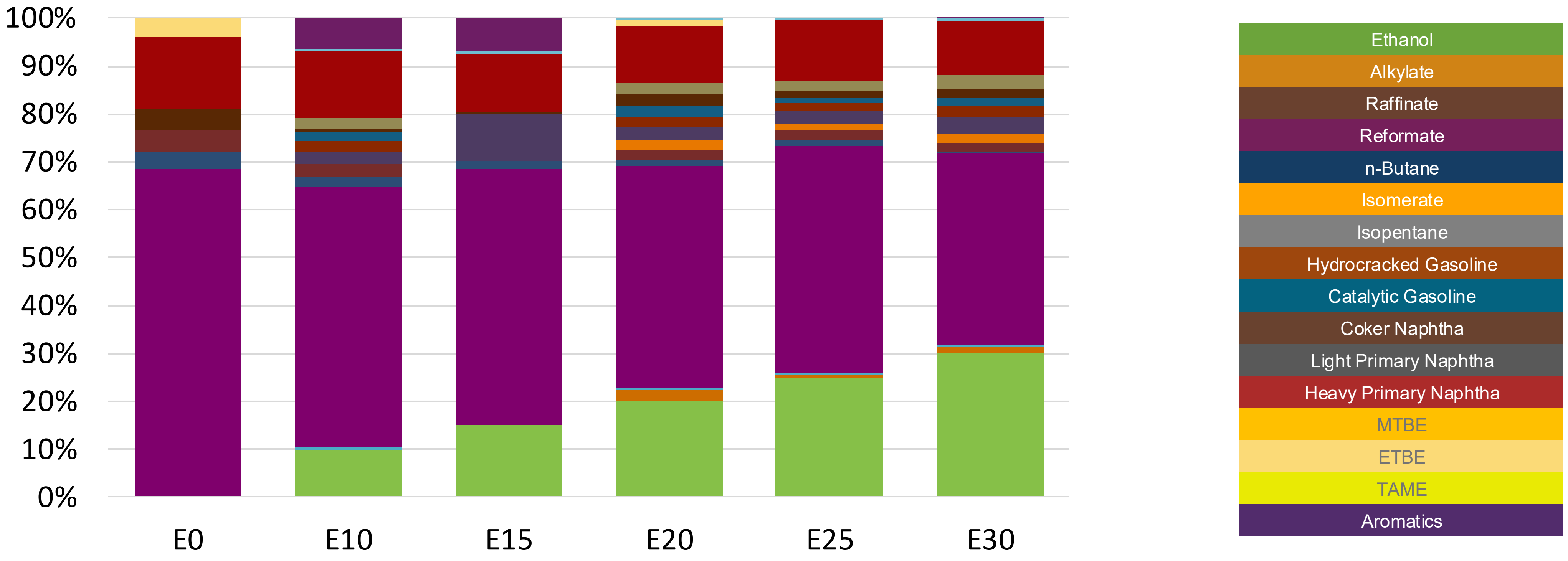


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.359	$ 2.235	$ 2.166	$ 2.091	$ 2.013	$ 1.934

Average CIF 2024 Prices

Source: Faro90

Portugal – Gasoline 95 – Increased Octane

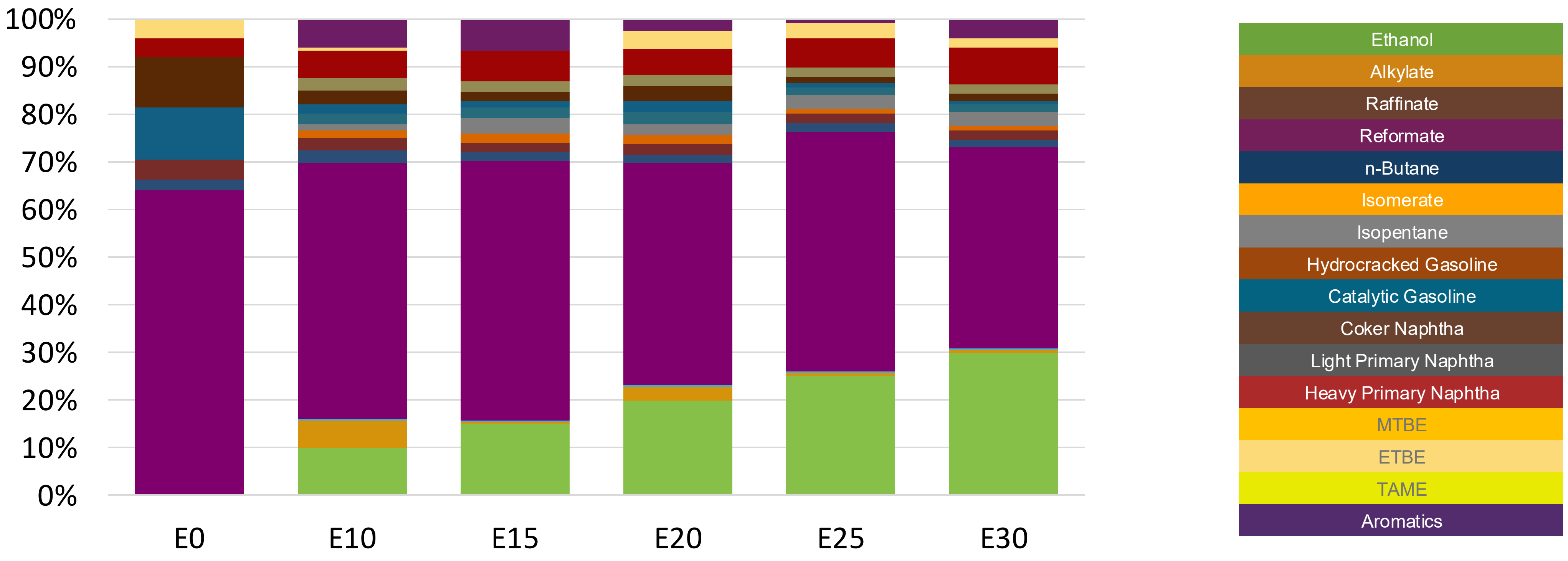


Octane (RON)	95.0	96.8	98.2	99.9	101.4	102.6
Price (US$/gal)	$ 2.222	$ 2.222	$ 2.222	$ 2.222	$ 2.222	$ 2.222

Average CIF 2024 Prices

Source: Faro90

Portugal – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.3	100.9	102.5	104.0	105.0
Price (US$/gal)	$ 2.359	$ 2.359	$ 2.359	$ 2.359	$ 2.359	$ 2.359

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Portugal

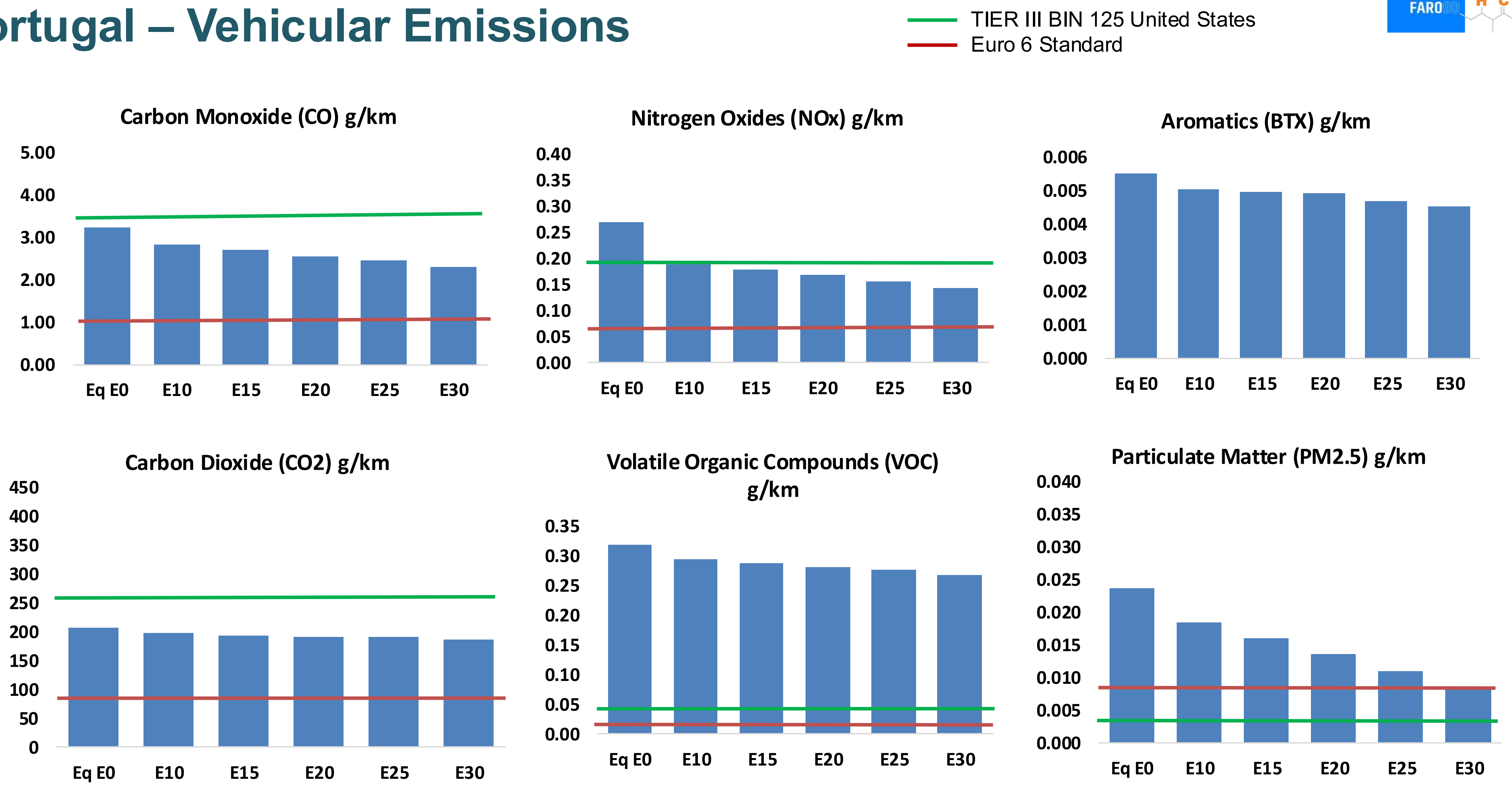
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	210
		Single FI	3-Ways		>8 years	48,136
		Multi FI	3-Ways		>8 years	610,603
			Euro III		>8 years	787,083
			Euro IV		>8 years	1,089,910
			Euro V		>8 years	241,125
			Euro 6		>8 years	626,653
					4-8 years	815,902
					<4 years	706,256
			Hybrid		All	47,501
		Electric			All	37,123
	Medium	Carburator	3-Ways	PCV Tank	>8 years	212
		Single FI	3-Ways		>8 years	34,282
		Multi FI	3-Ways		>8 years	207,932
			Euro III		>8 years	296,136
			Euro IV		>8 years	306,262
			Euro V		>8 years	49,391
			Euro 6		>8 years	108,733
					4-8 years	143,188
					<4 years	111,843
			Hybrid		All	5
		Electric			All	2,303
Sml Engines	Light	4-Cycle	None	PCV	>8 years	18,953
			Euro III		>8 years	102,702
			Euro IV		>8 years	657,528
			Euro V		>8 years	35,767
			Euro 6		>8 years	67,293
					4-8 years	120,172
					<4 years	162,209
			Hybrid		All	0
		Electric			All	13,219
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	34
		Fuel-Injection	3-Ways		>8 years	21,062
			Euro III		>8 years	17,672
			Euro IV		>8 years	24,223
			Euro V		>8 years	4,467
			Euro 6		>8 years	14,232
					4-8 years	22,182
					<4 years	25,555
			Hybrid		All	0
		Electric			All	153

0 200,000 400,000 600,000 800,000 1,000,000 1,200,000

Vehicle Fleet: 7,578,214
Gasoline Vehicle Fleet: 2,654,720

Source: Eurostat, ACEA

Portugal – Vehicular Emissions



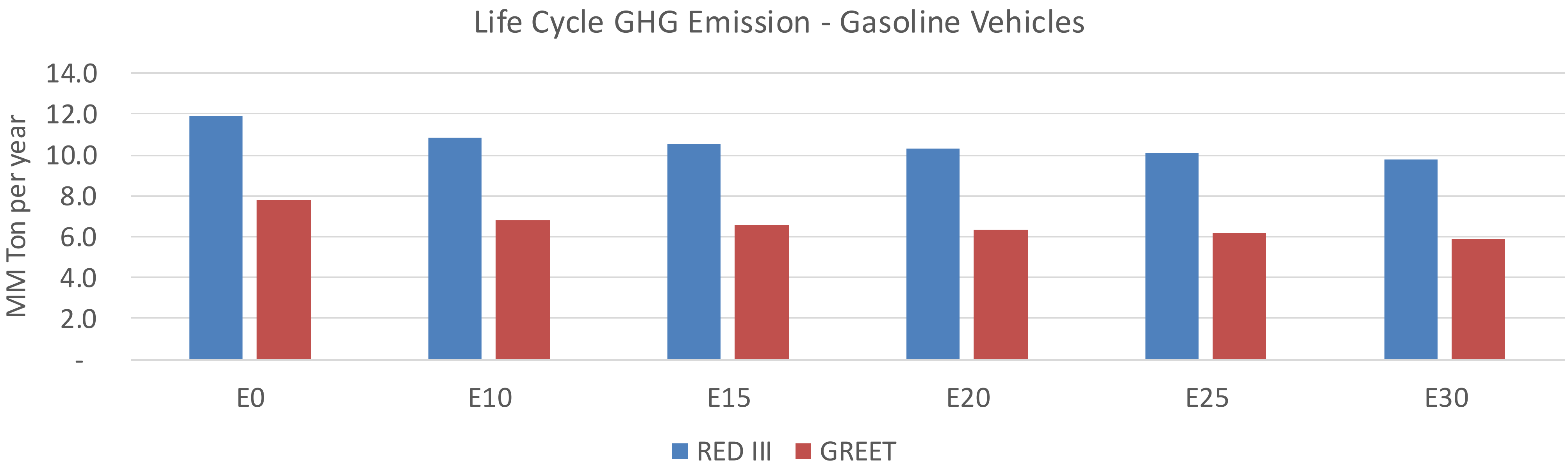
Source: Faro90

Portugal – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	209,074	196,659	184,244	174,785	165,594	158,626	150,367	-12%	-21%
CO2	13,460,947	13,114,808	12,787,900	12,530,796	12,403,570	12,331,260	12,103,959	-5%	-8%
VOC	20,658	19,868	19,077	18,586	18,162	17,858	17,377	-8%	-12%
NOx	17,465	13,098	12,225	11,522	10,866	10,143	9,379	-30%	-38%
SOx	160	148	136	125	116	105	94	-15%	-28%
NH3	4,599	4,553	4,508	4,529	4,580	4,616	4,646	-2%	0%
N2O	175	174	173	174	178	180	182	-1%	2%
CH4	4,900	4,900	4,900	4,974	5,082	5,175	5,263	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	198	193	187	184	182	181	178	-6%	-8%
Acetaldehyde	421	483	708	1,079	1,470	1,679	1,984	68%	249%
Formaldehyde	1,337	1,300	1,516	1,780	1,864	2,062	2,245	13%	39%
Benzene	358	344	326	321	319	305	295	-9%	-11%
PM2.5	839	745	651	566	480	390	295	-22%	-43%
PM10	690	613	536	465	395	320	243	-22%	-43%

Source: Faro90

Portugal – Gasoline Vehicle Fleet Life Cycle GHG Emissions



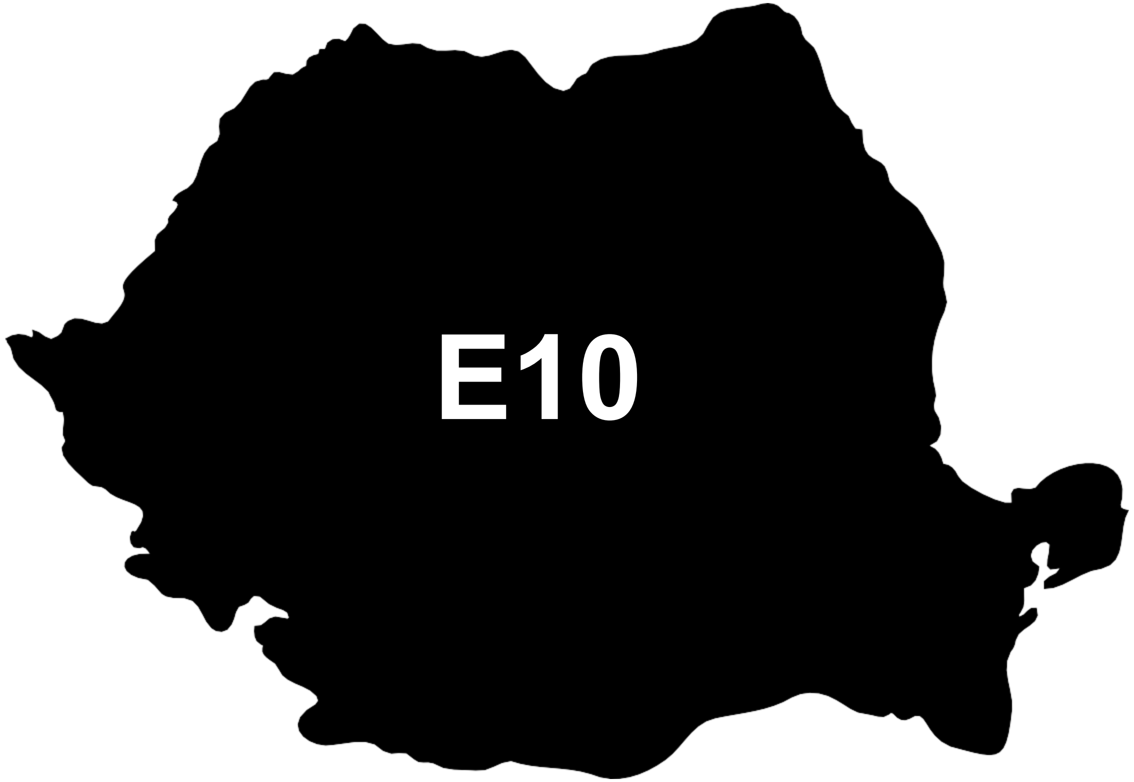
Country Emissions 2024 (MMT/año)	59.0
Target @2035 (MMT/year)	33.2
Delta	25.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.6%	15.8%	18.5%	21.0%	24.4%
RED II/III	0.0%	9.1%	11.6%	13.7%	15.7%	18.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.8%	4.8%	5.6%	6.4%	7.4%
RED II/III	0.0%	4.2%	5.4%	6.3%	7.3%	8.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Romania

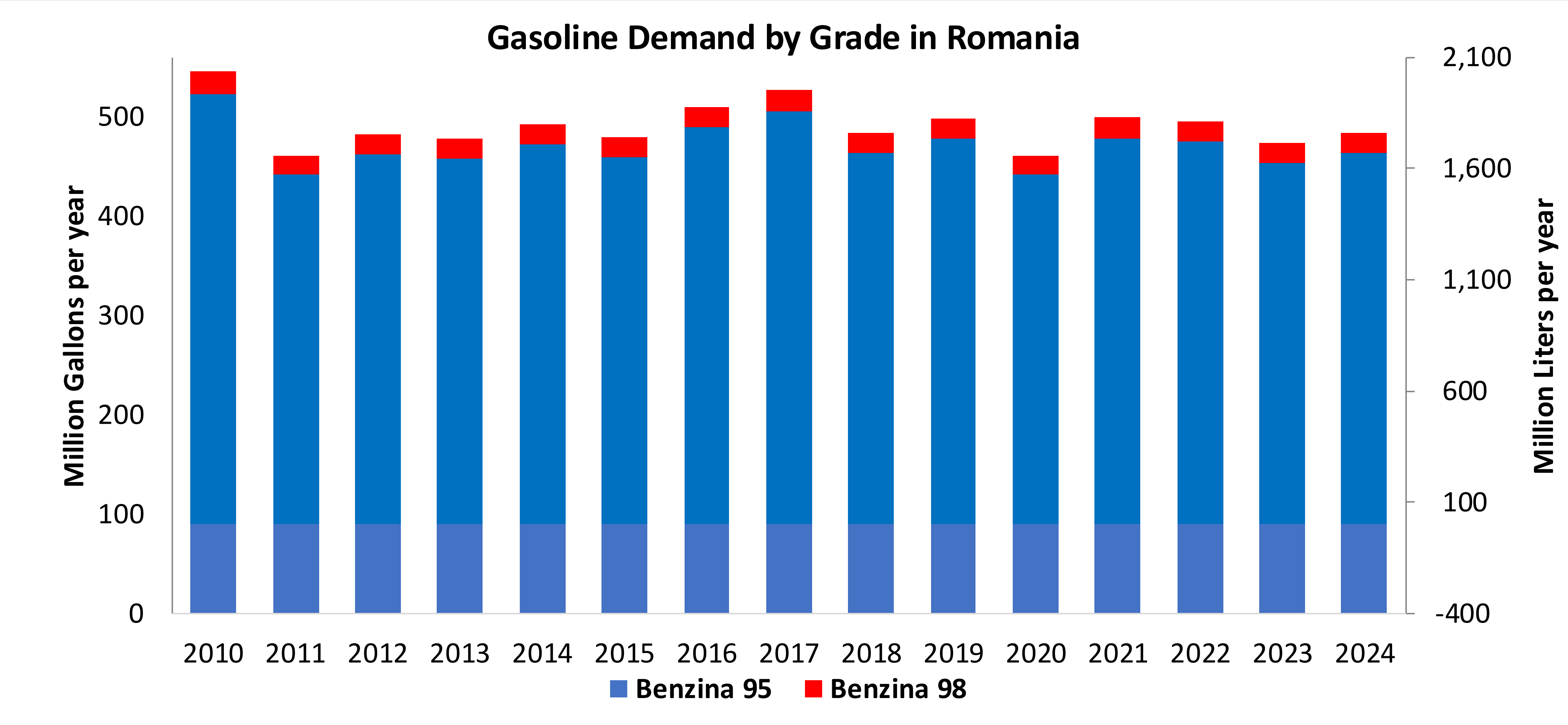


In 2024, gasoline consumption in Romania was 1,762 million liters (465 million gallons) and growth rate was -.7%. Gasoline production and net imports were 4,358 and -2,596 million liters (1,151 and -686 million gallons) correspondingly. Romania complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Romania was 144 million liters (38 million gallons) representing 8.2% of gasoline demand. Ethanol production and Net Imports were 107 and 37 million liters (38 and 10 million gallons) correspondingly. Ethanol sources are Corn 28% and Cereals 72%.

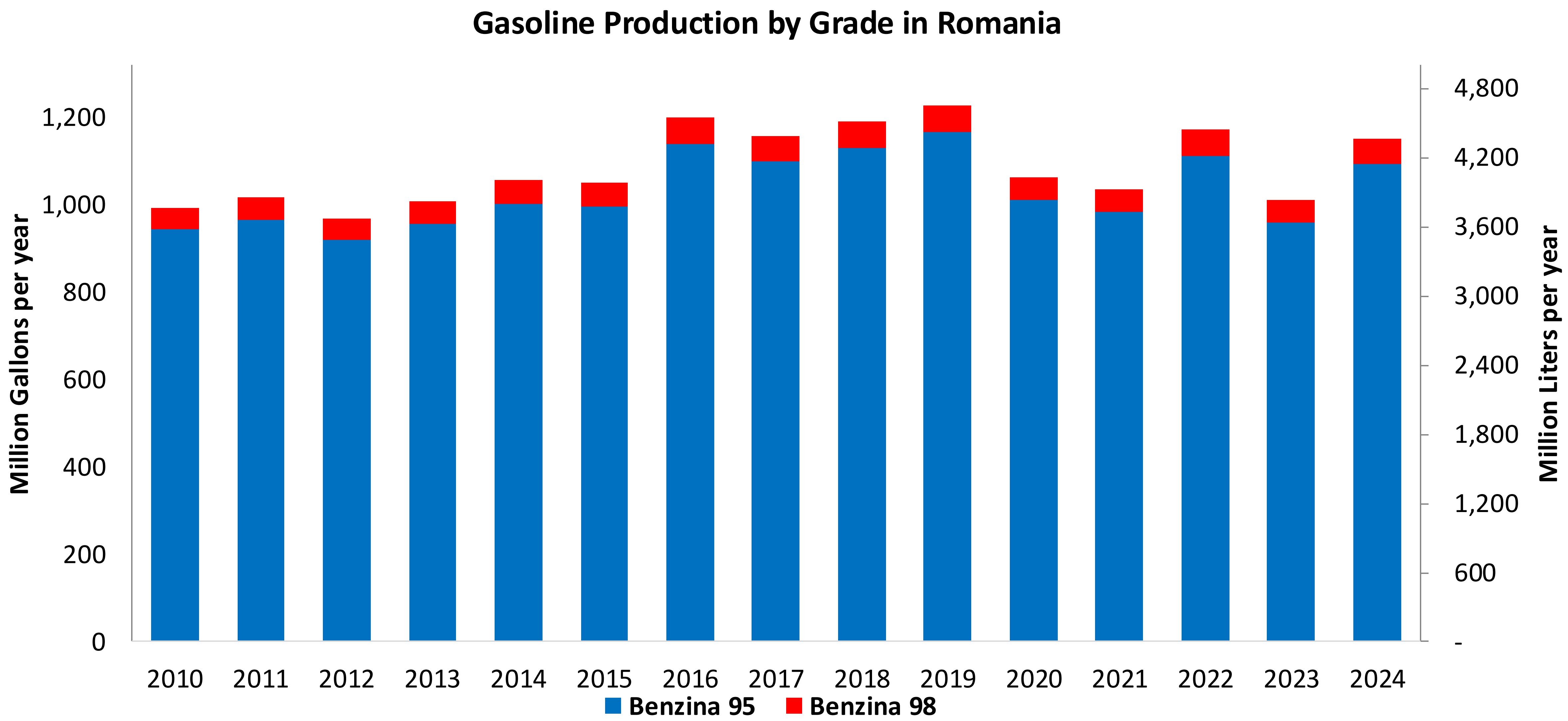
Source: Eurostat, European Commsion

Gasoline Demand in Romania



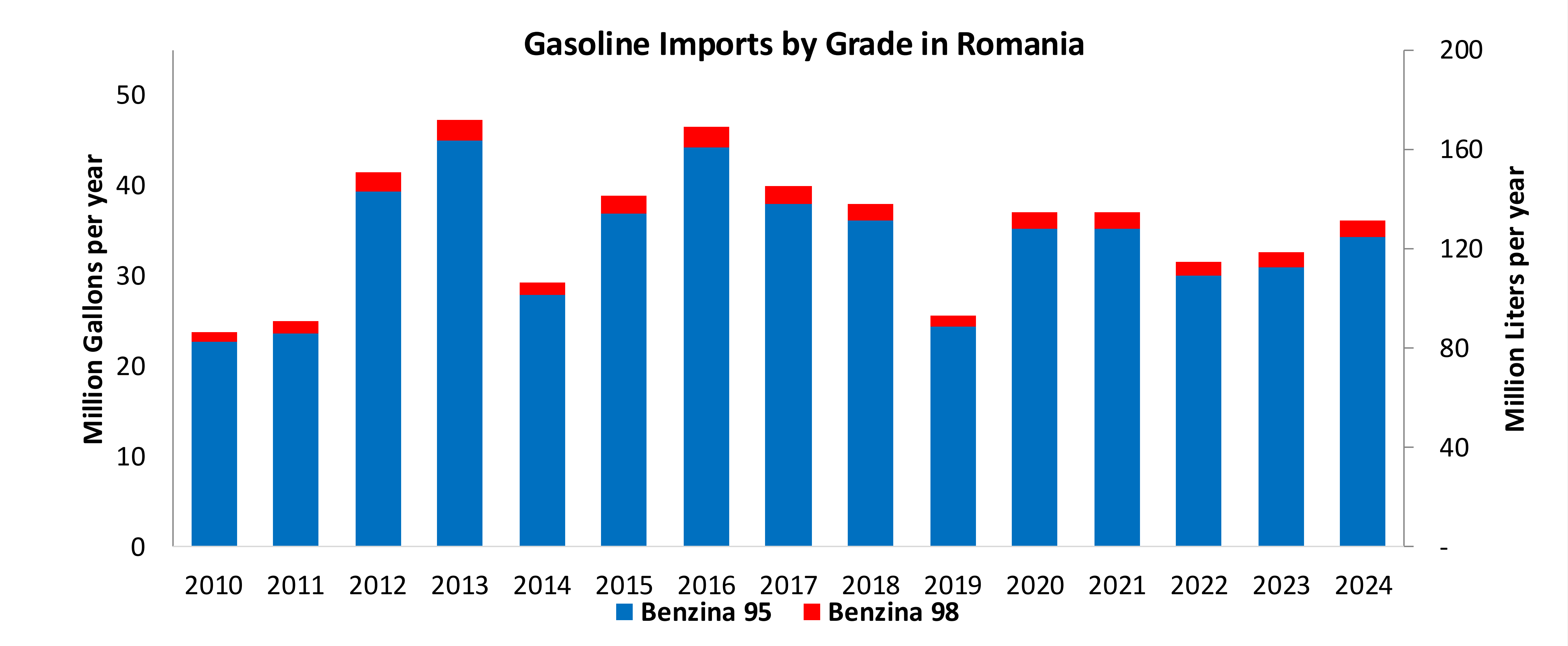
Source: Eurostat

Gasoline Production in Romania



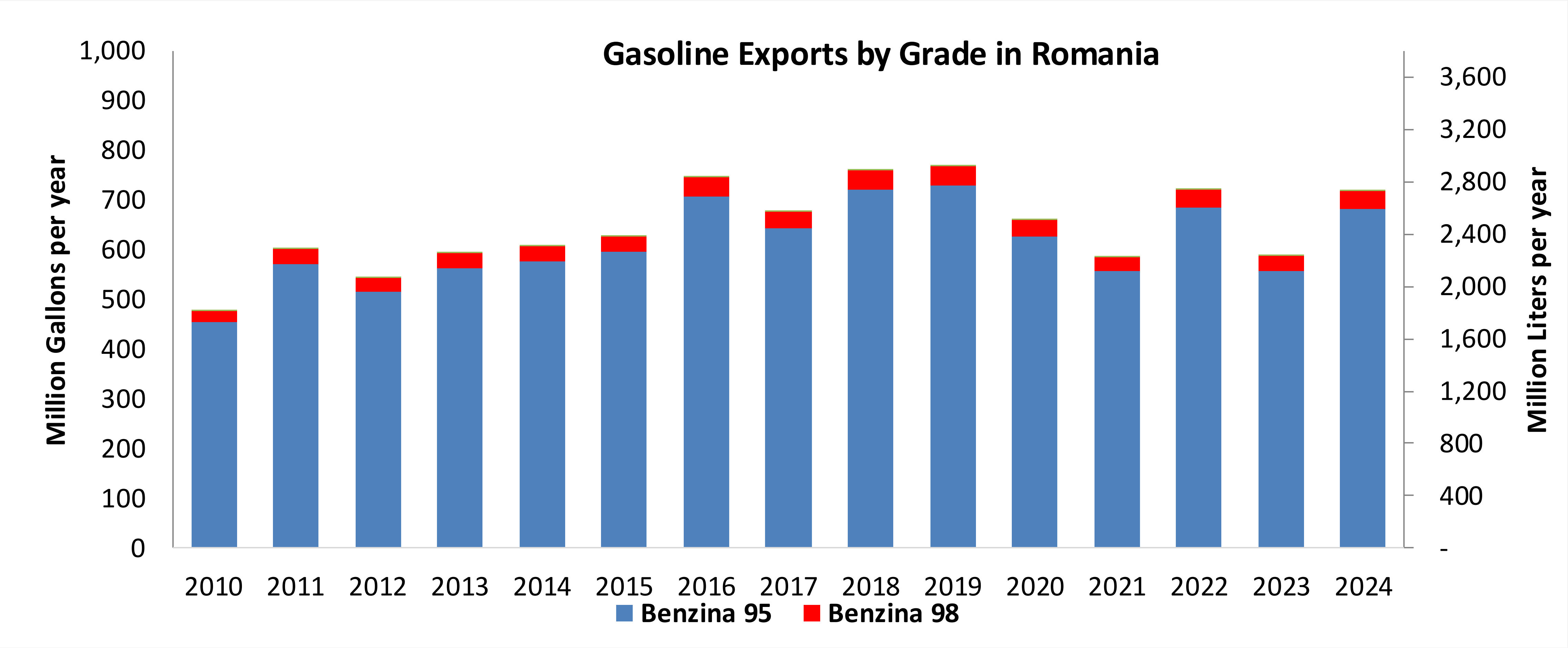
Source: Eurostat

Gasoline Imports in Romania



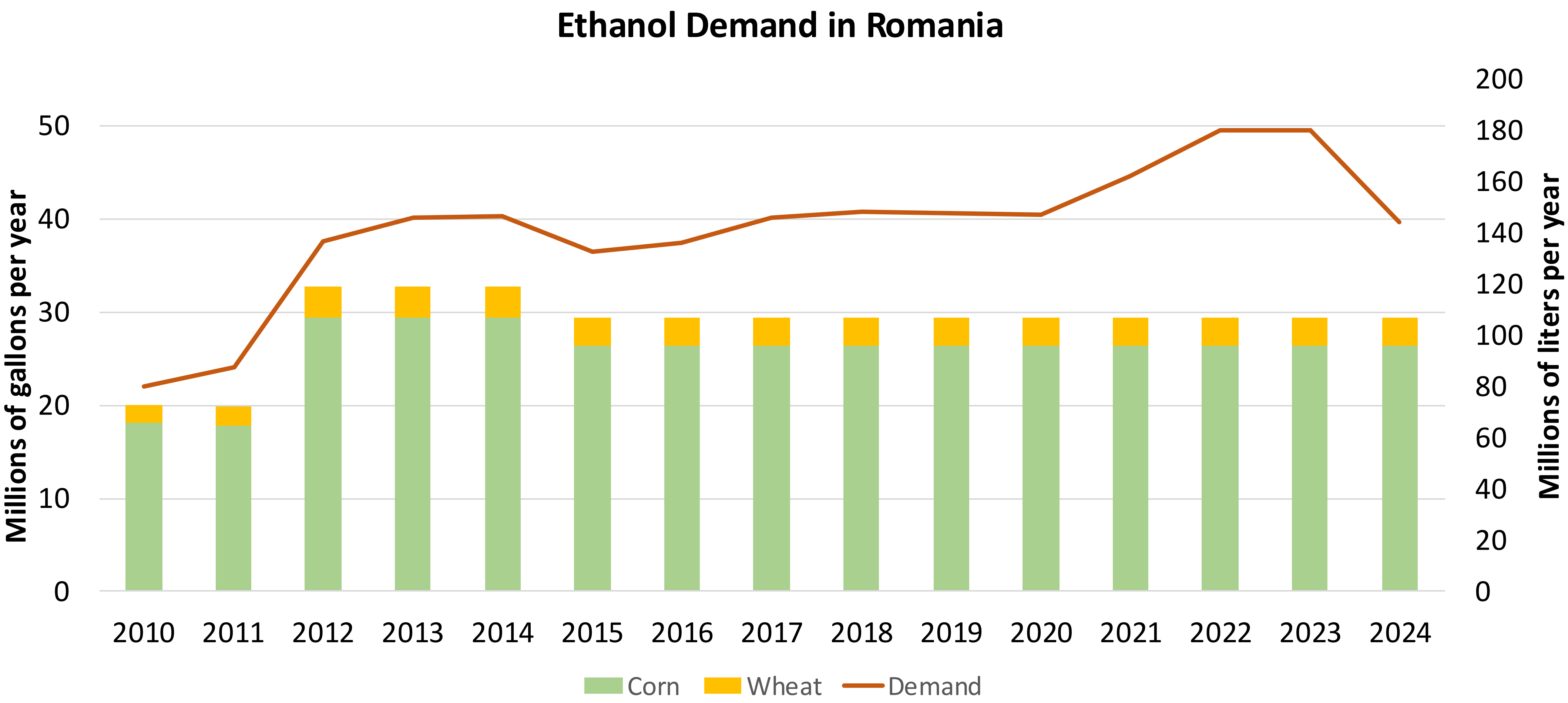
Source: Eurostat

Gasoline Exports in Romania



Source: Eurostat

Ethanol Balance in Romania



Source: Eurostat

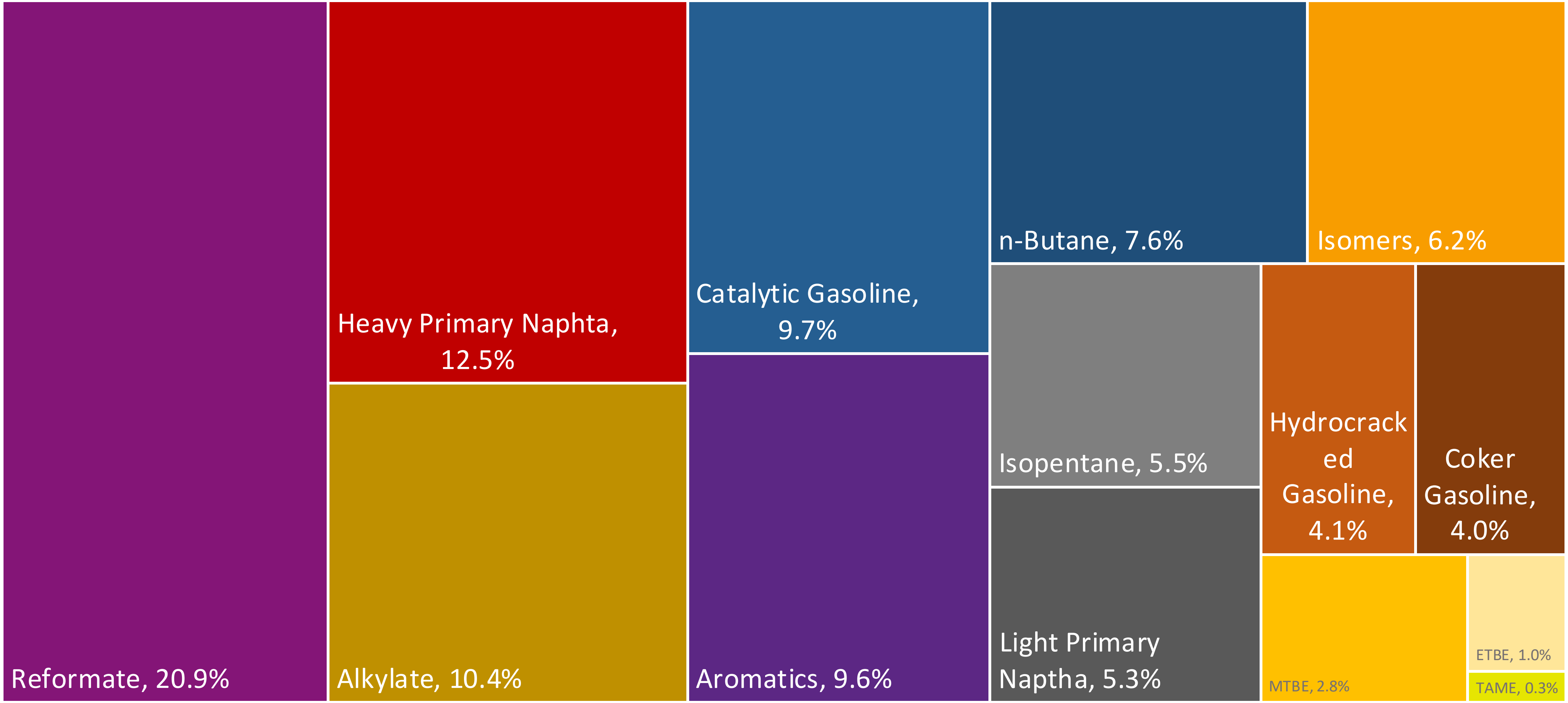
Gasoline Quality in Romania

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.0							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

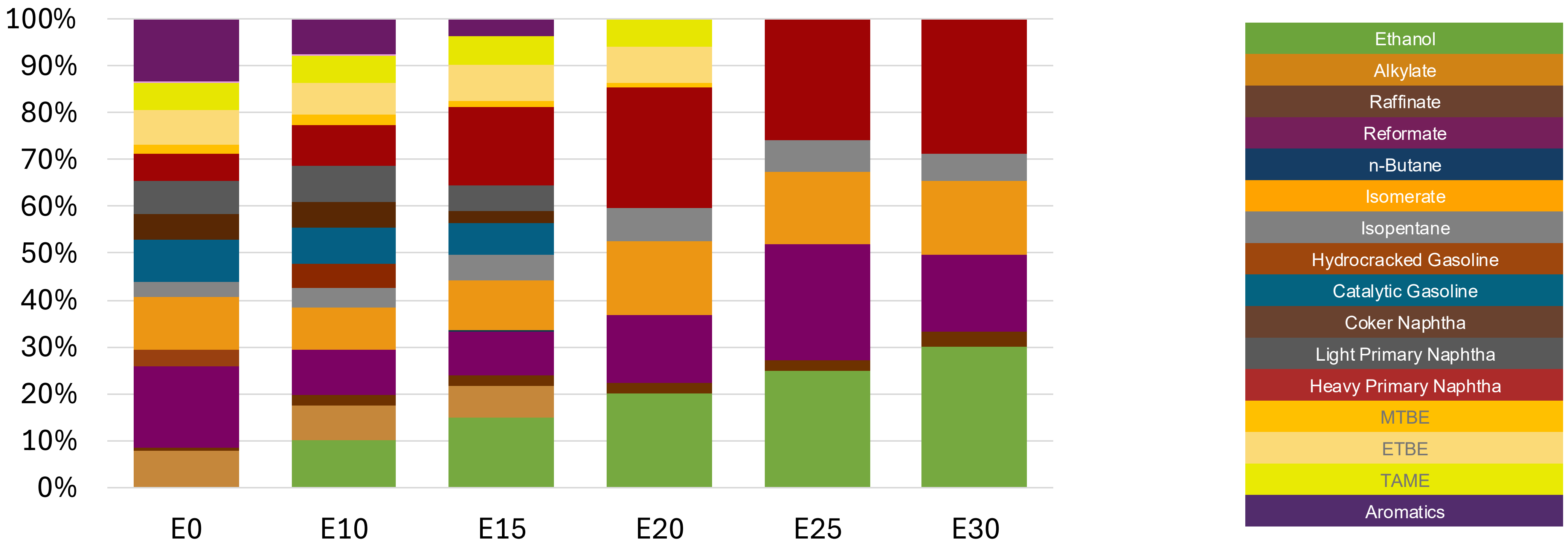
Source: European Commision

Romania Available Blending Components



Source: Faro90

Romania – Gasoline 95 – Constant Octane

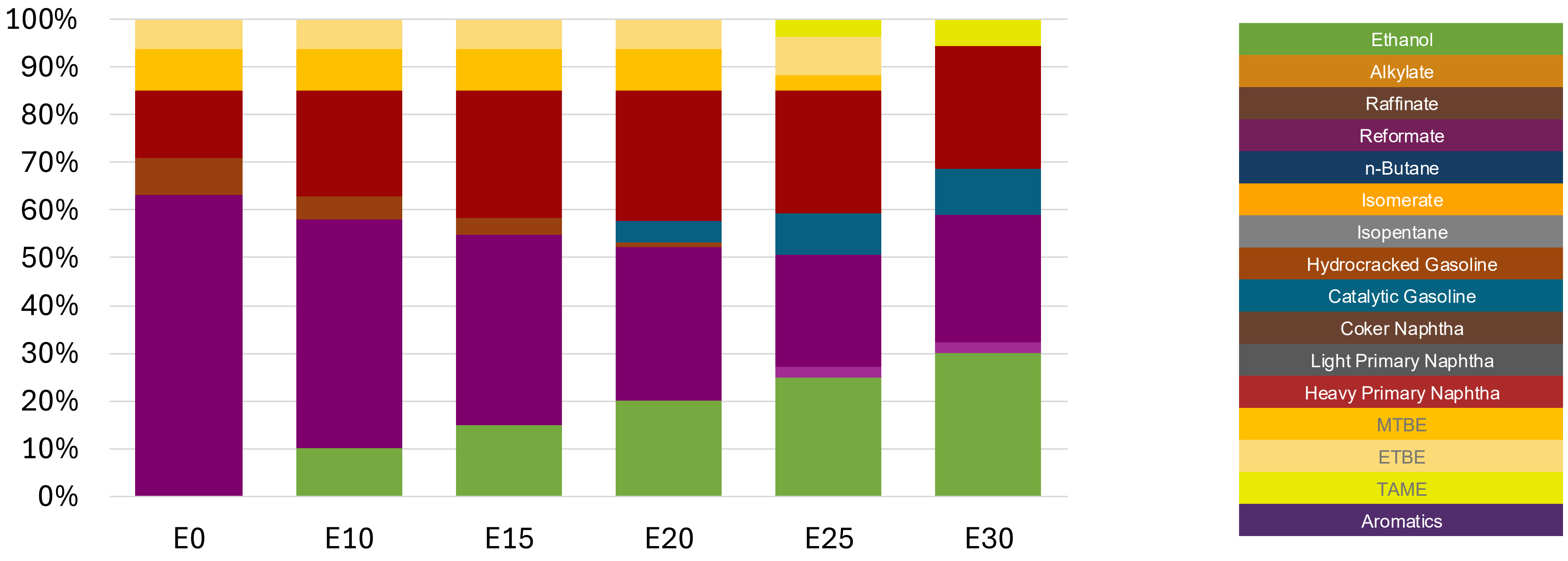


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.284	$ 2.146	$ 2.046	$ 1.931	$ 1.885	$ 1.832

Average CIF 2024 Prices

Source: Faro90

Romania - Gasoline 98 - Constant Octane

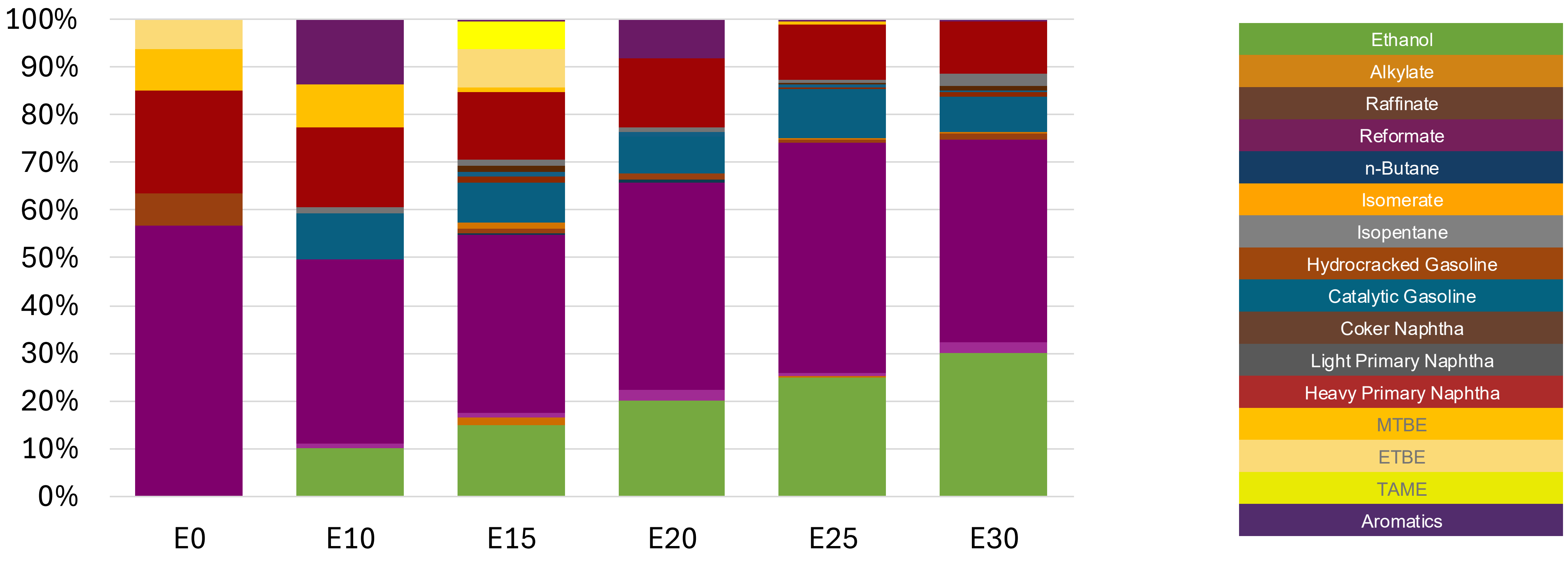


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.304	$ 2.187	$ 2.118	$ 2.046	$ 1.981	$ 1.931

Average CIF 2024 Prices

Source: Faro90

Romania – Gasoline 95 – Increased Octane

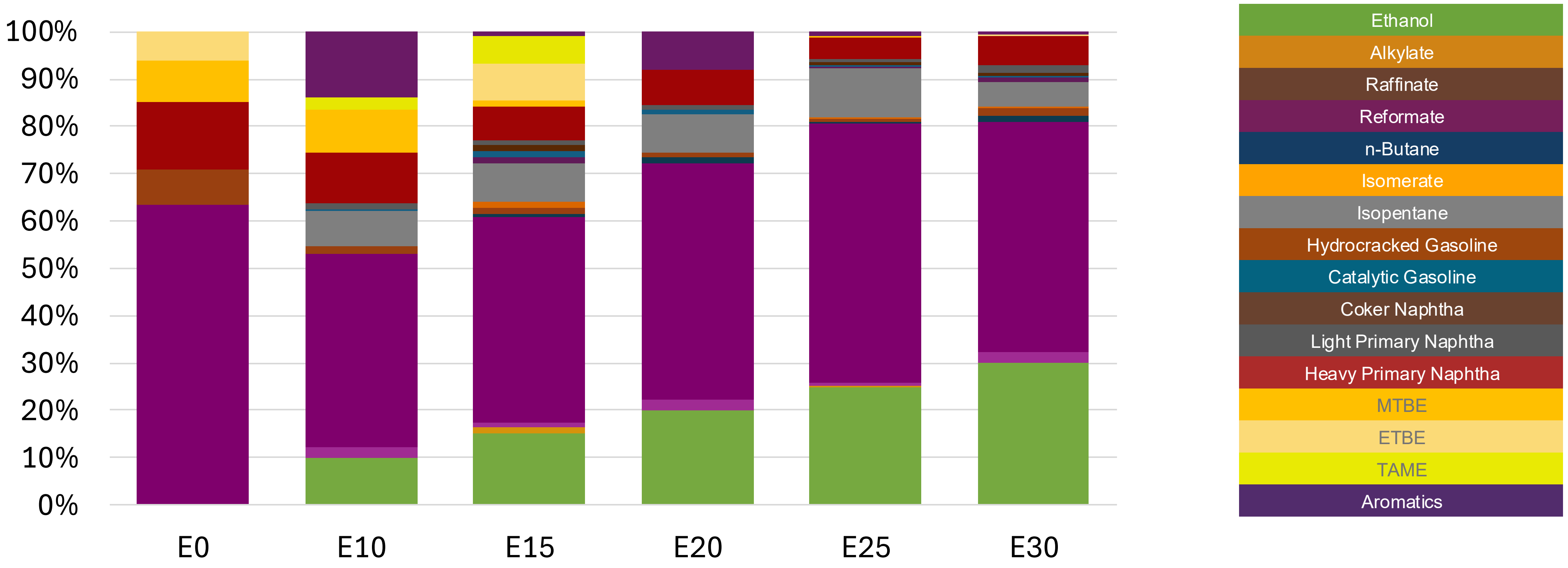


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.172	$ 2.172	$ 2.172	$ 2.172	$ 2.172	$ 2.172

Average CIF 2024 Prices

Source: Faro90

Romania – Gasoline 98 – Increased Octane

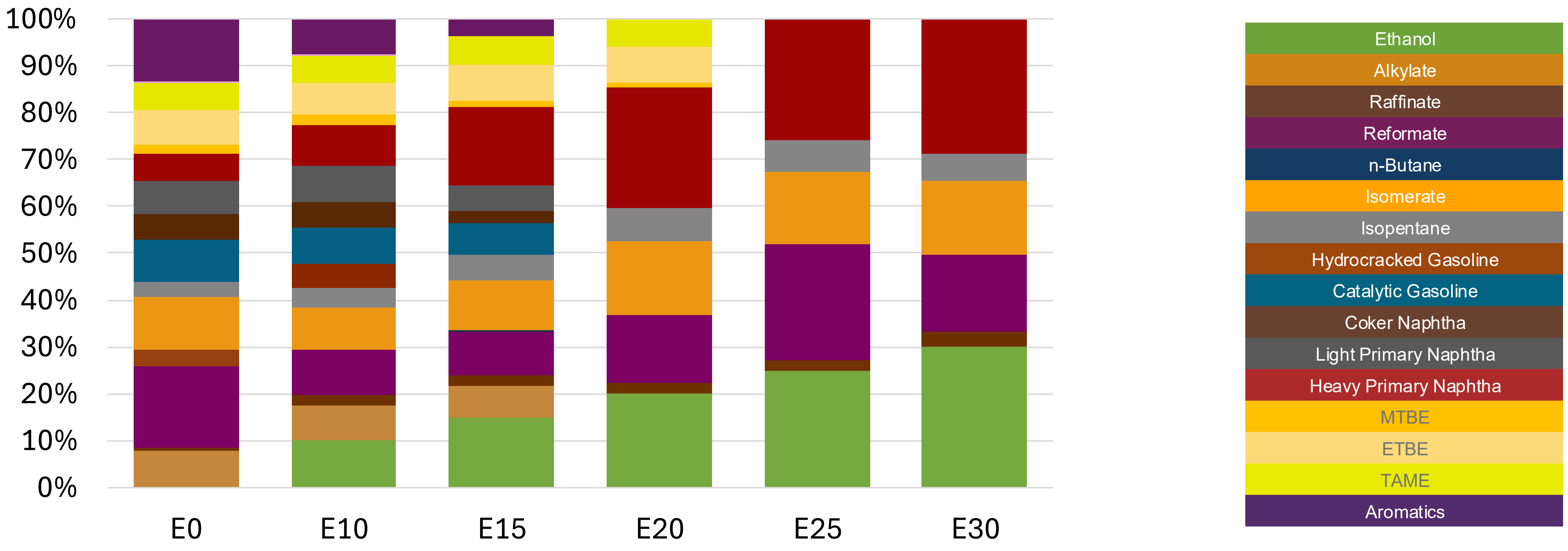


Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.304	$ 2.304	$ 2.304	$ 2.304	$ 2.304	$ 2.304

Average CIF 2024 Prices

Source: Faro90

Romania – Gasoline 95 – Constant Octane

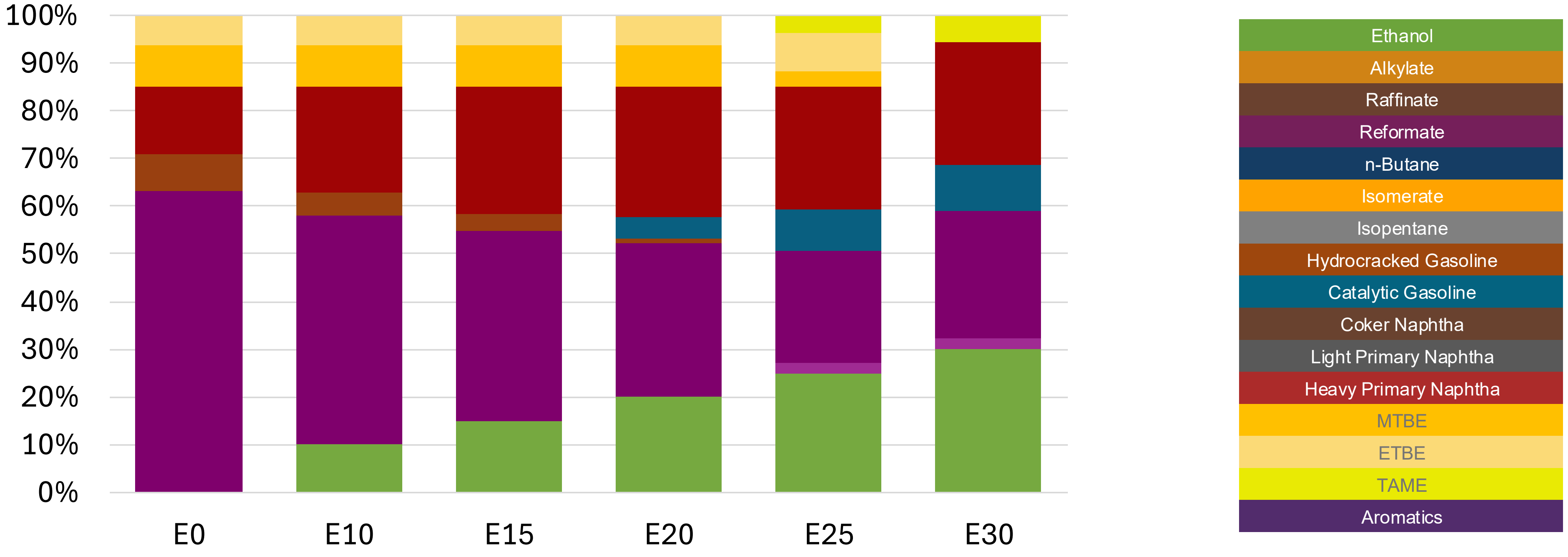


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.274	$ 2.136	$ 2.036	$ 1.921	$ 1.875	$ 1.822

Average CIF 2024 Prices

Source: Faro90

Romania - Gasoline 98 - Constant Octane

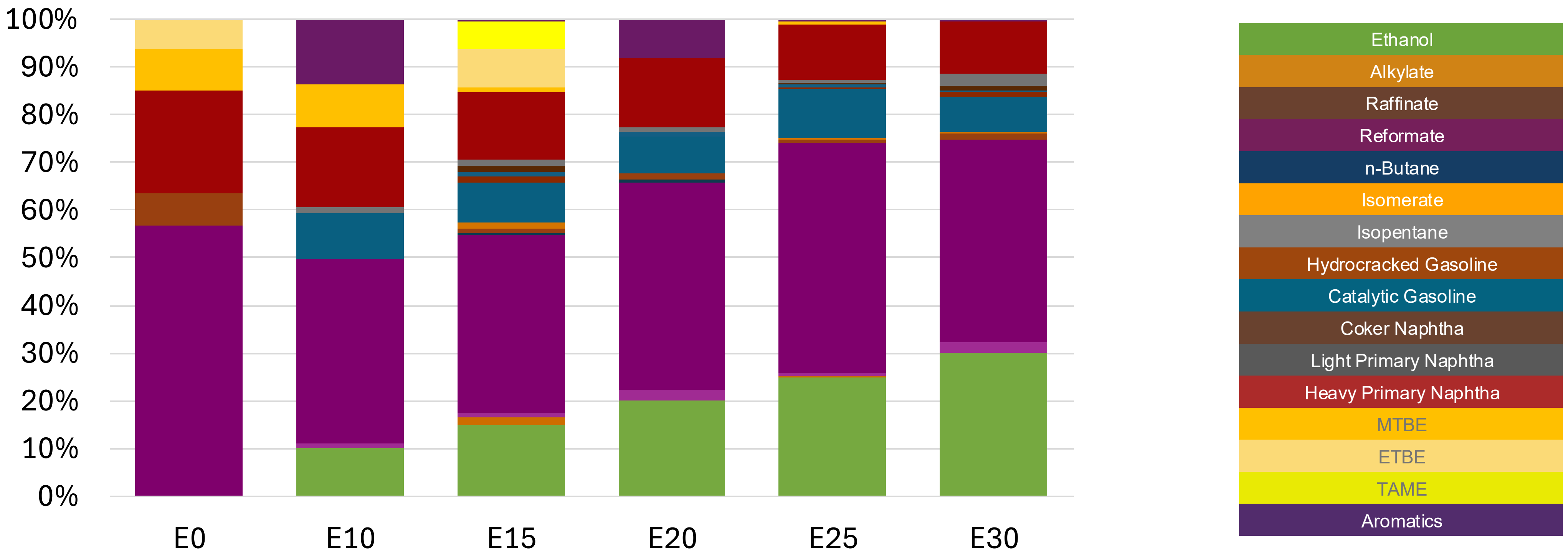


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.294	$ 2.177	$ 2.108	$ 2.036	$ 1.971	$ 1.921

Average CIF 2024 Prices

Source: Faro90

Romania – Gasoline 95 – Increased Octane

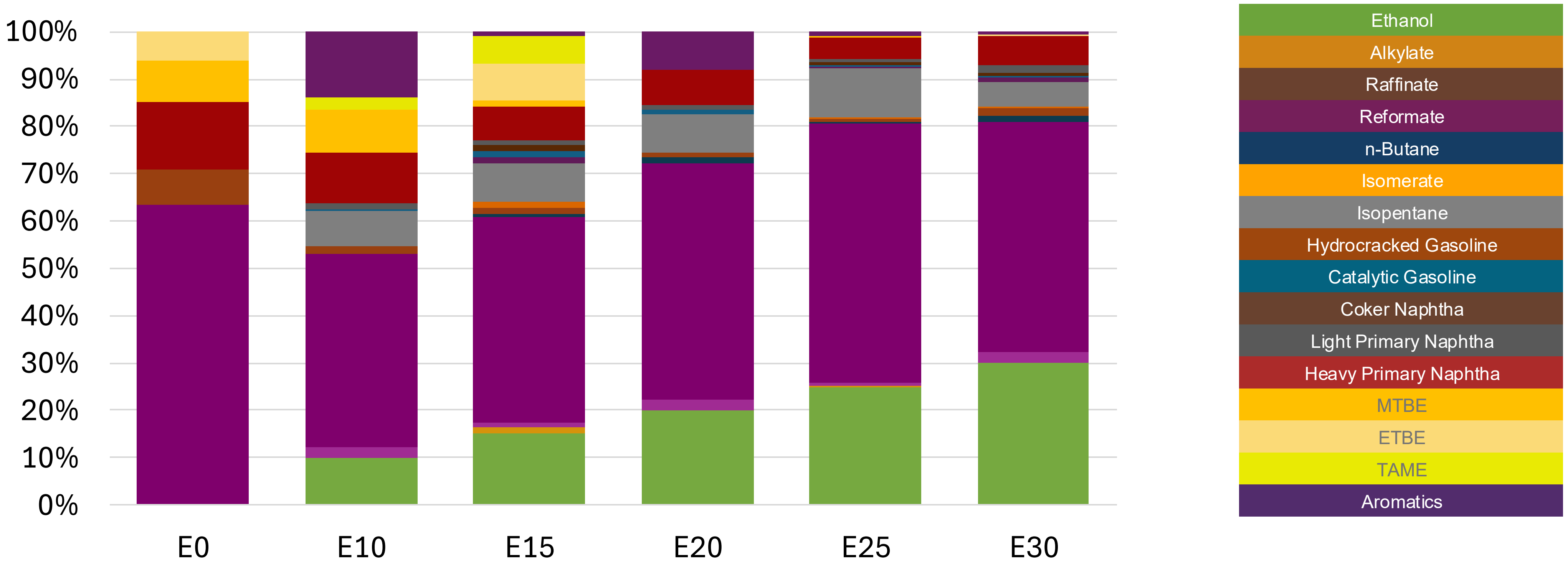


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162	$ 2.162

Average CIF 2024 Prices

Source: Faro90

Romania – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294	$ 2.294

Average CIF 2024 Prices

Source: Faro90

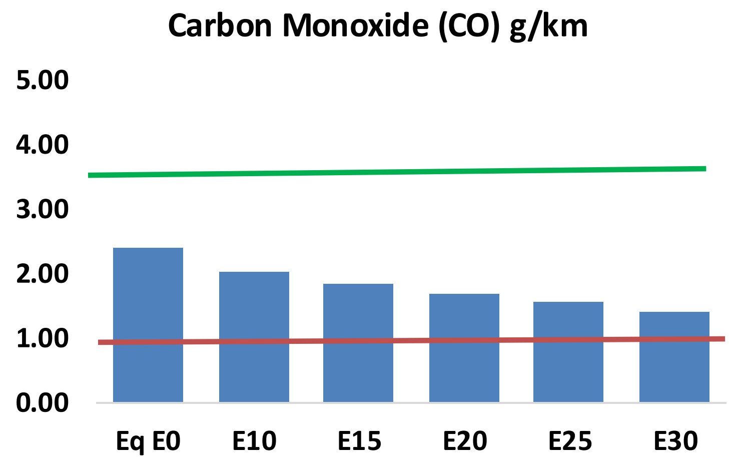
Gasoline Vehicle Fleet - Romania

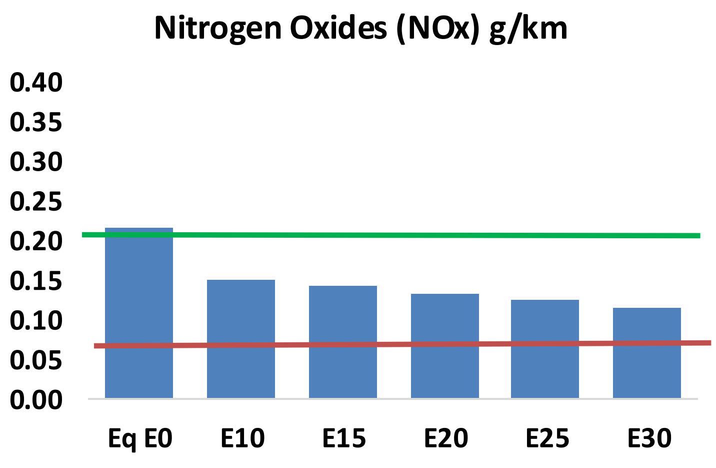
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	620
		Single FI	3-Ways		>8 years	19,893
		Multi FI	3-Ways		>8 years	200,796
			Euro III		>8 years	266,751
			Euro IV		>8 years	477,890
			Euro V		>8 years	211,220
			Euro 6		>8 years	596,719
					4-8 years	852,646
					<4 years	498,564
			Hybrid		All	61,401
		Electric			All	36,969
	Medium	Carburator	3-Ways	PCV Tank	>8 years	61
		Single FI	3-Ways		>8 years	582
		Multi FI	3-Ways		>8 years	24,332
			Euro III		>8 years	50,187
			Euro IV		>8 years	167,149
			Euro V		>8 years	65,014
			Euro 6		>8 years	198,906
					4-8 years	385,372
					<4 years	397,245
			Hybrid		All	49,317
		Electric			All	28,865
Sml Engines	Light	4-Cycle	None	PCV	>8 years	11,443
			Euro III		>8 years	11,973
			Euro IV		>8 years	71,783
			Euro V		>8 years	42,298
			Euro 6		>8 years	100,266
					4-8 years	89,645
					<4 years	203,058
			Hybrid		All	457,800
		Electric			All	165,200
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	8,287
		Fuel-Injection	3-Ways		>8 years	93,656
			Euro III		>8 years	52,526
			Euro IV		>8 years	60,548
			Euro V		>8 years	16,950
			Euro 6		>8 years	23,400
					4-8 years	11,032
					<4 years	1,433
			Hybrid		All	215
		Electric			All	450

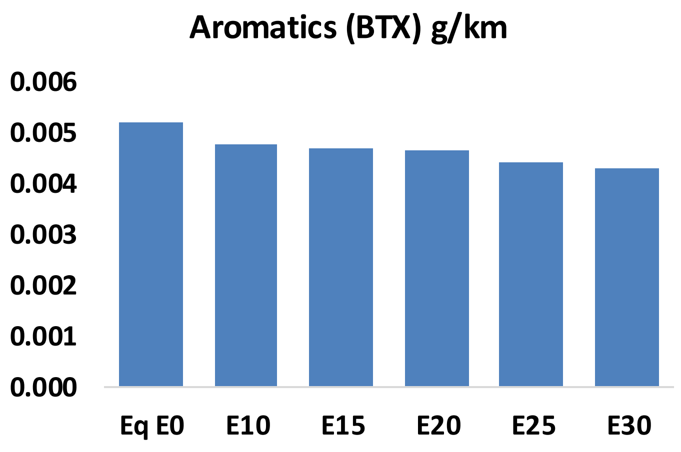
Vehicle Fleet: 6,012,464
Gasoline Vehicle Fleet: 2,699,932

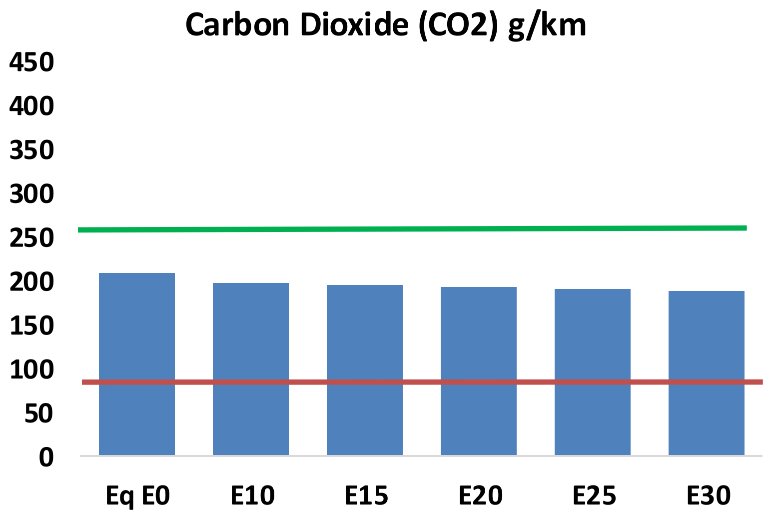
Source: Eurostat, ACEA

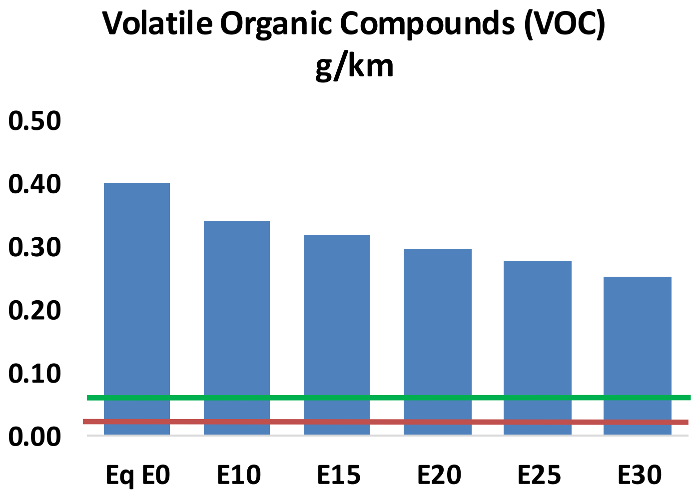
Romania – Vehicular Emissions

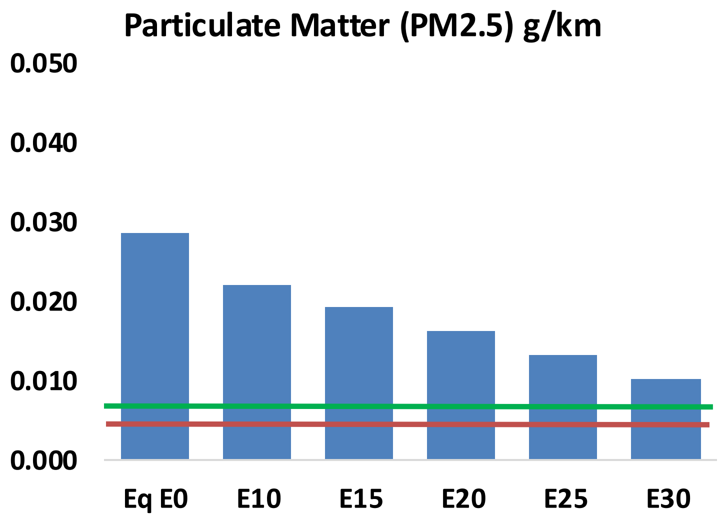












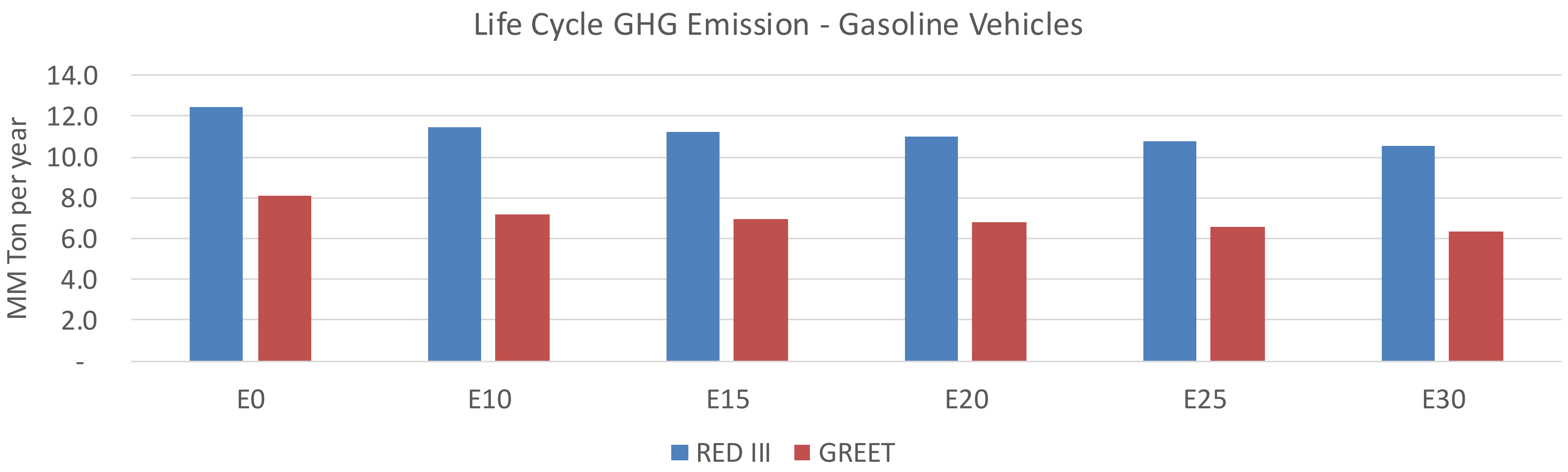
Source: Faro90

Romania – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	171,877	157,764	143,651	131,911	120,016	110,845	101,081	-16%	-30%
CO2	14,864,151	14,481,930	14,120,943	13,837,038	13,696,551	13,615,510	13,364,537	-5%	-8%
VOC	28,470	26,412	24,354	22,693	21,033	19,760	18,049	-14%	-26%
NOx	15,404	11,553	10,783	10,163	9,584	8,947	8,272	-30%	-38%
SOx	169	157	144	133	123	111	100	-15%	-28%
NH3	5,167	5,116	5,065	5,089	5,146	5,186	5,220	-2%	0%
N2O	247	246	245	246	251	254	257	-1%	2%
CH4	6,222	6,222	6,222	6,315	6,452	6,570	6,682	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	175	168	161	156	152	149	144	-8%	-14%
Acetaldehyde	489	561	823	1,253	1,707	1,950	2,305	68%	249%
Formaldehyde	1,766	1,717	2,002	2,350	2,462	2,724	2,965	13%	39%
Benzene	372	358	339	333	331	316	306	-9%	-11%
PM2.5	921	818	715	621	527	428	324	-22%	-43%
PM10	1,108	984	860	747	634	515	390	-22%	-43%

Source: Faro90

Romania – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	76.7
Target @2035 (MMT/year)	43.2
Delta	33.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.3%	14.4%	16.7%	18.9%	22.0%
RED II/III	0.0%	7.8%	10.0%	11.7%	13.3%	15.5%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.7%	3.5%	4.1%	4.6%	5.3%
RED II/III	0.0%	2.9%	3.7%	4.4%	5.0%	5.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

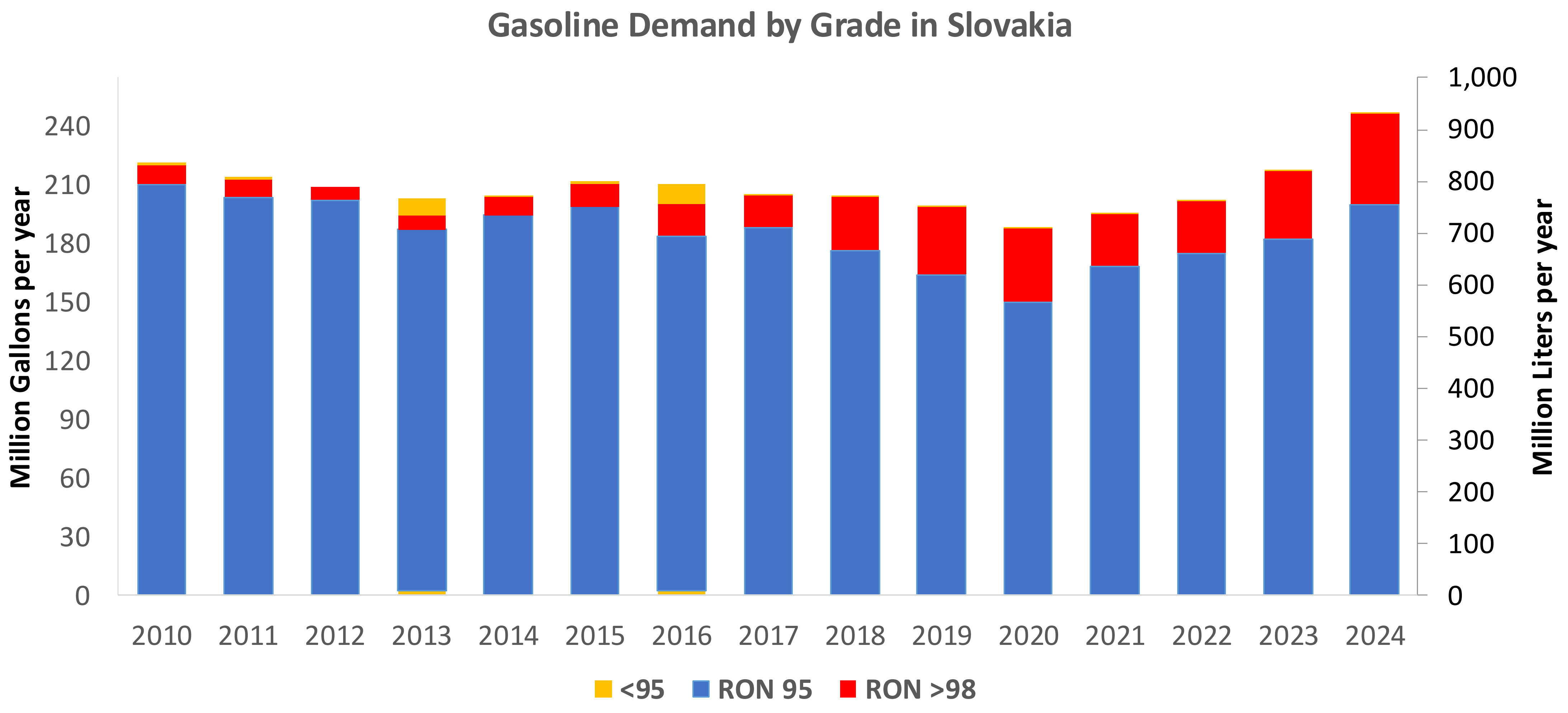
Ethanol Gasoline Blending in Slovakia

In 2024, gasoline consumption in Slovakia was 931 million liters (246 million gallons) and growth rate was 4.4%. Gasoline production and net imports were and 931 million liters (and 246 million gallons) correspondingly. Slovakia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.6 RON.

In 2024, ethanol consumption in Slovakia was 80 million liters (21 million gallons) representing 8.6% of gasoline demand. Ethanol production and Net Imports were 80 and million liters (21 and million gallons) correspondingly. Ethanol source is Cereals 100%.

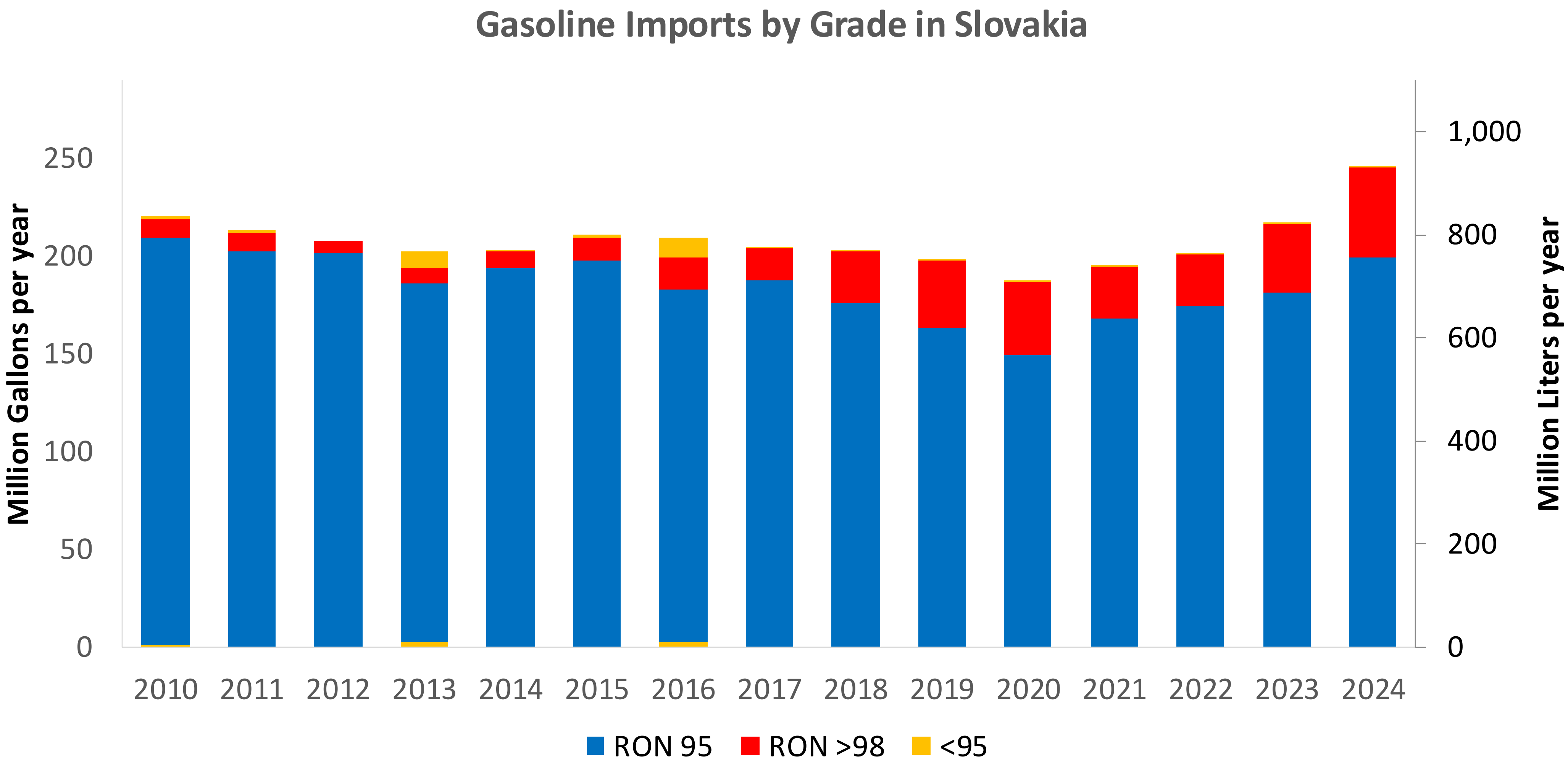
Source: Eurostat, European Commision

Gasoline Demand in Slovakia



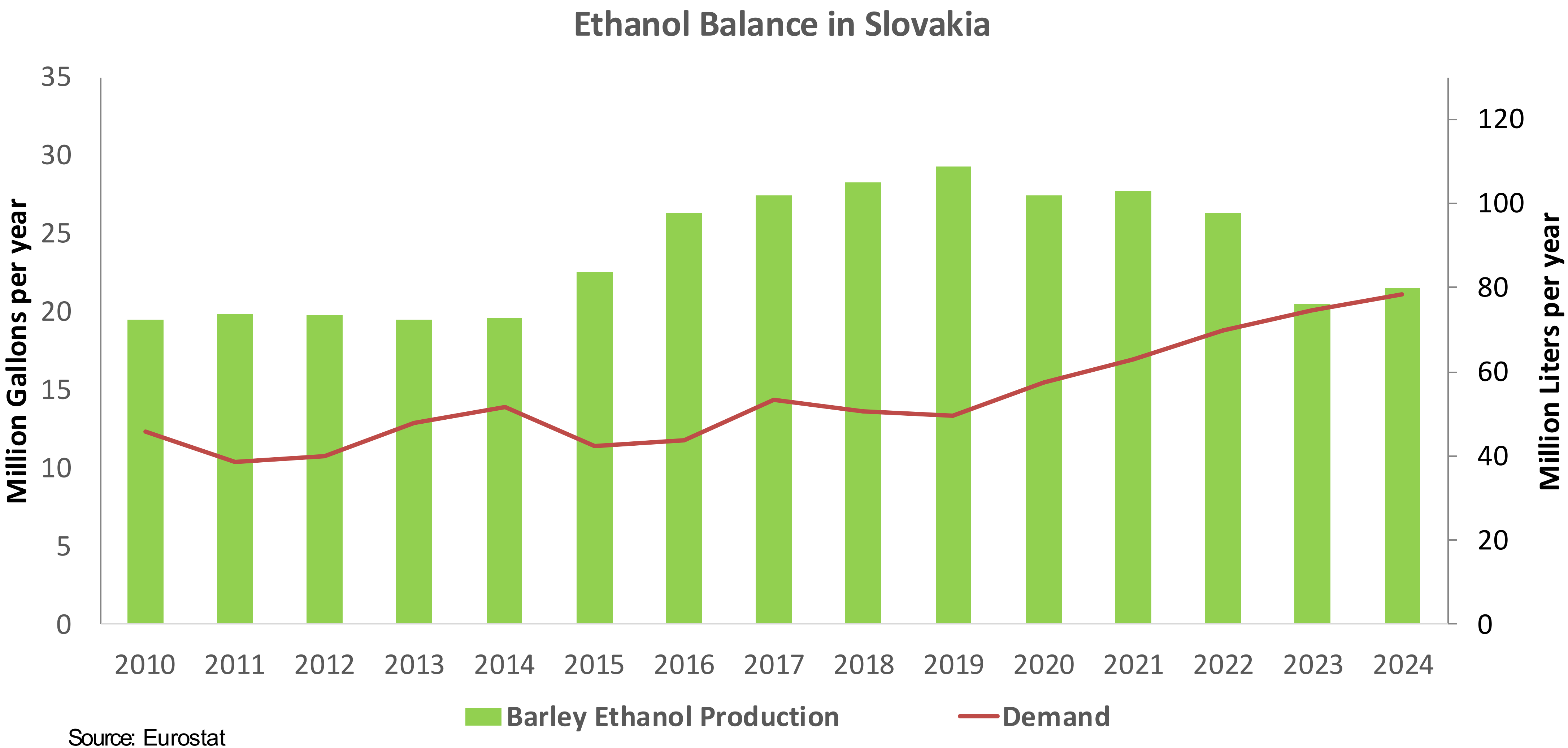
Source: Eurostat

Gasoline Imports in Slovakia



Source: Eurostat

Ethanol Balance in Slovakia



Source: Eurostat

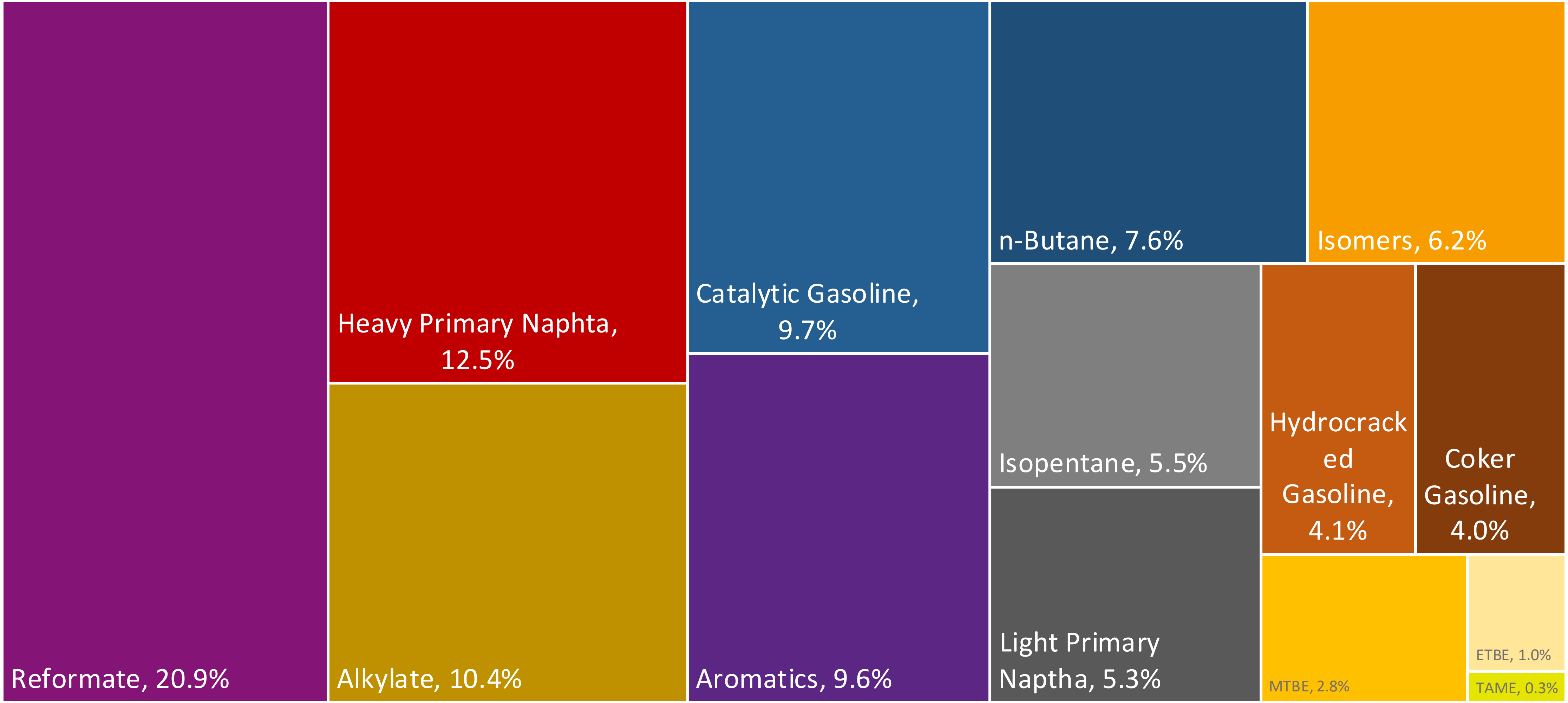
Gasoline Quality in Slovakia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	9.2							
Advanced Biofuels (%cal)	1.1							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

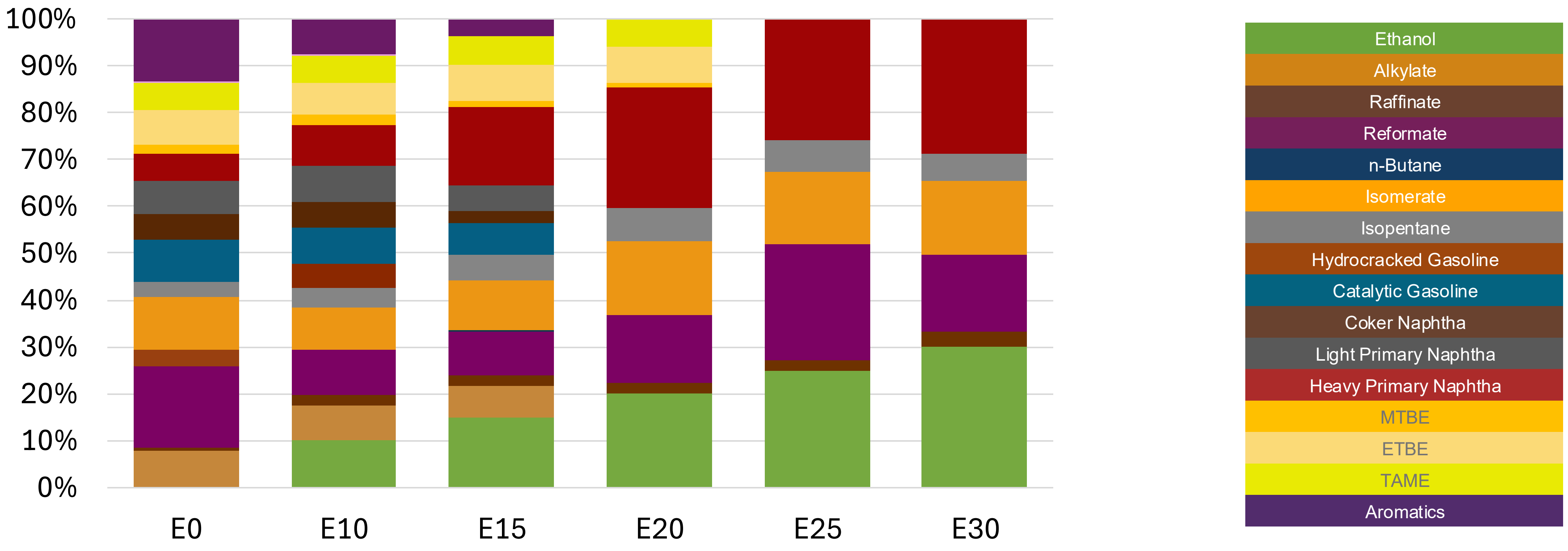
Source: European Commision

Slovakia Available Blending Components



Source: Faro90

Slovakia – Gasoline 95 – Constant Octane

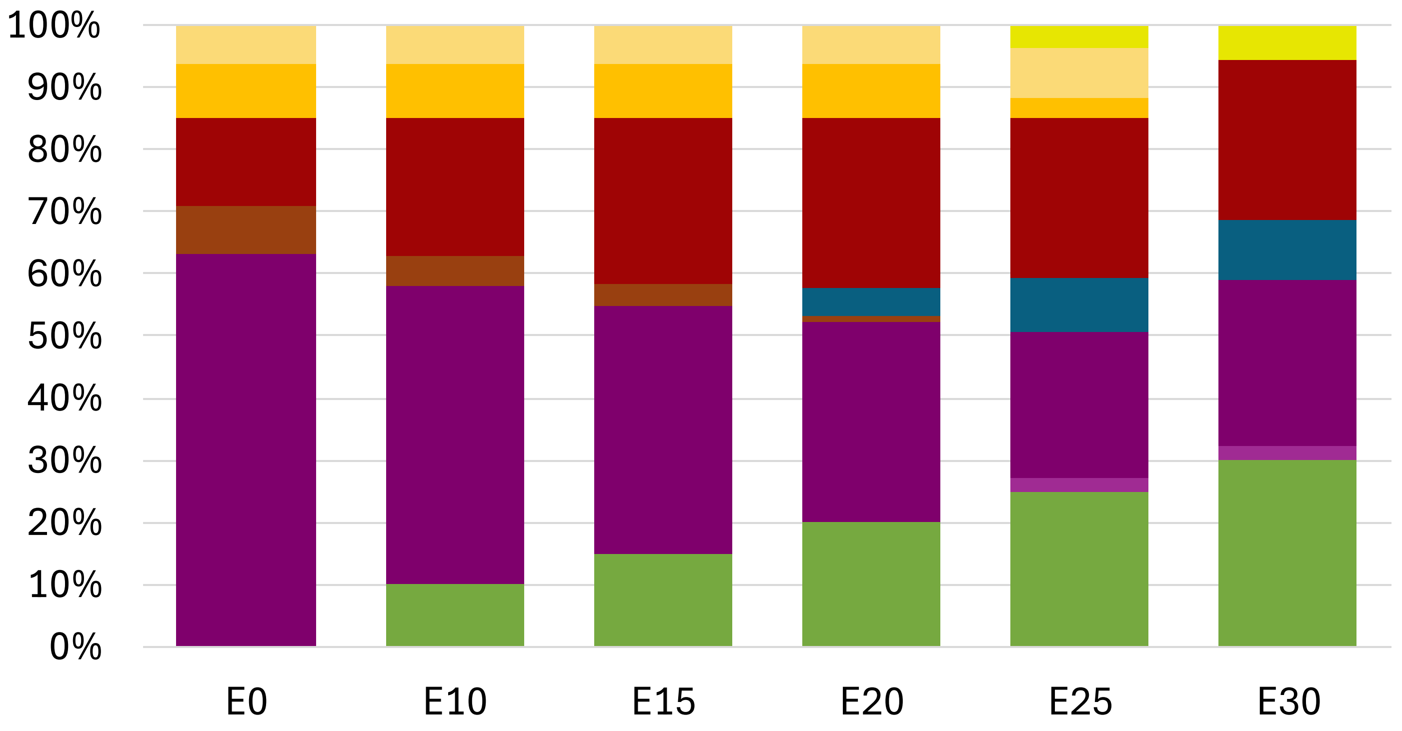


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.304	$ 2.166	$ 2.066	$ 1.951	$ 1.905	$ 1.852

Average CIF 2024 Prices

Source: Faro90

Slovakia - Gasoline 98 - Constant Octane

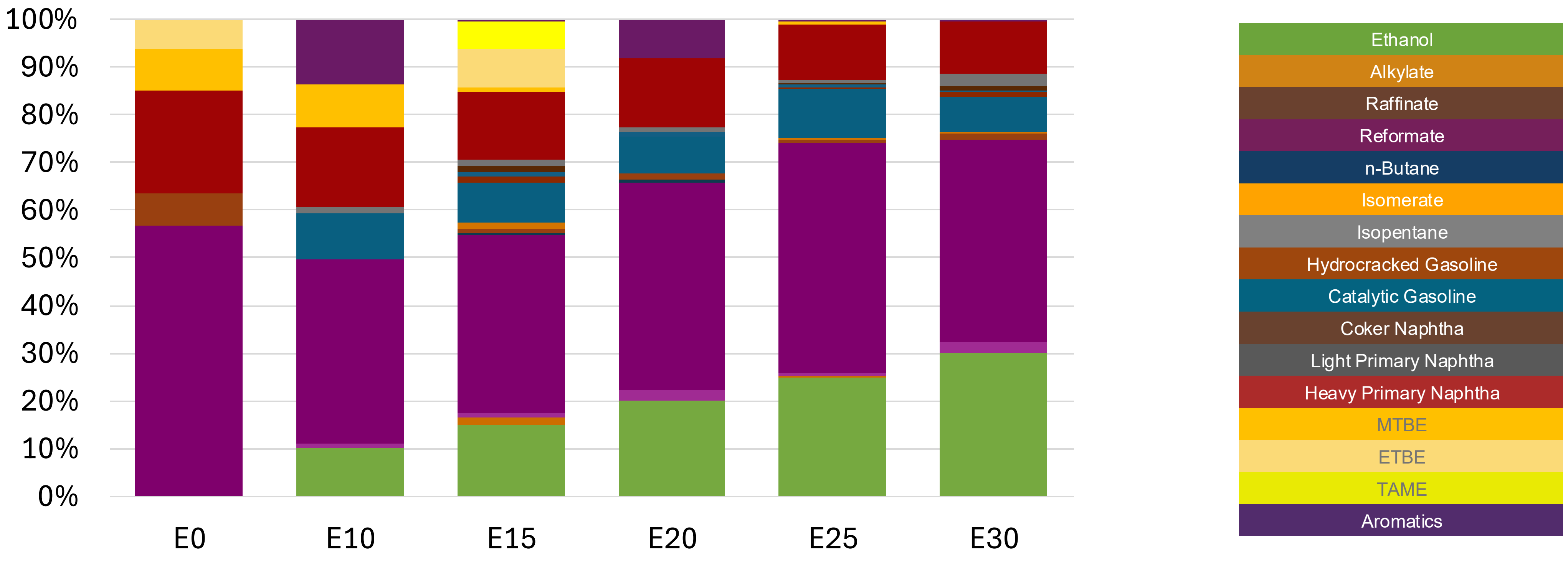


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.324	$ 2.207	$ 2.138	$ 2.066	$ 2.001	$ 1.951

Average CIF 2024 Prices

Source: Faro90

Slovakia – Gasoline 95 – Increased Octane

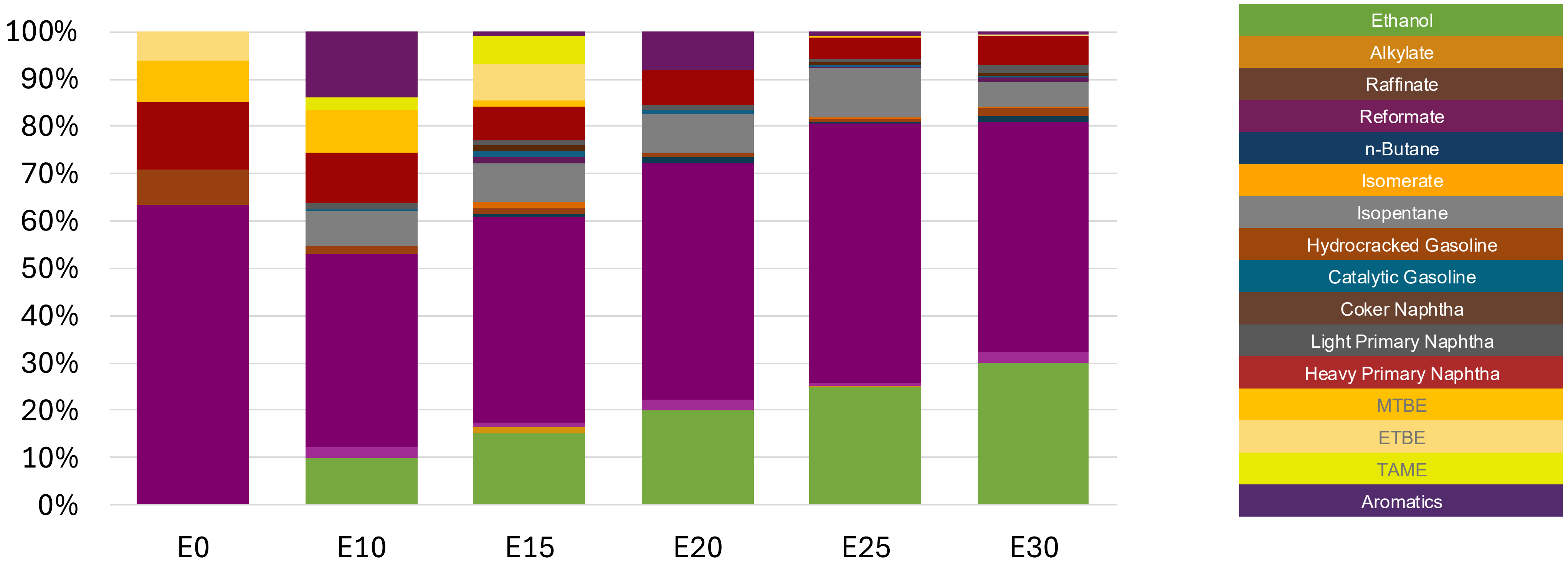


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.192	$ 2.192	$ 2.192	$ 2.192	$ 2.192	$ 2.192

Average CIF 2024 Prices

Source: Faro90

Slovakia – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.324	$ 2.324	$ 2.324	$ 2.324	$ 2.324	$ 2.324

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Slovakia

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	99
		Single FI	3-Ways		>8 years	42,839
		Multi FI	3-Ways		>8 years	1,490,963
			Euro III		>8 years	1,240,377
			Euro IV		>8 years	2,142,799
			Euro V		>8 years	891,502
			Euro 6		>8 years	1,814,851
					4-8 years	2,017,052
					<4 years	2,321,228
			Hybrid		All	45,708
		Electric			All	3,328
	Medium	Carburator	3-Ways	PCV Tank	>8 years	42
		Single FI	3-Ways		>8 years	3,555
		Multi FI	3-Ways		>8 years	189,014
			Euro III		>8 years	501,051
			Euro IV		>8 years	603,589
			Euro V		>8 years	202,535
			Euro 6		>8 years	372,100
					4-8 years	373,139
					<4 years	415,263
			Hybrid		All	475
		Electric			All	105
Sml Engines	Light	4-Cycle	None	PCV	>8 years	80,574
			Euro III		>8 years	188,128
			Euro IV		>8 years	219,981
			Euro V		>8 years	75,430
			Euro 6		>8 years	151,732
					4-8 years	145,987
					<4 years	157,298
			Hybrid		All	4
		Electric			All	1,131
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	35
		Fuel-Injection	3-Ways		>8 years	368,035
			Euro III		>8 years	154,164
			Euro IV		>8 years	223,237
			Euro V		>8 years	87,071
			Euro 6		>8 years	149,069
					4-8 years	152,370
					<4 years	168,360
			Hybrid		All	469
		Electric			All	103

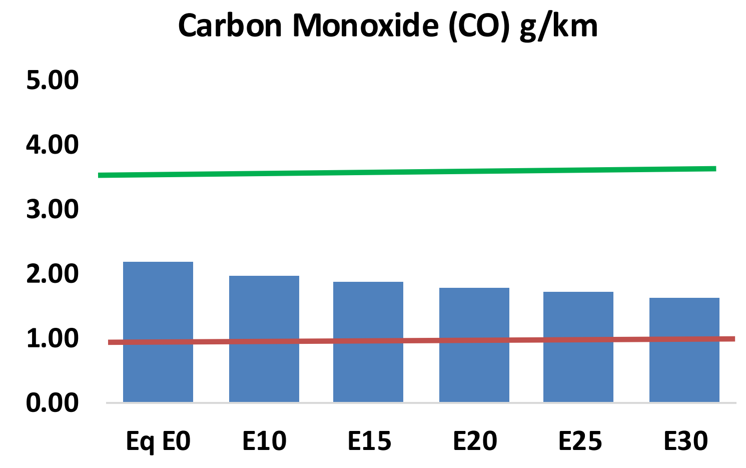
0 500,000 1,000,000 1,500,000 2,000,000 2,500,000

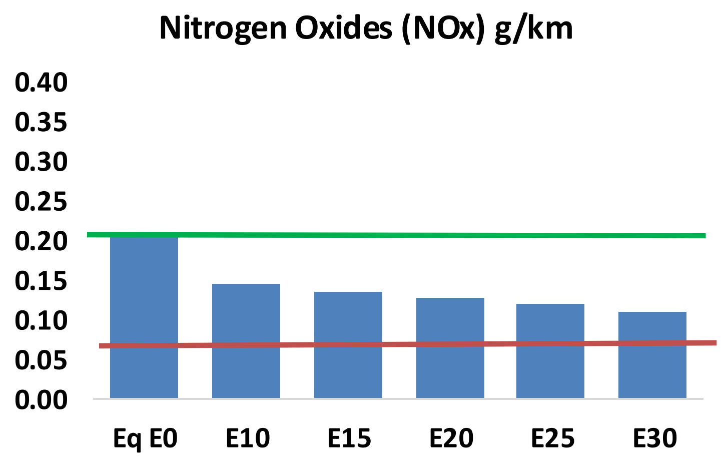
Vehicle Fleet: 5,792,410
Gasoline Vehicle Fleet: 1,843,374

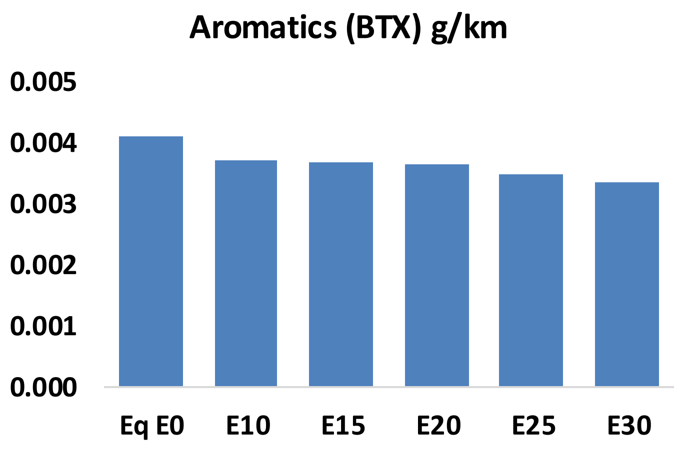
Source: Eurostat, ACEA

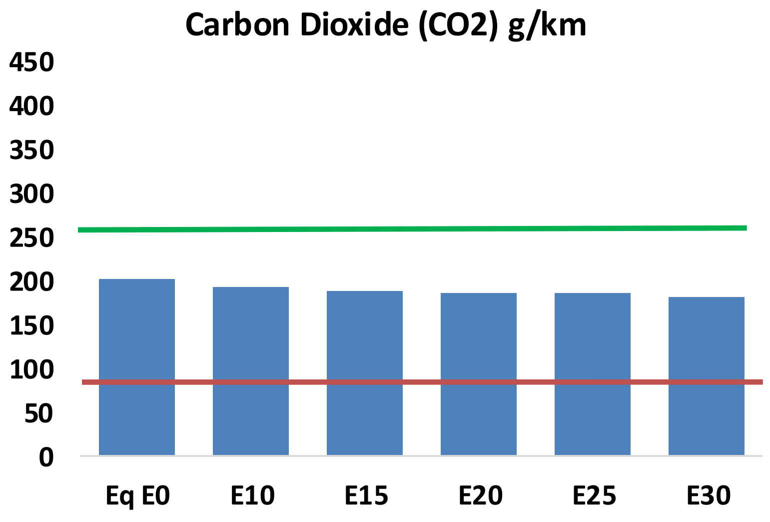
Slovakia – Vehicular Emissions

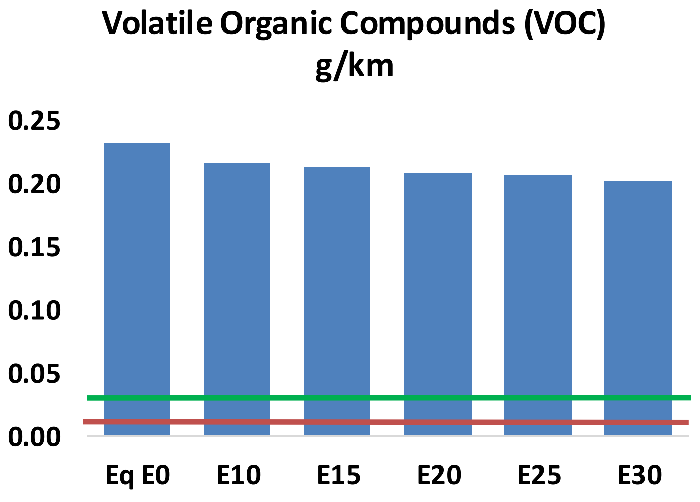
TIER III BIN 125 United States
Euro 6 Standard

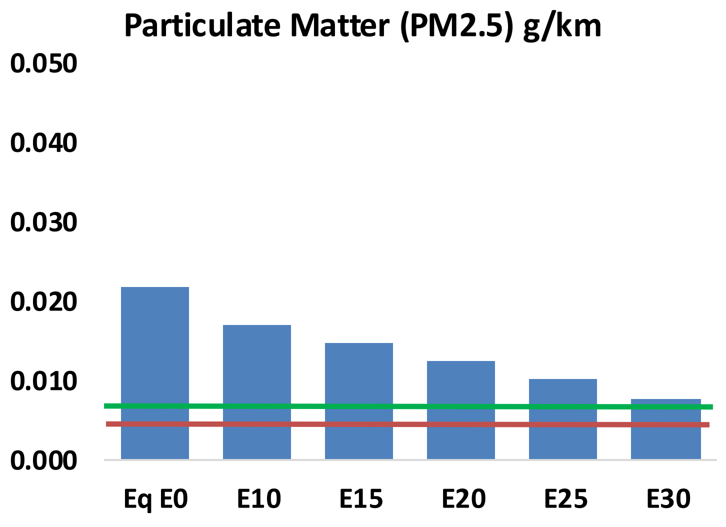












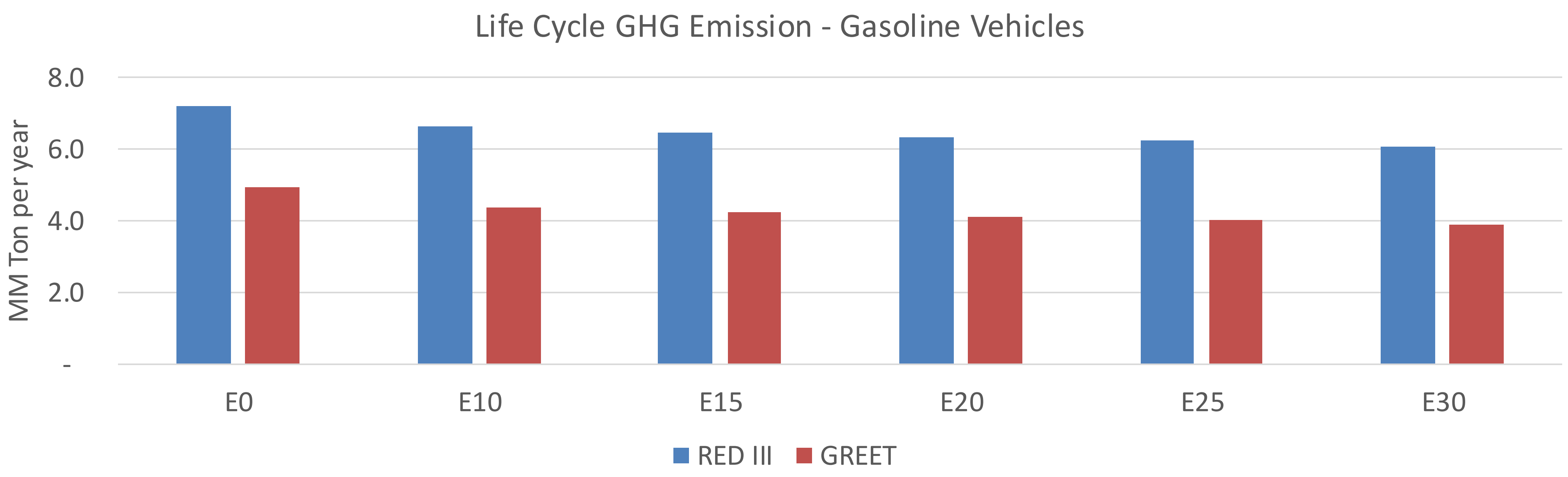
Source: Faro90

Slovakia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	99,865	94,329	88,793	84,675	80,722	77,741	74,129	-11%	-19%
CO2	9,214,229	8,977,292	8,753,517	8,577,526	8,490,438	8,440,370	8,284,790	-5%	-8%
VOC	10,576	10,223	9,870	9,671	9,511	9,401	9,210	-7%	-10%
NOx	9,439	7,079	6,608	6,228	5,873	5,482	5,069	-30%	-38%
SOx	104	96	88	81	75	68	61	-15%	-28%
NH3	3,205	3,174	3,142	3,157	3,192	3,217	3,238	-2%	0%
N2O	103	102	102	103	105	106	107	-1%	2%
CH4	2,586	2,586	2,586	2,625	2,681	2,731	2,777	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	109	106	103	102	102	101	100	-5%	-7%
Acetaldehyde	233	267	392	597	813	929	1,098	68%	249%
Formaldehyde	744	724	844	991	1,038	1,148	1,250	13%	39%
Benzene	187	179	170	167	166	159	153	-9%	-11%
PM2.5	590	524	458	398	338	274	208	-22%	-43%
PM10	403	358	313	272	231	187	142	-22%	-43%

Source: Faro90

Slovakia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	36.8
Target @ 2035 (MMT/year)	20.7
Delta	16.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.2%	14.2%	16.5%	18.6%	21.6%
RED II/III	0.0%	8.1%	10.3%	12.0%	13.5%	15.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.4%	4.4%	5.0%	5.7%	6.6%
RED II/III	0.0%	3.6%	4.6%	5.4%	6.1%	7.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

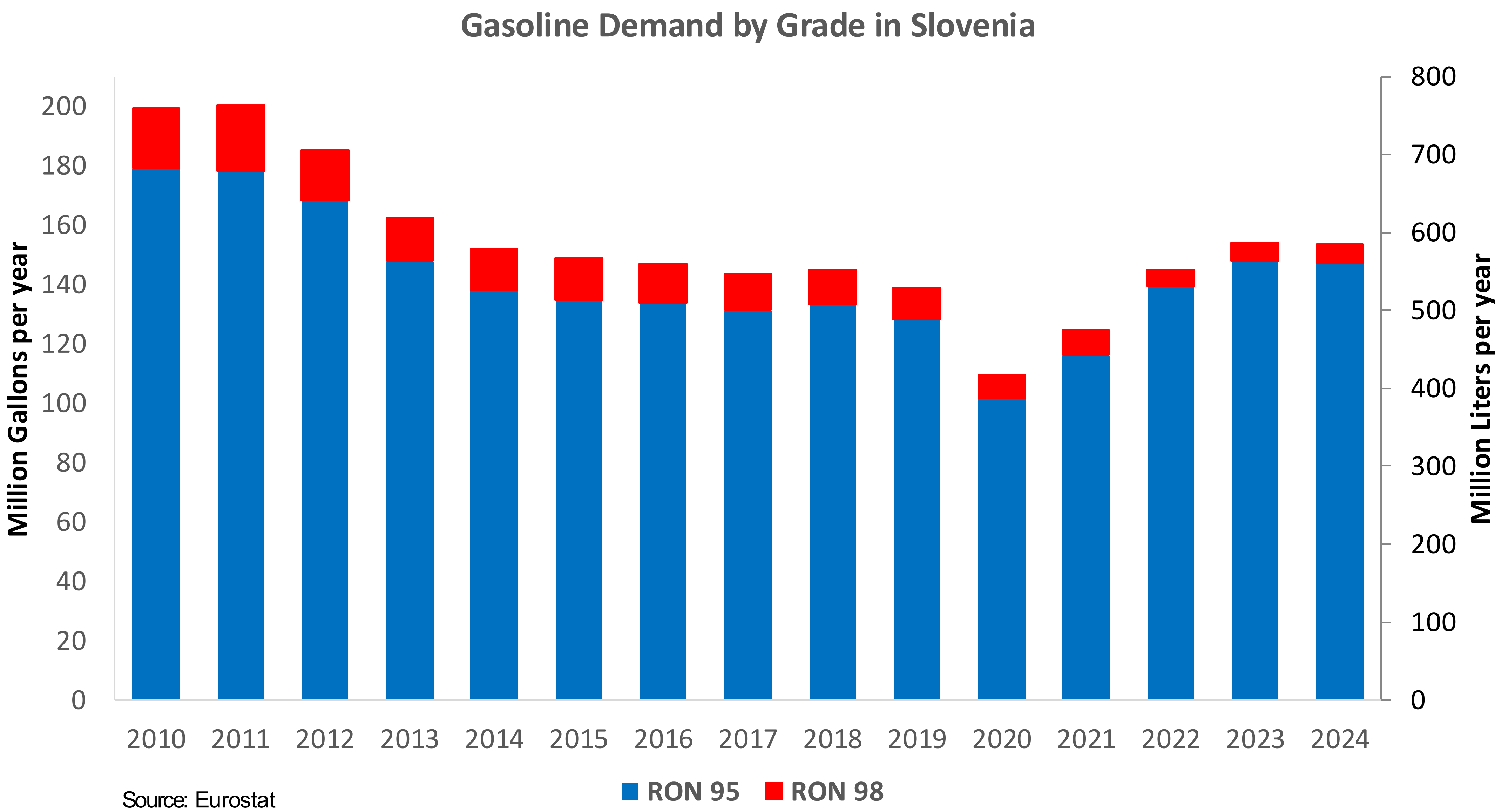
Ethanol Gasoline Blending in Slovenia

In 2024, gasoline consumption in Slovenia was 585 million liters (155 million gallons) and growth rate was 2.0%. Gasoline production and net imports were and 574 million liters (and 152 million gallons) correspondingly. Slovenia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.1 RON.

In 2024, ethanol consumption in Slovenia was 3 million liters (1 million gallons) representing .5% of gasoline demand. Ethanol production and Net Imports were and 3 million liters (1 and 1 million gallons) correspondingly. Ethanol source is Cereals 100%.

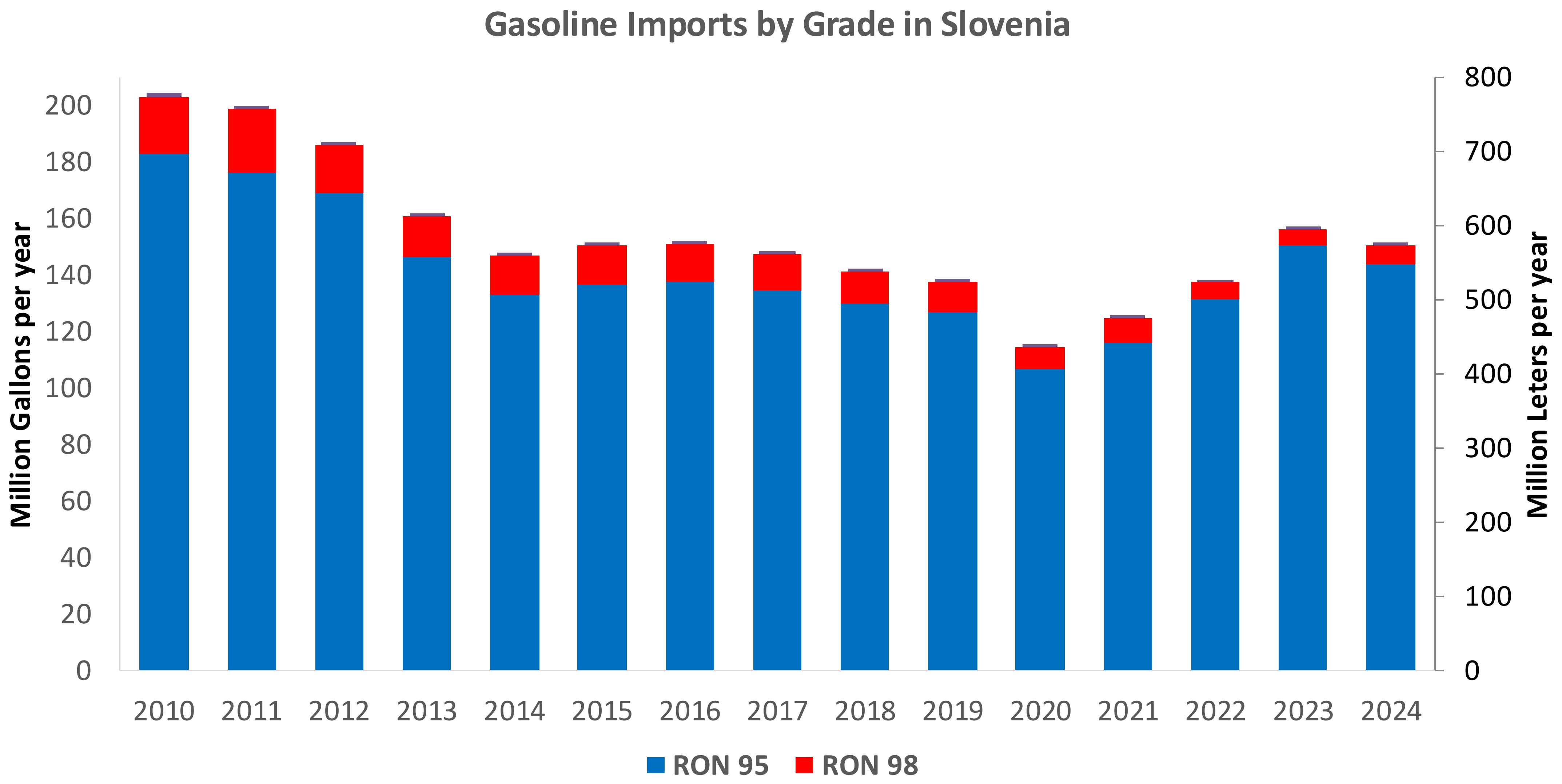
Source: Eurostat, European Commision

Gasoline Demand in Slovenia



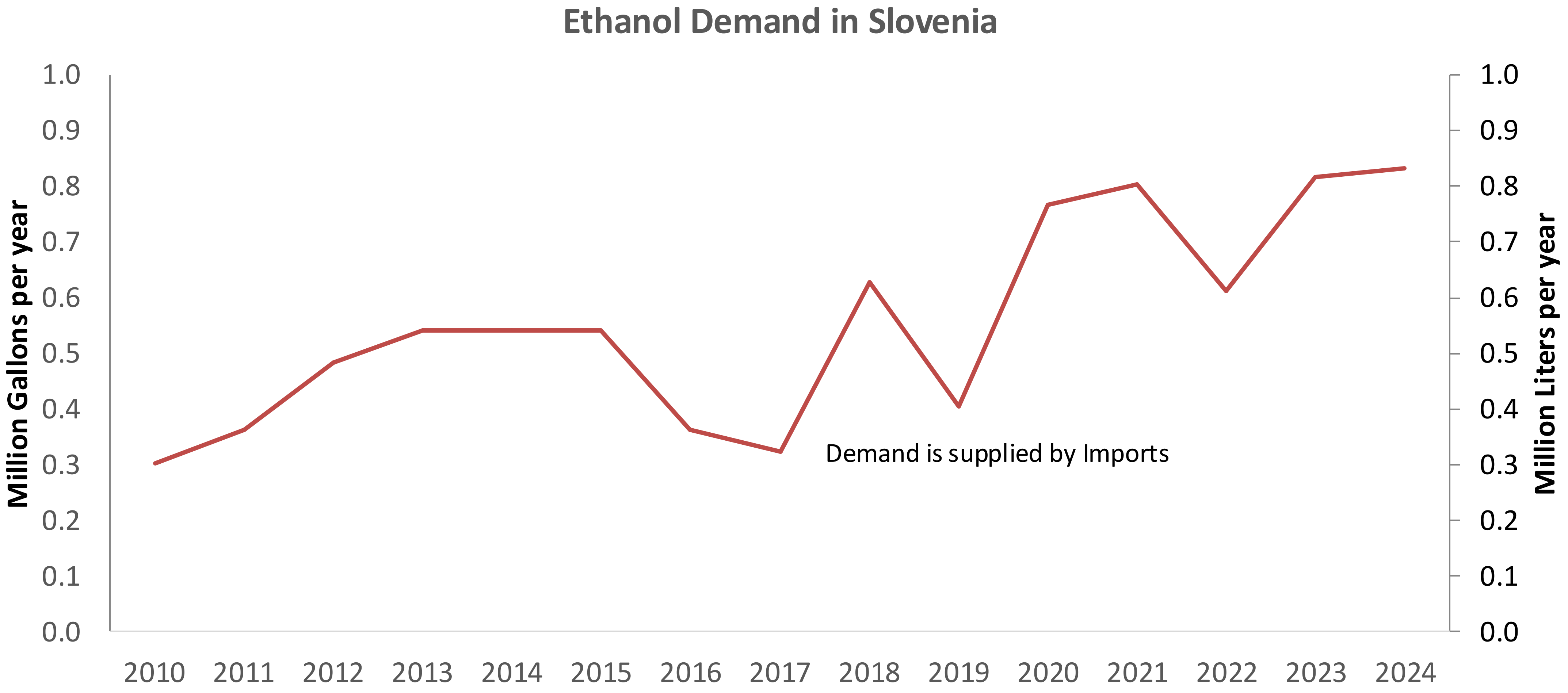
Source: Eurostat

Gasoline Imports in Slovenia



Source: Eurostat

Ethanol Balance in Slovenia



Source: Eurostat

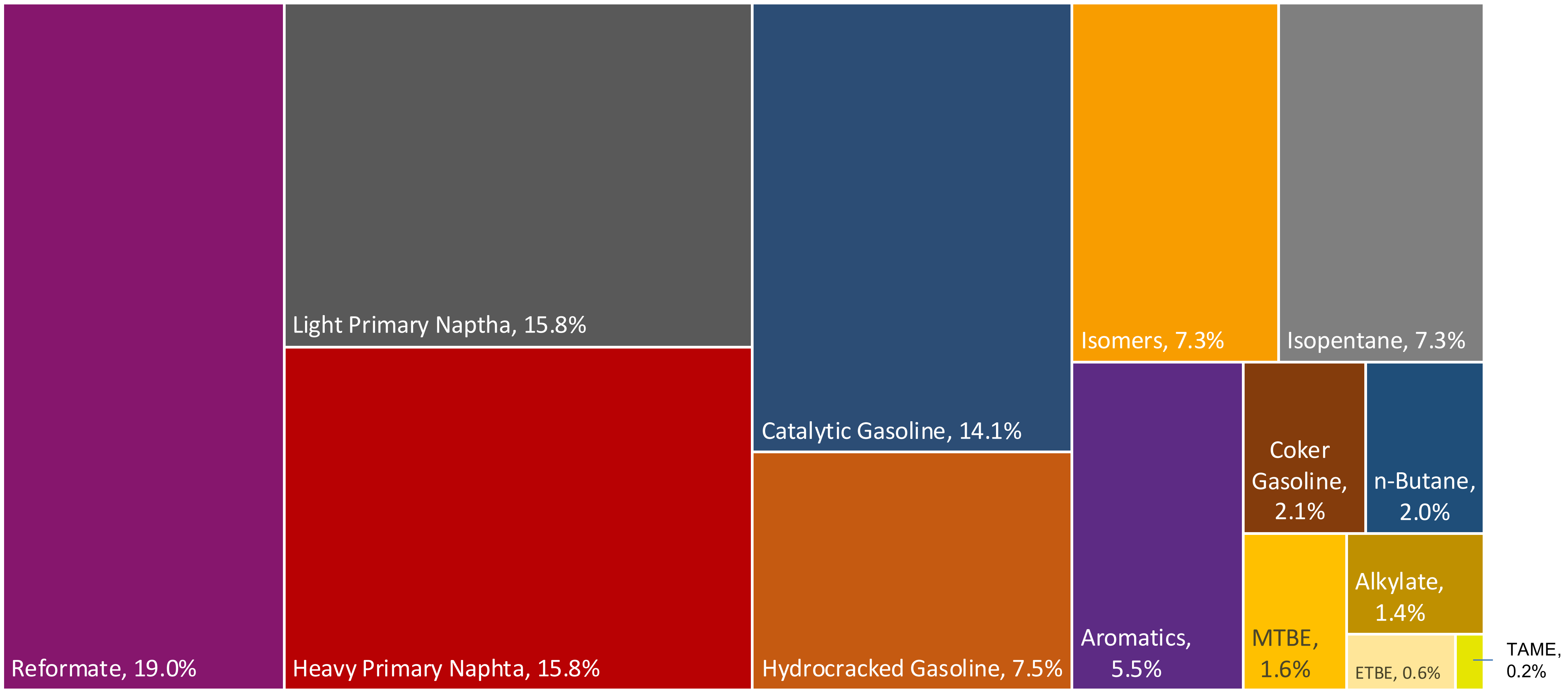
Gasoline Quality in Slovenia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.2							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

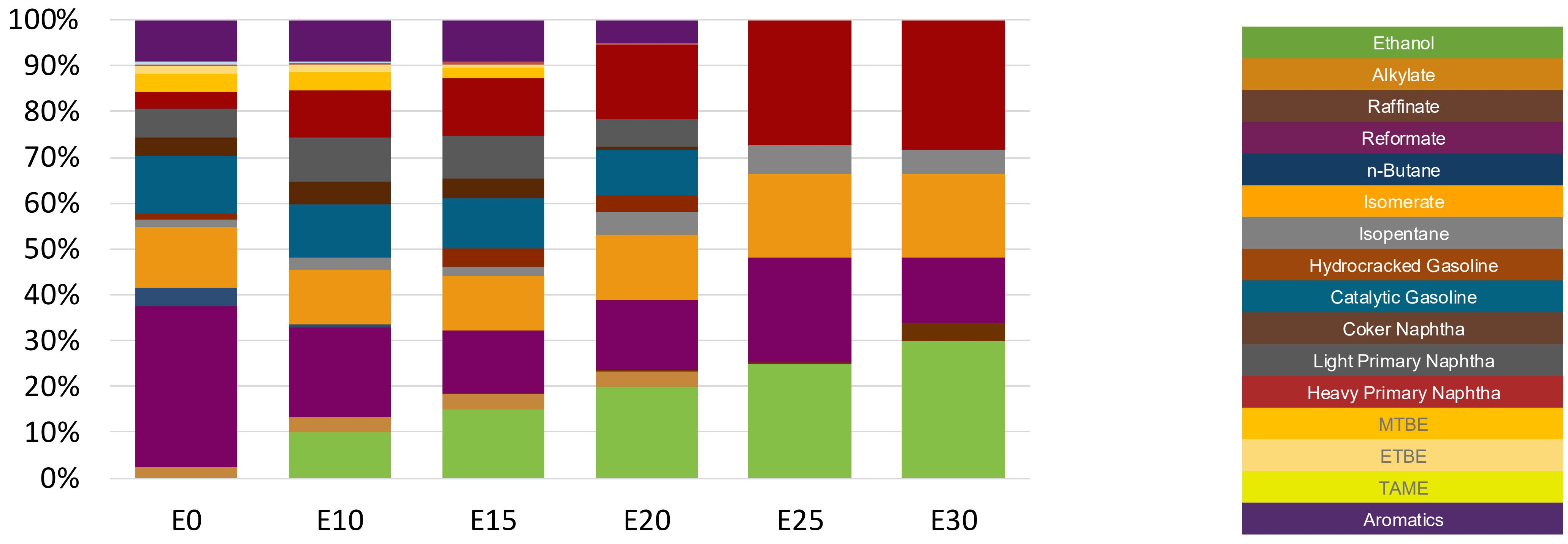
Source: European Commision

Slovenia Available Blending Components



Source: Faro90

Slovenia – Gasoline 95 – Constant Octane

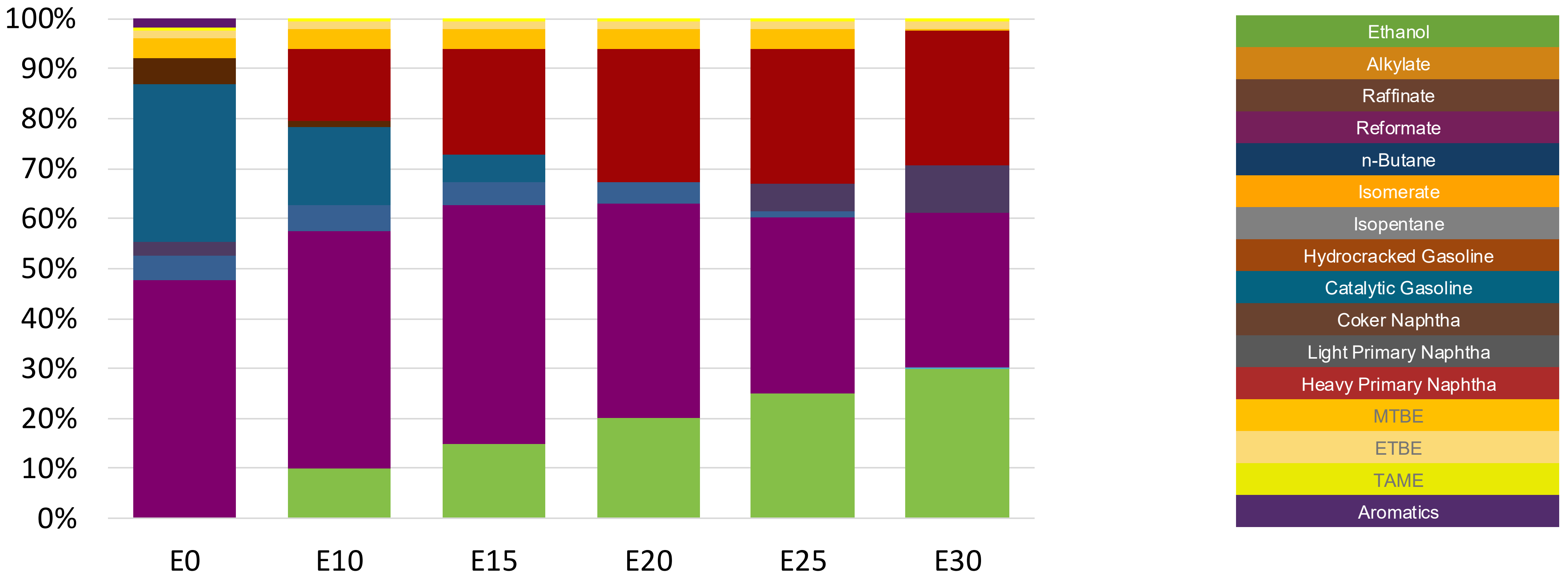


Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US$/gal)	$ 2.292	$ 2.164	$ 2.103	$ 2.022	$ 1.900	$ 1.834

Average CIF 2024 Prices

Source: Faro90

Slovenia - Gasoline 98 - Constant Octane

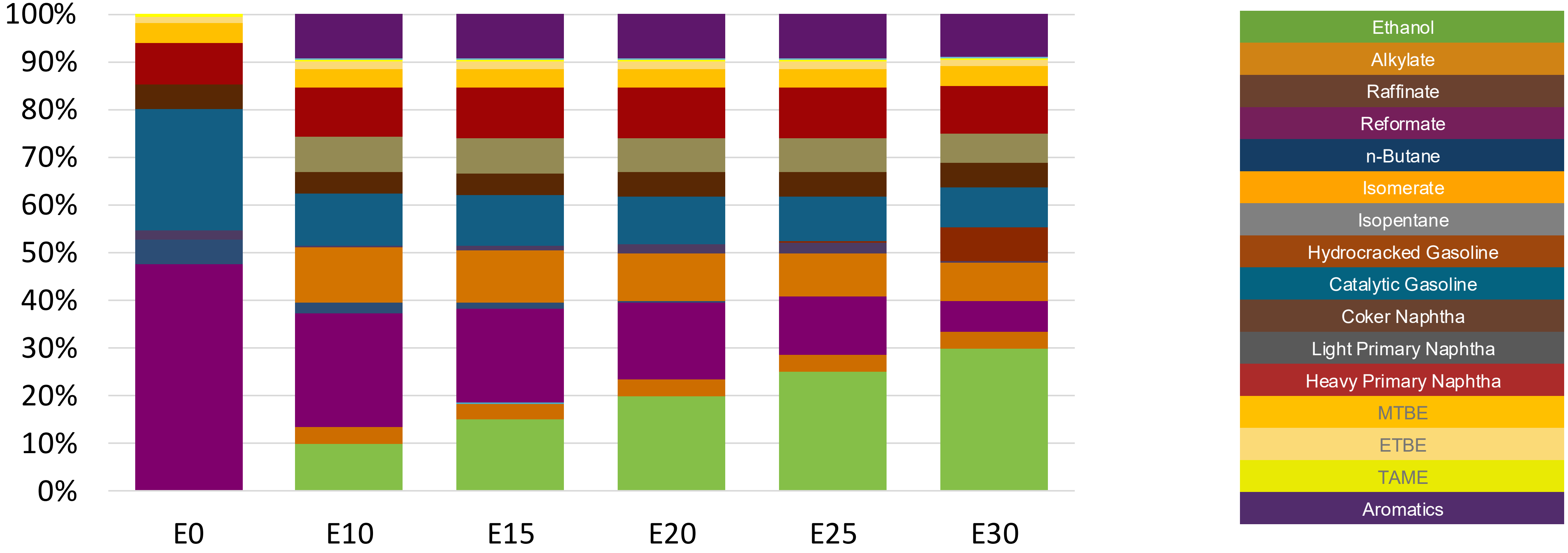


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.367	$ 2.238	$ 2.166	$ 2.089	$ 2.012	$ 1.945

Average CIF 2024 Prices

Source: Faro90

Slovenia – Gasoline 95 – Increased Octane

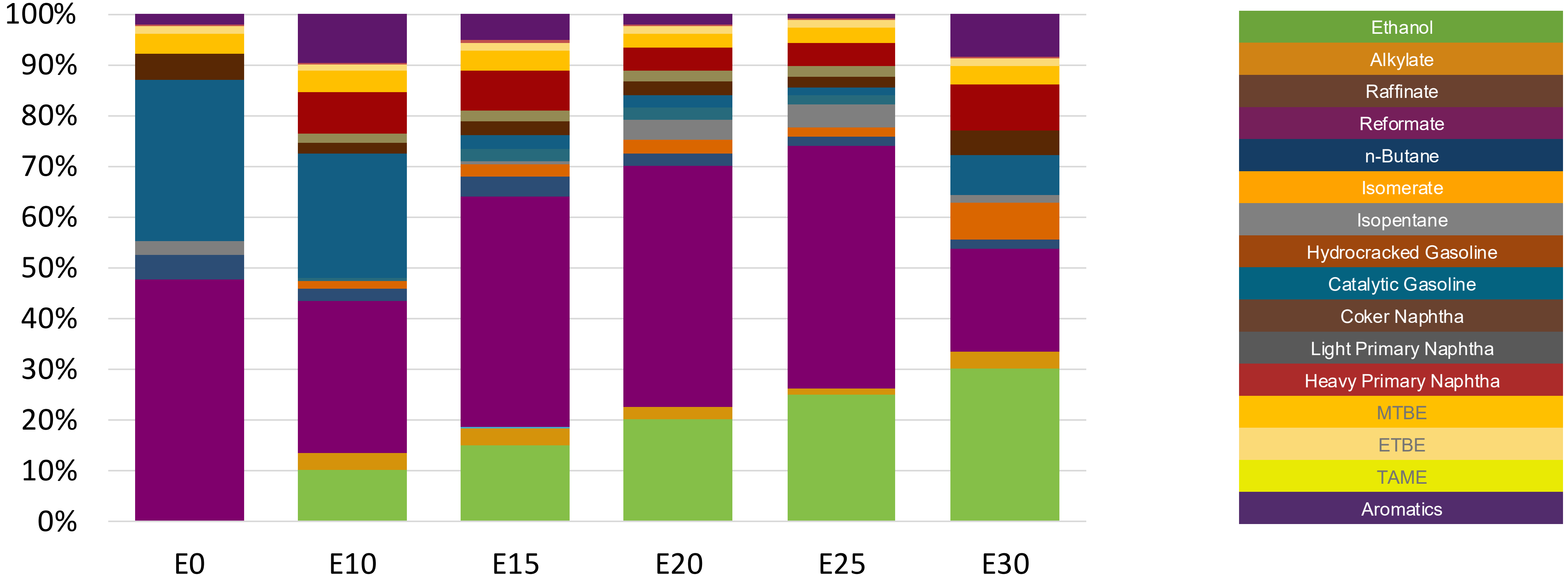


Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US$/gal)	$ 2.229	$ 2.229	$ 2.229	$ 2.229	$ 2.229	$ 2.229

Average CIF 2024 Prices

Source: Faro90

Slovenia – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US$/gal)	$ 2.367	$ 2.367	$ 2.367	$ 2.367	$ 2.367	$ 2.367

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Slovenia

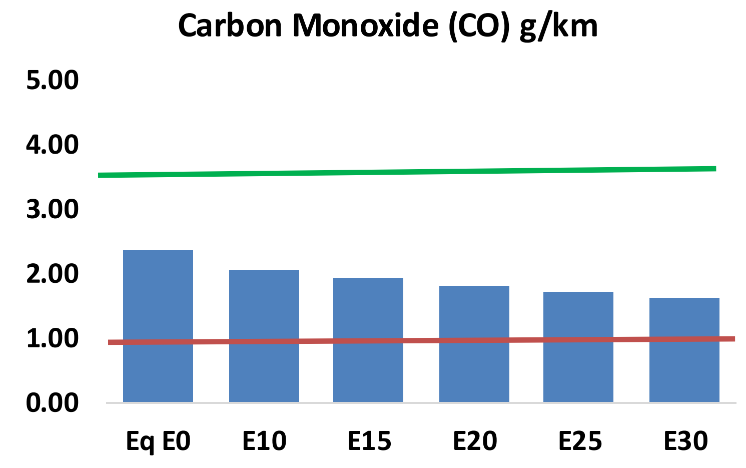
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	97
Auto/Sml Truck	Light	Single FI	3-Ways	PCV Tank	>8 years	6,881
Auto/Sml Truck	Light	Multi FI	3-Ways	PCV Tank	>8 years	166,459
Auto/Sml Truck	Light	Multi FI	Euro III	PCV Tank	>8 years	161,409
Auto/Sml Truck	Light	Multi FI	Euro IV	PCV Tank	>8 years	316,395
Auto/Sml Truck	Light	Multi FI	Euro V	PCV Tank	>8 years	90,173
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	>8 years	208,911
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	4-8 years	271,329
Auto/Sml Truck	Light	Multi FI	Euro 6	PCV Tank	<4 years	1,036,869
Auto/Sml Truck	Light	Multi FI	Hybrid	PCV Tank	All	7,870
Auto/Sml Truck	Light	Electric			All	11,534
Auto/Sml Truck	Medium	Carburator	3-Ways	PCV Tank	>8 years	0
Auto/Sml Truck	Medium	Single FI	3-Ways	PCV Tank	>8 years	1,394
Auto/Sml Truck	Medium	Multi FI	3-Ways	PCV Tank	>8 years	6,675
Auto/Sml Truck	Medium	Multi FI	Euro III	PCV Tank	>8 years	11,944
Auto/Sml Truck	Medium	Multi FI	Euro IV	PCV Tank	>8 years	26,760
Auto/Sml Truck	Medium	Multi FI	Euro V	PCV Tank	>8 years	9,211
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	>8 years	2,717
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	4-8 years	15,915
Auto/Sml Truck	Medium	Multi FI	Euro 6	PCV Tank	<4 years	32,026
Auto/Sml Truck	Medium	Multi FI	Hybrid	PCV Tank	All	28,790
Auto/Sml Truck	Medium	Electric			All	38
Sml Engines	Light	4-Cycle	None	PCV	>8 years	14,421
Sml Engines	Light	4-Cycle	Euro III	PCV	>8 years	8,960
Sml Engines	Light	4-Cycle	Euro IV	PCV	>8 years	23,316
Sml Engines	Light	4-Cycle	Euro V	PCV	>8 years	6,915
Sml Engines	Light	4-Cycle	Euro 6	PCV	>8 years	15,676
Sml Engines	Light	4-Cycle	Euro 6	PCV	4-8 years	36,500
Sml Engines	Light	4-Cycle	Euro 6	PCV	<4 years	144,108
Sml Engines	Light	4-Cycle	Hybrid	PCV	All	0
Sml Engines	Light	Electric			All	0
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	174
Truck	Heavy	Fuel-Injection	3-Ways	PCV	>8 years	133,707
Truck	Heavy	Fuel-Injection	Euro III	PCV	>8 years	11,114
Truck	Heavy	Fuel-Injection	Euro IV	PCV	>8 years	35,475
Truck	Heavy	Fuel-Injection	Euro V	PCV	>8 years	10,440
Truck	Heavy	Fuel-Injection	Euro 6	PCV	>8 years	45,655
Truck	Heavy	Fuel-Injection	Euro 6	PCV	4-8 years	50,135
Truck	Heavy	Fuel-Injection	Euro 6	PCV	<4 years	239,718
Truck	Heavy	Fuel-Injection	Hybrid	PCV	All	214
Truck	Heavy	Electric			All	372

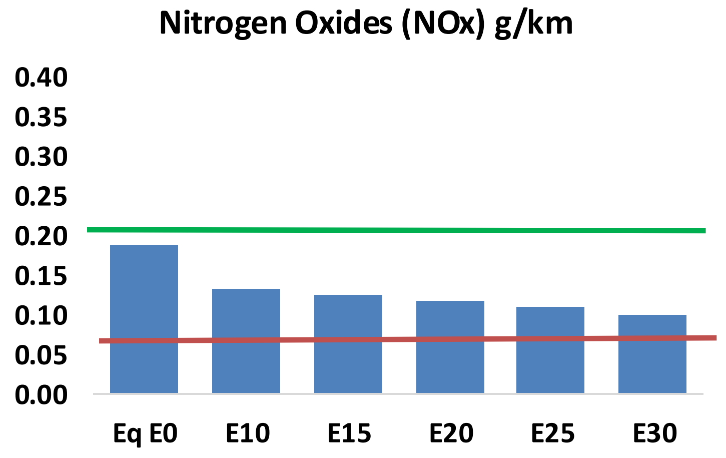
Vehicle Fleet: 3,190,297
Gasoline Vehicle Fleet: 1,163,863

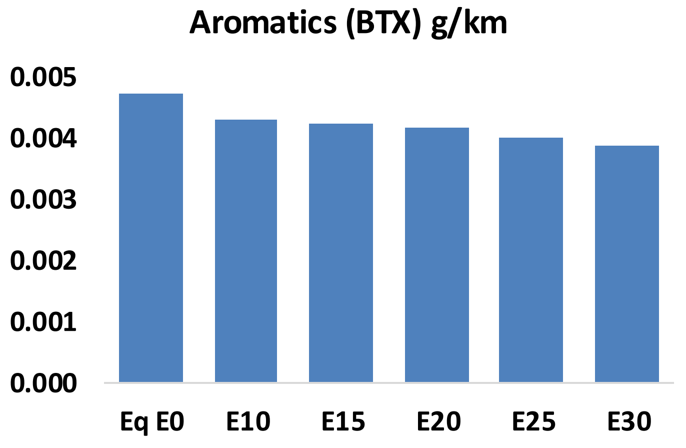
Source: Eurostat, ACEA

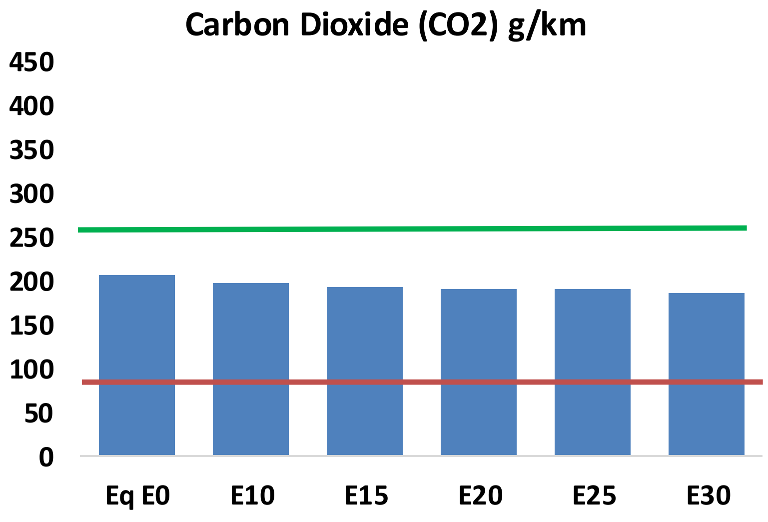
Slovenia – Vehicular Emissions

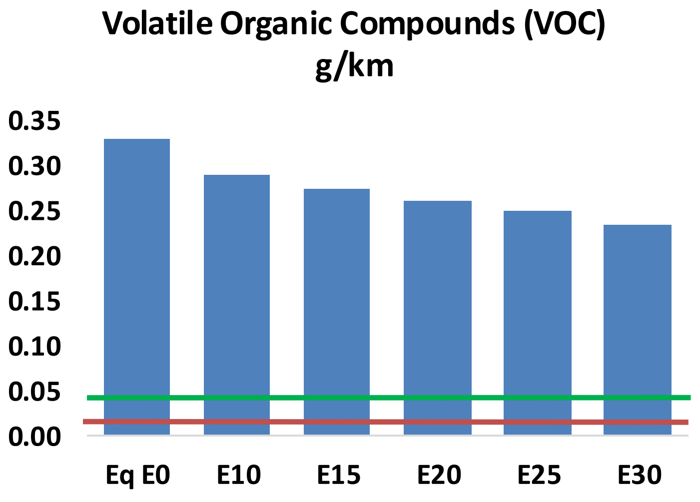
TIER III BIN 125 United States
Euro 6 Standard

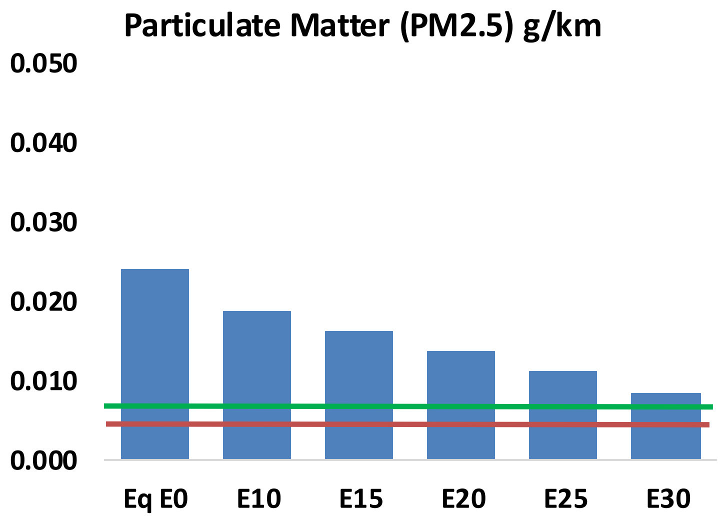












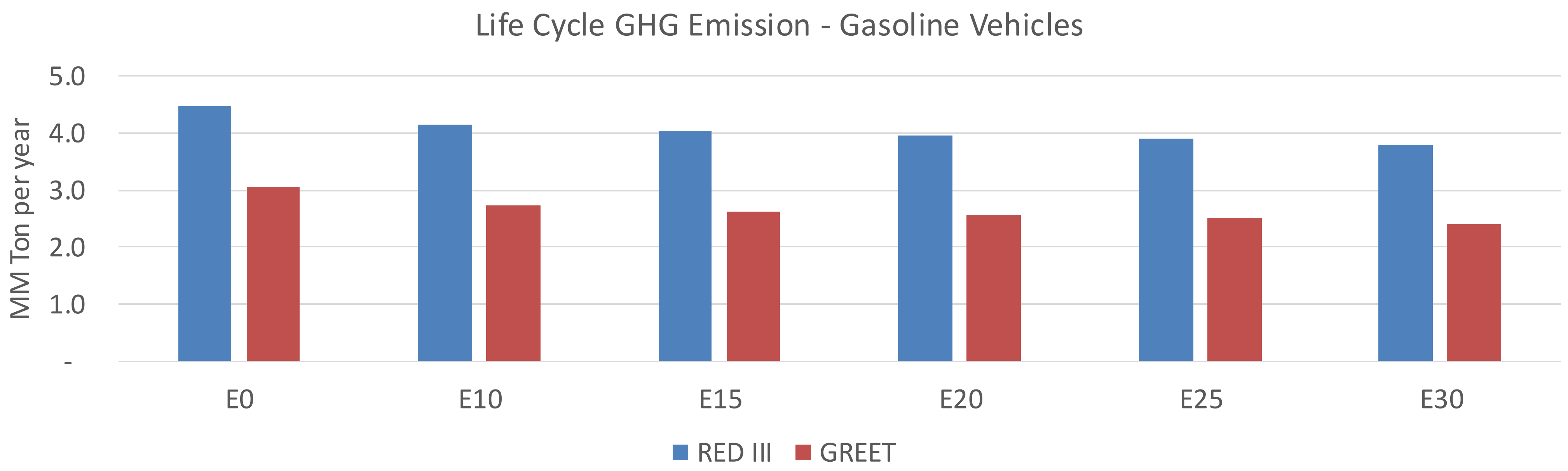
Source: Faro90

Slovenia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	66,460	62,046	57,631	54,148	50,707	48,079	45,138	-13%	-24%
CO2	5,814,866	5,665,340	5,524,122	5,413,058	5,358,100	5,329,401	5,231,165	-5%	-8%
VOC	9,254	8,696	8,137	7,709	7,293	6,976	6,539	-12%	-21%
NOx	5,327	3,996	3,729	3,515	3,315	3,094	2,861	-30%	-38%
SOx	65	60	55	51	47	43	38	-15%	-28%
NH3	2,043	2,023	2,003	2,012	2,035	2,051	2,064	-2%	0%
N2O	61	61	60	61	62	63	63	-1%	2%
CH4	2,030	2,030	2,030	2,061	2,105	2,144	2,180	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	63	61	59	57	57	56	55	-6%	-9%
Acetaldehyde	160	183	269	410	558	638	754	68%	249%
Formaldehyde	545	529	617	725	759	840	914	13%	39%
Benzene	132	127	120	119	118	112	109	-9%	-11%
PM2.5	371	329	288	250	212	172	130	-22%	-43%
PM10	303	269	235	204	173	141	107	-22%	-43%

Source: Faro90

Slovenia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	16.5
Target @2035 (MMT/year)	9.3
Delta	7.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.8%	13.7%	16.0%	18.0%	20.9%
RED II/III	0.0%	7.7%	9.9%	11.6%	13.1%	15.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.6%	5.8%	6.7%	7.6%	8.9%
RED II/III	0.0%	4.8%	6.2%	7.2%	8.1%	9.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

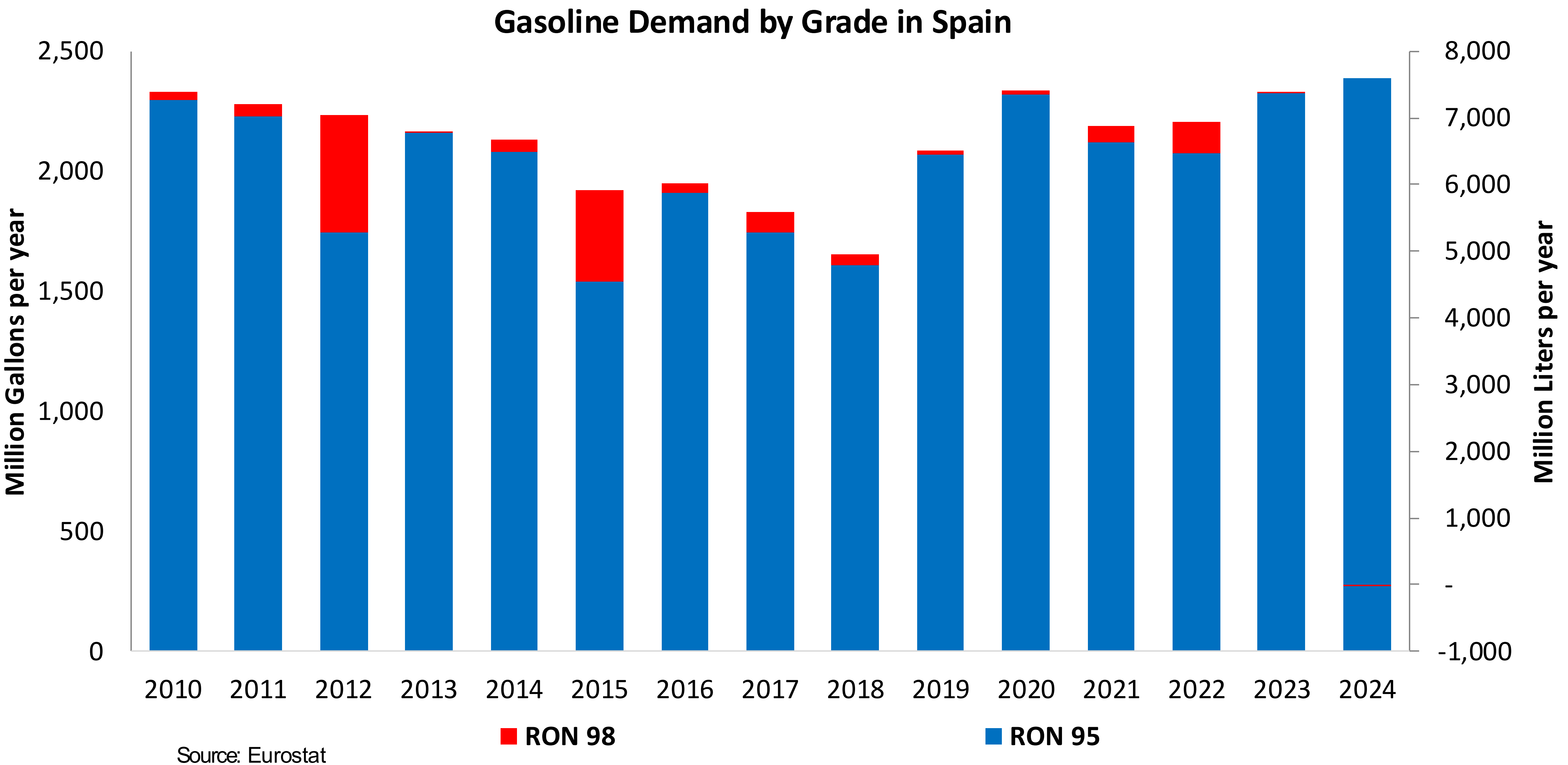
Ethanol Gasoline Blending in Spain

In 2024, gasoline consumption in Spain was 7,577 million liters (2,002 million gallons) and growth rate was 3.1%. Gasoline production and net imports were 11,890 and -3,071 million liters (3,141 and -811 million gallons) correspondingly. Spain complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.0 RON.

In 2024, ethanol consumption in Spain was 253 million liters (67 million gallons) representing 3.3% of gasoline demand. Ethanol production and Net Imports were 545 and -292 million liters (67 and -77 million gallons) correspondingly. Ethanol sources are Corn 60%, Cereals 25% and Sugar Beet/Wine 40%.

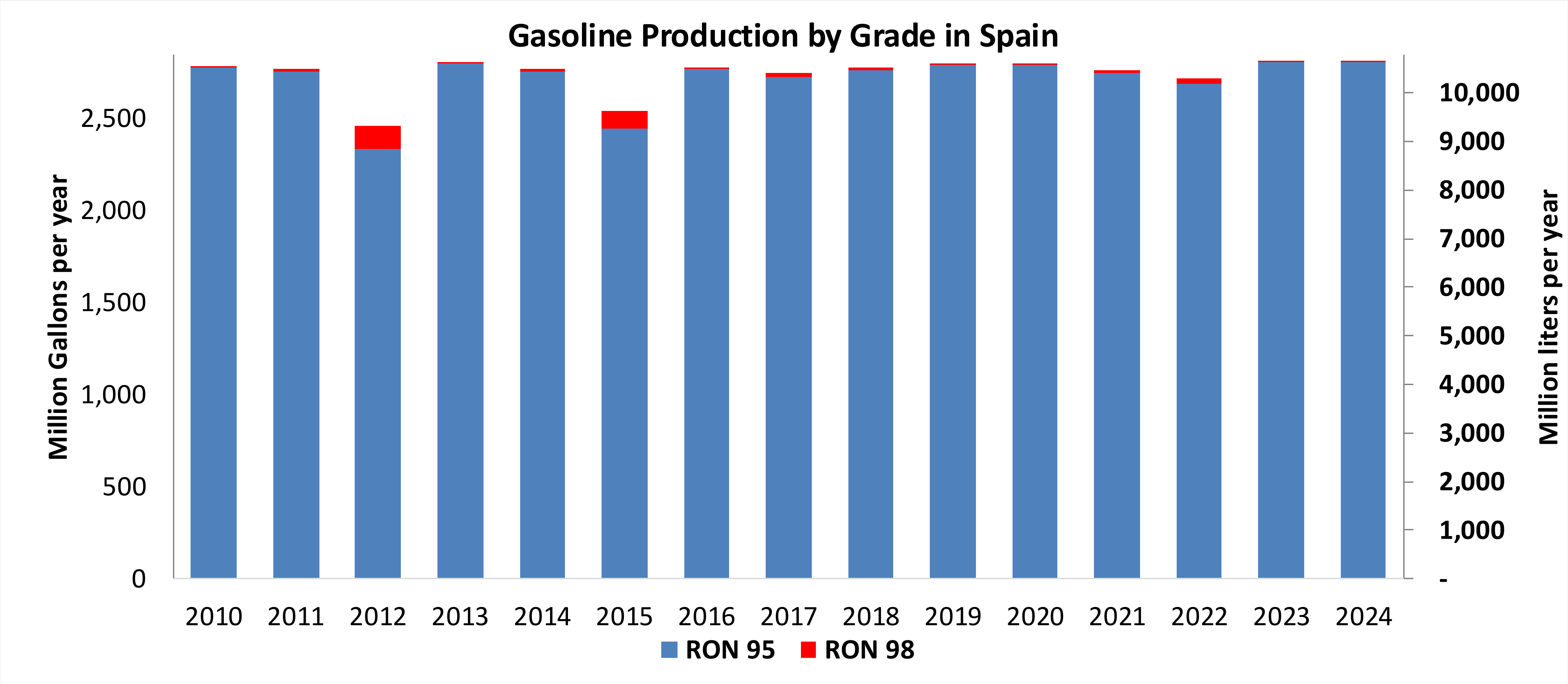
Source: Eurostat, European Commision

Gasoline Demand in Spain



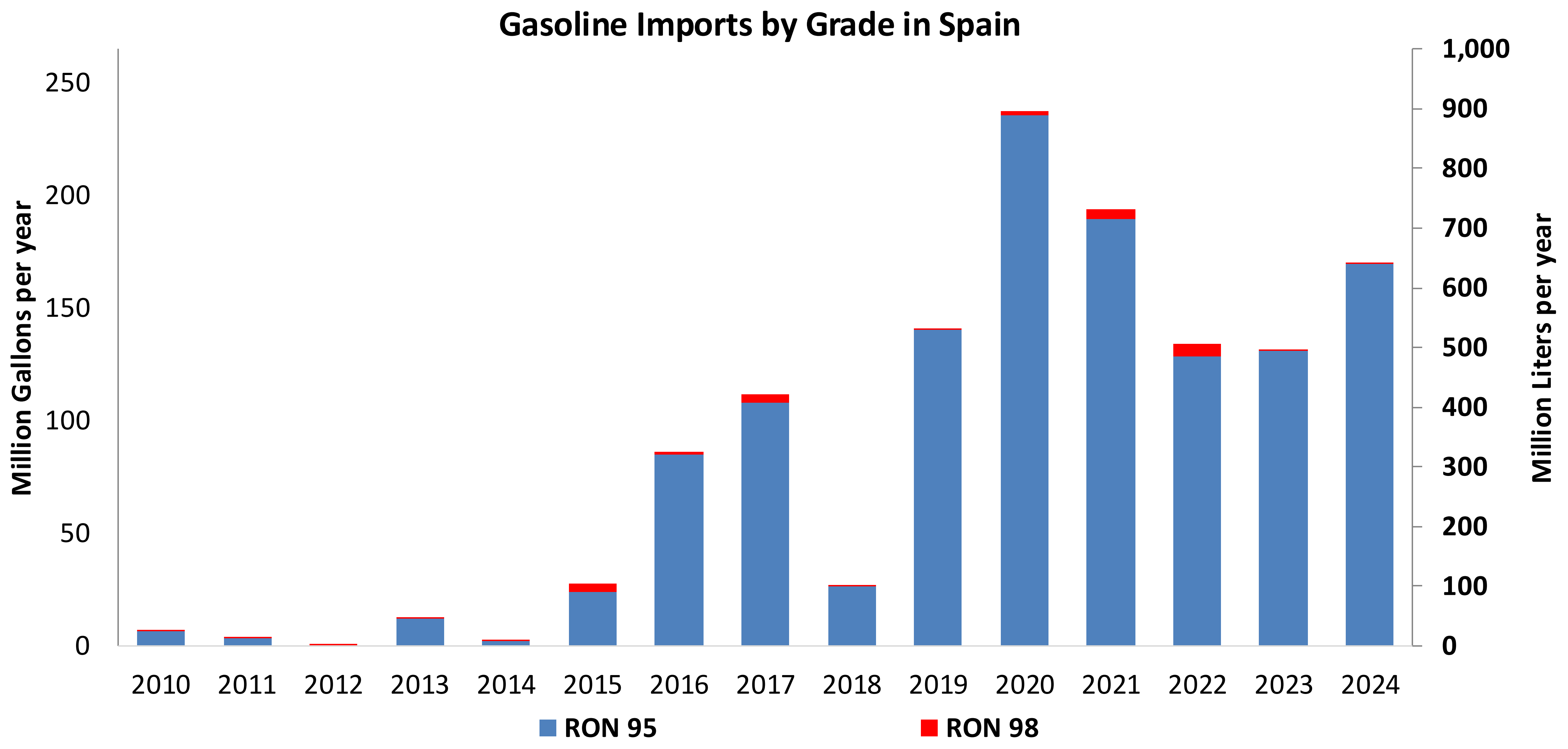
Source: Eurostat

Gasoline Production in Spain



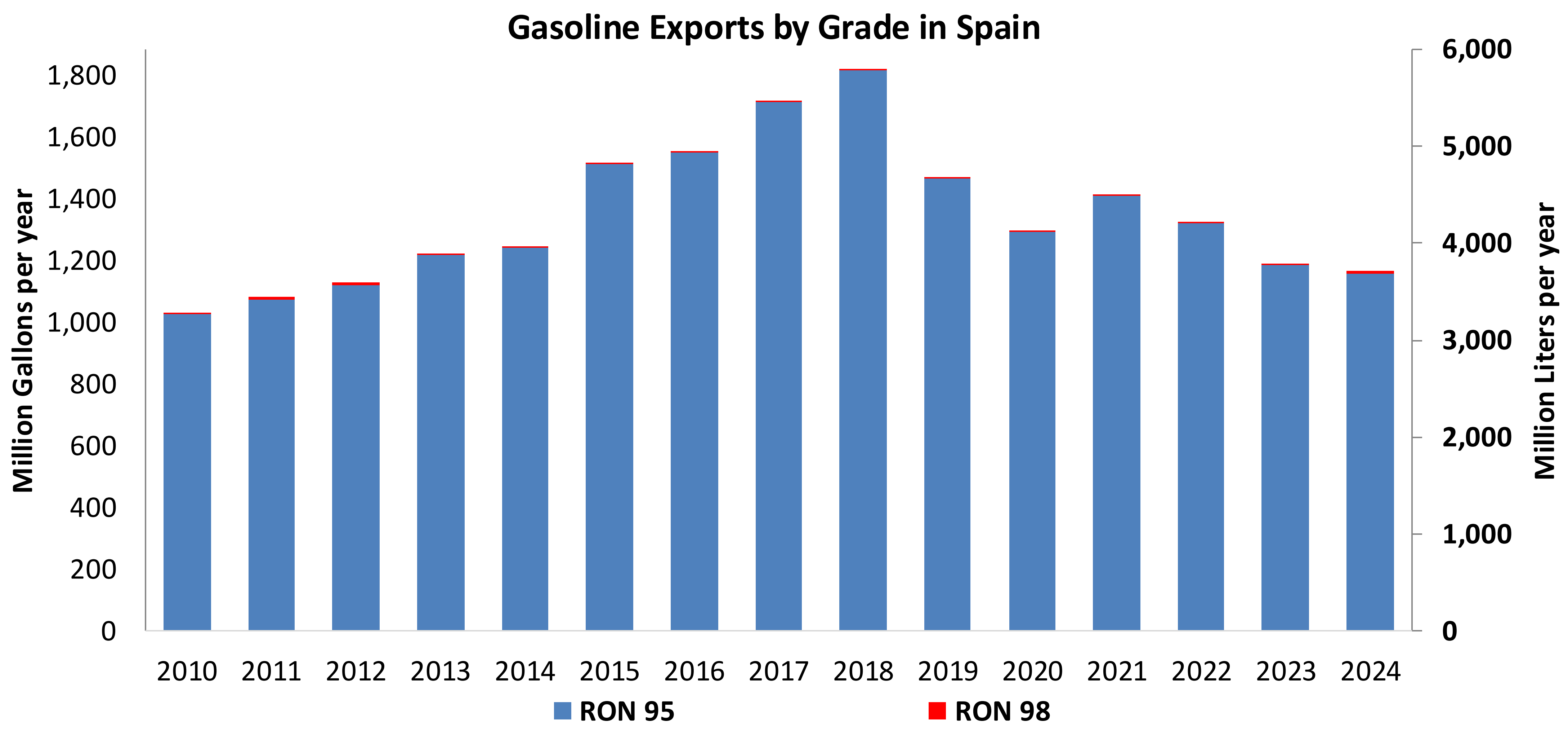
Source: Eurostat

Gasoline Imports in Spain



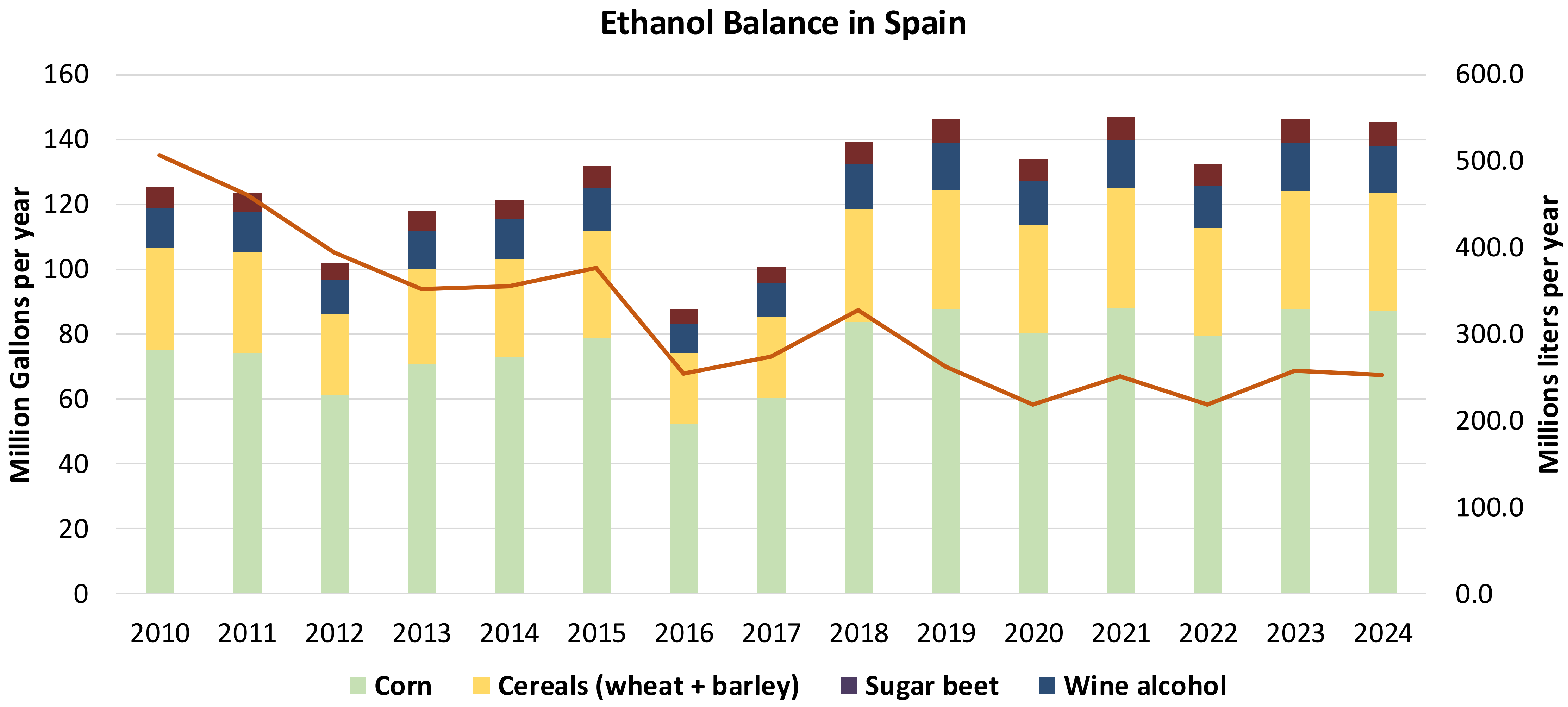
Source: Eurostat

Gasoline Exports in Spain



Source: Eurostat

Ethanol Balance in Spain



Source: Eurostat

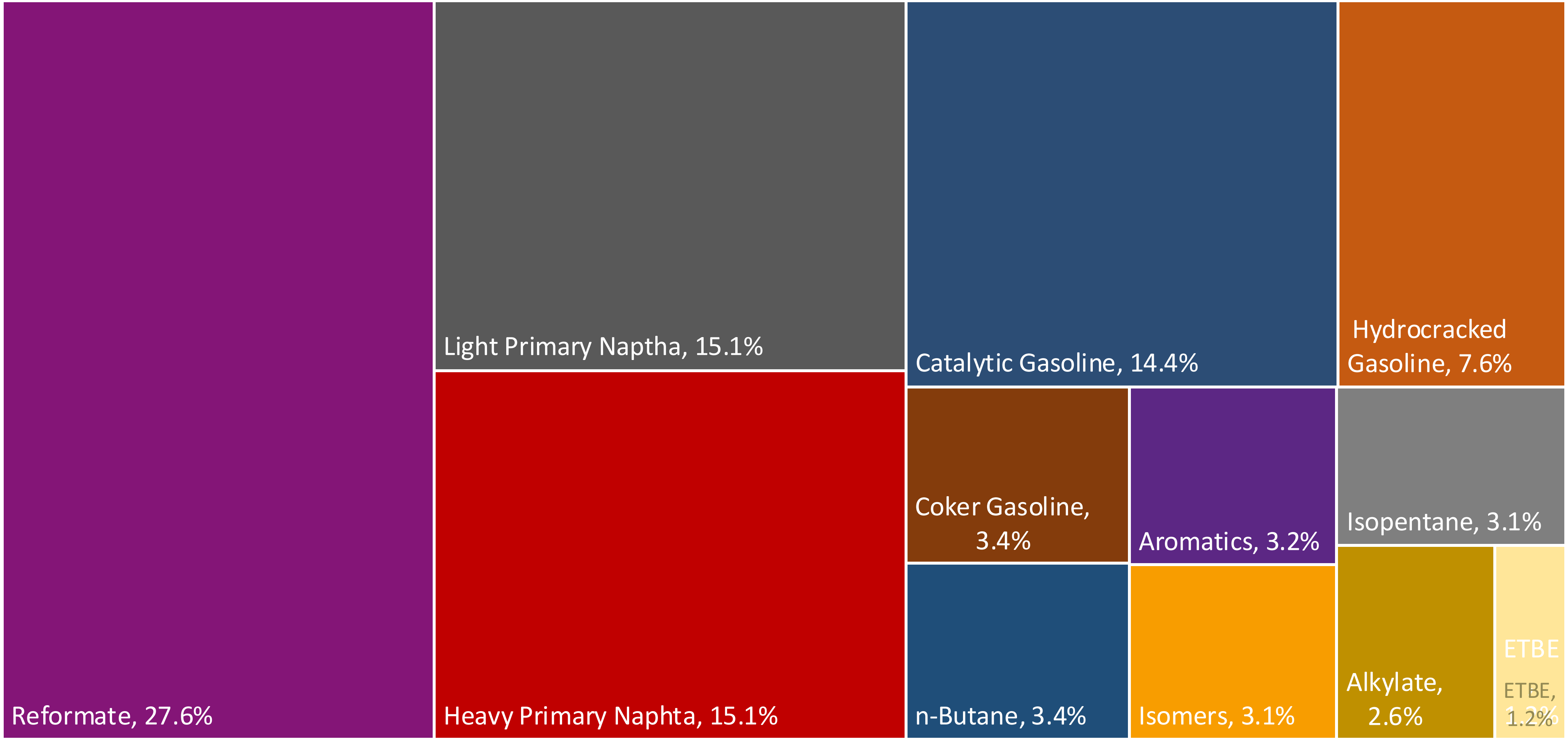
Gasoline Quality in Spain

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.5							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

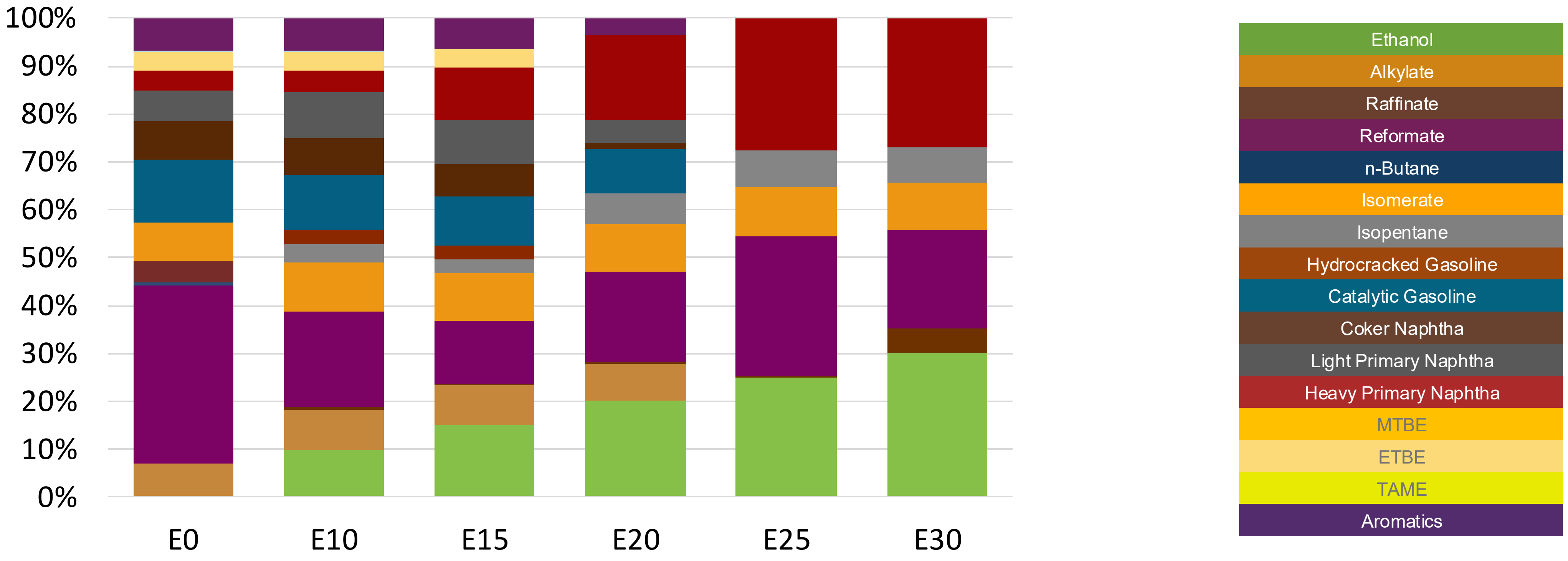
Source: European Commision

Spain Available Blending Components



Source: Faro90

Spain – Gasoline 95 – Constant Octane

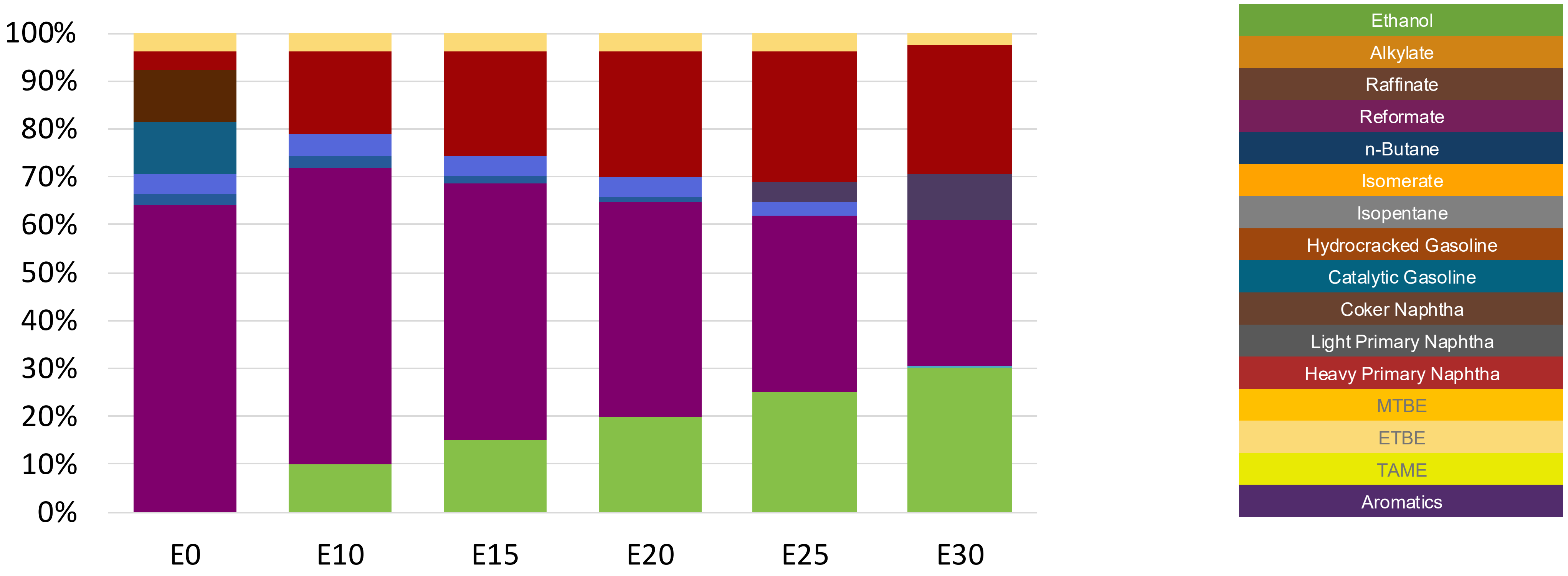


Octane (RON)	95.0	95.0	95.0	95.0	95.3	95.8
Price (US$/gal)	$ 2.280	$ 2.168	$ 2.088	$ 1.998	$ 1.883	$ 1.820

Average CIF 2024 Prices

Source: Faro90

Spain - Gasoline 98 - Constant Octane

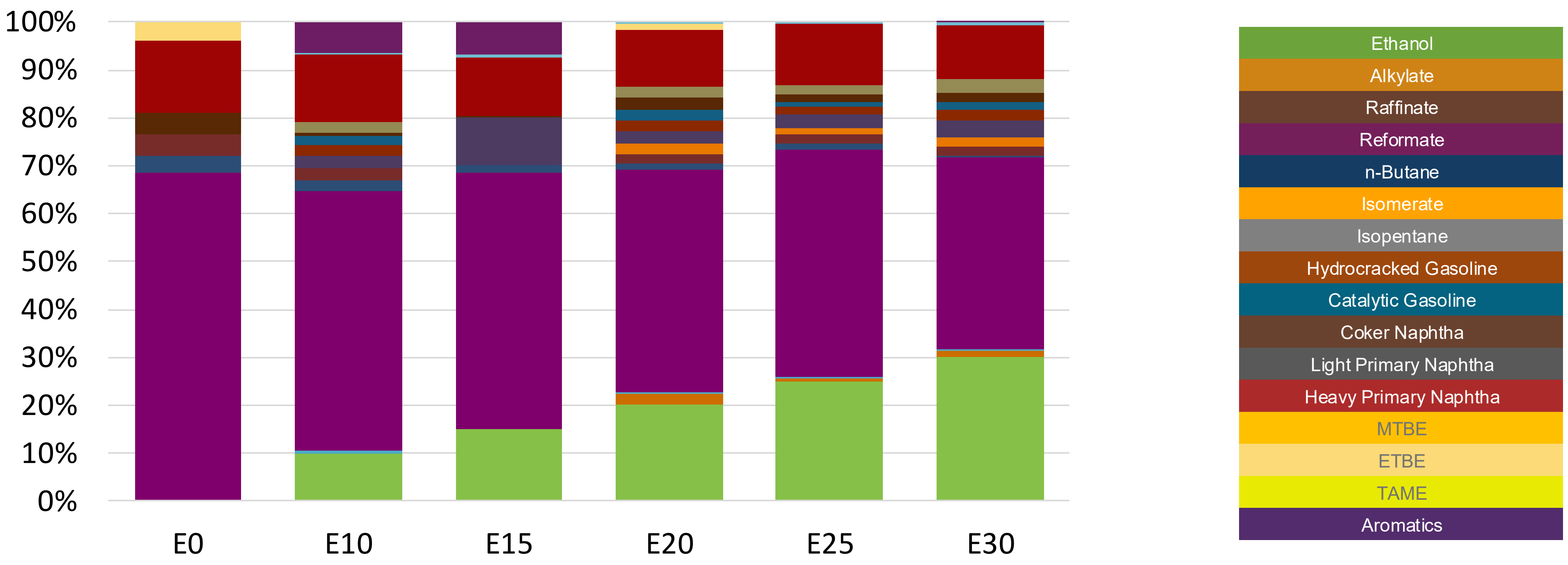


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.349	$ 2.225	$ 2.156	$ 2.081	$ 2.003	$ 1.924

Average CIF 2024 Prices

Source: Faro90

Spain – Gasoline 95 – Increased Octane

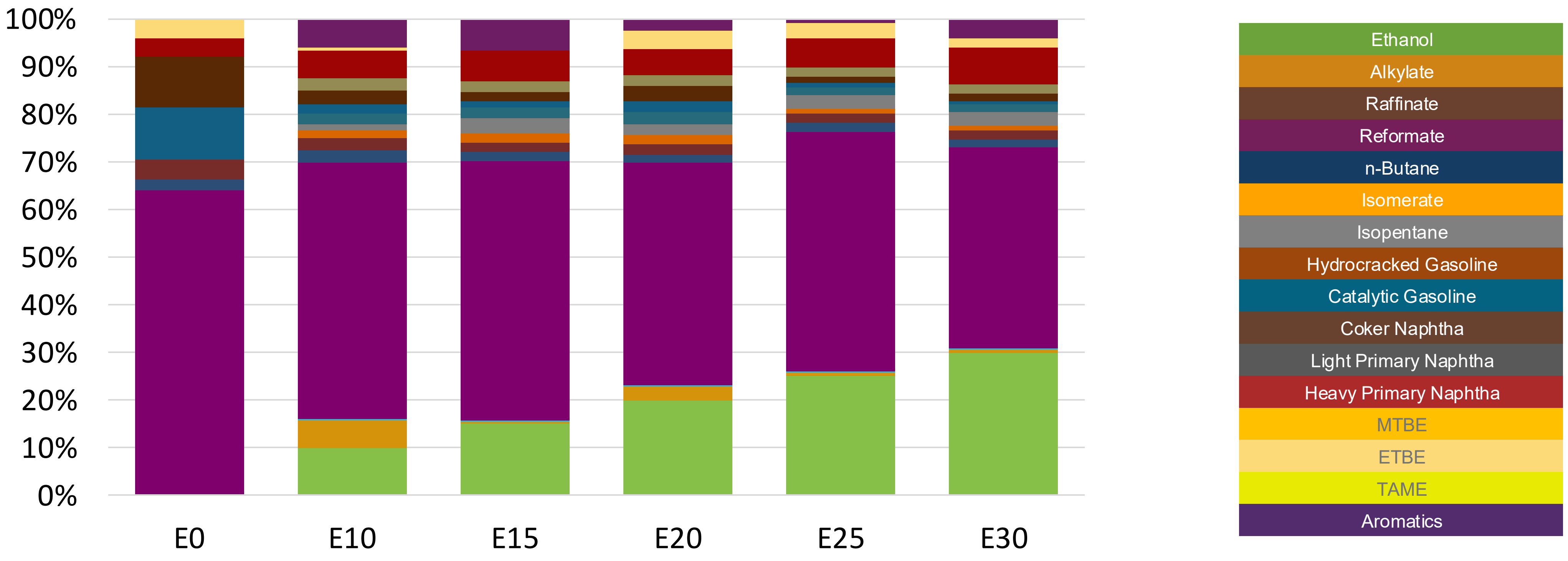


Octane (RON)	95.0	96.8	98.2	99.9	101.4	102.6
Price (US$/gal)	$ 2.212	$ 2.212	$ 2.212	$ 2.212	$ 2.212	$ 2.212

Average CIF 2024 Prices

Source: Faro90

Spain – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.3	100.9	102.5	104.0	105.0
Price (US$/gal)	$ 2.349	$ 2.349	$ 2.349	$ 2.349	$ 2.349	$ 2.349

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Spain

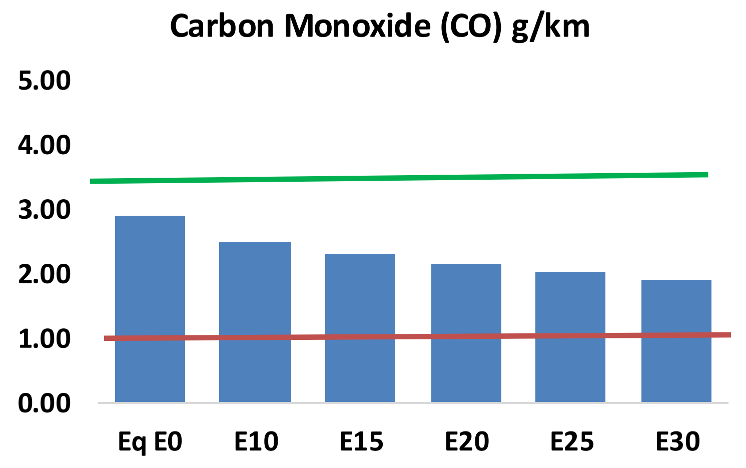
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	9,967
		Single FI	3-Ways		>8 years	184,079
		Multi FI	3-Ways		>8 years	1,986,509
			Euro III		>8 years	2,997,881
			Euro IV		>8 years	4,130,395
			Euro V		>8 years	1,048,814
			Euro 6		>8 years	2,531,411
					4-8 years	4,081,695
					<4 years	2,183,857
			Hybrid		All	771,450
		Electric			All	119,763
	Medium	Carburator	3-Ways	PCV Tank	>8 years	986
		Single FI	3-Ways		>8 years	5,454
		Multi FI	3-Ways		>8 years	291,511
			Euro III		>8 years	561,099
			Euro IV		>8 years	1,431,712
			Euro V		>8 years	321,320
			Euro 6		>8 years	843,804
					4-8 years	1,857,131
					<4 years	1,768,939
			Hybrid		All	538,150
		Electric			All	88,922
Sml Engines	Light	4-Cycle	None	PCV	>8 years	24,824
			Euro III		>8 years	80,432
			Euro IV		>8 years	603,608
			Euro V		>8 years	174,836
			Euro 6		>8 years	450,562
					4-8 years	623,207
					<4 years	754,983
			Hybrid		All	135,392
		Electric			All	21,575
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	-
		Fuel-Injection	3-Ways		>8 years	749,356
			Euro III		>8 years	951,622
			Euro IV		>8 years	1,363,585
			Euro V		>8 years	240,214
			Euro 6		>8 years	602,646
					4-8 years	792,932
					<4 years	632,043
			Hybrid		All	1,972
		Electric			All	141,946

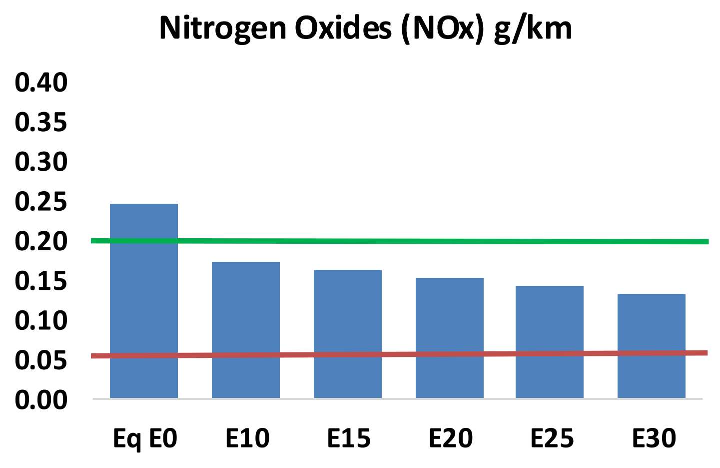
Vehicle Fleet: 36,100,582
Gasoline Vehicle Fleet: 11,736,365

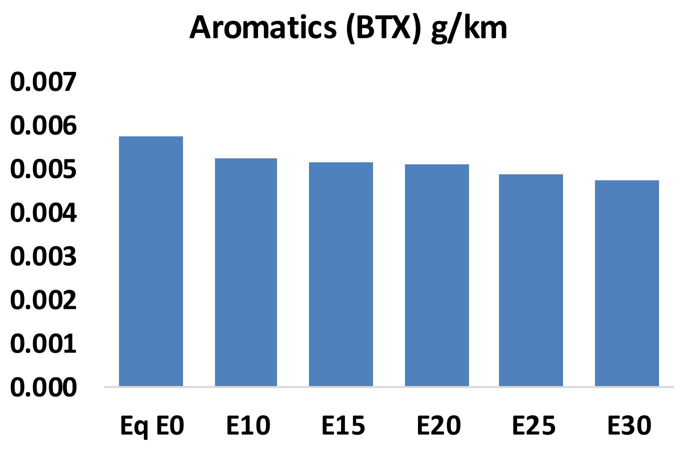
Source: Eurostat, ACEA

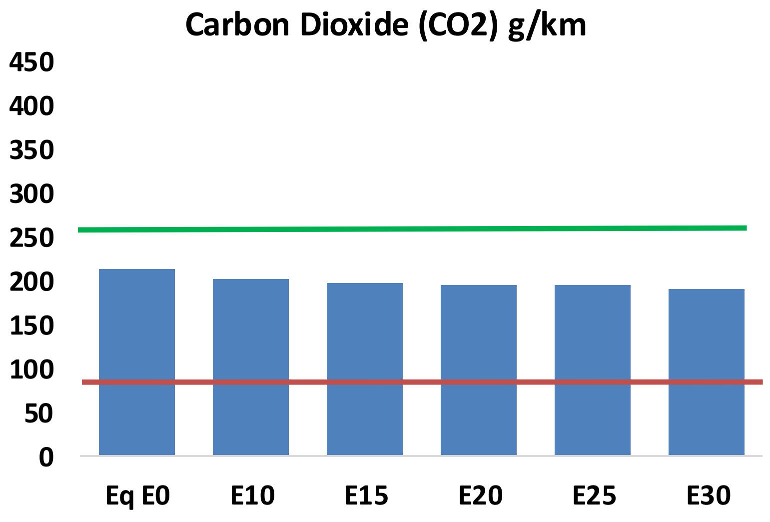
Spain – Vehicular Emissions

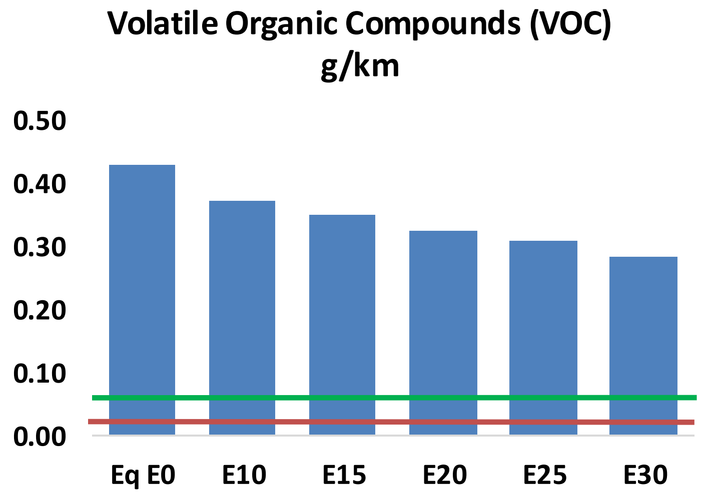
TIER III BIN 125 United States
Euro 6 Standard

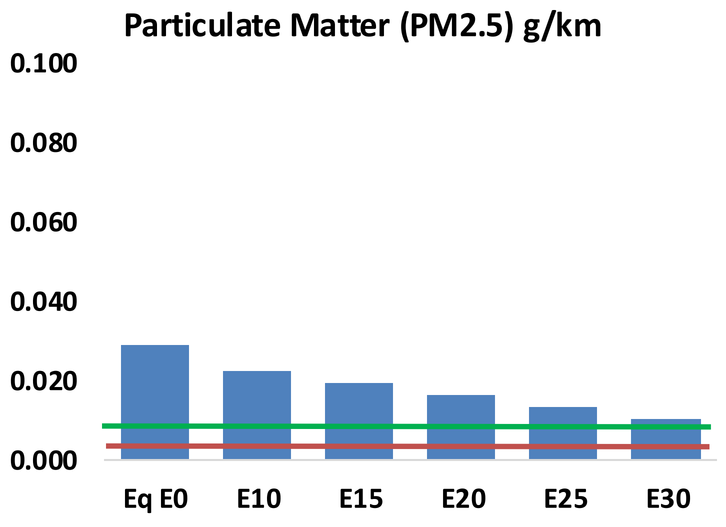












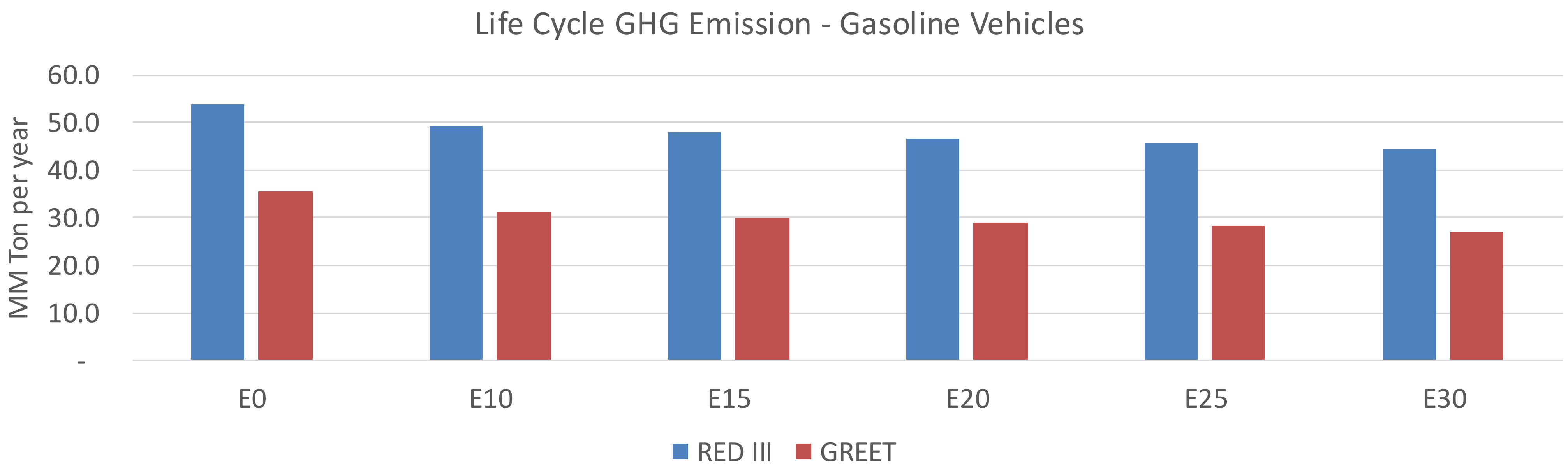
Source: Faro90

Spain – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	861,045	800,129	739,214	690,327	641,604	604,282	562,670	-14%	-25%
CO2	62,968,337	61,349,152	59,819,921	58,617,225	58,022,085	57,682,674	56,619,416	-5%	-8%
VOC	127,636	118,955	110,275	103,383	96,548	91,324	84,243	-14%	-24%
NOx	73,264	54,948	51,285	48,336	45,585	42,552	39,343	-30%	-38%
SOx	751	694	638	590	543	494	441	-15%	-28%
NH3	22,063	21,845	21,626	21,729	21,974	22,145	22,288	-2%	0%
N2O	1,015	1,010	1,006	1,011	1,031	1,043	1,055	-1%	2%
CH4	27,368	27,368	27,368	27,778	28,381	28,901	29,393	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	838	810	782	767	755	747	732	-7%	-10%
Acetaldehyde	2,231	2,561	3,757	5,722	7,796	8,908	10,526	68%	249%
Formaldehyde	7,772	7,556	8,808	10,342	10,835	11,987	13,049	13%	39%
Benzene	1,710	1,643	1,557	1,532	1,519	1,453	1,406	-9%	-11%
PM2.5	3,893	3,457	3,020	2,623	2,227	1,807	1,370	-22%	-43%
PM10	4,624	4,106	3,588	3,116	2,646	2,147	1,627	-22%	-43%

Source: Faro90

Spain – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	269.9
Target @2035 (MMT/year)	151.9
Delta	118.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.1%	15.3%	17.9%	20.3%	23.6%
RED II/III	0.0%	8.8%	11.3%	13.3%	15.3%	17.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.6%	4.6%	5.4%	6.1%	7.1%
RED II/III	0.0%	4.0%	5.2%	6.1%	7.0%	8.1%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

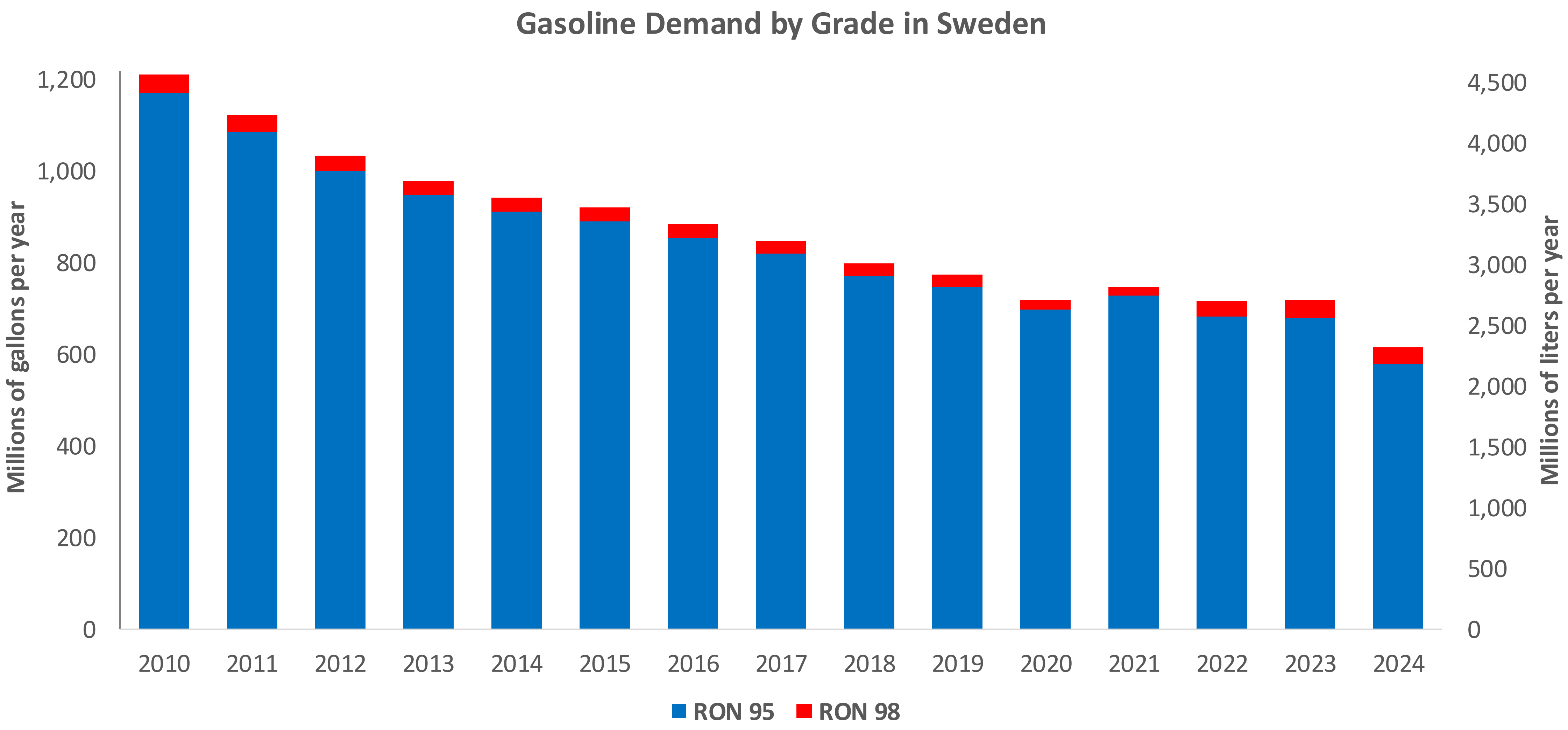
Ethanol Gasoline Blending in Sweden

In 2024, gasoline consumption in Sweden was 2,316 million liters (612 million gallons) and growth rate was -4.5%. Gasoline production and net imports were 6,510 and -4,194 million liters (1,720 and -1,108 million gallons) correspondingly. Sweden complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Sweden was 178 million liters (47 million gallons) representing 7.7% of gasoline demand. Ethanol production and Net Imports were 133 and 45 million liters (47 and 12 million gallons) correspondingly. Ethanol sources are Wood 25% and Cereals 75%.

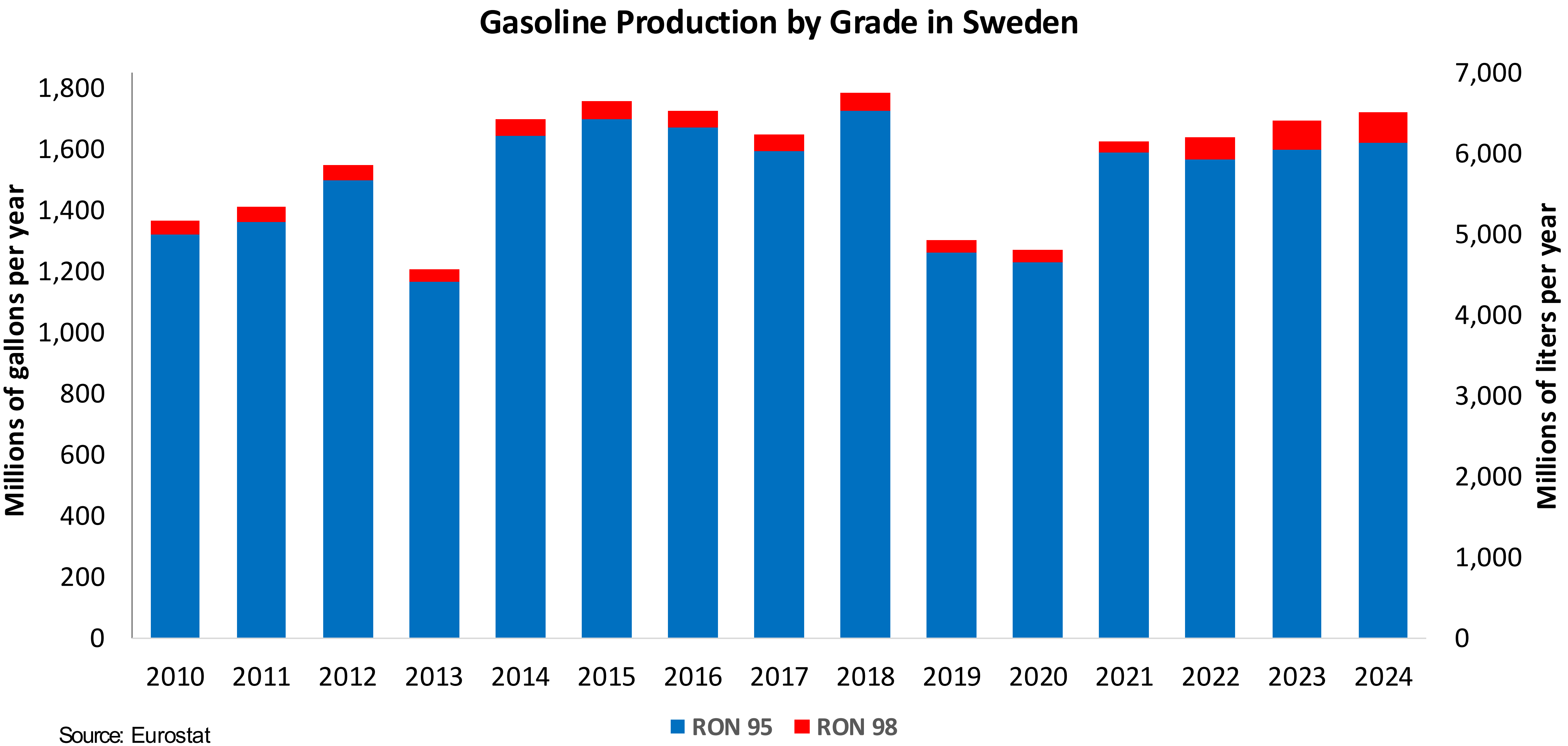
Source: Eurostat, European Commsion

Gasoline Demand in Sweden



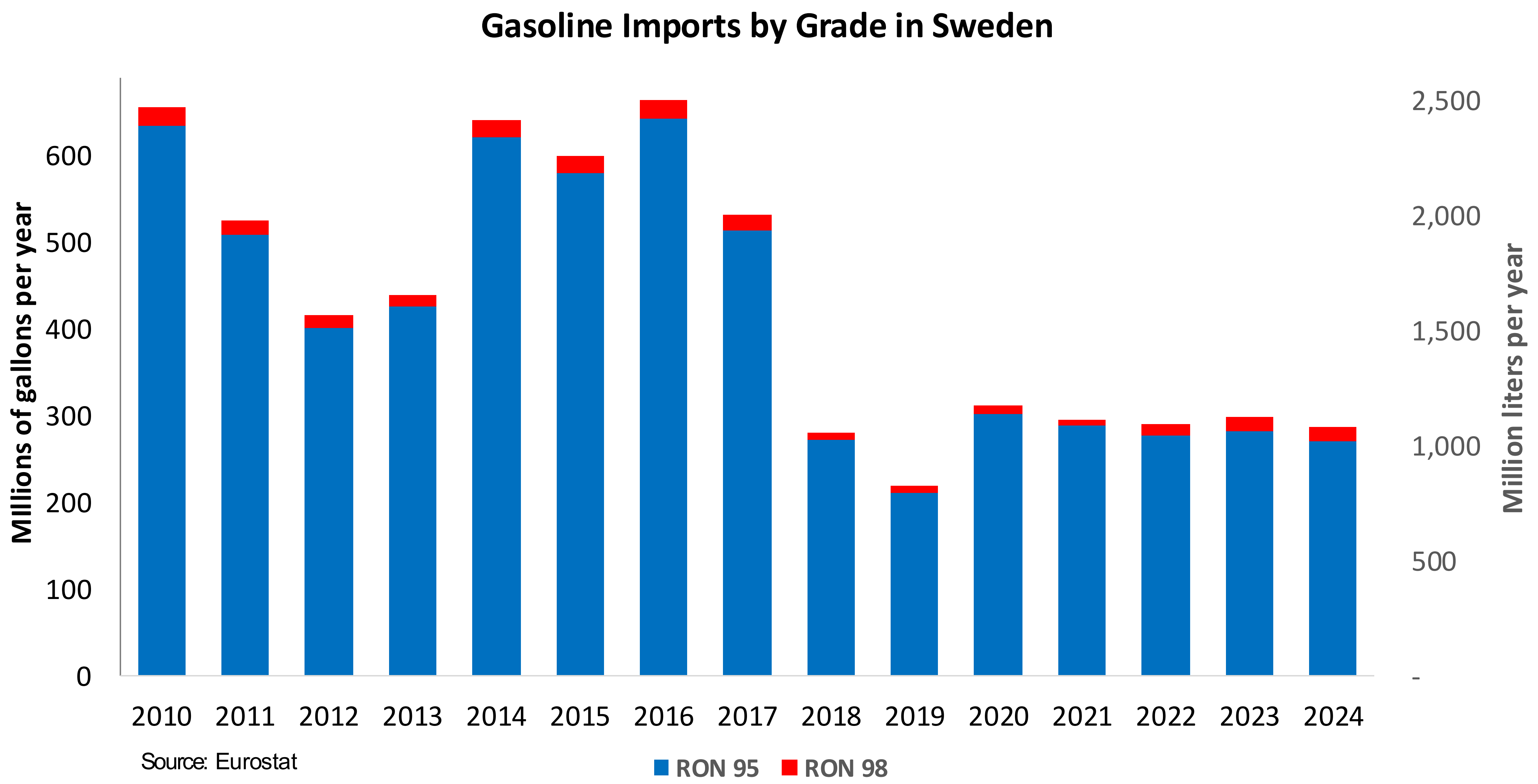
Source: Eurostat

Gasoline Production in Sweden

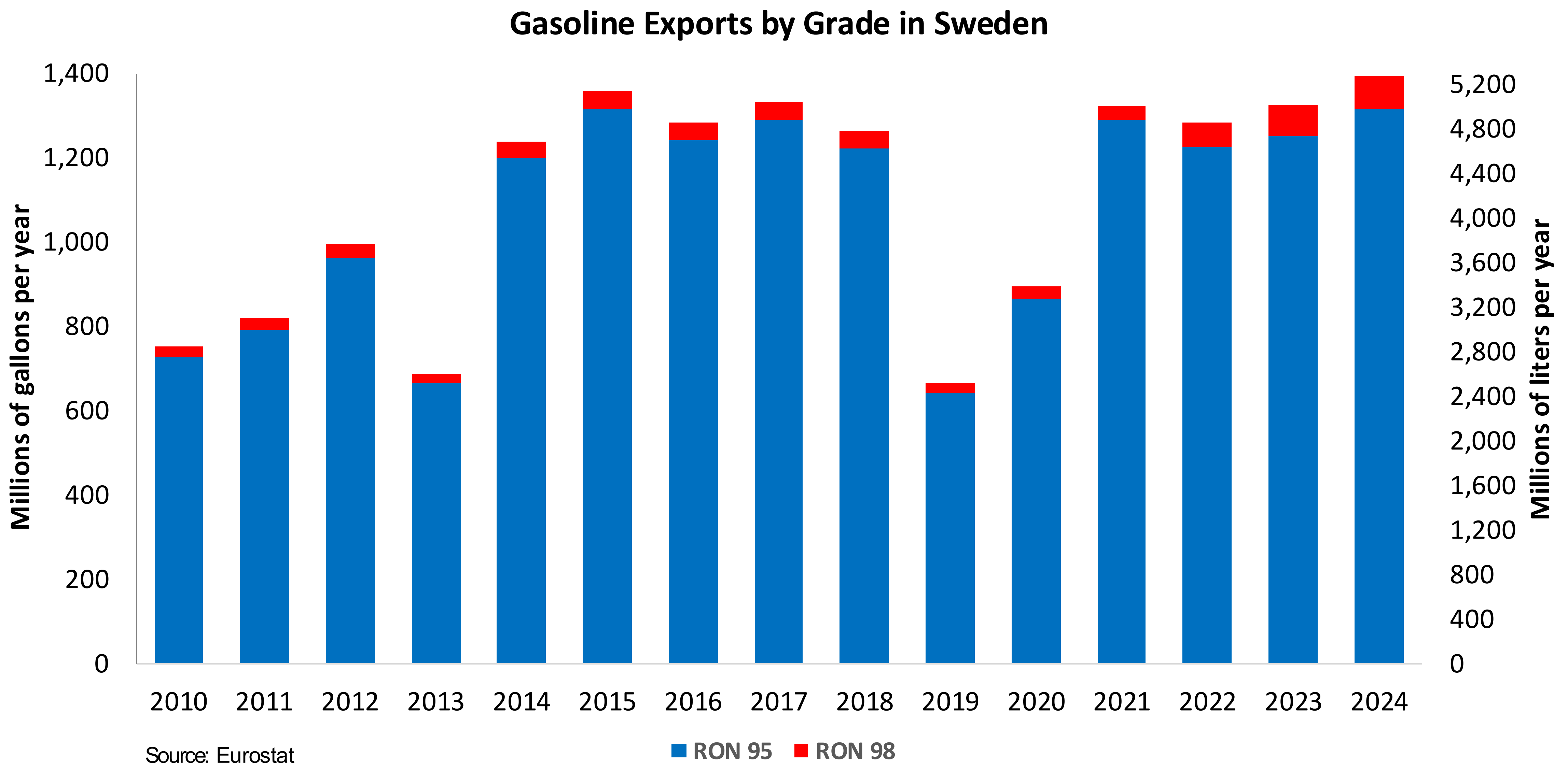


Source: Eurostat

Gasoline Imports in Sweden

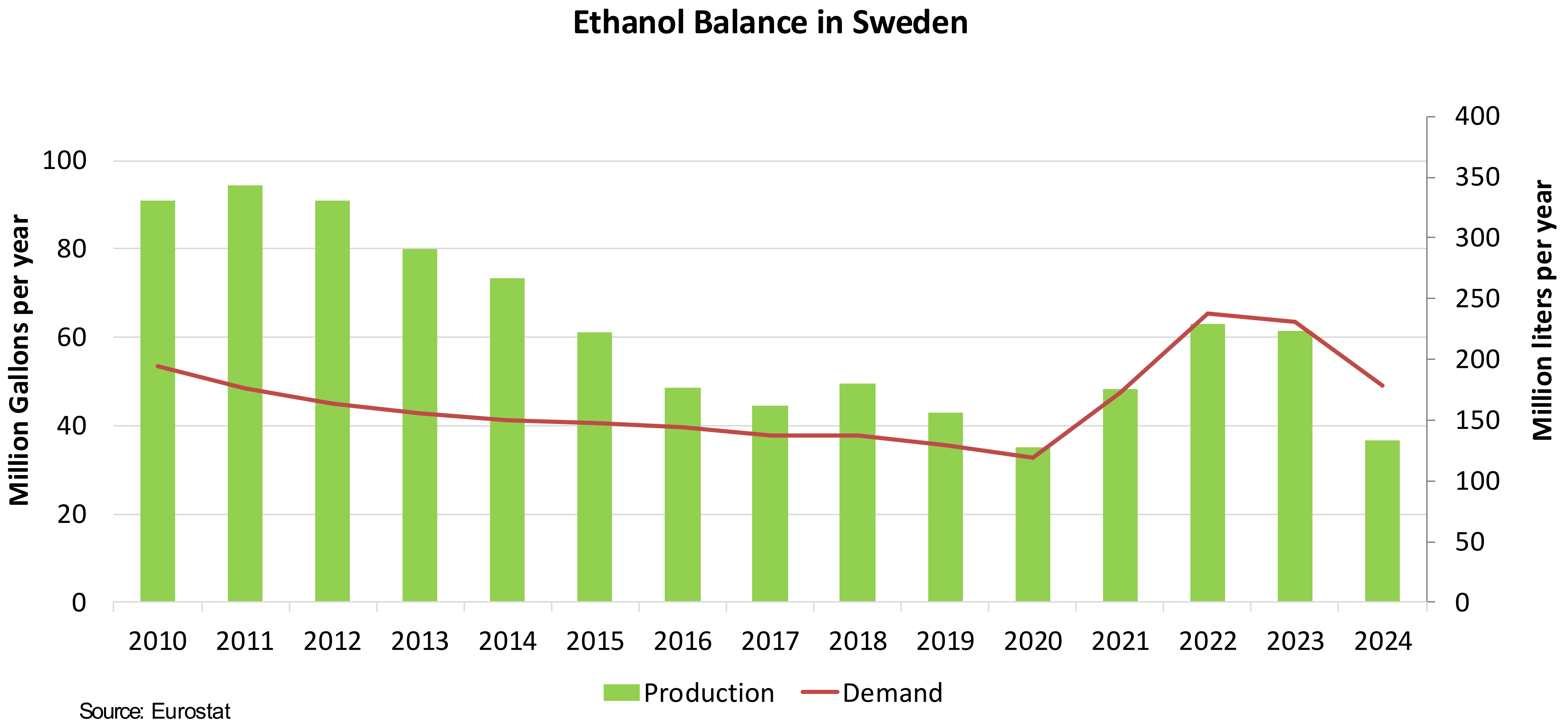


Gasoline Exports in Sweden



Source: Eurostat

Ethanol Balance in Sweden



Source: Eurostat

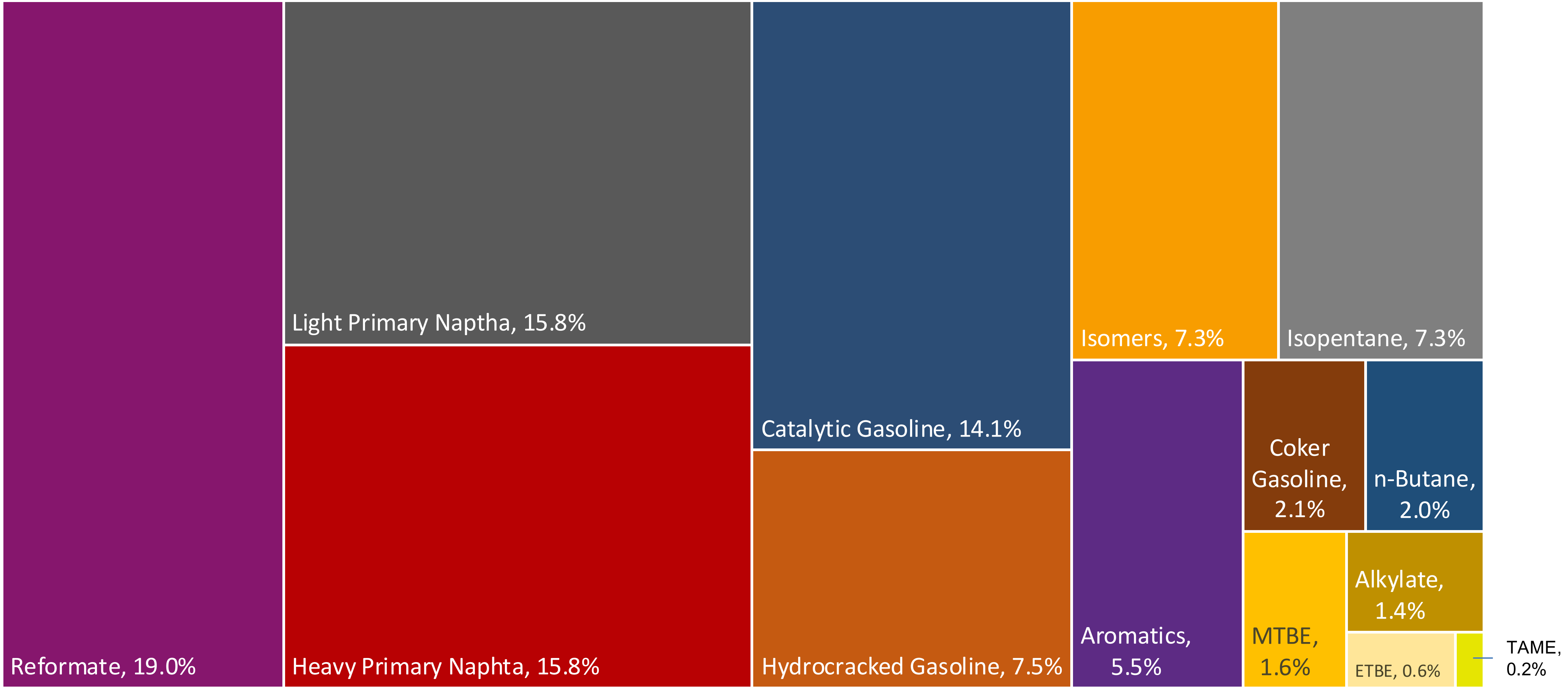
Gasoline Quality in Sweden

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

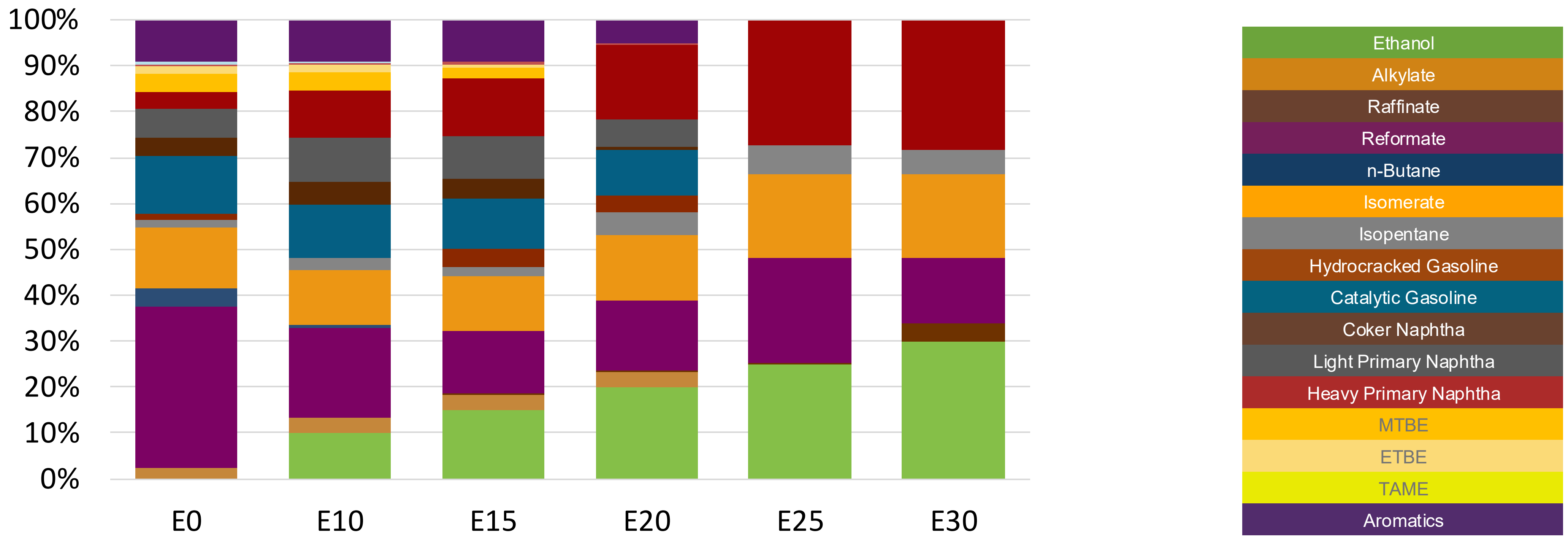
Source: European Commision

Sweden Available Blending Components



Source: Faro90

Sweden – Gasoline 95 – Constant Octane

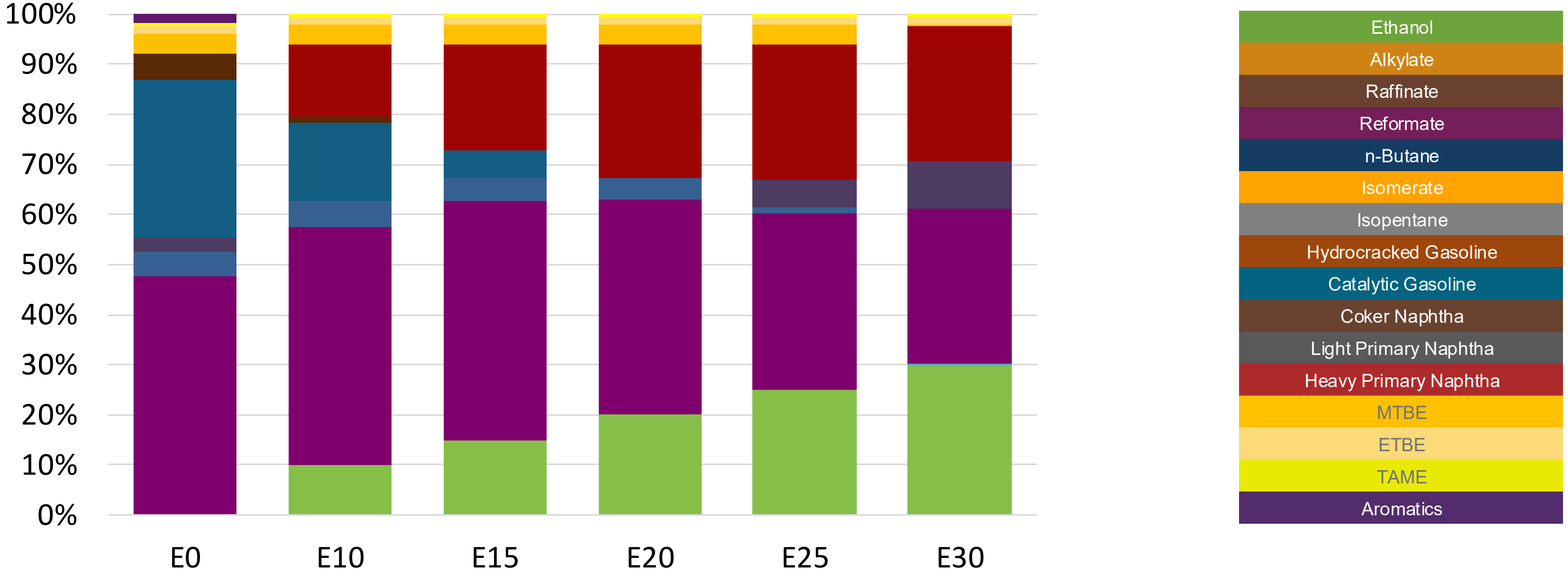


Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US$/gal)	$ 2.312	$ 2.184	$ 2.123	$ 2.042	$ 1.920	$ 1.854

Average CIF 2024 Prices

Source: Faro90

Sweden - Gasoline 98 - Constant Octane

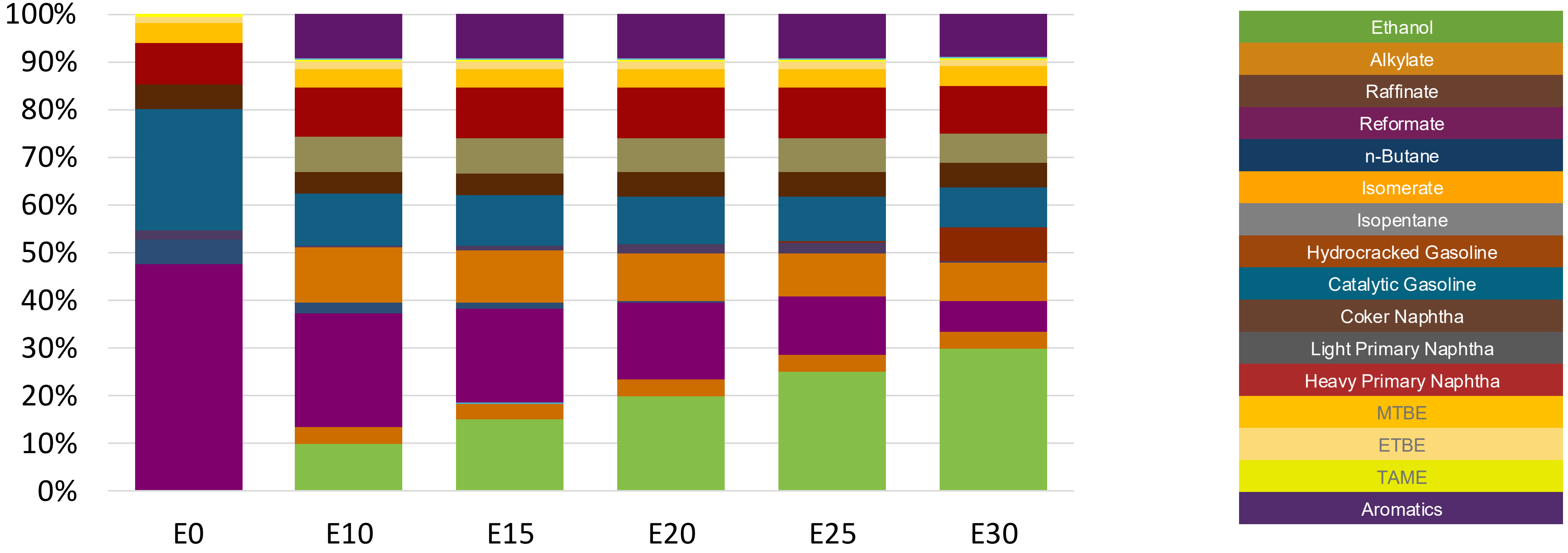


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.387	$ 2.258	$ 2.186	$ 2.109	$ 2.032	$ 1.965

Average CIF 2024 Prices

Source: Faro90

Sweden – Gasoline 95 – Increased Octane

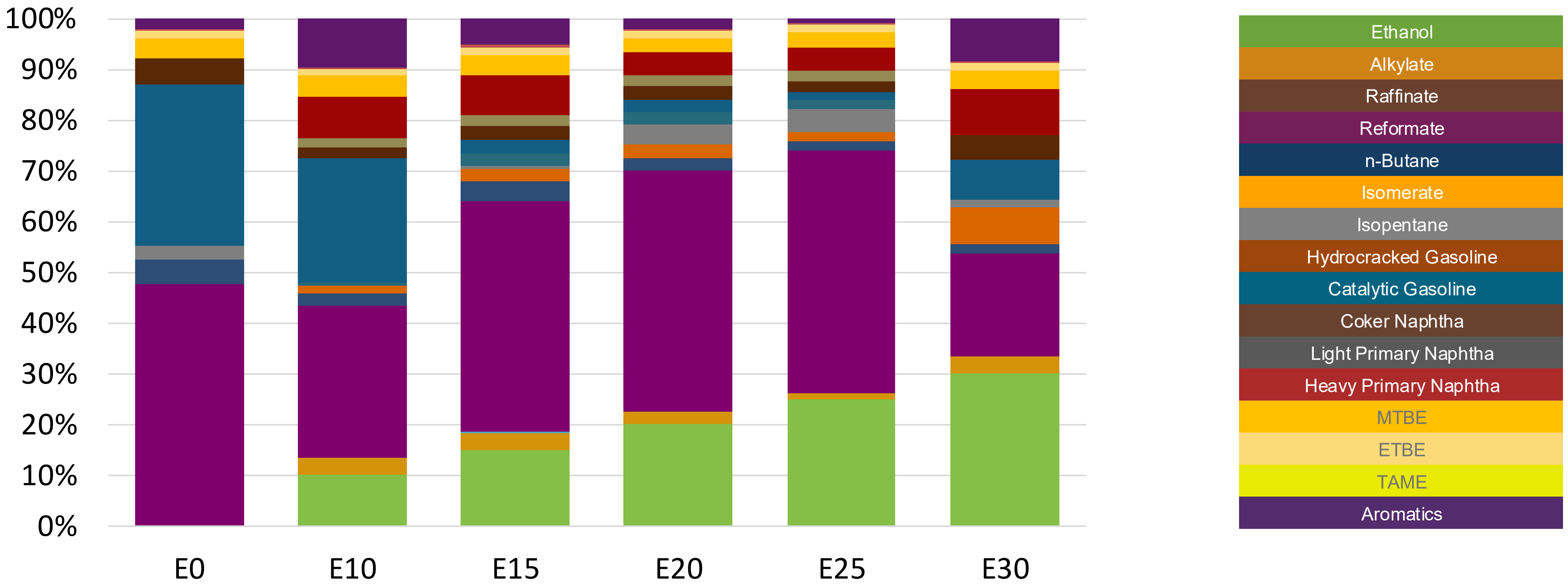


Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US$/gal)	$ 2.249	$ 2.249	$ 2.249	$ 2.249	$ 2.249	$ 2.249

Average CIF 2024 Prices

Source: Faro90

Sweden – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US$/gal)	$ 2.387	$ 2.387	$ 2.387	$ 2.387	$ 2.387	$ 2.387

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - Sweden

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	0
		Single FI	3-Ways		>8 years	98
		Multi FI	3-Ways		>8 years	18,243
			Euro III		>8 years	181,948
			Euro IV		>8 years	929,987
			Euro V		>8 years	329,428
			Euro 6		>8 years	827,899
					4-8 years	768,696
					<4 years	499,676
			Hybrid		All	339,754
		Electric			All	232,924
	Medium	Carburator	3-Ways	PCV Tank	>8 years	0
		Single FI	3-Ways		>8 years	53
		Multi FI	3-Ways		>8 years	9,823
			Euro III		>8 years	97,972
			Euro IV		>8 years	500,762
			Euro V		>8 years	177,384
			Euro 6		>8 years	445,792
					4-8 years	413,913
					<4 years	269,056
			Hybrid		All	182,944
		Electric			All	125,420
Sml Engines	Light	4-Cycle	None	PCV	>8 years	7,605
			Euro III		>8 years	75,445
			Euro IV		>8 years	317,335
			Euro V		>8 years	65,381
			Euro 6		>8 years	154,150
					4-8 years	148,589
					<4 years	94,621
			Hybrid		All	426
		Electric			All	17,244
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	51
		Fuel-Injection	3-Ways		>8 years	9,458
			Euro III		>8 years	94,334
			Euro IV		>8 years	476,616
			Euro V		>8 years	154,661
			Euro 6		>8 years	398,869
					4-8 years	425,972
					<4 years	261,164
			Hybrid		All	918
		Electric			All	44,201

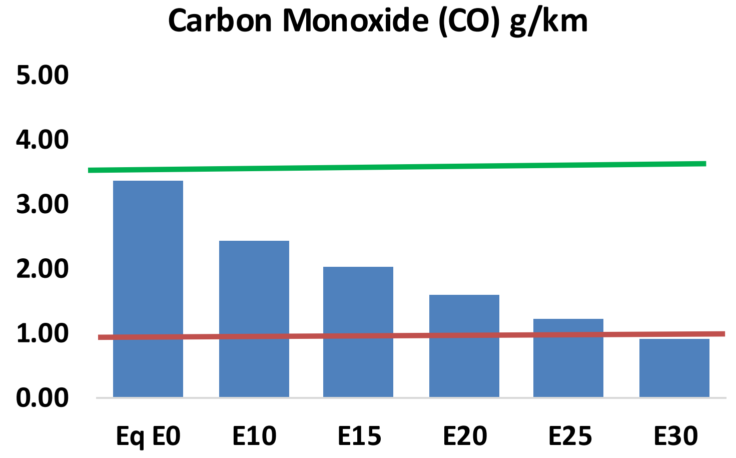
0 200,000 400,000 600,000 800,000 1,000,000

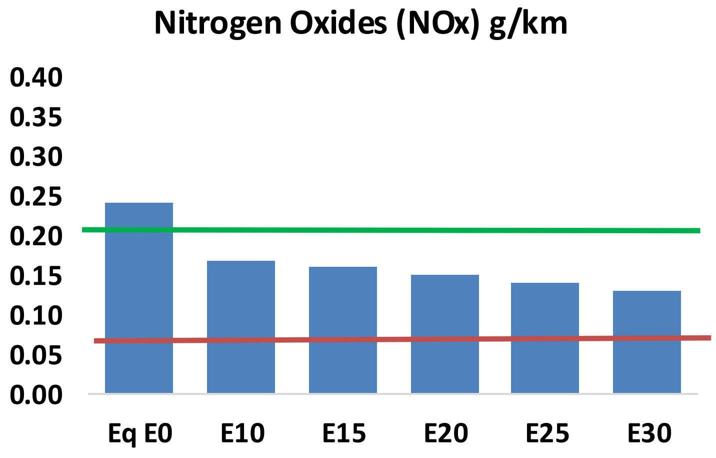
Vehicle Fleet: 9,098,811
Gasoline Vehicle Fleet: 3,553,018

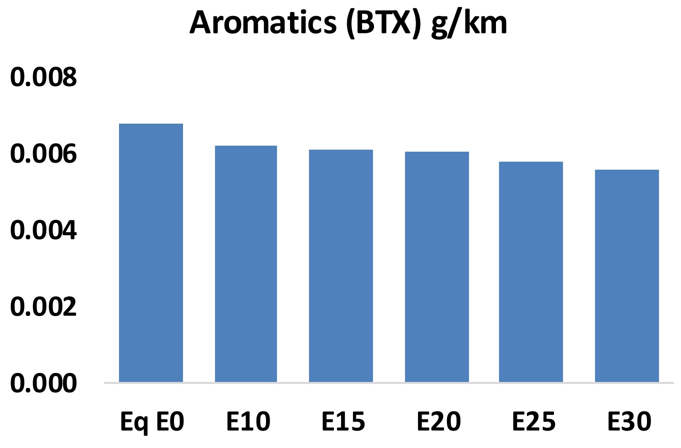
Source: Eurostat, ACEA

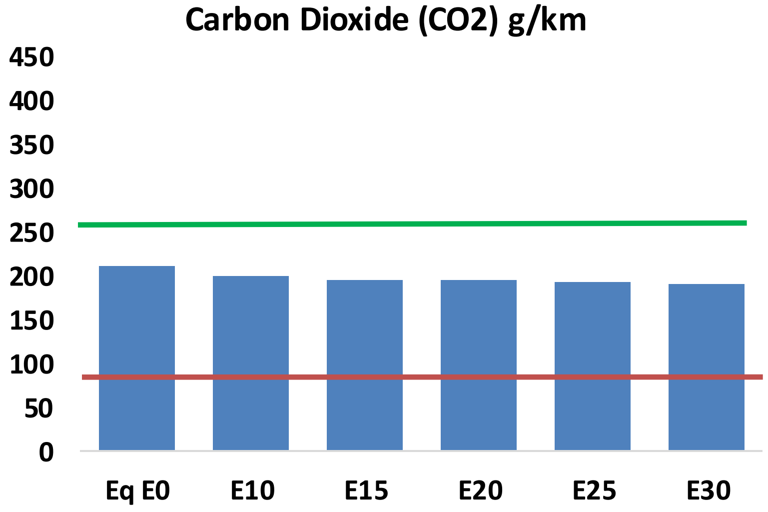
Sweden – Vehicular Emissions

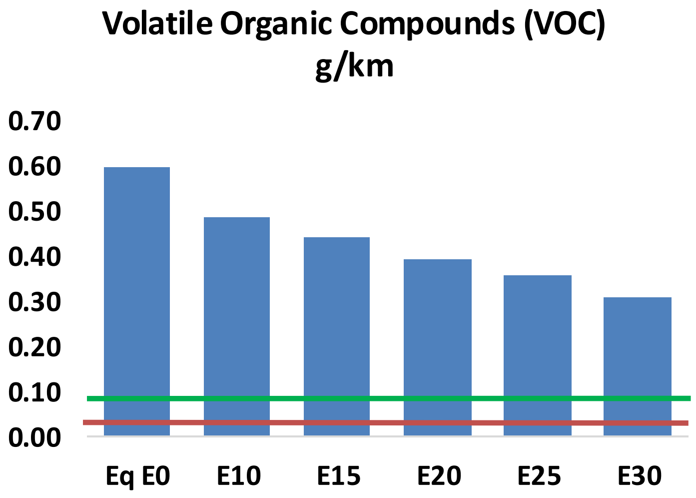
TIER III BIN 125 United States
Euro 6 Standard

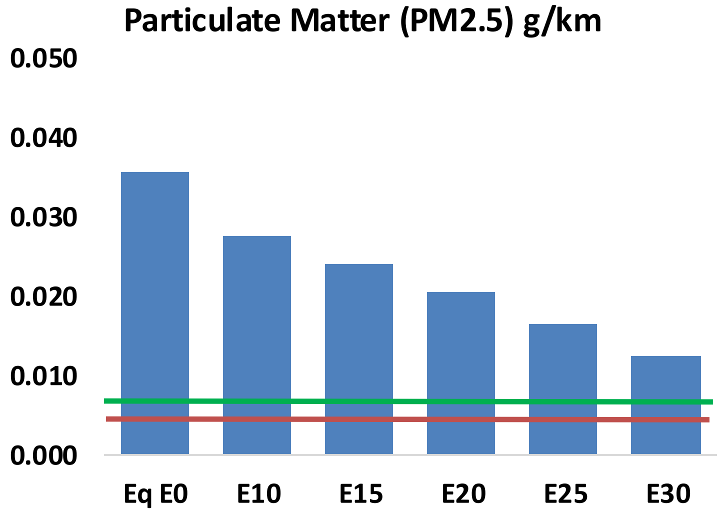












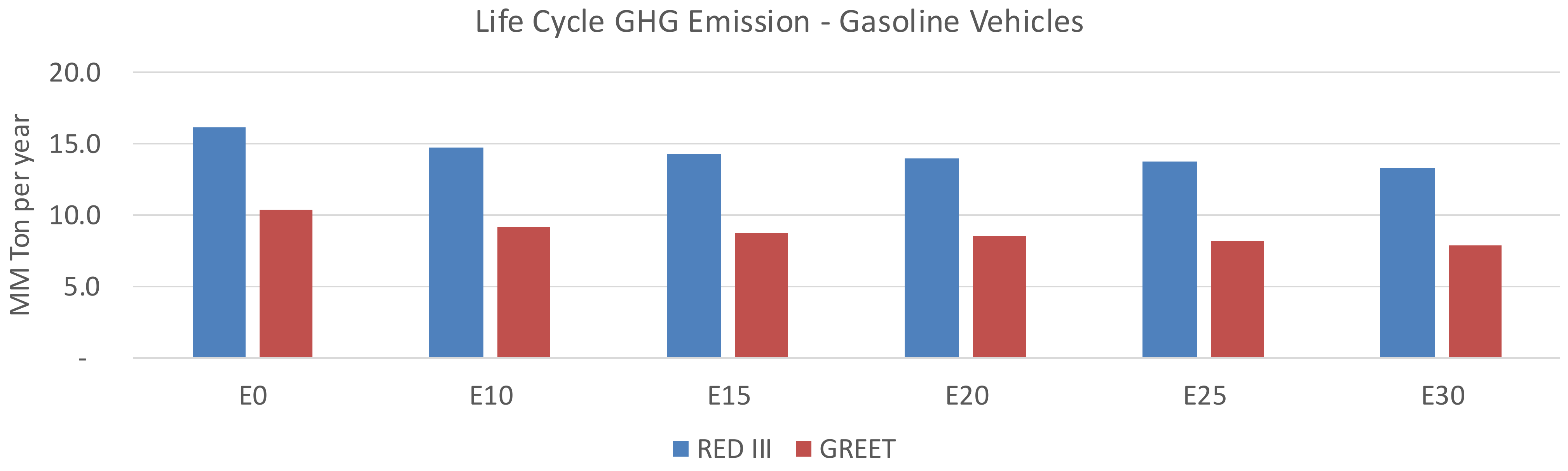
Source: Faro90

Sweden – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	294,225	253,630	213,035	176,261	137,640	107,455	77,868	-28%	-53%
CO2	18,387,085	17,914,274	17,467,730	17,116,537	16,942,753	16,846,077	16,535,555	-5%	-8%
VOC	51,886	47,146	42,405	38,374	34,251	31,059	26,872	-18%	-34%
NOx	21,156	15,867	14,809	13,958	13,163	12,288	11,361	-30%	-38%
SOx	212	196	180	166	153	139	124	-15%	-28%
NH3	6,619	6,553	6,488	6,519	6,592	6,644	6,686	-2%	0%
N2O	309	308	306	308	314	318	321	-1%	2%
CH4	10,736	10,736	10,736	10,897	11,133	11,337	11,530	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	261	242	224	210	195	184	169	-14%	-25%
Acetaldehyde	795	913	1,339	2,039	2,778	3,175	3,751	68%	249%
Formaldehyde	3,016	2,932	3,418	4,013	4,204	4,651	5,063	13%	39%
Benzene	594	570	541	532	528	505	488	-9%	-11%
PM2.5	1,127	1,001	874	759	645	523	397	-22%	-43%
PM10	1,979	1,757	1,535	1,333	1,132	919	696	-22%	-43%

Source: Faro90

Sweden – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	35.9
Target @2035 (MMT/year)	20.2
Delta	15.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.2%	15.5%	18.2%	20.8%	24.2%
RED II/III	0.0%	8.5%	10.9%	12.9%	14.8%	17.1%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.1%	10.3%	12.1%	13.7%	16.0%
RED II/III	0.0%	8.7%	11.2%	13.2%	15.1%	17.6%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

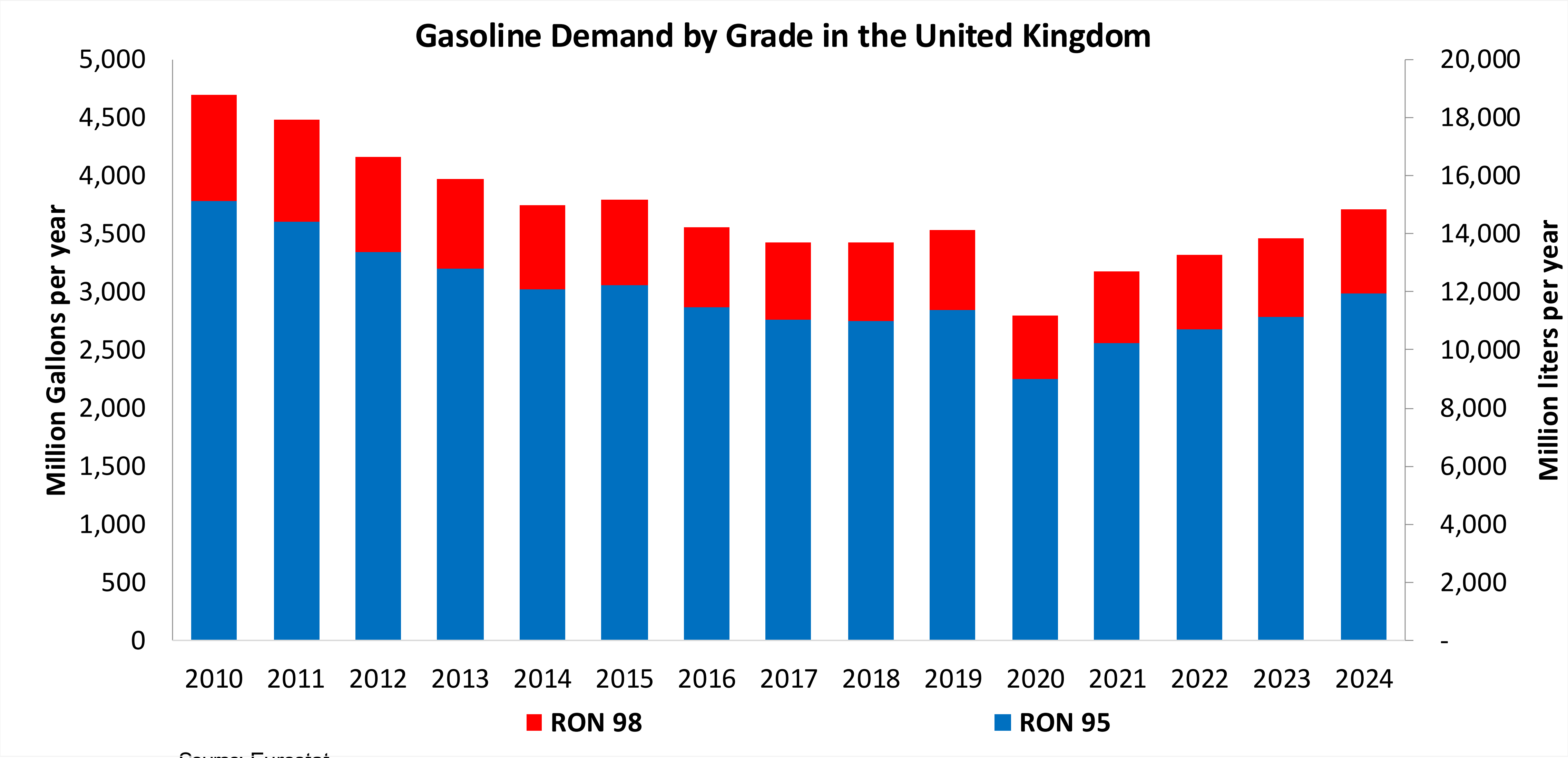
Ethanol Gasoline Blending in United Kingdom

In 2024, gasoline consumption in United Kingdom was 14,052 million liters (3,712 million gallons) and growth rate was 1.0%. Gasoline production and net imports were 19,997 and -5,945 million liters (5,283 and -1,571 million gallons) correspondingly. United Kingdom complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.5 RON.

In 2024, ethanol consumption in United Kingdom was 1,105 million liters (292 million gallons) representing 7.9% of gasoline demand. Ethanol production and Net Imports were 177 and 929 million liters (292 and 245 million gallons) correspondingly. Ethanol sources are Corn 84%, Cereals 13% and Sugar Beet 3%.

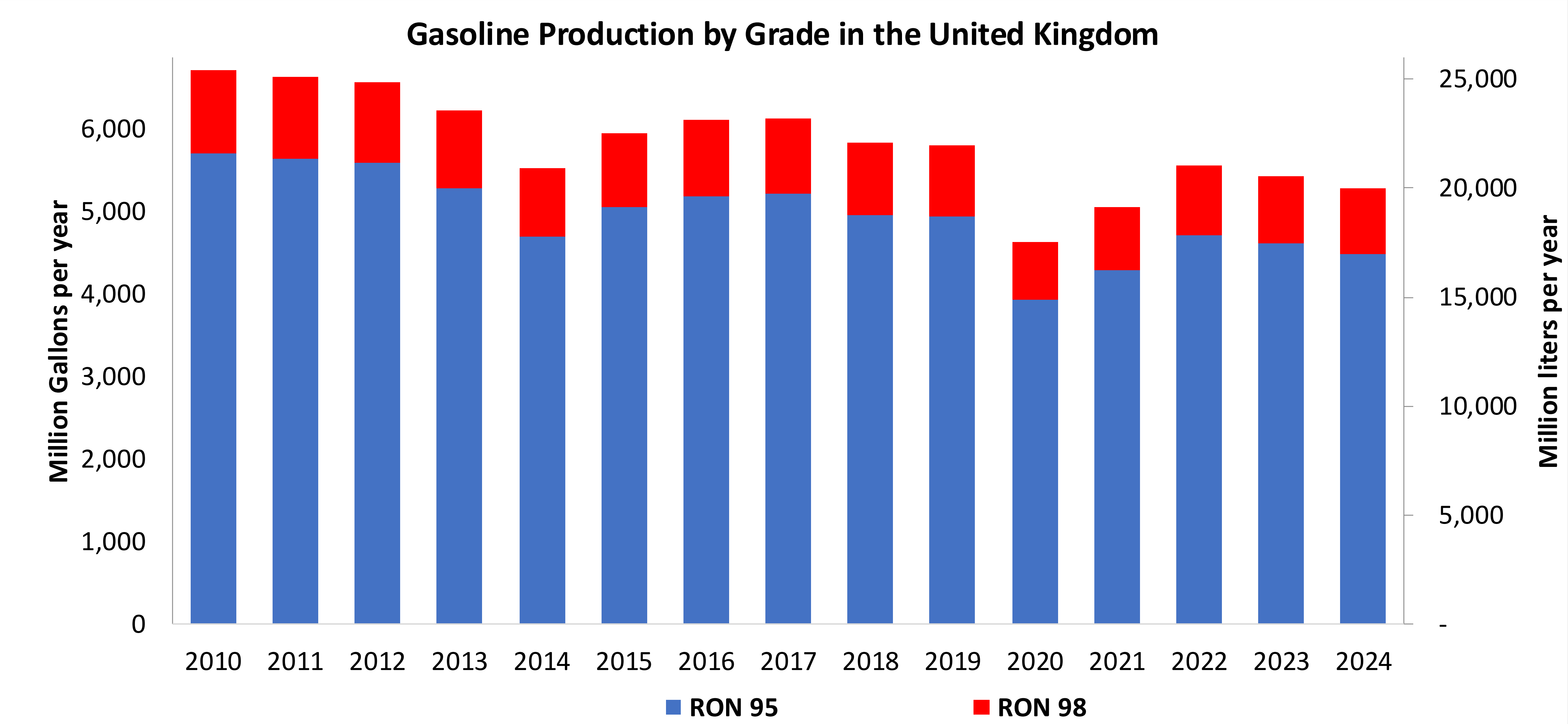
Source: Eurostat, European Commision

Gasoline Demand in United Kingdom



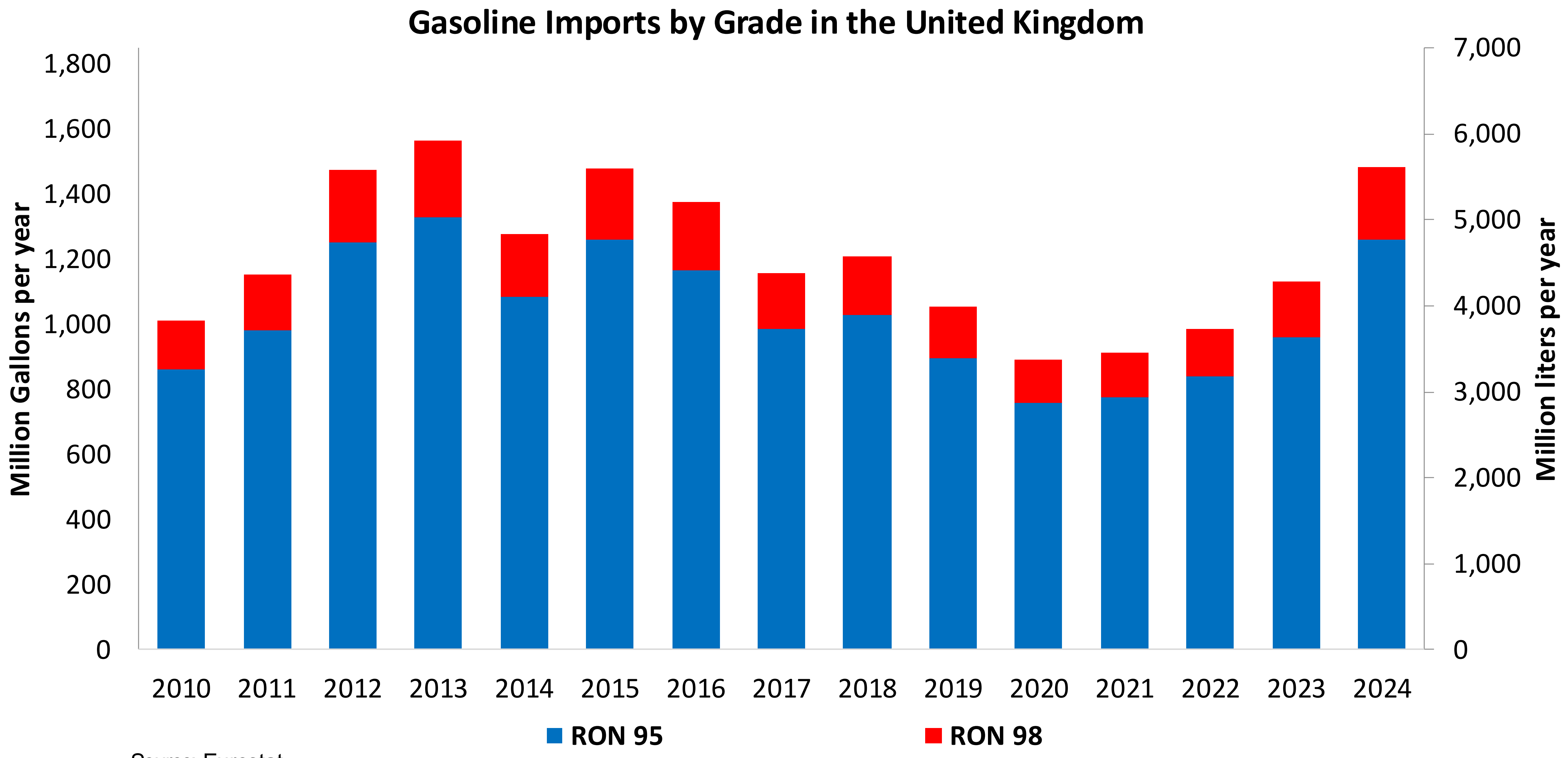
Source: Eurostat

Gasoline Production in United Kingdom



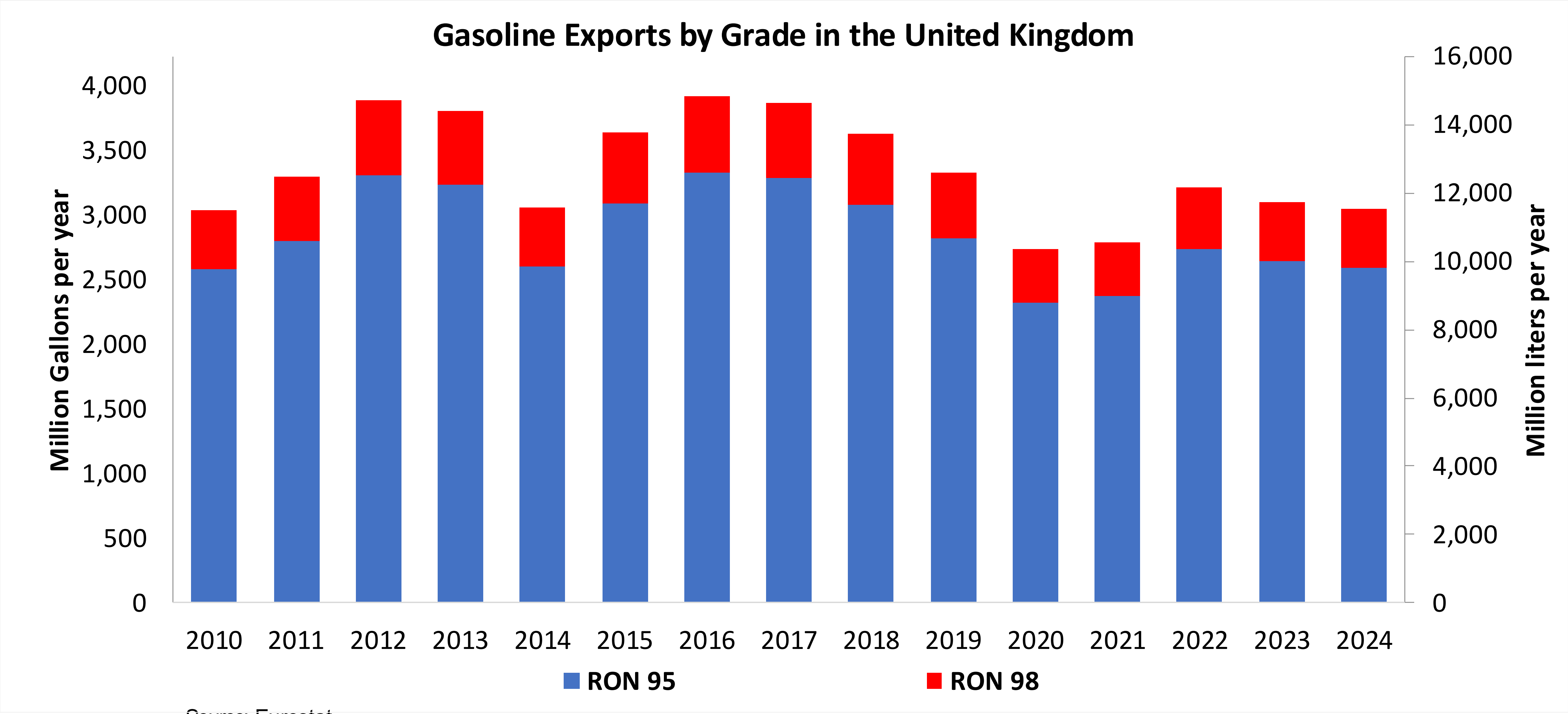
Source: Eurostat

Gasoline Imports in United Kingdom



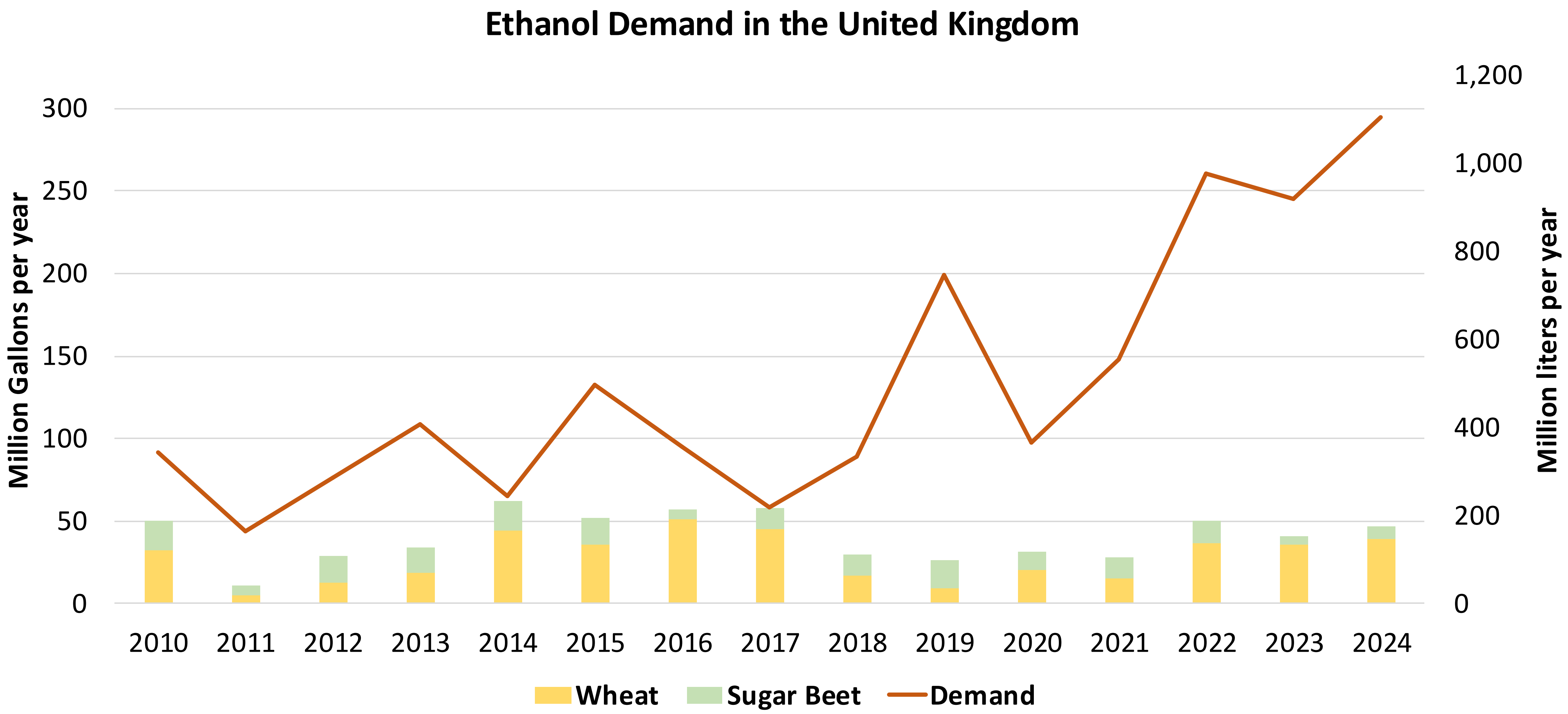
Source: Eurostat

Gasoline Exports in United Kingdom



Source: Eurostat

Ethanol Balance in United Kingdom



Source: Eurostat

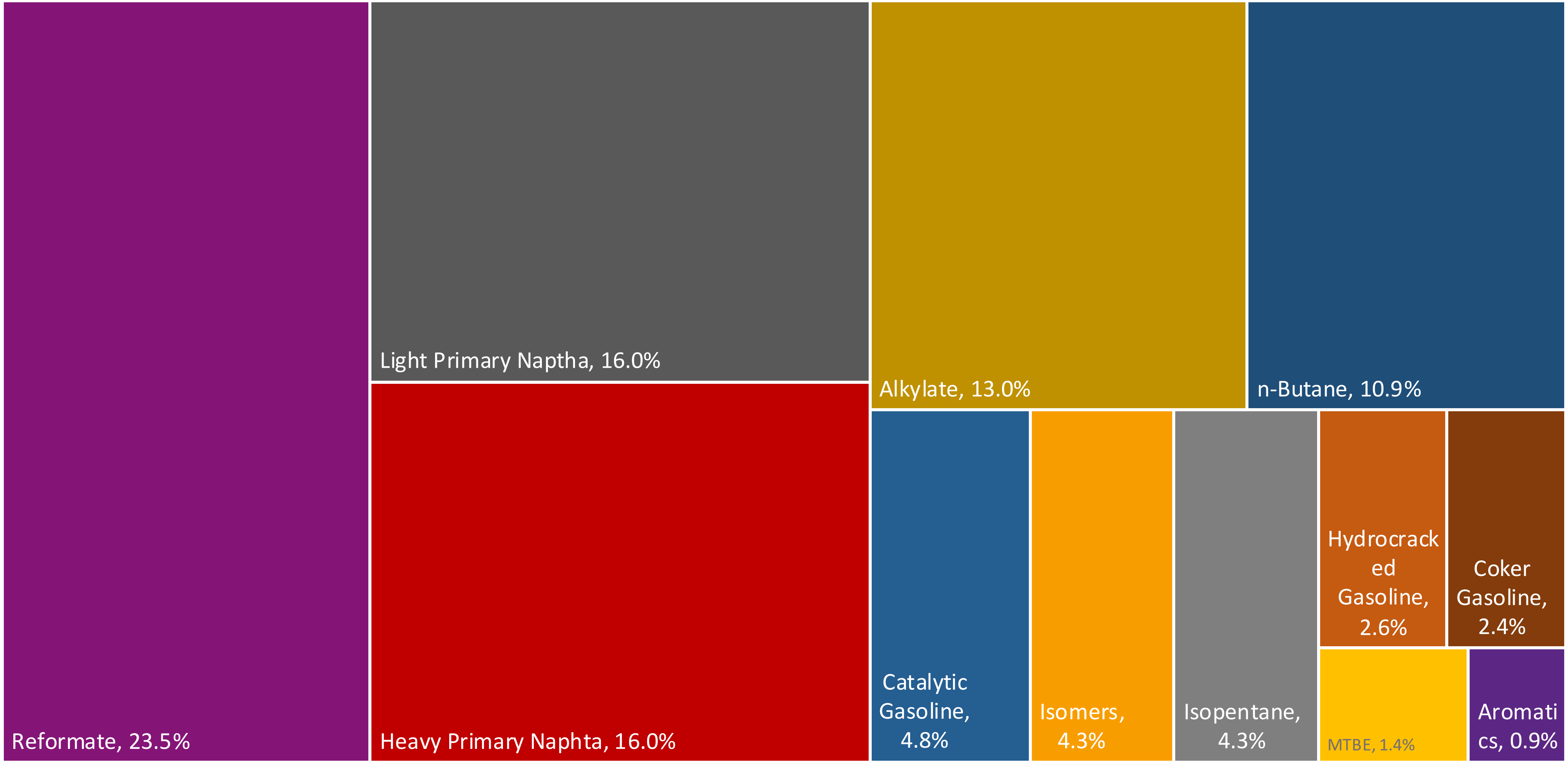
Gasoline Quality in United Kingdom

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

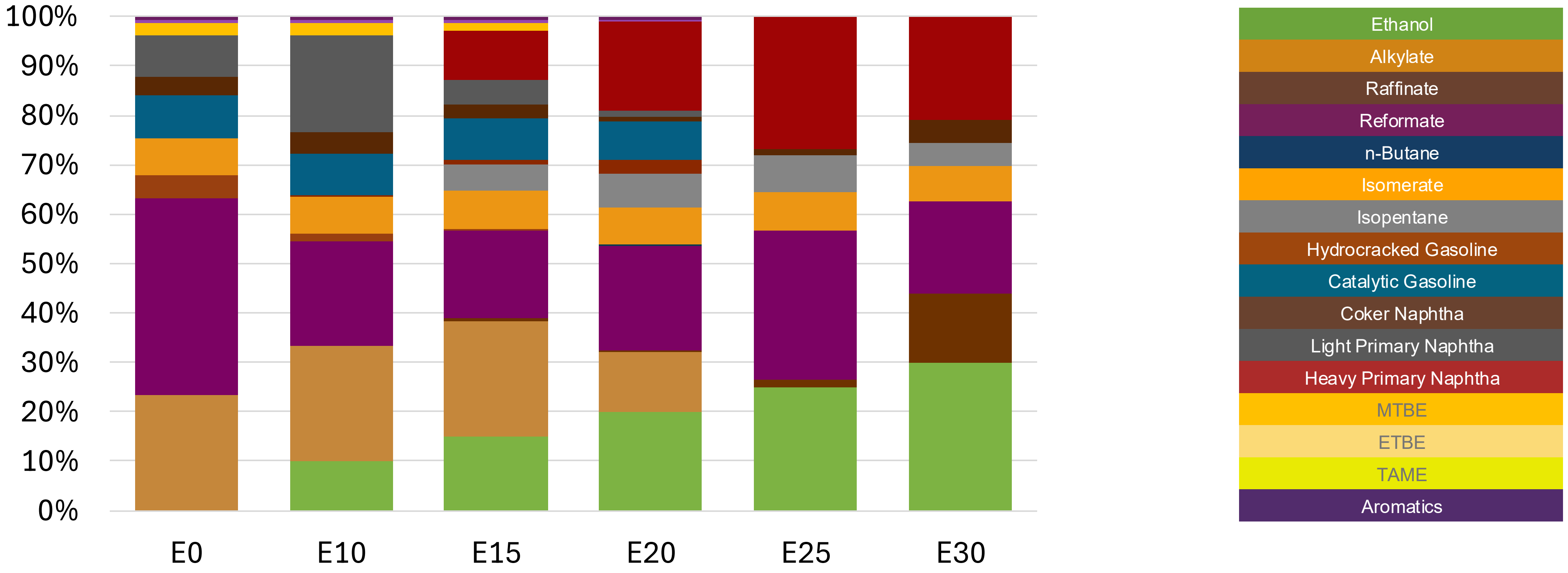
Source: European Commision

United Kingdom Available Blending Components



Source: Faro90

United Kingdom – Gasoline 95 – Constant Octane

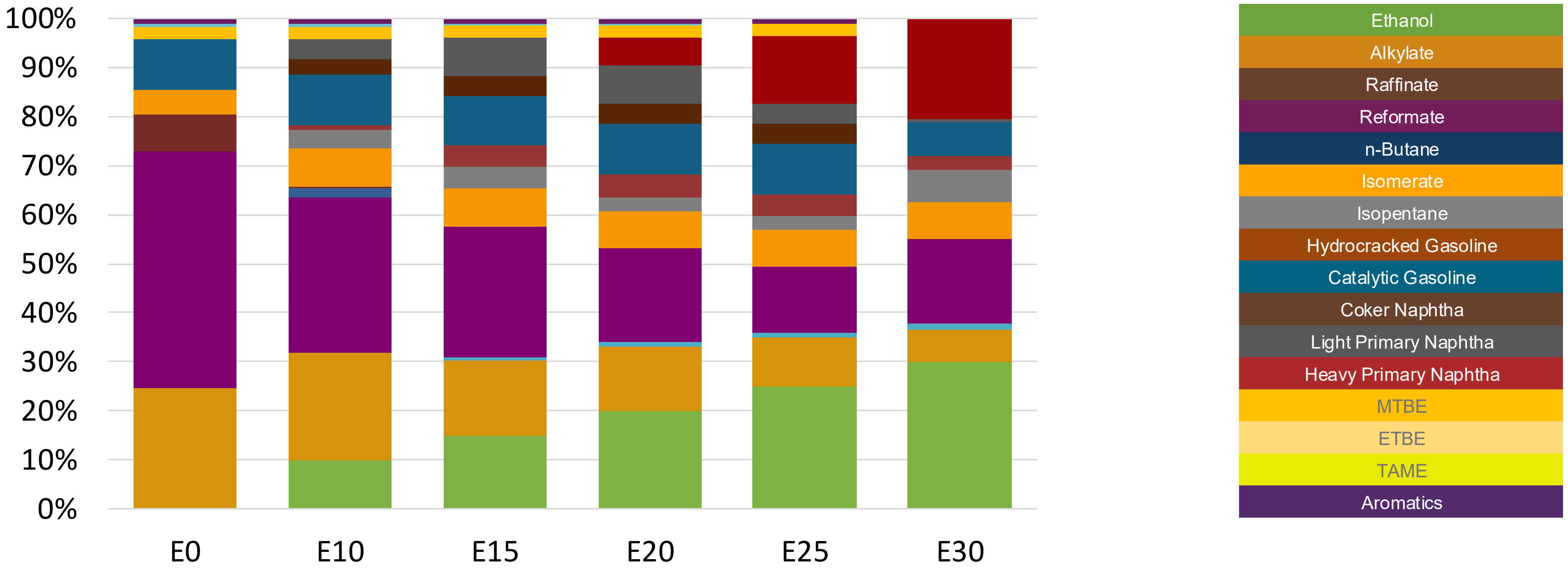


Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US$/gal)	$ 2.317	$ 2.187	$ 2.105	$ 1.996	$ 1.886	$ 1.821

Average CIF 2024 Prices

Source: Faro90

United Kingdom - Gasoline 98 - Constant Octane

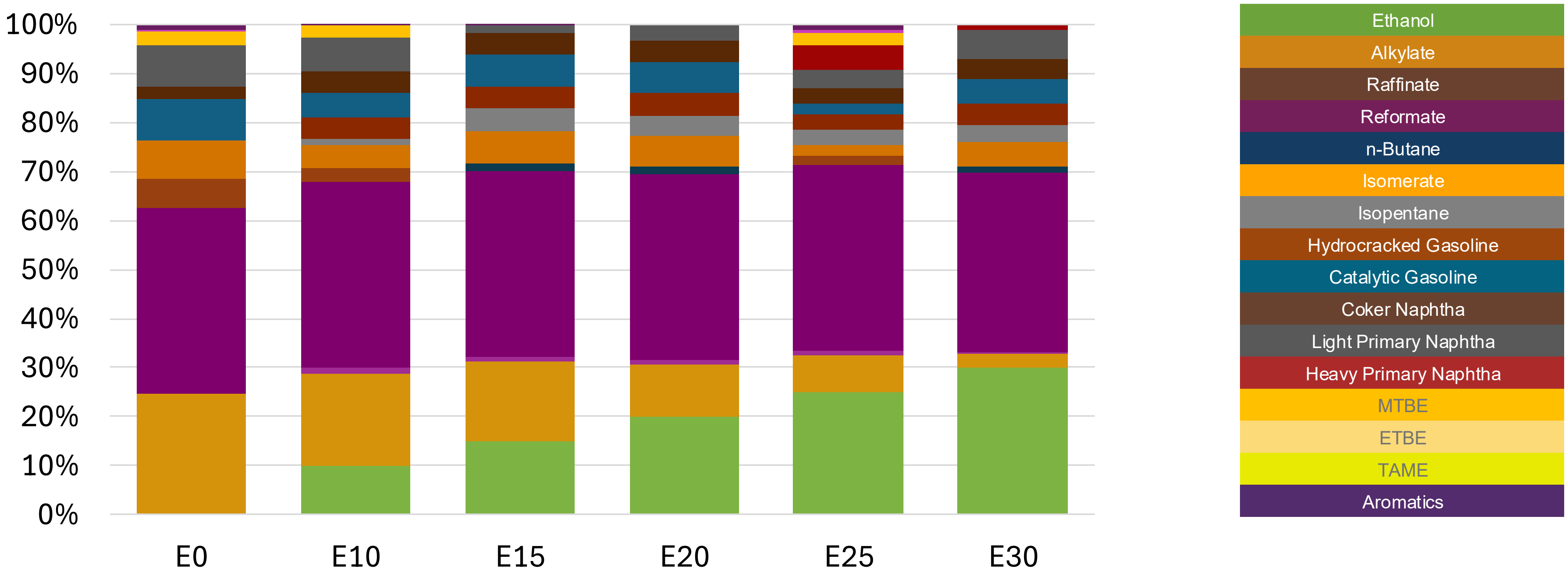


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.440	$ 2.317	$ 2.240	$ 2.157	$ 2.062	$ 1.967

Average CIF 2024 Prices

Source: Faro90

United Kingdom – Gasoline 95 – Increased Octane

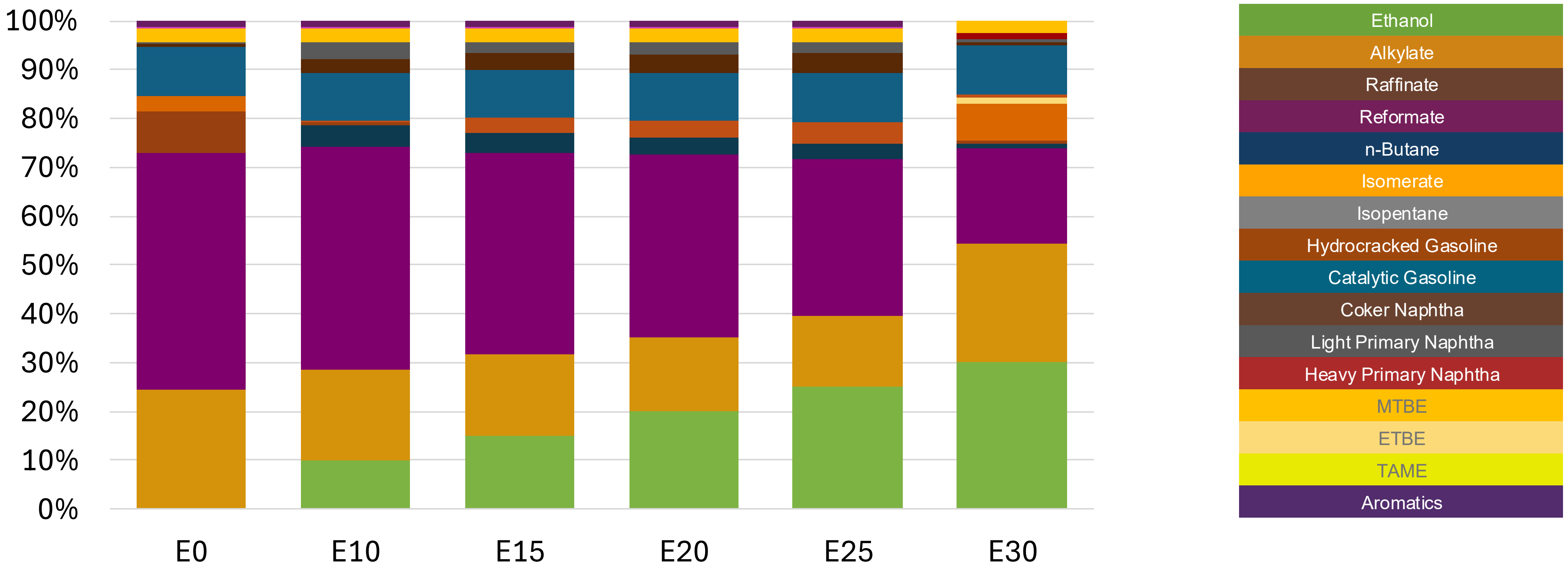


Octane (RON)	95.0	97.9	99.3	100.9	102.6	103.9
Price (US$/gal)	$ 2.301	$ 2.301	$ 2.301	$ 2.301	$ 2.301	$ 2.301

Average CIF 2024 Prices

Source: Faro90

United Kingdom – Gasoline 98 – Increased Octane



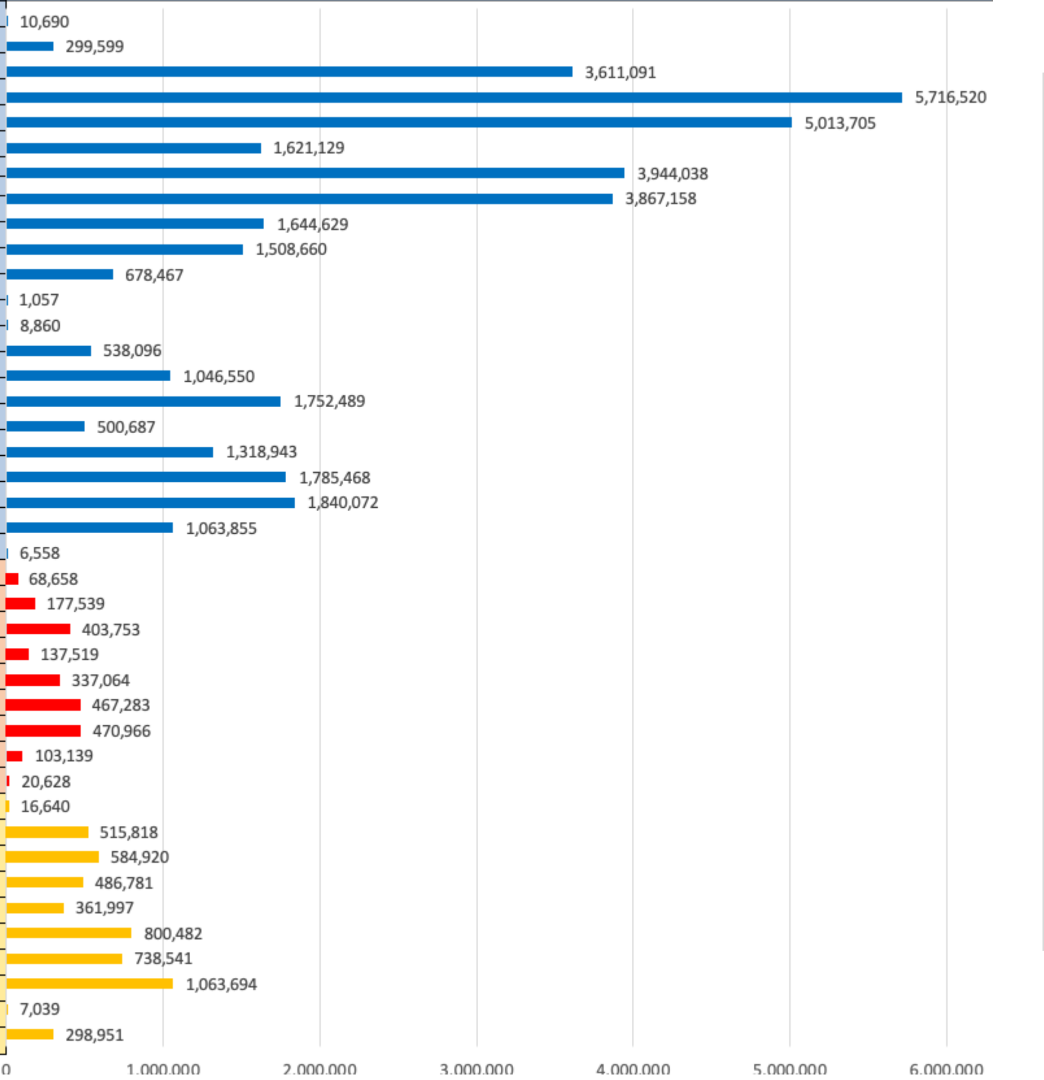
Octane (RON)	98.0	100.7	102.0	103.3	104.5	105.0
Price (US$/gal)	$ 2.425	$ 2.425	$ 2.425	$ 2.425	$ 2.425	$ 2.425

Average CIF 2024 Prices

Source: Faro90

Gasoline Vehicle Fleet - United Kingdom

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	10,690
		Single FI	3-Ways		>8 years	299,599
		Multi FI	3-Ways		>8 years	3,611,091
			Euro III		>8 years	5,716,520
			Euro IV		>8 years	5,013,705
			Euro V		>8 years	1,621,129
			Euro 6		>8 years	3,944,038
					4-8 years	3,867,158
					<4 years	1,644,629
			Hybrid		All	1,508,660
		Electric			All	678,467
	Medium	Carburator	3-Ways	PCV Tank	>8 years	1,057
		Single FI	3-Ways		>8 years	8,860
		Multi FI	3-Ways		>8 years	538,096
			Euro III		>8 years	1,046,550
			Euro IV		>8 years	1,752,489
			Euro V		>8 years	500,687
			Euro 6		>8 years	1,318,943
					4-8 years	1,785,468
					<4 years	1,840,072
			Hybrid		All	1,063,855
		Electric			All	6,558
Sml Engines	Light	4-Cycle	None	PCV	>8 years	68,658
			Euro III		>8 years	177,539
			Euro IV		>8 years	403,753
			Euro V		>8 years	137,519
			Euro 6		>8 years	337,064
					4-8 years	467,283
					<4 years	470,966
			Hybrid		All	103,139
		Electric			All	20,628
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	16,640
		Fuel-Injection	3-Ways		>8 years	515,818
			Euro III		>8 years	584,920
			Euro IV		>8 years	486,781
			Euro V		>8 years	361,997
			Euro 6		>8 years	800,482
					4-8 years	738,541
					<4 years	1,063,694
			Hybrid		All	7,039
		Electric			All	298,951

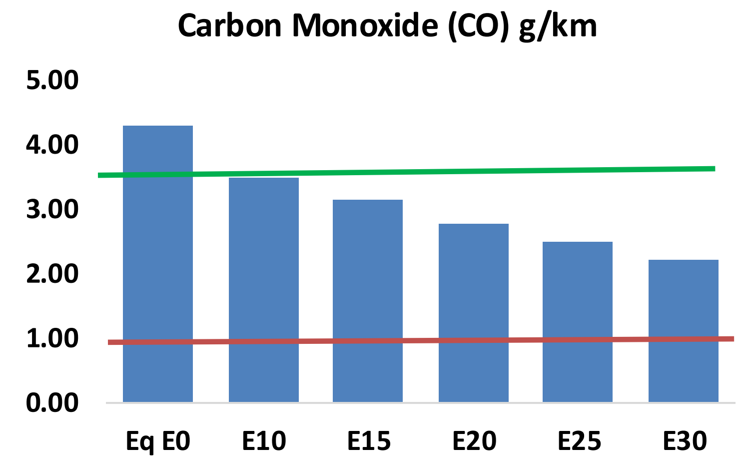


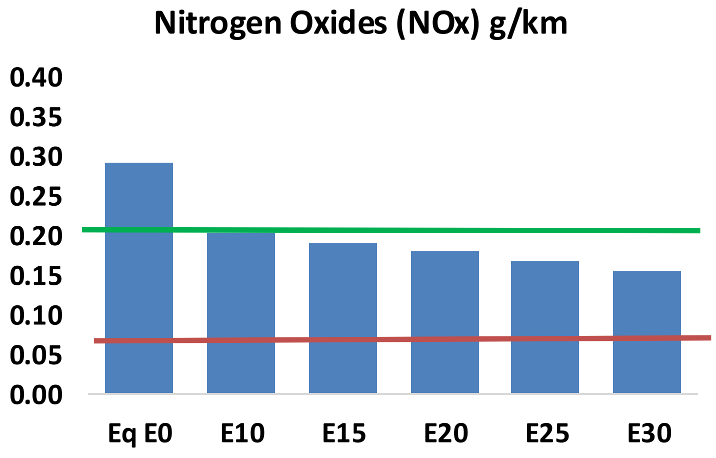
Vehicle Fleet: 44,839,734
Gasoline Vehicle Fleet: 22,834,680

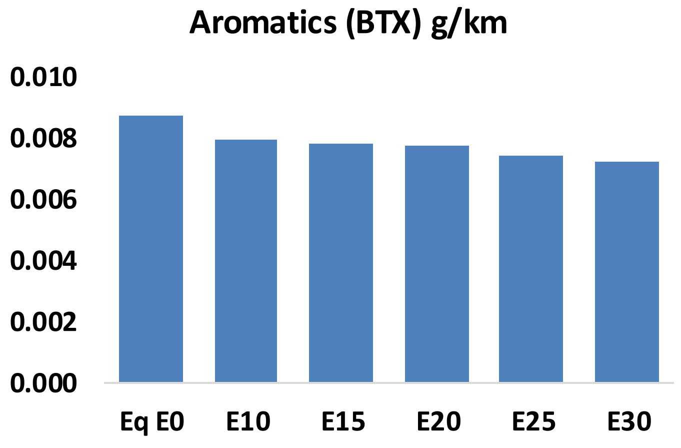
Source: Eurostat, ACEA

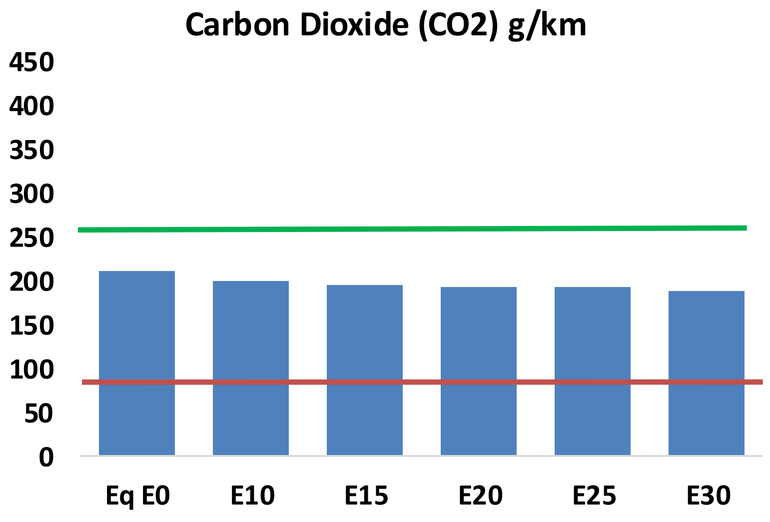
United Kingdom – Vehicular Emissions

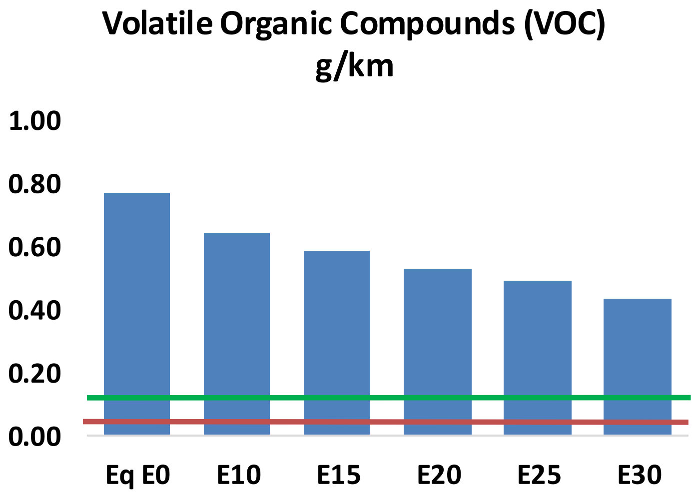
TIER III BIN 125 United States
Euro 6 Standard

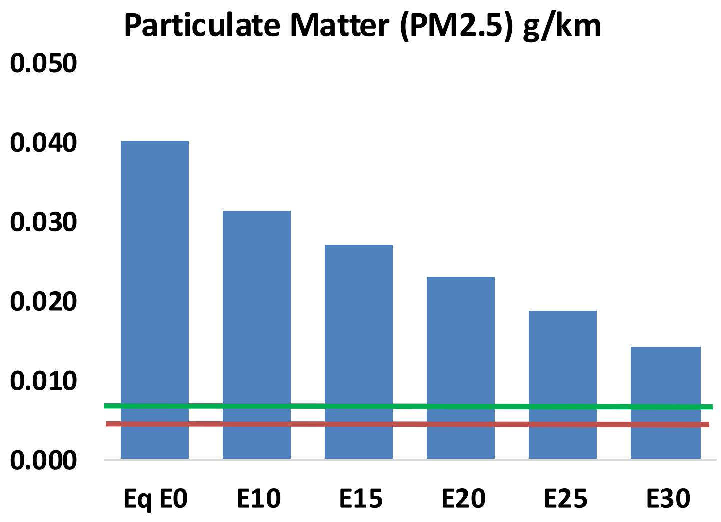












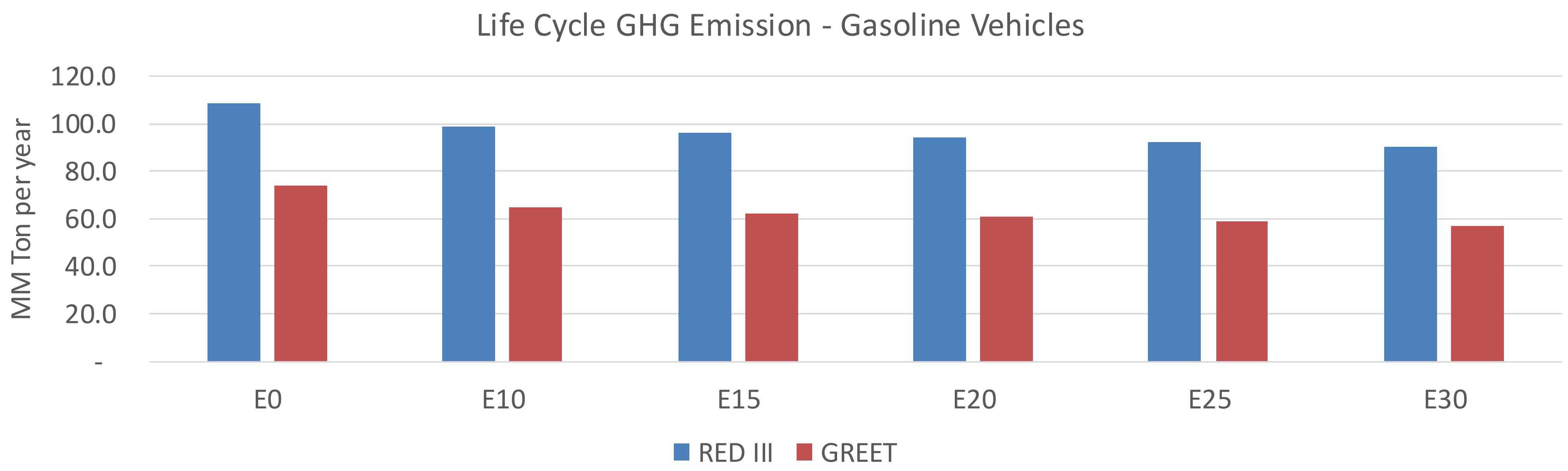
Source: Faro90

United Kingdom – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,592,464	2,347,600	2,102,737	1,893,191	1,678,226	1,511,697	1,337,359	-19%	-35%
CO2	126,780,565	123,520,493	120,441,537	118,020,028	116,821,771	116,113,706	113,973,396	-5%	-8%
VOC	464,487	425,388	386,288	353,587	320,373	294,742	260,863	-17%	-31%
NOx	176,401	132,301	123,481	116,381	109,757	102,454	94,728	-30%	-38%
SOx	1,449	1,339	1,230	1,137	1,048	952	850	-15%	-28%
NH3	46,556	46,095	45,634	45,851	46,366	46,729	47,031	-2%	0%
N2O	2,090	2,079	2,070	2,080	2,123	2,147	2,171	-1%	2%
CH4	93,659	93,659	93,659	95,064	97,125	98,904	100,590	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,440	2,325	2,211	2,131	2,058	2,003	1,924	-9%	-16%
Acetaldehyde	7,685	8,820	12,942	19,708	26,851	30,682	36,254	68%	249%
Formaldehyde	28,374	27,586	32,157	37,760	39,559	43,763	47,641	13%	39%
Benzene	5,291	5,084	4,818	4,741	4,702	4,497	4,352	-9%	-11%
PM2.5	7,836	6,958	6,080	5,280	4,483	3,638	2,757	-22%	-43%
PM10	16,450	14,606	12,763	11,083	9,411	7,637	5,787	-22%	-43%

Source: Faro90

United Kingdom – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	402.0
Target @2035 (MMT/year)	264.3
Delta	137.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.5%	16.5%	18.0%	20.3%	23.4%
RED II/III	0.0%	9.0%	11.3%	13.1%	14.9%	17.1%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.7%	8.9%	9.7%	10.9%	12.6%
RED II/III	0.0%	7.1%	8.9%	10.4%	11.7%	13.5%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90